Online 24th June Morning Shift

Physics

Q. 1 The bulk modulus of a liquid is $3 \times 10^{10}\ \mathrm{Nm}^{-2}$. The pressure required to reduce the volume of liquid by 2% is :

Option 1:
$3 \times 10^{8}\ \mathrm{Nm}^{-2}$

Option 2:
$9 \times 10^{8}\ \mathrm{Nm}^{-2}$

Option 3:
$6 \times 10^{8}\ \mathrm{Nm}^{-2}$

Option 4:
$12 \times 10^{8}\ \mathrm{Nm}^{-2}$

Correct Answer:
$6 \times 10^{8}\ \mathrm{Nm}^{-2}$

Solution:

$$\mathrm{B} = 3 \times 10^{10}\ \mathrm{N/m^2}$$

$$\left(\frac{\Delta \mathrm{v}}{\mathrm{v}}\right) = -0 \cdot 02\,(2\ \%\ \text{reduced})$$

We know that ,

$$\mathrm{B} = \frac{\mathrm{p}}{\frac{-\Delta \mathrm{v}}{\mathrm{v}}}$$

$$\mathrm{p} = \mathrm{B} \times \frac{-\Delta \mathrm{v}}{\mathrm{v}}$$

$$= 3 \times 10^{10} \times 0.02$$

$$\mathrm{B} = 6 \times 10^{8}\ \mathrm{N/m^2}$$

The correct option is (3)

Q. 2 Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A): In an uniform magnetic field, speed and energy remains the same for a moving charged particle.
Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

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Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A).

Option 2:
Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

Option 3:
(A) is true but (R) is false.

Option 4:
(A) is false but (R) is true.

Correct Answer:
Both (A) and (R) are true and (R) is the correct explanation of (A).

Solution:
Force acting on moving charge in uniform magnetic field is,
$\bar{F} = q\left(\bar{V} \times \bar{B}\right)$
As the force acting on charge is perpendicular to velocity direction, it will circular charge i.e, this for a provides nccessary centripetal force to charge so that it performs circular motion

$\therefore\ \overline{V} \perp \overline{F}\ (al\,way)$

Work done on charge by magnetic force is zero, therefore,
$\rightarrow$ There is no change in $K \cdot E.$
$\rightarrow$ No change in magnitude of momentum but the momentum changes.

The correct option is (1)

Q. 3 Two identical cells each of emf $1.5\ \mathrm{V}$ are connected in parallel across a parallel combination of two resistors each of resistance $20\ \Omega$. A voltmeter connected in the circuit measures $1.2\ \mathrm{V}$. The internal resistance of each cell is :

Option 1:
$2.5\ \Omega$

Option 2:
$4\ \Omega$

Option 3:
$5\ \Omega$

Option 4:
$10\ \Omega$

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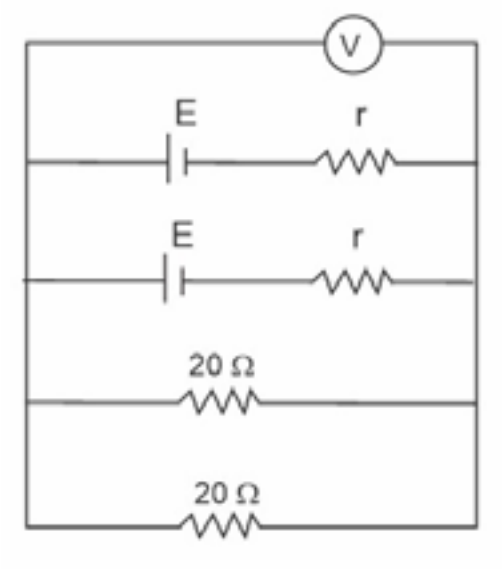
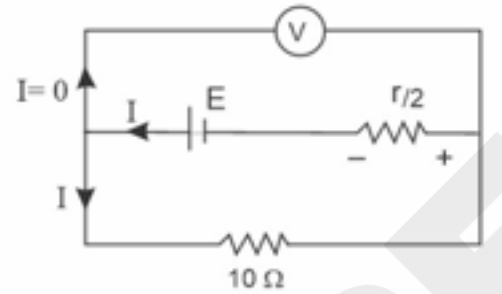
Correct Answer:

$5\,\Omega$

Solution:



Simplifying the above circuit,
We get,



For cells in parallel combination,

$$E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{E}{r} + \frac{E}{r}}{\frac{1}{r} + \frac{1}{r}}$$

$$= E$$

$$\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} \Rightarrow r_{eq} = \frac{r}{2}$$

$$\because \ E = 1.5\,V \ \ (\text{given})$$

$$I = \frac{E}{\frac{r}{2} + 10} = \frac{(1.5)}{r + 20} = \frac{3}{r + 20} \ \rightarrow (1)$$

$$E - I(r/2) = I(10) = V \ (\text{Reading of volume})$$

$$I(10) = 1.2\ V$$

I = 0.12 A

$$0.12A = \frac{3}{r + 20}$$

$$0.12\,r = 0.6$$

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$r = 5\Omega$

The correct option is (3)

Q. 4 Identify the pair of physical quantities which have different dimensions:

Option 1:
Wave number and Rydberg's constant

Option 2:
Stress and Coefficient of elasticity

Option 3:
Coercivity and Magnetisation

Option 4:
Specific heat capacity and Latent heat

Correct Answer:
Specific heat capacity and Latent heat

Solution:
SI unit of :

Specific heat capacity : $\frac{\mathrm{J}}{\mathrm{kg\ K}}$

Latent heat : $\frac{\mathrm{J}}{\mathrm{kg}}$

$\therefore$ Specific heat capacity and Latent heat has different dimensions

The correct option is (4)

Q. 5 A projectile is projected with velocity of $25\ \mathrm{m/s}$ at an angle of with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the valae of θ will be :
$\left[\text{use } \mathrm{g} = 10\,\mathrm{m/s^2}\right]$

Option 1:
$\frac{1}{2}\sin^{-1}\left[\frac{5\mathrm{t}^2}{4\mathrm{R}}\right]$

Option 2:
$\frac{1}{2}\sin^{-1}\left[\frac{4\mathrm{R}}{5\mathrm{t}^2}\right]$

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Option 3:

$\tan^{-1}\left[\frac{4t^2}{5R}\right]$

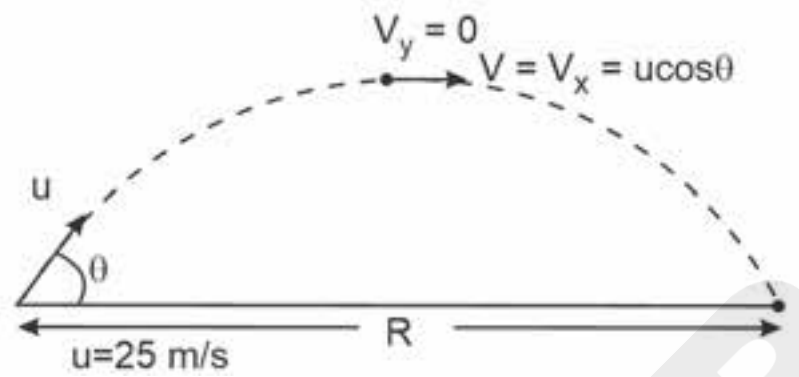
Option 4:

$\cot^{-1}\left[\frac{R}{20\,t^2}\right]$

Correct Answer:

$\cot^{-1}\left[\frac{R}{20\,t^2}\right]$

Solution:



The time at which the inclination of the projectile becomes zero with the horizontal is time of ascent (half of time of flight)

$$t = \frac{T}{2} = \frac{u\sin\theta}{g}$$

$$R = \frac{u^2 \times 2\sin\theta\cos\theta}{g}$$

$$\frac{R}{t^2} = \frac{u^2 \times 2\sin\theta\cos\theta}{\frac{u^2\sin^2\theta}{g}}$$

$$\frac{R}{t^2} = 20 \times \cot\theta$$

$$\cot\theta = \frac{R}{20\,t^2}$$

$$\theta = \cot^{-1}\left(\frac{R}{20\,t^2}\right)$$

The correct option is (4)

Q. 6 A block of mass $10\ \text{kg}$ starts sliding on a surface with an initial velocity of $9.8\ \text{ms}^{-1}$. The coefficient of friction between the surface and block is 0.5. The distance covered by the block bofore coming to rest is:

$\left[\text{use } g = 9.8\ \text{ms}^{-2}\right]$

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Option 1:
$4.9\,\mathrm{m}$

Option 2:
$9.8\,\mathrm{m}$

Option 3:
$12.5\,\mathrm{m}$

Option 4:
$19.6\,\mathrm{m}$

Correct Answer:
$9.8\,\mathrm{m}$

Solution:

$\mathrm{f} = \mu \mathrm{N} = \mu \mathrm{mg}$
$\mathrm{a} = \mu \mathrm{g} = 4.9\,\mathrm{m/s^2}$

$\mathrm{V}^2 = \mathrm{u}^2 + 2\mathrm{as}$
$0 = (9.8)^2 + 2\,(-4.9)\,\mathrm{s}$
$\mathrm{s} = 9.8\,\mathrm{m}$

The distance covered by thr block before coming to rest is $9.8\,\mathrm{m}$

The correct option is (2)

Q. 7 A boy ties a stone of mass $100\,\mathrm{g}$ to the end of a $2\,\mathrm{m}$ long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of $80\,\mathrm{N}$. If the maximum speed with which the stone can revolve is $\frac{\mathrm{K}}{\pi}\mathrm{rev./min.}$ The value of K is :

(Assume the string is massless and unstretchable)

Option 1:
400

Option 2:
300

Option 3:
600

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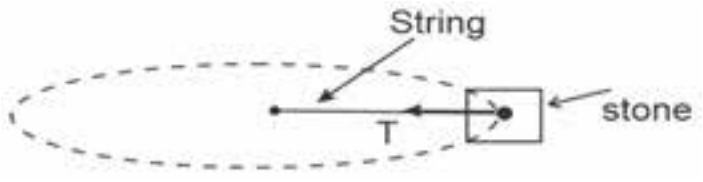
Option 4:
800

Correct Answer:
600

Solution:



Here tension (T) is providing centripetal force

$$\mathrm{T}=\frac{\mathrm{mV}^2}{\mathrm{R}}=\mathrm{MR}\omega^2$$

$$\omega=\frac{\mathrm{K}}{\pi}\left(\frac{\mathrm{rev}}{\mathrm{min}}\right)=\frac{\mathrm{K}\times 2\pi\,\mathrm{rad}}{\pi\times 60\,\mathrm{s}}$$

$$\mathrm{T}_{\max}=80=\mathrm{mR}\omega_{\max}^2$$

$$80=\left(10^{-1}\right)\times(2)\times\frac{\mathrm{K}^2}{900}$$

$$\mathrm{K}^2=400\times 900$$

$$\mathrm{K}=20\times 30=600$$

The correct option is (3)

Q. 8 A vertical electric field of magnitude $4.9\times10^5\mathrm{N/C}$ just prevents a water droplet of a mass $0.1\,\mathrm{g}$ from falling. The value of charge on the droplet will be :

(Given $\mathrm{g}=9.8\,\mathrm{m/s^2}$)

Option 1:
$1.6\times10^{-9}\mathrm{C}$

Option 2:
$2.0\times10^{-9}\mathrm{C}$

Option 3:
$3.2\times10^{-9}\mathrm{C}$

Option 4:
$0.5\times10^{-9}\mathrm{C}$

Correct Answer:
$2.0\times10^{-9}\mathrm{C}$

Solution:

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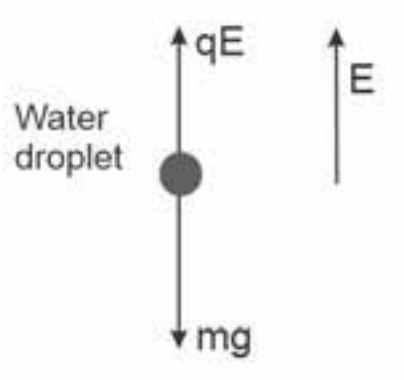

$$qE = mg$$
$$q\left(4.9 \times 10^{5}\right) = 10^{-4} \times 9.8$$
$$q = 2 \times 10^{-9} C$$

The correct option is (2)

Q. 9 A particle experiences a variable force $\vec{F} = \left(4x\,\hat{i} + 3y^2\,\hat{j}\right)$ in a horizontal $x - y$ plane. Assume distance in meters and force is newton. If the particle moves from point $(1, 2)$ to point $(2, 3)$ in the $x - y$ plane, then Kinetic Energy changes by :

Option 1:
$50.0\ J$

Option 2:
$12.5\ J$

Option 3:
$25.0\ J$

Option 4:
$0\ J$

Correct Answer:
$25.0\ J$

Solution:
Force experienced by particle in a honizontal X-Y plane is,
$$\bar{F} = 4x\left(\hat{i}\right) + 3y^2\left(\hat{j}\right)$$
Comparing with,
$$\bar{F} = F_x\hat{i} + F_y\hat{j}$$
$$F_x = 4x, \quad F_y = 3y^2$$

$$dW = \bar{F} \cdot d\bar{s}$$
$$= \left(F_x\hat{i} + F_y\hat{j}\right) \cdot \left(d_x\,\hat{i} + d_y\,\hat{j}\right)$$
$$dW = F_x\,dx + F_y\,dy$$

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$$W = \int dW = \int_{x=1}^{x=2} 4x\,dx + \int_{y=2}^{y=3} 3y^2\,dy = \Delta KE$$

(from Work-energy theorem)

$$W = 4\left[\frac{x^2}{2}\right]_1^2 + 3\left[\frac{y^3}{3}\right]_2^3$$
$$= 4\left[2 - \frac{1}{2}\right] + 3\left[9 - \frac{8}{3}\right]$$
$$W = 6 + 19 = 25J = \Delta KE$$

The correct option is (3)

Q. 10 The approximate height from the surface of earth at which the weight of the body becomes $\frac{1}{3}$ of its weight on the surface of earth is :

$\left[\text{Radius of earth } R = 6400\,\text{km and } \sqrt{3} = 1.732\right]$

Option 1:

$3840\,\text{km}$

Option 2:

$4685\,\text{km}$

Option 3:

$2133\,\text{km}$

Option 4:

$4267\,\text{km}$

Correct Answer:

$4685\,\text{km}$

Solution:

Acceleration due to gravity at height h from earth's surface is ,

$$g_h = \frac{GM}{(R+h)^2} = \frac{9R^2}{(R+h)^2}$$

$$W_h = \frac{Ws}{3} \quad (\text{Given})$$
$$mg_h = \frac{mg}{3}$$
$$g_h = \frac{9}{3} = \frac{9R^2}{(R+h)^2}$$

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$$R + h = \sqrt{3}R$$
$$h = \left(\sqrt{3} - 1\right) R$$
$$= 0.732 \times 6400$$
$$h = 4685\,\text{km}$$

The correct option is (2)

Q. 11 A resistance of $40\,\Omega$ is connected to a source of alternating current rated $220\,\text{V}, 50\,\text{Hz}$. Find the time taken by the current to change from its maximum value to the rms value :

Option 1:
$2.5\,\text{ms}$

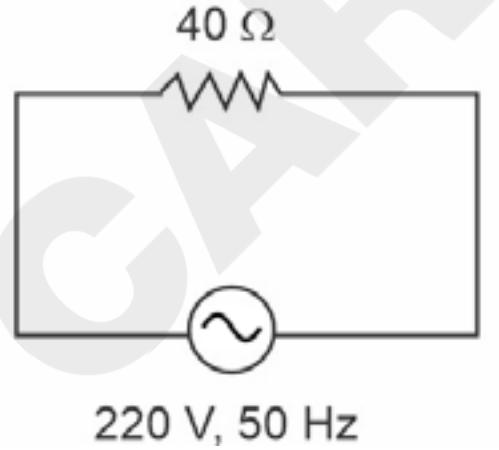
Option 2:
$1.25\,\text{ms}$

Option 3:
$2.5\,\text{s}$

Option 4:
$0.25\,\text{s}$

Correct Answer:
$2.5\,\text{ms}$

Solution:



V_{rms}=220 V
For pure resistive ckt ,
Current and voltage are in the same phase

$$\text{IF } V = V_0 \sin \omega t$$
$$\text{then } I = I_0 \sin \omega t$$

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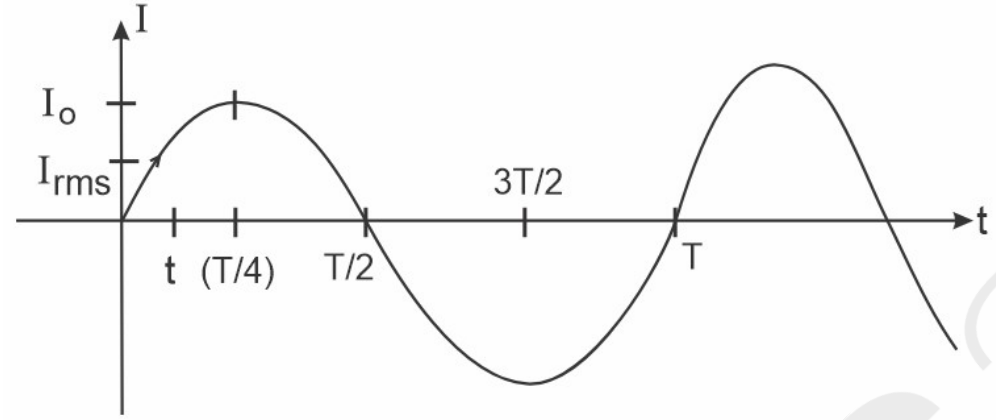
$$I = I_{rms} = \frac{I_0}{\sqrt{2}} = I_0 \sin \omega t$$
$$\omega t = \frac{\pi}{4}$$
$$\Rightarrow t = \frac{T}{8}$$



$$\Delta t = \left(\frac{T}{4}\right) - t = \frac{T}{8} = \frac{1}{8 \times F} = \frac{1}{400}$$
$$= 2.5\,ms$$

Time taken to change from its maximum value to rms value is same as time taken to change from rms value to maximum value

$$\therefore \Delta t = \frac{T}{8} = 2.5\,ms$$

The correct option is (1)

Q. 12 The equation of two waves are given by :

$$y_1 = 5 \sin 2\pi (x - vt)\ cm$$
$$y_2 = 3 \sin 2\pi (x - vt + 1.5)\ cm$$

These waves are simultaneously passing through a string. The amplitude of the resulting wave is :

Option 1:

$2\,cm$

Option 2:

$4\,cm$

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Option 3:
$5.8\,\mathrm{cm}$

Option 4:
$8\,\mathrm{cm}$

Correct Answer:
$2\,\mathrm{cm}$

Solution:
$\mathrm{y_1} = 5\sin(2\pi(\mathrm{x} - \mathrm{vt}))\mathrm{cm}$
$\mathrm{y_2} = 3\sin(2\pi(\mathrm{x} - \mathrm{vt} + 1.5))\mathrm{cm}.$

By the principle of superposition,the equation of resultant wave is

$\mathrm{y} = \mathrm{y_1} + \mathrm{y_2}$
$\mathrm{y} = \mathrm{R}\sin(\mathrm{Kx} - \omega\mathrm{t} + \theta)$
where ,
$\mathrm{R} = \sqrt{\mathrm{A_1^2} + \mathrm{A_2^2} + 2\mathrm{A_1}\mathrm{A_2}\cos\phi}$
$\phi = \text{ phase difference } = 3\pi$
$= \sqrt{5^2 + 3^2 + 2 \times 5 \times 3 \times (-1)}$
$\mathrm{R} = \sqrt{4} = 2\ \mathrm{cm}$

The amplitude of resultand wave is $2\ \mathrm{cm}$
The correct option is (1)

Q. 13 A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44.If magnitude of magnetic intensity is $4.5 \times 10^{-2}\,\mathrm{Am}^{-1}$at a point, what will be the approximate magnitude of electric field intensity at that point?

(Given : Permeability of free space $\mu_0 = 4\pi \times 10^{-7}\mathrm{NA}^{-2}$, speed of light in vacuum $\mathrm{c} = 3 \times 10^8\ \mathrm{ms}^{-1}$)

Option 1:
$16.96\,\mathrm{Vm}^{-1}$

Option 2:
$2.25 \times 10^{-2}\,\mathrm{Vm}^{-1}$

Option 3:
$8.48\,\mathrm{Vm}^{-1}$

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Option 4:

$6.75 \times 10^{6}\ \mathrm{Vm}^{-1}$

Correct Answer:

$8.48\ \mathrm{Vm}^{-1}$

Solution:

$\mu_r = 1.61$

$\xi_r = 6.44$

We know that ,

$$B = \mu H = \mu_0 \mu_r H$$

$$\frac{E}{B} = V = \frac{1}{\sqrt{\mu \xi}} = \frac{C}{\sqrt{\mu_r \xi_r}}$$

$$E = \frac{3 \times 10^{8} \times 1.61 \times (4\pi \times 10^{-7})\,(4.5 \times 10^{-2})}{\sqrt{1.61 \times 6.44}}$$

$$E = 8.478 \left(\frac{V}{m}\right)$$

The correct option is (3)

Q. 14 Choose the correct option from the following options given below :

Option 1:

In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.

Option 2:

An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in 'Thomson's model

Option 3:

A classical atom based on Rutherford's model is doomed to collapse.

Option 4:

The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

Correct Answer:

The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

Solution:

$\rightarrow$ Rutherford's performed $\alpha-$ particle scattering experiment

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$\rightarrow$ 1 out of 20000 $\alpha-$ particle had scattering angle equal to 180°

$\rightarrow$ Scattering of 180° is possible only when the mass is concentrated somewhere

$\rightarrow$ So, in the model he proposed that the positively changed part of the atom passesses most of the mass.

The correct option is (D)

Q. 15 Nucleus A is having mass number 220 and its binding energy per nucleon is $5.6\,\mathrm{MeV}$. It splits in two fragments $'\mathrm{B}'\,\mathrm{and}\,'\mathrm{C}'$ of mass numbers 105 and 115. The binding energy of nucleons in $'\mathrm{B}'\mathrm{and}\,'\mathrm{C}'\,\mathrm{is}\,6.4\,\mathrm{MeV}$ per nucleon. The energy Q released per fission will be :

Option 1:
$0.8\,\mathrm{MeV}$

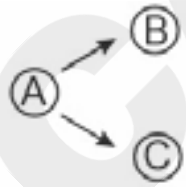
Option 2:
$275\,\mathrm{MeV}$

Option 3:
$220\,\mathrm{MeV}$

Option 4:
$176\,\mathrm{MeV}$

Correct Answer:
$176\,\mathrm{MeV}$

Solution:
$\mathrm{A_A} = 220$
$(\mathrm{BE})_{\mathrm{A}} = \mathrm{BE\ per\ nucleons} = 5.6\,\mathrm{MeV}$
For atom A



$$\mathrm{Total\ energy\ of\ A's\ nucleus} = -220\,(\mathrm{BE})_{\mathrm{A}}$$
$$(\mathrm{TE})_{\mathrm{A}} = -220 \times 5.6\,\mathrm{MeV}$$

$$\begin{aligned} \mathrm{Q-value} &= [(\mathrm{TE_A}) - (\mathrm{TE_B} + \mathrm{TE_C})] \\ &= [(-220 \times 5.6) - (-220 \times 6.4)] \\ &= 176\,\mathrm{MeV} \end{aligned}$$

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The correct option is (4)

Q. 16 A baseband signal of $3.5\,\mathrm{MHz}$ frequency is modulated with a carrier signal of $3.5\,\mathrm{GHz}$ frequency using amplitude modulation method. What should be the minimum size of antenna required to transmit the modulated signal ?

Option 1:
$42.8\,\mathrm{m}$

Option 2:
$42.8\,\mathrm{mm}$

Option 3:
$21.4\,\mathrm{mm}$

Option 4:
$21.4\,\mathrm{m}$

Correct Answer:
$21.4\,\mathrm{mm}$

Solution:
Frequency of signal $= \mathrm{f_m} = 3.5\,\mathrm{MH_z}$
Frequency of carrier signal $= \mathrm{f_c} = 3.5\,\mathrm{GH_z}$

$$\text{Minimum size of antenna} = \frac{\lambda}{4}$$

Frequency of modulating signal is $\mathrm{f_c}$

$$\because\ \mathrm{C} = \mathrm{f_c}\lambda$$
$$3 \times 10^8 = 3.5 \times 10^9 \times \lambda$$
$$\lambda = \frac{3}{3.5} \times 10^{-1}$$
$$\therefore \text{Minimum size of antenna} = \frac{\lambda}{4}$$
$$= \frac{3}{14} \times 10^{-1}$$
$$= 21.4\,\mathrm{mm}$$

The correct option is (3)

Q. 17 A carnot engine whose heat sinks at $27^\circ\mathrm{C}$, has an efficiency of 25%. By now many degress should be the temperature of the source be changed to increase the efficiency by 100% of the original efficiencey ?

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Option 1:
Increases by $18^{\circ}\mathrm{C}$

Option 2:
Increases by $200^{\circ}\mathrm{C}$

Option 3:
Increases by $120^{\circ}\mathrm{C}$

Option 4:
Increases by $73^{\circ}\mathrm{C}$

Correct Answer:
Increases by $200^{\circ}\mathrm{C}$

Solution:

$$\mathrm{T_{sink}} = 27^{\circ}\mathrm{C} = 300\mathrm{k}$$

$$\text{Initial efficiency} = \eta = 25\,\% = \frac{1}{4} = 1 - \frac{\mathrm{T_{sink}}}{\mathrm{T_{source}}}$$

$$\frac{\mathrm{T_{sink}}}{\mathrm{T_{source}}} = \frac{3}{4}$$

$$\mathrm{T_{source}} = 400\mathrm{k} = \mathrm{T}$$

$$\text{Final efficiency} = 2\eta = \frac{1}{2} = 1 - \frac{300}{(\mathrm{T} + \Delta\mathrm{T})}$$

$$\frac{1}{2} = \frac{300}{(\mathrm{T} + \Delta\mathrm{T})}$$

$$\mathrm{T} + \Delta\mathrm{T} = 600$$

$$\Delta\mathrm{T} = 200^{\circ}\mathrm{C} = 200\mathrm{k}$$

The correct option is (2)

Q. 18 A parallel plate capacitor is formed by two plates each of area $30\pi\ \mathrm{cm}^2$ seperated by $1\ \mathrm{mm}$. A material of dielectric strength $3.6 \times 10^7\ \mathrm{Vm}^{-1}$ is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is $7 \times 10^{-6}\ \mathrm{C}$, the value of dielectric constant of the material is :

$$\left[\text{Use } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \mathrm{Nm^2C^{-2}}\right]$$

Option 1:
1.66

Option 2:
1.75

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Option 3:
2.25

Option 4:
2.33

Correct Answer:
2.33

Solution:
$\mathrm{A} = 30\,\pi\,\mathrm{cm}^2 = 30\,\pi \times 10^{-4}\mathrm{m}^2$
$\mathrm{d} = 1\,\mathrm{mm} = 10^{-3}\mathrm{m}$
$\mathrm{E} = \text{Dielectric field strength} = 3.6 \times 10^7 \left(\frac{\mathrm{V}}{\mathrm{m}}\right)$
$\mathrm{Q_{max}} = 7 \times 10^{-6}\mathrm{C}$
$\mathrm{E} = \frac{\sigma}{\xi_0\,\mathrm{k}} = \frac{\mathrm{Q}}{\mathrm{A}\,\xi_0\,\mathrm{k}}$
$3.6 \times 10^7 = \frac{7 \times 10^{-6}}{30\,\pi \times 10^{-4} \times \frac{10^{-9}\mathrm{k}}{4\,\pi \times 9}}$

$\left(\frac{1}{4\,\pi\,\xi_0} = 9 \times 10^9 \;\Rightarrow\; \xi_0 = \frac{1}{4\,\pi \times 9 \times 10^9}\right)$

$\mathrm{k} = \frac{14 \times 9}{15 \times 3.6} = \frac{126}{15 \times 3.6}$
$\mathrm{k} = 2.33$

The correct option is (4)

Q. 19 The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distnace $\frac{\mathrm{r}}{2}$ from the centre is :

Option 1:
$\mathrm{B}/2$

Option 2:
$2\mathrm{B}$

Option 3:
$\left[\frac{2}{\sqrt{5}}\right]^3 \mathrm{B}$

Option 4:
$\left[\frac{2}{\sqrt{3}}\right]^3 \mathrm{B}$

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Correct Answer:

$$\left[\frac{2}{\sqrt{5}}\right]^3 \mathrm{B}$$

Solution:

$$\mathrm{B} = \mathrm{B}_{\text{circular coil}} = \frac{\mu_0 \mathrm{i}}{2\mathrm{r}}$$

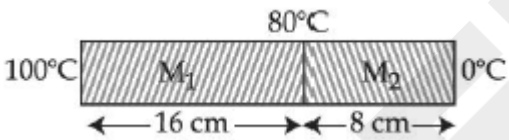
$$\mathrm{B}' = \mathrm{B}_{\text{axis}} = \frac{\mu_0 \mathrm{i}\, \mathrm{r}^2}{2\left(\mathrm{r}^2 + \mathrm{x}^2\right)^{3/2}} = \frac{\mu_0 \mathrm{i}\, \mathrm{r}^2}{2 \times \frac{5\sqrt{5}\,\mathrm{r}^3}{8}}$$

$$\frac{\mathrm{B}}{\mathrm{B}'} = \frac{5\sqrt{5}}{8}$$

$$\mathrm{B}' = \left(\frac{2}{\sqrt{5}}\right)^3 \mathrm{B}$$

The correct option is (3)

Q. 20 Two metalic blocks M_1 and M_2 of same area of cross-section are connected to each other(as shown in figure).If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be :
[Assume steady state heat conduction]



Option 1:

$10\ \mathrm{K}$

Option 2:

$8\ \mathrm{K}$

Option 3:

$12.5\ \mathrm{K}$

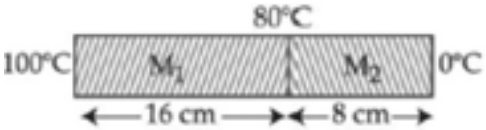
Option 4:

$2\ \mathrm{K}$

Correct Answer:

$8\ \mathrm{K}$

Solution:



Since, the two blocks are in series, the heat current through both of them will be same of steady state

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$$H_1 = H_2$$
$$\frac{\Delta T_1}{R_1} = \frac{\Delta T_2}{R_2}$$

$$\frac{(100-80)}{\left(\frac{l_1}{(k)A_1}\right)} = \frac{(80-0)}{\left(\frac{l_2}{k_1 A_2}\right)}$$

$\text{Here, } A_1 = A_2 = \text{Area of cross-section (given)}$
$k_1 \rightarrow \text{Thermal conductivity of block } M_1$
$l_1 = 16\,\text{cm}, l_2 = 8\,\text{cm}$

$$\frac{20}{\left(\frac{16}{k_1}\right)} = \frac{80}{\frac{8}{k_0}}$$
$$k_1 = 8\,k$$

The correct option is (2)

Q. 21 $0.056\ \text{kg}$ of Nitrogen is enclosed in a vessel at a temperature of $127\ \text{C}$. The amount of heat required to double the sped of its molecules is_____ k cal.
$(\text{Take } R = 2\,\text{cal mole}^{-1}\ \text{K}^{-1})$

Correct Answer:
4

Solution:
$m = 0.056\ \text{kg} = 56\,\text{gm}$
no of moles of $N_2 = n = 2\,\text{mol}$

$$T = 127\,\text{C} = 400\,\text{k}$$
$$V = \sqrt{\frac{3\ K_B\ T}{m}}$$
$$\frac{V_i}{V_f} = \sqrt{\frac{T_i}{T_f}}$$

$$V_f = 2V_i(\ \text{Given}\ )$$
$$\frac{1}{2} = \frac{\sqrt{400}}{\sqrt{T_f}}$$
$$T_f = 800\ \text{K} = 527^\circ\text{C}$$

Vessel is closed

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$\therefore W = 0$

By First law of thermodynamics,

$$Q = \frac{\Delta U}{\gamma - 1} = \frac{nR\Delta T}{\gamma - 1}$$
$$= \frac{2 \times 2 \times (400)}{\left(\frac{7}{5} - 1\right)}$$
$$Q = 4k \text{ cal}$$

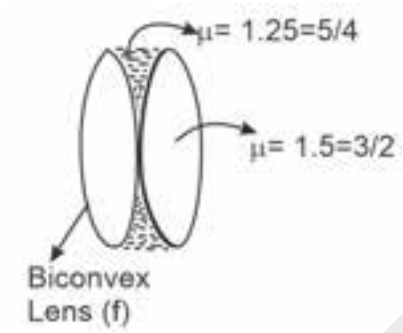
The amount of heat required to double the speed of its molecules is 4 k cal

---

Q. 22 Two identical thin biconvex lenses of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25.The focal length of the combination is _____ cm.

Correct Answer:
10

Solution:



$$\frac{1}{f} = (\mu - 1)\left(\frac{2}{R}\right) = \frac{1}{R} = \frac{1}{15 \text{ cm}}$$
$$R = 15 \text{ cm}$$

Let focal length of concave lens be f'

$$\frac{1}{f'} = (\mu - 1)\left(\frac{-2}{R}\right) = \frac{-1}{30 \text{ cm}}$$
$$f' = -30 \text{ cm}$$

$$\frac{1}{f_{eff}} = \frac{2}{f} + \frac{1}{f'} = \frac{2}{15} - \frac{1}{30} = \frac{1}{10 \text{ cm}}$$

The local length of combination is 10 cm

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Q. 23 A transistor is used in common-emitter mode in an amplifier circuit. When a signal of $10\,\mathrm{mV}$ is added to the base-emitter voltage, the base current changes by $10\,\mu\mathrm{A}$ and the collector current changes by $1.5\,\mathrm{mA}$. The load resistance is $5\,\mathrm{k}\Omega$.The voltage gain of the transistor will be_____.

Correct Answer:
750

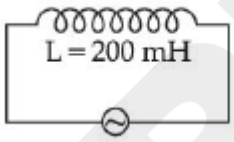
Solution:
$\Delta \mathrm{V_i} = 10\,\mathrm{mV}$
$\Delta \mathrm{I_B} = 10\,\mu\,\mathrm{A}$
$\Delta \mathrm{I_C} = 1.5\,\mathrm{m\,A}$
$\mathrm{R_L} = 5\,\mathrm{k}\,\Omega$

$$\text{Voltage gain } (\mathrm{A_v}) = \frac{\Delta \mathrm{V_0}}{\Delta \mathrm{V_i}} = \frac{\Delta \mathrm{I_C R_L}}{\Delta \mathrm{I_B R}}$$

$$(\mathrm{A_v}) = \frac{1.5 \times 10^{-3} \times 5 \times 10^{3}}{10 \times 10^{-3}}$$

$$\text{Voltage gain} = \mathrm{A_v} = 750$$

Q. 24 As shown in the figure an inductor of inductance $200\ \mathrm{mH}$ is connected to an AC source of emf $220\ \mathrm{V}$ and frequency $50\ \mathrm{Hz}$. The instantaneous voltage of the source is $0\mathrm{V}$ when the peak value of current is $\frac{\sqrt{\mathrm{a}}}{\pi}\mathrm{A}$. The value of a is _____.



Correct Answer:
242

Solution:
$\mathrm{L} = 200\,\mathrm{mH}$
$\mathrm{V_{rms}} = 220\,\mathrm{V}$
$\mathrm{f} = 50\,\mathrm{Hz}$

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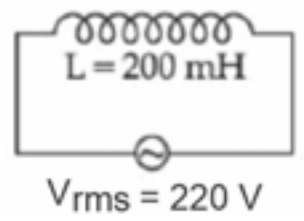

For pure inductive ckt ,

Instantaneous current & voltage has phase difference of $\frac{\pi}{2}\,\mathrm{rad}$

(Current lags voltage by $\frac{\pi}{2}\,\mathrm{rad}$)

$$\mathrm{I_0} = \text{ Peak value of current} = \frac{\sqrt{\mathrm{a}}}{\pi} = \frac{\mathrm{V_0}}{\mathrm{X_L}}$$

$$\frac{\sqrt{\mathrm{a}}}{\pi} = \frac{\sqrt{2} \times 220}{2\pi \times 50 \times 200 \times 10^{-3}}$$

$$\sqrt{\mathrm{a}} = 11\sqrt{2}$$

$$\sqrt{\mathrm{a}} = \sqrt{242}$$

$$\mathrm{a} = 242$$

Q. 25 Sodium light of wavelengths $650\ \mathrm{nm}$ and $655\ \mathrm{nm}$ is used to study diffraction at a single slit of aperture $0.5\ \mathrm{mm}$. The distance between the slit and the screen is $2.0\ \mathrm{m}$. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is _____ $10^{-5}\ \mathrm{m}$.

Correct Answer:
3

Solution:

$\mathrm{b} = 0.5 \times 10^{-3}\mathrm{m} = 5 \times 10^{-4}\mathrm{m}$

$\mathrm{D} = 2\mathrm{m}$

For maxima,

$$\beta = \frac{(2\mathrm{n}+1)}{2}\pi = \frac{\pi \mathrm{b} \sin\theta}{\lambda}$$

For First maxima, $\mathrm{n} = 1$

$$\frac{3\pi}{2} = \frac{\pi \mathrm{b} \sin\theta}{\lambda}$$

$$\sin\theta = \frac{3\lambda}{2\mathrm{b}}$$

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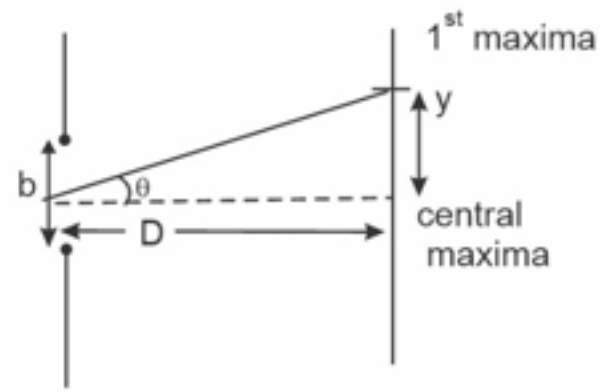

Since , λ is very small

$$\sin\theta \simeq \tan\theta = \frac{3\lambda}{2b} = \frac{y}{D}$$

$$y = \frac{3}{2}\frac{\lambda}{b}D$$

$$\text{For } \lambda_1 = 650\,\text{nm},$$

$$y_1 = \frac{3}{2} \times \frac{650 \times 10^{-9} \times 2}{5 \times 10^{-4}}$$

$$y_1 = 390 \times 10^{-5}\text{m}$$

$$\text{For } \lambda_2 = 655\,\text{nm}$$

$$y_2 = \frac{3}{2} \times \frac{655 \times 10^{-9} \times 2}{5 \times 10^{-4}}$$

$$y_2 = 393 \times 10^{-5}\text{m}$$

$$\Delta y = y_2 - y_1 = 3 \times 10^{-5}\text{m}$$

Q. 26 When light of frequency twice the threshold frequency is incident on the metal plate,the maximum velocity of emitted electron is v_1 . When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2 . If $v_2 = x\,v_1$, the value of x will be ____.

Correct Answer:
2

Solution:
Let f_0 be the threshold frequency
Initial condition : $f_1 = 2f_0$

$$(V_{max})_1 = V_1$$

Final condition : $f_2 = 5f_0$

$$(V_{max})_2 = V_2$$

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By Einstain's equation of photoelectric effect,

$$hf = hf_0 + \frac{1}{2}mv_{max}^2$$

$$\text{For } f = f_1 = 2f_0$$

$$2hf_0 = hf_0 + \frac{1}{2}m\,(V_{max})_1^2$$

$$hf_0 = \frac{1}{2}mv_1^2 \rightarrow (1)$$

$$\text{For } f = f_2 = 5f_0$$

$$4hf_0 = \frac{1}{2}mv_2^2 \rightarrow (2)$$

$$(1) \div (2)$$

$$\frac{1}{4} = \frac{V_1^2}{V_2^2}$$

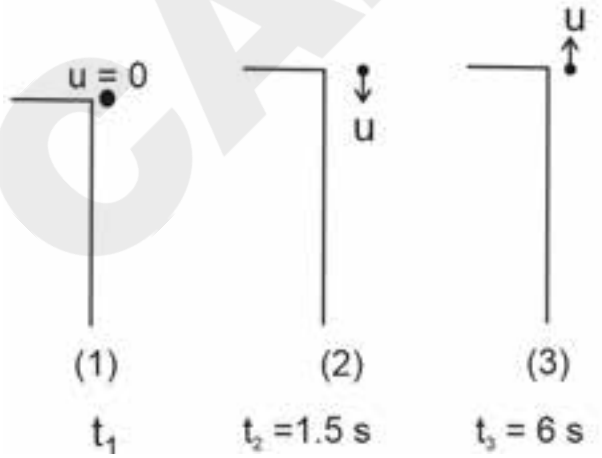
$$V_2 = 2V_1$$

$$\therefore x = 2$$

Q. 27 From the top of a tower, a ball is thrown vertically upward whcih reaches the ground in 6 s. A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location,will reach the ground in ____s.

Correct Answer:
3

Solution:



We know that ,

$$t_1 = \sqrt{t_2 t_3}$$

$$t_1 = \sqrt{9} = 3s$$

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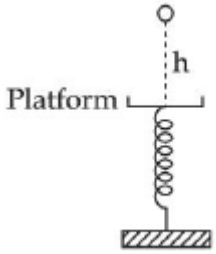
Ball released from the rest from the same location will reach ground after $t_1 = 3s$

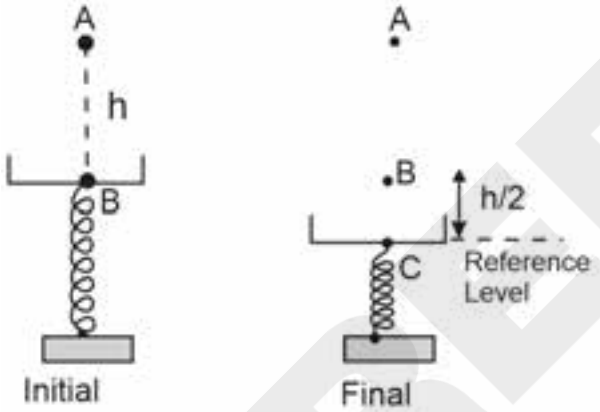
Q. 28 A ball of mass $100\ g$ is dropped from a height $h = 10\ cm$ on a platform fixed at the top of a vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance $\frac{h}{2}$. The spring constant is _____ Nm^{-1}.

$(\text{Use } g = 10\ ms^{-2})$



Correct Answer:
120

Solution:



Since the system is conservative ,

$$TE_A = TE_C$$

$$mg\left(h + \frac{h}{2}\right) + 0 = mg(0) + \frac{1}{2}kx^2$$

Compression in spring, $x = \frac{h}{2}$

$$mg \times \left(\frac{3h}{2}\right) = \frac{1}{2}k\left(\frac{h^2}{4}\right)$$

$$k = \frac{12mgh}{h^2} = \frac{12\,mg}{h}$$

$$= \frac{12\,(0.1) \times 10}{(0.1m)}$$

$$k = 120\,(N/m)$$

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Q. 29 In a potentiometer arrangement, a cell gives a balancing point at $75\ \text{cm}$ length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is $3:2$, the difference in the balancing length of the potentiometer wire in above two cases will be _____ cm.

Correct Answer:
25

Solution:
For potentiometer,

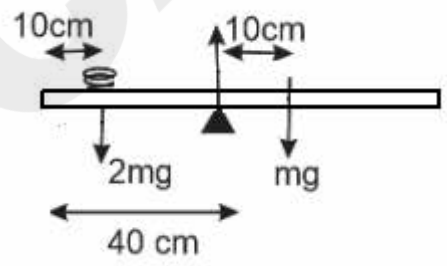
$$\begin{aligned}&E \alpha l\\&\frac{E_1}{E_2}=\frac{l_1}{l_2}\\&\frac{3}{2}=\frac{75\text{cm}}{l_2}\\&l_2=50\text{cm}\end{aligned}$$

The difference in the balancing Length of the potentiometer wire in above two cases is $\Delta l = l_1 - l_2$
$\Rightarrow \Delta l = 25\text{cm}$

Q. 30 A metre scale is balanced on a knife edge at its centre. When two coins, each of mass $10\ \text{g}$ are put one on the top of the other at the $10.0\ \text{cm}$ mark the scale is found to be balanced at $40.0\ \text{cm}$ mark. The mass of the metre scale is found to be $x \times 10^{-2}\text{kg}$. The value of x is _____.

Correct Answer:
6

Solution:



Let the mass of metre scale &
Coin be M and m respectively.

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For scale to be balanced ,

$\Sigma F = 0$ and $\Sigma \tau = 0$

R is the reaction force on a knife edge

$\Sigma F = 0$

$R = 2mg + Mg$

$\Sigma \tau = 0$

about knife edge

$(2\,mg)\,(30)\,(\leftarrow) + Mg \times 10\,(\rightarrow) = 0$

$m = \frac{M}{6} \Rightarrow M = 6M$

$= 609$

$M = 6 \times 10^{-2} kg$

$\therefore x = 6$

Chemistry

Q. 1 If a rocket runs on fuel $(C_{15}H_{30})$ and liquid oxygen,the weight of oxygen required and CO_2 released for every litre of fuel respectively are:
(Given: density of the fuel is $0.756\ g/mL$)

Option 1:
$1188\,g$ and $1296\,g$

Option 2:
$2376\,g$ and $2592\,g$

Option 3:
$2592\,g$ and $2376\,g$

Option 4:
$3429\,g$ and $3142\,g$

Correct Answer:
$2592\,g$ and $2376\,g$

Solution:
The combustion reaction is given as

$$C_{15}H_{30} + \frac{45}{2}O_2 \rightarrow 15\,CO_2 + 15\,H_2O$$

So,

$$1 \text{ mole fuel } \equiv \frac{45}{2} \text{ moles of } O_2 \equiv 15 \text{ mole of } CO_2$$

now,

1 lit of fuel weights $756\,g$

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$$\therefore \text{ moles in 1 lit of fuel} = \frac{756}{210} = 3.6$$

$$\therefore \text{ moles of } O_2 = 3.6 \times \frac{45}{2} = 81$$
$$\text{and moles of } CO_2 = 3.6 \times 15 = 54$$

$$\therefore \text{ weight of } O_2 = 81 \times 32 = 2592 \text{ g}$$
$$\text{weight of } CO_2 = 337.5 \times 44 = 2376 \text{ g}$$

Hence, the correct answer is Option (3)

Q. 2 Consider the following pairs of electrons

$$\text{(A)} \quad \text{(a)} \quad n = 3, 1 = 1, m_1 = 1, m_s = +\frac{1}{2}$$
$$\text{(b)} \quad n = 3, 1 = 2, \ m_1 = 1, \ m_s = +\frac{1}{2}$$

$$\text{(B)} \quad \text{(a)} \quad n = 3, 1 = 2, m_1 = -2, m_s = -\frac{1}{2}$$
$$\text{(b)} \quad n = 3, 1 = 2, m_1 = -1, m_s = -\frac{1}{2}$$

$$\text{(C)} \quad \text{(a)} \quad n = 4, 1 = 2, \ m_l = 2, \ m_s = +\frac{1}{2}$$
$$\text{(b)} \quad n = 3, 1 = 2, \ m_l = 2, \ m_s = +\frac{1}{2}$$

The pairs of electrons present in degenerate orbitals is / are :

Option 1:
Only (A)

Option 2:
Only (B)

Option 3:
Only (C)

Option 4:
(B) and (C)

Correct Answer:
Only (B)

Solution:
As we have learnt,

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Orbitals having same $(n+l)$ value are degenerate.

Since, only the pair of elections given in Option (B) have same value of $(n+l)$, they are degenerate.

Hence, the correct answer is Option (2)

Q. 3 Match **List - I** with **List - II** :

	List-I		List-II
(A)	$[PtCl_4]^{2-}$	(I)	sp^3d
(B)	BrF_5	(II)	d^2sp^3
(C)	PCl_5	(III)	dsp^2
(D)	$[Co(NH_3)_6]^{3+}$	(IV)	sp^3d^2

Choose the **most appropriate** answer trom the options given below :

Option 1:
$(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

Option 2:
$(A)-(III),(B)-(IV),(C)-(I),(D)-(II)$

Option 3:
$(A)-(III),(B)-(I),(C)-(IV),(D)-(II)$

Option 4:
$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

Correct Answer:
$(A)-(III),(B)-(IV),(C)-(I),(D)-(II)$

Solution:
$(A)\ [PtCl_4]^{2-}$: Square planar, dsp^2(III)

(B) B_rF_5 : octahedral, sp^3d^2(IV)

(C) PCl_5 : trigonal bipyramidal, sp^3d(I)

(D) $[Co(NH_3)_6]^{3+}$: octahedral, d^2sp^3(II)

Hence, the correct answer is Option (2)

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Q. 4 For a reaction at equilibrium

$$A(g) \rightleftharpoons B(g) + \frac{1}{2} C(g)$$

the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by :

Option 1:

$$K = \frac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$$

Option 2:

$$K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$$

Option 3:

$$K = \frac{(\alpha p)^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$$

Option 4:

$$K = \frac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$$

Correct Answer:

$$K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$$

Solution:

$$\begin{array}{lccccc} & A & \rightleftharpoons & B & + & \frac{C}{2} \\ t=0 & p_0 & & - & & - \\ teq & p_0(1-\alpha) & & p_0\alpha & & \frac{p_0\,\alpha}{2} \end{array}$$

$$\therefore P_{Total} = p = p_0\left(1+\frac{\alpha}{2}\right)$$

$$\Rightarrow p_0 = \frac{p}{\left(1+\frac{\alpha}{2}\right)}$$

$$\therefore K_P = \frac{p_B \cdot p_C^{1/2}}{p_A} = \frac{p_0\alpha \cdot \left(p_0\frac{\alpha}{2}\right)^{1/2}}{p_0(1-\alpha)}$$

$$= \frac{p_0^{1/2}\alpha^{3/2}}{(1-\alpha)2^{1/2}}$$

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$$= \frac{p^{1/2}\alpha^{3/2}}{2^{1/2}\left(1+\frac{\alpha}{2}\right)^{1/2}(1-\alpha)}$$
$$= \frac{p^{1/2}\alpha^{3/2}}{(2+\alpha)^{1/2}(1-\alpha)}$$

Hence, the correct answer is Option (2)

Q. 5 Given below are two statements:

Statement I : Emulsions of oil in water are unstable and sometimes they separate into two layers on standing.

Statement II : For stabilisation of an emulsion, excess of electrolyte is added.

In the light of the above statements, Choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
As we have learnt,

Emulsions of oil in water are not stable and sometimes, they separate into two layers on standing.
$\therefore$ Statement I is correct.

For stabilisation of an emulsion, usually an emulsifying agent is added.
$\therefore$ Statement II is incorrect.

Hence, the correct answer is Option (3)

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Q. 6 Given below are the oxides :

$Na_2O, As_2O_3, N_2O, NO \text{ and } Cl_2O_7$

Number of amphoteric oxides is:

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
1

Solution:
The nature of the oxides are given below:

Na_2O : Basic

As_2O_3 : Amphoteric

N_2O : Neutral

NO : Neutral

Cl_2O_7 : Acidic

Thus, only As_2O_3 is amphoteric in nature.

Hence, the correct answer is Option (2)

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Q. 7 Match **List -I** with **List- II** :

	List-I		List-II
(A)	Sphalerite	(I)	$FeCO_3$
(B)	Calamine	(II)	PbS
(C)	Galena	(III)	$ZnCO_3$
(D)	Siderite	(IV)	ZnS

Choose the **most appropriate** answer from the options given below :

Option 1:
$(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

Option 2:
$(A)-(IV),(B)-(I),(C)-(II),(D)-(III)$

Option 3:
$(A)-(II),(B)-(III),(C)-(I),(D)-(IV)$

Option 4:
$(A)-(III),(B)-(IV),(C)-(II),(D)-(I)$

Correct Answer:
$(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

Solution:
The correct matching is given below:

$(A)\ \text{Sphalerite}: ZnS\ (IV)$

$(B)\ \text{Calamine}: ZnCO_3\ (III)$

$(C)\ \text{Galena}: PbS (II)$

$(D)\ \text{Siderite}: FeCO_3 (I)$

Hence, the correct answer is Option (1)

Q. 8 The highest industrial consumption of molecular hydrogen is to produce compounds of element :

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Option 1:
Carbon

Option 2:
Nitrogen

Option 3:
Oxygen

Option 4:
Chlorine

Correct Answer:
Nitrogen

Solution:
The highest industrial consumption of molecular hydrogen is in the production of Ammonia

$$N_2 + 3H_2 \rightarrow 2NH_3$$

Hence, the correct answer is Option (2)

Q. 9 Which of the following statements are correct ?

(A) Both $LiCl$ and $MgCl_2$ are soluble in ethanol.

(B) The oxides Li_2O and MgO combine with excess of oxygen to give superoxide.

(C) LiF is less soluble in water than other alkali metal fluorides.

(D) Li_2O is more soluble in water than other alkali metal oxides.

Choose the **most appropriate** answer from the options given below :

Option 1:
(A) and (C) only

Option 2:
(A) , (C) and (D) only

Option 3:
(B) and (C) only

Option 4:
(A) and (D) only

Correct Answer:
(A) and (C) only

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Solution:
As we have learnt,

(A) Both LiCl and $\mathrm{MgCl_2}$ have significant covalent character and are soluble in ethanol

(B) Li and Mg do not form their superoxide

(C) LiF is less soluble in water than other alkali metal fluorides because of high lattice energy

(D) $\mathrm{Li_2O}$ is less soluble in water than other alkali metal oxides because of high lattice energy

Thus, only statements (A) and (C) are correct in the given question.

Hence, the correct answer is Option (A)

Q. 10 Identify the correct statement for $\mathrm{B_2H_6}$ from those given below.

(A) In $\mathrm{B_2H_6}$, all $\mathrm{B-H}$ bonds are equivalent.

(B) In $\mathrm{B_2H_6}$, there are four 3-centre-2-electron bonds.

(C) $\mathrm{B_2H_6}$ is a Lewis acid.

(D) $\mathrm{B_2H_6}$ can be synthesized from both $\mathrm{BF_3}$ and $\mathrm{NaBH_4}$.

(E) $\mathrm{B_2H_6}$ is a planar molecule.

Choose the **most appropriate** answer from the options given below :

Option 1:
$\mathrm{(A)\ and\ (E)\ only}$

Option 2:
$\mathrm{(B)\ ,\ (C)\ and\ (E)\ only}$

Option 3:
$\mathrm{(C)\ and\ (D)\ only}$

Option 4:
$\mathrm{(C)\ and\ (E)\ only}$

Correct Answer:
$\mathrm{(C)\ and\ (D)\ only}$

Solution:
As we have learnt,
Diborane is a non planar molecule with $\mathrm{sp^3}$ hybridised Boron centres.

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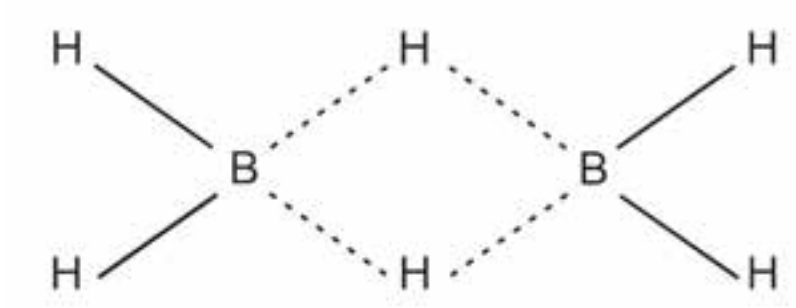
It is electron deficient and contains two $3c-2e$ bridge bonds. The structure iss given below :



All the B-H bonds are not equivalent as the two bridged bonds are electron deficient and weak.

Diborane can be synthesised from both BF_3 and $NaBH_4$. The reactions are given below:

$$BF_3 + LiAlH_4 \rightarrow B_2H_6 + LiF + AlF_3$$

$$NaBH_4 + I_2 \longrightarrow B_2H_6 + NaI + H_2$$

Hence, the correct answer is Option (3)

Q. 11 The most stable trihalide of nitrogen is:

Option 1:
NF_3

Option 2:
NCl_3

Option 3:
NBr_3

Option 4:
NI_3

Correct Answer:
NF_3

Solution:
The most stable trihalide of nitrogen is NF_3

Hence, the correct answer is Option (1)

Q. 12 Which one of the following elemental forms is not present in the enamel of the teeth ?

Option 1:
Ca^{2+}

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Option 2:
P^{3+}

Option 3:
F^{-}

Option 4:
P^{5+}

Correct Answer:
P^{3+}

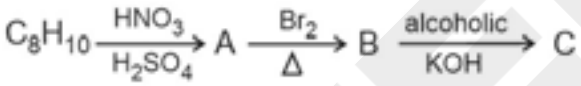
Solution:
The enamel of teeth contains hydroxyapatite $(3Ca_3(PO_4)_2 \cdot Ca(OH)_2)$ and fluorapatite $(3Ca_3(PO_4)_2 \cdot CaF_2)$

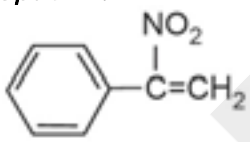
Thus, the species present in the enamel of teeth are Ca^{2+}, P^{5+} and F^{-}.

P^{3+} is not present in enamel of teeth.

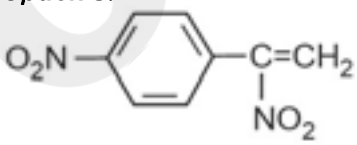
Hence, the correct answer is Option (2)

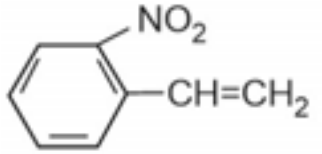
Q. 13 In the given reaction sequence, the major product $'C'$ is :



Option 1:


Option 2:

Option 3:


Option 4:


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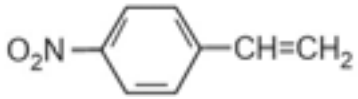
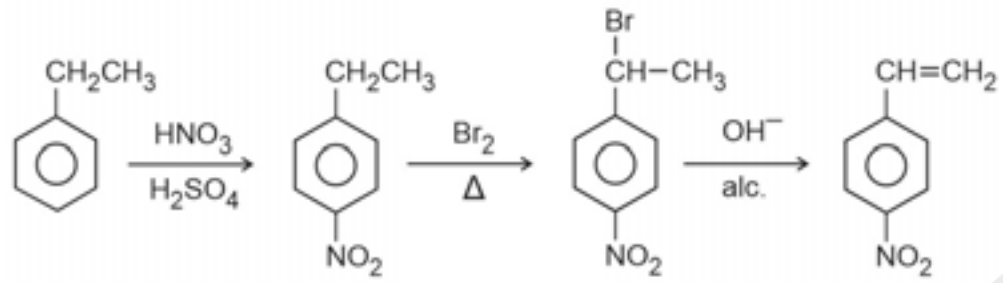
Correct Answer:



Solution:
The given reaction occurs as



Hence, the correct answer is Option (2)

Q. 14 Two statements are given below :

Statement I : The melting point of monocarboxylic acid with even number of carbon atoms is higher than that of with odd number of carbon atoms acid immediately below and above it in the series.

Statement II : The solubility of monocarboxylic acids in water decreases with increase in molar mass.

Choose the **most appropriate** option :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

Solution:
The melting point of monocarboxylic acid with even number of carbon atoms is higher than that of acids with odd number of carbon atoms inmediately below and above it in the series.
$\therefore$ Statement I is correct.

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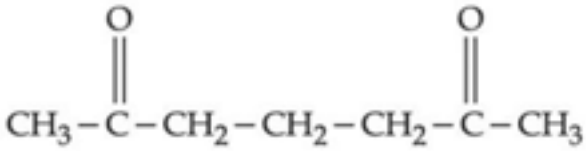
The solubility of monocarboylic acids in water decreases with increase in molar mass.
∴ Statement II is correct.

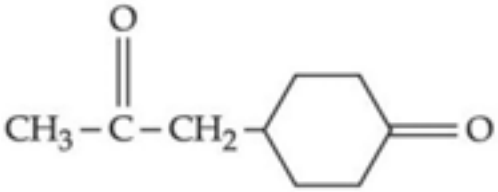
Hence, the correct answer is Option (1)

Q. 15 Which of the following is an example of conjugated diketone ?

Option 1:

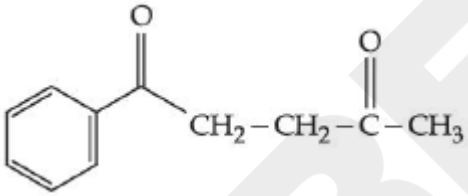


Option 2:



Option 3:

Option 4:



Correct Answer:

Solution:
An example of a conjugated diketone among the given compounds is Quinone.

Hence, the correct answer is Option (3)

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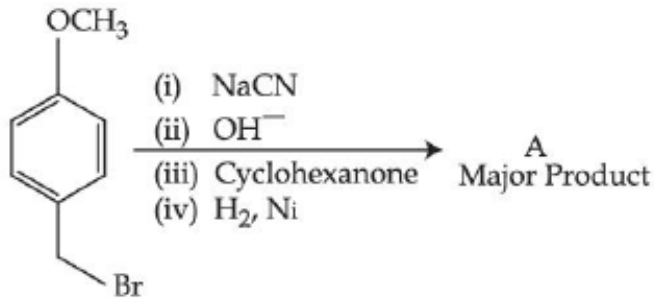
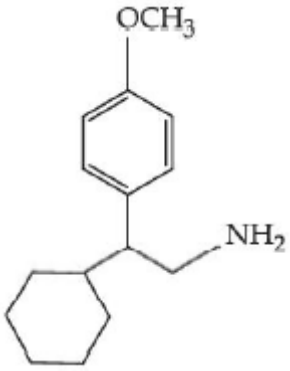
Q. 16

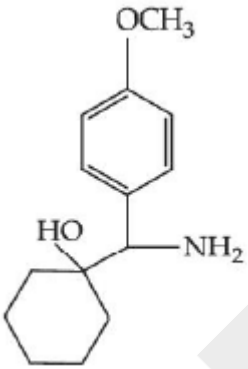


The major product of the above reactions is :

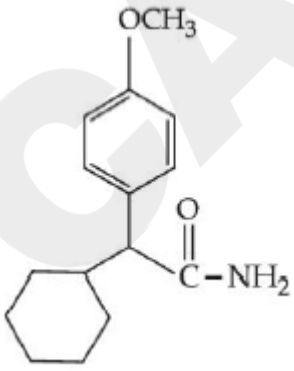
Option 1:



Option 2:



Option 3:



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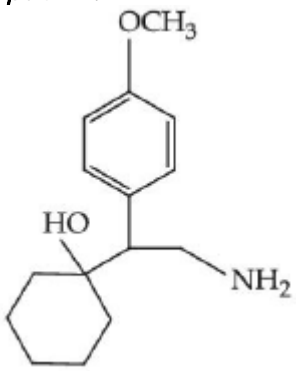
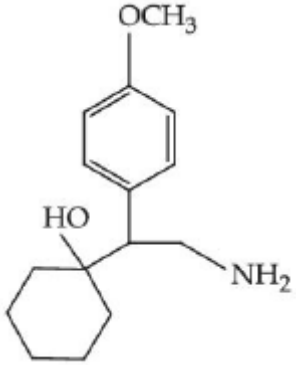
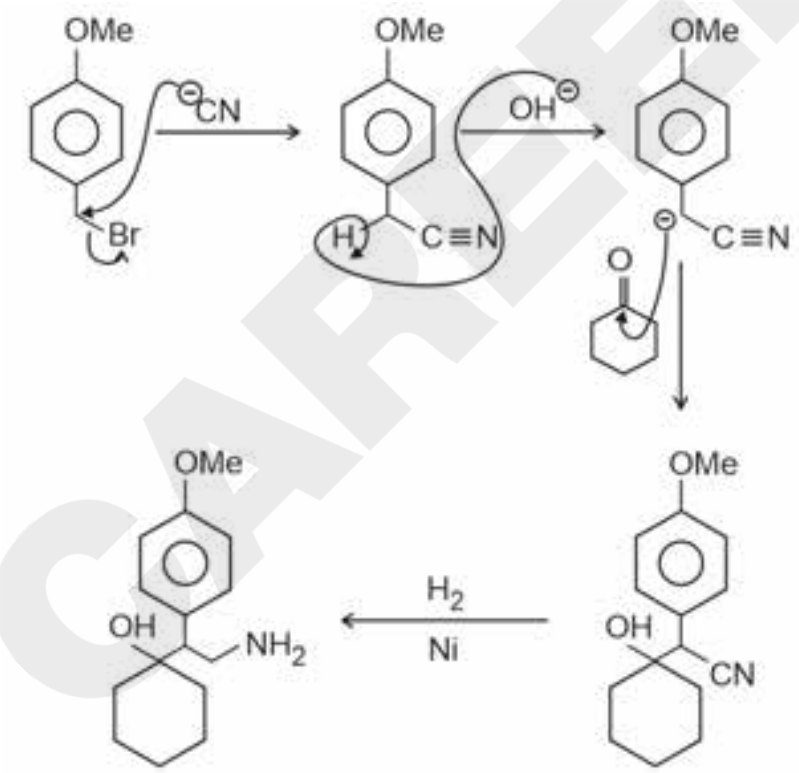
Option 4:



Correct Answer:



Solution:
The given reaction sequence occurs as



Hence, the correct answer is Option (4)

Q. 17 Which of the following is an example of polyester?

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Option 1:
Butadiene-styrene copolymer

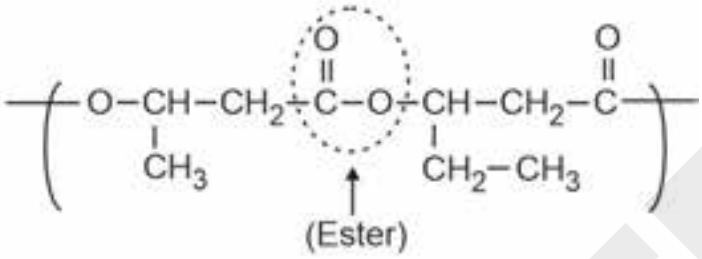
Option 2:
Melamine polymer

Option 3:
Neoprene

Option 4:
Poly- β-hydroxybutyrate-co-β-hydroxyvalerate

Correct Answer:
Poly- β-hydroxybutyrate-co-β-hydroxyvalerate

Solution:
Poly $-\beta-$ hydroxybutyrate $-\mathrm{co}-\beta-$ hydroxyvalerate (PHBV) is an example of a polyester.



Hence, the correct answer is Option (D)

Q. 18 A polysaccharide $'\mathrm{X}'$ on boiling with $\mathrm{dil}\ \mathrm{H_2SO_4}$ at $393\ \mathrm{K}$ under 2-3 atm pressure yielde $'\mathrm{Y}'$.

$'\mathrm{Y}'$on treatment with bromine water gives gluconic acid. $'\mathrm{X}'$ contains β-glycosidic linkages only. Compound $'\mathrm{X}'$ is :

Option 1:
starch

Option 2:
cellulose

Option 3:
amylose

Option 4:
amylopectin

Correct Answer:
cellulose

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Solution:
According to the question, $'\mathrm{X}'$ is a polymer of only $\beta-\mathrm{D}-\mathrm{Glucose}$.

Hence, $'\mathrm{X}'$ must be Cellulose.

Thus, the correct answer is Option (2)

Q. 19 Which of the following is **not** a broad spectrum antibiotic?

Option 1:
Vancomycin

Option 2:
Ampicillin

Option 3:
Ofloxacin

Option 4:
Penicillin G

Correct Answer:
Penicillin G

Solution:
Fact based.

Penicillin G is an example of a narrow spectrum antibiotic.

Hence, the correct answer is Option (D)

Q. 20 During the qualitative analysis of salt with cation Y^{2+}, addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation $\left(\mathrm{Y}^{2+}\right)$ present respectively are:

Option 1:
Dimethylglyoxime and Ni^{2+}

Option 2:
Dimethylglyoxime and Co^{2+}

Option 3:
Nessler's reagent and Hg^{2+}

Option 4:
Nessler's reagent and Ni^{2+}

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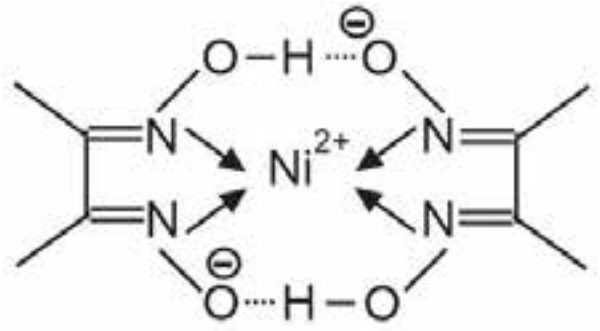
Correct Answer:
Dimethylglyoxime and Ni^{2+}

Solution:
Ni^{2+} ions on reaction with dimethyl glyoxime (DMG) forms a bright red complex.



Hence, the correct answer is Option (1)

Q. 21 Atoms of element X form hcp lattice and those of element Y occupy $\frac{2}{3}$ of its tetrahedral Voids. The percentage of element X in the lattice is _____. (Nearest integer)

Correct Answer:
43

Solution:
As we have learnt,

If the number of atoms forming an HCP lattice be 'n', then the number of tetratedral voids is $2n$.

$$\therefore \frac{X}{Y} = \frac{n}{\frac{2}{3} \times 2n} = \frac{3}{4}$$

$$\therefore \% \text{ of } X = \frac{3}{7} \times 100 \cong 43\,\%$$

Hence, the answer is 43

Q. 22 $2O_3\,(g) \rightleftharpoons 3O_2\,(g)$

At $300\,K$, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is (-)_____ $J\,mol^{-1}$. (Nearest integer)

[Given : $\ln 1.35 = 0.3$ and $R = 8.3\,J\,K^{-1}\,mol^{-1}$]

Correct Answer:
747

Solution:

$$2O_3 \rightleftharpoons 3O_2$$

$t = 0 \qquad P_0 \qquad -$

$t = teq \quad \frac{P_0}{2} \qquad \frac{3P_0}{4}$

$\text{Given, } \frac{P_0}{2} + \frac{3P_0}{4} = 1 \Rightarrow P_0 = \frac{4}{5}$

$\therefore P_{O_3} = \frac{2}{5} \text{ and } P_{O_2} = \frac{3}{5}$

$\therefore K_p = \frac{P_{O_2}^3}{P_{O_3}^2} = \frac{(3/5)^3}{(2/5)^2} = \frac{27}{20} = 1.35$

$$\therefore \Delta G = -RT \ln K = -8 \cdot 3 \times 300 \times \ln(1 \cdot 35)$$
$$= -8 \cdot 3 \times 300 \times 0 \cdot 3$$
$$= -747 \text{ J mol}^{-1} \text{ K}^{-1}$$

Hence, the answer is 747

Q. 23 The osmotic pressure of blood is 7.47 bar at $300\,\text{K}$. To inject glucose to a patient intravenously,it has to be isotonic with blood. The concentration of glucose solution in gL^{-1} is _____.
(Molar mass of glucose $= 180\,\text{g mol}^{-1}$, $\text{R} = 0.083\,\text{L bar K}^{-1}\text{mol}^{-1}$) (Nearest integer)

Correct Answer:
54

Solution:
Let the concentration of Glucose be C M

$\pi = \text{CRT}$

$\Rightarrow 7.47 = \text{C}(0.083)(300)$

$\Rightarrow \text{C} = 0.3\,\text{M}$

$\therefore$ The concentration of Glucose in gL^{-1} is 54

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Q. 24 The cell potential for the following cell

$$\mathrm{Pt}\left|\mathrm{H_2\,(g)\,H^+\,(aq)}\right|\left|\mathrm{Cu^{2+}\,(0.01\,M)}\right|\mathrm{Cu(s)}$$

is $0.576\,\mathrm{V}$ at $298\,\mathrm{K}$. The pH of the solution is _____ (Nearest integer)

(Given : $\mathrm{E^o_{Cu^{2+}/Cu}} = 0.34\ \mathrm{V}$ and $\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}$)

Correct Answer:
5

Solution:
The cell reaction is

$$\mathrm{H_2 + Cu^{2+} \longrightarrow 2H^+ + Cu}$$

$$\mathrm{Now, E^0 = 0.34 - 0 = 0.34\ V}$$

Applying Nernst equation, we get

$$\mathrm{E = E^0 - \frac{0.06}{2}\ log\ \frac{[H^+]^2}{0.01}}$$

$$\Rightarrow 0.576 = 0.34 - 0.06\ \log\{10[\mathrm{H^+}]\}$$

$$\Rightarrow 0.236 = 0.06\{\mathrm{pH} - 1\}$$

$$\Rightarrow \quad \mathrm{pH} = 4.93 \simeq 5$$

Hence, the answer is 5

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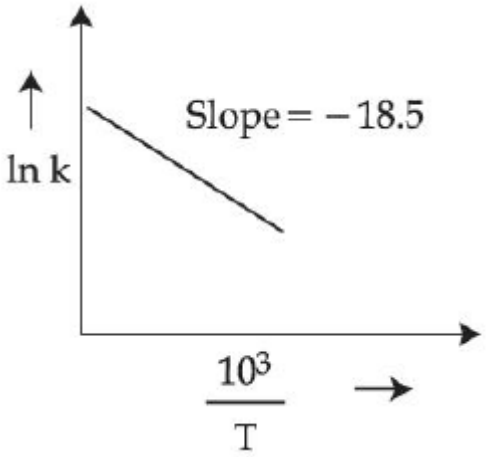
Q. 25 The rate constants for decomposition of acetaldehyde have been measured over the temperature range $700 - 1000\,\mathrm{K}$. The data has been analysed by ploting $\ln k$ vs $\frac{10^3}{T}$ graph.

The value of activation energy for the reaction is _____ $\mathrm{KJ\,mol^{-1}}$.(Nearest integer)

(Given : $\mathrm{R = 8.31\,J\,K^{-1}\,mol^{-1}}$)



Correct Answer:

154

Solution:

As we have learnt,

$$\mathrm{K = Ae^{-Ea/RT}}$$

$$\Rightarrow \ln \mathrm{K} = \ln \mathrm{A} - \frac{\mathrm{Ea}}{\mathrm{R}}\left(\frac{1}{\mathrm{T}}\right)$$

$$\Rightarrow \ln \mathrm{K} = \left(\frac{-\mathrm{Ea}}{\mathrm{R}}\right)\frac{1}{\mathrm{T}} + \ln \mathrm{A}$$

$$\therefore \ \frac{\mathrm{Ea}}{10^3\mathrm{R}} = 18.5$$

$$\Rightarrow \ \mathrm{Ea} = 18.5\mathrm{R} \times 10^3$$
$$= 153.75\ \mathrm{kJ\,mol^{-1}}$$

Hence, the answer is 154 .

Q. 26 The difference in oxidation state of chromium in chromate and dichromate salts is _____.

Correct Answer:

0

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Solution:
In both Chromate as well as Dichromate, the oxidation state of Chromium is $(+6)$.

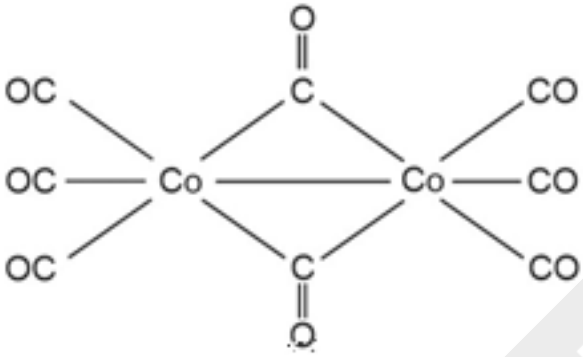
Thus, the difference in Oxidation State is 0 .

Hence, the arswer is 0 .

Q. 27 In the cobalt-carbonyl complex : $[Co_2(CO)_8]$, number of $Co-Co$ bonds is "X" and terminal CO ligands is "Y". $X+Y=$ _____.

Correct Answer:
7

Solution:
The structure of $Co_2(CO)_8$ is given as



Thus, $X=1$
$Y=6$

Hence, the answer is 7

Q. 28 A $0.388\ g$ sample of an organic compound was digested with $conc.H_2SO_4$ and then distilled with $NaOH$. The ammonia gas evolved was passed through $50.0\ mL$ of $0.5\ N\ H_2SO_4$. The used acid required $30.0\ mL$ of $0.25\ N\ NaOH$ for complete neutralization. The mass percentage of nitrogen in the organic ccompound is _____.

Correct Answer:
63

Solution:
Let m moles of NH_3 obtained be x

$\therefore$ From law of Equivalence,

$$\text{meq. of } H_2SO_4 = \text{meq of } NH_3 + \text{meq of } NaOH$$

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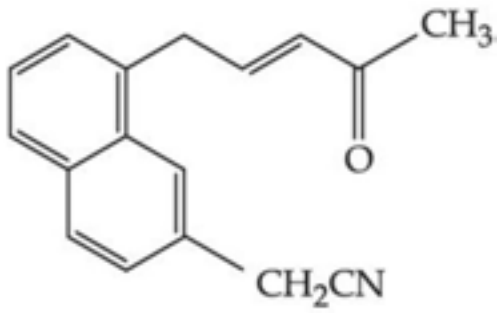
$$\Rightarrow 50 \times \frac{1}{2} = x + 30 \times \frac{1}{4}$$
$$\Rightarrow x = 17.5$$

$$\therefore \text{moles of N in organic compound} = 17.5 \times 10^{-3}$$
$$\therefore \text{mass of N} = 17.5 \times 10^{-3} \times 14 = 0.245$$
$$\therefore \% \text{ of N} = \frac{0.245}{0.388} \times 100 = 63.14\%$$

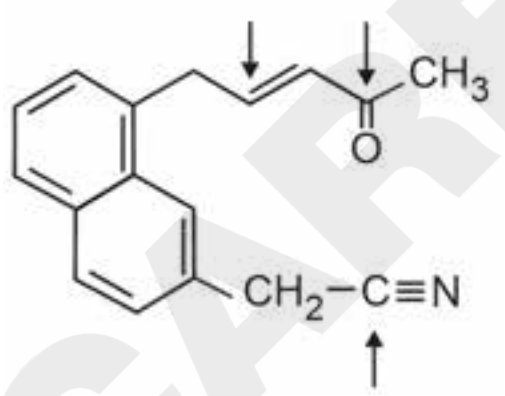
Hence, the answer is 63

Q. 29 Number of electrophilic centres in the given compound is _____.



Correct Answer:
3

Solution:
The number of electrophilic centres in the given compound is 3 as shown below:



Hence the answer is 3

Q. 30 The major product $'A'$ of the following given reaction has _____ sp^2 hybridized carbon atoms:

$$2,7-\text{Dimethyl}-2,6-\text{octadiene} \xrightarrow{H^+} \underset{\text{Major Product}}{A}$$

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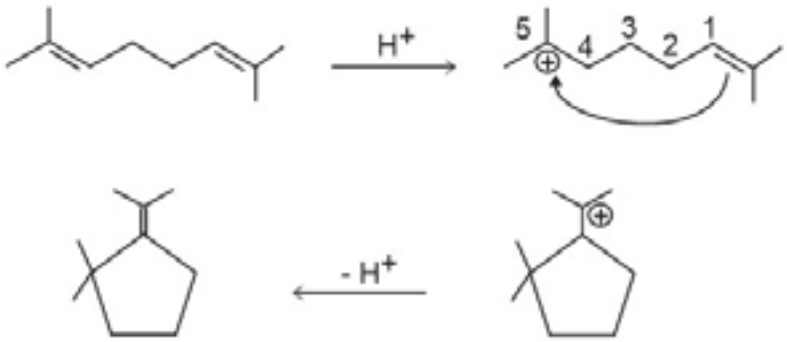
Correct Answer:
2

Solution:
The given reaction leads to isomerisation and cyclisation



The product contains 2 sp^2 hybrid carbon atoms.

Hence, the correct answer is 2

Maths

Q. 1 Let $\mathrm{A}=\{\mathrm{z} \in \mathrm{C}: 1 \leqslant|\mathrm{z}-(1+\mathrm{i})| \leqslant 2\}$ and $\mathrm{B}=\{\mathrm{z} \in \mathrm{A}:|\mathrm{z}-(1-\mathrm{i})|=1\}$. Then, B:

Option 1:
is an empty set

Option 2:
contains exactly two elements

Option 3:
contains exactly three elements

Option 4:
is an infinite set

Correct Answer:
is an infinite set

Solution:
$\mathrm{A}=\{\mathrm{z} \in \mathrm{C}: 1 \leqslant|\mathrm{z}-(1+\mathrm{i})| \leqslant 2\}$
$\mathrm{B}=\{\mathrm{z} \in \mathrm{A}:|\mathrm{z}-(1-\mathrm{i})|=1\}$
$\mathrm{A}: 1 \leqslant(\mathrm{x}-1)^2+(\mathrm{y}-1)^2 \leqslant 4$

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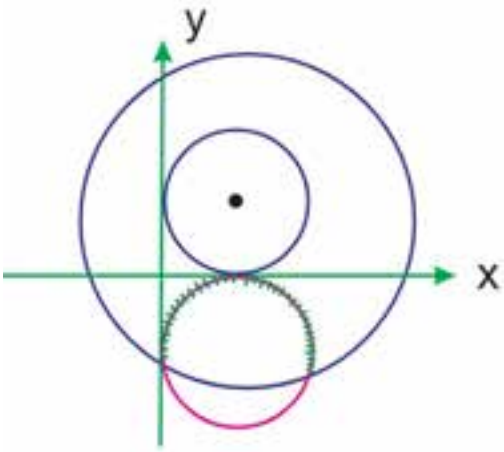

Hence B is an infinite set
The correct answer is option (D)

Q. 2 The remainder when 3^{2022} is divided by 5 is :

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
4

Solution:

$$3^{2022} = (10-1)^{1011} = 10\,\mathrm{k} - 1$$
$$= 5\lambda + 4$$

$$\therefore \text{ Remainder} = 4$$

The correct answer is option (D)

Q. 3 The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is:

Option 1:
9

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Option 2:
10

Option 3:
11

Option 4:
12

Correct Answer:
9

Solution:
Let r : radius

$$s = 4\pi r^2$$

$$\frac{ds}{dt} = 8\pi\, r \left(\frac{dr}{dt}\right) = \lambda\,(\text{constant})$$

$$4\pi r^2 = \lambda\, t + c$$

$$\text{at } t = 0,\ r = 3 \quad \Rightarrow \quad c = 36\,\pi$$
$$\text{at } t = 5,\ r = 7 \quad \Rightarrow \quad \lambda = 32\,\pi$$

$$\therefore \lambda^2 = 8\, t + 9$$

$$\therefore \text{at } t = 9$$
$$r^2 = 81$$
$$r = 9 \text{ unit}$$

The correct answer is option (A)

Q. 4 Bag A contains 2 white, 1 black and 3 red balls and bag B contains 3 black, 2 red and n white balls. One bag is chosen at random and 2 balls drawn from it at random, are found to be 1 red and 1 black. If the probability that both balls come from Bag A is $\frac{6}{11}$, then n is equal to _____.

Option 1:
13

Option 2:
6

Option 3:
4

Option 4:
3

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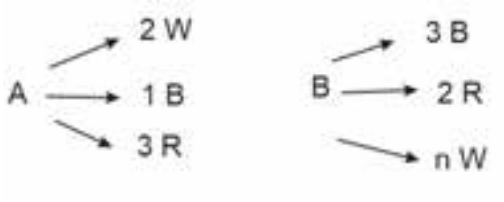
Correct Answer:
4

Solution:



$$A \xrightarrow{1R,1B} \frac{{}^{3}C_1 \times {}^{1}C_1}{{}^{6}C_2}$$

$$B \xrightarrow{1R,1B} \frac{{}^{2}C_1 \times {}^{3}C_1}{{}^{n+5}C_2}$$

$$P\left(\frac{A}{1R,1B}\right) = \frac{\frac{1}{2} \times \frac{{}^{3}C_1 \times {}^{1}C_1}{{}^{6}C_2}}{\frac{1}{2} \times \frac{{}^{3}C_1 \times {}^{1}C_1}{{}^{6}C_2} + \frac{1}{2} \times \frac{{}^{2}C_1 \times {}^{3}C_1}{{}^{n+5}C_2}}$$

$$\frac{6}{11} = \frac{\frac{3}{15}}{\frac{3}{15} + \frac{6}{(n+5)_{C_2}}}$$

$$6\left[\frac{1}{15} + \frac{2 \times 2}{(n+5)(n+4)}\right] = \frac{1}{15} \times 11$$

$$\frac{24}{(n+5)(n+4)} = \frac{5}{15}$$

$$n^2 + 9n + 20 = 72$$

$$n^2 + 9n - 52 = 0$$

$$(n+13)(n-4) = 0$$

$$n = -13, n = 4$$

$$\therefore \quad n = 4$$

The correct answer is option (c)

Q. 5 Let $x^2 + y^2 + Ax + By + C = 0$ be a circle passing through $(0, 6)$ and touching the parabola $y = x^2$ at $(2, 4)$. Then $A + C$ is equal to _____.

Option 1:
16

Option 2:
88/5

Option 3:
72

Option 4:
-8

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Correct Answer:
16

Solution:
$0+36+6B+C=0$
$6B+C=-36 \quad \cdots(i)$

$4+16+2A+4B+C=0$
$2A+4B+C=-20 \quad \cdots(ii)$

Also,
$2x+2y\,y'+A+By'=0$
$y')_{(2,4)}=\dfrac{-A-4}{8+B}=(2x)_{x=2}$
$\dfrac{-(A+4)}{B+8}=4$
$A+4B=-36 \quad \cdots(iii)$

$A=\dfrac{-4}{5},\ B=\dfrac{-44}{5},\ C=\dfrac{84}{5}$

$\therefore \quad A+C=16$

The correct answer is option (A)

Q. 6 The number of values of α for which the system of equations:
$x+y+z=\alpha$
$\alpha x+2\alpha y+3z=-1$
$x+3\alpha y+5z=4$
is inconsistent, is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
1

Solution:
$x+y+z=\alpha$

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$\alpha x + 2\alpha y + 3z = -1$
$x + 3\alpha y + 5z = 4$

For inconsistant system,

$D = 0$ & atleast one of $D_1, D_2, D_3 \neq 0$

$$D = \begin{vmatrix} 1 & 1 & 1 \\ \alpha & 2\alpha & 3 \\ 1 & 3\alpha & 5 \end{vmatrix} = \alpha + 3 - 5\alpha + 3\alpha^2 - 2\alpha$$
$$= 3\alpha^2 - 6\alpha + 3$$
$$= 3(\alpha - 1)^2$$

$$D_1 = \begin{vmatrix} \alpha & 1 & 1 \\ -1 & 2\alpha & 3 \\ 4 & 3\alpha & 5 \end{vmatrix} = \alpha^2 + 17 - 3\alpha - 8\alpha$$
$$= \alpha^2 - 11\alpha + 17$$

$$D_2 = \begin{vmatrix} 1 & \alpha & 1 \\ \alpha & -1 & 3 \\ 1 & 4 & 5 \end{vmatrix} = -17 + 3\alpha - 5\alpha^2 + 4\alpha + 1$$
$$= -5\alpha^2 + 7\alpha - 16$$

$$D_3 = \begin{vmatrix} 1 & 1 & \alpha \\ \alpha & 2\alpha & -1 \\ 1 & 3\alpha & 4 \end{vmatrix} = 11\alpha - 1 - 4\alpha + 3\alpha^3 - 2\alpha^2$$
$$= 3\alpha^3 - 2\alpha^2 + 7\alpha - 1$$

Hence inconsistant solutions for $\alpha = 1$
The correct answer is option (B)

Q. 7 If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6\left(\alpha^3 + \beta^3\right)^2$ is equal to:

Option 1:
18

Option 2:
24

Option 3:
36

Option 4:
96

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Correct Answer:
24

Solution:

$3x^2 + \lambda x - 1 = 0$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 15$$
$$\alpha^2 + \beta^2 = \frac{15}{9} = \frac{5}{3}$$

$$\therefore \ (\alpha + \beta)^2 - 2\alpha\beta = \frac{5}{3}$$
$$\frac{\lambda^2}{9} = 1 \quad \Rightarrow \quad \lambda^2 = 9$$

$$\therefore \ 6\left(\alpha^3 + \beta^3\right)^2 = 6\left[(\alpha + \beta)\left(\alpha^2 + \beta^2 - \alpha\beta\right)\right]^2$$
$$= 6\left[\left(\frac{-\lambda}{3}\right)\left(\frac{5}{3} + \frac{1}{3}\right)\right]^2$$
$$= 6 \times 4 \times \frac{\lambda^2}{9}$$
$$= 24$$

The correct answer is option (B)

Q. 8 The set of all values of k for which $\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\pi^3, x \in R$, is the interval :

Option 1:
$\left[\frac{1}{32}, \frac{7}{8}\right)$

Option 2:
$\left(\frac{1}{24}, \frac{13}{16}\right)$

Option 3:
$\left[\frac{1}{48}, \frac{13}{16}\right]$

Option 4:
$\left[\frac{1}{32}, \frac{9}{8}\right)$

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Correct Answer:

$\left[\frac{1}{32}, \frac{7}{8}\right)$

Solution:

$\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\,\pi^3$

Using complementary formulae for inverse trigonometric functions

$\frac{\pi}{2}\left[\left(\frac{\pi}{2}\right)^2 - 3\tan^{-1} x\ \cot^{-1} x\right] = k\,\pi^3$

$\frac{\pi^2}{2} - 3\tan^{-1} x\left(\frac{\pi}{2} - \tan^{-1} x\right) = 2k\,\pi^2$

$3\left(\tan^{-1} x\right)^2 - \frac{3\pi}{2}\left(\tan^{-1} x\right) + \pi^2\left(-2k + \frac{1}{4}\right) = 0$

$\left(\tan^{-1} x - \frac{\pi}{4}\right)^2 = \left(\frac{2k}{3} - \frac{1}{48}\right)\pi^2$

$0 \le \left(\frac{2k}{3} - \frac{1}{48}\right)\pi^2 < \frac{9\pi^2}{16}$

$\frac{1}{48} \le \frac{2k}{3} < \frac{9}{16} + \frac{1}{48}$

$\frac{1}{32} \le k < \frac{7}{8}$

The correct answer is (A)

Q. 9 Let $S = \left\{\sqrt{n} : 1 \leqslant n \leqslant 50 \text{ and } n \text{ is odd}\right\}$.

Let $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$

If $\sum_{a \epsilon S} \det(\text{adj } A) = 100\lambda$, then λ is equal to :

Option 1:
218

Option 2:
221

Option 3:
663

Option 4:
1717

Correct Answer:
221

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Solution:

$S = \left\{\sqrt{n} : 1 \leqslant n \leqslant 50 \text{ and } n \text{ is odd}\right\}$

$$A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$$

$\text{Det}\,(A) = 1 + a\,(0 + a) = a^2 + 1$

$$\sum \det\,(\text{Adj } A) = 100\,\lambda$$
$$\sum_{a\,\varepsilon\,s} \left(a^2 + 1\right)^2 = 100\,\lambda$$
$$\sum_{n=\text{odd}}^{a\,\varepsilon\,s} (n+1)^2 = 100\,\lambda$$

$$2^2 + 4^2 + 6^2 + \cdots 50^2 = 100\,\lambda$$
$$2^2\left(1^2 + 2^2 + \cdots + 25^2\right) = 100\,\lambda$$
$$\frac{4 \times 25 \times 26 \times 51}{6} = 100\,\lambda$$
$$\lambda = 221$$

The correct answer is option (B)

Q. 10 For the function, $f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5, x > 1$, which one of the following is NOT correct?

Option 1:
f is increasing in $(1, 2)$ and decreasing in $(2, \infty)$

Option 2:
$f(x) = -1$ has exactly two solutions

Option 3:
$f'(e) - f''(2) < 0$

Option 4:
$f(x) = 0$ has a root in the interval $(e, e+1)$

Correct Answer:
$f'(e) - f''(2) < 0$

Solution:

$f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5, x > 1$

$$f'(x) = \frac{4}{x-1} - 4x + 4 = 4\frac{[1 - x^2 + x + x - 1]}{x-1}$$

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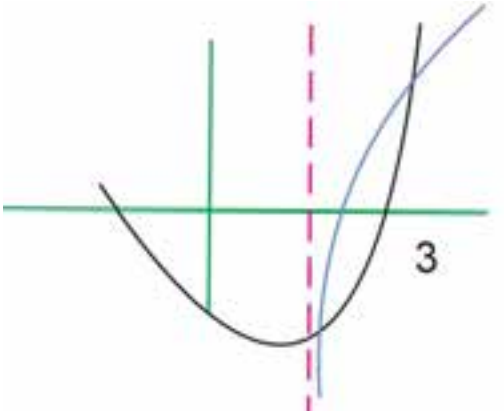
$$= -4x\frac{(x-2)}{x-1}$$

increasing in $(1, 2)$ & decreasing in $(2, \infty)$

$4\log_e(x-1) - 2x^2 + 4x + 5 = -1$
$4\log_e(x-1) = 2x^2 - 4x - 6$
$2\log_e(x-1) = x^2 - 2x - 3$



Hence, two solutions for $f(x) = -1$

Now,

$$f''(x) = \frac{-4}{(x-1)^2} - 4$$
$$f'(e) - f''(2) = \frac{4}{(e-1)^2} - 4e + 4 + 4 + 4 = {}' + {}' \text{ ve}$$

Now
$f(e) = {}' + {}' \text{ ve}$
$f(e+1) = {}' - {}' \text{ ve}$
Hence a root lies in $(e, e+1)$

The correct answer is option (C)

Q. 11 If the tangent at the point (x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve :

Option 1:
$$x^2 + \frac{y^2}{81} = 2$$

Option 2:
$$\frac{y^2}{9} - x^2 = 8$$

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Option 3:

$y = 4x^2 + 5$

Option 4:

$\frac{x}{3} - y^2 = 2$

Correct Answer:

$\frac{x}{3} - y^2 = 2$

Solution:

$y_1 = {x_1}^3 + 3{x_1}^2 + 5 \quad \cdots (i)$

$\frac{dy}{dx} = 3{x_1}^2 + 6x_1$

$y - y_1 = \left(3{x_1}^2 + 6x_1\right)(x - x_1)$

Passing through origin :

$y_1 = \left(3{x_1}^2 + 6x_1\right)(x_1) \quad \cdots (ii)$

From equation (i) & (ii),

${x_1}^3 + 3{x_1}^2 + 5 = 3{x_1}^3 + 6{x_1}^2$

$2{x_1}^3 + 3{x_1}^2 - 5 = 0$

$(x_1 - 1)\left(2{x_1}^2 + 5x_1 + 5\right) = 0$

$x_1 = 1$

$\therefore \ \text{point} : (1, 9)$

The correct answer is option (D)

Q. 12 The sum of absolute maximum and absolute minimum values of the function $f(x) = \left|2x^2 + 3x - 2\right| + \sin x \cos x$ in the interval $[0, 1]$ is:

Option 1:

$3 + \frac{\sin(1)\cos^2(1/2)}{2}$

Option 2:

$3 + \frac{1}{2}(1 + 2\cos(1))\sin(1)$

Option 3:

$5 + \frac{1}{2}(\sin(1) + \sin(2))$

Option 4:

$2 + \sin\left(\frac{1}{2}\right)\cos\left(\frac{1}{2}\right)$

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Correct Answer:

$3+\frac{1}{2}(1+2\cos(1))\sin(1)$

Solution:

$\mathrm{f}(\mathrm{x})=\left|2\mathrm{x}^2+3\mathrm{x}-2\right|+\sin\mathrm{x}\cos\mathrm{x}$

$$\mathrm{f}(\mathrm{x})=\begin{cases}-(2x^2+3x-2) & +\sin x\cos x & ;0\le x\le\frac{1}{2}\\ 2x^2+3x-2 & +\sin x\cos x & ;\frac{1}{2}<x<1\end{cases}$$

$$\mathrm{f}'(\mathrm{x})=\begin{cases}-4x-3 & +\frac{\cos 2x}{4} & ;0\le x<\frac{1}{2}\\ 4x+3 & +\frac{\cos 2x}{4} & ;\frac{1}{2}<x\le 1\end{cases}$$

$\mathrm{f}'(\mathrm{x})<0 \quad \forall\,\mathrm{x}\in\left[0,\frac{1}{2}\right)$

$\mathrm{f}'(\mathrm{x})>0 \quad \forall\,\mathrm{x}\in\left(\frac{1}{2},1\right]$

$\mathrm{f}(0)=2$

$\mathrm{f}(1)=3+\sin 1\,\cos 1$

$\mathrm{f}\left(\frac{1}{2}\right)=\sin\frac{1}{2}\,\cos\frac{1}{2}$

$\therefore$ Sum of Absolute maximum & minimum

$$=\mathrm{f}(1)+\mathrm{f}\left(\frac{1}{2}\right)$$
$$=3+\sin\frac{1}{2}\,\cos\frac{1}{2}(1+2\,\cos 1)$$
$$=3+\frac{1}{2}\sin 1\,(1+2\,\cos 1)$$

The correct answer is option (B)

Q. 13 If $\{\mathrm{a_i}\}_{\mathrm{i=1}}^{\mathrm{n}}$, where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{\mathrm{i=1}}^{\mathrm{n}}\mathrm{a_i}=192,\ \sum_{\mathrm{i=1}}^{\mathrm{n/2}}\mathrm{a_{2i}}=120$, then n is equal to:

Option 1:

48

Option 2:

96

Option 3:

92

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Option 4:
104

Correct Answer:
96

Solution:

$$\sum_{i=1}^{n} a_i = 192 \quad , \sum_{i=1}^{n/2} a_{2i} = 120$$

$$\frac{n}{2}\left[2\,a_1 + (n-1)\right] = 192 \quad ; \quad \frac{n}{4}\left[2\,a_2 + \left(\frac{n}{2} - 1\right)2\right] = 120$$

$$2\,a_1 + (n-1) = \frac{384}{n} \quad ; \quad 2\,a_1 + n = \frac{480}{n}$$

$$1 = \frac{96}{n}$$

$$n = 96$$

The correct answer is option (B)

Q. 14 If $x = x(y)$ is the solution of the differential equation $y\frac{dx}{dy} = 2x + y^3(y+1)e^y, x(1) = 0$; then $x(e)$ is equal to:

Option 1:
$e^3(e^e - 1)$

Option 2:
$e^e(e^3 - 1)$

Option 3:
$e^2(e^e - 1)$

Option 4:
$e^e(e^2 - 1)$

Correct Answer:
$e^3(e^e - 1)$

Solution:

$$y\frac{dx}{dy} = 2x + y^3(y+1)e^y \quad , \quad x(1) = 0$$

$$\frac{dx}{dy} - \left(\frac{2}{y}\right)x = y^2(y+1)e^y$$

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$$I \cdot F = e^{\int \frac{-2}{y}\,dy} = e^{-2\ln y} = \frac{1}{y^2}$$

$$\therefore \quad x\left(\frac{1}{y^2}\right) = \int (y+1)\,e^y\,dy$$

$$\frac{x}{y^2} = (y+1)\,e^y - e^y + c$$

$$\text{at } y = 1, \quad x = 0$$
$$0 = 2e - e + c \quad \Rightarrow \quad c = -e$$

$$\therefore \quad x(e) = e^2(e+1)\,e^e - e^{e+2} - e^3$$
$$x(e) = e^{e+3} - e^3$$
$$x(e) = e^3(e^e - 1)$$

The correct answer is option (A)

Q. 15 Let $\lambda x - 2y = \mu$ be a tangent to the hyperbola $a^2x^2 - y^2 = b^2$. Then $\left(\frac{\lambda}{a}\right)^2 - \left(\frac{\mu}{b}\right)^2$ is equal to:

Option 1:
-2

Option 2:
-4

Option 3:
2

Option 4:
4

Correct Answer:
4

Solution:

$$\frac{x^2}{\left(\frac{b^2}{a^2}\right)} - \frac{y^2}{b^2} = 1$$

$$\text{Tangent}: \quad y = \frac{\lambda}{2}x - \frac{\mu}{2}$$

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Using condition for tangency for hyperbola

$$\frac{\mu^2}{4} = \frac{b^2}{a^2} \times \frac{\lambda^2}{4} - b^2$$

$$\frac{\mu^2}{4b^2} = \frac{1}{4}\left(\frac{\lambda^2}{a^2} - 4\right)$$

$$\frac{\lambda^2}{a^2} - \frac{\mu^2}{b^2} = 4$$

The correct answer is option (D)

Q. 16 Let $\hat{a}, \hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$, then $|6\vec{c}|^2$ is equal to :

Option 1:
$6(3 - \sqrt{3})$

Option 2:
$3 + \sqrt{3}$

Option 3:
$6(3 + \sqrt{3})$

Option 4:
$6(\sqrt{3} + 1)$

Correct Answer:
$6(3 + \sqrt{3})$

Solution:

$$\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$$

$$\hat{b}.\hat{b} = [\vec{c} + 2(\vec{c} \times \hat{a})].[\vec{c} + 2(\vec{c} \times \hat{a})]$$

$$1 = |\bar{c}|^2 + 4|\bar{c} \times \bar{a}|^2 + 4\,\bar{c} \cdot (\bar{c} \times \bar{a})$$

$$1 = |\bar{c}|^2 + 4|\bar{c}|^2 \sin^2\frac{\pi}{12} + 0$$

$$|\bar{c}|^2\left(1 + \frac{4(3 + 1 - 2\sqrt{3})}{8}\right) = 1$$

$$|\bar{c}|^2 = \frac{8}{24 - 8\sqrt{3}}$$

$$36|\bar{c}|^2 = \frac{36}{3 - \sqrt{3}}$$

$$|6\,\bar{c}|^2 = 6(3 + \sqrt{3})$$

The correct answer is option (C)

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Q. 17 If a random variable X follows the Binomial distribution $B(33, p)$ such that $3P(X = 0) = P(X = 1)$, then the value of $\frac{P(X=15)}{P(X=18)} - \frac{P(X=16)}{P(X=17)}$ is equal to:

Option 1:
1320

Option 2:
1088

Option 3:
$\frac{120}{1331}$

Option 4:
$\frac{1088}{1089}$

Correct Answer:
1320

Solution:

$3P(X = 0) = P(X = 1)$

$3 \cdot {}^{33}C_0 (q)^{33} = {}^{33}C_1 p (q)^{32}$

$q^{33} = 11 \cdot p\, q^{32}$

$\frac{q}{P} = 11$

As p + q = 1

$\therefore \ P = \frac{1}{12}, \ q = \frac{11}{12}$

$\therefore \ \frac{P(X=15)}{P(X=18)} - \frac{P(X=16)}{P(X=17)}$

$\frac{{}^{33}C_{15}\ p^{15}\ q^{18}}{{}^{33}C_{18}\ p^{18}\ q^{15}} - \frac{{}^{33}C_{16}\ p^{16}\ q^{17}}{{}^{33}C_{17}\ p^{17}\ q^{16}}$

$\left(\frac{q}{p}\right)^3 - \left(\frac{q}{p}\right) = 1331 - 11$

$= 1320$

The correct answer is option (A)

Q. 18 The domain of the function $f(x) = \frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e (x^2 - 3x + 2)}$ is :

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Option 1:
$(-\infty, 1) \cup (2, \infty)$

Option 2:
$(2, \infty)$

Option 3:
$[-1/2, 1) \cup (2, \infty)$

Option 4:
$[-1/2, 1) \cup (2, \infty) - \left\{\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}, 3\right\}$

Correct Answer:
$[-1/2, 1) \cup (2, \infty) - \left\{\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}, 3\right\}$

Solution:

$$f(x) = \frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$$

$$-1 \le \frac{x^2-5x+6}{x^2-9} \le 1 \quad \& \quad x^2-3x+2 > 0 \quad \& \quad x^2-3x+2 \ne 1$$

$$-1 \le \frac{x-2}{x+3} \le 1 \quad \& \; x \ne 3 \; \& \quad (x-2)(x-1) > 0 \; \& \; x^2-3x+1 \ne 0$$

$$-1 \le \; 1 - \frac{5}{x+3} \le 1$$

$$0 \le \frac{5}{x+3} \le 2$$

$$0 \le \frac{5}{x+3} \quad \& \quad \frac{5}{x+3} - 2 \le 0 \quad \& \quad x \in (-\infty, 1) \cup (2, \infty) \quad \& \quad x \ne \frac{3 \pm \sqrt{5}}{2} \quad \& \quad x \ne 3$$

$$x \in (-3, \infty) \quad \& \quad \frac{2x+1}{x+3} \ge 0 \quad \& \quad x \in (-\infty, 1) \cup (2, \infty) \quad \& \quad x \ne \left\{\frac{3 \pm \sqrt{5}}{2}, 3\right\}$$

On taking intersection

$$x \in \left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}, 3\right\}$$

The correct answer is option (D)

Q. 19 Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$. If $T = \sum_{\theta \in S} \cos 2\theta$, then $T + n(S)$ is equal to :

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Option 1:
$7+\sqrt{3}$

Option 2:
9

Option 3:
$8+\sqrt{3}$

Option 4:
10

Correct Answer:
9

Solution:
For S:

$$\tan\theta(\sin\theta+1)=\frac{2\tan\theta}{1+\tan^2\theta}$$

$$\tan\theta=0, \quad \sin\theta+1=2\cos^2\theta$$

$$\sin\theta=-1, \quad \sin\theta=\frac{1}{2}$$

rejected

$$\theta=0,-\pi,\pi \qquad \theta=\frac{\pi}{6},\frac{5\pi}{6}$$

$$\therefore \mathrm{T}=\cos 0+\cos(-2\pi)+\cos(2\pi)+\cos\left(\frac{\pi}{3}\right)+\cos\left(\frac{5\pi}{3}\right)$$

$$\mathrm{T}=3+\frac{1}{2}+\frac{1}{2}=4$$

$$\therefore \quad \mathrm{T}+\mathrm{n(S)}=4+5=9$$

The correct answer is option (B)

Q. 20 The number of choices for $\Delta\in\{\wedge,\vee,\Rightarrow,\Leftrightarrow\}$, such that $(\mathrm{p}\,\Delta\,\mathrm{q})\Rightarrow((\mathrm{p}\,\Delta\sim\mathrm{q})\vee((\sim\mathrm{p})\,\Delta\mathrm{q}))$ is a tautology, is :

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

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Correct Answer:
2

Solution:
$(p \Delta q) \Rightarrow (p \Delta \sim q) \vee (\sim p \Delta q)$ is tautology
when $\Delta \equiv \vee, \Rightarrow$

The correct answer is option (B)

Q. 21 The number of one-one functions $f : \{a, b, c, d\} \rightarrow \{0, 1, 2, \cdots, 10\}$ such that $2f(a) - f(b) + 3f(c) + f(d) = 0$ is _____.

Correct Answer:
31

Solution:
$f : \{a, b, c, d\} \rightarrow \{0, 1, 2, 3, \cdots 10\}$

$2f(a) - f(b) + 3f(c) + f(d) = 0$
$2f(a) + 3f(c) = f(b) - f(d)$

Total number of possibilities = 31

Q. 22 In an examination, there are 5 multiple choice questions with 3 choices, out of which exactly one is correct. There are 3 marks for each correct answer, -2 marks for each wrong answer and 0 mark if the question is not attempted. Then, the number of ways a student appearing in the examination gets 5 marks is_____.

Correct Answer:
40

Solution:
$x_1 + x_2 + x_3 + x_4 + x_5 = 5$

$x_i = 3 \text{ or } -2 \text{ or } 0$

Possible only when 3,3,3,-2,-2

$$\text{Total ways } = \frac{5!}{3!\,2!} \times 2 \times 2 = 40$$

(2 x 2 as there are two wrong options for each of the question marked incorrectly)

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Q. 23 Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$, $a > 0$, be a fixed point in the $xy-$plane. The image of A in $y-$axis be B and the image of B in $x-$axis be C. If $D(3\cos\theta, a\sin\theta)$ is a point in the fourth quadrant such that the maximum area of $\triangle ACD$ is 12 square units, then a is equal to _____.

Correct Answer:
8

Solution:

$A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$

$B\left(\frac{-3}{\sqrt{a}}, \sqrt{a}\right)$

$C\left(\frac{-3}{\sqrt{a}}, -\sqrt{a}\right)$ $\qquad D : (3\cos\theta, a\sin\theta)$

$$\text{Ar}(\triangle ACD) = \frac{1}{2}\left[\frac{3}{\sqrt{a}}\left(-\sqrt{a} - a\sin\theta\right) + \left(\frac{-3}{\sqrt{a}}\right)\left(a\sin\theta - \sqrt{a}\right) + 3\cos\theta\left(\sqrt{a} + \sqrt{a}\right)\right]$$
$$= 3\sqrt{a}\left(\cos\theta - \sin\theta\right)$$

Maximum area $= 12 = 3\sqrt{a}\,\sqrt{2}$

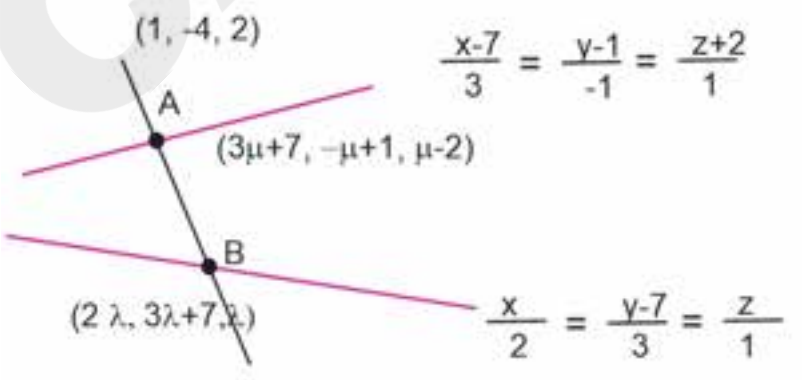
$2\sqrt{a} = \sqrt{a}$

$a = 8$

Q. 24 Let a line having direction ratios $1, -4, 2$ intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the points A and B. Then $(AB)^2$ is equal to _____.

Correct Answer:
84

Solution:



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Direction ratio's of AB:

$$\frac{3\mu-2\lambda+7}{1}=\frac{-\mu-3\lambda-6}{-4}=\frac{\mu-\lambda-2}{2}$$

On solving $\mu=-5$

$\lambda=-3$

$$\therefore\ (\mathrm{AB})^2=(-15+6+7)^2+(5+9-6)^2+(-5+3-2)^2$$
$$=4+64+16$$
$$=84$$

Q. 25 The number of points where the function

$$\mathrm{f(x)}=\begin{cases}|2x^2-3x-7| & \text{if } x\le -1\\ [4x^2-1] & \text{if } -1<x<1\\ |x+1|+|x-2| & \text{if } x\geqslant 1,\end{cases}$$

$[\mathrm{t}]$ denotes the greatest integer $\leqslant \mathrm{t}$, is discontinuous is _____.

Correct Answer:
7

Solution:

$$\mathrm{f(x)}=\begin{cases}|2x^2-3x-7| & x\le -1\\ [4x^2-1] & -1<x<1\\ |x+1|+|x-2| & x\geqslant 1\end{cases}$$

As modulus function is continuous.

$\therefore\ \mathrm{f(x)}$ can be discontinuous at $\mathrm{x}=-1,1$ and at all those points where $4\mathrm{x}^2-1$ is integer

$$4x^2-1=0\ ;4x^2-1=-1\ ,4x^2-1=1,\ \ 4x^2-1=2$$
$$x=\pm\tfrac{1}{2}\qquad x=0\qquad x=\pm\tfrac{1}{\sqrt{2}}\qquad x=\pm\tfrac{\sqrt{3}}{2}$$

Hence discontinuous at

$$\mathrm{x}=\frac{1}{2},\frac{-1}{2},\frac{1}{\sqrt{2}},\frac{-1}{\sqrt{2}},\frac{\sqrt{3}}{2},\frac{-\sqrt{3}}{2},1$$
$$=7$$

Q. 26 $\mathrm{Let\ f}(\theta)=\sin\theta+\int_{-\pi/2}^{\pi/2}(\sin\theta+\mathrm{t}\cos\theta)\mathrm{f(t)dt}.\mathrm{Then\ the\ value\ of}\left|\int_0^{\pi/2}\mathrm{f}(\theta)\mathrm{d}\theta\right|$ is_____.

Correct Answer:
1

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Solution:

$$f(\theta) = \sin\theta + \int_{-\pi/2}^{\pi/2} (\sin\theta + t\cos\theta) f(t)dt$$

$$f(\theta) = \sin\theta + \sin\theta \int_{-\pi/2}^{\pi/2} f(t)dt + \cos\theta \int_{-\pi/2}^{\pi/2} t\, f(t)dt$$

$$\text{let } P = \int_{-\pi/2}^{\pi/2} f(t)dt \quad , \quad Q = \int_{-\pi/2}^{\pi/2} t\, f(t)dt$$

$$\therefore\ f(\theta) = \sin\theta + P\sin\theta + Q\cos\theta$$

$$P = \int_{-\pi/2}^{\pi/2} ((P+1)\sin t + Q\cos t)\, dt \; P = 2Q \quad \cdots (i)$$

$$\therefore\ Q = \int_{-\pi/2}^{\pi/2} t\,((P+1)\sin t + Q\cos t)\, dt$$

$$2P + 2 - Q = 0 \quad \cdots (ii)$$

Solving (i) & (ii)

$$P = \frac{-4}{3},\ Q = \frac{-2}{3}$$

$$\therefore \left| \int_0^{\pi/2} f(\theta)\, d\theta \right| = \left| \int_0^{\pi/2} \left(\frac{-1}{3}\sin\theta - \frac{2}{3}\cos\theta \right) d\theta \right|$$
$$= 1$$

Q. 27 Let $\text{Max}_{0\le x\le 2}\left\{\frac{9-x^2}{5-x}\right\} = \alpha$ and $\text{Min}_{0\le x\le 2}\left\{\frac{9-x^2}{5-x}\right\} = \beta$.

$If \int_{\beta-\frac{8}{3}}^{2\alpha-1} \text{Max}\left\{\frac{9-x^2}{5-x}, x\right\} dx = \alpha_1 + \alpha_2 \log_e\left(\frac{8}{15}\right)$ *then* $\alpha_1 + \alpha_2$ *is equal to*

Correct Answer:
34

Solution:

$$\text{Max}_{0\le x\le 2}\left\{\frac{9-x^2}{5-x}\right\} = \alpha\ ;\ \text{Min}_{0\le x\le 2}\left\{\frac{9-x^2}{5-x}\right\} = \beta$$

$$\text{let } y = \frac{9-x^2}{5-x} = x + 5 + \frac{16}{x-5}$$

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$$y' = 1 - \frac{16}{(x-5)^2} \quad \Rightarrow y' = 0 \quad \Rightarrow x = 1, 9 \text{ 9 is rejected}$$

$$\begin{matrix} max\,(y) & = 2 & = \alpha \\ min\,(y) & = \frac{5}{3} & = \beta \end{matrix} \quad \begin{cases} Since, y\,(1) & = 2 \\ y\,(0) & = \frac{9}{5} \\ y\,(2) & = \frac{5}{3} \end{cases}$$

$$\therefore \quad \int_{\beta-\frac{8}{3}}^{2\alpha-1} \max\left\{\frac{9-x^2}{5-x}, x\right\} dx$$

$$= \int_{-1}^{3} \max\left\{\frac{9-x^2}{5-x}, x\right\} dx$$

$$= \int_{-1}^{\frac{9}{5}} \frac{9-x^2}{5-x}\, dx + \int_{\frac{9}{5}}^{3} x\, dx$$

$$= 14 + \frac{28}{25} + 16 \ln\left(\frac{8}{15}\right) + \frac{72}{25}$$

$$= 18 + 16 \ln\left(\frac{8}{15}\right)$$

$$\therefore\ \alpha_1 = 18,\ \ \alpha_2 = 16$$

$$\therefore\ \alpha_1 + \alpha_2 = 34$$

Q. 28 If two tangents drawn from a point (α, β) lying on the ellipse $25x^2 + 4y^2 = 1$ to the parabola $y^2 = 4x$ are such that the slope of one tangent is four times the other, then the value of $(10\alpha + 5)^2 + (16\beta^2 + 50)^2$ equals_____.

Correct Answer:
2929

Solution:

$$25x^2 + 4y^2 = 1$$

$$\frac{x^2}{\left(\frac{1}{5}\right)^2} + \frac{y^2}{\left(\frac{1}{2}\right)^2} = 1$$

Let point on ellipse $P : \left(\frac{1}{5}\cos\theta, \frac{1}{2}\sin\theta\right)$

$$y = mx + \frac{1}{m} \quad (\text{Tangent of parabola } \ y^2 = 4x)$$

$$m\frac{\sin\theta}{2} = \frac{n^2\cos\theta}{5} + 1$$

$$(2\cos\theta)\,m^2 - (5\sin\theta)m + 10 = 0$$

$$m_1 + 4m_1 = \frac{5\sin\theta}{2\cos\theta} \quad \& \quad 4{m_1}^2 = \frac{5}{\cos\theta}$$

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$$m_1 = \frac{\sin\theta}{2\cos\theta}$$

$$\therefore \frac{4\sin^2\theta}{4\cos^2\theta} = \frac{5}{\cos\theta}$$

$$\sin^2\theta = 5\cos\theta$$
$$\cos^2\theta + 5\cos\theta - 1 = 0$$
$$\cos\theta = \frac{-5 \pm \sqrt{29}}{2}$$

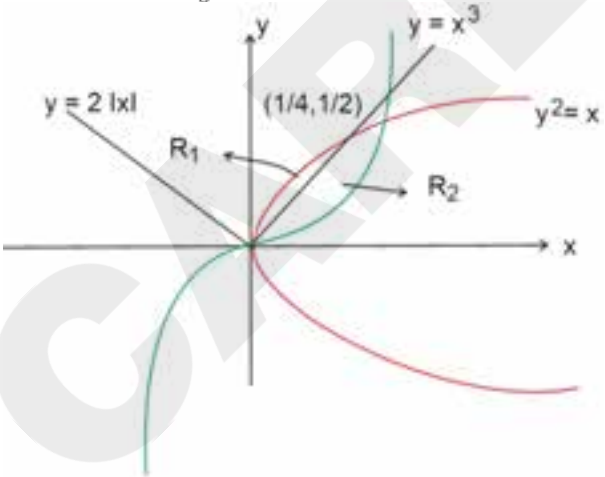
$$\alpha = \frac{1}{5}\cos\theta = \frac{-5 \pm \sqrt{29}}{10}$$
$$(10\alpha + 5)^2 = 29$$
$$16\beta^2 = -50 \pm 10\sqrt{29}$$
$$\therefore \left(16\beta^2 + 50\right)^2 = 2900$$
$$\therefore (10\alpha + 5)^2 + \left(16\beta^2 + 50\right)^2 = 2929$$

Q. 29 Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1 and R_2.
If $\max\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____.

Correct Answer:
19

Solution:

$$R_1 + R_2 = \int_0^1 \left(\sqrt{x} - x^3\right) dx = \frac{5}{12}$$



$$R_1 = \int_0^{\frac{1}{4}} \left(\sqrt{x} - 2x\right) dx = \frac{1}{48}$$
$$R_2 = \frac{5}{12} - \frac{1}{48} = \frac{19}{48}$$

$$\therefore \frac{R_2}{R_1} = 19$$

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Q. 30 If the shortest distance between the lines $\vec{r}=\left(-\hat{i}+3\hat{k}\right)+\lambda\left(\hat{i}-a\hat{j}\right)$ and $\vec{r}=\left(-\hat{j}+2\hat{k}\right)+\mu\left(\hat{i}-\hat{j}+\hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____.

Correct Answer:
2

Solution:

$\vec{r}=\left(-\hat{i}+3\hat{k}\right)+\lambda\left(\hat{i}-a\hat{j}\right)$
$\vec{r}=\left(-\hat{j}+2\hat{k}\right)+\mu\left(\hat{i}-\hat{j}+\hat{k}\right)$

$$\text{shortest distance}=\frac{\left[(\bar{a}-\bar{b})\ \bar{c}\ \bar{d}\right]}{\left|\bar{c}\times\bar{d}\right|}$$

$$\begin{vmatrix}-1 & 1 & 1\\ 1 & -a & 0\\ 1 & -1 & 1\end{vmatrix}=a-1-1+a=2a-2$$

$$\left(\bar{c}\times\bar{d}\right)=-a\hat{i}-\hat{j}+(a-1)\hat{k}$$

$$\therefore\ \sqrt{\frac{2}{3}}=\pm\frac{2a-2}{\sqrt{a^2+1+(a-1)^2}}$$

$$\frac{2}{3}=\frac{4a^2+4-8a}{a^2+1+a^2+1-2a}$$

$$4a^2+4-4a=12a^2+12-24a$$
$$8a^2-20a+8=0$$
$$2a^2-5a+2=0$$
$$a=\frac{1}{2},2$$
$\therefore$ integral value of $a=2$

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Online 24th June Evening Shift

Physics

Q. 1 Identify the pair of physical quantities that have same dimensions :

Option 1:
velocity gradient and decay constant

Option 2:
wien's constant and Stefan constant

Option 3:
angular frequency and angular momentum

Option 4:
wave number and Avogadro number

Correct Answer:
velocity gradient and decay constant

Solution:

SI unit of velocity gradient $\left(\frac{\mathrm{dV}}{\mathrm{dx}}\right)$ and decay constanct $\left(\lambda=\frac{\ln (2)}{\mathrm{T}_{1 / 2}}\right)$ is

$(\text{ second })^{-1}$

$\therefore\left[\frac{dv}{dx}\right]=[\lambda]=\left[T^{-1}\right]$

The correct option is (1)

Q. 2 The distance between Sun and Earth is R. The duration of year if the distance between Sun and Earth becomes 3R will be :

Option 1:
$\sqrt{3}$ years

Option 2:
3 years

Option 3:
9 years

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Option 4:
$3\sqrt{3}$ years

Correct Answer:
$3\sqrt{3}$ years

Solution:
From Kepter's law,
$T^2 \propto \gamma^3$
$T \rightarrow$ Time period of revolution

$$\frac{T_1^2}{T_2^2} = \frac{\text{t } R^3}{(3R)^3} = \frac{1}{27}$$
$$T_2 = T_1(\sqrt{27})$$
$$= 3\sqrt{3}T_1$$
$$T_1 = 1year$$
$$\Rightarrow \pi_2 = \sqrt[3]{3}years$$

The correct option is (4)

Q. 3 A stone of mass m, tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is

Option 1:
The same throughout the motion.

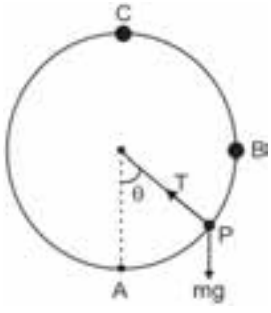
Option 2:
minimum at the highest position of the circular path.

Option 3:
minimum at the lowest position of the circular path.

Option 4:
minimum when the rope is in the horizontal position.

Correct Answer:
minimum at the highest position of the circular path.

Solution:



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Tension at any general point P

$$T_P = T = \frac{mv^2}{R} + mg\cos\theta$$

$$\text{at } \theta = 0$$

$$\text{(Lowest poition)}$$

$$T_{\max} = \frac{mv^2}{R} + mg \Rightarrow \text{ maximum}$$

$$\theta = 180^\circ, \cos\theta = -1$$

$$T_{min} = \frac{mv^2}{R} - mg$$

The correct option is (2)

Q. 4 Two identical charged particles each having a mass 10 g and charge $2.0 \times 10^{-7}\mathrm{C}$ are placed on a horizontal table with a separation of L between them such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L. (Use $g = 10 ms^{-2}$]

Option 1:
12 cm

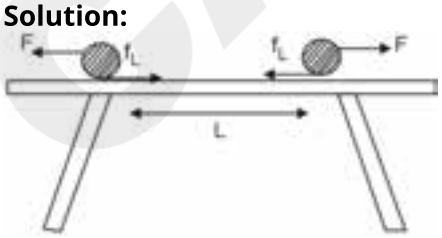
Option 2:
10 cm

Option 3:
8 cm

Option 4:
5 cm

Correct Answer:
12 cm

Solution:



$f_L \rightarrow$ Limiting friction
$F \rightarrow$ Electrostatic repulcion fora

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$$F = \frac{kq^2}{L^2} = \mu mg$$

$$f_L = \mu N = \mu\ mg$$

for limited equlibrikm,

$$F = f_L$$

$$\frac{kq^2}{L^2} = \mu \mathrm{mg}$$

$$\frac{9 \times 10^9 \times (2 \times 10^{-7})^2}{L^2} = \left(\frac{1}{4}\right) \times 10^{-2} \times 10$$

$$L^2 = 9 \times 16^{-4} \times 10^{-4}$$

$$L = 12 \times 10^{-2}\ \mathrm{m} = 12\ \mathrm{cm}$$

The correct option is (1)

Q. 5 A Carnot engine takes 5000 kcal of heat from a reservoir at $727^\circ C$ and gives heat to a sink at $127^\circ C$. The work done by the engine is

Option 1:

$3 \times 10^6 J$

Option 2:

Zero

Option 3:

$12.6 \times 10^6 J$

Option 4:

$8.4 \times 10^6 J$

Correct Answer:

$12.6 \times 10^6 J$

Solution:

$Q_{\text{source}} = Q_H = 5000\mathrm{kcal}$

$T_{\text{source}} = T_H = 1000\ \mathrm{K} = 727^\circ\mathrm{C}$

$T_{\sin k} = T_c = 400\ \mathrm{K} = 127^\circ\mathrm{c}$

$\eta = \frac{kl}{Q_{\text{source}}} = 1 - \frac{T_C}{T_H}$

$\frac{W_1}{Q_{\text{source}}} = 1 - \frac{400}{1000}$

$W = \frac{6}{10} \times 5000\mathrm{kcal}$

$k = 3000\mathrm{kcal}$

$\mathrm{Ical} = 4.2\ \mathrm{J}$

$w = 12.6 \times 10^6\ \mathrm{J}$

The correct option is (3)

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Q. 6 Two massless springs with spring constants 2 k and 9 k, carry 50 g and 100 g masses at their free ends. These two masses oscillate vertically such that their maximum velocities are equal. Then, the ratio of their respective amplitudes will be :

Option 1:
1:2

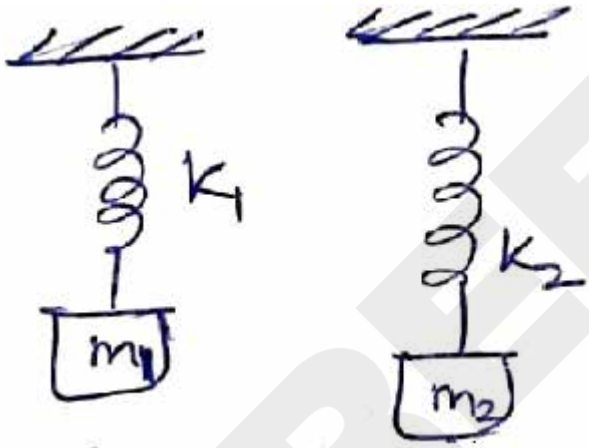
Option 2:
3:2

Option 3:
3:1

Option 4:
2:3

Correct Answer:
3:2

Solution:



$k_1 = 2k$
$K_2 = 9k$

For particle performing oscillation,

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$$V_{\max} = \omega A$$
$$(V_1)_{\max} = (V_2)_{\max}$$
$$\omega_1 A_1 = \omega_2 A_2$$
$$\sqrt{\frac{k_1}{m_1}} A_1 = \sqrt{\frac{k_2}{m_2}} A_2$$
$$\sqrt{\frac{2k}{50 \times 10^{-3}}} A_1 = \sqrt{\frac{9k}{100 \times 10^{-3}}} \times A_2$$
$$\frac{A_1}{A_2} = \frac{3}{2}$$

The correct option is (2)

Q. 7 What will be the most suitable combination of three resistors $\mathrm{A} = 2\Omega, \mathrm{B} = 4\Omega, \mathrm{C} = 6\Omega$ so that $\left(\frac{22}{3}\right)$ is equivalent resistance of combination?

Option 1:
Parallel combination of A and C connected in series with B.

Option 2:
Parallel combination of A and B connected in series with C.

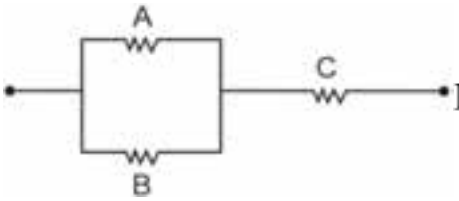
Option 3:
Series combination of A and C connected in parallel with B.

Option 4:
Series combination of B and C connected in parallel with A.

Correct Answer:
Parallel combination of A and B connected in series with C.

Solution:
$$A = 2\Omega \quad B = 4\Omega \quad C = 6\Omega$$
$$R_{eq} = \left(\frac{22}{3}\right)\Omega$$

It is possilde when $\mathrm{A\&B}$ are in parallel combination and then in series combination with C



$$\mathrm{R_{eq}} = \frac{\mathrm{A} \times \mathrm{B}}{\mathrm{A} + \mathrm{B}} + \mathrm{C} = \frac{22}{3}$$

he correct option is (2)

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Q. 8 The soft-iron is a suitable material for making an electromagnet. This is because soft-iron has

Option 1:
low coercivity and high retentivity.

Option 2:
low coercivity and low permeability.

Option 3:
high permeability and low retentivity.

Option 4:
high permeability and high retentivity.

Correct Answer:
high permeability and low retentivity.

Solution:
$\rightarrow$ Soff iron has high susceptiolity to magnetism
$\rightarrow$ Since, $\mu_r = (1 + \mathrm{X})$
$\rightarrow$ Therefore, μ_r is high
$\rightarrow$ But, $\mu = \mu_0 \mu_r$
$\rightarrow$ So, soff iron has high permeability and it has low retentivity i.e., residual mognetism is very low when the current flow or source field is taken off
The correct option is (3)

Q. 9 A proton, a deuteron and an C-particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

Option 1:
$1 : \sqrt{2} : \sqrt{2}$

Option 2:
$1 : 1 : \sqrt{2}$

Option 3:
$\sqrt{2} : 1 : 1$

Option 4:
$1 : \sqrt{2} : 1$

Correct Answer:
$1 : \sqrt{2} : 1$

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Solution:
It is given that,
$\bar{V} \perp \bar{B}$ (for all particles)

$$KE_\alpha = KE_D = KE_p$$
$$m_\alpha = 4amu;\ q_\alpha = 2e$$
$$m_D = 2amu;\ q_D = 1e$$
$$m_p = 1amp;\ q_p = 1e$$

We know that,

$$\gamma = \frac{mv}{q_B} = \frac{\sqrt{2mkE}}{qB}$$
$$\gamma_p : \gamma_D : \gamma_\alpha = \frac{\sqrt{m_p}}{q_p} : \frac{\sqrt{m_D}}{q_D} : \frac{\sqrt{m_\alpha}}{q_\alpha}$$
$$= 1 : \sqrt{2} : 1$$

The correct option is (4)

Q. 10 Given below are two statements :

Statement-I: The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor

Statement-II: In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true and Statement II is false

Option 4:
Statement I is false and Statement II is true

Correct Answer:
Statement I is true and Statement II is false

Solution:
statement 1
the reactance of an ac circuit is zero event though the circuit contain a capacitor and an inductor at series resonance

$$\left|x_L - x_c\right| = 0 \Rightarrow x_L = x_c$$

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$\rightarrow$ Therefore the first first statement is true.

statement 2

$\rightarrow$ The average power delivered by the Source becomes zero for pure inductive or capacitive ckt

$\rightarrow$ The sccond statement is false

The correct option is (3).

Q. 11 Potential energy as a function of r is given by $\mathrm{U}=\frac{\mathrm{A}}{\mathrm{r}^{10}}-\frac{\mathrm{B}}{\mathrm{r}^{5}}$ where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be:

Option 1:

$\left(\frac{\mathrm{A}}{\mathrm{B}}\right)^{\frac{1}{5}}$

Option 2:

$\left(\frac{\mathrm{B}}{\mathrm{A}}\right)^{\frac{1}{5}}$

Option 3:

$\left(\frac{2\mathrm{A}}{\mathrm{B}}\right)^{\frac{1}{5}}$

Option 4:

$\left(\frac{\mathrm{B}}{2\mathrm{A}}\right)^{\frac{1}{5}}$

Correct Answer:

$\left(\frac{2\mathrm{A}}{\mathrm{B}}\right)^{\frac{1}{5}}$

Solution:

$U=\frac{A}{r^{10}}-\frac{B}{r^{5}}$

$F=-\frac{dU}{dr}$

$F=-10Ar^{-11}+5Br^{-6}$

At equilibrium, F=0

$0=-10A\left(r^{-11}\right)+5B\left(r^{-6}\right)$

$0=-10Ar^{-5}+5B$

$\frac{10A}{5B}=r^{5}$

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$\Rightarrow \left(\frac{2A}{B}\right)^{\frac{1}{5}} = r$

The corredt option is (3)

Q. 12 An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to : [Use g = 10ms^{-2}].

Option 1:
1:1

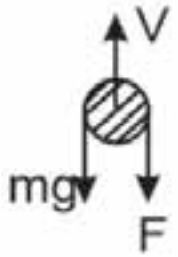
Option 2:
$\sqrt{2} : \sqrt{3}$

Option 3:
$\sqrt{3} : \sqrt{2}$

Option 4:
2:3

Correct Answer:
$\sqrt{2} : \sqrt{3}$

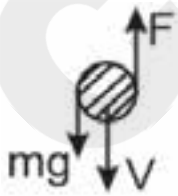
Solution:



Total force cownwards =$\mathrm{mg} + \mathrm{F} = 60$

a_1=retardation during ascent=$12\ \mathrm{m/s^2}$

During descent:



During descent

Total force downward =$\mathrm{mg} - \mathrm{F}$ = 40

a_2= refartation during descent =$8\ \mathrm{m/s}$

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Time of ascent or descent is

$$T = \sqrt{\frac{2h}{a}}$$

$$\therefore \frac{T_a}{T_d} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{2}{3}}$$

The correct opfion is (2)

Q. 13 A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be:

Option 1:
7.5 rad

Option 2:
15 rad

Option 3:
20 rad

Option 4:
30 rad

Correct Answer:
15 rad

Solution:
For uniform acceleration,

Angle rotated by wheel in n^{th} second is

$$\theta_{\text{n}} = \left(\frac{\alpha}{2}\right)(2\text{n} - 1)$$

$\text{Q}_1 \rightarrow$ Angle rotated in 1st secend
$\theta_2 \rightarrow$ Angle rotated in 2nd second

$$\frac{\theta_1}{\theta_2} = \frac{1}{3} \text{ rad}$$

$$\theta_2 = 15$$

The corred option is (2)

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Q. 14 A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of $60ms^{-1}$ What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail ?

[Specific heat capacity of iron = $0.42\ Jg^{-1}\ ^{\circ}C^{-1}$]

Option 1:
$675^{\circ}C$

Option 2:
$1600^{\circ}C$

Option 3:
$16.07^{\circ}C$

Option 4:
$6.75^{\circ}C$

Correct Answer:
$16.07^{\circ}C$

Solution:

$KE \text{ of hammer } = \frac{1}{2}mv^2$

$= \frac{1}{2} \times 1.5 \times (60)^2$

$= \frac{3}{4} \times 3600$

$= 2700 \text{ J}$

$\text{Energy given to nail } = \frac{1}{4}(\text{KE of hammer})$

$mC\Delta T = \frac{2700}{4}$

$100 \times 0.42 \times \Delta T = \frac{2700}{4}$

$\Delta T = \frac{2700}{4 \times 42}$

$\Delta T = 16.07^{\circ}\text{C}$

$\text{The correct option is}(3)$

Q. 15 If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C)

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Option 1:
10

Option 2:
20

Option 3:
30

Option 4:
40

Correct Answer:
10

Solution:

Let the initial and final charge of capacitor be Q_i and Q_f respectively

$Q_f = Q_i + 2 \rightarrow$ (Given)

$E_f = E_i + \frac{44}{100} \times E_i$ (Given)

$E_f = 1.44E_i$

Where,

E_f and E_i are the final and initial energy stored in the capacitor

$\frac{1}{2}\frac{Q_f^2}{C} = (1.44) \times \frac{1}{2}\frac{Q_i^2}{C}$

$Q_f = 1.2Q_i$

But $Q_f = Q_i + 2$

$\therefore Q_i 2Q_i = 2$

$\therefore Q_i = 10\text{C}$

$\therefore Q_f = 12\text{C}$

The original change on the Capacitor is10 C

The correct option is (1)

Q. 16 A long cylindrical volume contains a uniformly distributed charge of density p. The radius of cylindrical volume is R. A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is :

Option 1:
$\frac{\text{pqR}^2}{4\epsilon_0}$

Option 2:
$\frac{\text{pqR}^2}{2\epsilon_0}$

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Option 3:

$\frac{qp}{4\epsilon_0 R^2}$

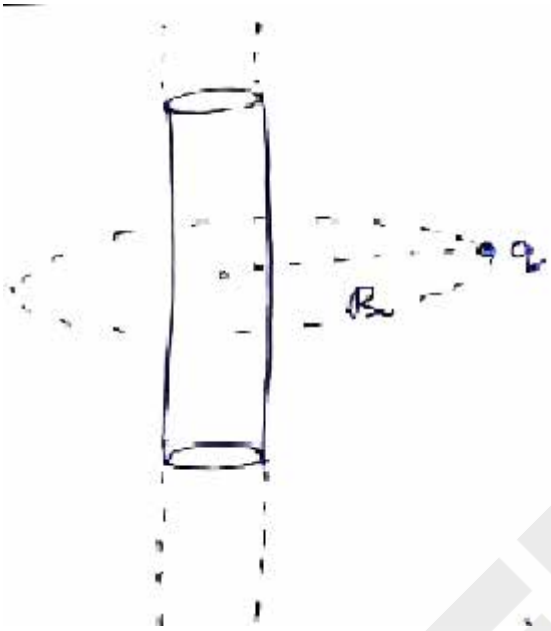
Option 4:

$\frac{4\epsilon_0 R^2}{qp}$

Correct Answer:

$\frac{pqR^2}{4\epsilon_0}$

Solution:



Electric field due to cylinder at Separation r (change q location)

$$E = \frac{\lambda}{2\pi\varepsilon_0 R} = \frac{\rho\,(\pi R^2)}{2\pi\varepsilon_0 R}$$

$$E = \frac{\rho R}{2\varepsilon_0}$$

Electrostatic force provides centripetal force

$$F = qE = \frac{mv^2}{R}$$

$$\text{KE of charge } = \frac{1}{2}mv^2 = \frac{\rho qR^2}{4\varepsilon_0}$$

The corred option is (1)

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Q. 17 An electric bulb is rated as 200 W. What will be the peak magnetic field at 4 m distance produced by the radiations coming from this bulb? Consider this bulb as a point source with 3.5% efficiency.

Option 1:

$1.19 \times 10^{-8} T$

Option 2:

$1.71 \times 10^{-8} T$

Option 3:

$0.84 \times 10^{-8} T$

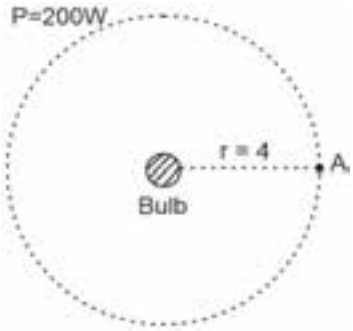
Option 4:

$3.36 \times 10^{-8} T$

Correct Answer:

$1.71 \times 10^{-8} T$

Solution:



Power emitted from source $=3.5\%$ of $200\ \mathrm{W}$

$$= \frac{3.5}{100} \times 200$$

$P_0 = 7w$

Intensity at pt. A

$$\mathrm{A} = \mathrm{I} = \frac{1}{2} \frac{\mathrm{B}_0^2}{\mu_0} \mathrm{C}$$

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$$\frac{P_0}{4\pi r^2} = \frac{1}{2}\frac{B_0^2}{\mu_0}C$$

$\because \mu_0 = 4\pi \times 10^{-7}$

$$\frac{7}{16} = \frac{B_0^2 \times 3 \times 10^8}{2 \times 10^{-7}}$$

$$B_0^2 = \frac{14}{48} \times 10^{-15}$$

$$B_0^2 = \frac{140}{48} \times 10^{-16}$$

$$\Rightarrow \quad B_0 = 1.71 \times 10^{-8}\ \text{T}$$

The correct option is (2)

Q. 18 The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectivly will be :

Option 1:
1:1

Option 2:
2:1

Option 3:
4:1

Option 4:
1:4

Correct Answer:
2:1

Solution:
Work function =$\phi_0 = 0.6\text{eV}$
$hf_1 = 3.8\text{eV}$
$hf_2 = 1.4\text{eV}$

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From Einstein equation of photoelectric effect,

$hf = \phi_0 + \frac{1}{2}mv_{\max}^2$

$\therefore hf_1 - \phi_0 = \frac{1}{2}mv_1^2 \rightarrow (1)$

$hf_2 - \phi_0 = \frac{1}{2}mv_2^2 \rightarrow (2)$

$(1) \div (2)$

$\frac{3 \cdot 2}{0 \cdot 8} = \frac{v_1^2}{v_2^2}$

$\frac{v_1}{v_2} = 2$

The correct option is (2)

Q. 19 Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be:

Option 1:
2:3

Option 2:
16:81

Option 3:
25:169

Option 4:
25:1

Correct Answer:
25:1

Solution:

$\frac{I_1}{I_2} = \frac{9}{4}$

$\frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{25}{1}$

The correct option is (4)

Q. 20 In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :

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Option 1:
All K, P and E increase.

Option 2:
K decreases, P and E increase.

Option 3:
P decreases, K and E increase.

Option 4:
K increases, P and E decrease.

Correct Answer:
K decreases, P and E increase.

Solution:
In Bohr's orbit,

$$\mathrm{T.E} = -\mathrm{KE} = \frac{\mathrm{PE}}{2}$$

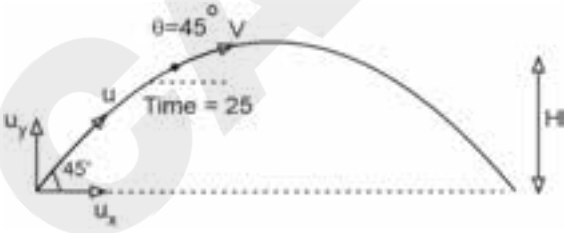
$$\mathrm{TE\ in\ n^{th}\ orbit} = -13.6\frac{\mathrm{z^2}}{\mathrm{n^2}}\mathrm{eV}$$

As we go to higher level,
$\mathrm{TE\ \&\ PE}$ increases whereas KE decrease.
The correct option is (2)

Q. 21 A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is $20\ ms^{-1}$. The maximum height reached by the body during its motion is _____ m. (use $g = 10\ ms^{-2}$)

Correct Answer:
40m

Solution:



$$\tan 45^\circ = \frac{u_y}{u_x} = 1$$

$$u_y = u_x \rightarrow \ (1)$$

$$V = \sqrt{V_x^2 + v_y^2}$$

$$\because V_x = v_x = \ \text{conotant}$$

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$$V_y = u_y - gt$$
$$(20)^0 = \sqrt{(u_x)^2 + (u_y - gt)^2}$$
$$(20) = \sqrt{(u_x)^2 + (u_x - 20)^2}$$
$$(20)^2 = u_x^2 + u_x^2 - 40u_x + 400$$
$$0 = 2u_x^2 - 40u_x$$
$$0 = u_x\,(2u_x - 40)$$
$$u_x = 0 \text{ OR } u_x = 20 \text{ m/s}$$
$$\therefore U_x = U_y = 20 \text{ m/s}$$
$$H = \frac{u_y^2}{2g} = \begin{matrix}\text{Maximum heiglt} \\ \text{reaches by the body}\end{matrix}$$
$$\Rightarrow H = 20 \text{ m}$$

Q. 22 An antenna is placed in a dielectric medium of dielectric constant 6.25. If the maximum size of that antenna is 5.0 mm, it can radiate a signal of minimum frequency _____ of GHz.

(Given μ_r = 1 for dielectric medium)

Correct Answer:
6GHz

Solution:

$$\varepsilon_r = k = 6.25 = \frac{100}{16}(\text{ Given })$$
$$\text{Maximum size of antenna} = \frac{\lambda}{4} = 5 \times 10^{-3}$$
$$\mu_r = 1$$
$$\text{Specd of signal} = v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$
$$V = 1.2 \times 10^8 \text{ m/s}$$
$$V = f\lambda$$
$$1.2 \times 10^8 = f\,(20 \times 10^{-3})$$
$$f = 6 \times 10^9 \text{ Hz} = 6\text{GHz}$$

Q. 23 A potentiometer wire of length 10 m and resistance 20Ω is connected in series with a 25 V battery and an external resistance 30Ω. A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E in (volt) is $\frac{x}{10}$. The value of is_____

Correct Answer:
2.5

Solution:

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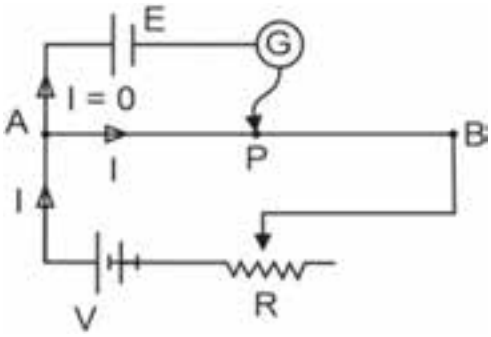

P $\longrightarrow$ Balancing point (null poind)

$\mathrm{v} = 25\mathrm{v}$
$R = 30\Omega$
$\mathrm{l_{AB}} = 10m$
$\mathrm{R_{AB}} = 20\Omega$

$\phi =$ potential gradient

$\phi = \dfrac{V_{AB}}{l_{AB}} = \dfrac{IR_{AB}}{l_{AB}}$

For p to be the null point

$$\mathrm{E = V_{AP} = IR_{AP} = \phi l_{AP}}$$

$$E = \frac{I\,(R_{AB})}{(l_{AB})} \times (l_{AP})$$

$$= \left(\frac{V}{R + R_{AB}}\right) \times \left(\frac{R_{AB}}{l_{AB}}\right) \times l_{AP}$$

$$E = 2.5V$$

the value of x is 2.5

Q. 24

Two travelling waves of equal amplitudes and equal frequencies move in opposite directions along a string. They interfere to produce a stationary wave whose equation is given by

$$\mathrm{y} = \left(10\cos\pi\ \mathrm{x}\sin\frac{2\pi\mathrm{t}}{\mathrm{T}}\right)cm$$

The amplitude of the particle at $\mathrm{x} = \dfrac{4}{3}\mathrm{cm}$ will be _____ cm.

Correct Answer:
-5

Solution:

$$\mathrm{y} = 10\cos(\pi\mathrm{x})\sin\left(\frac{2\pi\mathrm{t}}{\mathrm{T}}\right)$$

Comparing with general equation

$$\mathrm{y} = \underbrace{A\cos(Kx)\sin(\omega t)}_{\text{Amplitude}}$$

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Amplitude of the particle at $x = \frac{4}{3}\ cm = 10\cos(\pi x)$

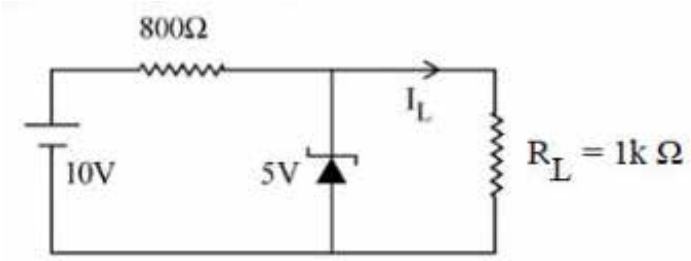
Amplitude $= A = 10\cos\left(\frac{4\pi}{3}\right)$

$A = -5$ cm

Negative sign denotes downward direction

Q. 25 In the given circuit, the value of current I will be___mA.

(When $R_L = 1k\ ;\Omega$)

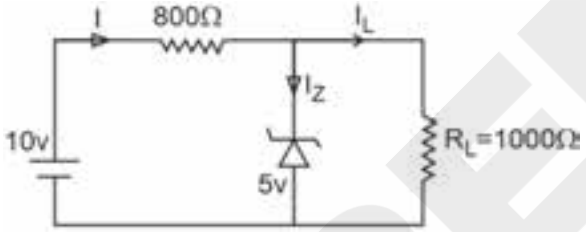


Correct Answer:

5

Solution:

There is breakdown of zener diode



$I = I_Z + I_L$

$V_Z = I_L R_L = 5$ V

$I_L = 5$ mA

Q. 26 A sample contains $10^{-2}kg$ each of two substances A and B with half lives 4 s and 8 s respectively. The ratio of their atomic weights is 1:2. The ratio of the amounts of A and B after 16 s is $\frac{x}{100}$. The value of x is_____

Correct Answer:

50

Solution:

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$$m_A = m_B = 10^{-2}\ \text{kg}$$

At time $t = 0$

$$\frac{(N_A)_0}{(N_B)_0} = \frac{m_A/M_A}{m_B/M_B} = \frac{M_B}{M_A} = \frac{2}{1}$$

$$(T_A)_{1/2} = 4s$$

$$(T_B)_{1/2} = 8\ \text{s}$$

$$N_A = (N_A)_0\, 2^{-t/(T_{1/2})_A}$$

$$N_B = (N_B)_0\, 2^{-t/(T_{1/2})_B}$$

$$\frac{N_A}{N_B} = \frac{(N_A)_0}{(N_B)_0} \times \frac{2^{-t/(T_{1/2})_A}}{2^{-t/(T_{1/2})_B}}$$

$$= 2 \times \frac{2^{-16/4}}{2^{-16/8}}$$

$$= 2 \times \frac{2^{-4}}{2^{-2}}$$

$$\frac{N_A}{N_B} = \frac{1}{2} = \frac{50}{100}$$

$$x = 50$$

Q. 27 A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is _____cm.

Correct Answer:
12

Solution:

$i = 60^\circ$, $\mu_2 = \sqrt{3}$

By snell's law

$\mu_1 \sin i = \mu_2 sinr$

$1 \times \sin 60^\circ = \sqrt{3} sinr$

$r = 30^\circ$

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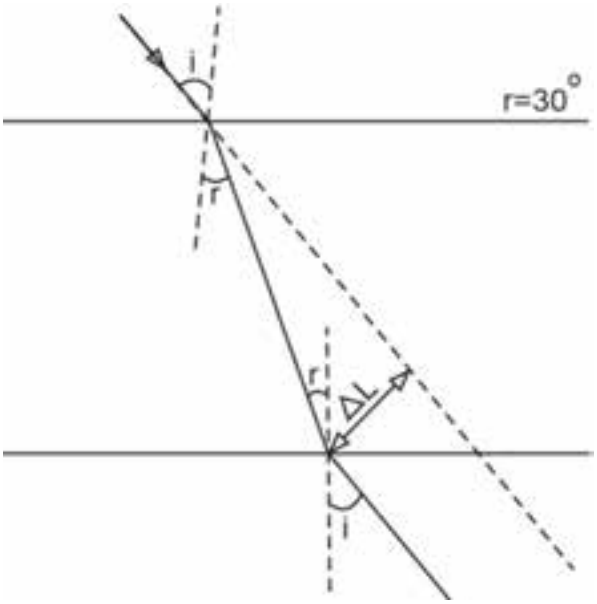

$$\Delta L = \text{ Lateral shift } = 4\sqrt{3} \text{ cm}$$

$$4\sqrt{3} = \frac{t \sin(i-r)}{\cos r}$$

$$4\sqrt{3} = \frac{t \sin 30^\circ}{\cos 30^\circ}$$

$$t = 12 \text{ cm}$$

The thickness of the glass slab is 12 cm

Q. 28 A circular coil of 1000 turns each with area $1m^2$ is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T. The maximum voltage generation will be

Correct Answer:
70v

Solution:

$$e = e_0 \sin \omega t$$

$$e_0 = nAB\omega$$

$$= 1000 \times 1 \times 0.07 \times \left(\frac{2\pi \text{rad}}{1 \text{ s}}\right)$$

$$e_0 = 70 \text{ V}$$

Q. 29 A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be____ R during this transformation. Where R is the gas constant.

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Correct Answer:

$2\mathrm{R}$

Solution:

$w = \frac{Q}{4}$

$\gamma = \frac{5}{3}$

By First law of thermodynamics,

$Q = W + \Delta U$

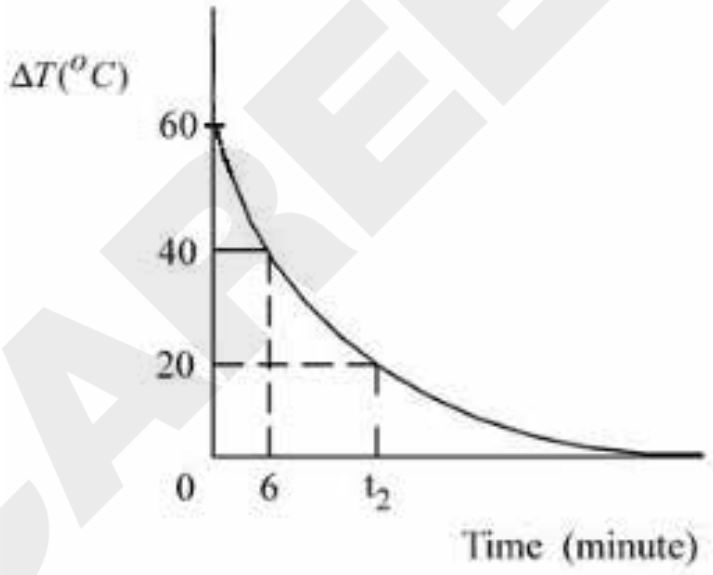
$\Delta U = Q - W = \frac{3Q}{4} = nC_v\Delta T$

$\Delta U = \frac{3Q}{4} = \frac{nR\Delta T}{(2/3)}$

$\frac{nc\Delta T}{2} = \frac{Q}{2} = nR\Delta T$

$C = 2R$

Q. 30 In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference $(\Delta \mathrm{T})$ of the water and surroundings and time as shown in figure. The initial temperature of water is taken as $80^{\circ}C$. The value of t_2, as mentioned in the graph will be_____.



Correct Answer:

$10\mathrm{mm}$

Solution:

Rate of cooling: $\mathrm{R} = \frac{-\Delta \mathrm{T}}{\Delta \mathrm{t}}$

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At $t = 0 \quad = \mathrm{k}\left(\mathrm{T}_{\mathrm{avg}} - \mathrm{T}_0\right)$

$\mathrm{T}_1 = 80^{\circ}\mathrm{C} =$ Temperature of water

$T_1 - T_{\text{surrounding}} = 60$

$T_0 = T_{\text{surrounding}} = 20^{\circ}\mathrm{C}$

At $t = 6$ s

$T^t_{\text{temperatune of water}} = T_2$

$T_2 - T_{\text{surrounding}} = 40$

$T_2 = 60$

At time $= t_2$

Temperature of water $= T_3$

$T_3 - T_{surrounding} = 20$

$T_3 = 40^{\circ}\mathrm{C}$

$from\ t = 0\ to\ t = 6$ s

$$-\frac{\Delta T}{\Delta t} = -\frac{(T_2 - T_1)}{\Delta t} = K\left(\frac{T_1 + T_2}{2} - T_0\right)$$

$$\frac{20}{6} = K(70 - 20)$$

$$k = \frac{1}{15} \rightarrow (1)$$

From $t = 0$ to $t = t_2$

$$-\frac{\Delta t}{\Delta t} = \frac{-(t_3 - t_2)}{2 + (t_2 - 0)} = k\left(\frac{t_2 + T_3}{2} - T_0\right)$$

$$\frac{20}{t_2} = \left(\frac{1}{15}\right)(30)$$

$$t_2 = 10 \text{ min}$$

Chemistry

Q. 1 120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2, and 270 g of water on complete combustion, The percentage of carbon and hydrogen, respectively are

Option 1:
25 and 75

Option 2:
40 and 60

Option 3:
60 and 40

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Option 4:
75 and 25

Correct Answer:
75 and 25

Solution:

$$C_xH_y + \left(x + \frac{y}{4}\right) O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$$

$$120\ g \qquad\qquad 330\ g \quad 270\ g$$

$$\text{So, moles of } CO_2 = \frac{330}{44} = \frac{15}{2}\ \text{mol} = x$$

$$\text{and, moles of } H_2O = \frac{270}{18} = 15\ \text{mol} = \frac{y}{2}$$

$\therefore$ The hydrocarbon is $C_{15}H_{60}$ $\left(\because C_{\frac{15}{2}}H_{30} = C_{15}H_{60}\right)$

Now,

$$\text{mass \% of C} = \frac{180}{240} \times 100 = 75\% \text{mass \% of H} = \frac{60}{240} \times 100 = 25\%$$

Hence, the correct **option (4)**

Q. 2 The energy of one mole of photons of radiation of wavelength 300 nm is

(Given : $h = 6.63 \times 10^{-34} Js,\ \ N_A = 6.02 \times 10^{23} mol^{-1},\ \ c = 3 \times 10^8 ms^{-1}$)

Option 1:
$235\ kJ\ mol^{-1}$

Option 2:
$325\ kJ\ mol^{-1}$

Option 3:
$399\ kJ\ mol^{-1}$

Option 4:
$435\ kJ\ mol^{-1}$

Correct Answer:
$399\ kJ\ mol^{-1}$

Solution:
We know,

$$\text{Energy of one photon, } E = h\nu = \frac{hc}{\lambda}$$

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$$\text{Energy of 1 mole photon} = \mathrm{E} \times \mathrm{N_A} = \frac{\mathrm{hc}}{\lambda} \times \mathrm{NA}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{300 \times 10^{-2}}$$

$$= 399 \text{ kJ mol}^{-2}$$

Hence, the correct answer is **Option (3)**

Q. 3 The correct oredr of bond orders of $\mathrm{C_2}^{2-}$, $\mathrm{N_2}^{2-}$ and $\mathrm{O_2}^{2-}$ is respectively

Option 1:
$\mathrm{C_2^{2-}} < \mathrm{N_2^{2-}} < \mathrm{O_2^{2-}}$

Option 2:
$\mathrm{O_2^{2-}} < \mathrm{N_2^{2-}} < \mathrm{C_2^{2-}}$

Option 3:
$\mathrm{C_2^{2-}} < \mathrm{O_2^{2-}} < \mathrm{N_2^{2-}}$

Option 4:
$\mathrm{N_2^{2-}} < \mathrm{C_2^{2-}} < \mathrm{O_2^{2-}}$

Correct Answer:
$\mathrm{O_2^{2-}} < \mathrm{N_2^{2-}} < \mathrm{C_2^{2-}}$

Solution:
According to MOT ,

$$\mathrm{B.O.} = \frac{\mathrm{N_B} - \mathrm{N_A}}{2}$$

$$\mathrm{C_2^{2-}}, \mathrm{B.O.} = \frac{6-0}{2} = 3$$

$$\mathrm{N_2^{2-}}, \mathrm{B.O.} = \frac{6-2}{2} = 2$$

$$\mathrm{O_2^{2-}}, \mathrm{B.O.} = \frac{6-4}{2} = 1$$

So, order of Bond order is $\mathrm{O_2^{2-}} < \mathrm{N_2^{2-}} < \mathrm{C_2^{2-}}$

Hence, the correct answer is **option (2)**

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Q. 4 At $25^{\circ}C$ and 1 atm pressure, the enthalpies of combustion are as given below:

Substance	H_2	C (graphite)	$C_2H_6(g)$
$\frac{\Delta_c H^{\Theta}}{kJ\ mol^{-1}}$	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is

Option 1:
$+54.0\ kJ\ mol^{-1}$

Option 2:
$-68.0\ kJ\ mol^{-1}$

Option 3:
$-86.0\ kJ\ mol^{-1}$

Option 4:
$+97.0\ kJ\ mol^{-1}$

Correct Answer:
$-86.0\ kJ\ mol^{-1}$

Solution:
Given,

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \quad \Delta_c H = -286\ kJ/mol \ ----(1)$$
$$C_{(s)} + O_2 \rightarrow CO_2 \quad \Delta_c H = -394\ kJ/mol ------(2)$$
$$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O \quad \Delta_C H = -1560\ kJ\ /mol ----- \ (3)$$

The enthapy of formation of ethane :

$$2C_{(s)} + 3H_2 \rightarrow C_2H_6 \quad \Delta_C H = \ ?$$

The formation reaction of ethane can be obtained by $2 \times (2) + 3 \times (1) - (3)$

Then ,

$$\begin{aligned}\Delta_f H \text{ of } C_2H_6 &= 2 \times (-394) + 3 \times (-286) - (-1560) \\ &= -1646 + 1560 \\ &= -86\ kJ/mol\end{aligned}$$

Hence, the correct answer is **Option (3)**

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Q. 5 For a first order reaction, the time required for completion of 90% reaction is 'x' times the half life of the reaction. The value of 'x' is (Given: $\ln 10 = 2.303$ and $\log 2 = 0.3010$)

Option 1:
1.12

Option 2:
2.43

Option 3:
3.32

Option 4:
33.31

Correct Answer:
3.32

Solution:
First order Reaction,

$$\ln \frac{[A]}{[A]_0} = -kt, t_{1/2} = \frac{\ln 2}{K}$$

Now,

$$90\% \text{ Reaction completed} \Rightarrow [A] = \frac{10}{100}[A]_0 \Rightarrow [A] = 0.1[A]_0$$

Putting this in the intgrated rate equation, we have

$$\ln\left(\frac{0.1[A]_0}{[A]_0}\right) = -kt_{90\%}$$

So, $t_{90\%} = \frac{\ln\left(\frac{10}{1}\right)}{k} = x \times t_{50\%} = x\frac{\ln 2}{k}$

$$\Rightarrow \quad x = \frac{\ln\left(\frac{10}{1}\right)}{\ln 2} = \frac{\ln 10}{\ln 2} = 3.32$$

Hence, the correct answer is **Option (3)**

Q. 6 Metals generally melt at vey high temperature. Amongst the following, the metal with the highest melting point will be

Option 1:
Hg

Option 2:
Ag

Option 3:
Ga

Option 4:
Cs

Correct Answer:
Ag

Solution:
The melting point data is given below:

$$\mathrm{Hg} = \text{ liquid at Room temperature}$$
$$\mathrm{Ag} = \text{ Above } 900^{\circ}\mathrm{K}$$
$$\mathrm{Cs} = 302 \text{ K}$$
$$\mathrm{Ga} = 303 \text{ K}$$

So, Ag has highest melting point among the given metals.

Hence, the correct answer is **Option (2)**

Q. 7 Which of the following chemical reactions represents Hall-Heroult Process?

Option 1:
$\mathrm{Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr}$

Option 2:
$\mathrm{2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2}$

Option 3:
$\mathrm{FeO + CO \rightarrow Fe + CO_2}$

Option 4:
$\mathrm{2[Au(CN)_2]^-_{(aq)} + Zn(s) \rightarrow 2Au(s) + [Zn(CN_4)]^{2-}}$

Correct Answer:
$\mathrm{2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2}$

Solution:
Hall - Heroult process is used in the electrolytic refining of Aluminium.

A steel vessel with the lining of carbon and graphite rods is used in the process. The carbon lining acts as cathode while graphite acts as anode.

The overall reaction occuring in the process can be represented as

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$2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

Option (2) is correct.

Q. 8 In the industrial production of which of the following, molecular hydrogen is obtained as a byproduct?

Option 1:
$NaOH$

Option 2:
$NaCl$

Option 3:
$Na \text{ metal}$

Option 4:
Na_2CO_3

Correct Answer:
$NaOH$

Solution:
H_2 is obtained as by product is the manufacture of $NaOH \text{ and } Cl_2$ by the electrolysis of Brine solution.

Option(1) is correct

Q. 9 Which one of the following compounds is used as a chemical in certain type of fire extinguishers?

Option 1:
Baking soda

Option 2:
Soda ash

Option 3:
Washing soda

Option 4:
Caustic Soda

Correct Answer:
Baking soda

Solution:
Baking soda, $NaHCO_3$ is used as a chemical in certain type of fire extinguishers.

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Option (1) is correct.

Q. 10 PCl_5 is well known, but NCl_5 is not. Because,

Option 1:
nitrogen is less reactive than phosphorous.

Option 2:
nitrogen doesn't have d-orbitals in its valence shell.

Option 3:
catenation tendency is weaker in nitrogen than phosphorous.

Option 4:
size of phosphorous is larger than nitrogen.

Correct Answer:
nitrogen doesn't have d-orbitals in its valence shell.

Solution:
PCl_5 well known, but NCl_5 is not because nitrogen doesn't have d-orbitals in its valence shell. So nitrogen cannot make 5 bonds and cannot expand its octet.

Option (2) is correct.

Q. 11 Transition metal complex with highest value of crystal field splitting (Δ_0) will be

Option 1:
$[Cr(H_2O)_6]^{3+}$

Option 2:
$[Mo(H_2O)_6]^{3+}$

Option 3:
$[Fe(H_2O)_6]^{3+}$

Option 4:
$[Os(H_2O)_6]^{3+}$

Correct Answer:
$[Os(H_2O)_6]^{3+}$

Solution:
As, Os belongs to the 5d transition series of elements, it is bigger than the others. So, it will have a greater spliting and correspondingly, it will have highest value of Δ_o among them.

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Option (4) is correct.

Q. 12 Some gases are responsible for heating of atmosphere (green house effect). Identify from the following the gaseous species which does not cause it.

Correct Answer:
N_2

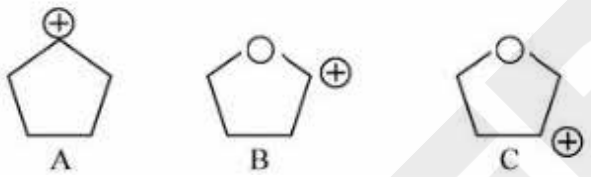
Solution:
Green house gases are

$CO_2, CH_4, H_2O_{(g)}, CFCs, N_2O$ and Ozone

Thus, N_2 is not responsible for greenhouse effect and global warming

Option (4) is correct.

Q. 13 Arrange the following carbocations in decreasing order of stability



Option 1:
$B > A > C$

Option 2:
$A > B > C$

Option 3:
$C > B > A$

Option 4:
$C > A > B$

Correct Answer:
$B > A > C$

Solution:
The given Carbocations can be arranged in order of their stability as

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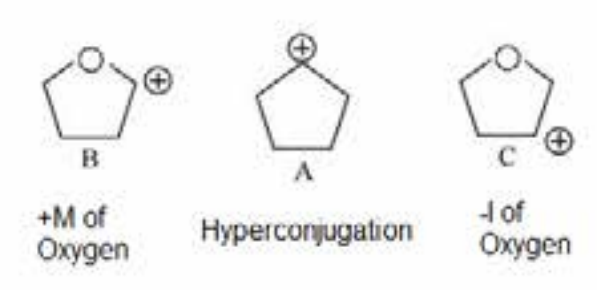

Thus, decreasing order of stability of carbocations is

$$B > A > C$$

Option (1) is correct.

Q. 14 Given below are two statements.

Statement I: The presence of weaker π-bonds make alkenes less stable than alkanes.

Statement II: The strength of the double bond is greater than that of carbon-carbon single bond.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

Solution:
As we have learnt,

The presence of weaker π-Bonds make alkenes less stable than alkanes, this is true.
The strength of double bond is greater than that of carbon - carbon single bond, this is also correct.

So, Both statement I and statement II are correct.

Option (1) is correct.

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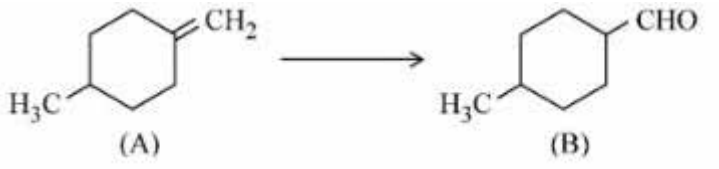
Q. 15 Which of the following reagents / reactions will convert 'A' to 'B'?



Option 1:
PCC oxidation

Option 2:
Ozonolysis

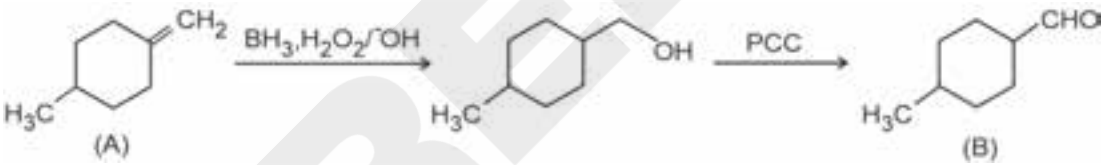
Option 3:
$BH_3, H_2O_2/^-OH$ followed by PCC oxidation

Option 4:
HBr, hydrolysis followed by oxidation by $K_2Cr_2O_7$

Correct Answer:
$BH_3, H_2O_2/^-OH$ followed by PCC oxidation

Solution:
$BH_3, H_2O_2/^-OH$ will convert double to alcohol.

The PCC convert alcohol to Aldehyde.



Option (3) is correct.

Q. 16 Hex-4-ene-2-ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is

Option 1:
2-pentene

Option 2:
proponaldehyde

Option 3:
2-butene

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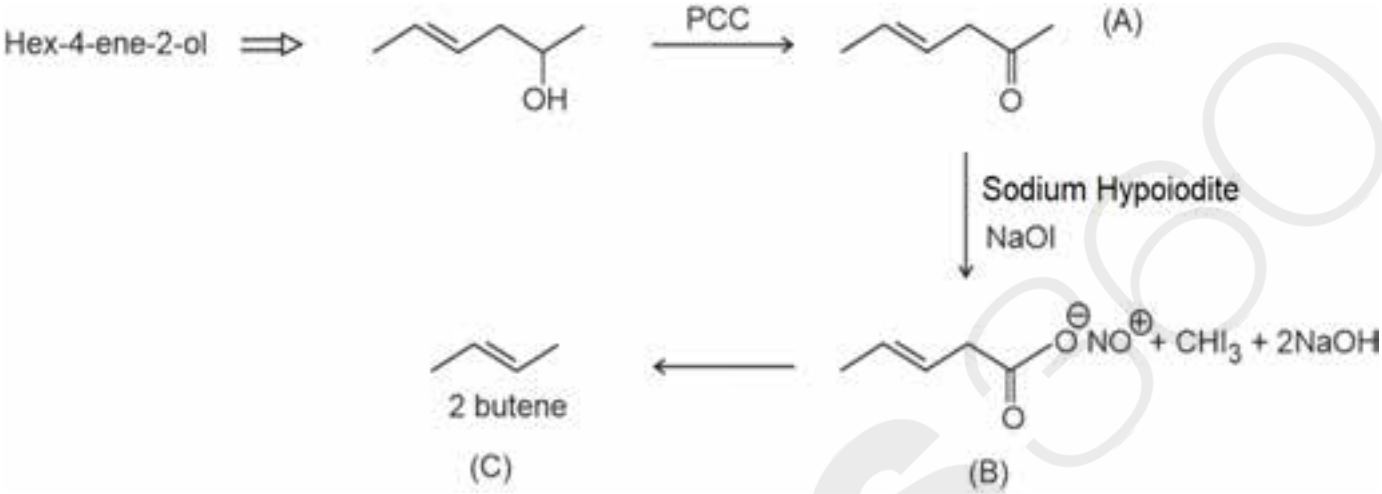
Option 4:
4-methylpent-2-ene

Correct Answer:
2-butene

Solution:



Option (3) is correct.

Q. 17 The conversion of propan-1-ol to n-butylamine involves the sequential addition of reagents. The correct sequential order of reagents is

Option 1:
$(\mathrm{i})\mathrm{SOCl_2}\ (\mathrm{ii})\mathrm{KCN}\ (\mathrm{iii})\mathrm{H_2/Ni,\ Na(Hg)/C_2H_5OH}$

Option 2:
$(\mathrm{i})\mathrm{HCl}\ (\mathrm{ii})\mathrm{H_2/Ni,\ Na(Hg)/C_2H_5OH}$

Option 3:
$(\mathrm{i})\mathrm{SOCl_2}\ (\mathrm{ii})\mathrm{KCN}\ (\mathrm{iii})\mathrm{CH_3NH_2}$

Option 4:
$(\mathrm{i})\mathrm{HCl}\ (\mathrm{iii})\mathrm{CH_3NH_2}$

Correct Answer:
$(\mathrm{i})\mathrm{SOCl_2}\ (\mathrm{ii})\mathrm{KCN}\ (\mathrm{iii})\mathrm{H_2/Ni,\ Na(Hg)/C_2H_5OH}$

Solution:

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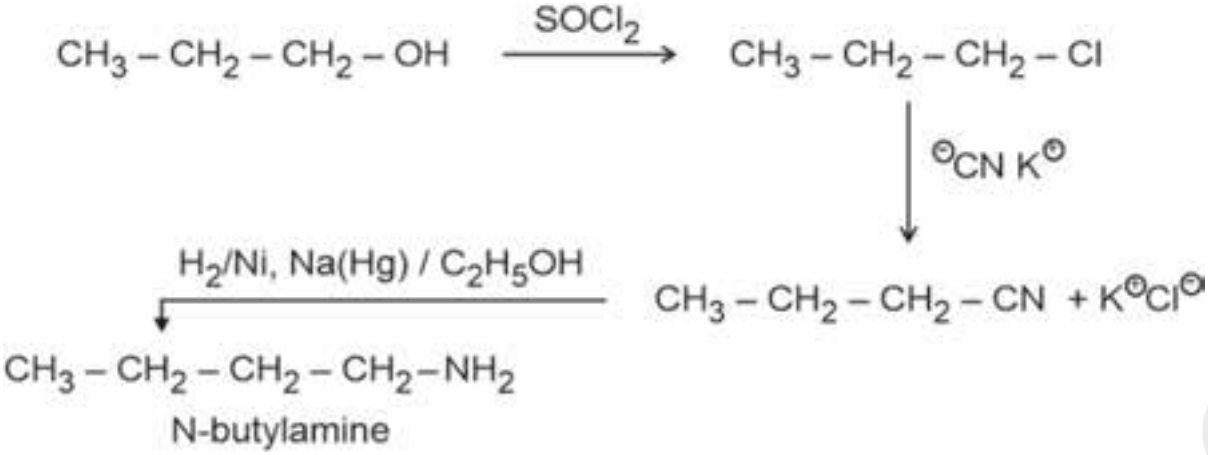

Option (1) is correct.

Q. 18 Which of the following is **not** an example of a condensation polymer?

Option 1:
Nylon 6,6

Option 2:
Dacron

Option 3:
Buna-N

Option 4:
Silicone

Correct Answer:
Buna-N

Solution:
The given polymers can be classified as

$$\text{Addition polymers : Buna - N}$$

$$\text{Condensation polymers : Nylon} - 6,6, \text{Dacron, Silicone}$$

So, Buna - N is not condensation polymer out of the given polymers

Option (3) is correct.

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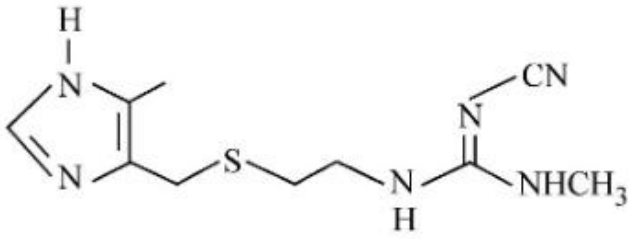
Q. 19 The structure shown below is of which well-known drug molecule ?



Option 1:
Ranitidine

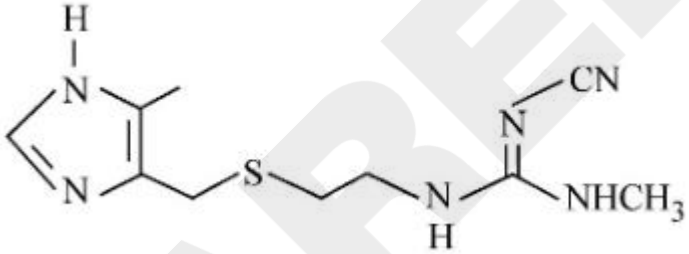
Option 2:
Seldane

Option 3:
Cimetidine

Option 4:
codeine

Correct Answer:
Cimetidine

Solution:
The given structure is of Cimetidine which is well known drug molecule.



Option (3) is correct.

Q. 20 In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present?

Option 1:
Cu^{2+}

Option 2:
Sr^{2+}

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Option 3:
Ba^{2+}

Option 4:
Ca^{2+}

Correct Answer:
Cu^{2+}

Solution:
In the flame test,

Cu^{2+} : green flame with blue centre

Sr^{2+} : crimson red

Ba^{2+} : Apple green

Ca^{2+} : Brick Red

So, **Option (1)** is correct.

Q. 21 At 300 K, a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is_____g mol^{-1}. (nearest integer)
Assume that the behaviour of gases as ideal.

(Given: The molar mass of hydrogen (H_2) gas is 2.0 g mol^{-1}.)

Correct Answer:
45

Solution:
Let molar mass of $A = M_A$

Given, A and H_2 occupy same volume at the same pressure.

$$V_A = V_{H_2}$$

So, from the ideal gas equation, we have

$$\frac{n_A RT_1}{P} = \frac{n_{H_2} RT_2}{P}$$

$$\Rightarrow 300 \times n_A = n_{H_2} \times 200$$

$$\Rightarrow 300 \times \frac{3.0\ g}{M_A} = \frac{0.2\ g}{2.0\ g/mol} \times 200$$

$$\Rightarrow M_A = 30 \times \frac{3}{2}\ g/mol$$

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$\Rightarrow \mathrm{M_A} = 45\ \mathrm{g/mol}$

Hence, the answer is 45

Q. 22 A company dissolves 'x' amount of $\mathrm{CO_2}$ at 298 K in 1 litre of water to prepare soda water. $\mathrm{X} =$_____$\times 10^{-3}$. (nearest integer)

(Given: partial pressure of $\mathrm{CO_2}$, at 298 K = 0.835 bar.

Henry's law constant for $\mathrm{CO_2}$ at 298K = 1.67 kbar.

Atomic mass of H, C and O is 1, 12, and 16 g mol^{-1}, respectively)

Correct Answer:
1221

Solution:
We have to find amount of $\mathrm{CO_2}$ dissolved, in 1 litre of $\mathrm{H_2O}$

Given,

$\mathrm{P_{CO_2}} = 0.835\ \mathrm{bar}$,

Henry's law constant, $\mathrm{K_H} = 1.67\ \mathrm{kbar}$

Using Henry's Law, we have

$\mathrm{P} = \mathrm{K_H X}, \quad \mathrm{X\ is\ mole\ fraction}$

Let the mass of $\mathrm{CO_2}$ dissolved be x. So we can write

$$0.835 = 1.67 \times 10^3 \times \left(\frac{\frac{x}{M_{CO_2}}}{\frac{x}{M_{CO_2}} + \frac{1000}{18}} \right) \quad (\because \mathrm{1ml = 1g\ of\ H_2O})$$

$$0.835 = 1.67 \times 10^3 \left(\frac{\frac{x}{44}}{\frac{x}{44} + 55.5} \right) = 1.67 \times 10^3 \left(\frac{\frac{x}{44}}{55.5} \right)$$

Now, assuming a very dilute solution, we can approximate as

$$55.5 >> \frac{x}{44} \Rightarrow 55.5 + \frac{x}{44} \simeq 55.5$$

Then,

$$\frac{x}{44} = \frac{0.835 \times 55.5}{1.67 \times 10^3} \Rightarrow x \approx 1221 \times 10^{-3}\ \mathrm{g}$$

Correct answer is 1221

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Q. 23 PCl_5, dissociates as

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2, and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p, for the dissociation of PCl_5 is _____ $\times 10^{-3}$. (nearest integer)

(Given: $R = 0.082\ L\ atm\ K^{-1}mol^{-1}$: Assume ideal gas behaviour)

Correct Answer:
1107

Solution:

$$\begin{array}{ccccc} PCl_5 & \rightleftharpoons & PCl_3 & + & Cl_2 \\ P_\circ - x & & x & & x \end{array}$$

$$K_p = \frac{x^2}{P_0 - x}$$

Now, initially PCl_5 and N_2 were added to the container. N_2 is inert to the system and will not react under the given conditions

$$P_{N_2} = \frac{nRT}{V} = \frac{2 \times R \times 600}{200} = 6R$$

$$P_0 = P_{PCl_5} = \frac{nRT}{V} = \frac{5 \times R \times 600}{200} = 15R$$

Given,

$$P_{eq} = P_0 - x + x + x + P_{N_2}$$

$$2.46 = P_0 + x + P_{N_2} = 15R + x + 6R$$

$$2.46 = 21R + x$$

$$x = 2.46 - 21R = 2.46 - 21 \times 0.082$$

$$x = 2.46 - 1.722 = 0.738$$

and,

$$P_0 - x = 15R - 0.738 = 1.23 = 0.492$$

Now, $K_p = \frac{x^2}{P_0 - x} = \frac{(0.738)^2}{0.492} = 1.107$

$$K_p = 1107 \times 10^{-3}$$

Hence, the answer is 1107

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Q. 24 The resistance of a conductivity cell containing 0.01 M KCl solution at 298 K is 1750 Ω. If the conductivity of 0.01M KCl solution at 298 K is $0.152 \times 10^{-3}\ \mathrm{S\ cm^{-1}}$, then the cell constant of the conductivity cell is _____$\times 10^{-3}\ \mathrm{cm^{-1}}$

Correct Answer:
266

Solution:
We know the formula

$$\mathrm{Conductivity(K) = Conductance(G)\ \times\ cell\ constant(G^*)}$$

$$\mathrm{K = G \times \left(\frac{\ell}{A}\right)}$$

$$\mathrm{Resistance\ R = \frac{1}{G}}$$

Therefore, we can write

$$\mathrm{K = \frac{1}{R}\left(\frac{\ell}{A}\right)}$$

$$0.152 \times 10^{-3} = \frac{1}{1750} \times \left(\frac{\ell}{A}\right)$$

$$\left(\frac{\ell}{A}\right) = 266 \times 10^{-3}\ \mathrm{cm^{-1}}$$

Hence, the answer is 266

Q. 25 When 200 mL of 0.2 M acetic acid is shaken with 0.6 g of wood charcoal, the final concentration of acetic acid after adsorption is 0.1 M. The mass of acetic acid adsorbed per gram of carbon is_____ g.

Correct Answer:
2

Solution:
$$\mathrm{moles\ of\ CH_3COOH = 0.2M \times 0.2(L) = 0.04\ mole}$$

Given due to adsorption, Molarity changes from 0.2 M to 0.1 M

So, moles of $\mathrm{CH_3COOH}$ remaining in solution= 0.02

So, 0.02 moles of are consumed by 0.06 of wood

Now,

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0.6 g has adsorbed 0.02 mole of CH_3COOH

0.6 g has adsorbed 0.02×60 gram of CH_3COOH

1 g of charcoal has adsorbed $\frac{0.02 \times 60}{0.6}$ gram of CH_3COOH

1 g of charcoal has adsorbed 2 gram of CH_3COOH

So, 2 gram of acetic acid is adsorbed per gram of carbon.

Hence, the answer is 2

Q. 26 (a) Baryte, (b) Galena, (c) Zinc blende and (d) Copper pyrites. How many of these minerals are sulphide based?

Correct Answer:
3

Solution:
The ores and their formula are given below:

$a)\text{Baryte} - BaSO_4(\text{sulphate})$

$b)\text{Galena} - PbS(\text{sulphide})$

$c)\text{ZincBlende} - ZnS(\text{sulphide})$

$d)\text{Copper pyrites} - CuFeS_2(\text{sulphide})$

So, b,c and d are sulphide ores

Hence, the answer is 3.

Q. 27 Manganese (VI) has ability to disproportionate in acidic solution. The difference in oxidation states of two ions it forms in acidic solution is_____.

Correct Answer:
3

Solution:
Manganese (VI) is not stable in acidic medium and undergoes disproportionation reaction.

The reaction is given below:

$$\underset{(VI)}{3MnO_4^{2-}} + 4H^+ \longrightarrow \underset{(VII)}{2MnO_4^-} + \underset{(IV)}{MnO_2} + 2H_2O$$

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The difference in oxidation state of two lons

$= 7 - 4 = 3$

Hence, the answer is 3

Q. 28 0.2 g of an organic compound was subjected to estimation of nitrogen by Dumas method in which volume of N_2, evolved (at STP) was found to be 22.400 mL. The percentage of nitrogen in the compound is_____[nearest integer] (Given: Molar mass of N_2, is $28\ g\ mol^{-1}$ Molar volume of N_2 at STP : 22.4L)

Correct Answer:
14

Solution:
The estimation by Duma's method involves the reaction:

$$C_xH_yN_z + (2x + \frac{y}{2})CuO \rightarrow xCO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + (2x + \frac{y}{2})Cu$$

$$\text{mole of } N_2 \text{ Volume } = 22.400\ mL = 22.4 \times 10^{-3} L$$

$$\text{mole of } N_2 = \frac{22.4 \times 10^{-3}}{22.4}$$

$$\text{mole of } N_2 = 10^{-3} \text{ mole}$$

$$\text{mass of } N_2 = 28 \times 10^{-3} \text{ gram}$$

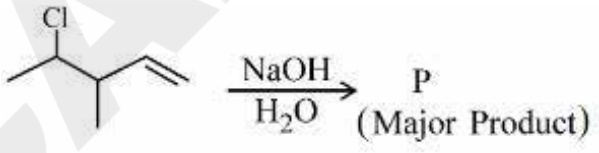
% of nitrogen in the compound

$$= \frac{28 \times 10^{-3}}{0.2} \times 100$$

$$= 14\%$$

Hence, the answer is 14

Q. 29



Consider the above reaction. The number of π electrons present in the product P is _____

Correct Answer:
2

Solution:
Substitution reaction will happen as the solvent is H_2O which is a polar protic solvent.

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Only one double bond will be present and alcohol will be formed.

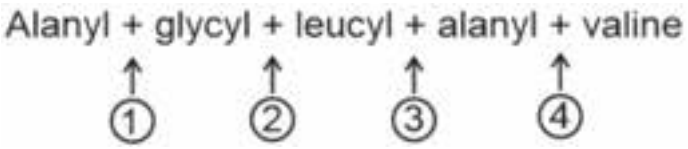
So, number of π electrons present in product is 2 .

Hence, the answer is 2

Q. 30 In alanylglycylleucylalanylvaline, the number of peptide linkages is_____.

Correct Answer:
4

Solution:
In the given polypeptide, alanylglycylleucylalanylvaline the amino acids are



It is a pentapeptide with four peptide linkages combining the amino acids

Thus, there are four linkages present

Hence, the answer is 4

Maths

Q. 1 Let $\mathrm{x} * \mathrm{y}=x^2+y^3$ and $(\mathrm{x} * 1) * 1=\mathrm{x} *(1 * 1)$

Then a value of $2 \sin^{-1}\left[\frac{\mathrm{x}^4+\mathrm{x}^2-2}{\mathrm{x}^4+\mathrm{x}^2+2}\right]$ is

Option 1:
$\frac{\pi}{4}$

Option 2:
$\frac{\pi}{3}$

Option 3:
$\frac{\pi}{2}$

Option 4:
$\frac{\pi}{6}$

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Correct Answer:

$\frac{\pi}{3}$

Solution:

As $(x * 1) * 1 = x * (1 * 1)$

$\therefore (x^2 + 1) * 1 = x * 2$

$(x^2 + 1)^2 + 1 = x^2 + 8$

$x^4 + x^2 - 6 = 0$

$\therefore x^2 = 2$

$2\sin^{-1}\left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2}\right) = 2\sin^{-1}\left(\frac{4}{8}\right) = \frac{\pi}{3}$

Q. 2 The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^{x} + 1) = 0$ is

Option 1:

$\log_e 3$

Option 2:

$-\log_e 3$

Option 3:

$\log_e 6$

Option 4:

$-\log_e 6$

Correct Answer:

$-\log_e 3$

Solution:

$\left(e^{2x} - 4\right)\left(6e^{2x} - 5e^{x} + 1\right) = 0$

$\left(e^{x} - 2\right)\left(e^{x} + 2\right)\left(3e^{x} - 1\right)\left(2e^{x} - 1\right) = 0$

$e^{x} = 2, \frac{1}{3}, \frac{1}{2}$; e^{x} can not be -2

$\therefore \quad x = \ln 2, -\ln 3, -\ln 2$

$\therefore$ sum of all real solutions = $-\log_e 3$

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Q. 3 Let the system of linear equations

$$x + y + \alpha z = 2$$
$$3x + y + z = 4$$
$$x + 2z = 1$$

Have a unique solution (x^*, y^*, z^*) If $(\alpha, x^*), (y^*, \alpha)$ and $(x^*, -y^*)$ are collinear points, then the sum of absolute values of all possible values of α is

Option 1:
4

Option 2:
3

Option 3:
2

Option 4:
1

Correct Answer:
2

Solution:

$x + y + \alpha z = 2$

$3x + y + z = 4$

$x + 2z = 1$

For unique solution:

$$D = \begin{vmatrix} 1 & 1 & \alpha \\ 3 & 1 & 1 \\ 1 & 0 & 2 \end{vmatrix} \neq 0$$

$$2 - 5 - \alpha \neq 0$$
$$\alpha \neq -3$$

$$D_1 = \begin{vmatrix} 2 & 1 & \alpha \\ 4 & 1 & 1 \\ 1 & 0 & 2 \end{vmatrix} = -\alpha - 3$$

$$D_2 = \begin{vmatrix} 1 & 2 & \alpha \\ 3 & 4 & 1 \\ 1 & 1 & 2 \end{vmatrix} = -\alpha - 3$$

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$$D_3 = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 4 \\ 1 & 0 & 1 \end{vmatrix} = 0$$

Hunce $\alpha \neq -3, \quad x = 1 = y, z = 0$

Non

$$\begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & -1 & 1 \end{vmatrix} = 0$$

$$\alpha^2 = 1$$

$$\alpha = \pm 1$$

Q. 4 Let $x, y > 0$. If $x^3y^2 = 2^{15}$, then the least value of $3x + 2y$ is

Option 1:
30

Option 2:
32

Option 3:
36

Option 4:
40

Correct Answer:
40

Solution:
$x, y > 0$

$x^3y^2 = 2^{15}$

we know that $A.m \geq G.m$

$$\frac{x + x + x + y + y}{5} \geq \left(x^3y^2\right)^{\frac{1}{5}}$$

$$\frac{3x + 2y}{5} \geq 2^3$$

$$3x + 2y \geq 0$$

$\therefore$ least value of $3x + 2y = 40$

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Q. 5

Let $f(x)=\begin{cases}\frac{\sin(x-[x])}{x-[x]} & ,x\in(-2,-1)\\ \max\{2x,3[|x|]\} & ,|x|<1\\ 1 & ,\text{ otherwise}\end{cases}$

where [t] denotes greatest integer $\leq t$. If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is:

Option 1:
(3,3)

Option 2:
(2,4)

Option 3:
(2,3)

Option 4:
(3,4)

Correct Answer:
(2,3)

Solution:

$$f(x)=\begin{cases}\frac{\sin(x-[x])}{(x-[x])}; & x\in(-2,7)\\ \max\{2x,3[|x|]\};|x|<1\end{cases}$$

$$f(x)=\begin{cases}\frac{\sin(x+2)}{x+2}; & x\in(-2,-1)\\ 0 & ;(-1,0]\\ 2x & ;[0,1)\\ 1 & ;\text{ atherisise}\end{cases}$$

Discontinuous at $x=-1,1$

Non differentiable at $x=-1,1,-2$

$\therefore m=2$

$n=3$

$(m,n)\equiv(2,3)$

Q. 6

The value of the integral $\int_{-\pi/2}^{\pi/2}\frac{dx}{(1+e^x)\left(\sin^6 x+\cos^6 x\right)}$ is equal to

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Option 1:

2π

Option 2:

0

Option 3:

π

Option 4:

$\frac{\pi}{2}$

Correct Answer:

π

Solution:

$$I = \int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)\left(\sin^6 x + \cos^6 x\right)} \text{.....(i)}$$

also

$$I = \int_{-\pi/2}^{\pi/2} \frac{e^x dx}{(1+e^x)\left(\sin^6 x + \cos^6 x\right)} \text{.....(ii)}$$

on adding qeuation (i) and (ii)

$$2I = \int_{-\pi/2}^{\pi/2} \frac{dx}{\left(\sin^6 x + \cos^6 x\right)}$$

$$2I = 2\int_{0}^{\pi/2} \frac{dx}{\left(\sin^4 x - \sin^2 x\cos^2 x + \cos^4 x\right)}$$

$$I = \int_{0}^{\pi/2} \frac{(1+\tan^2 x)\sec^2 x dx}{(\tan^4 x - \tan^2 x + 1)}$$

Let $\tan x = t$

$$I = \int_{0}^{\infty} \frac{1+t^2}{(t+-t^2+1)} dt$$

$$I = \int_{0}^{\infty} \frac{\left(1+\frac{1}{t^2}\right)dt}{\left(t-\frac{1}{t}\right)^2+1}$$

let $t - \frac{1}{t} = z$

$$I = \int_{-\infty}^{\infty} \frac{dz}{z^2+1} = \left[\tan^{-1} z\right]_{-\infty}^{\infty}$$

$$= \pi$$

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Q. 7 $\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+\frac{n^2}{(n^2+9)(n+3)}+\ldots+\frac{n^2}{(n^2+n^2)(n+n)}\right)$

Option 1:
$\frac{\pi}{8}+\frac{1}{4}\log_e 2$

Option 2:
$\frac{\pi}{4}+\frac{1}{8}\log_e 2$

Option 3:
$\frac{\pi}{4}-\frac{1}{8}\log_e 2$

Option 4:
$\frac{\pi}{8}+\log_e\sqrt{2}$

Correct Answer:
$\frac{\pi}{8}+\frac{1}{4}\log_e 2$

Solution:

$$\lim_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+\cdots+\frac{n^2}{(n^2+n^2)(n+n)}\right)$$

$$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{n^2}{(n^2+r^2)(n+r)}$$

$$=\int_0^1\frac{1}{(1+x^2)(1+x)}dx$$

$$=\frac{1}{2}\int_0^1\frac{1}{1+x}dx+\frac{1}{2}\int_b^1\frac{dx}{1+x^2}-\frac{1}{4}\int_0^1\frac{2x}{(1+x^2)}dx$$

$$=\frac{1}{2}[\ln(1+x)]_0^1+\frac{1}{2}\left[\tan^{-1}x\right]_0^1-\frac{1}{4}\left[\ln\left(1+x^2\right)\right]_0^1$$

$$=\frac{1}{2}\ln 2+\frac{\pi}{8}-\frac{1}{4}\ln 2$$

$$=\frac{\pi}{8}+\frac{1}{4}\log e^2$$

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Q. 8 A particle is moving in the xy-plane along a curve C passing through the point (3, 3). The tangent to the curve C at the point P meets the x-axis at Q. If the y-axis bisects the segment PQ, then C is a parabola with

Option 1:
length of latus rectum 3

Option 2:

length of latus rectum 6

Option 3:
focus $\left(\frac{4}{3}, 0\right)$

Option 4:
focus $\left(0, \frac{3}{4}\right)$

Correct Answer:
length of latus rectum 3

Solution:
Let point $\mathrm{p} : (\mathrm{x}_1, \mathrm{y}_1)$

equation of tangent at p:

$$\mathrm{y} - \mathrm{y}_1 = \mathrm{m}\,(\mathrm{x} - \mathrm{x}_1)$$

$$\mathrm{Q} : \left(\mathrm{x}_1 - \frac{\mathrm{y}_1}{\mathrm{m}}, 0\right)$$

$\because$ y-axis bisexts the segment PQ

$$\therefore \frac{\mathrm{x}_1 + \mathrm{x}_1 - \frac{\mathrm{y}_1}{\mathrm{m}}}{2} = 0$$

$$\mathrm{y}_1 = 2\mathrm{m}\mathrm{x}_1$$

$$\therefore \frac{\mathrm{dy}}{\mathrm{y}} = \frac{\mathrm{dx}}{2\mathrm{x}}$$

On Integrating both the sides

$$2\ln \mathrm{y} = \ln \mathrm{x} + \ln \mathrm{c}$$

$$\mathrm{y}^2 = \mathrm{cx}$$

passes through $(3, 3)$

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$\therefore c = 3$

$\therefore$ equation of parabola: $y^2 = 3x$

$\therefore$ length of latus rectum = 3

Q. 9 Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1, a > 2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is:

Option 1:
$\frac{\sqrt{3}}{2}$

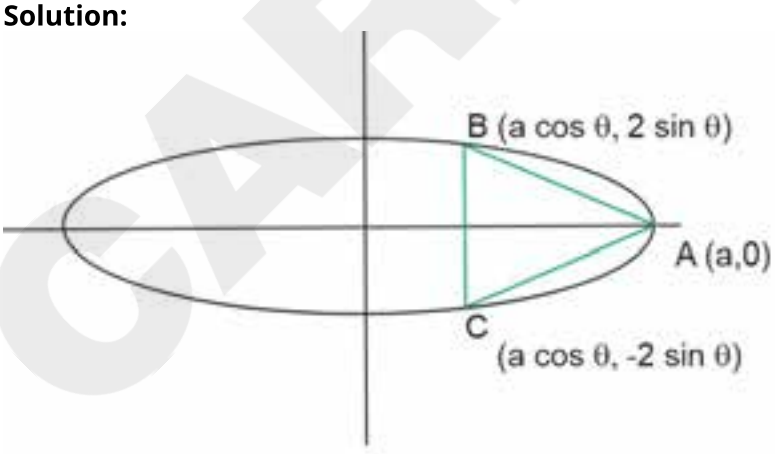
Option 2:
$\frac{1}{2}$

Option 3:
$\frac{1}{\sqrt{2}}$

Option 4:
$\frac{\sqrt{3}}{4}$

Correct Answer:
$\frac{\sqrt{3}}{2}$

Solution:



Area of $\underset{(A)}{\triangle ABC} = \frac{1}{2} \times 4\sin\theta \times a(1 - \cos\theta)$

for maximum area,

$\frac{dA}{d\theta} = 0$

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$$2a\cos\theta(1-\cos\theta)+2a\sin\theta(\sin\theta)=0$$

$$2a\left(\sin^2\theta-\cos^2\theta+\cos\theta\right)=0$$

$$2a\left(\sin^2\theta-\cos^2\theta+\cos\theta\right)=0$$

$$-2\cos^2\theta+\cos\theta+1=0$$

$$\cos\theta=1,\cos\theta=-\frac{1}{2}$$

$$\theta=0\quad\theta=\frac{2\pi}{3}$$

(not possibla)

$$\frac{d^2A}{d\theta^2}=2a(4\sin\theta\cos\theta-\sin\theta)$$

at $\theta=\frac{2\pi}{3};\quad\frac{d^2A}{d\theta^2}=\text{negative}$

Hence at $\theta=\frac{2\pi}{3}$, Area is maximum

maximum Area = $=\frac{1}{2}\times4\times\frac{\sqrt{3}}{2}\times a\left(1+\frac{1}{2}\right)=6\sqrt{3}$

$$a\sqrt{3}\left(\frac{3}{2}\right)=6\sqrt{3}$$

$$a=4$$

$$\therefore b^2=a^2\left(1-e^2\right)$$

$$4=16\left(1-e^2\right)$$

$$e=\frac{\sqrt{3}}{2}$$

Q. 10 Let the area of the triangle with vertices $A(1,\alpha),B(\alpha,0)$ and $C(0,\alpha)$ be 4 sq. units. If the points $(\alpha,-\alpha),(-\alpha,\alpha)$ and are collinear, then ß is equal to

Option 1:
64

Option 2:
-8

Option 3:
-64

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Option 4:
512

Correct Answer:
-64

Solution:

$$\text{Area } = \left|\frac{1}{2}[1(-\alpha)+\alpha(\alpha-\alpha)+o(\alpha-0)]\right| = 4$$

$$\left|\frac{\alpha}{2}\right| = 4$$

$$|\alpha| = 8$$

$$\alpha = \pm 8$$

$$also\ (\alpha,-\alpha),(-\alpha,\alpha),\left(\alpha^2,\beta\right)\ are\ collincar$$

$$\therefore \quad \frac{\alpha+\alpha}{-\alpha-\alpha} = \frac{\beta-\alpha}{\alpha^2+\alpha}$$

$$-1 = \frac{\beta-\alpha}{64+\alpha}$$

$$\beta = -64$$

Q. 11 The number of distinct real roots of the equation $x^7 - 7x - 2 = 0$ is

Option 1:
5

Option 2:
7

Option 3:
1

Option 4:
3

Correct Answer:
3

Solution:

$$x^7 - 7x - 2 = 0$$

let $f(x) = x^7 - 7x - 2$

$$f'(x) = 7\left(x^6 - 1\right) = 7\left(x^3 - 1\right)\left(x^3 + 1\right)$$

$$= 7(x-1)(x+1)\left(x^2+x+1\right)\left(x^2-x+1\right)$$

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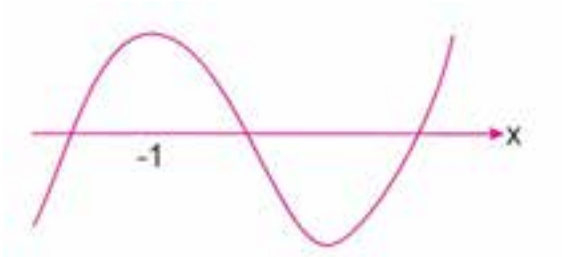
at $x = 1; f(x) = 1 + 7 - 2 = -8$, $x = -1; f(x) = -1 + 7 - 2 = 4$



Hence 3 distinct solutions

Q. 12 A random variable X has the following probability distribution:

X	0	1	2	3	4
P(X)	k	2k	4k	6k	8k

The value of $P(1 < X < 4 \mid X \leq 2)$ is equal to:

Option 1:
$\frac{4}{7}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{3}{7}$

Option 4:
$\frac{4}{5}$

Correct Answer:
$\frac{4}{7}$

Solution:

X	0	1	2	3	4
P(X)	k	2k	4k	6k	8k

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$$k + 2k + 4k + 6k + 8k = 1$$

$$k = \frac{1}{21} = \frac{p(x = 2)}{p(x \leq 2)}$$

$$P(1 < x < 4/x \leq 2) = \frac{4k}{k + 2k + 4k}$$

$$= \frac{4}{7}$$

Q. 13 The number of solution of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x\epsilon\left[-3\pi, 3\pi\right]$ is

Option 1:
8

Option 2:
5

Option 3:
6

Option 4:
7

Correct Answer:
7

Solution:

$$\cos\left(x + \frac{\pi}{3}\right) \quad \cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$$

$$\cos^2\frac{\pi}{3} - \sin^2 x = \frac{1 - 2\sin^2 x}{4}$$

$$\frac{1}{4} - \sin^2 x = \frac{1}{4} - \frac{\sin^2 x}{2}$$

$$\sin^2 x = 0$$

$$\sin x = 0 \quad \therefore \quad x = -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, 3\pi$$

7 solution

Q. 14 If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$ then the sum of all possible values of λ is

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Option 1:
16

Option 2:
6

Option 3:
12

Option 4:
15

Correct Answer:
16

Solution:

$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{\lambda}$ $\quad \tilde{a}=(1,2,3)$
$\quad \tilde{c}=(2,3,\lambda)$
$\frac{x-2}{1}=\frac{y-4}{4}=\frac{z-5}{5}$ $\quad \tilde{b}=(2,4,5)$
$\quad \tilde{d}=(1,4,5)$

Shortest distancee=$\frac{[(\bar{a}-\bar{b})\quad \bar{c}\bar{d}]}{|\bar{c}\times\bar{d}|}$

$$\left[(\tilde{a}-\tilde{b})\quad \tilde{c}\quad \tilde{d}\right]=\begin{vmatrix}-1 & -2 & -2\\ 2 & 3 & \lambda\\ 1 & 2 & 5\end{vmatrix}$$

$$=4\lambda-15+20-2\lambda-10$$

$$=2\lambda-5$$

$$\bar{c}\times\bar{d}=(15-4\lambda)\hat{u}+j(\lambda-10)+\hat{x}(5)$$

$$\therefore |\tilde{c}\times\tilde{d}|=\sqrt{225+16x^2-120\lambda+\lambda^2+100-20\lambda+25}$$

$$=\sqrt{17\lambda^2-140\lambda+350}$$

$$\therefore \frac{1}{\sqrt{3}}=\pm\frac{2\lambda-5}{\sqrt{17\lambda^2-140\lambda+350}}$$

$$17\lambda^2-140\lambda+350=3\left(4\lambda^2+25-20\lambda\right)$$

$$5\lambda^2-80\lambda+275=0$$

$$\lambda=5,11$$

$\therefore$ sum of all values of λ =16

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Q. 15 Let the points on the plane P be equidistant from the points (-4,2,1)$ and (2,-2,3). Then the acute angle between the plane p and the plane $2\mathrm{x}+\mathrm{y}+3\mathrm{z}$ is

Option 1:

$\frac{\pi}{6}$

Option 2:

$\frac{\pi}{4}$

Option 3:

$\frac{\pi}{3}$

Option 4:

$\frac{5\pi}{12}$

Correct Answer:

$\frac{\pi}{3}$

Solution:

$\mathrm{A}:(-4,2,1) \quad \mathrm{B}:(2,-2,3)$

mid point of A & B=$(-1,0,2)$

Normal vector : AB=$(6,-4,2) \therefore \quad \mathrm{P}': \quad 6(\mathrm{x}+1)+(-4)(\mathrm{y}-0)+2(\mathrm{z}-2)=0$

$3\mathrm{x}-2\mathrm{y}+\mathrm{z}+1=0$

$\mathrm{P}: \quad 2\mathrm{x}+\mathrm{y}+3\mathrm{z}-1=0$

Angle between $\mathrm{p}\&\mathrm{P}'=(\theta)=\cos^2\left(\frac{6-2+3}{14}\right)$

$=\frac{\pi}{3}$

Q. 16 Let $\hat{\mathrm{a}}$ and $\hat{\mathrm{b}}$ be two unit vectors such that $|(\hat{\mathrm{a}}+\hat{\mathrm{b}})+2(\hat{\mathrm{a}}\times\hat{\mathrm{b}})|=2$. if $\theta\in(0,\pi)$ is the angle between $\hat{\mathrm{a}}$ and $\hat{\mathrm{b}}$ then among the statements:

$(\mathrm{S1}): 2|\hat{\mathrm{a}}\times\hat{\mathrm{b}}|=|\hat{\mathrm{a}}-\hat{\mathrm{b}}|$

$(\mathrm{S2})$: The projection of $\hat{\mathrm{a}}$ on $(\hat{\mathrm{a}}+\hat{\mathrm{b}})$ is $\frac{1}{2}$

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Option 1:
Only (S1) is true.

Option 2:
Only (S2) is true.

Option 3:
Both (S1) and (S2) are true.

Option 4:
Both (S1) and (S2) are false.

Correct Answer:
Both (S1) and (S2) are true.

Solution:

$|(\hat{a}+\hat{b})+2(\hat{a}\times\hat{b})|=2$

$\{(\hat{a}+\hat{b})+2(\hat{a}\times\hat{b})\}\cdot\{(\hat{a}+\hat{b})+2(\hat{a}\times\hat{b})\}=4$

$1+1+2\hat{a}\cdot\hat{b}+2(\hat{a}+\hat{b})\cdot(\hat{a}\times\hat{b})+2(\hat{a}\times\hat{b})\cdot(\hat{a}+\hat{b})+4(\hat{a}\times\hat{b})^2=4$

$2\cos\theta+0+4\sin^2\theta=4-2$

$4-4\cos^2\theta+2\cos\theta=2$

$2\cos^2\theta-\cos\theta-1=0$

$(2\cos\theta+1)(\cos\theta-1)=0$

$\cos\theta=1,\quad \cos\theta=-\frac{1}{2}$

(rejectad)

$\theta=\frac{2\pi}{3}$

$S_1:\quad 2|\hat{a}\times\hat{b}|=|\hat{a}-\hat{b}|$

$\text{LHS}: 2\sin\frac{2\pi}{3}=\sqrt{3};\quad \text{RHS}: \sqrt{1+1-2\left(-\frac{1}{2}\right)}=\sqrt{3}$

$\therefore S_1$ is true

S_2 : Projection of $\tilde{a}$ on$(\tilde{a}+\tilde{b})$= $\frac{(\hat{a})\cdot(\hat{a}+\hat{b})}{|\hat{a}+\hat{b}|}$

$=\frac{1+\left(-\frac{1}{2}\right)}{\sqrt{1+1-2\left(\frac{1}{2}\right)}}$

$=\frac{1}{2}$

$\therefore S_2$ is true

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Q. 17 If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right), \frac{\pi}{2} < x^3 < \frac{3\pi}{2}$,, then

Option 1:
$xy'' + 2y' = 0$

Option 2:
$x^2y'' - 6y + \frac{3\pi}{2} = 0$

Option 3:
$x^2y'' - 6y + 3\pi = 0$

Option 4:
$xy'' - 4y' = 0$

Correct Answer:
$x^2y'' - 6y + \frac{3\pi}{2} = 0$

Solution:

$$y = \tan^{-1}\left(\sec x^3 - \tan x^3\right)$$

$$= \tan^{-1}\left(\frac{1 - \sin x^3}{\cos x^3}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x^3}{2}\right)\right) \quad \left\{\text{As } \frac{\pi}{4} - \frac{x^3}{2} \in \left(-\frac{\pi}{2}, 0\right)\right.$$

$$y = \frac{\pi}{4} - \frac{x^3}{2}$$

$$y' = -\frac{3x^2}{2}$$

$$4y = \pi - 2x^2\left(-\frac{y''}{3}\right)$$

$$y'' = -3x$$

$$x^2y'' - 6y + \frac{3\pi}{2} = 0$$

Q. 18 Consider the following statements:
A : Rishi is a judge.
B : Rishi is honest.
C: Rishi is not arrogant.
The negation of the statement "if Rishi is a judge and he is not arrogant, then he is Question: honest" is

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Option 1:
$B \rightarrow (A \vee C)$

Option 2:
$(\sim B) \wedge (A \wedge C)$

Option 3:
$B \rightarrow ((\sim A) \vee (\sim C))$

Option 4:
$B \rightarrow (A \wedge C)$

Correct Answer:
$(\sim B) \wedge (A \wedge C)$

Solution:
$\sim ((A \wedge C) \longrightarrow B)$
$(A \wedge C) \wedge (\sim B)$

Q. 19 The slope of normal at any point $(x, y), x > 0, y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$. If the curve passes through the point $(1, 1)$, then $e.y(e)$, is equal to

Option 1:
$\frac{1 - \tan(1)}{1 + \tan(1)}$

Option 2:
$\tan(1)$

Option 3:
1

Option 4:
$\frac{1 + \tan(1)}{1 - \tan(1)}$

Correct Answer:
$\frac{1 + \tan(1)}{1 - \tan(1)}$

Solution:
$$\frac{-1}{\left(\frac{dy}{dx}\right)} = \frac{x^2}{xy - x^2y^2 - 1}$$

$$\frac{dy}{dx} = \frac{1 + x^2y^2 - xy}{x^2}$$

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$$x^2dy = dx + x^2y^2dx - xydx$$
$$x(xdy + ydx) = dx + x^2y^2dx$$
$$\frac{d(xy)}{1+x^2y^2} = \frac{dx}{x}$$

$$\tan^{-1}(xy) = \ln x + c$$

Pass through(1,1)

$$\frac{\pi}{4} = c$$
$$\therefore \tan^{-1}(xy) = \ln x + \frac{\pi}{4}$$
$$x = c$$
$$\therefore \quad \tan^{-1}(ey) = 1 + \frac{\pi}{4}$$
$$ey = \frac{1+\tan 1}{1-\tan 1}$$

Q. 20 let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all $x \in \mathbb{R}$. Then ion: $f_{\lambda^*}(1) + f_{\lambda_**}(-1)$ is equal to :

Option 1:
36

Option 2:
48

Option 3:
64

Option 4:
72

Correct Answer:
72

Solution:
$f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$
$f'_\lambda(x) = 12\lambda x^2 - 72\lambda x + 36 \geq 0$
$D \leq 0$

$(72\lambda)^2 - 4(12\lambda)(36) \leq 0$
$144\lambda^2 - 48\lambda \leq 0$
$\lambda(3\lambda - 1) \leq 0$
$\lambda \in \left[0, \frac{1}{3}\right]$

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$$\lambda^* = \frac{1}{3}$$

$$\therefore f_{\left(\frac{1}{3}\right)}1 + f_{\left(\frac{1}{3}\right)}{}^{-1} = -36\left(\frac{1}{3}\right)(1+1) + 48(1+1)$$

$$= -24 + 96 = 72$$

Q. 21 Let $S = \{z \in \mathbb{C} : |z-3| \leq 1$ and $z(4+3i) + \bar{z}(4-3i) \leq 24\}$. If $\alpha + i\beta$ is the point in S which is closest to $4i$, then $25(\alpha+\beta)$ is equal to

Correct Answer:
80

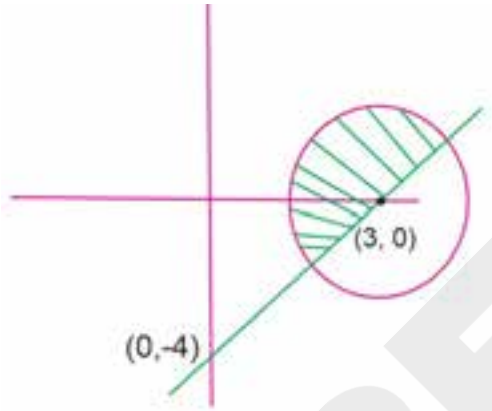
Solution:

$$(x-3)^2 + y^2 \leq 1$$

$$(x+iy)(4+3i) + (x-iy)(4-3i) \leq 24$$

$$8x - 6y \leq 24$$

$$4x - 3y \leq 12$$



Let (x_1, y_1) be a point which is closest to $(0, 4)$

$$y - 4 = \frac{4 - y_1}{-x_1}(x - 0)$$

Passes through (3,0)

$$-4 = \frac{4 - y_1}{-x_1}(3)$$

$$4x_1 = 12 - 3y_1$$

also

$$(x_1 - 3)^2 + (y_1)^2 = 1$$

$$\left(3 - \frac{3}{4}y_1 - 3\right)^2 + y_1^2 = 1$$

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$$9y_1^2 + 16{y_1}^2 = 16$$

$$y_1 = \frac{4}{5}$$

$$\therefore x_1 = \frac{12 - \frac{12}{5}}{4} = \frac{12}{5}$$

$$\therefore (\alpha + i\beta) = \frac{12}{5} + i\left(\frac{4}{5}\right)$$

$$\therefore 25(\alpha + \beta) =$$

$$\therefore (16) = 80$$

Q. 22 Let $\mathrm{S} = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; \mathrm{a}, \mathrm{b} \in \{1, 2, 3, \ldots 100\} \right\}$ and let $\mathrm{T_n} = \left\{ \mathrm{A} \in \mathrm{S} : \mathrm{A}^{\mathrm{n(n+1)}} = \mathrm{I} \right\}$. Then the number of elements in $\cap_{\mathrm{n=1}}^{100} \mathrm{T_n}$ is

---

Correct Answer:
100

Solution:

$$\mathrm{S} = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix}; \quad \mathrm{a}, \mathrm{b} \in \{1, 2, 3, \ldots 100\}$$

$$\mathrm{T_n} = \left\{ \mathrm{A} \in \mathrm{s} : \mathrm{A}^{\mathrm{n(n+1)}} = \mathrm{I} \right\}$$

$$\mathrm{A}^2 = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} = \begin{bmatrix} 1 & -a + ab \\ 0 & b^2 \end{bmatrix}$$

b should be 1

$\mathrm{A}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ hence a can take any value

$\therefore$ Total number of elements = 100

Q. 23 The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1,2,3,4,5,7 and 9 is_____

Correct Answer:
576

Solution:

$$\text{digits} \equiv 1, 2, 3, 4, 5, 7, 9$$

let number=$\mathrm{a\ \ b\ \ c\ \ d\ \ e\ \ f\ \ g}$

$$(\mathrm{a + c + e + g}) \sim (\mathrm{b + d + f}) = 11\mathrm{k}$$

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$a+c+e+g=21 \quad , b+d+f=10$

Case-1 $(9,7,1,4),(9,7,2,3)$

$(9,5,3,4)$

Total number in this case $=(4! \times 3!) \times 3=432$

case-2

$a+c+e+g=10 \quad , b+d+f=21$

$(1,2,3,4)$

Total number in this case $=432+144=576$

Q. 24 { The sum of all the elements of the set $\{\alpha \in \{1,2,\ldots..100\} : \mathrm{HCF}(\alpha,24)=1\}$ is

Correct Answer:
1633

Solution:

$\alpha \in \{1,2,3,\ldots,100\} \quad ; \mathcal{HCF}(2,24)=1$

$\text{Sum } =[1+2+3+\ldots.100]-[2+4+\ldots 100]-[3+6+\ldots-99]+[6+12+\ldots.96]$

$=\frac{100 \times 101}{2}-\frac{2 \times 50 \times 51}{2}-3 \times \frac{33 \times 34}{2}+\frac{6 \times 16 \times 17}{2}$

$=5050-2550-(99)(17)+(48)(17)$

$=2500-1683+816$

$=3316-1683$

$= \quad 1633$

Q. 25 The remainder on dividing $1+3+3^2+3^3+\ldots.+3^{2021}$ by 50 is

Correct Answer:
4

Solution:

$1+3+3^2+\cdots \quad 3^{2021}$ is divided by 50

$1+3+3^2+\cdots+3^{2021}$

$=\frac{3^{2022}-1}{2}$

$\{3^{2022}=(10-1)^{1011}=10\mathrm{k}+9$

$\frac{10k+9-1}{2}$

$=5k+4 \text{ type}$

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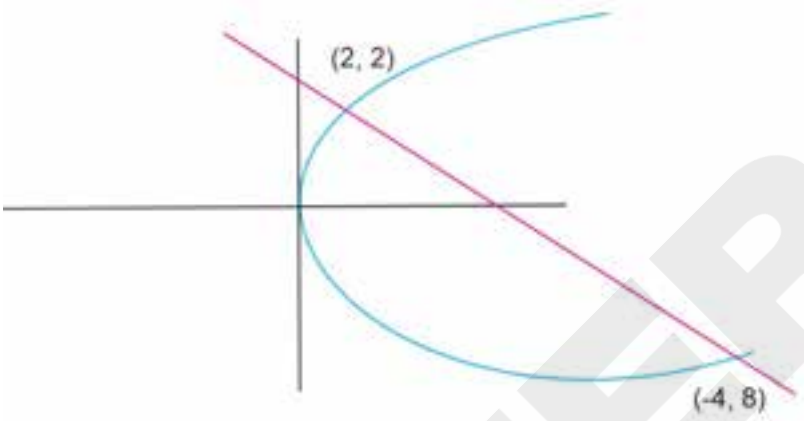
Hence remainder=4

Q. 26 The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is

Correct Answer:
18

Solution:
$y^2 = 2x, \quad x + y = 4$
$y^2 = 2(4 - y)$
$y^2 + 2y - 8 = 0$
$y = -4, 2$



$$\text{Area} = \int_{-4}^{2} \left[(4 - y) - \frac{y^2}{2}\right] dy$$
$$= \left[4y - \frac{y^2}{2} - \frac{y^3}{6}\right]_{-4}^{2}$$
$$= \left(8 - 2 - \frac{8}{6}\right) - \left(-16 - \frac{16}{2} + \frac{64}{6}\right)$$
$$= \frac{28}{6} - \left(-24 + \frac{32}{3}\right)$$
$$= \frac{14}{3} + 24 - \frac{32}{3} = 18$$

Q. 27 Let a circle $C : (x - h)^2 + (y - k)^2 = r^2, k > 0$, touch the x-axis at $(1, 0)$. If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2. then the value of $h + k + r$ is equal to

Correct Answer:
7

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Solution:

$(x-h)^2+(y-k)^2=r^2; k>0$

$(x-1)^2+y^2-2ky+k^2=k^2$

$x^2+y^2-2x-2ky+1=0$

$k^2-\left(\frac{1+k}{\sqrt{2}}\right)^2=1$

$2k^2-1-k^2-2k=2$

$k^2-2k-3=0$

$k=3,-1$

$h+k+r=1+3+3=7$

Q. 28 In an examination, there are 10 true-false type questions. Out of 10 , a student can guess the answer of 4 questions correctly with probability $\frac{3}{4}$ and the remaining 6 questions correctly with probability $\frac{1}{4}$. If the probability that the student guesses the answers of exactly 8 questions correctly out of 10 is $\frac{27\mathrm{k}}{4^{10}}$, then k is equal to

---

Correct Answer:

479

Solution:

$$P(x=8)=(2 from\ first\ 4\ and\ 6\ from\ remaening\ 6)+(3\ from\ first\ 4\ and\ 5\ from\ ramoining\ 6)+(4\ from\ first\ 4\ and\ 4\ from\ remaining\ 6)$$

$$=\left[4c_2\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right)^2\times{}^6c_6\left(\frac{1}{4}\right)^6\right]+\left[4c_3\left(\frac{3}{4}\right)^3\left(\frac{1}{4}\right)^1\times{}^6c_5\left(\frac{1}{4}\right)^5\left(\frac{3}{4}\right)^1\right]+\left[4c_4\left(\frac{3}{4}\right)^4\times{}^6c_4\left(\frac{1}{41}\right)^4\left(\frac{3}{4}\right)^2\right]$$

$$=\frac{1}{4^{10}}[54+1944+10935]$$

$$=\frac{(27)(479)}{4^{10}}$$

$\therefore K=479$

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Q. 29 Let the hyperbola $\mathrm{H}: \frac{\mathrm{x}^2}{\mathrm{a}^2}-\mathrm{y}^2=1$ and the ellipse $\mathrm{E}: 3\mathrm{x}^2+4\mathrm{y}^2=12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If and $\mathrm{e_H}$are the eccentricities of H and E respectively, then the value of $12\left(\mathrm{e}_{\mathrm{H}}^2+\mathrm{e}_{\mathrm{E}}^2\right)$ is equal to_____

Correct Answer:
42

Solution:

$H: \frac{x^2}{a^2}-y^2=1$

E: $3x^2+4y^2=12$

$(LR)_H=(LR)_E$

$2\times\frac{1}{a}=2\times\frac{3}{2}$

$a=\frac{2}{3}$

AlSO

$1=a^2\left(e_H^2-1\right)$

$e_H{}^2=\frac{13}{4}$

&

$3=4\left(1-e_E{}^2\right)$

$e_{E^2}=\frac{1}{4}$

$12(e_H^2+e_E^2)$

$=12\left(\frac{14}{4}\right)=42$

Q. 30 Let P_1 be a parabola with vertex $(3,2)$ and focus $(4,4)$ and P_2 be its mirror image with respect to the line $\mathrm{x}+2\mathrm{y}=6$. Then the directrix of P_2 is $\mathrm{x}+2\mathrm{y}=$____

Correct Answer:
10

Solution:

For P_1 :

slope of axis $=\frac{4-2}{4-3}=2$

Distance between vertex and focus: $\sqrt{5}$

Now line is perpendicular to axis

focus of $\mathrm{P}_2: \frac{\mathrm{x}-4}{1}=\frac{\mathrm{y}-4}{2}=\frac{-2(4+8-6)}{5}$

$\mathrm{x}=\frac{-12}{5}+4=\frac{8}{5}$

$\mathrm{y}=\frac{-24}{5}+4=\frac{-4}{5}$

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Let directrix of $P_2 : x + 2y = \lambda$

$$\therefore \left| \frac{\frac{8}{5} - \frac{8}{5} - \lambda}{\sqrt{5}} \right| = 2\sqrt{5}$$

$$|-\lambda| = 10$$

$$\lambda = \pm 10$$

$\lambda = -10$ is rejected because directrix can't cut parabola

$\therefore \lambda = 10$ is the answer.

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Online 25th June Morning Shift

Physics

Q. 1 If $Z=\frac{A^2B^3}{C^4}$, then the relative error in Z will be :

Option 1:
$\frac{\Delta A}{A}+\frac{\Delta B}{B}+\frac{\Delta C}{C}$

Option 2:
$\frac{2\Delta A}{A}+\frac{3\Delta B}{B}-\frac{4\Delta C}{C}$

Option 3:
$\frac{2\Delta A}{A}+\frac{3\Delta B}{B}+\frac{4\Delta C}{C}$

Option 4:
$\frac{\Delta A}{A}+\frac{\Delta B}{B}-\frac{\Delta C}{C}$

Correct Answer:
$\frac{2\Delta A}{A}+\frac{3\Delta B}{B}+\frac{4\Delta C}{C}$

Solution:

$$Z=\frac{A^2B^3}{C^4}=A^2B^3C^{-4}$$

$$\frac{\Delta Z}{Z}=\frac{2\Delta A}{A}+\frac{3\Delta B}{B}+4\frac{\Delta C}{C}$$

The correct option is (3)

Q. 2 $\overrightarrow{A}$ is a vector quantity such that $|\overrightarrow{A}|=\text{non - zero constant}$. Which of the following expression is true for $\vec{A}$?

Option 1:
$\overrightarrow{A}\cdot\overrightarrow{A}=0$

Option 2:
$\overrightarrow{A}\times\overrightarrow{A}<0$

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Option 3:
$\overrightarrow{A} \times \overrightarrow{A} = 0$

Option 4:
$\vec{A} \times \vec{A} > 0$

Correct Answer:
$\overrightarrow{A} \times \overrightarrow{A} = 0$

Solution:
$|\bar{A}| = \text{non-zero constant}$

$\bar{A} \cdot \bar{A} = (A)\,(A) \cos 0^\circ = A^2\,(1) = A^2$
$\bar{A} \times \bar{A} = (A)\,(A) \sin 0^\circ = 0$

$\therefore$ The correct option is (3)

Q. 3 Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other ?

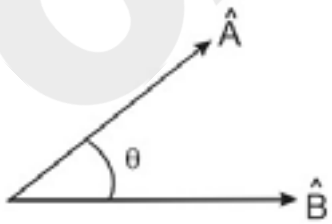
Option 1:
$|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \theta/2$

Option 2:
$|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan \theta/2$

Option 3:
$|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \cos \theta/2$

Option 4:
$|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \cos \theta/2$

Correct Answer:
$|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan \theta/2$

Solution:


$$\begin{aligned}\left|\hat{A} + \hat{B}\right|^2 &= \left(\hat{A}\right)^2 + \left(\hat{B}\right)^2 + 2\hat{A}\,\hat{B}\,\cos\theta \\ &= 2 + 2\cos\theta \\ &= 2\,(1 + \cos\theta)\end{aligned}$$

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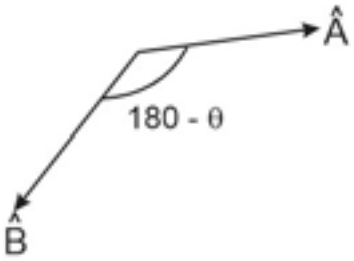
$$\left|\hat{A}+\hat{B}\right|^2 = 4\cos^2\left(\frac{\theta}{2}\right)$$
$$\therefore \left|\hat{A}+\hat{B}\right| = 2\cos\left(\frac{\theta}{2}\right) \rightarrow (1)$$

$$\begin{aligned}\left|\hat{A}-\hat{B}\right|^2 &= \left(\hat{A}\right)^2+\left(\hat{B}\right)^2+2\hat{A}\,\hat{B}\cos\left(180-\hat{\theta}\right)\\ &= 1+1-2\cos\theta\\ &= 2\left(1-\cos\theta\right)\\ &= 4\sin^2\left(\frac{\theta}{2}\right)\end{aligned}$$



$$\left|\hat{A}-\hat{B}\right| = 2\sin\left(\frac{\theta}{2}\right) \rightarrow (2)$$

$(1) \div (2)$

$$\frac{\left|\hat{A}+\hat{B}\right|}{\left|\hat{A}-\hat{B}\right|} = \cot\left(\frac{\theta}{2}\right) = \frac{1}{\tan\left(\frac{\theta}{2}\right)}$$
$$\left|\hat{A}-\hat{B}\right| = \tan\left(\frac{\theta}{2}\right)\left|\hat{A}+\hat{B}\right|$$

The correct option is (2)

Q. 4 If force $\overrightarrow{F} = 3\hat{i}+4\hat{j}-2\hat{k}$ acts on a particle having position vector $2\hat{i}+\hat{j}+2\hat{k}$ then, the torque about the origin will be :

Option 1:
$3\hat{i}+4\hat{j}-2\hat{k}$

Option 2:
$-10\hat{i}+10\hat{j}+5\hat{k}$

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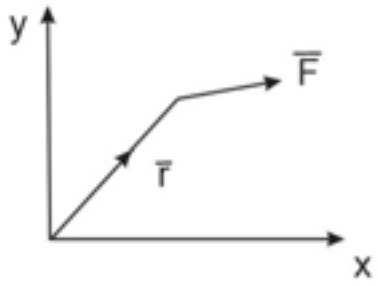
Option 3:
$10\hat{i}+5\hat{j}-10\hat{k}$

Option 4:
$10\hat{i}+\hat{j}-5\hat{k}$

Correct Answer:
$-10\hat{i}+10\hat{j}+5\hat{k}$

Solution:
$\bar{\mathrm{F}}=3\hat{\mathrm{i}}+4\hat{\mathrm{j}}-2\hat{\mathrm{k}}$
$\bar{\mathrm{r}}=2\hat{\mathrm{i}}+\hat{\mathrm{j}}+2\hat{\mathrm{k}}$



$$\begin{aligned}\bar{\tau}&=\bar{\mathrm{r}}\times\bar{\mathrm{F}}\\&=\left(2\hat{\mathrm{i}}+\hat{\mathrm{j}}+2\hat{\mathrm{k}}\right)\times\left(3\hat{\mathrm{i}}+4\hat{\mathrm{j}}-2\hat{\mathrm{k}}\right)\\&=\left(0+8\hat{\mathrm{k}}+4\hat{\mathrm{j}}\right)+\left(-3\hat{\mathrm{k}}+0-2\hat{\mathrm{i}}\right)+\left(6\hat{\mathrm{j}}-8\hat{\mathrm{i}}+0\right)\\\bar{\tau}&=-10\hat{\mathrm{i}}+10\hat{\mathrm{j}}+5\hat{\mathrm{k}}\end{aligned}$$

The correct option is (2)

Q. 5 The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be : (Given g = acceleration due to gravity at the surface of earth).

Option 1:
$\mathrm{g}/2$

Option 2:
$\mathrm{g}/4$

Option 3:
$\mathrm{g}/3$

Option 4:
$\mathrm{g}/9$

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Correct Answer:
$\mathrm{g}/4$

Solution:
At $\mathrm{h}=\mathrm{R}$,

$$\mathrm{g_h}=\frac{\mathrm{gR}^2}{(\mathrm{R}+\mathrm{h})^2}=\frac{\mathrm{gR}^2}{(2\mathrm{R})^2}=\frac{\mathrm{g}}{4}$$

The acceleration due to gravity at point P will be $\left(\frac{\mathrm{g}}{4}\right)$

The correct answer is (2)

Q. 6 The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as :

Option 1:
$\mathrm{r}^{1/2}$

Option 2:
r

Option 3:
r^2

Option 4:
r^3

Correct Answer:
r^2

Solution:
For the spherical raindrops, the terminal velocity is equal to

$$\mathrm{V_t}=\frac{2\mathrm{r}^2\left(\rho_{\mathrm{drop}}-\rho_{\mathrm{air}}\right)\mathrm{g}}{9\eta}$$

$$\mathrm{V_t}\propto\mathrm{r}^2$$

The correct option is (3)

Q. 7 The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of $300\ \mathrm{K}$ will be :

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Option 1:

$v_{\mathrm{rms}}=\sqrt{\frac{2}{3}}v_{\mathrm{p}}$

Option 2:

$v_{\mathrm{rms}}=\sqrt{3/2}v_{\mathrm{p}}$

Option 3:

$v_{\mathrm{rms}}=v_{\mathrm{p}}$

Option 4:

$v_{\mathrm{rms}}=\sqrt{1/3}v_{\mathrm{p}}$

Correct Answer:

$v_{\mathrm{rms}}=\sqrt{3/2}v_{\mathrm{p}}$

Solution:

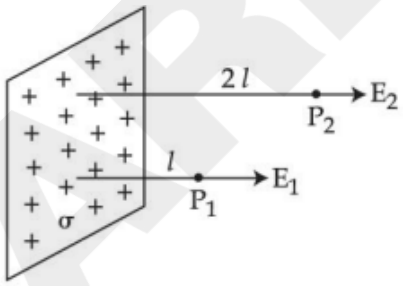
$$\mathrm{V_{rms}}=\sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}}$$

$$\mathrm{V_p}=\sqrt{\frac{2\mathrm{RT}}{\mathrm{M}}}$$

$$\frac{\mathrm{V_{rms}}}{\mathrm{V_p}}=\sqrt{\frac{3}{2}}$$

The correct option is (2)

Q. 8 In the figure, a very large plane sheet of positive charge is shown.$\mathrm{P_1}$ and $\mathrm{P_2}$ are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



Option 1:

$\mathrm{E_1}=\sigma/\varepsilon_{\mathrm{o}},\mathrm{E_2}=\sigma/2\varepsilon_{\mathrm{o}}$

Option 2:

$\mathrm{E_1}=2\sigma/\varepsilon_{\mathrm{o}},\mathrm{E_2}=\sigma/\varepsilon_{\mathrm{o}}$

Option 3:

$\mathrm{E_1}=\mathrm{E_2}=\sigma/2\varepsilon_{\mathrm{o}}$

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Option 4:

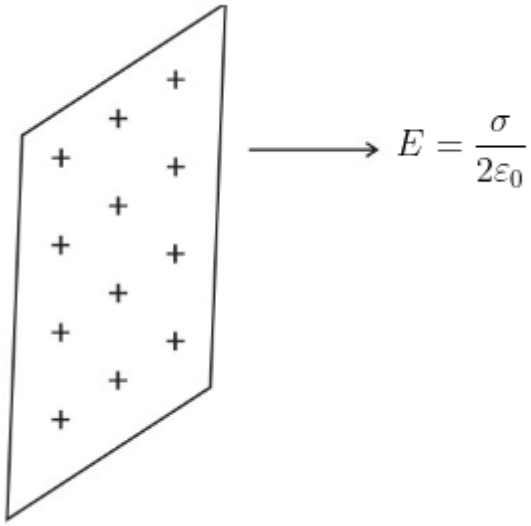
$E_1 = E_2 = \sigma/\varepsilon_o$

Correct Answer:

$E_1 = E_2 = \sigma/2\varepsilon_o$

Solution:

For very large plane sheet of positive charge



$$\longrightarrow E = \frac{\sigma}{2\varepsilon_0}$$

The electric field at any point due to a very large sheet is $\left(\frac{\sigma}{2\varepsilon_0}\right)$

$$\therefore E_1 = E_2 = \frac{\sigma}{2\varepsilon_0}$$

The correct option is (3)

Q. 9 Match List -I with List -II.

List - I	**List - II**
(A) AC generator	(I) Detects the presence of current in the circuit
(B) Galvanometer	(II) Converts mechanical energy into electrical energy
(C) Transformer	(III) Works on the principle of resonance in AC circuit
(D) Metal detector	(IV) Changes an alternating voltage for smaller or greater value

Choose the correct answer from the options given below :

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Option 1:
(A)-(II),(B)-(I),(C)-(IV),(D)-(III)

Option 2:
(A)-(II),(B)-(I),(C)-(III),(D)-(IV)

Option 3:
(A)-(III),(B)-(IV),(C)-(II),(D)-(IV)

Option 4:
(A)-(III),(B)-(I),(C)-(II),(D)-(IV)

Correct Answer:
(A)-(II),(B)-(I),(C)-(IV),(D)-(III)

Solution:
AC generator $\longrightarrow$ Convert mechanical energy into elecrical energy

Galvanometer $\longrightarrow$ Detects the presence of current in the circuit

Transformer $\longrightarrow$ Changes an alternating voltage for smaller or greater value

Metal detector $\longrightarrow$ Works on the principle of resonance in AC circuit

The correct option is (1)

Q. 10 A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance $r\ \ (\ r < R)$ from its centre will be :

Option 1:
$B \propto r^2$

Option 2:
$B \propto r$

Option 3:
$B \propto \frac{1}{r^2}$

Option 4:
$\mathrm{B} \propto \frac{1}{\mathrm{r}}$

Correct Answer:
$B \propto r$

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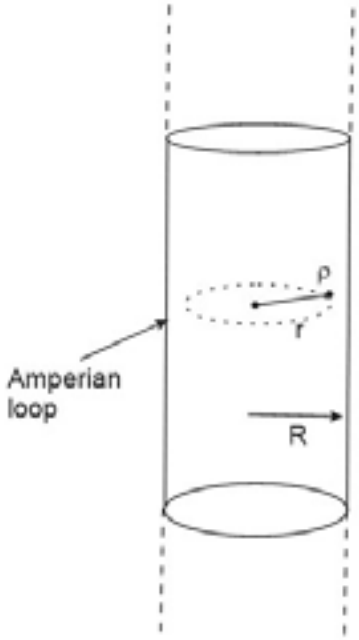
Solution:



By Ampere's law,

$$\oint \bar{B}.d\bar{l} = \mu_0 I_{enc}$$

$$= \mu_0 J A_{enc}$$

$$B \int dl \cos 0^\circ = \frac{\mu_0 I}{\pi R^2} \times \pi r^2$$

$$B (2\pi r) = \frac{\mu_0 I r^2}{R^2}$$

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

Here pt .p lies at $r < R$

$$\therefore B \alpha r \quad (\text{for } r < R)$$

The correct option is (2)

Q. 11 If wattless current flows in the AC circuit, then the circuit is :

Option 1:
Purely Resistive circuit

Option 2:
Purely Inductive circuit

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Option 3:
LCR series circuit

Option 4:
RC series circuit only

Correct Answer:
Purely Inductive circuit

Solution:
If wattless current flows in the AC circuit, that means the phase difference between current voltage is 90°

It is possible only for pure inductive or capacitive circuit

∴ The correct option is (2)

Q. 12 The electric field in an electromagnetic wave is given by $\mathrm{E} = 56.5 \sin \omega(\mathrm{t} - x/\mathrm{c}) \mathrm{NC}^{-1}$. Find the intensity of the wave if it is propagating along $x - axis$ in the free space. (Given $\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C}^2\ \mathrm{N}^{-1}\ \mathrm{m}^{-2}$)

Option 1:
$5.65 \mathrm{Wm}^{-2}$

Option 2:
$4.24 \mathrm{Wm}^{-2}$

Option 3:
$1.9 \times 10^{-7} \mathrm{Wm}^{-2}$

Option 4:
$56.5 \mathrm{Wm}^{-2}$

Correct Answer:
$4.24 \mathrm{Wm}^{-2}$

Solution:
$$\mathrm{E} = 56.5\ \sin\left(\omega\left(\mathrm{t} - \frac{\mathrm{x}}{\mathrm{c}}\right)\right)$$

$$\text{Intensity} = \frac{1}{2}\varepsilon_0 \mathrm{E}_0^2\, \mathrm{c}$$

$$= \frac{1}{2} \times \frac{1}{4\pi \times 9 \times 10^9} \times (56.5)^2 \times \left(3 \times 10^8\right)$$

$$\left(\because\ \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\right)$$

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$\mathrm{I} = 4.24\ \mathrm{W/m^2}$

The correct option is (2)

Q. 13 The two light beams having intensities I and $9I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

Option 1:
$2I$

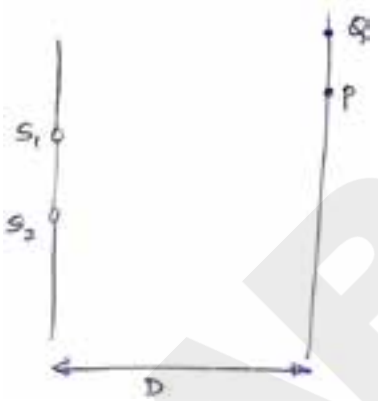
Option 2:
$6I$

Option 3:
$5I$

Option 4:
$7I$

Correct Answer:
$6I$

Solution:



$$\mathrm{I_P} = \mathrm{I_1} + \mathrm{I_2} + 2\sqrt{\mathrm{I_1}}\sqrt{\mathrm{I_2}}\cos\left(\frac{\pi}{2}\right)$$
$$\mathrm{I_P} = 10\,\mathrm{I} \rightarrow (1)$$

$$\mathrm{I_Q} = \mathrm{I} + 9\mathrm{I} + 2\sqrt{\mathrm{I}}\sqrt{9\mathrm{I}}\cos(\pi)$$
$$\mathrm{I_Q} = 4\mathrm{I} \rightarrow (2)$$

$$\mathrm{I_P} - \mathrm{I_Q} = 6\mathrm{I}$$

The correct option is (2)

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Q. 14 A light wave travelling linearly in a medium of dielectric constant 4, incidents on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :
(Given : relative permeability of medium $\mu_r = 1$)

Option 1:
10°

Option 2:
20°

Option 3:
30°

Option 4:
60°

Correct Answer:
30°

Solution:
$k = \varepsilon_r = 4$

$$\mu_{med} = \frac{C}{V} = \sqrt{\mu_r \varepsilon_r} = 2$$

$$\left(\because V = \frac{C}{\sqrt{\mu_r \varepsilon_r}} \right)$$

When the angle of incidence is equal to the critical angle the total intensity of the incident wave will be reflected back into the same medium

$$\sin i_C = \frac{1}{\mu_{med}} = \frac{1}{2}$$
$$i_C = 30^\circ$$

The correct option is (3)

Q. 15 Given below are two statements:
Statement I : Davisson-Germer experiment establishes the wave nature of electrons.
Statement II : If electrons have wave nature, they can interfere and show diffraction.
In the light of the above statements choose the **correct answer** from the option given below :

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Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is true but **Statement II** is false.

Option 4:
Statement I is false but **Statement II** is true.

Correct Answer:
Both **Statement I** and **Statement II** are true.

Solution:
$\rightarrow$ Davisson -Germer experiment gives experimental proof of de-broglie hypothesis

$\rightarrow$ Wave exhibits both the phenomenon of interference and diffraction

$\rightarrow$ The correct option is (1)

Q. 16 The ratio for the speed of the electron in the 3^{rd} orbit of $\text{He}^{+}$to the speed of the electron in the 3^{rd} orbit of hydrogen atom will be :

Option 1:
$1:1$

Option 2:
$1:2$

Option 3:
$4:1$

Option 4:
$2:1$

Correct Answer:
$2:1$

Solution:
Velocity of electron in n^{th} orbit is

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$$V_n = V_0\left(\frac{Z}{n}\right)$$

For He^+,

$$V_1 = V_0\left(\frac{2}{3}\right) \rightarrow (1)$$

For Hydrogen atom ,

$$V_2 = V_0\left(\frac{1}{3}\right) \rightarrow (2)$$

$$\frac{V_1}{V_2} = \frac{2}{1}$$

The correct option is (4)

Q. 17 The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because :

Option 1:
fractional change in majority carriers produce higher forward bias current

Option 2:
fractional change in majority carriers produce higher reverse bias current

Option 3:
fractional change in minority carriers produce higher forward bias current

Option 4:
fractional change in minority carriers produce higher reverse bias current

Correct Answer:
fractional change in minority carriers produce higher reverse bias current

Solution:
$\rightarrow$ A photodiode is a special purpose p-n junction diode fabricated with a transparent window to allow light to fall on the diode

$\rightarrow$ It is operated under reverse bias

$\rightarrow$ The fractional change due to the photo-effects on the minority carrier dominated reverse bias current is more easily measurable than the fractional change in the forward bias current.

The correct option is (4)

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Q. 18 A signal of $100\ \mathrm{THz}$ frequency can be transmitted with maximum efficiency by :

Option 1:
Coaxial cable

Option 2:
Optical fibre

Option 3:
Twisted pair of copper wires

Option 4:
Water

Correct Answer:
Optical fibre

Solution:
$\rightarrow$ A signal of $100\ \mathrm{THz}$ frequency can be transmitted with maximum efficiency by optical fibre

$\rightarrow$ Since, the optical communication using fibres is performed in the frequency range
$1\ \mathrm{THz}\ \text{to}\ 100\ \mathrm{THz}$

$\rightarrow$ The correct option is (2)

Q. 19 The difference of speed of light in the two media A and $\mathrm{B}\ (v_{\mathrm{A}} - v_{\mathrm{B}})$ is $2.6 \times 10^{7}\ \mathrm{m/s}$. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is: (Given : speed of light in vacuum $\mathrm{c} = 3 \times 10^{8}\ \mathrm{ms}^{-1}$)

Option 1:
1.303

Option 2:
1.318

Option 3:
1.13

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Option 4:
0.12

Correct Answer:
1.13

Solution:

$$\mu_B = 1.47 = \frac{C}{V_B} = \frac{3 \times 10^8}{V_B}$$

$$V_B = \frac{3}{1.47} \times 10^8$$
$$V_B = 2.041 \times 10^8 \, m/s \rightarrow (1)$$

$$\because \; V_A - V_B = 2.6 \times 10^7 \, m/s \, (Given)$$

$$\therefore \; V_A = V_B + 2.6 \times 10^7 \, m/s$$
$$V_A = 2.301 \times 10^8 \, m/s$$

$$\mu_A = \frac{C}{V_A} = 1.304 \rightarrow (2)$$
$$\frac{\mu_B}{\mu_A} = \frac{1.47}{1.304} = 1.13$$

The correct option is (3)

Q. 20 A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G ?

Option 1:
$\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

Option 2:
$\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).

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Option 3:

$\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).

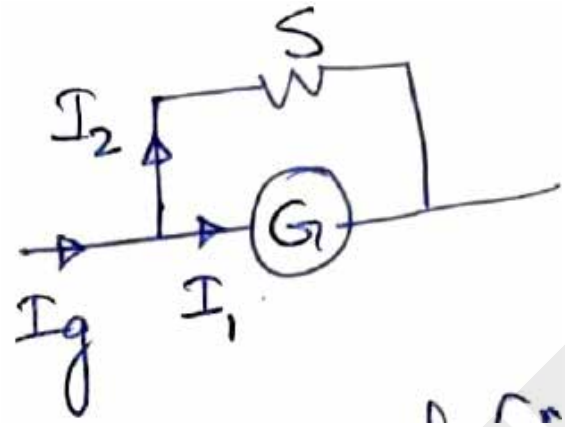
Option 4:

$\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistance(s).

Correct Answer:

$\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).

Solution:



For full deflection , $I_1 = I_g$

and $I_2 = 0$ (No shunt resistance or $s \rightarrow \infty$)

For $\frac{1}{3}$ deflection , $I_1 = \frac{I_g}{3}$

$$I_2 = \frac{2I_g}{3}$$

$I_1 G = I_2 S \,(\text{parallel})$

$G = 2\,S$

The correct option is (2)

Q. 21 A uniform chain of $6\ \text{m}$ length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is_____ m.

Correct Answer:

2

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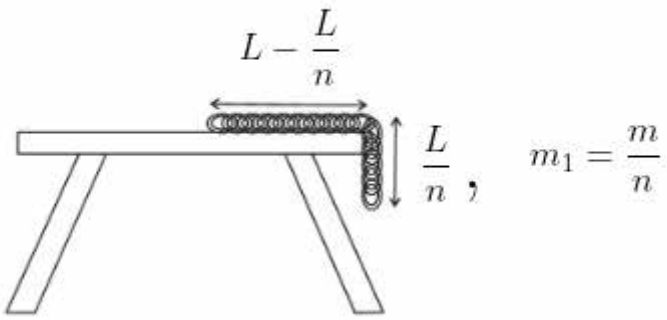
Solution:

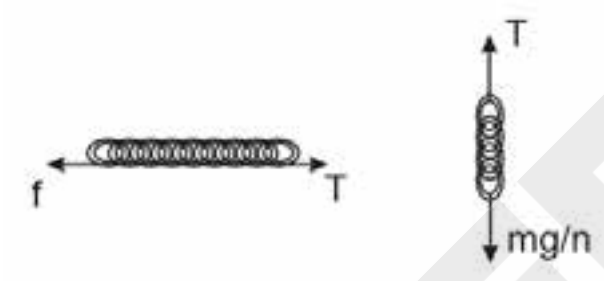
$$m_2 = m(1 - \frac{1}{n})$$



Let say , $\left(\frac{1}{n}\right)^{th}$ of chain length is hanging from the table

$$\text{Mass of hanging length} = m_1 = \frac{m}{n}$$

$$\text{Mass of chain on table } = m_2 = m\left(1 - \frac{1}{n}\right)$$



At maximum hanging length,

$$T = f = \frac{mg}{n}$$

$$\mu\left(m\left(1 - \frac{1}{n}\right)9\right) = \frac{m9}{n}$$

$$0.5\left(1 - \frac{1}{n}\right) = \frac{1}{n}$$

$$1 - \frac{1}{n} = \frac{2}{n}$$

$$1 = \frac{3}{n}$$

$$\Rightarrow n = 3$$

Maximum Length of the chain hanging from the table

$$= \frac{l}{n} = \frac{6}{3} = 2\,m$$

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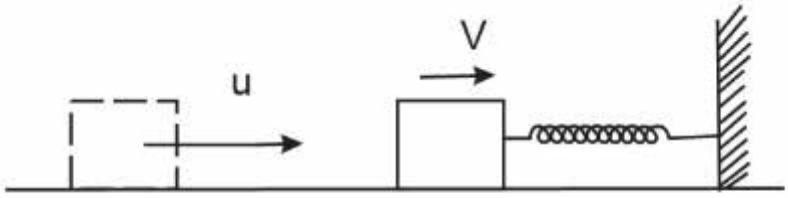
Q. 22 A $0.5\ \text{kg}$ block moving at a speed of $12\ \text{ms}^{-1}$ compresses a spring through a distance $30\ \text{cm}$ when its speed is halved. The spring constant of the spring will be_____ Nm^{-1}.

Correct Answer:
600

Solution:



$$v = \frac{u}{2}$$
$$x = 30\,\text{cm} = 0.3\,\text{m}$$

$$TE_i = TE_f$$

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$
$$\frac{3}{4} \times \frac{1}{2}mu^2 = \frac{1}{2}k(0.3)^2$$
$$\frac{3}{8} \times 144 = k\left(9 \times 10^{-2}\right)$$

$$k = 600\left(\frac{N}{m}\right)$$

Q. 23 The velocity of upper layer of water in a river is 36kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3}Nm^{-2}. Depth of the river is_____m. (Co-efficient of viscosity of water is $10^{-2}\ \text{Pa.s}$)

Correct Answer:
100

Solution:

$$V = 36\frac{\text{km}}{\text{hr}} = 10\,\text{m/s}$$

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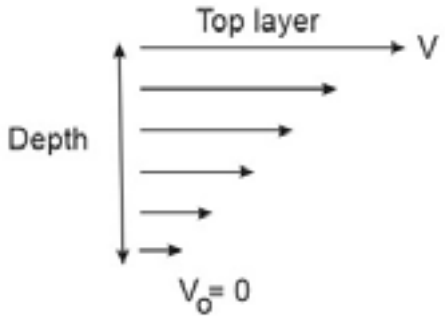

By Newton's law for laminar flow ,

$$F = \eta A \left(\frac{dv}{dx}\right)$$

$$\text{Shearing stress} = \left(\frac{F}{A}\right) = \eta \left(\frac{\Delta v}{\Delta x}\right)$$

$$10^{-3} = 10^{-2} \frac{(v - 0)}{(\text{Depth})}$$

$$\text{Depth of river} = 100\,\text{m}$$

Q. 24 A steam engine intakes $50\ \text{g}$ of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is $540\ \text{cal g}^{-1}$, then the heat rejected by the steam engine per minute is_____ $\times 10^3 \text{cal}$.
(Given : specific heat capacity of water : $1\ cal\ \text{g}^{-1}{}^\circ\text{C}^{-1}$)

Correct Answer:
31

Solution:
By principal of calorimeter,
Heat lost = Heat gained

In one minute,

$$mL_v + mC\Delta T = \text{Heat gained}$$

$$50 \times 540 + 50 \times 1 \times (80) = \text{Heat gained}$$

Heat gained = Heat rejected by the steam engine

$$= 31000\,\text{cal}$$

$$= 31 \times 10^3\,\text{cal}$$

Q. 25 The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is $20\ \text{cm}$. The length of the open organ pipe is _____cm.

Correct Answer:
80

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Solution:

For open organ pipe ,

$$f_n = \frac{nv}{2l} \rightarrow (1)$$

For closed organ pipe

$$f_n = \frac{(2n-1)\,v}{4l} \rightarrow (2)$$

For first overtone of an open organ pipe

$$n = 2\,(2^{nd}\,\text{mode})$$

$$f_1 = \frac{2\,v}{2l_1} \rightarrow (3)$$

Fundamental frequency of closed organ pipe, $n = 1$

$$f_2 = \frac{v}{4l_2} \rightarrow (4)$$

$$f_1 = f_2 \quad (\text{Given})$$

$$\frac{2v}{2l_1} = \frac{v}{4l_2}$$

$$l_1 = 4l_2$$

$$= 4(20)$$

$$= 80\ \text{cm}$$

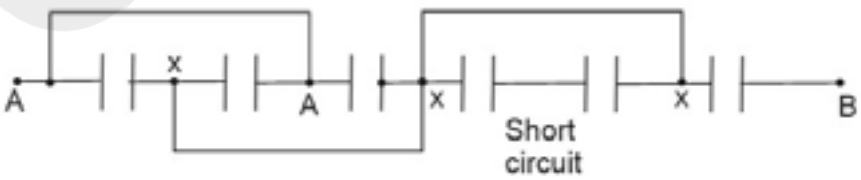
The length of the open organ pipe is $80\ \text{cm}$.

Q. 26 The equivalent capacitance between points A and B in below shown figure will be_____ μF.

Correct Answer:

6

Solution:



Let $C = 8\,\mu F$

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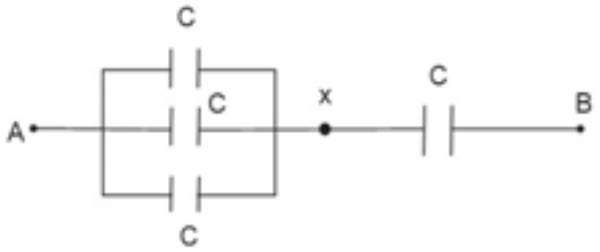


$$C_{AB} = \frac{3C}{3C + C} = \frac{3C}{4} = \frac{3 \times 8}{4}$$
$$C_{AB} = 6\,\mu F$$

The equivalent capacitance between points A and B is $(6\,\mu F)$

Q. 27 A resistor develops $300\ J$ of thermal energy in $15\ s$, when a current of $2\ A$ is passed through it. If the current increases to $3\ A$, the energy developed in $10\ s$ is _____J.

Correct Answer:
450

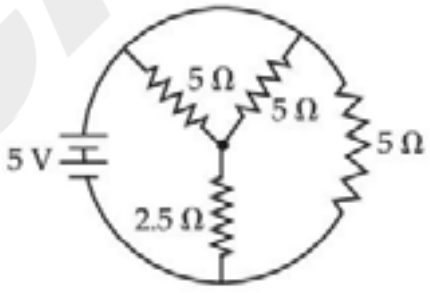
Solution:
$$H_1 = 300\,J = I_1^2\,R\,t_1$$
$$H_2 = I_2^2\,R\,t_2$$

$$\frac{H_1}{H_2} = \frac{I_1^2\,t_1}{I_2^2\,t_2}$$

$$\frac{300}{H_2} = \frac{4}{9} \times \frac{15}{10} = \frac{2}{3}$$
$$H_2 = 450\,J$$

The energy developed in 10 s is $450\,J$

Q. 28 The total current supplied to the circuit as shown in figure by the $5\ V$ battery is _____A.



Correct Answer:
2

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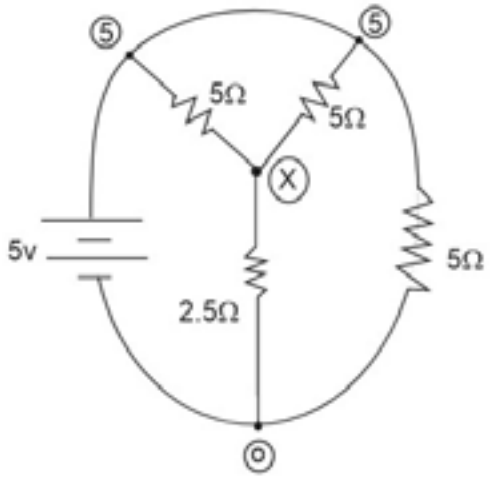
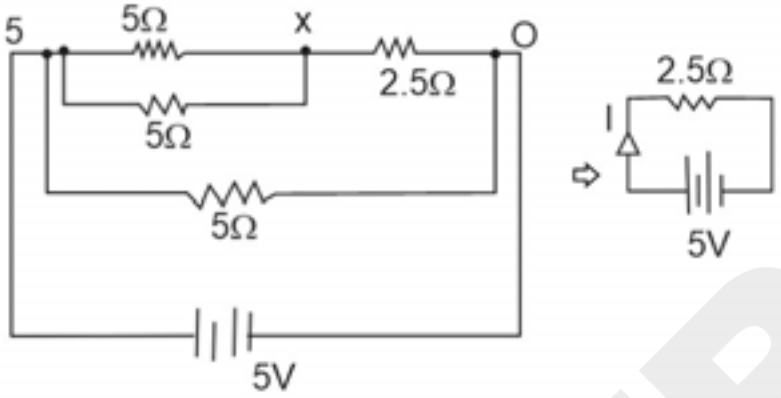
Solution:



Redrawing the circuit we get ,



$$\mathrm{I}=\frac{5\mathrm{V}}{2.5\,\Omega}=2\mathrm{A}$$

The total current supplied $5\mathrm{V}$ battery is $2\mathrm{A}$

Q. 29 The current in a coil of self inductance $2.0\mathrm{H}$ is increasing according to $I=2\sin\left(t^2\right)A$. The amount of energy spent during the period when current changes from $0\ to\ 2\ \mathrm{A}$ is_____J.

Correct Answer:
4

Solution:

$$\mathrm{e}=\left|\mathrm{L}\left(\frac{\mathrm{di}}{\mathrm{dt}}\right)\right|=2\times2\left[\cos\left(\mathrm{t}^2\right)\right]2\mathrm{t}$$
$$\mathrm{e}=8\left[\cos\left(\mathrm{t}^2\right)\right]\mathrm{t}$$

$$\mathrm{dH}=\mathrm{ei}\,\mathrm{dt}$$
$$\mathrm{dH}=16\mathrm{t}\,\sin\left(\mathrm{t}^2\right)\cos\left(\mathrm{t}^2\right)\mathrm{dt}$$
$$\mathrm{dH}=8\mathrm{t}\,\sin\left(2\mathrm{t}^2\right)\mathrm{dt}$$

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$$\begin{array}{c|c} & When \\ Put, 2t^2 = x & I = 0, t^2 = 0 \\ 4t\ dt = dx & x = 0 \\ \int dH = \int_0^{\pi} 2\sin(x)\,dx & I = 2, t^2 = \frac{\pi}{2} \\ & x = \pi \end{array}$$

$$\begin{aligned} \mathrm{H} &= 2\left[-\cos \mathrm{x}\right]_0^{\pi} \\ &= 2\left(-\cos\pi - (-\cos 0^{\circ})\right) \\ \mathrm{H} &= 4\,\mathrm{J} \end{aligned}$$

Q. 30 A force on an object of mass $100\ \mathrm{g}$ is $(10\hat{i} + 5\hat{j})\mathrm{N}$. The position of that object at $\mathrm{t} = 2\ \mathrm{s}$ is $(\mathrm{a}\hat{i} + \mathrm{b}\hat{j})\mathrm{m}$ after starting from rest. The value of $\frac{\mathrm{a}}{\mathrm{b}}$ will be_____.

Correct Answer:
2

Solution:

$$\bar{\mathrm{F}} = \left(10\,\hat{\mathrm{i}} + 5\,\hat{\mathrm{j}}\right)$$

$$\mathrm{m} = 0.1\,\mathrm{kg}$$

$$\bar{\mathrm{a}} = \left(100\,\hat{\mathrm{i}} + 50\,\hat{\mathrm{j}}\right)$$

$$\mathrm{u} = 0$$

$$\bar{\mathrm{s}} = \bar{\mathrm{u}} + \frac{1}{2}\,\bar{\mathrm{a}}\,\mathrm{t}^2$$

$$\left(\mathrm{a}\,\hat{\mathrm{i}} + \mathrm{b}\,\hat{\mathrm{j}}\right) = 0 + \frac{1}{2}\left(100\,\hat{\mathrm{i}} + 50\,\hat{\mathrm{j}}\right)(4)$$

$$\mathrm{a}\,\hat{\mathrm{i}} + \mathrm{b}\,\hat{\mathrm{j}} = 200\,\hat{\mathrm{i}} + 100\,\hat{\mathrm{j}}$$

$$\mathrm{a} = 200$$

$$\mathrm{b} = 100$$

$$\frac{\mathrm{a}}{\mathrm{b}} = 2$$

Chemistry

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Q. 1 Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?

(A) NO

(B) N_2

(C) O_2

(D) C_2

(E) B_2

Choose the most appropriate answer from the options given below:

Option 1:
(A), (B), (C) only

Option 2:
(B), (C), (E) only

Option 3:
(A), (C) only

Option 4:
(D) only

Correct Answer:
(A), (C) only

Solution:
If the highest occupied molecular orbital (HOMO) is an antibonding molecular orbital, removal of an electron leads to an increase in Bond order thereby increasing the bond strength.

Among the given species, the ones having Antibonding Molecular orbital as the HOMO are NO, O_2

Hence, the correct answer is **Option (3)**

Q. 2 **Incorrect** statement for Tyndall effect is :

Option 1:
The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude.

Option 2:
The diameter of the dispersed particles is much smaller than the wavelength of the light used.

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Option 3:
During projection of movies in the cinemas hall, Tyndall effect is noticed.

Option 4:
It is used to distinguish a true solution from a colloidal solution.

Correct Answer:
The diameter of the dispersed particles is much smaller than the wavelength of the light used.

Solution:
As we have learnt,

For Tyndall effect, the refractive indices of the dispersed phase and the dispersion medium should differ greatly in magnitude and the diameter of the dispersed phase and the wavelength of light used showed be similar.

Thus, statement given in Option (2) is incorrect.

Hence, the answer is **Option (2)**

Q. 3 The pair, in which ions are isoelectronic with Al^{3+} is :

Option 1:
Br^{-} and Be^{2+}

Option 2:
Cl^{-} and Li^{+}

Option 3:
S^{2-} and K^{+}

Option 4:
O^{2-} and Mg^{2+}

Correct Answer:
O^{2-} and Mg^{2+}

Solution:
As we have learnt,

Isoelectronic species have same number of electrons.

The set of ions having same electrons as Al^{3+} (10 electrons) are O^{2-} and Mg^{2+}

Hence, the answer is **Option (4)**

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Q. 4 Leaching of gold with dilute aqueous solution of NaCN in presence of oxygen gives complex [A], which on reaction with Zinc forms the elemental gold and another complex [B]. [A] and [B] respectively are :

Option 1:
$[\mathrm{Au(CN)_4}]^-$ and $[\mathrm{Zn(CN)_2(OH)_2}]^{2-}$

Option 2:
$[\mathrm{Au(CN)_2}]^-$ and $[\mathrm{Zn(OH)_4}]^{2-}$

Option 3:
$[\mathrm{Au(CN)_2}]^-$ and $[\mathrm{Zn(CN)_4}]^{2-}$

Option 4:
$[\mathrm{Au(CN)_4}]^{2-}$ and $[\mathrm{Zn(CN)_6}]^{4-}$

Correct Answer:
$[\mathrm{Au(CN)_2}]^-$ and $[\mathrm{Zn(CN)_4}]^{2-}$

Solution:
Leaching of Gold occurs as

$$\mathrm{Au + CN^- + O_2 \rightarrow \underset{[A]}{[Au(CN)_2]^-} + OH^-}$$

$$\mathrm{[Au(CN)_2]^- + Zn \rightarrow Au + \underset{[B]}{[Zn(CN)_4]^{2-}}}$$

Thus, the complexes $[\mathrm{A}]$ and $[\mathrm{B}]$ are respectively $[\mathrm{Au(CN)_2}]^-$ and $[\mathrm{Zn(CN)_4}]^{2-}$

Hence, the correct answer is **Option (3)**

Q. 5 Number of electrons deficient molecules among the following

$\mathrm{PH_3,\ B_2H_6, CCl_4, NH_3, LiH}$ and $\mathrm{BCl_3}$ is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

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Correct Answer:
2

Solution:
The electron deficient species among the given ones are B_2H_6 and BCl_3.

Hence, the answer is **Option (3)**

Q. 6 Which one of the following alkaline earth metal ions has the highest ionic mobility in its aqueous solution?

Option 1:
Be^{2+}

Option 2:
Mg^{2+}

Option 3:
Ca^{2+}

Option 4:
Sr^{2+}

Correct Answer:
Sr^{2+}

Solution:
Greater the charge density on an ion, greater is the extent of hydration and lesser is the ionic mobility in aqueous solution.

Order of Ionic mobility is given as

$$Be^{2+} < Mg^{2+} < Ca^{2+} < Sr^{2+}$$

Hence, the correct answer is **Option (4)**.

Q. 7 White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of:

Option 1:
$[Ag(NH_3)_4]Cl_2$

Option 2:
$[Ag(Cl)_2(NH_3)_2]$

Option 3:
$[Ag(NH_3)_2]Cl$

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Option 4:
$[Ag\ (NH_3)\ Cl]\ Cl$

Correct Answer:
$[Ag\ (NH_3)_2]\ Cl$

Solution:
AgCl dissolves in aq. Ammonia due to the formation of Diamminesilver(I) chloride.

$$AgCl + 2NH_3 \longrightarrow [Ag\ (NH_3)_2]\ Cl$$

Hence, the answer is **Option (3)**

Q. 8 Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?

Option 1:
It will not prefer to undergo redox reactions.

Option 2:
It will prefer to gain electron and act as an oxidizing agent

Option 3:
It will prefer to give away an electron and behave as reducing agent

Option 4:
It acts as both, oxidizing and reducing agent.

Correct Answer:
It will prefer to gain electron and act as an oxidizing agent

Solution:
Ce has two stable oxidation states of +3 and +4.

Ce^{+4} prefers to gain an electron and acts as an oxidising agent.

Hence, the answer is **Option (2)**

Q. 9 Among the following, which is the strongest oxidizing agent?

Option 1:
Mn^{3+}

Option 2:
Fe^{3+}

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Option 3:
Ti^{3+}

Option 4:
Cr^{3+}

Correct Answer:
Mn^{3+}

Solution:
As we have learnt,

Reduction potential denotes the tendency of any species to get reduced and thus act as an Oxidising agent

The reduction potential $E^0_{M^{3+}|M^{2+}}$ of the given ions are given below

$$\begin{aligned} Mn^{3+}: &\quad +1.57\ V \\ Fe^{3+}: &\quad +0.77\ V \\ Ti^{3+}: &\quad -0.37\ V \\ Cr^{3+}: &\quad -0.41\ V \end{aligned}$$

Thus, Mn^{3+} has the greatest reduction potential and is the strongest oxidising agent.

Hence, the correct answer is **option (1)**

Q. 10 The eutrophication of water body results in :

Option 1:
loss of Biodiversity.

Option 2:
breakdown of organic matter.

Option 3:
increase in biodiversity.

Option 4:
decrease in BOD.

Correct Answer:
loss of Biodiversity.

Solution:
Eutrophication refers to the overpopulation of water bodies by aquatic plants which depreoves the animal life of oxygen and thereby kills animal life.

Thus, eutrophication leads to a loss of biodiversity.

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Hence, the answer is **Option (1)**

Q. 11 Phenol on reaction with dilute nitric acid, gives two products. Which method will be most efficient for large scale separation ?

Option 1:
Chromatographic separation

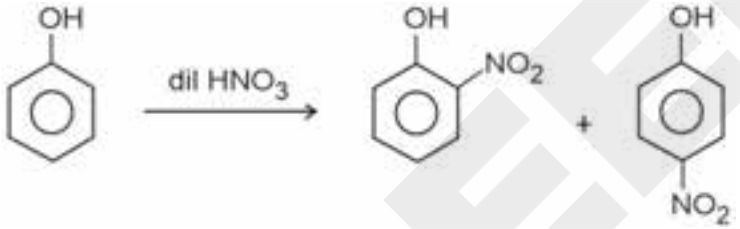
Option 2:
Fractional Crystallisation

Option 3:
Steam distillation

Option 4:
Sublimation

Correct Answer:
Steam distillation

Solution:
Phenol reacts with $\mathrm{dil.HNO_3}$ to form ortho and para nitrophenols.



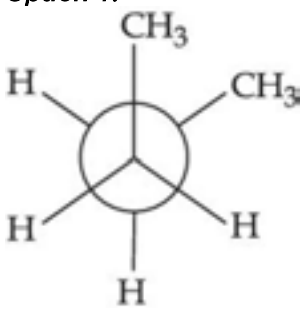
o - Nitrophenols are steam volatile while p - Nitrophenols are not.

Hence, they can be separated by steam distillation.

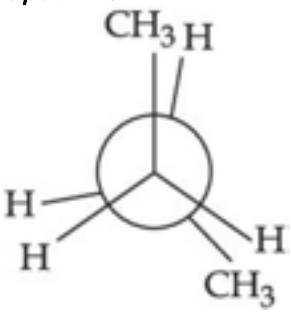
Hence, the correct answer is **Option (3)**

Q. 12 In the following structures, which one is having staggered conformation with maximum dihedral angle?

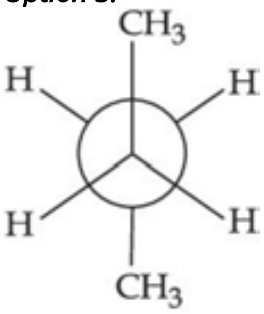
Option 1:



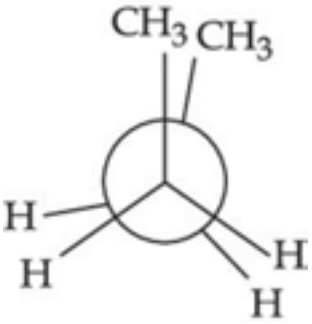
Option 2:



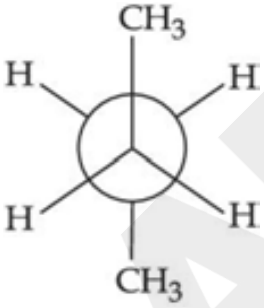
Option 3:



Option 4:



Correct Answer:



Solution:

The staggered form having max dihedral angle is the anti form. It has a dihedral angle of 180° between the two bulky methyl groups.

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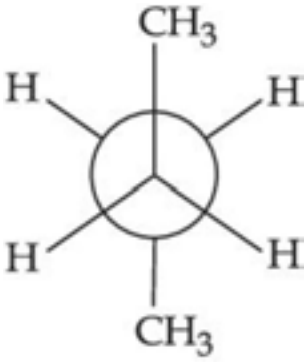
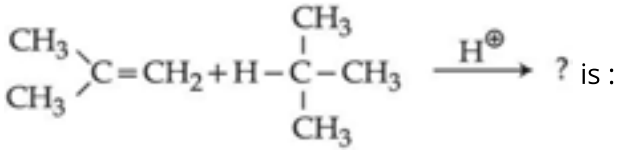
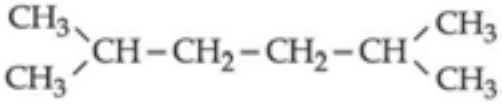


Hence, the correct answer is **option (3)**

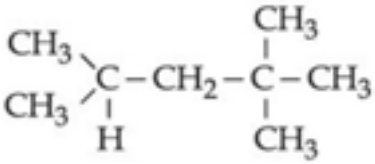
Q. 13 The product formed in the following reaction.



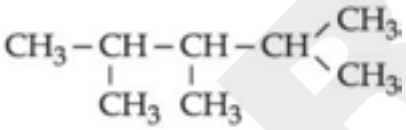
Option 1:



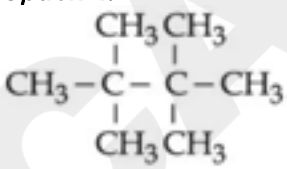
Option 2:



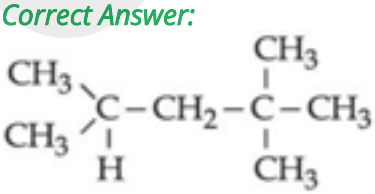
Option 3:



Option 4:



Correct Answer:



Solution:
The reaction occurs as

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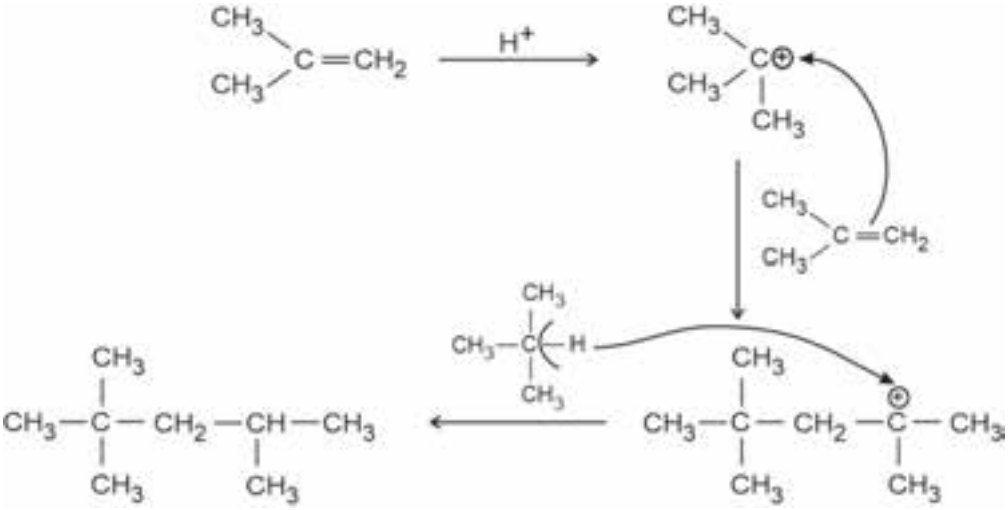


Hence, the correct answer is **Option (2)**

Q. 14 The IUPAC name of ethylidene chloride is :

Option 1:
1-Chloroethene

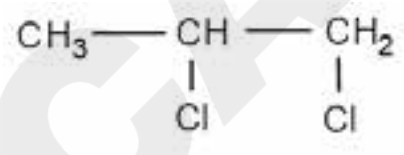
Option 2:
1-Chloroethyne

Option 3:
1,2-Dichloroethane

Option 4:
1,1-Dichloroethane

Correct Answer:
1,1-Dichloroethane

Solution:
The structure of ethylidene chloride is



Its IUPAC name is 1,2-Dichloroethane

Hence, the answer is **Option (4)**

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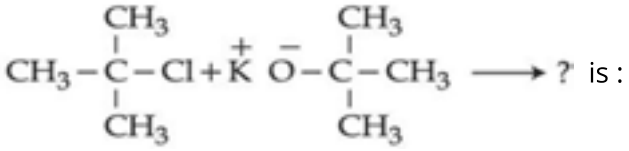
Q. 15 The major product in the reaction



Option 1:
t-Butyl ethyl ether

Option 2:
2,2 -Dimethyl butane

Option 3:
2-Methyl pent-1-ene

Option 4:
2-Methyl prop-1-ene

Correct Answer:
2-Methyl prop-1-ene

Solution:
Bulky base on reaction with a tertiary halide causes dehydrohalogenation and forms an alkene.

Hence, the correct answer is **Option (4).**

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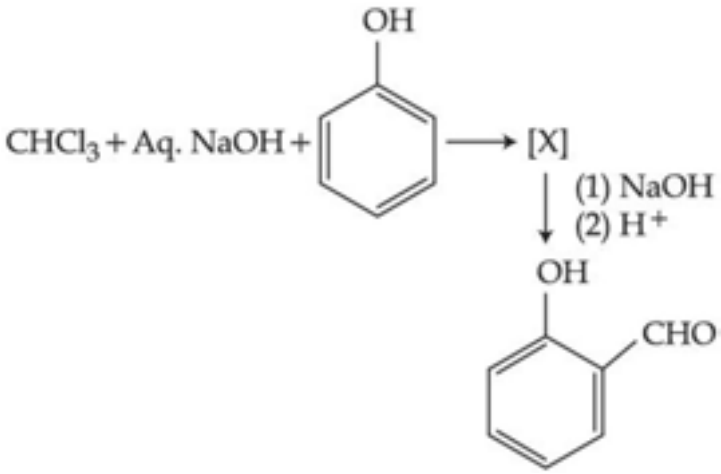
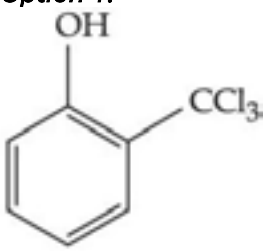
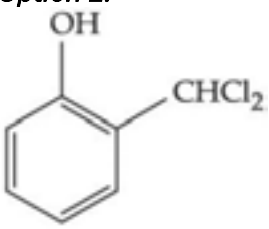
Q. 16 The intermediate X, in the reaction is :



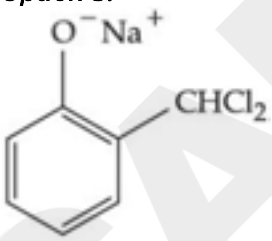
Option 1:



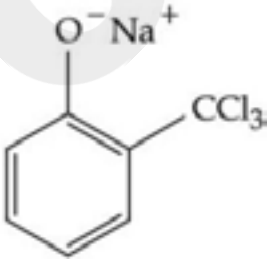
Option 2:



Option 3:



Option 4:



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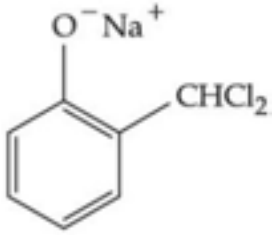
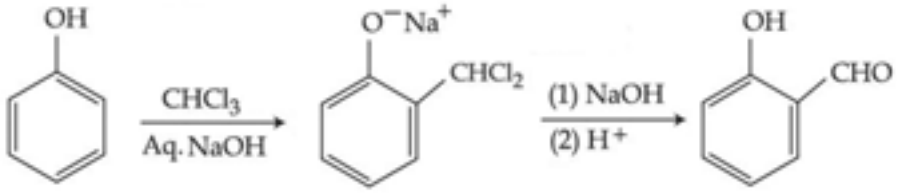
Correct Answer:



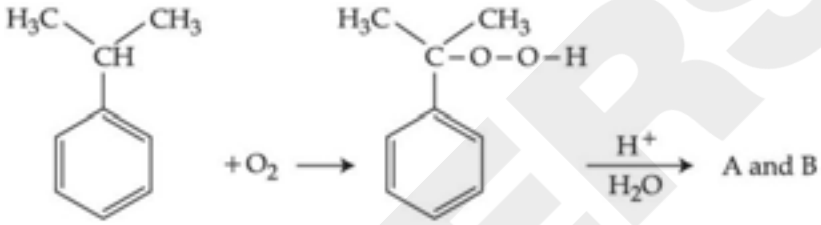
Solution:

The given reaction is known as Reimer Tiemann reaction and it occurs as



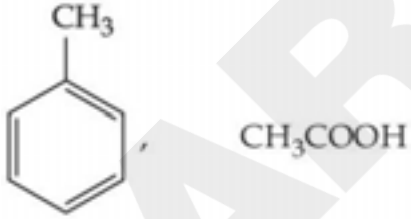
Hence, the answer is **Option (3)**

Q. 17 In the following reaction :

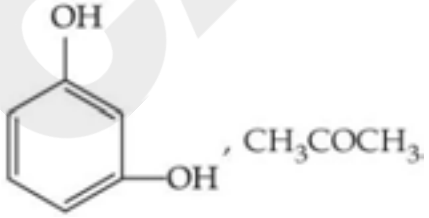


The compounds A and B respectively are :

Option 1:



Option 2:



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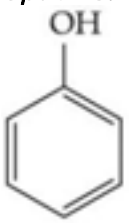
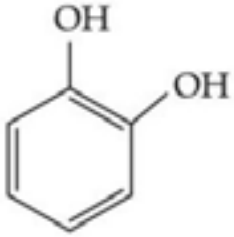
Option 3:



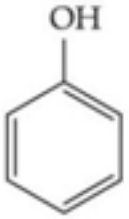
, CH_3COCH_3

Option 4:



, CH_3COCH_3

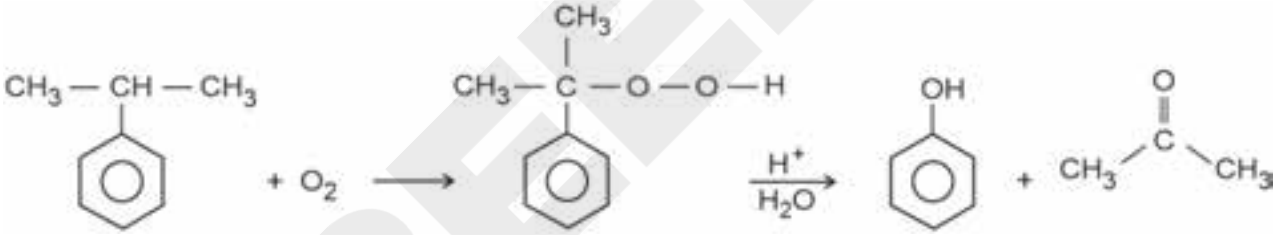
Correct Answer:



, CH_3COCH_3

Solution:

The given reaction is Cumene hydroperoxide synthesis of phenol.



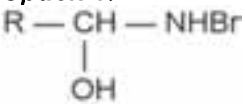
Hence, the correct answer is **Option (3)**

Q. 18

The reaction of $R-\underset{\underset{O}{\|}}{C}-NH_2$ with bromine and KOH gives RNH_2 as the end product.

Which one of the following is the intermediate product formed in this reaction ?

Option 1:



Option 2:

$R-NH-Br$

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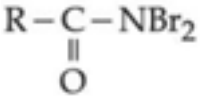
Option 3:

Option 4:

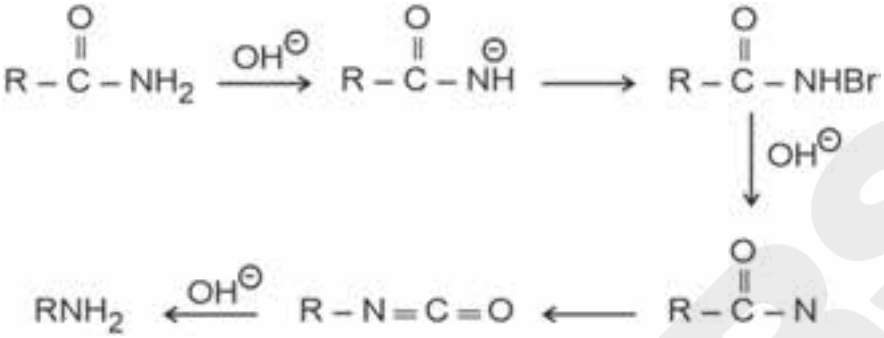


Correct Answer:

Solution:

The given reaction is called the Hoffman Bromamide reaction.

The mechanism of the reaction is



Hence, the correct answer is **Option (3)**

Q. 19 Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes, because:

Option 1:

soap particles remain floating in water as ions.

Option 2:

the hydrophobic part of soap is not able to take away grease.

Option 3:

the micelles are not formed due to concentration of soap, below its CMC value.

Option 4:

colloidal structure of soap in water is completely disturbed.

Correct Answer:

the micelles are not formed due to concentration of soap, below its CMC value.

Solution:

Cleansing action of Soap is due to micelle formation. If the concentration of Soap is very little, the micelles will not be formed as there is a requirement of a minimum concentration of soap for micelle formation called as the critical micelle concentration(CMC).

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Hence, the answer is **Option (3)**

Q. 20 Which one of the following is an example of artificial sweetener?

Option 1:
Bithional

Option 2:
Alitame

Option 3:
Salvarsan

Option 4:
Lactose

Correct Answer:
Alitame

Solution:
Fact Based.

Alitame is an example of artificial sweetener.

Hence, the answer is **Option (2)**

Q. 21 The number of N atoms in 681 g of $C_7H_5N_3O_6$ is x $\times 10^{21}$. The value of x is _____.

$\left(N_A = 6.02 \times 10^{23}\ mol^{-1}\right)$ (Nearest Integer)

Correct Answer:
5418

Solution:

$$\text{Molar mass of } C_7H_5N_3O_6 = 227$$

$$\therefore \text{Moles of } C_7H_5N_3O_6 = \frac{681}{227} = 3$$

$$\begin{aligned}\therefore \text{ number of N atoms } &= 3 \times 3 \times N_A \\ &= 9 \times 6.02 \times 10^{23} \\ &= 5418 \times 10^{21}\end{aligned}$$

Hence, the answer is 5418

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Q. 22 The distance between $\mathrm{Na}^{+}$and Cl^{-}ions in solid NaCl of density 43.1 g cm^{-3} is_____ $\times 10^{-10} \mathrm{m}$. (Nearest Integer)

$(\text{Given : } \mathrm{N_A} = 6.02 \times 10^{23} \text{ mol}^{-1})$

Correct Answer:
1

Solution:

$$\text{Density } = \frac{ZM_0}{N_A\ a^3} = 43.1$$

$$\Rightarrow \quad \frac{4 \times 58.5}{6.02 \times 10^{23}\ a^3} = 43.1$$

$$\Rightarrow \quad a^3 = 9 \times 10^{-24}$$

$$\Rightarrow \quad a = 2.08 \times 10^{-8} \text{ cm}$$

$$\Rightarrow \quad a = 2.08 \times 10^{-10} \text{ m}$$

Now in NaCl type of solid

$$a = 2\left(r_+ + r_-\right)$$

$$\Rightarrow r_+ + r_- = \frac{a}{2} = 1.04 \times 10^{-10} \text{ m}$$

Thus, the distance between Na^{+} and Cl^{-} is $1.04 \times 10^{-10} \mathrm{m}$

Hence , the answer is 1

Q. 23 The longest wavelength of light that can be used for the ionisation of Li^{2+} ion in its ground state is $\mathrm{x} \times 10^{-8}$ m. The value of x is_____ . (Nearest Integer)

(Given : Energy of the electron in the first shell of the hydrogen atom is $-2.2 \times 10^{-18} \mathrm{J}; \mathrm{h} = 6.63 \times 10^{34} \mathrm{Js}$ and $\mathrm{c} = 3 \times 10^{8} \mathrm{ms}^{-1}$)

Correct Answer:
1

Solution:

$$\mathrm{IE} = \mathrm{E}_0 \times \frac{\mathrm{Z}^2}{\mathrm{n}^2} = 2.2 \times 10^{-18} \times 9$$

$$= 19.8 \times 10^{-18} \text{ J}$$

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$$\therefore \frac{hc}{\lambda} = 19.8 \times 10^{-18}$$

$$\Rightarrow \lambda = \frac{hc}{19.8 \times 10^{-18}} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{19.8 \times 10^{-18}}$$

$$= 1 \times 10^{-8}$$

Hence, the answer is 1

Q. 24 The standard entropy change for the reaction

$4\mathrm{Fc(s)} + 3\mathrm{O_2(g)}, 2\mathrm{Fe_2O_3(\ s)}$ is $-550\ \mathrm{J\ K^{-1}}$ at $298\ \mathrm{K}$
[Given : The standard enthalpy change for the reaction is $-165\ \mathrm{kJ\ mol^{-1}}$]. The temperature in K at which the reaction attains equilibrium is (Nearest Integer)

Correct Answer:
300

Solution:
At equilibrium,

$$\Delta G = 0$$

$$\Rightarrow \Delta H = T\Delta S$$

$$\Rightarrow T = \frac{\Delta H}{\Delta S}$$

$$\Rightarrow T = \frac{-165 \times 10^{3}}{-550} = 300\ \mathrm{K}$$

Hence, the answer is 300

Q. 25 1 L aqueous solution of $\mathrm{H_2SO_4}$ contains 0.02 m mol $\mathrm{H_2SO_4}$.50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of $\mathrm{H_2SO_4}$ are added. Total m mols of $\mathrm{H_2SO_4}$ in the final solution is _____$\times 10^{-3}\mathrm{m\ moles.}$

Correct Answer:
20

Solution:
0.5 lit solution (A) initially contains 0.01 mmole. The solution is then diluted and further 0.01 mmole are added.

$$\therefore \text{Total mmoles} = 0.02\ \text{mmoles}$$

$$= 20 \times 10^{-3}\ \text{mmoles}$$

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Hence, the answer is 20

Q. 26 The standard free energy change (ΔG^{o}) for 50% dissociation of N_2O_4 into NO_2 at $27^{o}C$ and 1 atm pressure is $-x$ J mol^{-1}. The value of x is _____.(Nearest Integer)

$[\text{Given } : R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}, \log 1.33 = 0.1239 \ln 10 = 2.3]$

Correct Answer:
710

Solution:

$N_2O_4 \rightleftharpoons 2NO_2$

P_o

$\frac{P_o}{2}$ $\quad P_o$

Given,

$$P_0 + \frac{P_0}{2} = 1$$

$$\Rightarrow \frac{3P_0}{2} = 1 \Rightarrow P_0 = \frac{2}{3}$$

$$\therefore P_{NO_2} = \frac{2}{3}; \quad P_{N_2O_4} = \frac{1}{3}$$

$$K_p = \frac{P_{NO_2}{}^2}{P_{N_2O_4}} = \frac{(2/3)^2}{(1/3)} = \frac{4}{3}$$

$$\therefore \Delta G^0 = -RT \ln K = -8.31 \times 300 \times \ln\left(\frac{4}{3}\right)$$

$$= -8.31 \times 300 \times 2.3 \times 0.1239$$

$$= -710.4 \text{ J mol}^{-1}$$

Hence, the answer is 710

Q. 27 In a cell, the following reactions take place

$$Fe^{2+} \rightarrow Fe^{3+} + e^- \quad E^{\circ}_{Fe^{3+}|Fe^{2+}=0.77 \text{ V}}$$

$$2I^- \rightarrow I_2 + 2e^- \quad E^{\circ}_{I_2|I^-} = 0.54 \text{ V}$$

The standard electrode potential for the spontaneous reaction in the cell is $x \times 10^{-2}$ V at 298 K. The value of x is (Nearest Integer)

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Correct Answer:
23

Solution:
The spontaneous reaction is

$$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$$

$$\begin{aligned} E^0 &= E^0_{Fe^{3+}|Fe^{2+}} - E^0_{I_2|I^-} \\ &= 0.77 - 0.54 \\ &= 0.23\ V \end{aligned}$$

Thus, the electrode potential of the spontaneous reaction is $23 \times 10^{-2} V$

Hence, the answer is 23.

Q. 28 For a given chemical reaction

$$\gamma_1\ A + \gamma_2\ B \rightarrow \gamma_3 C + \gamma_4 D$$

Concentration of C changes from $10\ mmol\ dm^{-3}$ to $20\ mmol\ dm^{-3}$ in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be $9\ mmol\ dm^{-3}\ s^{-1}$. Therefore the rate of reaction is _____$mmol\ dm^{-3}\ s^{-1}$.

(Nearest Integer)

Correct Answer:
1

Solution:
$$\frac{\Delta[C]}{\Delta t} = 1 mmol\ dm^{-3}\ s^{-1}$$

Also,

$$\frac{\Delta[D]}{\Delta t} = -1.5\frac{\Delta[B]}{\Delta t}$$

$$\text{and } \frac{\Delta[B]}{\Delta t} = -2\frac{\Delta[A]}{\Delta t}$$

$$\text{and } \frac{\Delta[D]}{\Delta t} = 9\frac{\Delta[C]}{\Delta t}$$

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$$\therefore -3\frac{\Delta[A]}{\Delta t} = -1.5\frac{\Delta[B]}{\Delta t} = 9\frac{\Delta[C]}{\Delta t} = \frac{\Delta[D]}{\Delta t}$$

$$\Rightarrow \frac{-\Delta[A]}{3\Delta t} = -\frac{\Delta[B]}{6\Delta t} = \frac{\Delta[C]}{\Delta t} = \frac{\Delta[D]}{9\Delta t}$$

Thus, the reaction is

$$3\mathrm{A} + 6\mathrm{B} \rightarrow \mathrm{C} + 9\mathrm{D}$$

$$\therefore \text{ Rate } = \frac{\Delta[C]}{\Delta t} = 1$$

Hence, the answer is 1

Q. 29 If $\left[\mathrm{Cu}\left(\mathrm{H_2O}\right)_4\right]^{2+}$ absorbs a light of wavelength 600 nm for d-d transition, then the value of octahedral crystal field splitting energy for $\left[\mathrm{Cu}\left(\mathrm{H_2O_6}\right]^{2+}$ will be _____ $\times 10^{-21}$ J [Nearest integer]

$(\text{Given : } h = 6.63 \times 10^{-34}\text{Js and c} = 3.08 \times 10^{8} \text{ ms}^{-1})$

Correct Answer:
766

Solution:
Given,

$$\lambda = 600 \times 10^{-9} \text{ m} = 6 \times 10^{-7}\text{m}$$

$$\therefore \Delta_t = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.08 \times 10^{-8}}{6 \times 10^{-7}}$$

$$= 3.403 \times 10^{-19}$$

Now,

$$\Delta_0 = \frac{9}{4}\Delta_t = \frac{9}{4} \times 3.403 \times 10^{-19}$$
$$= 7.658 \times 10^{-19}$$
$$= 766 \times 10^{-21} \text{ J}$$

Hence, the answer is 766

Q. 30 Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g.

$(\text{Atomic mass of Br} = 80 \text{ g/mol})$ [Nearest integer]

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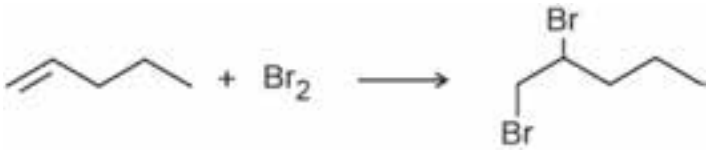
Correct Answer:
1143

Solution:
The reaction is



1 mole of pent-1-ene reects with 1 mole of $\mathrm{Br_2}$

$$\text{Now, moles of Alkene } = \frac{5}{70} = \text{ moles of } \mathrm{Br_2}$$

$$\therefore \text{ Mass of } \mathrm{Br_2} = \frac{5}{70} \times 160$$

$$= 11.43 \text{ g}$$

$$= 1143 \times 10^{-2} \text{ g}$$

Hence, the answer is 1143

Maths

Q. 1 Let a circle C touch the lines $\mathrm{L_1} : 4x - 3y + \mathrm{K_1} = 0$ and $\mathrm{L_2} : 4x - 3y + \mathrm{K_2} = 0,\ \mathrm{K_1},\ \mathrm{K_2} \in \mathbf{R}$. If a line passing through the center of the circle C intersects $\mathrm{L_1}$ at $(-1, 2)$ and L_2 at $(3, -6)$, then the equation of the circle C is:

Option 1:
$(\mathrm{x} - 1)^2 + (\mathrm{y} - 2)^2 = 4$

Option 2:
$(\mathrm{x} + 1)^2 + (\mathrm{y} - 2)^2 = 4$

Option 3:
$(\mathrm{x} - 1)^2 + (\mathrm{y} + 2)^2 = 16$

Option 4:
$(\mathrm{x} - 1)^2 + (\mathrm{y} - 2)^2 = 16$

Correct Answer:
$(\mathrm{x} - 1)^2 + (\mathrm{y} + 2)^2 = 16$

Solution:
Given,

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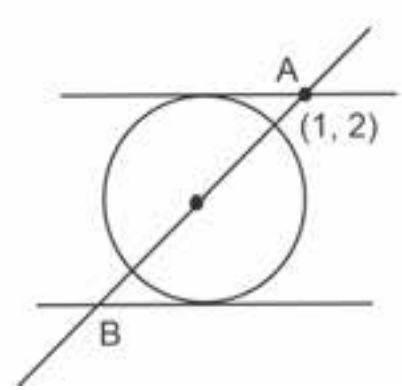
$l_1 = 4x - 3y + k_1 = 0 \quad l_2 = 4x - 3y + k_2 = 0$
Here line $l_1 \text{ and } l_2$ are parallel.
So,



Now point $A(-1, 2)$ will satisfy $l_1 \ \text{so,} \ -4 - 3 \times 2 + k_1 = 0 \quad \Rightarrow k_1 = 10$

Also point $B(3, -6)$ will satisfy $l_2 \ \Rightarrow k_2 = -30$

Now Distance b/w two parallel line will be diameter

$$\text{Diameter} = \left|\frac{k_1 - k_2}{\sqrt{4^2 + 3^2}}\right| = \left|\frac{10 + 30}{5}\right| = 8$$

$$\text{So Radius} = \frac{8}{2} = 4$$

Now mid-point of AB will give us the centre of circle (by symmetry), so by midpoint formula in AB we get centre $C \equiv (1, -2)$
Now eqn of circle will be $(x - 1)^2 + (y + 2)^2 = 4^2$

Q. 2 The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1 + \cos^2 x)\left(e^{\cos x} + e^{-\cos x}\right)} dx$ is equal to:

Option 1:
$\frac{\pi^2}{4}$

Option 2:
$\frac{\pi^2}{2}$

Option 3:
$\frac{\pi}{4}$

Option 4:
$\frac{\pi}{2}$

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Correct Answer:

$\frac{\pi}{4}$

Solution:

$$I = \int_0^{\pi} \frac{\sin x \cdot e^{\cos x}}{(1+\cos^2 x)\left(e^{\cos x}+e^{-\cos x}\right)}\,dx \quad ----(i)$$

$$\text{using property of integral } \int_a^b f(a+b-x) = \int_a^b f(x)$$

$$\text{we get } I = \int_0^{\pi} \frac{\sin x \cdot e^{-\cos x}}{(1+\cos^2 x)\left(e^{-\cos x}+e^{\cos x}\right)}\,dx \quad ----(ii)$$

Adding eqn (i) & (ii) we get

$$2I = \int_0^{\pi} \frac{\sin x}{(1+\cos^2 x)} \cdot \frac{\left(e^{\cos x}+e^{-\cos x}\right)}{e^{\cos x}+e^{-\cos x}}\,dx$$

$$2I = \int_0^{\pi} \frac{\sin x\,dx}{1+\cos^2 x} \Rightarrow 2I = 2\int_0^{\pi/2} \frac{\sin x}{1+\cos^2 x}\,dx$$

$$\text{Let } \cos x = t \Rightarrow -\sin x\,dx = dt$$

$$I = \int_{-1}^{0} \frac{dt}{1+t^2} \Rightarrow I = \left[\tan^{-1} t\right]_0^{-1} = \frac{\pi}{4}$$

Q. 3 Let a,b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$

If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to:

Option 1:

$\frac{5}{2}$

Option 2:

2

Option 3:

$\frac{3}{2}$

Option 4:

1

Correct Answer:

$\frac{5}{2}$

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Solution:

Let $\frac{a+b}{7}=\frac{b+c}{8}=\frac{c+a}{9}=k$

So $a+b=7k, \quad b+c=8k, \quad c+a=9k$

Solving this we get $a=4k, b=3k, c=5k$ and calculating semi-perimeter

$s=\frac{a+b+c}{2} \Rightarrow s=6k$

we know that $\frac{r}{R}=4 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \sin \frac{C}{2}$

$\Rightarrow \frac{r}{R}=4 \sqrt{\frac{(s-b)(s-c)}{bc}} \times \sqrt{\frac{(s-a)(s-c)}{ac}} \times \sqrt{\frac{(s-a)(s-b)}{ab}}$

$\Rightarrow \frac{r}{R}=4 \cdot \frac{(s-a)(s-b)(s-c)}{abc}$

$\Rightarrow \frac{r}{R}=4 \cdot \frac{(2k)(3k)(k)}{(4k)(3k)(5k)} \Rightarrow \frac{r}{R}=\frac{2}{5}$

So the value of $\frac{R}{r}=\frac{5}{2}$

Q. 4 Let $f: \mathbf{N} \rightarrow \mathbf{R}$ be a function such that $f(x+y)=2f(x)f(y)$ for natural numbers x and y. If $f(1)=2$, then the value of α for which

$\sum_{k=1}^{10} f(\alpha+k)=\frac{512}{3}\left(2^{20}-1\right)$ holds, is:

Option 1:

2

Option 2:

3

Option 3:

4

Option 4:

6

Correct Answer:

4

Solution:

Given $f(x+y)=2f(x) \cdot f(y) \ \& \ f(1)=2$

Now putting $x=1 \ \& \ y=1$ in $f(x+y)=2f(x) \cdot f(y)$.

We get

$f(1+1)=2f(1) \cdot f(1)=2 \times 2^2=2^3$

So $f(2)=2^3$

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Similarly, $f(3) = 2^5, f(4) = 2^7 \cdots$

Now

$$\sum_{k=1}^{10} f(a+k) = \sum_{k=1}^{10} 2f(a) \cdot f(k) = \frac{512}{3}\left(2^{20}-1\right).$$

$$\Rightarrow 2f(a)\sum_{k=1}^{10} f(k) = \frac{512}{3}\left(2^{20}-1\right)$$

$$\Rightarrow 2f(a)\{f(1)+f(2)+\cdots f(10)\} = \frac{512}{3}\left(2^{20}-1\right)$$

$$\Rightarrow 2f(a)\left(2+2^3+2^5+\cdots\right) = \frac{512}{3}\left(2^{20}-1\right)$$

$$\Rightarrow 2f(a)\frac{2\left((2^2)^{10}-1\right)}{2^2-1} = \frac{512}{3}\left(2^{20}-1\right)$$

$$\Rightarrow 2\times 2f(a)\times\frac{2^{20}-1}{3} = \frac{512}{3}\left(2^{20}-1\right)$$

Now comparing both side we get

$$4f(a) = 512 \Rightarrow f(a) = 128$$
$$\Rightarrow f(a) = 2^7 \Rightarrow a = 4$$

Q. 5 Let A be a 3×3 real matrix such that

$$A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}; A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix} \text{ and } A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}.$$

If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A-2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$ has:

Option 1:
no solution

Option 2:
infinitely many solutions

Option 3:
unique solution

Option 4:
exactly two solutions

Correct Answer:
infinitely many solutions

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Solution:

$$\mathrm{A} = \begin{bmatrix} a & b & c \\ l & m & n \\ p & q & r \end{bmatrix}$$

$$\mathrm{A}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c \\ n \\ r \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \mathrm{c} = 1,\ \mathrm{n} = 1,\ \mathrm{r} = 2$$

$$\mathrm{A}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c+a \\ n+l \\ r+p \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \mathrm{a} = -2,\ \mathrm{l} = -1,\ \mathrm{p} = -1$$

$$\mathrm{A}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a+b \\ l+m \\ p+q \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \mathrm{b} = 3,\ \mathrm{m} = 2,\ \mathrm{q} = 1$$

$$\mathrm{A} = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow \mathrm{A} - 2\mathrm{I} = \begin{bmatrix} -4 & 3 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$|\mathrm{A} - 2\mathrm{I}| = 0$$

Now,

$$, \begin{bmatrix} -4 & 3 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$$-4\mathrm{x}_1 + 3\mathrm{x}_2 + \mathrm{x}_3 = 4 \ ----(1)$$
$$-\mathrm{x}_1 + \mathrm{x}_3 = 1 \ ----(2)$$
$$-\mathrm{x}_1 + \mathrm{x}_2 = 1 \ ----(3)$$

$$\text{From eqn}(1) - [\text{eqn}(2) + 3 \times \text{eqn}(3)]$$
$$0 = 0 \Rightarrow \text{infinite solution}$$

Q. 6 Let $f : \mathbf{R} \to \mathbf{R}$ be defined as

$$f(x) = x^3 + x - 5$$

If $g(x)$ is a function such that $f(g(x)) = x, \quad \forall\ 'x' \in \mathbf{R}$, then $g'(63)$ is equal to _____ .

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Option 1:
$\frac{1}{49}$

Option 2:
$\frac{3}{49}$

Option 3:
$\frac{43}{49}$

Option 4:
$\frac{91}{49}$

Correct Answer:
$\frac{1}{49}$

Solution:
Here $g(f(x)) = x$ means $f(x)$ and $g(x)$ are inverse of each other.

$$\text{Now}, g'(f(x))f'(x) = 1.$$
$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)} - - - -(1)$$
$$\text{Now } f(x) = 63 \Rightarrow x^3 + x - 5 = 63.$$
$$\Rightarrow x^3 + x - 68 = 0$$
So $x = 4$ Satisfies the above eqn

$$g'(63) = \frac{1}{f'(4)} \text{ from eq}^{n} - (1)$$
$$= \frac{1}{3(4)^2 + 1} = \frac{1}{49}$$

Q. 7 Consider the following two propositions:

$$P1 : \sim (p \rightarrow \sim q)$$
$$P2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then:

Option 1:
P1 is TRUE and P2 is FALSE

Option 2:
P1 is FALSE and P2 is TRUE

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Option 3:
Both P1 and P2 are FALSE

Option 4:
Both P1 and P2 are TRUE

Correct Answer:
Both P1 and P2 are FALSE

Solution:

p	q	$\sim p$	$\sim q$	$\sim p \vee q$	$p \rightarrow (\sim p \vee q)$	$p \rightarrow \sim q$	$P_1 : \sim (p \rightarrow \sim q)$	$p \wedge \sim q$	$P_2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$
T	T	F	F	T	T	F	T	F	F
T	F	F	T	F	F	T	F	T	F
F	T	T	F	T	T	T	F	F	F
F	F	T	T	T	T	T	F	F	F

$p \rightarrow (\sim p \vee q)$ is F when p is true and q is false from table.
Hence, P_1 & P_2 both are false

Q. 8 If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \ldots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$' then the remainder when K is divided by 6 is:

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
5

Correct Answer:
5

Solution:

$$\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \cdots + \frac{1}{2^9 \cdot 3^2} + \frac{1}{2^{10} \cdot 3}$$

This is a GP with common ration = 3/2 and n = 10

So, Sum $= \frac{3^{10} - 2^{10}}{2^{10} \cdot 3^{10}}$

$\text{So, k} = 3^{10} - 2^{10}$

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Since we need the remainder when k is divided by 6

$\text{So, } 3^{10} = 6q_1 + 3 \text{ and } 2^{10} = 6q_2 + 4$

Now k will be of the form $(6q_1 + 3) - (6q_2 + 4)$
$= 6(q_1 - q_2) - 1$
Hence, when k is divided by 6 , we get the remainder as $6 - 1 = 5$

Q. 9 Let $f(x)$ be a polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then the value of $\lim\limits_{x \to 1} \dfrac{f(x)}{x - 1}$ is equal to:

Option 1:
-15

Option 2:
-60

Option 3:
60

Option 4:
15

Correct Answer:
-15

Solution:
$f(x) = (x - 1)\left(x^4 + ax^3 + bx^2 + cx + d\right)$
Hence,
$f(x) + f'(x) + f''(x) = x^5 + (a + 4)x^4 + (b + 3a + 16)x^3 + (9a + 2b + c - 12)x^2 + (d + c + 4b - 6a)x + (c - 2b)$

$\because f(x) + f'(x) + f''(x) = x^5 + 64 \Rightarrow a + 4 = 0, b + 3a + 16 = 0, 9a + 2b + c - 12 = 0$
$, d + c + 4b - 6a = 0$
$\Rightarrow a = -4, b = -4, c = 56, d = -64$

$$\text{So } \lim_{x \to 1} \frac{f(x)}{(x - 1)} = \lim_{x \to 1} \left(x^4 - 4x^3 - 4x^2 + 56x - 64\right)$$
$$= 1 - 4 - 4 + 56 - 64$$
$$= -15$$

Q. 10 Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1 \mid E_2) = \dfrac{1}{2}$, $P(E_2 \mid E_1) = \dfrac{3}{4}$ and $P(E_1 \cap E_2) = \dfrac{1}{8}$. Then:

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Option 1:
$P\left(E_1 \cap E_2\right)=P\left(E_1\right) \cdot P\left(E_2\right)$

Option 2:
$\mathrm{P}\left(\mathrm{E}_1^{\prime} \cap \mathrm{E}_2^{\prime}\right)=\mathrm{P}\left(\mathrm{E}_1^{\prime}\right) \cdot \mathrm{P}\left(\mathrm{E}_2\right)$

Option 3:
$P\left(E_1 \cap E_2^{\prime}\right)=P\left(E_1\right) \cdot P\left(E_2\right)$

Option 4:
$P\left(E_1^{\prime} \cap E_2\right)=P\left(E_1\right) \cdot P\left(E_2\right)$

Correct Answer:
$P\left(E_1 \cap E_2^{\prime}\right)=P\left(E_1\right) \cdot P\left(E_2\right)$

Solution:
$\frac{\mathrm{P}\left(\mathrm{E}_1 \cap \mathrm{E}_2\right)}{\mathrm{P}\left(\mathrm{E}_2\right)}=\frac{1}{2}, \frac{\mathrm{P}\left(\mathrm{E}_2 \cap \mathrm{E}_1\right)}{\mathrm{P}\left(\mathrm{E}_1\right)}=\frac{3}{4}$
$\mathrm{P}\left(\mathrm{E}_1 \cap \mathrm{E}_2\right)=\frac{1}{8}$

$\Rightarrow \mathrm{P}\left(\mathrm{E}_2\right)=\frac{1}{4}, \mathrm{P}\left(\mathrm{E}_1\right)=\frac{1}{6}, \mathrm{P}\left(\mathrm{E}_1 \cup \mathrm{E}_2\right)=\frac{1}{6}+\frac{1}{4}-\frac{1}{8}=\frac{7}{24}$

Clearly, option (A) is wrong

$\mathrm{P}\left(\mathrm{E}_1^{\prime} \cap \mathrm{E}_2^{\prime}\right)=\mathrm{P}\left(\mathrm{E}_1 \cup \mathrm{E}_2\right)^{\prime}=1-\frac{7}{24}=\frac{17}{24}$

So option B is also wrong

$P(E_1 \cap E_2{}') = P(E_1) - P(E_1 \cap E_2) = \frac{1}{24} = P(E_1).P(E_2)$

Option C is correct

Q. 11 Let $A=\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M=\sum_{k=1}^{10} A^{2k}$ and $N=\sum_{k=1}^{10} A^{2k-1}$. Then MN^2 is:

Option 1:
a non-identity symmetric matrix

Option 2:
a skew-symmetric matrix

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Option 3:
neither symmetric nor skew-symmetric matrix

Option 4:
an identity matrix

Correct Answer:
a non-identity symmetric matrix

Solution:

$$A^2 = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = -4I \text{ : symmetric}$$

$$\& \ A^3 = -4A(\text{Skew Symmetric})$$

$$\Rightarrow M = \sum_{k=1}^{10} A^{2k} = \left[(-4) + (-4)^2 + (-4)^3 + \cdots + (-4)^{10}\right] I$$

$$= -4\,\lambda\, I \text{ is Symmetric}$$

$$\Rightarrow N = \sum_{k=1}^{10} A^{2k-1} = A\left[1 + (-4) + (-4)^3 + \cdots + (-4)^9\right] I$$

$$= \lambda\, A \text{ is Skew Symmetric}$$

$$\text{where } \lambda = \left\{1 + (-4) + (-4)^3 + \cdots + (-4)^9\right\}$$

$$MN^2 = -4\lambda^3 A^2$$

$$\Rightarrow MN^2 \text{ is Symmetric matrix}$$

Q. 12 Let $g : (0, \infty) \to \mathbf{R}$ be a differentiable function such that

$$\int \left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)\,(e^x + 1 - xe^x)}{(e^x + 1)^2}\right) dx = \frac{xg(x)}{e^x + 1} + c, \text{ for all } x > 0,$$

where c is an arbitrary constant. Then:

Option 1:
g is decreasing in $\left(0, \frac{\pi}{4}\right)$

Option 2:
g' is increasing in $\left(0, \frac{\pi}{4}\right)$

Option 3:
$g + g'$ is increasing in $\left(0, \frac{\pi}{2}\right)$

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Option 4:

g-g' is increasing in $\left(0, \frac{\pi}{4}\right)$

Correct Answer:

g-g' is increasing in $\left(0, \frac{\pi}{4}\right)$

Solution:

$$\int \frac{x}{e^x+1}(\cos x - \sin x)dx + \int g(x)\frac{e^x+1-xe^x}{(-e^x+1)^2}dx$$

Integrating by parts in first integral

$$= \frac{x}{e^x+1}(\sin x + \cos x) - \int \frac{e^x+1-xe^x}{(e^x+1)^2}(\sin x + \cos x)dx + \int gx\frac{e^x+1-xe^x}{(e^x+1)^2}dx$$

By comparison, we get $g(x) = \sin x + \cos x$

$g(x) - g'(x) = 2sinx$

This is increasing in $(0, \frac{\pi}{4})$

Q. 13 Let $f : \mathbf{R} \to \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be two functions defined by $f(x) = \log_e\left(x^2+1\right) - e^{-x} + 1$ and $g(x) = \frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

Option 1:

$(2, 3)$

Option 2:

$(-2, -1)$

Option 3:

$(1, 2)$

Option 4:

$(-1, 1)$

Correct Answer:

$(2, 3)$

Solution:

$$f(x) = \log_e\left(x^2+1\right) - e^{-x} + 1 \Rightarrow f'(x) = \frac{2x}{x^2+1} + e^{-x} > 0 \,\forall\, x \in R$$

So f(x) increasing

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$g(x) = e^{-x} - 2e^{x} \Rightarrow g'(x) = -\left(e^{-x} + 2e^{x}\right) < 0 \; \forall \, x \in R.$
So $g(x)$ decreasing

$\Rightarrow f(g(x))$is decreasing
$\Rightarrow f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$
$\Rightarrow \frac{(\alpha-1)^2}{3} < \alpha - \frac{5}{3}$
$\Rightarrow (\alpha-2)(\alpha-3) < 0 \Rightarrow \alpha \in (2,3)$

Q. 14 Let $\overrightarrow{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k},\ a_i > 0, i = 1,2,3$ be a vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\overrightarrow{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\overrightarrow{b}$ be a vector obtained by rotating $\overrightarrow{a}$ with $90°$. If $\vec{a}, \vec{b}$ and $x-$axis are coplanar, then projection of a vector $\overrightarrow{b}$ on $3\hat{i} + 4\hat{j}$ is equal to:

Option 1:
$\sqrt{7}$

Option 2:
$\sqrt{2}$

Option 3:
2

Option 4:
7

Correct Answer:
$\sqrt{2}$

Solution:
$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
$\vec{a} = \lambda\left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\right) = \frac{\lambda}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

Now projection of $\vec{a}$ on given vector = 7
$\frac{\lambda}{\sqrt{3}} \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (3\hat{i} + 4\hat{j})}{5} = 7$
$\lambda = 5\sqrt{3}$

$\vec{a} = 5(\hat{i} + \hat{j} + \hat{k})$
$now\, \vec{b} = 5\alpha(\hat{i} + \hat{j} + \hat{k}) + \beta(\hat{i})$
$\vec{a} \cdot \vec{b} = 0$
$\Rightarrow 25\alpha(3) + 5\beta = 0$
$\Rightarrow 15\alpha + \beta = 0 \Rightarrow \beta = -15\alpha$
$\vec{b} = 5\alpha(-2\hat{i} + \hat{j} + \hat{k})$

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Also $|\vec{b}| = 5\sqrt{3}$

$\Rightarrow \alpha = \pm\frac{1}{\sqrt{2}}$

$\vec{b} = \pm\frac{5}{\sqrt{2}}(-2\hat{i} + \hat{J} + \hat{k})$

$Projection\ of\ \vec{b}\ on\ 3\hat{i} + 4\hat{j}\ is$

$\frac{\vec{b}\cdot(3\hat{i}+4\hat{j})}{5} = \pm\frac{5}{\sqrt{2}}\left(\frac{-6+4}{5}\right) = \pm\sqrt{2}$

Q. 15 Let $y = y(x)$ be the solution of the differential equation $(x+1)y' - y = e^{3x}(x+1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve $y = y(x)$ is :

Option 1:
not a critical point

Option 2:
a point of local minima

Option 3:
a point of local maxima

Option 4:
a point of inflection

Correct Answer:
a point of local minima

Solution:

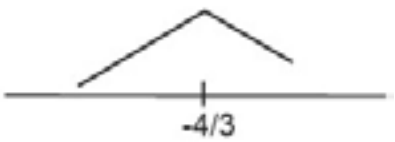
$(x+1)\,dy - y\,dx = e^{3x}(x+1)^2$

$\frac{(x+1)\,dy - y\,dx}{(x+1)^2} = e^{3x}$

$d\left(\frac{y}{x+1}\right) = d(e^{3x}) \Rightarrow \frac{y}{x+1} = \frac{e^{3x}}{3} + c$

$\left(0, \frac{1}{3}\right) \Rightarrow c = 0 \Rightarrow y = \frac{(x+1)e^{3x}}{3}$

$\frac{dy}{dx} = \frac{1}{3}\left((x+1)3e^{3x} + e^{3x}\right) = \frac{e^{3x}}{3}(3x+4)$



At x = -4/3, f'(x) changes sign from negative to positive
Hence , $x = -4/3$ is point of local minima.

Q. 16 If $y = m_1x + c_1$ and $y = m_2x + c_2,\ m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8\left|\ m_1\ m_2\right|$ is equal to:

Option 1:
$3 + 4\sqrt{2}$

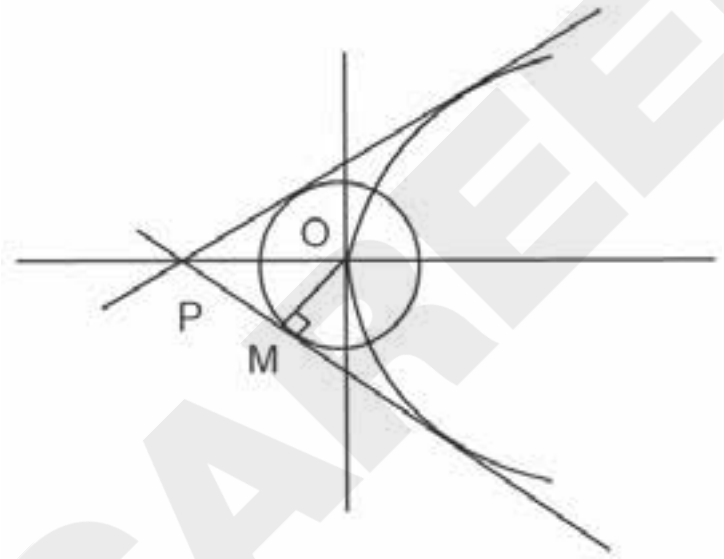
Option 2:
$-5 + 6\sqrt{2}$

Option 3:
$-4 + 3\sqrt{2}$

Option 4:
$7 + 6\sqrt{2}$

Correct Answer:
$-4 + 3\sqrt{2}$

Solution:



Equation of the tangent to the parabola can be written as

$$y = mx + \left(\frac{1}{4m}\right) \text{ or } y - mx - \frac{1}{4m} = 0$$

Now, the perpendicular distance from $(0, 0)$ to the tangent is equal to the radius of the given circle, so

$$OM = \sqrt{2} \Rightarrow \left|\frac{-\frac{1}{4m}}{\sqrt{1+m^2}}\right| = \sqrt{2}$$

$$\Rightarrow\ 2\left(16m^2\right)\left(1+m^2\right) = 1 \Rightarrow 32m^4 + 32m^2 - 1 = 0$$

$$\Rightarrow m^2 = \frac{-2^5 + 2^3\sqrt{16+2}}{2 \times 2^5}$$

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(Ignoring the negative sign as it is square function)

$$\text{Now } |8m_1m_2| = |8m^2|$$
$$= \left|\frac{-2^5 + 2^3\sqrt{16+2}}{2^3}\right| = 3\sqrt{2} - 4$$

Q. 17 Let Q be the mirror image of the point P$(1, 0, 1)$ with respect to the plane $S : x + y + z = 5$. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R, then QR^2 is equal to:

Option 1:
2

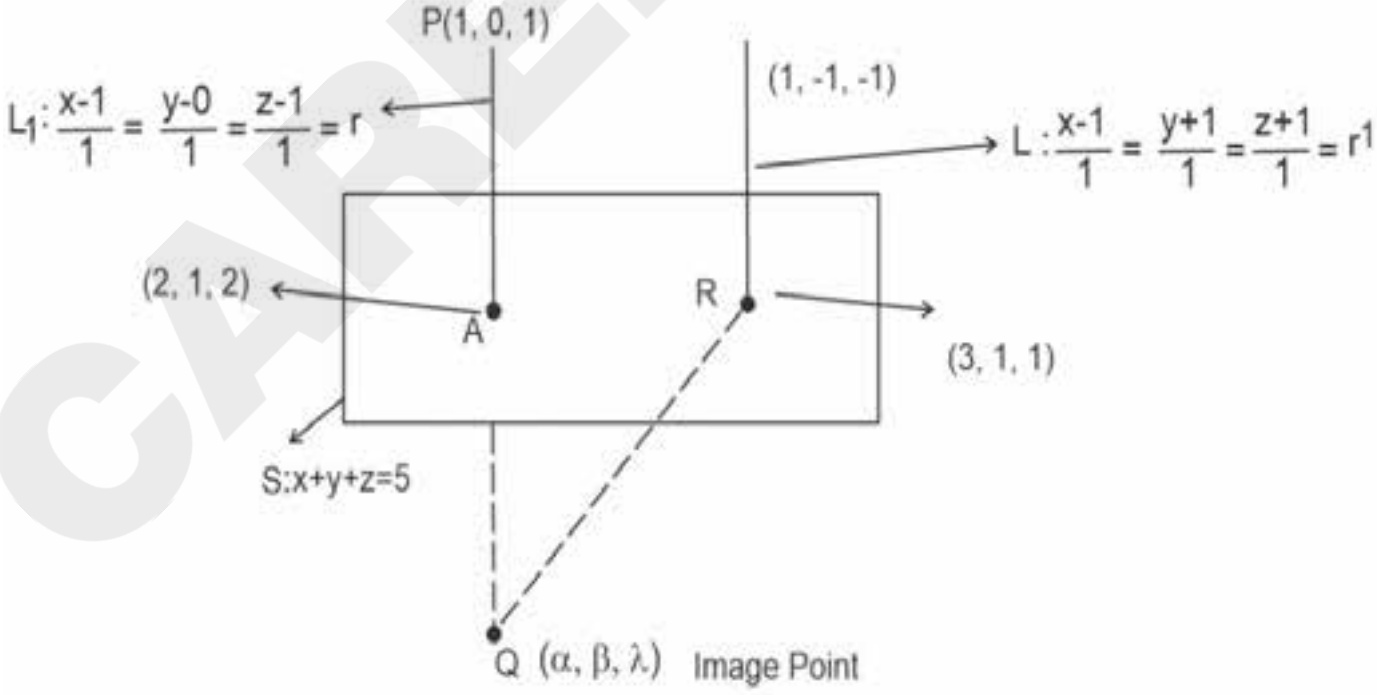
Option 2:
5

Option 3:
7

Option 4:
11

Correct Answer:
5

Solution:



By symmetry, QR = PR
The eqn of the plane $S : x + y + z = 5 - - - (1)$
Eqn of the line L passing through $(1, -1, -1)$ and parallel to the line PQ

$\Rightarrow L: \frac{x-1}{1}=\frac{y+1}{1}=\frac{z+1}{1}=r'$

General point $R\left[r'+1, r'-1, r'-1\right]$

$\because$ Point R lies on the plane (1)

$\therefore r'+1+r'-1+r'-1=5$

$3r'=6 \Rightarrow r'=2$

$R=(3,1,1)$

$QR^2=PR^2$

$=(4+1+0)=5$

Q. 18 If the solution curve $y=y(x)$ of the differential equation $y^2dx+\left(x^2-xy+y^2\right)dy=0$, which passes through the point $(1,1)$ and intersects the line $y=\sqrt{3}x$ at the point $\left(\alpha, \sqrt{3}\alpha\right)$, then value of $\log_e(\sqrt{3}\alpha)$ is equal to:

Option 1:

$\frac{\pi}{3}$

Option 2:

$\frac{\pi}{2}$

Option 3:

$\frac{\pi}{12}$

Option 4:

$\frac{\pi}{6}$

Correct Answer:

$\frac{\pi}{12}$

Solution:

given $y^2dx+\left(x^2-xy+y^2\right)dy=0$

$\Rightarrow \frac{dx}{dy}=\frac{-\left(x^2-xy+y^2\right)}{y^2}---(1)$

Now let $x=vy \Rightarrow v+y\frac{dv}{dy}=\frac{dx}{dy}$

Now putting in eqn (i) we get

$\Rightarrow v+y\frac{dv}{dy}=-\left(v^2-v+1\right)$

$\Rightarrow y\frac{dv}{dy}=-v^2-1$

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$$\Rightarrow -\frac{dv}{1+v^2} = \frac{dy}{y}$$

Now Integrating both side

$$\Rightarrow -\int \frac{dv}{1+v^2} = \int \frac{dy}{y}$$

$$\Rightarrow -\tan^{-1} v = \log|y| + c \Rightarrow -\tan^{-1}\left(\frac{x}{y}\right) = \log|y| + c$$

Also this curve passes through $(1, 1)$

$$\Rightarrow -\tan^{-1} 1 = \log|1| + c \Rightarrow c = -\frac{\pi}{4}$$

$$\Rightarrow -\tan^{-1}\frac{x}{y} = \log|y| - \frac{\pi}{4}$$

Now putting $y = \sqrt{3}x$ and $x = \alpha$ we get.

$$= -\tan^{-1}\frac{\alpha}{\sqrt{3}\alpha} = \log|\sqrt{3}\alpha| - \frac{\pi}{4}$$

$$\Rightarrow \log|\sqrt{3}\alpha| = \frac{\pi}{12}$$

Q. 19 Let $x = 2t, y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that SA $\perp$ BA, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim_{t\to 1} k$ is equal to :

Option 1:

$\frac{17}{18}$

Option 2:

$\frac{19}{18}$

Option 3:

$\frac{11}{18}$

Option 4:

$\frac{13}{18}$

Correct Answer:

$\frac{13}{18}$

Solution:

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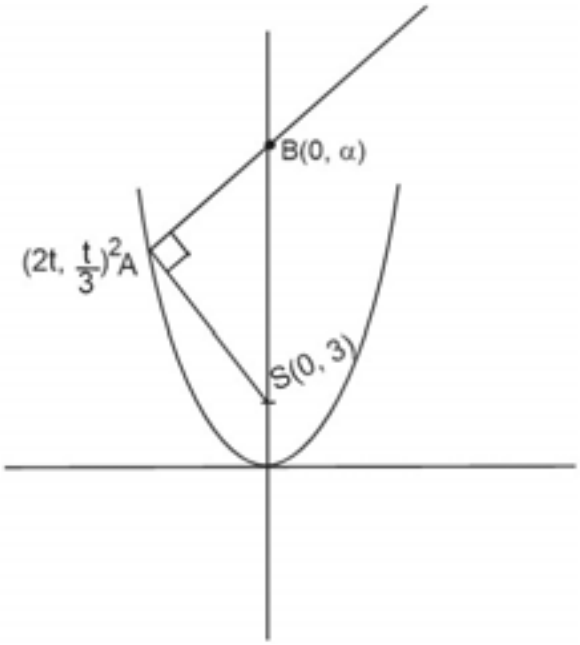

Parabola $x^2 = 12y$

$SA \perp SB$

So, $m_{AS} \cdot m_{AB} = -1$

$$\frac{\left(3-\frac{t^2}{3}\right)\left(\alpha-\frac{t^2}{3}\right)}{(0-2t)} \cdot \frac{\left(\alpha-\frac{t^2}{3}\right)}{(0-2t)} = -1$$

by solving

$$3\alpha = \frac{27t^2+t^4}{t^2-9}$$

$$\text{ordinate of centroid of } \triangle SAB = k = \frac{\alpha+\frac{t^2}{3}+3}{3}$$

$$k = \frac{9+3\alpha+t^2}{9}$$

$$\lim_{t\to 1} k = \lim_{t\to} \frac{1}{9}\left(9+t^2+\frac{27t^2+t^4}{(t^2-9)}\right) = \left(\frac{13}{18}\right)$$

Q. 20 Let a circle C in complex plane pass through the points $z_1 = 3+4i, z_2 = 4+3i$ and $z_3 = 5i$.If $z\,(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg (z) is equal to:

Option 1:

$$\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$$

Option 2:

$$\tan^{-1}\left(\frac{24}{7}\right) - \pi$$

Option 3:

$$\tan^{-1}(3) - \pi$$

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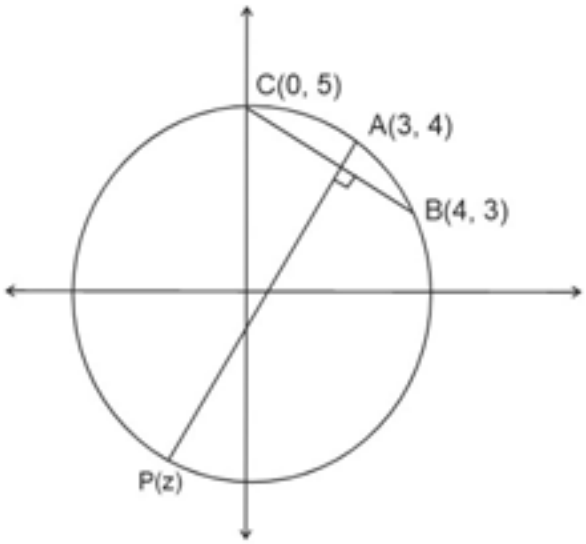
Option 4:

$\tan^{-1}\left(\frac{3}{4}\right)-\pi$

Correct Answer:

$\tan^{-1}\left(\frac{24}{7}\right)-\pi$

Solution:



$\text{Slope of BC } = \frac{3-5}{4-0} = \frac{-1}{2}$

Slope of AP $= 2$

$\text{eqn of AP} : y-4=2(x-3)$

$\Rightarrow y=2(x-1)$

P lies on circle $x^2+y^2=25$

$\Rightarrow x=\frac{-7}{5} \text{ and } y=\frac{-24}{5}$

$\Rightarrow \arg(z)=\tan^{-1}\left(\frac{24}{7}\right)-\pi$

Q. 21 Let C_r denote the binomial coefficient of x^r in the expansion of $(1+x)^{10}$. If for $\alpha, \beta \epsilon R$, $C_1+3\cdot 2C_2+5\cdot 3C_3+\ldots$.upto 10 terms $=\frac{\alpha\times 2^{11}}{2^{\beta}-1}\left(C_0+\frac{C_1}{2}+\frac{C_2}{3}+\ldots \text{ upto 10 terms}\right)$, then the value of $\alpha+\beta$ is equal to _____.

Correct Answer:

286

Solution:

$(1+x)^{10}=C_0+C_1x+C_2x^2+\cdots+C_{10}x^{10}$

Differentiating both side w.r.t x

$10(1+x)^9=C_1+2C_2x+3C_3x^2+\cdots+^{10}C_{10}x^9$

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Replace $x \to x^2$

$10\left(1+x^2\right)^9 = C_1 + 2C_2x^2 + 3C_3x^4 + \cdots + {}^{10}C_{10}x^{18}$

$10 \cdot x\left(1+x^2\right)^9 = C_1x + 2C_2x^3 + 3C_3x^5 + \cdots + {}^{10}C_{10}x^{19}$

Differentiating both side w.r.t. x

$10\left(\left(1+x^2\right)^9 \cdot 1 + x \cdot 9\left(1+x^2\right)^8 \cdot 2x\right)$

$= C_1x + 2C_2 \cdot 3x^3 + 3 \cdot 5 \cdot C_3x^4 + \cdots + 10 \cdot 19.C_{10}x^{18}$

Putting x = 1

$10\left(2^9 + 18 \cdot 2^8\right)$

$= C_1 + 2 \cdot 3 \cdot C_2 + 3 \cdot 5 \cdot C_3 + \cdots + 19 \cdot 10 \cdot C_{10}$

$= 10\left(2^9 + 2 \cdot 9 \cdot 2^8\right) = 10 \cdot 2^9(1+9)$

$= 100 \cdot 2^9$

Now

$\because\ (1+x)^{10} = C_0 + C_1x + C_2x^2 + \cdots + C_{10}x^{10}$

integrating both side under. limit 0 to 1 .

$\int_0^1 (1+x)^{10}dx = \int_0^1 \left(C_0 + C_1x + C_2x^2 + \cdots\cdots + C_{10}x^{10}\right)dx$

$\left[\frac{(1+x)^{11}}{11}\right]_0^1 = \left[C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + \cdots + \frac{C_{10}x^{11}}{11}\right]_0^1$

$\frac{2^{11}}{11} - \frac{1}{11} = c_0 + \frac{c_1}{2} + \frac{c_2}{3} + \cdots + \frac{c_{10}}{11}$

$\because 100 \times 2^9 = \frac{\alpha \cdot 2^{11}}{(2^\beta - 1)} \cdot \left(\frac{2^{11}-1}{11}\right) \Rightarrow \frac{100 \times 2^9 \times 11}{2^{11}-1} = \frac{\alpha.2^9 \cdot 2^2}{2^\beta - 1}$

So $\alpha = 25 \times 11 = 275, \beta = 11$

therefore $\alpha + \beta = 286$

Q. 22 The number of 3-digit odd numbers, whose sum of digits is a multiple of 7, is_____.

Correct Answer:

63

Solution:

Largest digit in number $= 9$

Sum of three digit maximum can be $= 27$

Hence sum of digit can be $= 7, 14, 21$

$a_1 + a_2 + a_3 = 7, 14, 21$

$\text{case } 1 : a_1 + a_2 + a_3 = 7 \quad a_3 = \text{odd}$

$a_3 = 1 \quad (a_1, a_2) \equiv (1,5), (2,4), (3,3), (4,2), (5,1), (6,0) : \text{Total number} = 6$

$a_3 = 3 \quad (a_1, a_2) \equiv (1,3), (2,2), (3,1), (4,0) : \text{Total number} = 4$

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$a_3 = 5 \quad (a_1, a_2) \equiv (1,1),(2,0) : \text{Total number} = 2$

$\text{case} - 2 \quad a_1 + a_2 + a_3 = 14$
$a_3 = 1 \quad (a_1, a_2) \equiv (4,9),(5,8),\ldots.(9,4) : \text{Total number} = 6$
$a_3 = 3 \quad (a_1, a_2) \equiv (2,9),(3,8)\cdots(9,2) : \text{Total number} = 8$
$a_3 = 5 \quad (a_1, a_2) \equiv (1,8)\cdots\cdots(8,1),(9,0) : \text{Total number} = 9$
$a_3 = 7 \quad (a_1, a_2) \equiv (1,6)\ldots(6,1),(7,0) : \text{Total number} = 7$
$a_3 = 9 \quad (a_1, a_2) \equiv (1,4)\ldots(4,1),(5,0) : \text{Total number} = 5$

$\text{case3} \Rightarrow \quad a_1 + a_2 + a_3 = 21$
$a_3 = 3 \quad (a_1, a_2) \equiv (9,9) : \text{Total number} = 1$
$a_3 = 5 \quad (a_1, a_2) \equiv (7,9),(8,8),(9,7) : \text{Total number} = 3$
$a_3 = 7 \quad (a_1, a_2) \equiv (5,9)\ldots.(9,5) : \text{Total number} = 5$
$a_3 = 9 \quad (a_1, a_2) \equiv (3,9)\cdots(9,3) : \text{Total number} = 7$

$\text{Total} = 63$

Q. 23 Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4, |\vec{b}| = 3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $|(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})|^2 + 4(\vec{a} \cdot \vec{b})^2$ is equal to_____.

Correct Answer:
576

Solution:
$|(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})|^2 + 4(\vec{a} \cdot \vec{b})^2$
$= |-\vec{b} \times \vec{b} - \vec{b} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{a}|^2 + 4(\vec{a} \cdot \vec{b})^2$
$= 4\left[|\vec{a} \times \vec{b}|^2 + (\vec{a}.\vec{b})^2\right] = 4|\vec{a}|^2|\vec{b}|^2$
$= 4 \times 9 \times 16$
$= 576$

Q. 24 Let the abscissae of the two points P and Q be the roots of $2x^2 - rx + p = 0$ and the ordinates of P and Q be the roots of $x^2 - sx - q = 0$. If the equation of the circle described on PQ as diameter is $2\left(x^2 + y^2\right) - 11x - 14y - 22 = 0$, then $2\text{r} + \text{s} - 2\text{q} + \text{p}$ is equal to _____.

Correct Answer:
7

Solution:
Let co-ordinate of P and Q are (α, β) and (γ, δ) respectively
$\alpha + \gamma = \frac{\text{r}}{2}, \alpha\gamma = \frac{\text{p}}{2}$
and $\beta + \delta = \text{s}, \beta\delta = -\text{q}$

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Equation of circle with PQ as diameter

$$(x-\alpha)(x-\gamma)+(y-\beta)(y-\delta)=0$$
$$x^2+y^2-x(\alpha+\gamma)-y(\beta+\delta)+\alpha\gamma+\beta\delta=0$$
$$x^2+y^2-x\left(\frac{r}{2}\right)-y(S)+\frac{P}{2}-q=0 \quad ---(I)$$

$$S: 2\left(x^2+y^2\right)-11x-14y-22=0$$
$$S: x^2+y^2-\frac{11}{2}x-7y-11=0 \quad ----(II)$$

$\because$ eq(I) & (II)represent same circle

$$\therefore \begin{matrix} \frac{r}{2}=\frac{11}{2} \\ r=11 \end{matrix} \Bigg| \; S=7 \Bigg| \begin{matrix} \frac{P}{2}-q=-11 \\ \frac{P-2q}{2}=-11 \\ P-2q=-22 \end{matrix}$$

$$\therefore \; 2r+S-2q+P$$
$$2\times 11+7-22=7$$

Q. 25 The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which $14\,\mathrm{cosec}^2 x-2\sin^2 x=21-4\cos^2 x$ holds, is_____.

Correct Answer:
4

Solution:

$$x\in\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$$
$$14\,\mathrm{cosec}^2 x-2\sin^2 x=21-4\cos^2 x$$
$$=21-4\left(1-\sin^2 x\right)$$
$$=17+4\sin^2 x$$

$$14\,\mathrm{cosec}^2 x-6\sin^2 x=17$$
$$\text{Let } \sin^2 x=p$$
$$\frac{14}{p}-6p=17\Rightarrow 14-6p^2=17p$$
$$6p^2+17p-14=0$$
$$p=-3\cdot 5, \frac{2}{3}\Rightarrow \sin^2 x=\frac{2}{3}$$
$$\sin x=\pm\sqrt{\frac{2}{3}}$$

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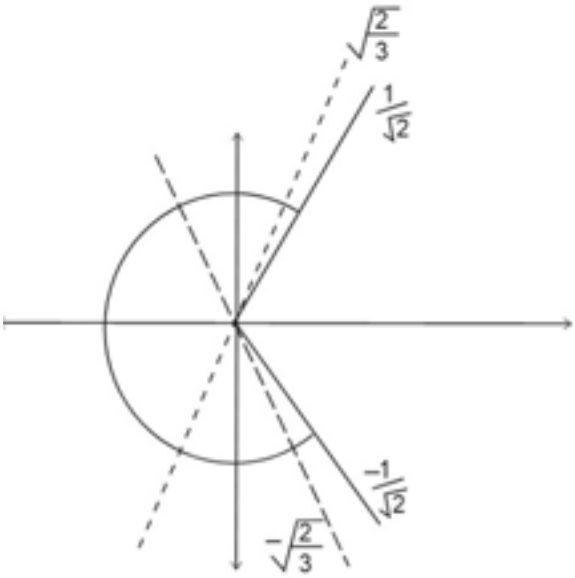


In the given interval, there are 4 solutions

Q. 26 For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is_____.

Correct Answer:
4

Solution:
given $a_n = 19^n - 12^n$
So $\frac{31a_9 - a_{10}}{57a_8}$
$\Rightarrow \frac{31\,(19^9 - 12^9) - (19^{10} - 12^{10})}{57\,(19^8 - 12^8)}$
$\Rightarrow \frac{19^9(12) - 12^9(19)}{57\,(19^8 - 12^8)}$
$\Rightarrow \frac{19 \times 12\,(19^8 - 12^8)}{57\,(19^8 - 12^8)}$
$\Rightarrow 4$

Q. 27 Let $f : \mathbf{R} \to \mathbf{R}$ be a function defined by $f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to $g(1)$ is_____.

Correct Answer:
2

Solution:

$$f(x) = \left[2\left(1 - \frac{x^{25}}{2}\right)\left(2 + x^{25}\right)\right]^{1/50}$$
$$f(x) = \left[\left(2 - x^{25}\right)\left(2 + x^{25}\right)\right]^{1/50}$$
$$= \left(4 - x^{50}\right)^{1/50}$$
$$f\left(f\left(x\right)\right) = \left(4 - \left(\left(4 - x^{50}\right)^{1/50}\right)^{50}\right)^{1/50} = x$$
$$g(x) = f(f(f\left(x\right))) + f(f(x))$$
$$= f(x) + x$$

$$g(1) = f(1) + 1 = 3^{1/50} + 1$$
$$[g(1)] = \left[3^{1/50} + 1\right] = 2$$

Q. 28 Let the lines

$$L_1 : \overrightarrow{\mathbf{r}} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbf{R}$$

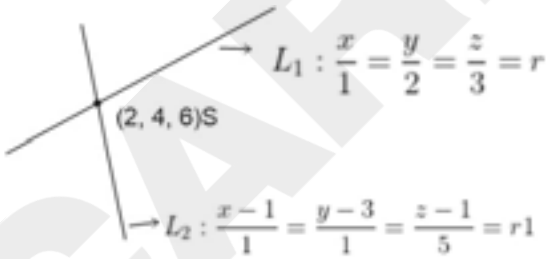
$$L_2 : \overrightarrow{\mathrm{r}} = (\hat{i} + 3\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k}); \mu \in \mathbf{R}$$

intersect at the point S. If a plane ax + by - z + d = 0 passes through S and is parallel to both the lines L_1 and L_2, then the value of a+b+d is equal to_____.

Correct Answer:
5

Solution:
$L_1 : \vec{r} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in R$

$L_2 : \vec{r} = (\hat{i} + 3\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k})$



From the given conditions, the plane contains both the lines

Normal vector is cross product of direction vectors of both the lines

$\vec{n} = (i + 2j + 3k) \times (i + j + 5k)$
$\vec{n} = (7i - 2j - k)$

It also passes through point (0,0,0) lying on line 1

So the plane is 7(x - 0) - 2(y - 0) - (z - 0) = 0

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Hence a = 7, b = -2, c = -1, d = 0

a + b + d = 5

Q. 29 Let A be 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is_____.

Correct Answer:
414

Solution:

$$\text{Given matrix } 3 \times 3 = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

Here we can see we have to find nine elements whose sum is 5.

Case-1 when five 1's are there & four 0's are there.
Again by division & distribution method

$$\text{we get } \Rightarrow \frac{9!}{5!4!} = 126 \cdots\cdots \text{(i)}$$

Case-2 when six 1's are there & one $\{-1\}$ & two 0's are there.
Again by division & distribution method

$$\text{we get } \Rightarrow \frac{9!}{6!2!1!} = 252 \cdots\cdots \text{(ii)}$$

Case-3 when seven 1's are there are two $\{-1\}$ are there,
By division & distribution we get

$$\Rightarrow \frac{9!}{7!2!} = 36 \cdots\cdots \text{(iii)}$$

Now adding eqn (i)+(ii)+(iii)

$$\text{we get } 126 + 252 + 36 = 414$$

So total 414 ways will be there.

Q. 30 The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \ldots$. is equal to _____.

Correct Answer:
98

Solution:

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Let $S = \frac{1}{3} + \frac{5}{9} + \frac{19}{27} + \cdots$ upto 100 terms

$$S = \frac{3-2}{3^1} + \frac{3^2-2^2}{3^2} + \frac{3^3-2^3}{3^3} + \cdots \text{ upto 100 term}$$

So, $T_n = 1 - \left(\frac{2}{3}\right)^n \Rightarrow S_{100} = 100 - \left[\frac{\frac{2}{3}\left[1 - \left(\frac{2}{3}\right)^{100}\right]}{1 - \frac{2}{3}}\right]$

$$= 98 + 2\left(\frac{2}{3}\right)^{100} \Rightarrow S_{100} = 98$$

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Online 25th June Evening Shift

Physics

Q. 1 Given below are two statements:One is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Two identical balls A and B thrown with same velocity $'u'$ at two different angles with horizontal attained the same $h_1$range R. If A and B reached the maximum height h_1 and h_2 respectively,then $R = 4\sqrt{h_1 h_2}$

Reason R: Product of said heights.

$$h_1 h_2 = \left(\frac{u^2 \sin^2\theta}{2g}\right) \cdot \left(\frac{u^2 \cos^2\theta}{2g}\right)$$

Question: Choose the correct answer:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is the NOT the correct explanation of A.

Option 3:
A is true but R is false.

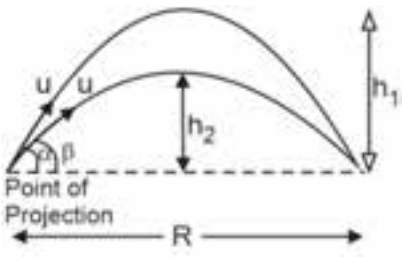
Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
For two balls $A \text{ and } B$ thrown with the same velocity $'u'$ at two different angles with horizontal obtained the same range R then their angles of projection let say $\alpha \text{ and } \beta$ respectively will have a relation

$$\alpha + \beta = 90^\circ$$

If $\alpha = \theta$, then $\beta = 90 - \alpha = 90 - \theta$



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$$h_1 = \frac{u^2 \sin\theta}{2g}; h_2 = \frac{u^2 \sin(90-\theta)}{2g}$$

$$h_2 = \frac{u^2 \cos\theta}{2g}$$

$$R = \frac{2u \sin\theta \cos\theta}{g} = 4\sqrt{h_1 h_2}$$

Hence the correct answer is option 1.

Q. 2 Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = at + \beta t^2$ and $X_Q(t) = ft - r^2$. At what time, both the buses have sam velocity?

Option 1:
$\frac{\alpha - f}{1+\beta}$

Option 2:
$\frac{\alpha + f}{2(\beta - 1)}$

Option 3:
$\frac{\alpha + f}{2(1+\beta)}$

Option 4:
$\frac{f - \alpha}{2(1+\beta)}$

Correct Answer:
$\frac{f - \alpha}{2(1+\beta)}$

Solution:
$x_p(t) = \alpha t + \beta t^2$

$x_Q(t) = ft - t^2$

Velocity of bus $P = V_p = \alpha + 2\beta t$

Velocity of bus $Q = V_Q = f - 2t$

If $V_p = V_Q$

$\alpha + 2\beta t = f - 2t$

$t(2\beta + 2) = f - \alpha$

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$$t = \frac{f - \alpha}{2(\beta + 1)}$$

Hence the correct answer is option 4.

Q. 3 A disc with a flat small bottm beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if:

Option 1:
$R \leq \frac{\mu g}{2\omega^2}$

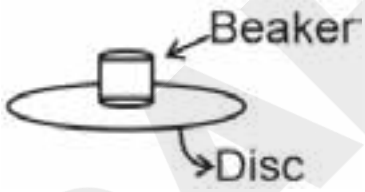
Option 2:
$R \leq \frac{\mu g}{\omega^2}$

Option 3:
$R \geq \frac{\mu g}{2\omega^2}$

Option 4:
$R \geq \frac{\mu g}{\omega^2}$

Correct Answer:
$R \leq \frac{\mu g}{\omega^2}$

Solution:
The beaker will revolve with the disc if friction force is able to provide necessary centripetal force



$$f \leq \mu mg$$

$$\frac{mv^2}{R} = mR\omega^2 \leqslant \mu mg$$

$$R \leqslant \frac{\mu g}{\omega^2}$$

The correct option is (2)

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Q. 4 A solid metallic cube having total surface area $24\mathrm{m}^2$ is uniformly heated. If it's temperature is increased by $10^{\circ}\mathrm{C}$,calculate the increase in volume of the cube. (Given $\alpha = 5.0 \times 10^{-4^{\circ}}C^{-1}$)

Option 1:
$2.4 \times 10^6\mathrm{cm}^3$

Option 2:
$1.2 \times 10^5\mathrm{cm}^3$

Option 3:
$6.0 \times 10^4\mathrm{cm}^3$

Option 4:
$4.8 \times 10^5\mathrm{cm}^3$

Correct Answer:
$1.2 \times 10^5\mathrm{cm}^3$

Solution:
Area of Surface $= 24\mathrm{m}^2 = 6\mathrm{l}^{2*}$

Side length $(\mathrm{l}) = 2\mathrm{m}$

Volume of cube $= 8\mathrm{m}^3$

$\gamma = \alpha = 15 \times 10^{-4\circ}\mathrm{C}^{-1}$

$\Delta\mathrm{T} = 10^{\circ}\mathrm{C}$

For Volume expansion,

$$\Delta\mathrm{V} = \mathrm{V}\gamma\Delta\mathrm{T}$$
$$= 8 \times 15 \times 10^{-4} \times 10$$
$$= 1200 \times 10^{-4}$$
$$= 0.12\mathrm{m}^3$$
$$\Delta\mathrm{V} = 1.2 \times 10^5\mathrm{cm}^3$$

The correct answer is option 2.

Q. 5 A copper block of mass $5.0\mathrm{kg}$ is heated to a temperature of $500^{\circ}\mathrm{C}$ and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper : $0.39\mathrm{Jg}^{-1}{}^{\circ}\mathrm{C}^{-1}$ and latent heat of fusion of water : $335\mathrm{Jg}^{-1}$]

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Option 1:
1.5kg

Option 2:
5.8kg

Option 3:
2.9kg

Option 4:
3.8kg

Correct Answer:
2.9kg

Solution:
$\text{m} = 5\text{kg}$

$\text{C}_{\text{cu}} = 0.39\text{Jg}^{-1\circ}\text{C}^{-1}$

$\text{L}_{\text{water}} = 335\text{Jg}^{-1}$

By the principle of calorimeter,

Heat lost by copper =Heat gained by ice

$\text{m}_1\text{C}_{\text{cu}}\,\Delta\text{T} = \text{m}_2\text{L}_{\text{water}}$

$5000 \times 0.39 \times (500 - 0) = \text{m}_2 \times 335$

$\text{m}_2 = \dfrac{9.75 \times 10^5}{335}\text{g}$

$\text{m}_2 = 2.9\ \text{kg}$

Hence the correct answer is option 3

Q. 6 The ratio of specific heats $\left(\dfrac{\text{C}_\text{p}}{\text{C}_\text{v}}\right)$ in terms of degree of freedom (f) is given by :

Option 1:
$\left(1 + \dfrac{\text{f}}{3}\right)$

Option 2:
$\left(1 + \dfrac{2}{\text{f}}\right)$

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Option 3:

$\left(1+\frac{f}{2}\right)$

Option 4:

$\left(1+\frac{1}{f}\right)$

Correct Answer:

$\left(1+\frac{2}{f}\right)$

Solution:

$\frac{C_p}{C_v}=\gamma$

$C_V=\frac{f}{2}R$

$C_p=C_v+R\ (\text{By Mayer's relation})$

$C_p=\left(\frac{f}{2}+1\right)R$

$\frac{C_p}{C_v}=\frac{2+f}{f}=\gamma$

Hence the correct answer is option 2.

Q. 7 For a particle in uniform circular motion, the acceleration $\underset{a}{\rightarrow}$ at ay point $P\left(R,\theta\right)$ on the circular path of radius R is (when θ is measured from the positive x-axis and v is uniform speed):

Option 1:

$-\frac{v^2}{R}\sin\theta\hat{i}+\frac{v^2}{R}\cos\theta\hat{j}$

Option 2:

$-\frac{v^2}{R}\cos\theta\hat{i}+\frac{v^2}{R}\sin\theta\hat{j}$

Option 3:

$-\frac{v^2}{R}\cos\theta\hat{i}-\frac{v^2}{R}\sin\theta\hat{j}$

Option 4:

$-\frac{v^2}{R}\hat{i}+\frac{v^2}{R}\hat{j}$

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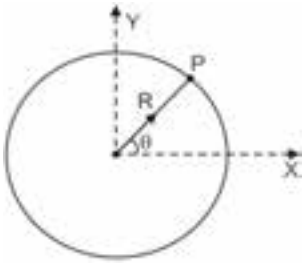
Correct Answer:

$$-\frac{v^2}{R}\cos\theta\hat{i}-\frac{v^2}{R}\sin\theta\hat{j}$$

Solution:



$$\bar{R}=R\cos\theta\hat{i}+R\sin\theta\hat{j}$$

$$\bar{a}=d^2\bar{R}/dt^2=-\left(\frac{d^2\theta}{dt^2}\right)R[\cos\theta\hat{i}+\sin\theta\hat{j}]$$

$$\bar{a}=-\omega^2R[\cos\theta\hat{i}+\sin\theta\hat{j}]$$

$$\bar{a}=-\frac{V^2}{R}[\cos\theta\hat{i}+\sin\theta\hat{j}]$$

Hence the correct answer is option 3.

Q. 8 Two metallic plates form a parallel plate capacitor. The distance between the plate is $'d'$. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be the ratio of the new capacitance to the original capacitance of the capacitor?

Option 1:
$2:1$

Option 2:
$1:2$

Option 3:
$1:4$

Option 4:
$4:1$

Correct Answer:
$2:1$

Solution:
Let the new and original capacitance be C and C_0 respectively

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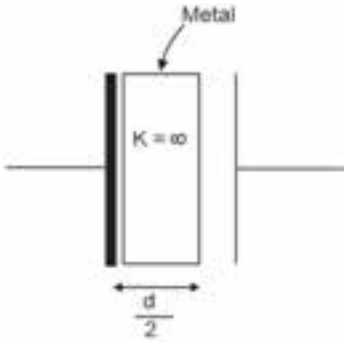

$$C_0 = \frac{A\epsilon_0}{d}$$

$$C = \frac{A\epsilon_0}{\frac{t_1}{k_1} + \frac{t_2}{k_2}}$$

$$= \frac{A\epsilon_0}{\frac{d/2}{\infty} + \frac{d/2}{1}}$$ (Dielectric constant of metal is ∞)

$$C = \frac{2A\epsilon_0}{d} = 2C_0$$

$$\frac{C}{C_0} = \frac{2}{1}$$

Hence the correct answer is option 1.

Q. 9 Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R. The value of resistance R.The value of resistance R,for which the potential difference across second cell is zero, is :

Option 1:
$r_2 - r_1$

Option 2:
$r_1 - r_2$

Option 3:
r_1

Option 4:
r_2

Correct Answer:
$r_2 - r_1$

Solution:

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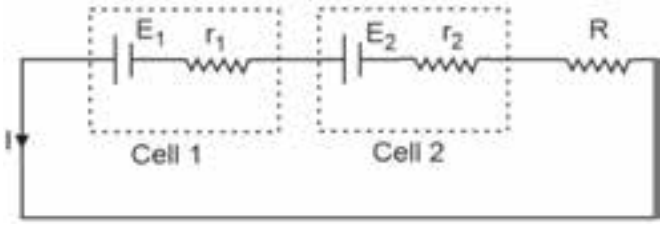
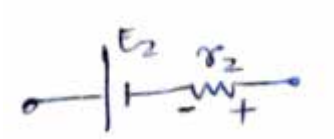


$$I = \frac{E_1 + E_2}{r_1 + r_2 + R} = \frac{2E_1}{r_1 + r_2 + R} \quad \text{.....(1)}$$

$$(\because E_1 = E_2 \cdots (\text{ given }))$$



The potential drop across the second cell is zero

$$\therefore V = E_2 - Ir_2 = 0 \ldots \ (\text{given})$$

$$E_2 = E_1 = Ir_2 \quad \text{.....(2)}$$

From eq (1) and (2)

$$\frac{E_1}{r_2} = \frac{2E_1}{r_1 + r_2 + R}$$

$$r_1 + r_2 + R = 2r_2$$

$$r_1 + R = r_2$$

$$R = r_2 - r_1$$

Hence the correct answer is option 1.

Q. 10 Given below are two statements :

Statement-I : Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.

Statement-II : Diamagetic is a result of orbital motions of electrons developing magnetic moments opposite given below.

Choose the correct answer from the options given below :-

Option 1:
Both Statement - I and Statement - II are true.

Option 2:
Both Statement - I and Statement - II are false.

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Option 3:
Statement - I is true but Statement - II is false.

Option 4:
Statement - I is false but Statement - II is true.

Correct Answer:
Statement - I is false but Statement - II is true.

Solution:
For paramagnetic substances,

$$\chi \propto \frac{1}{T}$$

For ferromagnetic substances,

$$\chi \propto \frac{1}{T - T_c}$$

$T_c \rightarrow$ curie temperature

Statement I is false but Statement II is true.

Hence the correct answer is option 4.

Q. 11 A long solenoid carrying a current produces a magnetic field B along it's axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

Option 1:
B

Option 2:
$2B$

Option 3:
$4B$

Option 4:
$\frac{B}{2}$

Correct Answer:
B

Solution:
Magnetic field at the axis of soleoid $= B = \mu_0 ni$

Initial : $B_1 = \mu_0 n_1 i_1 = B$

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Final : $B_2 = \mu_0 n_2 i_2$

$$n_2 = \frac{n_1}{2}, i_2 = 2i_1$$

$$\therefore B_1 = B_2 = B$$

Hence the correct answer is option 1.

Q. 12 A sinusoidal voltage $V(t) = 210 \sin 3000\, t$ volt is applied to a series LCR circuit in which $L = 10 mH$, $C = 25 \mu F$ and $R = 100 \Omega$. The phase difference (ϕ) between the applied voltage and resultant current will be :

Option 1:
$\tan^{-1}(0.17)$

Option 2:
$\tan^{-1}(9.46)$

Option 3:
$\tan^{-1}(0.30)$

Option 4:
$\tan^{-1}(13.33)$

Correct Answer:
$\tan^{-1}(0.17)$

Solution:
$V(t) = 210 \sin(3000t)$

$L = 10 \times 10^{-3} H$

$C = 25 \times 10^{-6}\ F$

$R = 100 \Omega$

$$\tan \phi = \frac{|X_L - X_C|}{R}$$

$$X_L = \omega L = 2\pi f L$$

$$= 3000 \times 10 \times 10^{-3}$$

$$X_L = 30 \Omega$$

$$X_c = \frac{1}{\omega c} = \frac{1}{3000 \times 25 \times 10^{-6}}$$

$$X_C = \frac{40 \Omega}{3}$$

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$$\tan\phi = \frac{\left|30 - \frac{40}{3}\right|}{100} = \frac{1}{6}$$

$$\phi = \tan^{-1}\left(\frac{1}{6}\right) = \tan^{-1}(0.166)$$

Hence the correct answer is option 1.

Q. 13 The electromagnetic waves travel in a medium at a speed of $2.0 \times 10^8 \mathrm{m/s}$. The relative permeability of the medium is 1.0 . The relative permitivity of the medium will be :

Option 1:
2.25

Option 2:
4.25

Option 3:
6.25

Option 4:
8.25

Correct Answer:
2.25

Solution:
$\mathrm{V} = 2 \times 10^8 \ \mathrm{m/s}$

$\mu_\mathrm{r} = 1$

$\varepsilon_\mathrm{r} = ?$

$$\frac{\mathrm{C}}{\sqrt{\mu_\mathrm{r}\varepsilon_\mathrm{r}}} = \mathrm{V}$$

$$\frac{3 \times 10^8}{\sqrt{\mu_\mathrm{r}\varepsilon_\mathrm{r}}} = 2 \times 10^8$$

$$\frac{3}{2} = \sqrt{\mu_\mathrm{r}\varepsilon_\mathrm{r}}$$

$$\frac{9}{4} = 1 \times \varepsilon_\mathrm{r}$$

$$\varepsilon_\mathrm{r} = 2.25$$

Hence the correct answer is option 1.

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Q. 14 The interference pattern is obtained with two coherent light sources of intensity ratio $4:1$. And the ratio $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

Option 1:
3

Option 2:
4

Option 3:
2

Option 4:
1

Correct Answer:
4

Solution:

$$\frac{I_1}{I_2}=\frac{4}{1}$$

$$I_{max}=\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$$

$$I_{min}=\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$$

$$I_{max}+I_{min}=2\left(I_1+I_2\right)$$

$$I_{max}-I_{min}=4\sqrt{I_1}\sqrt{I_2}$$

$$\frac{I_{max}+I_{min}}{I_{max}-I_{min}}=\frac{2\left(I_1+I_2\right)}{4\sqrt{I_1}\sqrt{I_2}}$$

$$=\frac{2(5)}{4(2)}=\frac{5}{4}$$

$$\therefore x=4$$

Hence the correct answer is option 2.

Q. 15 A light whose electric field hectors are completely removed by using a good polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

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Option 1:
Reflected and refracted rays will be perpendicular to each other.

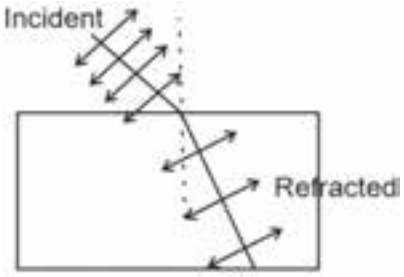
Option 2:
Wave will propogate along the surface of prism.

Option 3:
No refraction, and there will be total reflection of light.

Option 4:
No reflection, and there will be total transmission of light.

Correct Answer:
No reflection, and there will be total transmission of light.

Solution:



$\longrightarrow$ Vibration will be limited only in one plane (in the plane of incidence)

$\longrightarrow$ Reflected light is polarised as the other vibration (perpendicular to the plane of incidence is not present)

$\longrightarrow$ Refracted light is partially polarized

Hence the correct answer is option 4.

Q. 16 A proton, a neutron, an electron and α-particle have same energy. If $\lambda_{\mathrm{p}}, \lambda_{\mathrm{n}}, \lambda_{\mathrm{e}}$ and λ_{α} are the de Broglie's wavelengths of prorton, neutron, electron and α particle respectively, then choose the correct relation from the following :

Option 1:
$\lambda_{\mathrm{p}}=\lambda_{\mathrm{n}}>\lambda_{\mathrm{e}}>\lambda_{\alpha}$

Option 2:
$\lambda_{\alpha}<\lambda_{\mathrm{n}}<\lambda_{\mathrm{p}}<\lambda_{\mathrm{e}}$

Option 3:
$\lambda_{\mathrm{e}}<\lambda_{\mathrm{p}}=\lambda_{\mathrm{n}}>\lambda_{\alpha}$

Option 4:
$\lambda_{\mathrm{e}}=\lambda_{\mathrm{p}}=\lambda_{\mathrm{n}}=\lambda_{\alpha}$

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Correct Answer:

$\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

Solution:

$\mathrm{KE}_\alpha = \mathrm{KE}_p = \mathrm{KE}_e = \mathrm{KE}_n$

De Broglie wavelength $= \lambda = \dfrac{h}{\sqrt{2m(KE)}}$

We know that

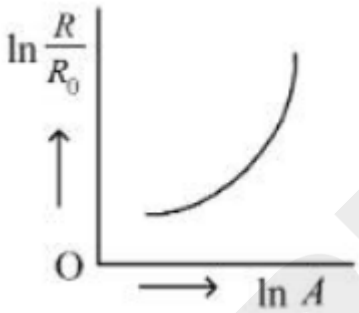
$m_\alpha > m_n > m_p > m_e$

$\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

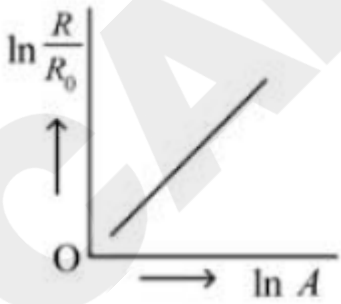
Hence the correct answer is option 2.

Q. 17 Which of the following figure represents the variable of $\ln\left(\dfrac{R}{R_0}\right)$ with $\ln A$ (if R= radius of nucleus and A=it's mass number)

Option 1:



Option 2:



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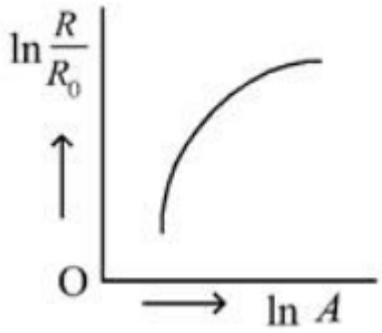
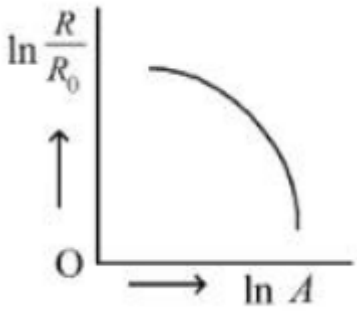
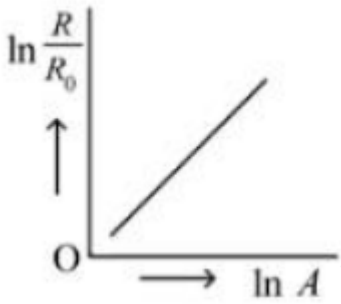
Option 3:



Option 4:



Correct Answer:



Solution:
We know that,

$$\mathrm{R} = \mathrm{R_0 A^{1/3}}$$

$$\frac{\mathrm{R}}{\mathrm{R_0}} = \mathrm{A^{1/3}}$$

$$\ln\left(\frac{\mathrm{R}}{\mathrm{R_0}}\right) = \ln \mathrm{A^{1/3}}$$

$$\ln\left(\frac{\mathrm{R}}{\mathrm{R_0}}\right) = \frac{1}{3}\ln \mathrm{A}$$

Slope of graph $\ln\left(\frac{\mathrm{R}}{\mathrm{R_0}}\right)$ vs $\ln(\mathrm{A})$ is $\left(\frac{1}{3}\right)$.

Hence the correct answer is option 2.

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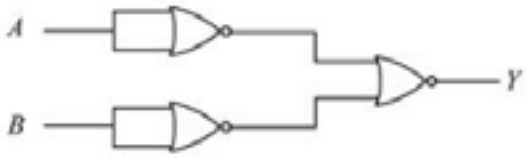
Q. 18 Identify the logic operation performed by given circuit :



Option 1:
AND gate

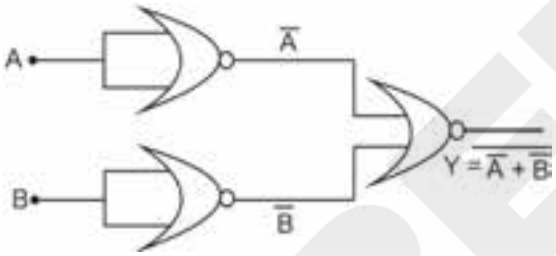
Option 2:
OR gate

Option 3:
NOR gate

Option 4:
NAND gate

Correct Answer:
AND gate

Solution:



$$Y = \overline{\bar{A} + \bar{B}}$$

By De-Morgan's law

$$Y = A \cdot B$$

Hence the correct answer is option 1

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Q. 19 Match List I with List II

	List I		**List II**
A	Facsimile	I	Static Document Image
B	Guided media Channel	II	Local Broadcast Radio
C	Frequency Modulation	III	Rectangular wave
D	Digital Signal	IV	Optical Fiber

Choose the correct answer from the following options :

Option 1:
$\mathrm{A-IV, B-III, C-II, D-I}$

Option 2:
$\mathrm{A-I, B-IV, C-II, D-III}$

Option 3:
$\mathrm{A-IV, B-II, C-III, D-I}$

Option 4:
$\mathrm{A-I, B-II, C-III, D-IV}$

Correct Answer:
$\mathrm{A-I, B-IV, C-II, D-III}$

Solution:
Facsimile - Static document image.

Guided media channel - Optical Fiber.

Frequency Modulation - Local broadcast radio.

Digital Signal - Rectangular wave.(Output $0 \text{ or } 1$)

Hence the correct answer is option 2.

Q. 20 If n represents the actual number of reflections in a converted galvanometer of resistance G and shunt resistance S. Then the total current I when its figure of merit is K will be :

Option 1:
$\frac{\mathrm{KS}}{(\mathrm{S+G})}$

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Option 2:

$\frac{(G+S)}{nKS}$

Option 3:

$\frac{nKS}{G+S}$

Option 4:

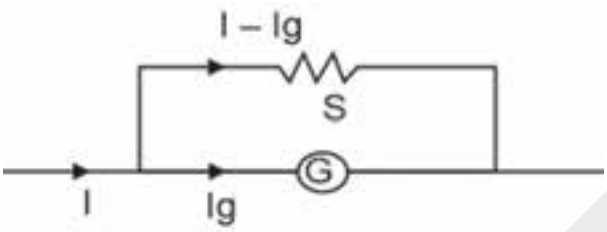
$\frac{nK\,(G+S)}{S}$

Correct Answer:

$\frac{nK\,(G+S)}{S}$

Solution:

Figure of merit $= K = \frac{I_g}{\theta}$



When the current through Galvanomter is I_g then the reflection is on

$K = \frac{I_g}{n}; I_g = nK$

$(I - I_g)\,S = I_g G$

$I = Ig\frac{(G+S)}{S}$

$I_g = \frac{I(S)}{(G+S)}$

$\frac{I\,(S)}{(G+S)} = nK$

$\Rightarrow I = \frac{nK(G+S)}{S}$

Hence the correct answer is option 4.

Q. 21 For $z = a^2x^3y^{\frac{1}{2}}$, where $'a'$ is a constant. If percentage error in measurement of $'x'$ and $'y'$ are 4% and 12% respectively, then the percentage error for $'z'$ will be _____$\%$.

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Correct Answer:
18

Solution:
$z = a^2x^3y^{\frac{1}{2}}$

$$\frac{\Delta z}{z} \times 100 = \frac{3\Delta x}{x} \times 100 + \frac{1}{2}\frac{\Delta y}{y} \times 100$$

$$\frac{\Delta z}{z} \times 100 = 3(4\%) + \frac{1}{2} \times (12\%)$$

$$= 18\%$$

The percentage error in measurement of z is 18%

Q. 22 A curved in a level road has radius 75m. The maximum speed of a car turning this curved road can be 30m/s without skidding. If radius of curved road is chamged to 48m and the coefficient of friction between the tyres and the road reamins same, then maximum allowed speed woud be _____m/s

Correct Answer:
24m/s

Solution:
For level road

$$V = \sqrt{\mu Rg}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{R_1}{R_2}}$$

$$\frac{30}{V_2} = \sqrt{\frac{75}{48}} = \sqrt{\frac{25}{16}}$$

$$V_2 = 24 \text{ m/s}$$

The maximum allowed speed for the current level road of the radius 48m is 24m/s

Hence the answer is 24m/s.

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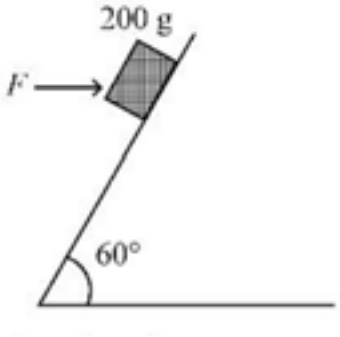
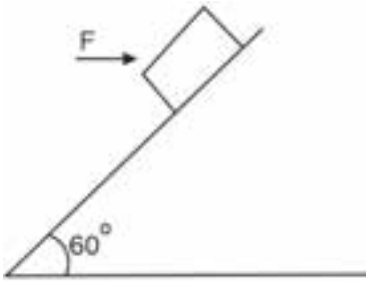
Q. 23 A block of mass $200\mathrm{g}$ is kept stationary on a smooth inclined plane by applying a minimum horizontal force $\mathrm{F} = \sqrt{\mathrm{xN}}$ as shown in figure.

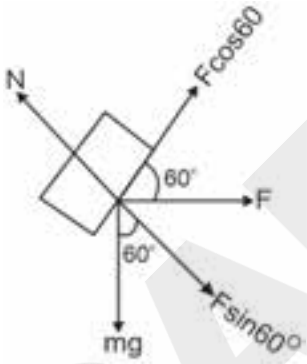


The value of $\mathrm{x} =$_____.

Correct Answer:
12

Solution:





$$\mathrm{N} = \mathrm{mg}\cos 60^\circ + \mathrm{F}\sin 60^\circ$$

For stationary block

$$\mathrm{F}\cos 60^\circ = \mathrm{mg}\sin 60^\circ$$

$$\sqrt{\mathrm{x}} = 2 \times \sqrt{3}$$

$$\mathrm{x} = 12$$

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Q. 24 Momet of Inertia (M.I) of four bodies having same mass $'\mathrm{M}'$ and radius $'2\mathrm{R}'$ are os follows :

$\mathrm{I}_1 = \mathrm{M.I}$ of solid spehere about it's diameter

$\mathrm{I}_2 = \mathrm{M.I}$ of solid cylinder about it's axis

$\mathrm{I}_3 = \mathrm{M.I}$ of solid circular disc about it's daimeter

$\mathrm{I}_4 = \mathrm{M.I}$ of thin circular ring about it's diameter

If $2\left(\mathrm{I}_2 + \mathrm{I}_3\right) + \mathrm{I}_4 = \mathrm{x} \cdot \mathrm{I}_1$ then the value of x will be_____.

Correct Answer:
5

Solution:

$$\mathrm{I}_1 = \frac{2}{5}\mathrm{M}(2\mathrm{R})^2 = \frac{8\mathrm{MR}^2}{5}$$

$$\mathrm{I}_2 = \frac{\mathrm{M}(2\mathrm{R})^2}{2} = 2\mathrm{MR}^2$$

$$\mathrm{I}_3 = \frac{\mathrm{M}(2\mathrm{R})^2}{4} = \mathrm{MR}^2$$

$$\mathrm{I}_4 = \frac{\mathrm{M}(2\mathrm{R})^2}{2} = \frac{4\mathrm{MR}^2}{2} = 2\mathrm{MR}^2$$

$$2\left(\mathrm{I}_2 + \mathrm{I}_3\right) + \mathrm{I}_4 = \mathrm{xI}_1$$

$$6\mathrm{MR}^2 + 2\mathrm{MR}^2 = \mathrm{x}\frac{(8\mathrm{MR}^2)}{5}$$

$$8\mathrm{MR}^2 = \mathrm{x}\frac{(8\mathrm{MR}^2)}{5}$$

$$\mathrm{x} = 5$$

Q. 25 Two satellites S_1 and S_2 are revolving in circular orbits around a planet with radius $\mathrm{R}_1 = 3200\mathrm{km}$ and $\mathrm{R}_2 = 800\mathrm{km}$ respectively. The ratio of speed of satellitte S_1 to the speed of satellite S_2 in their respective orbits would be $\frac{1}{\mathrm{x}}$ where x=_____

Correct Answer:
2

Solution:
Let the speed of satellite S_1 & S_2 be V_1 & V_2 respectively

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$$V_1 = \sqrt{\frac{GM}{R_1}}$$

$$V_2 = \sqrt{\frac{GM}{R_2}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{R_2}{R_1}} = \frac{1}{2} = \frac{1}{x}$$

$$x = 2$$

Q. 26 When a gas filled in a closed vessel is heated by raising the temperature by $1°C$, it's pressure increases by 0.4%. The initial temperature of the gas is _____K.

Correct Answer:
250

Solution:
For a closed vessel,

$V =$ constant

$$\therefore \quad P \alpha T$$

$$\frac{\Delta P}{P} = \frac{\Delta T}{T}$$

$$\frac{0.4}{100} = \frac{1}{T}$$

$$T = 250k$$

Q. 27 27 identical drops are charged at $22V$ each. They combine to form a bigger drop. The potential of the bigger drop will be _____V.

Correct Answer:
198

Solution:
For sphere capacitor,

Capacitance $= C = 4\pi\varepsilon_0 r$

Let the charge and radius of each drop be q and r respectively

Potential of drop $= V_1 = 22V$

$$\frac{q}{V_1} = C_1 = 4\pi\varepsilon_0 r$$

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$q = V_1 (4\pi\varepsilon_0 r)$ .....(1)

For bigger drop,let the charge and radius be $Q \text{ and } R$ respectively

$Q = 27q = 27 (V_1 (4\pi\varepsilon_0 r)$

$Q = 27q = V_2 (4\pi\varepsilon_0 R)$ .....(2)

By Volume conversation

$V_{initial} = V_{final}$

$27 \left(\frac{4}{3}\pi r^3\right) = \left(\frac{4}{3}\pi R^3\right)$

$R = 3r$ .....(3)

Form eqn (2) and (3)

$27 (V_1 (4\pi\varepsilon_0 r)) = V_2 (4\pi\varepsilon_0 (3r))$

$27V_1 = 3V_2$

$9V_1 = V_2$

$V_2 = 198 \text{ V}$

Q. 28 The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be_____$\%$

Correct Answer:
200

Solution:
$l_f = 2l_i$

Condition for strectching of wire is ,

$R \propto l^2$

$\frac{\Delta R}{R} = \frac{2\Delta l}{l} = 2$

$\frac{\Delta R}{R} \times 100 = 200\%$

Hence the answer is 200

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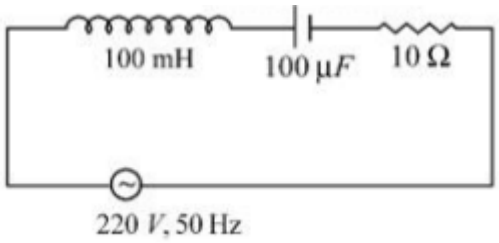
Q. 29 In a series LCR circuit, the inductance, capacitance and resistance are $\mathrm{L} = 100\mathrm{mH}, \mathrm{C} = 100\mu\mathrm{F}$ and $\mathrm{R} = 10\Omega$ respectively. They are connected to an AC source of voltage $220\mathrm{V}$ and frequency of $50\mathrm{Hz}$. The approximate value of current in the ircuit will be _____A



Correct Answer:
22

Solution:

$$\mathrm{X_L} = \omega\mathrm{L} = (2\pi\mathrm{fL}) = 100\pi \times 10^{-1}$$

$$\mathrm{X_L} = 10\pi \quad \text{.....(1)}$$

$$\mathrm{X_C} = \frac{1}{\omega\mathrm{c}} = \frac{1}{2\pi\mathrm{fC}} = \frac{1}{100 \times 10^{-4}(\pi)}$$

$$\mathrm{X_C} = \frac{100}{\pi} \quad \text{.....(2)}$$

$$\mathrm{R} = 10\Omega$$

$$\mathrm{Z} = \sqrt{\mathrm{R}^2 + (\mathrm{X_L} - \mathrm{X_C})^2}$$

$$= \sqrt{100 + \left(10\pi - \frac{100}{\pi}\right)^2}$$

$$\simeq \sqrt{100} \simeq 10$$

$$\mathrm{I_{rms}} \frac{\mathrm{V_{rms}}}{\mathrm{Z}} = \frac{220}{10} = 22\ \mathrm{A}$$

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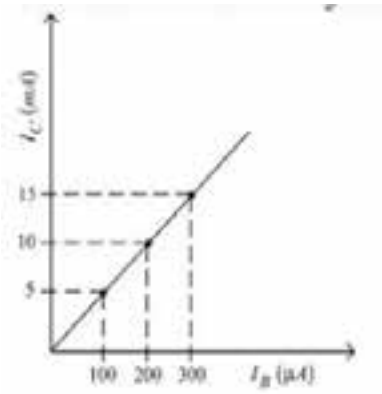
Q. 30 In an experiment of CE configuration of $\mathrm{n-p-n}$ transistor, the transistor characteristics are observed as given in figure:



If the input resistance is 200Ω and output resistance is 60Ω, the voltage gain in this experiment will be_____.

Correct Answer:
15

Solution:
$\mathrm{R_{input}} = 200\Omega$

$\mathrm{R_{output}} = 60\Omega$

Slope $= \dfrac{\Delta \mathrm{I_C}}{\Delta \mathrm{I_B}} = \dfrac{1}{20} \times 10^3$

Voltage Gain $(\mathrm{A_1}) = \dfrac{\Delta \mathrm{I_C}}{\Delta \mathrm{I_B}} \times \dfrac{\mathrm{R_0}}{\mathrm{R_i}}$

$= \dfrac{1000}{20} \times \dfrac{6}{20}$

Voltage Gain $= 15$

Chemistry

Q. 1 The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is:

[Given: The threshold frequency of platinum is $1.3 \times 10^{15}\mathrm{s}^{-1}$ and $\mathrm{h} = 6.6 \times 10^{-34}\mathrm{Js}$.]

Option 1:
$3.21 \times 10^{-14} J$

Option 2:
$6.24 \times 10^{-16} J$

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Option 3:

$8.58 \times 10^{-19} J$

Option 4:

$9.76 \times 10^{-20} J$

Correct Answer:

$8.58 \times 10^{-19} J$

Solution:

$$\begin{aligned}\text{Minimum Energy} &= \text{h} \times \text{threshold frequency} \\ &= \text{h} \times \nu_0 \\ &= 6.6 \times 10^{-14} \times 1.3 \times 10^{15}\ \text{J} \\ &= 8.58 \times 10^{-19}\ \text{J}\end{aligned}$$

Hence, the correct answer is **Option (3)**

Q. 2 At $25^{\circ}C$ and 1 atm pressure, the enthalpy of combustion of benzene (l) and acetylene (g) are $-3268\ \text{kJ mol}^{-1}$ and $-1300\ \text{kJ mol}^{-1}$, respectively. The change in enthalpy for the reaction $3\text{C}_2\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{l})$, is

Option 1:

$+324\ \text{kJ mol}^{-1}$

Option 2:

$+632\ \text{kJ mol}^{-1}$

Option 3:

$-632\ \text{kJ mol}^{-1}$

Option 4:

$-732\ \text{kJ mol}^{-1}$

Correct Answer:

$-632\ \text{kJ mol}^{-1}$

Solution:

Given,

$$\underset{\text{Benzene}}{\text{C}_6\text{H}_6(l)} + \left(\frac{15}{2}\right)\text{O}_2 \rightarrow 6\text{CO}_2 + 3\text{H}_2\text{O}, \Delta_c\text{H} = -3268\ \text{kJ/mol} \quad (1)$$

$$\underset{\text{(Acetylene)}}{\text{C}_2\text{H}_2(\text{g})} + \left(\frac{5}{2}\right)\text{O}_2 \rightarrow 2\text{CO}_2 + \text{H}_2\text{O}, \Delta_C\text{H} = -1300\ \text{kJ/mol} \quad (2)$$

We need to find the enthallpy change of the following reaction

$$3\text{C}_2\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6\ (\text{l})\ , \Delta_r\text{H} = \ ?$$

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So, by doing $3 \times (2) - (1)$, we get the required reaction.

Thus, enthalpy change of the reaction is given as

$$\Delta_r H = 3 \times (-1300) - (-3268)$$
$$\Delta_r H = -632 \text{ kJ/mol}$$

Hence, the correct answer is **Option (3)**

Q. 3 Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water,it depresses the freezing point by $0.2^{\circ}C$. The percentage association of solute A in water is:

[Given: Molar mass of $A = 93 \text{ g mol}^{-1}$. Molal depression constant of water is $1.86 \text{ K kg mol}^{-1}$.]

Option 1:
50%

Option 2:
60%

Option 3:
70%

Option 4:
80%

Correct Answer:
80%

Solution:
Solute A associates in water.

So, $\underset{1}{2A} \rightleftharpoons \underset{0}{A_2}$

$1-\alpha \qquad \frac{\alpha}{2}, \alpha = \%$ association of solute A in water

The van't Hoff factor can be calculated as

$$i = 1 - \alpha + \frac{\alpha}{2}$$
$$i = 1 - \frac{\alpha}{2} \qquad \rightarrow (1)$$

Now given,

$$\Delta T_f = 0.2^{\circ}C = 0.2 \text{ K}, \quad K_f = 1.86 \text{ K Kg mol}^{-1}$$

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$$n_A = \frac{W_A}{M_A} = \frac{0.7}{93}$$

$$W_B = 42.0\ g = \frac{42.0}{1000}\ kg$$

We know,

$$\Delta T_f = i\ kf\ m$$

$$\Rightarrow 0.2 = i \times (1.86) \times \frac{0.7}{93} \times \frac{1000}{42.0}$$

$$\Rightarrow i = 0.60 \quad \rightarrow (2)$$

Putting value of i in (1), we have

$$0.60 = 1 - \frac{\alpha}{2} \Rightarrow \alpha = 0.80$$

$$\therefore \%\ \alpha = 80\%$$

Hence, the correct answer is **Option (4)**

Q. 4 The K_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in $mol L^{-1}$ at 298 K is

Option 1:
1.0×10^{-15}

Option 2:
2.7×10^{-12}

Option 3:
3.2×10^{-10}

Option 4:
4.2×10^{-8}

Correct Answer:
1.0×10^{-15}

Solution:
Let, the solubility of Bi_2S_3 be $s\ mol\ L^{-1}$

So,

$$Bi_2\ S_3 \rightleftharpoons 2Bi^{3+} + 3S^{2-}$$

$$2s \qquad 3s$$

$$k_{sp} = (2s)^2(3s)^3 \Rightarrow 108s^5 = 1.08 \times 10^{-73}$$

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$$\Rightarrow s^5 = \frac{1.08 \times 10^{-73}}{108}$$
$$\Rightarrow s = 10^{-15} = 1.0 \times 10^{-15}$$

Hence, the correct answer is **Option (1)**

Q. 5 Match **List I** with **List II.**

List I	List II
A.Zymase	I. Stomach
B.Diastase	II. Yeast
C.Urease	III. Malt
D.Pepsin	IV. Soyabean

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-II, C-IV, D-I

Option 4:
A-III, B-II, C-I, D-IV

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
As we have learnt,

(1) Zymase naturally occurs in yeast.

(2) Diastase is present in malt

(3) Urease is found in soyabean

(4) Pepsin and hydrochloric acid are found in stomach

So, the correct match is

$$A - II, B - III, C - IV, D - I$$

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Hence, the correct answer is **Option (2)**

Q. 6 The correct order of electron gain enthalpies of Cl, F, Te and Po is

Option 1:
$\mathrm{F < Cl < Te < Po}$

Option 2:
$\mathrm{Po < Te < F < Cl}$

Option 3:
$\mathrm{Te < Po < Cl < F}$

Option 4:
$\mathrm{Cl < F < Te < Po}$

Correct Answer:
$\mathrm{Cl < F < Te < Po}$

Solution:
We know, Chlorine has the maximum negative value of electron gain enthpy.

The electron gain enthalpy data for the elements are given below :

$\mathrm{F \Rightarrow \Delta_{eg}H = -328\ kJ/mol}$

$\mathrm{Cl \Rightarrow \Delta_{eg}\ H = -349\ kJ/mol}$

$\mathrm{Te \Rightarrow \Delta_{eg}H = -190\ kJ/mol}$

$\mathrm{Po \Rightarrow \Delta_{eg}H = -174\ kJ/mol}$

Hence, the correct answer is Option (4)

Q. 7 Given below are two statements.

Statement I: During electrolytic refining, blister copper deposits precious metals.

Statement II: In the process of obtaining pure copper by electrolysis method, copper blister is used to make the anode.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are **true**.

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Option 2:
Both Statement I and Statement II are **false.**

Option 3:
Statement I is **true** but Statement II is **false**.

Option 4:
Statement I is **false** but Statement II is **true**.

Correct Answer:
Both Statement I and Statement II are **true**.

Solution:
During electrolytic refining of copper, impure Blister copper is taken as the anode and deposits anode mud which contains antimony, selenium, tellurium, silver, gold and platinum. These doposits are precious metals.

So, Both statement I and II are correct.

Hence, the correct answer is **Option (1)**

Q. 8 Given below are two statements one is labelled as **Assertion A** and the other is labelled as **Reason R:**

Assertion A: The amphoteric nature of water is explained by using Lewis acid/base concept.

Reason R: Water acts as an acid with $\mathrm{NH_3}$ and as a base with $\mathrm{H_2S}$

In the light of the above statements choose the ***correct*** answer from the options given below:

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true

Correct Answer:
A is false but **R** is true

Solution:
Water has the ability to act as an acid as well as a base, it behaves as an amphoteric substance. In the Bronsted sense , it acts as an acid with $\mathrm{NH_3}$ and a base with $\mathrm{H_2S}$.

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$$H_2O(\ell) + NH_3(\text{ aq }) \rightleftharpoons OH^{\ominus}_{(aq)} + NH^{\oplus}_4(aq)$$

$$\text{acid} \qquad \text{Base} \qquad \text{CB} \qquad \text{CA}$$

$$H_2O(\ell) + H_2S(\text{ aq }) \rightleftharpoons H_3O^{\oplus}_{(aq)} + HSO^{\ominus}_4\ (aq)$$

$$\text{Base} \qquad \text{Acid} \qquad \text{CA} \qquad \text{CB}$$

A is false, as the Amphoteric nature cannot be explained by using Lewis acid/base concept. It is explained by the Bronsted Lowry theory of acid and base

R is true.

Hence, the correct answer is **Option (4)**

Q. 9 The correct order of reduction potentials of the following pairs is

A. $Cl_2|Cl^-$

B. $I_2|I^-$

C. $Ag^+|Ag$

D. $Na^+|Na$

E. $Li^+|Li$

Option 1:
$A > C > B > D > E$

Option 2:
$A > B > C > D > E$

Option 3:
$A > C > B > E > D$

Option 4:
$A > B > C > E > D$

Correct Answer:
$A > C > B > D > E$

Solution:
Fact based on the reduction potential values.

The correct order will be

$$Cl_2|Cl^{\ominus} > Ag^+|Ag > I_2|I^{\ominus} > Na^{\oplus}|Na > Li^{\oplus}|Li$$

Thus, the given electrode couple can be arranged in order of their reduction potential values as

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$A > C > B > D > E$

Hence, the correct answer is **option(1)**

Q. 10 The number of bridged oxygen atoms present in compound B formed from the following reactions is

$$Pb(NO_3)_2 \xrightarrow{673\ K} A + PbO + O_2$$

$$A \xrightarrow{Dimerise} B$$

Option 1:
0

Option 2:
1

Option 3:
2

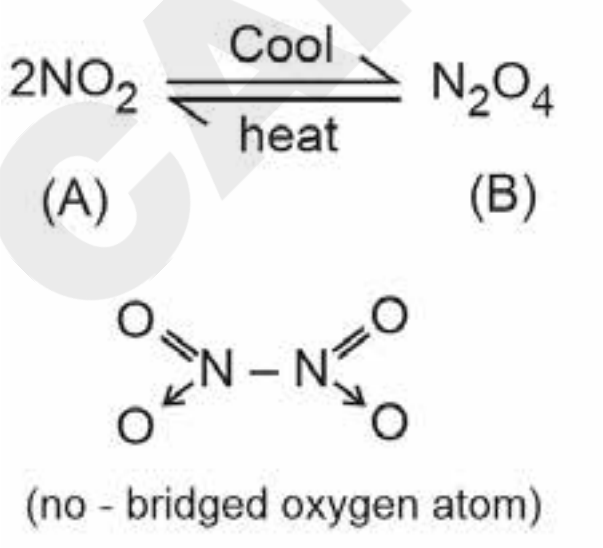
Option 4:
3

Correct Answer:
0

Solution:
As we have learnt,

Heating of $PbNO_3$ produces NO_2 apart from PbO and O_2 as given below

$$Pb(NO_3)_2 \xrightarrow{673\ K} \underset{(A)}{4NO_2} + 2PbO + O_2$$



So, **Option (1)** is correct

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Q. 11 The metal ion (in gaseous state) with lowest spin-only magnetic moment value is

Option 1:
V^{2+}

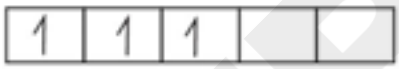
Option 2:
Ni^{2+}

Option 3:
Cr^{2+}

Option 4:
Fe^{2+}

Correct Answer:
Ni^{2+}

Solution:
The metal ion with their electronic configuration

$V^{2+} : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3,$  unpaired $e^- = 3$

$Ni^{2+} : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8,$ unpaired $e^- = 2$

$Cr^{2+} : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^4,$ unpaired $e^- = 4$

$Fe^{2+} : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6,$ unpaired $e^- = 4$

Lowest spin-only magnetic moment value will be for the ion having the least number of unpaired electron.

Thus, Ni^{2+} has the lowest spin only magnetic moment out of the given ions.

So, **option (2)** is correct.

Q. 12 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: Polluted water may have a value of BOD of the order of 17ppm.

Reason R: BOD is a measure of oxygen required to oxidise both the bio-degradable and non- biodegradable organic material in water.

In the light of the above statements, choose the most appropriate answer from the options given below.

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Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

Option 4:
A is not correct but **R** is correct.

Correct Answer:
A is correct but **R** is not correct.

Solution:
The amount of BOD in the water is a measure of the amount of organic material in the water, in terms of how much oxygen will be required to break it down biologically by bacteria.

Clean water would have BOD value of less than 5ppm whereas highly polluted water could have a BOD value of 17ppm or more.

So, A is correct but R is not correct.

Option (3) is correct.

Q. 13 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: A mixture contains benzoic acid and napthalene. The pure benzoic acid can be separated out by the use of benzene.

Reason R: Benzoic acid is soluble in hot water.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true.

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Correct Answer:
A is false but **R** is true.

Solution:
Pure benzoic acid cannot be separated out by the use of Benzene from a mixture of Benzoic acid and Naphthalene as both the compounds will be soluble in Benzene.

So, A is false.

Benzoic acid is soluble in hot water. It is a weak acid which dissolves in water upon heating.

So, R is true.

Option (4) is correct.

Q. 14 During halogen test, sodium fusion extract is boiled with concentrated HNO_3 to

Option 1:
remove unreacted sodium

Option 2:
decompose cyanide or sulphide of sodium

Option 3:
extract halogen from organic compound

Option 4:
maintain the pH of extract.

Correct Answer:
decompose cyanide or sulphide of sodium

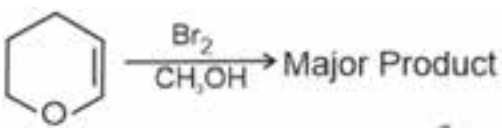
Solution:
The sodium fusion extract is acidified with nitric acld and then treated with silver nitrate to test the halogen presence.

If nitrogen or sulphur is also present in the compound, the sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test.

These ions, if not removed, would interfere with silver nitrate test for halogens.

Option (2) is correct.

Q. 15 Amongst the following, the major product of the given chemical reaction is



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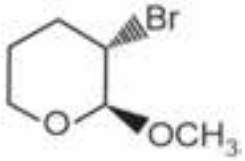
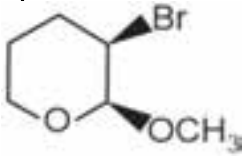
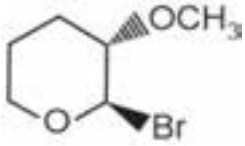
Option 1:



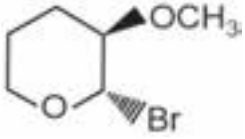
Option 2:



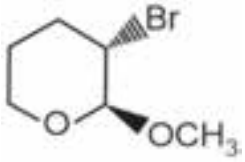
Option 3:



Option 4:



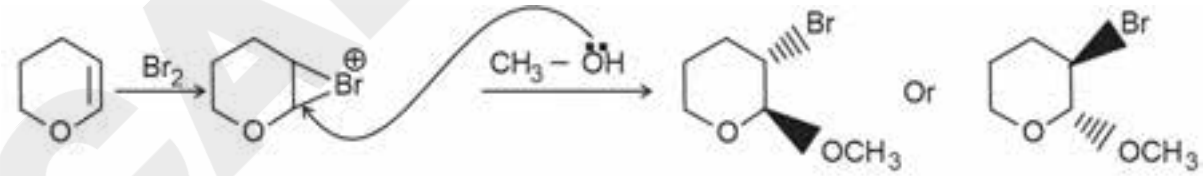
Correct Answer:



Solution:

The given reaction proceeds by the formation of a Cyclic Bromonium ion on which the attack of the alcohol takes place.

It is to be noted that the attack of the alcohol takes place on the carbon atom adjacent to the Oxygen atom.



Option (1) is correct.

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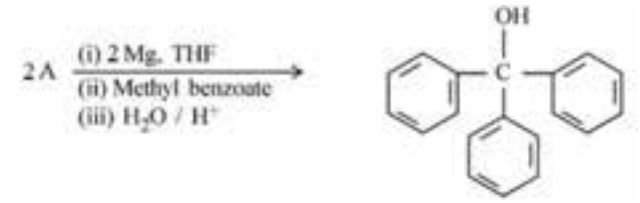
Q. 16 In the given reaction



'A' can be

Option 1:
benzyl bromide

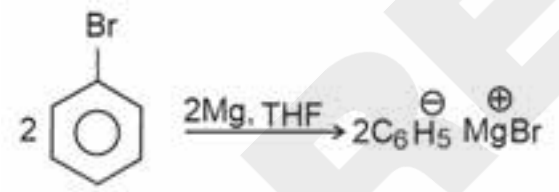
Option 2:
bromobenzene

Option 3:
cyclohexyl bromide

Option 4:
methyl bromide

Correct Answer:
bromobenzene

Solution:
For making the given product, we need $C_6H_5^{\ominus}$, which will come from Bromobenzene.



Next, the reaction of Phenyl Magnesium bromide with Methyl benzoate $C_6H_5-COOCH_3$, occurs as

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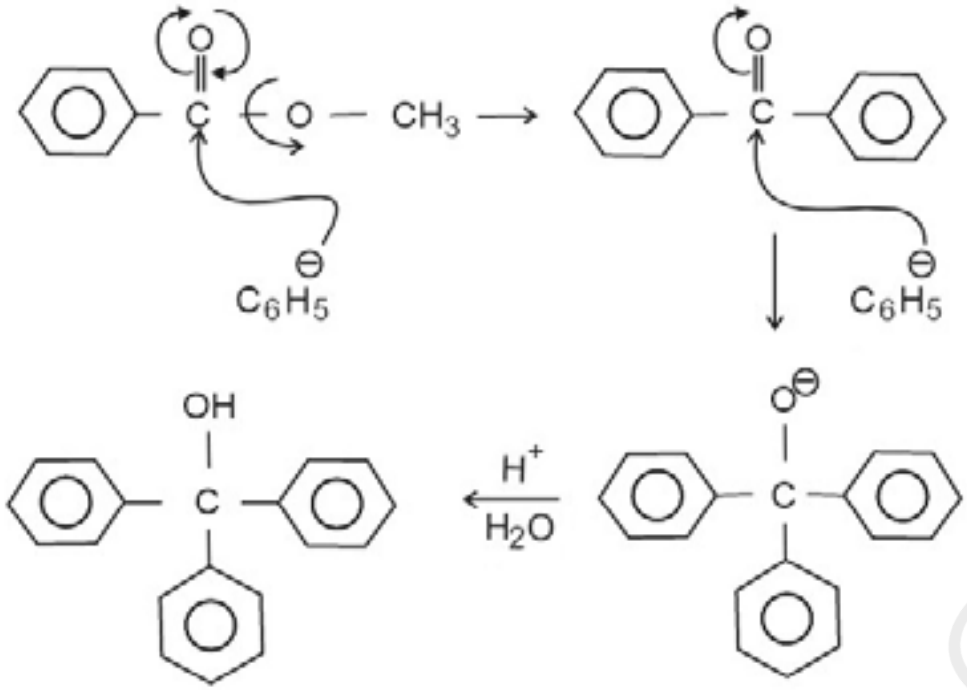


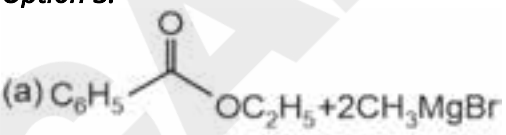
Hence, **Option (2)** is correct.

Q. 17 Which of the following conditions or reaction sequence will **NOT** give acetophenone as the major product?

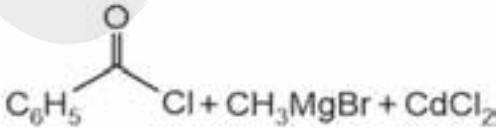
Option 1:

Option 2:

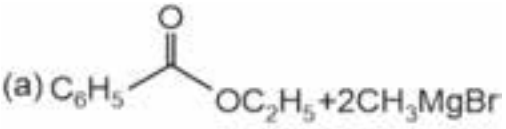
Option 3:



Option 4:



Correct Answer:



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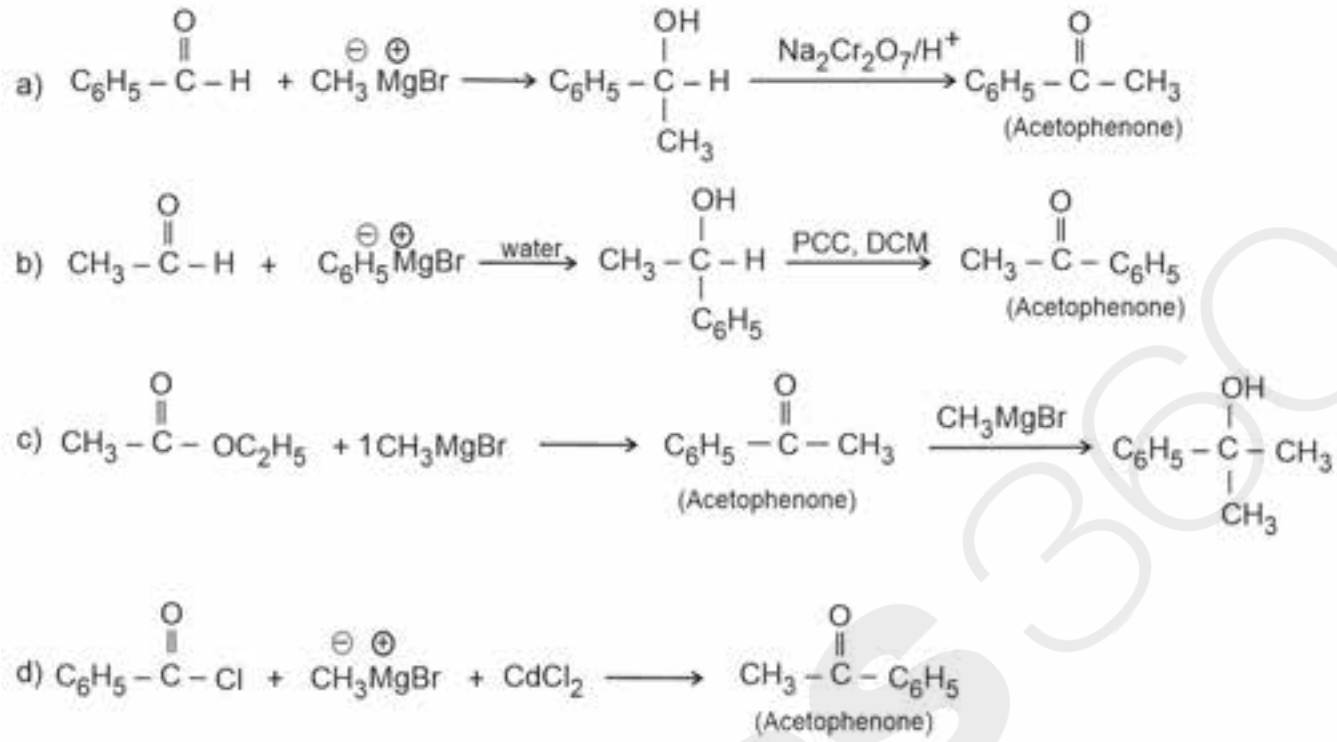
Solution:

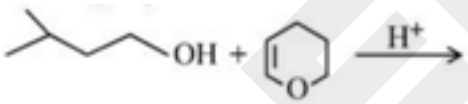
$\text{Acetophenone} \Rightarrow C_6H_5 - CO - CH_3$



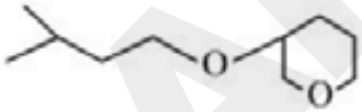
Thus, the reaction given in Option(3) does not give Acetophenone as a final Product.

Option (3) is correct.

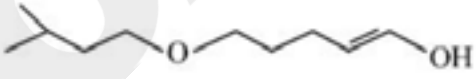
Q. 18 The major product formed in the following reaction, is



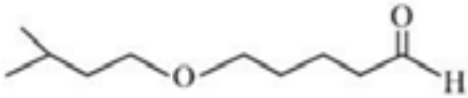
Option 1:



Option 2:



Option 3:



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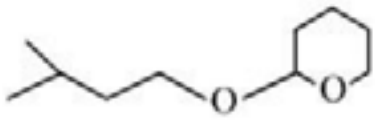
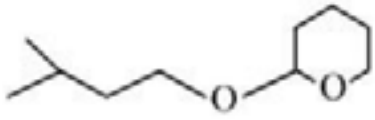
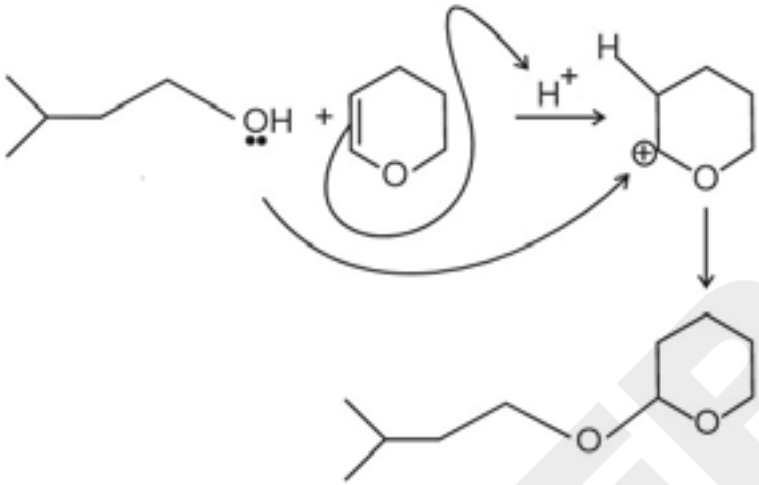
Option 4:



Correct Answer:



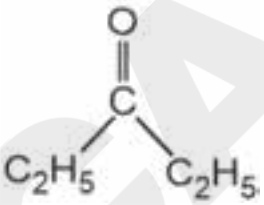
Solution:
The reaction will be



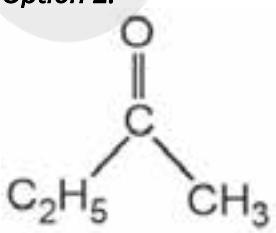
Option (4) is correct.

Q. 19 Which of the following ketone will **NOT** give enamine on treatment with secondary amines?
[where t-Bu is $-\mathrm{C}(\mathrm{CH}_3)_3$]

Option 1:



Option 2:



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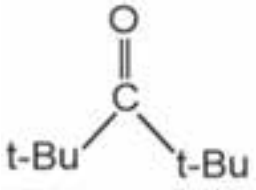
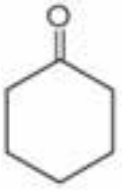
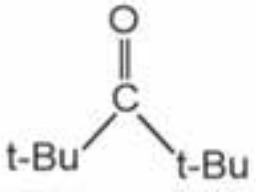
Option 3:



Option 4:

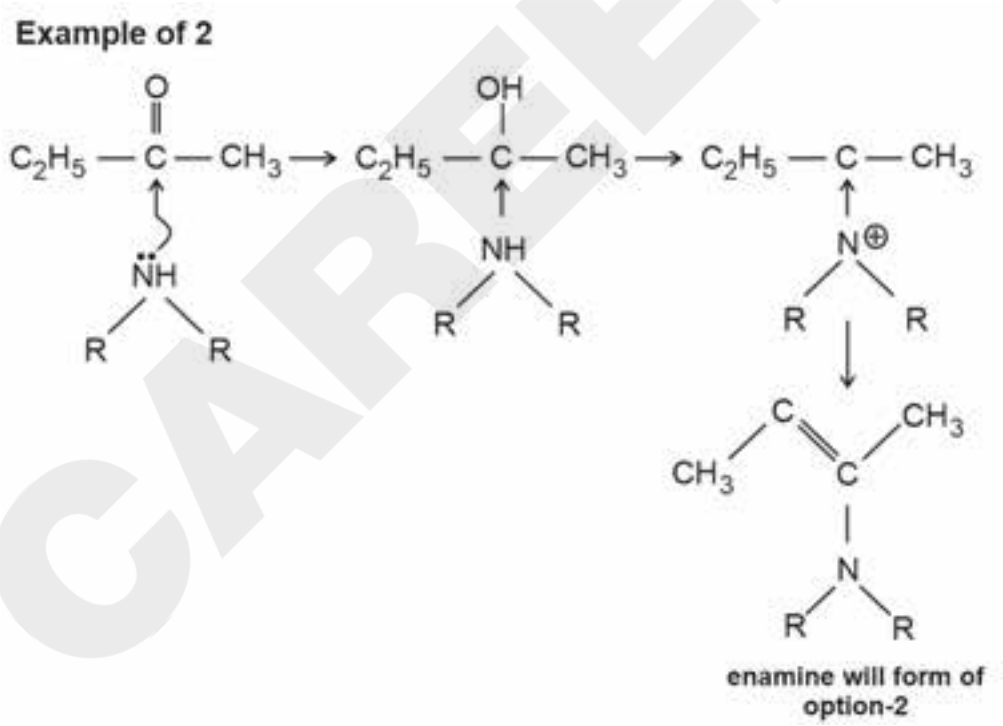


Correct Answer:



Solution:

Enamine $\Rightarrow \mathrm{C}=\mathrm{C}-\mathrm{N}$ this group should be present. will not able to add on carbonyl group of ketone on option (3) compound due to presense of Bulky group $(\mathrm{t\text{-}Bu})$, also elimination will not possible.



Hence,**option (3)** is correct.

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Q. 20 An antiseptic dettol is a mixture of two compounds 'A' and 'B' where A has 6π electrons and B has 2π electrons. What is 'B'?

Option 1:
Bithionol

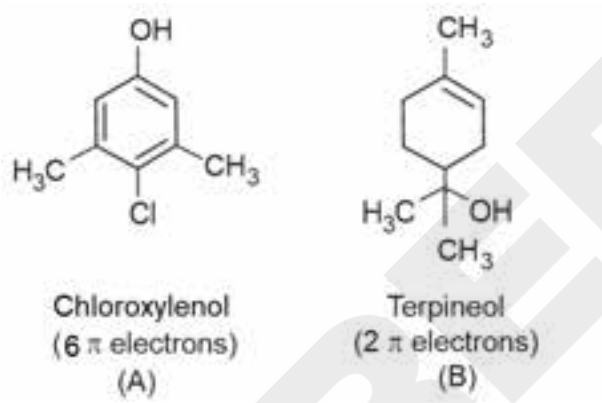
Option 2:
Terpineol

Option 3:
Chloroxylenol

Option 4:
Chloramphenicol

Correct Answer:
Terpineol

Solution:
Dettol is a mixture of chloroxylenol and terpineol.



Hence,**Option (2)** is correct.

Q. 21 A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is _____$\times 10^3 \mathrm{gmol}^{-1}$ [nearest interger]

Correct Answer:
25

Solution:
A protein 'A' contain 0.30% of glycine which has 75 molecular weight.

Let the molar mass of the Protein be M

We have,

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$$\frac{75}{M} \times 100 = 0.3$$

$$M = \frac{7500}{0.3} = 25 \times 10^3 \text{ g mol}^{-1}$$

Hence, the correct answer is 25

Q. 22 A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is $27^\circ C$. At 03:00 pm, when the temperature is $45^\circ C$, the pressure in the tank will be _____atm. [nearest integer]

Correct Answer:
32

Solution:
A rigid tank means volume and number of moles of the gas are constant

So, from Gay Lussac's law, we can write

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{30}{300} = \frac{P_2}{318}$$

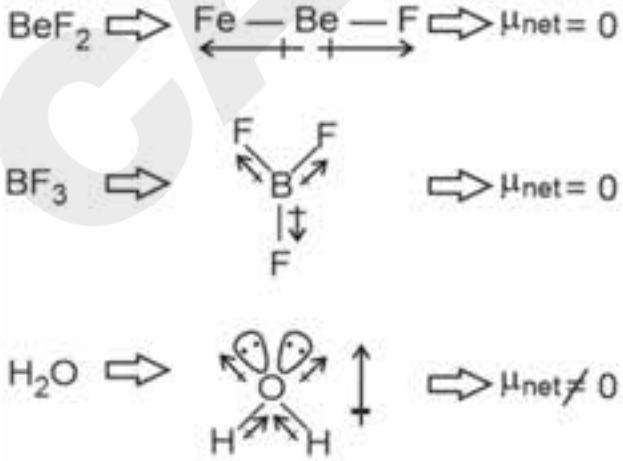
$$P_2 = 31.8 \approx 32 \text{ (nearest Integer)}$$

Hence, the correct answer is 32

Q. 23 Amongst $BeF_2, BF_3, H_2O, NH_3, CCl_4$ and HCl, the number of molecules with non-zero net dipole moment is_____.

Correct Answer:
3

Solution:



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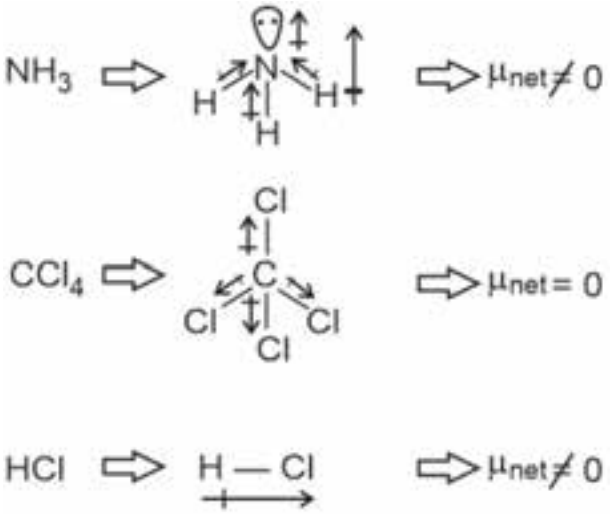

Hence, the answer is 3

Q. 24 At 345 K, the half life for the decomposition of a sample of a sample of a gaseous compound initially at 55.5 kPa was 340 s. When the pressure was 27.8kPa, the half life was found to be 170 s. The order of the reactionis _____. [integer answer]

Correct Answer:
0

Solution:
As we have learnt,

The half life is related to the order of the reaction as

$$t_{1/2} \propto [A_0]^{1-n}$$

The above relation can be modified in terms of pressure as

$$t_{1/2} \propto [P_0]^{1-n}$$

$$t_{1/2} \propto \frac{1}{[P_0]^{n-1}}$$

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \frac{(P_2)^{n-1}}{(P_1)^{n-1}} \Rightarrow \frac{340}{170} = \left(\frac{27.8}{55.5}\right)^{n-1}$$

$$\Rightarrow 2 = \left(\frac{1}{2}\right)^{n-1} \Rightarrow n-1 = -1 \Rightarrow n = 0$$

So, order of the reaction is Zero.

Hence, the correct answer is 0.

Q. 25 A solution of $Fe_2(SO_4)_3$ is electrolyzed for 'x' min with a current of 1.5 A to deposit $0.3482\ g$ of Fe. The value of x is _____.[nearest integer]

Given: 1F = $96500 C mol^{-1}$

$\text{Atomic mass of Fe} = 56\ g\ mol^{-1}$

Correct Answer:
20

Solution:
Reduction Reaction of $Fe_2(SO_4)_3$ will be

$$Fe^{3+} + 3e^- \rightarrow Fe$$

So, 3 moles of e^- will be required to deposit 1 mol of Fe

$\Rightarrow \quad 56\ g\ Fe \text{ requires } 3 \text{ mole of } e^-$

$\Rightarrow \quad 56\ g\ Fe \text{ requires } 3 \times 96500\ C$

$\Rightarrow \quad 0.3482\ g\ Fe \text{ requires } \frac{3 \times 96500}{56} \times 0.3482\ C$

$\Rightarrow \quad 0.3482\ g\ Fe \text{ requires } 1800.06\ C$

Formula,

$Q = it \Rightarrow 1800.06 = 1.5 \times t$

$t = 1200.04\ sec$

$t = 20\ min$

Hence, the correct answer is **20**

Q. 26 Consider the following reactions:

$$PCl_3 + H_2O \rightarrow A + HCl$$
$$A + H_2O \rightarrow B + HCl$$

The number of ionisable protons present in the product B is_____.

Correct Answer:
2

Solution:
Reaction will be :

$$PCl_3 + H_2O \rightarrow PCl_2(OH) + HCl$$
$$PCl_2(OH) + H_2O \rightarrow PCl(OH)_2 + HCl$$
$$(A) \qquad\qquad (B)$$

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In B, there are two ionisable protons $\left(H^{+}\right)$ present.

Hence , the answer is **(2)**

Q. 27 Amongst $FeCl_3.3H_2O$, $K_3\left[Fe(CN)_6\right]$ and $\left[Co\left(NH_3\right)_6\right]Cl_3$, the spin-only magnetic moment value of the inner-orbital complex that absorbs light at shortest wavelength is _____B.M [nearest integer]

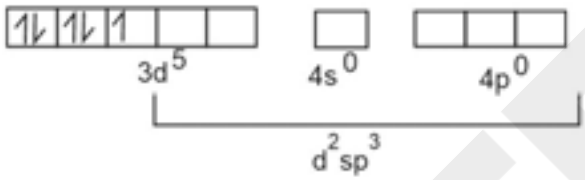
Correct Answer:
2

Solution:
$[Fe(H_2O)_3Cl_3] \rightarrow \text{outer orbital complex}$
$K_3\left[Fe(CN)_6\right] \rightarrow \text{inner orbital complex}$
$\left[Co\left(NH_3\right)_6\right]Cl_3 \rightarrow \text{inner orbital complex}$

but $K_3\left[Fe(CN)_6\right]$ will have more spliting than $\left[Co\left(NH_3\right)_6\right]$, because $CN^{\ominus}$ more stronger ligand.

$\text{more spliting} \rightarrow \text{more } \Delta_0 \rightarrow \text{lesser wavelength absorbed.}$

So, spin -only magnetic moment value of $[Fe(CN)_6]^{3-}$



$\mu = \sqrt{n(n+2)}BM \quad [\because 1 \text{ unpaired } e^{\ominus}]$
$\mu = \sqrt{3}\ BM$
$\mu = 1.732\ BM \approx 2\ BM$

Hence, the correct answer is 2

Q. 28 The Novolac polymer has mass of 963 g. The number of monomer units present in it are

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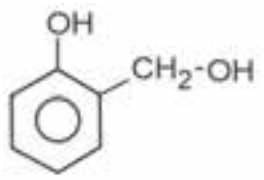
Correct Answer:
9

Solution:
Monomer unit of Novolac is $C_7H_8O_2$ or



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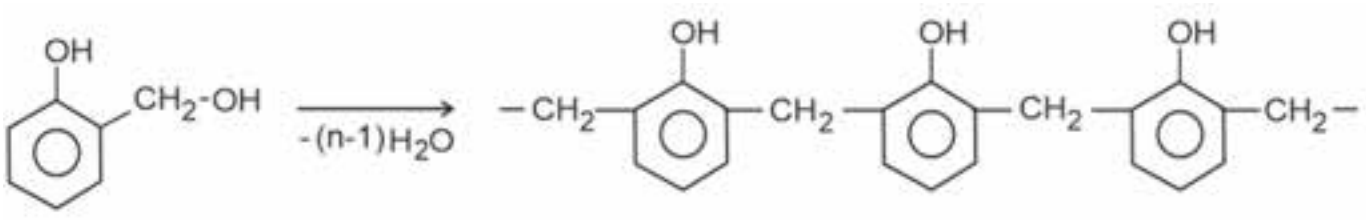
molar mass of monomer = 124 g/mol



$$\text{n(monomer) weight} = \text{Novolac} + (\text{n}-1)\text{H}_2\text{O} \text{ weight}$$

$$\text{n} \times 124 = 963 + (\text{n}-1) \times 18 \Rightarrow \text{n} = 9$$

Hence, the correct answer is 9.

Q. 29 How many of the given compounds will give a positive Biuret test _____?

Glycine, Glycylalanine, Tripeptide, Biuret

Correct Answer:
2

Solution:
Biuret test is a chemical test used for detecting the presence of two or more peptide bonds.

Glycine : 0 Peptide Bond

Glycylalanine : 1 peptide Bond

Tripeptide: 3 peptide Bond

Biuret : 2 peptide Bond

Thus, tripeptide and Biuret give the Biuret test

Hence, the correct answer is 2

Q. 30 The neutralization occurs when 10 mL of 0.1M acid 'A' is allowed to react with 30 mL of 0.05 M base M(OH)_2. The basicity of the acid 'A' is _____.

[M is a metal]

Correct Answer:
3

Solution:
From law of equivalence, we have

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$$\text{meq of (A)} = \text{meq of } M(OH)_2$$
$$10 \times 0.1 \times x = 30 \times 0.05 \times 2$$
$$x = 3$$

So, the Basicity of the acid 'A' is 3

Hence, the correct answer is 3

Maths

Q. 1 Let $A = \{x \in \mathbb{R} : |x+1| < 2\}$ and $B = \{x \in \mathbb{R} : |x-1| \geq 2\}$. Then which one of the following statements is NOT true ?

Option 1:
$A - B = (-1, 1)$

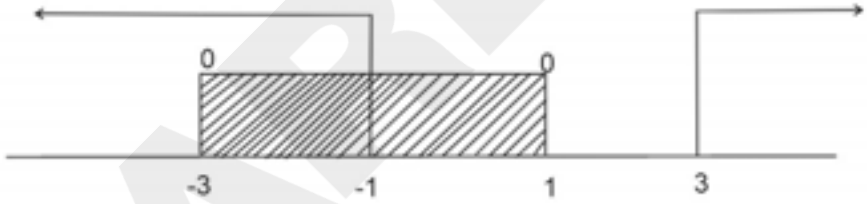
Option 2:
$B - A = \mathbb{R} - (-3, 1)$

Option 3:
$A \cap B = (-3, -1]$

Option 4:
$A \cup B = \mathbb{R} - [1, 3)$

Correct Answer:
$B - A = \mathbb{R} - (-3, 1)$

Solution:



$|x+1| < 2 \Rightarrow -2 < x+1 < 2 \Rightarrow x \in (-3, 1) = A$

$|x-1| \geq 2 \Rightarrow x - 1 \leq -2 \vee x - 1 \geq 2$

$\Rightarrow x \leq -1 \cup x \geq 3 \Rightarrow x \in (-\infty - 1] \cup [3, \infty)$

$A = (-3, 1) \quad B = (-\infty, -1] \cup [3, \infty)$

$A' = (-\infty, -3] \cup [1, \infty) \quad B' = (-1, 3)$

$A - B = A \cap B' = (-1, 1)$

$B - A = A' \cap B = (-\infty, -3] \cup [3, \infty)$

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$A \cup B = (-\infty, 1) \cup [3, \infty)$

$A \cap B = (-3, -1]$

Q. 2 Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root a. If a and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $a^2 + \beta^2$ is equal to:

Option 1:
37

Option 2:
58

Option 3:
68

Option 4:
92

Correct Answer:
58

Solution:
Given

$ax^2 - 2bx + 15 = 0$

Has repeated roots so $D = 0$

$4b^2 - 4 \times 15 \times a = 0$

$\Rightarrow b^2 = 15a$

Also Given

$x^2 - 2bx + 21 = 0$

now α will satisfy both qudratic

$ax^2 - 2bx + 15 = 0$ and $x^2 - 2bx + 21 = 0$

Putting the value we get

$a\alpha^2 - 2b\alpha + 15 = 0$

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$-\alpha^2 - 2b\alpha + 21 = 0$

$(a-1)\alpha^2 = 6$

$\alpha^2 = \frac{6}{a-1}$

Now in eqn (1) product of root $\alpha^2 = \frac{15}{a}$

so $\frac{15}{a} = \frac{6}{a-1} \Rightarrow 2a = 5a - 5 \Rightarrow a = \frac{5}{3}$

Now $b^2 = 15a \Rightarrow b^2 = 15 \times \frac{5}{3} \Rightarrow b^2 = 25$

so $b = \pm 5$

Now in quadratic $x^2 - 2bx + 21 = 0$

Putting the value of b we get

$x^2 - 10x + 21 = 0 \Rightarrow (x-7)(x-3) = 0$

so $x = 3$ or $x = 7$

or $x^2 + 10x + 21 = 0 \Rightarrow x = -3$ or $x = -7$

so $\alpha = \pm 3$ & $\beta = \pm 7$

so $\alpha^2 + \beta^2 = 3^2 + 7^2 = 9 + 49 = 58$

Hence the correct answer is option 2

Q. 3 Let z_1 and z_2 be two complex number such that $\bar{z}_1 = i\bar{z}_2$ and arg $\left(\frac{z_1}{\bar{z}_2}\right) = \pi$. Then

Option 1:
$\arg z_2 = \frac{\pi}{4}$

Option 2:
$\arg z_2 = \frac{3\pi}{4}$

Option 3:
$\arg z_1 = \frac{\pi}{4}$

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Option 4:

$\arg z_1 = \frac{3\pi}{4}$

Correct Answer:

$\arg z_1 = \frac{\pi}{4}$

Solution:

$\overline{z_1} = i \cdot \overline{z_2} \Rightarrow z_1 = -iz_2 \Rightarrow iz_1 = z_2 \Rightarrow e^{\frac{i\pi}{2}} z_1 = z_2$

$\text{given}, \arg\left(\frac{z_1}{\bar{z}_2}\right) = \pi$

$\text{Let } z_1 = re^{i\theta} \Rightarrow \bar{z}_2 = r \cdot e^{-i\left(\frac{\pi}{2}+\theta\right)}$

$\text{so } \arg\left(e^{i\left(2\theta+\frac{\pi}{2}\right)}\right) = \pi$

$\Rightarrow \quad 2\theta + \frac{\pi}{2} = \pi$

$\Rightarrow \quad \theta = \frac{\pi}{4}$

Hence the correct answer is option 3

Q. 4 The system of equations

$-kx + 3y - 14z = 25$

$-15x + 4y - kz = 3$

$-4x + y + 3z = 4$

is consistent for all k in the set

Option 1:

$\mathbb{R}$

Option 2:

$\mathbb{R} - \{-11, 13\}$

Option 3:

$\mathbb{R} - \{13\}$

Option 4:

$\mathbb{R} - \{-11, 11\}$

Correct Answer:

$\mathbb{R} - \{-11, 11\}$

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Solution:
For cosistance solutions $\Delta \neq 0$

$$\Rightarrow \begin{vmatrix} -k & 3 & -14 \\ -15 & 4 & -k \\ -4 & 1 & 3 \end{vmatrix} \neq 0$$

$$\Rightarrow -\mathrm{k}(12+\mathrm{k})-3(-45-4\mathrm{k})-14(-15+16) \neq 0$$

$$\Rightarrow -12\mathrm{k}-\mathrm{k}^2+135+12\mathrm{k}-14 \neq 0$$

$$\Rightarrow \mathrm{k}^2 \neq 121$$

$$\mathrm{k} \neq 11,-11$$

Hence the correct answer is option 4

Q. 5 $\lim\limits_{x \to \frac{\pi}{2}} \left(\tan^2 \mathrm{x}\left(\left(2\sin^2 \mathrm{x}+3\sin \mathrm{x}+4\right)^{\frac{1}{2}}-\left(\sin^2+6\sin \mathrm{x}+2\right)^{\frac{1}{2}}\right)\right)$ is equal to

Option 1:
$\frac{1}{12}$

Option 2:
$-\frac{1}{18}$

Option 3:
$-\frac{1}{12}$

Option 4:
$\frac{1}{6}$

Correct Answer:
$\frac{1}{12}$

Solution:
$$\lim_{\mathrm{x} \to \frac{\pi}{2}} \left[\tan^2 \mathrm{x}\left(\left(2\sin^2 \mathrm{x}+3\sin \mathrm{x}+4\right)^{1/2}\left(\sin^2 \mathrm{x}+6\sin \mathrm{x}+2\right)^{1/2}\right]$$

$$\lim_{\mathrm{x} \to \frac{\pi}{2}} \frac{\sin^2 \mathrm{x}\left(\sqrt{2\sin^2 \mathrm{x}+3\sin \mathrm{x}+4}-\sqrt{\sin^2 \mathrm{x}+6\sin \mathrm{x}+2}\right)}{\cos^2 \mathrm{x}}$$

$$\lim_{\mathrm{x} \to \frac{\pi}{2}} \frac{\sin^2 \mathrm{x}\left[\left(2\sin^2 \mathrm{x}+3\sin \mathrm{x}-4\right)-\left(\sin^2 \mathrm{x}+6\sin \mathrm{x}+2\right)\right]}{\cos^2 \mathrm{x}\left[\sqrt{2\sin^2 \mathrm{x}+3\sin \mathrm{x}+4}+\sqrt{\sin^2 \mathrm{x}+6\sin \mathrm{x}+2}\right]}$$

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$$\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x\left[\sin^2 x-3\sin x+2\right]}{\cos^2 x\left[\sqrt{2\sin^2 x+3\sin x+4}+\sqrt{\sin^2 x+6\sin x+2}\right]}$$

$$\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x(\sin x-1)(\sin x-2)}{(1-\sin^2 x)\left(\sqrt{2\sin^2 x+3\sin x+4}+\sqrt{\sin^2 x+6\sin x+2}\right]}$$

$$\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x(\sin x-2)(\sin x-1)}{(1+\sin x)(1-\sin x)\left(\sqrt{2\sin^2 x+3\sin x+4}+\sqrt{\sin^2 x+6\sin x+2}\right)}$$

$$\frac{1}{2[3+3]}=\frac{1}{12}$$

Hence the correct answer is option 1

Q. 6 The area of the region eclosed between the parabolas $y^2=2x-1$ and $y^2=4x-3$ is

Option 1:
$\frac{1}{3}$

Option 2:
$\frac{1}{6}$

Option 3:
$\frac{2}{3}$

Option 4:
$\frac{3}{4}$

Correct Answer:
$\frac{1}{3}$

Solution:
We have, $y^2=2x-1$ .....(1)

and $y^2=4x-3$ .....(2)

We first find the point of inter section of the given parabolas by solving the eqn (1) and (2) simultaneously.

This gives, $2x-1=4x-3$

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$\Rightarrow 2x = 2 \Rightarrow x = 1$

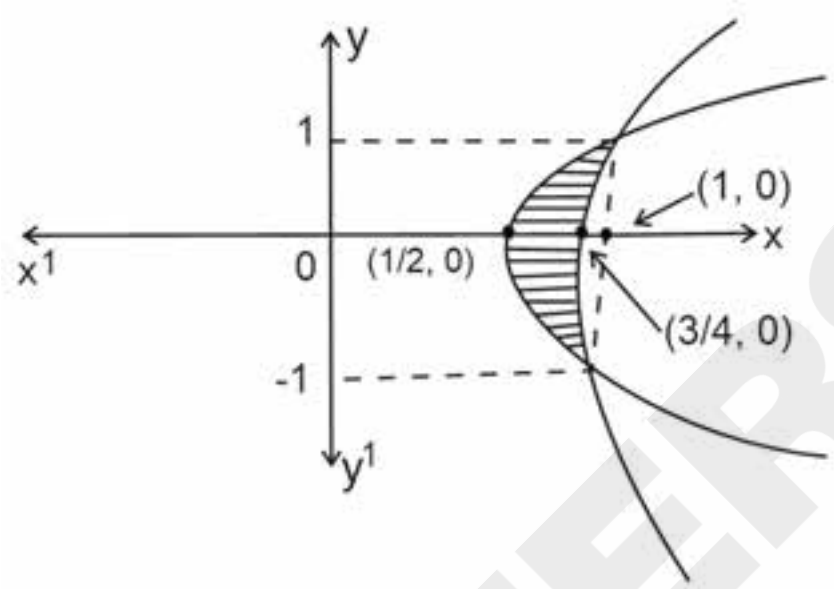
Now, put $x = 1$ in eqn (1), we get $y = \pm 1$

Now the eqn(1) and (2) can be written as

$$y^2 = 2\left(x - \frac{1}{2}\right)$$

$y^2 = 4\left(x - \frac{3}{4}\right)$ respectively.

The graphs of these two curves are



Now, the eqn (1) and (2) can be written as

$x = \frac{y^2 + 1}{2}$ & $x = \frac{y^2 + 3}{4}$ respectively

So $A = 2\int_0^1 (x_2 - x_1)\, dy$

($\because$ Area is symmetrical about x-axis)

Where $x_1 = \frac{y^2 + 1}{2}$ and $x_2 = \frac{y^2 + 3}{4}$

$$A = 2\int_0^1 \left(\frac{y^2 + 3}{4} - \frac{y^2 + 1}{2}\right) dy$$

$$= \frac{1}{2}\left[y - \frac{y^3}{3}\right]_0^1 = \frac{1}{2}\left[1 - \frac{1}{3}\right]$$

$$= \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \text{ sq.units}$$

Hence he correct answer is option 1.

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Q. 7 The coefficient of x^{101} in the expression $(5+x)^{500}+x(5+x)^{499}+x^2(5+x)^{498}+-------+x^{500}, x>0$, is

Option 1:
$^{501}C_{101}\,(5)^{399}$

Option 2:
$^{501}C_{101}\,(5)^{400}$

Option 3:
$^{501}C_{100}\,(5)^{400}$

Option 4:
$^{500}C_{101}\,(5)^{399}$

Correct Answer:
$^{501}C_{101}\,(5)^{399}$

Solution:

$$\frac{(5+x)^{500}\left[\left(\frac{x}{5+x}\right)^{501}-1\right]}{\left(\frac{x}{5+x}\right)-1}=\frac{x^{501}-(5+x)^{501}}{-5}$$

$$=\frac{1}{5}(5+x)^{501}-\frac{x^{501}}{5}$$

Now coefficient of x^{101} in $\frac{1}{5}(5+x)^{50}$ will be

$$\frac{1}{5}\left(^{501}C_{101}\right)5^{501-101}=5^{399}\left(^{501}C_{101}\right)$$

Hence the correct answer is option 1.

Q. 8 The sum $1+2\cdot3+3\cdot3^2+.....+10\cdot3^9$ is equal to :

Option 1:
$\frac{2\cdot3^{12}+10}{4}$

Option 2:
$\frac{19\cdot3^{10}+1}{4}$

Option 3:
$5\cdot3^{10}-2$

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Option 4:

$\frac{9 \cdot 3^{10}+1}{2}$

Correct Answer:

$\frac{19 \cdot 3^{10}+1}{4}$

Solution:

$S_n=1+2 \cdot 3^1+3 \cdot 3^2+4 \cdot 3^3+\ldots .+10 \cdot 3^9$

$3 S_n=1 \cdot 3^1+2 \cdot 3^2+3 \cdot 3^2+\ldots . .+10 \cdot 3^{10}$

$-2 S_n=1+3+3^2+\ldots . .+3^9-10 \cdot 3^{10}$

$=\frac{1\left(3^{10}-1\right)}{3-1}-10 \cdot 3^{10}$

$=\frac{1}{2} \cdot 3^{10}-\frac{1}{2}-3^{10} \cdot 10$

$-2 S_n=\frac{-1}{2}-\frac{19}{2} \cdot 3^{10}$

$S_n=\frac{19}{4} \cdot 3^{10}+\frac{1}{4}$

Hence the correct answer is option 2

Q. 9 Let P be the plane passing through the intersectio of th planes $\tilde{r} \cdot(\hat{i}+3 \hat{j}-\hat{k})=5$ and $\tilde{r} \cdot(2 \hat{i}-\hat{j}+\hat{k})=3$, and the point $(2,1,-2)$. Let the position vectors of the points X and Y be $\hat{i}-2 \hat{j}+4 \hat{k}$ and $5 \hat{i}-\hat{j}+2 \hat{k}$ respectively. Then the points

Option 1:

X and $X+Y$ are on the same side of P

Option 2:

Y and $Y-X$ are on the opposite side of P

Option 3:

X and Y are on the opposite side of P

Option 4:

$X+Y$ and $X-Y$ are on the same side of P

Correct Answer:

X and Y are on the opposite side of P

Solution:

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$P_1 : \tilde{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) = 5$

$x + 3y - z = 5$

$P_2 : \tilde{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 3$

$2x - y + z = 3$

Equation of a plane passing through the intersection of P_1 and P_2

$P : \quad P_1 + kP_2 = 0$

$P : (x + 3y - z - 5) + k(2x - y + z - 3) = 0$

$\because$ P is passing through $(2, 1; -2)$

$\therefore P : (2 + 3 + 2 - 5) + k(4 - 1 - 2 - 3) = 0$

$\because \quad 2 - 2k = 0 \quad k = 1$

$3x + 2y - 8 = 0$

$X = (1, -2, 4) X = (1, -2, 4)$

$Y = (5, -1, 2)$

$\therefore$ X and Y are on opposite sides of P

Hence the correct answer is option 3.

Q. 10 A circle touches both the y-axis and the line $x + y = 0$. Then the locus of its center is :

Option 1:
$y = \sqrt{2x}$

Option 2:
$x = \sqrt{2y}$

Option 3:
$y^2 - x^2 = 2y$

Option 4:
$x^2 - y^2 = 2xy$

Correct Answer:
$x^2 - y^2 = 2xy$

Solution:

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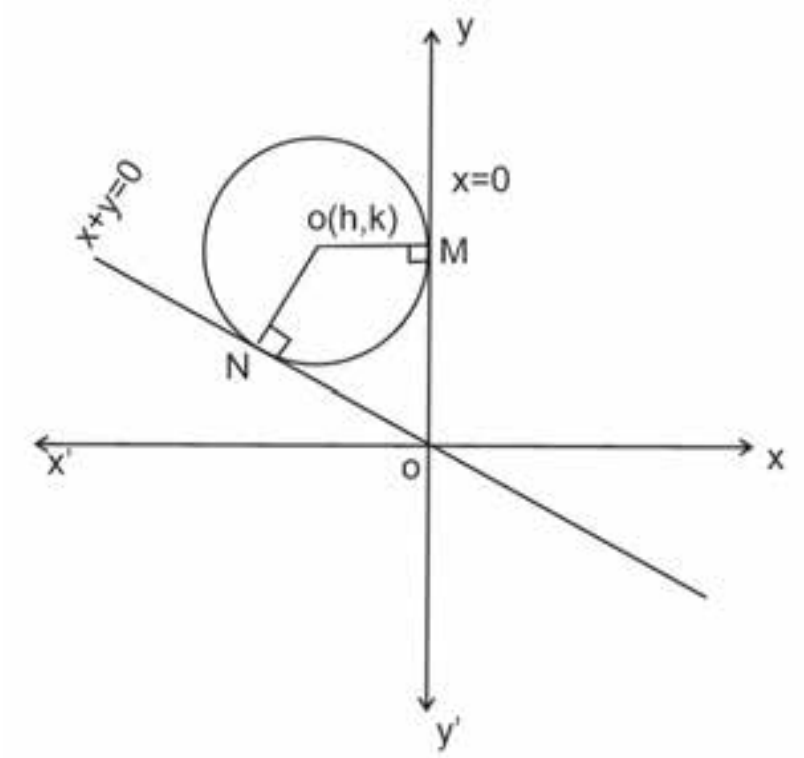

To find locus, let centre be h, k

$$|\mathrm{OM}| = |\mathrm{ON}| \text{distance}$$

$$\Rightarrow |\mathrm{h}| = \frac{|\mathrm{h}+\mathrm{k}|}{\sqrt{2}}$$

Squaring both side we get

$$\Rightarrow 2\mathrm{h}^2 = (\mathrm{h}+\mathrm{k})^2$$

$$\Rightarrow 2\mathrm{h}^2 = \mathrm{h}^2 + \mathrm{k}^2 + 2\mathrm{kh}$$

$$2\mathrm{x}^2 = \mathrm{x}^2 + \mathrm{y}^2 + 2\mathrm{xy}$$

$$\Rightarrow \mathrm{x}^2 - \mathrm{y}^2 - 2\mathrm{xy} = 0$$

Hence the correct answer is option 4

Q. 11 Water is being filled at the rate of $1\mathrm{cm}^3/\mathrm{sec}$ in a right circular conical vessel (vertex downwards) of height $35\mathrm{cm}$ and diameter $14\mathrm{cm}$. Whe the height of the water level is $10\mathrm{cm}$, the rate (in $\mathrm{cm}^2/\mathrm{sec}$) at which the wet coical surface area of the vessel increases is

Option 1:

5

Option 2:

$\frac{\sqrt{21}}{5}$

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Option 3:

$\frac{\sqrt{26}}{5}$

Option 4:

$\frac{\sqrt{26}}{10}$

Correct Answer:

$\frac{\sqrt{26}}{5}$

Solution:

Let the volume of the cone be $V\ cm^3$

Given $\frac{dV}{dt} = 1cm^3/sec, h = 35cm, r = 7cm$

i.e $\frac{h}{r} = 5$

We know for a cone $l^2 = r^2 + h$

Lateral Surface area,

$S = \pi r\sqrt{r^2 + h^2}$

$S = \pi\frac{h}{5}\sqrt{\frac{h^2}{25} + h^2} = \pi\frac{\sqrt{26}}{25}h^2$

$V = \frac{1}{3}\pi r^2 h\frac{1}{3}\pi\left(\frac{h}{5}\right)^2 h = \frac{\pi}{75}h^3$

$\Rightarrow \frac{dV}{dt} = \frac{\pi}{25}h^2\frac{dh}{dt} \Rightarrow \frac{\pi}{25}h^2\frac{dh}{dt} = 1$

$\Rightarrow \frac{dh}{dt} = \frac{25}{\pi h^2}$ .....(1)

$\frac{ds}{dt} = \frac{\pi\sqrt{26}}{25} \times 2h\frac{dh}{dt} = \frac{2\sqrt{26}}{h}$ .....from eqn(1)

$\left(\frac{ds}{dt}\right)_{h=10} = \frac{\sqrt{26}}{5}$

Hence the correct answer is option 3

Q. 12 If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x}dx, n \in \mathbb{N}$, then

Option 1:

$b_3-b_2, b_4-b_3, b_5-b_4$ are in an A.P. with common difference -2

Option 2:

$\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P with common difference 2

Option 3:

$b_3-b_2, b_4-b_3, b_5-b_4$ are in a G.P

Option 4:

$\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P with common difference -2

Correct Answer:

$\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P with common difference -2

Solution:

$$b_n-b_{n-1}=\int_0^{\pi/2}\frac{\cos^2 nx-\cos^2(n-1)x}{\sin x}$$

$$=\int_0^{\pi/2}\frac{\sin(2n-1)x\sin(-x)}{\sin x}$$

$$=-\int_{\pi}^{\pi/2}\sin(2n-1)xdx$$

$$=\frac{1}{2n-1}\left[\cos(2n-1)x\right]_0^{\frac{\pi}{2}}=-\frac{1}{2n-1}$$

$$\frac{1}{b_3-b_2}=-5, \frac{1}{b_4-b_3}=-7, \frac{1}{b_5-b_4}=-9$$

Common Difference $=-2$

Hence the correct answer is option 4

Q. 13 If $y=y(x)$ is the solution of the dieffrential equation $2x^2\frac{dy}{dx}-2xy+3y^2=0$ such that $y(e)=\frac{e}{3}$, then $y(1)$ is equal to

Option 1:

$\frac{1}{3}$

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Option 2:

$\frac{2}{3}$

Option 3:

$\frac{3}{2}$

Option 4:

3

Correct Answer:

$\frac{2}{3}$

Solution:

$2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$

$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{xy} + \frac{3}{2x^2} = 0 \quad \text{Let } \frac{1}{y} = t$

$-\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$-\frac{dt}{dx} - \frac{1}{x}t + \frac{3}{2x^2} = 0$

$\frac{dt}{dx} + \frac{1}{x}t = \frac{3}{2x^2}$

If $= e^{\int \frac{1}{x}dx} = x$

$x \cdot t = \int x \cdot \frac{3}{2x^2}dx$

$x \cdot t = \frac{3}{2}(\ln x) + c$

$\frac{x}{y} = \frac{3}{2}\ln x + c$

$x = e, y = \frac{e}{3}$

$\frac{e}{e/3} = \frac{3}{2} \cdot 1 + c$

$\Rightarrow c = \frac{3}{2}$

$\frac{x}{y} = \frac{3}{2}\ln x + \frac{3}{2}$

$\Rightarrow x = 1$

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$$\Rightarrow \frac{1}{y} = \frac{2}{3}$$

$$y = \frac{2}{3} \qquad \text{.....} \because y = y(x)$$

$$y = y(1)$$

Hence the correct answer is option (2)

Q. 14 If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12\,(t + \sin t \cos t)$, $y = 12\,(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to :

Option 1:
$6\left(3 + 2\sqrt{2}\right)$

Option 2:
$3\left(7 + 4\sqrt{3}\right)$

Option 3:
27

Option 4:
48

Correct Answer:
27

Solution:
Given,

Curve $x = 12\,(t + \sin t \cos t)$

$$y = 12\,(1 + \sin t)^2$$

Now $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$$\tan(\pi/3) = \frac{\frac{d}{dt}\left(12(1 + \sin t)^2\right)}{\frac{d}{dt}(12(t + \sin t \cos t))}$$

$$\sqrt{3} = \frac{12 \times 2(1 + \sin t)\cos t}{12(1 + \cos 2t)}$$

$$\sqrt{3} = \frac{2(1 + \sin t)\cos t}{2\cos^2 t}$$

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$$\sqrt{3}=\frac{1+\sin t}{\cos t}=\frac{(\cos t/2+\sin t/2)^2}{\left(\cos^2 t/2-\sin^2 t/2\right)}$$

$$\sqrt{3}=\tan(t/2+\pi/4)$$

$$\pi/3=\frac{t}{2}+\frac{\pi}{4}\Rightarrow\frac{t}{2}=\frac{\pi}{12}\Rightarrow t=\pi/6$$

Now putting $t=\frac{\pi}{6}$ in $y=12\,(1+\sin t)^2$ as (x_0,y_0) lie on curve

So $y_0=12\left(1+\sin\frac{\pi}{6}\right)^2$

$$=12\left(\frac{3}{2}\right)^2=27$$

Hence the correct answer is option 3

Q. 15 The value of $2\sin(12^\circ)-\sin(72^\circ)$ is :

Option 1:
$\frac{\sqrt{5}\left(1-\sqrt{3}\right)}{4}$

Option 2:
$\frac{1-\sqrt{5}}{8}$

Option 3:
$\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{2}$

Option 4:
$\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{4}$

Correct Answer:
$\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{4}$

Solution:
To find

$\Rightarrow 2\sin 12^\circ-\sin 72^\circ$

$\Rightarrow \sin 12^\circ+\sin 12^\circ-\sin 72^\circ$

$\Rightarrow \sin 12^\circ+2\cos 42^\circ\sin(-30^\circ)\Rightarrow \sin 12^\circ-\cos 42^\circ$

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$\Rightarrow \sin 12^\circ - \sin 48^\circ$

$\Rightarrow 2\cos(30^\circ)\sin(-18^\circ)$

$\Rightarrow -\sqrt{3}\sin 18^\circ$

$\Rightarrow -\sqrt{3}\left(\frac{\sqrt{5}-1}{4}\right) = \sqrt{3}\left(\frac{1-\sqrt{5}}{4}\right)$

Hence the correct answer is option 4

Q. 16 A biased die is marked with numbers $2, 4, 8, 16, 32, 32$ on its faces and the probability of getting a face with mark n is $\frac{1}{n}$. If the die is thrown thrice, the the probability,that the sum of the numbers obtained is 48 is :

Option 1:

$\frac{7}{2^{11}}$

Option 2:

$\frac{7}{2^{12}}$

Option 3:

$\frac{3}{2^{10}}$

Option 4:

$\frac{13}{2^{12}}$

Correct Answer:

$\frac{13}{2^{12}}$

Solution:

$P(2) = \frac{1}{2}, P(4) = \frac{1}{4}, P(8) = \frac{1}{8}, P(16) = \frac{1}{16}, P(32) = \frac{1}{32}$.....

When we get the faces as $8, 8, 32$ the the probability will be $3\left(\frac{1}{8} \times \frac{1}{8} \times \frac{2}{32}\right) = \frac{3}{1024}$

Whe we get the faces as$16, 16, 16$ then the proability will be $\frac{1}{16} \times \frac{1}{16} \times \frac{1}{16} = \frac{1}{4096}$

Hece, the required proaility will be

$\frac{3}{1024} + \frac{1}{4096} = \frac{13}{4096} = \frac{13}{2^{12}}$

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Hence the correct answer is option 4

Q. 17 The negation of the Boolean expression $((\sim q) \wedge p) \Rightarrow ((\sim p) \vee q)$ is logically equivalent to :

Option 1:
$p \Rightarrow q$

Option 2:
$q \Rightarrow p$

Option 3:
$\sim (p \Rightarrow q)$

Option 4:
$\sim (q \Rightarrow p)$

Correct Answer:
$\sim (p \Rightarrow q)$

Solution:
To find negation of $((\sim q \wedge p) \rightarrow ((\sim p) \vee q)$

Let $\sim q \wedge p = A \& \sim p \forall q = B$

$$\sim (A \rightarrow B) = \sim (\sim A \vee B) = A \wedge \sim B.$$

Using abve formulae we get

$$(\sim q \wedge p) \wedge \sim (\sim p \vee q) = (\sim q \wedge p) \wedge (p \wedge \sim q)$$

$$= A \wedge A = A = \sim q \wedge p = p \wedge \sim q \text{ or } \sim (p \rightarrow q)$$

Hence the correct answer is option 3

Q. 18 If the line $y = 4 + kx, k > 0$, is the tangent to the parabola $y = x - x^2$ at the point P and V is the vertex of the parabola,then the slope of the line through P and V is :

Option 1:
$\frac{3}{2}$

Option 2:
$\frac{26}{9}$

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Option 3:

$\frac{5}{2}$

Option 4:

$\frac{23}{6}$

Correct Answer:

$\frac{5}{2}$

Solution:

$y = kx + 4$

$y = x - x^2$

$kx + 4 = x - x^2$ .....(1)

$x^2 + (k-1)x + 4 = 0$

$(k-1)^2 - 4 \cdot 4 = 0$

$k - 1 = \pm 4$

if $k = 5$

Now put the value of $k = 5$

$5x + 4 = x - x^2$

$x^2 + 4x + 4 = 0$

$(x+2)^2 = 0$

$x = -2$

$y = -6$

if $k = -3$

Now put the value of $k = -3$ in eqn (1)

$-3x + 4 = x - x^2$

$x^2 - 4x + 4 = 0$

$x = 2 \quad y = -2$

Then the point of P is $(2, -2)$ and $(-2, -6)$ and vertex of parabola $'O' = y - \frac{1}{4} = -\frac{1}{4} + x - x^2$

$y - \frac{1}{4} = -\left(x - \frac{1}{2}\right)^2$

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Point P is $(2, 2)$

Slope of $\mathrm{OP} = \dfrac{-2 - \frac{1}{4}}{2 - \frac{1}{2}} = \dfrac{-3}{2}$

Point P is $(-2, -6)$ slope of $\mathrm{OP} = \dfrac{-6 - \frac{1}{4}}{-2 - \frac{1}{2}} = \dfrac{5}{2}$

Hence the correct answer is option 3

Q. 19 The value of $\tan^{-1}\left(\dfrac{\cos\left(\frac{15\pi}{4}\right) - 1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to :

Option 1:

$\dfrac{-\pi}{4}$

Option 2:

$\dfrac{-\pi}{8}$

Option 3:

$\dfrac{-5\pi}{12}$

Option 4:

$\dfrac{-4\pi}{9}$

Correct Answer:

$\dfrac{-\pi}{8}$

Solution:
Given,

$$\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right) - 1}{\sin\frac{\pi}{4}}\right)$$

$$= \tan^{-1}\left(\frac{\cos\left(4\pi - \frac{\pi}{4}\right) - 1}{\sin\frac{\pi}{4}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{\sqrt{2}} - 1}{1/\sqrt{2}}\right)$$

$$= \tan^{-1}(1 - \sqrt{2}) = -\tan -1(\sqrt{2} - 1)$$

$$= \frac{-\pi}{8}$$

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Hence the correct answer is option 2

Q. 20 The line $y = x + 1$ meets the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to :

Option 1:
20

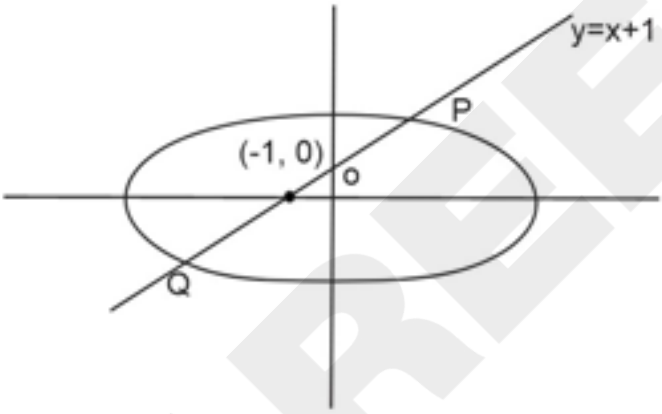
Option 2:
12

Option 3:
11

Option 4:
8

Correct Answer:
20

Solution:



Let P be the point $\left(-1 + \frac{p}{\sqrt{2}}, \frac{p}{\sqrt{2}}\right)$

Since it lies o the point $\frac{x^2}{4} + \frac{y^2}{2} = 1$ we get

$$\frac{\left(-1 + \frac{p}{\sqrt{2}}\right)^2}{4} + \frac{\left(\frac{p}{\sqrt{2}}\right)^2}{2} = 1 \Rightarrow \frac{3p^2}{8} - \frac{p}{2\sqrt{2}} - \frac{3}{4} = 0$$

Here $(P_1 - P_2)^2 = \left(\frac{2\sqrt{2}}{3}\right)^2 - 4(-2) = \frac{8}{9} + 8 = \frac{80}{9}$

We know, $(P_1 - P_2)^2 = PQ^2$

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$\Rightarrow PQ^2 = \frac{80}{9}$

Now $r = \frac{PQ}{2} \Rightarrow 9r^2 = 9\frac{(PQ)^2}{4} = \frac{9}{4} \times \frac{80}{9} = 20$

Hence the correct answer is option 4.

Q. 21 Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set $\{(n, m) : n, m \in \{1, 2, \ldots\ldots, 10\}$ and $nA^n + mB^m = I\}$ is _____.

Correct Answer:
One set

Solution:
$A^2 = A$ and $B^2 = B$

Therefore equation $nA^n + mB^m = I$ becomes $nA + mB = I$, which gives $m = n = I$

So, only one set is possible.

Hence answer is one set.

Q. 22 Let $f(x) = [2x^2 + 1]$ and $g(x) = \begin{cases} 2x - 3, & x < 0 \\ 2x + 3, & x \geq 0 \end{cases}$, where $[t]$ is the greatest integer $\leq t$. Then, in the open interval $(-1, 1)$,the number of points where fog is discontinuous is equal to _____.

Correct Answer:
62

Solution:
$f(g(x)) = [2g^2(x)] + 1$

$= \begin{cases} [2(2x-3)^2] + 1, x < 0 \\ [2(2x+3)^2] + 1, x \geq 0 \end{cases}$

$\therefore$ Fog is discontinuous whenever $2(2x - 3)^2$ or $2(2x + 3)^2$ belongs to integer except $x = 0$

$\therefore 62$ points of discontinuity.

Hence the answer is 62.

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Q. 23 The value of $b > 3$ for which $12\int_3^b \frac{1}{(x^2-1)(x^2-4)}dx = \log_e\left(\frac{49}{40}\right)$, is equal to _____.

Correct Answer:
b=6

Solution:
Given

$$12 \times \int_3^b \frac{dx}{(x^2-4)(x^2-1)} = \ln\left[\frac{49}{40}\right] \ \& \ b > 3$$

By using Partial fraction

$$\frac{1}{(x^2-4)(x^2-1)} = \frac{1}{3}\left(\frac{1}{x^2-4} - \frac{1}{x^2-1}\right)$$

$$\Rightarrow \frac{12}{3}\int_3^b \left[\frac{1}{x^2-4} - \frac{1}{x^2-1}\right] = \ln\left(\frac{49}{40}\right)$$

$$\Rightarrow 4\left[\frac{1}{4}\log\left(\frac{x-2}{x+2}\right) - \frac{1}{2}\log\left(\frac{x-1}{x+1}\right)\right]_3^b = \ln\left(\frac{49}{40}\right)$$

$$\Rightarrow \log\left(\frac{(b-2)(b+1)^2}{(b+2)(b-1)^2} \times \frac{5}{4}\right) = \log\left(\frac{49}{40}\right)$$

$$\Rightarrow \frac{(b-2)(b+1)^2}{(b+2)(b+1)^2} \times \frac{5}{4} = \frac{49}{40}$$

$$\Rightarrow \frac{(b-2)(b+1)^2}{(b+2)(b-1)^2} = \frac{49}{50} \quad \text{.....(1)}$$

Comparing by hit and trial taking

$$(b+1)^2 = 49$$

$$\Rightarrow b+1 = 7$$

$$b = 6$$

Now putting $b = 6$ in eqn (1) cross check we get

$\frac{4 \times 7^2}{8 \times 5^2} = \frac{49}{50}$ which is true

So $b = 6$ is correct answer.

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Q. 24 If the sum of the co-efficients of all the positive even powers of x in the binomial expansion of $\left(2x^3+\frac{3}{x}\right)^{10}$ is $5^{10}-\beta\cdot 3^9$, then β is equal to _____

Correct Answer:
83

Solution:

$$T_{r+1}={}^{10}C_r\left(2x^3\right)^{10-r}\cdot\left(\frac{3}{x}\right)^r$$

$$={}^{10}C_r 2^{10-r}\cdot 3^r\cdot x^{30-4r}$$

Put $r=0,1;2,.....7$ and we get

$$\beta=83$$

Q. 25 If the maean deviation about the mean of the numbers $1,2,3,.....,n$, where n is odd,is $\frac{5\,(n+1)}{n}$, then n is equal to _____.

Correct Answer:
21

Solution:

$$\bar{x}=\frac{1+2+3+\cdots+n}{n}=\frac{\frac{n(n+1)}{2}}{n}=\left(\frac{n+1}{2}\right)$$

.....(Which is middle term)

$$M.D=\frac{\sum|x_i-\bar{x}|}{n}=\frac{5(n+1)}{n}$$

$$\Rightarrow\frac{2\left(1+2+3+\cdots+\frac{(n-1)}{2}\right)}{n}=\frac{5(n+1)}{n}$$

$$\Rightarrow\frac{2\left(\frac{(n-1)}{2}\left(\frac{n-1}{2}+1\right)\right)}{2}=\frac{5\,(n+1)}{n}$$

$$\Rightarrow\frac{(n-1)(n+1)}{4n}=\frac{5(n+1)}{n}$$

$$n=21$$

Hence the correct answer is $n=21$

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Q. 26 Let $\tilde{b} = \hat{i} + \hat{j} + \lambda\hat{k}, \lambda \in \mathbb{R}$. If $\tilde{a}$ is a vector such that $\tilde{a} \times \tilde{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\tilde{a} \cdot \tilde{b} + 21 = 0$, then $(\tilde{b} - \tilde{a}) \cdot (\hat{k} - \hat{j}) + (\tilde{b} + \tilde{a}) \cdot (\hat{i} - \hat{k})$ is equal to _____.

Correct Answer:
14

Solution:
$(\tilde{a} \times \tilde{b}) \cdot \tilde{b} = 0$

$\Rightarrow 13 - 1 - 4\lambda = 0 \Rightarrow \lambda = 3$

$\Rightarrow \tilde{b} = \hat{i} + \hat{j} + 3\hat{k} \Rightarrow \tilde{a} \times \tilde{b} = 13i - \hat{j} - 4\hat{k}$

$\Rightarrow (\tilde{a} \times \tilde{b}) \times \tilde{b} = (13i - \hat{j} - 4\hat{k}) \times (\hat{i} + \hat{j} + 3\hat{k})$

$\Rightarrow -21\tilde{b} - 11\tilde{a} = \hat{i} - 43\hat{j} + 14\hat{k}$

$\Rightarrow \tilde{a} = -2\hat{i} + 2\hat{j} - 7\hat{k}$

$\text{Now } (\tilde{b} - \tilde{a}) \cdot (\hat{k} - \hat{j}) + (\tilde{b} + \tilde{a}) \cdot (\hat{i} - \hat{k}) = 14$

Q. 27 The total number of three digit numbers,with one digit repeated exactly two times, is _____

Correct Answer:
243

Solution:
since three digit number can not start with 0 at hundred's place, therefore number of ways to write it will be = $9_{c_1} \times 2$

later, for remaining non zero digits taken two at a time can be written in $9_{c_1} \times 1$ ways

now, the other ways of making three digit numbers can be written in $9_{c_2} \times 2_{c_1}$ and finally as there is repetition of two numbers in three digit number the total possible ways will be multiplied by $\frac{3!}{2!}$

$0\text{aa} + 100\text{a} + \text{aab or abb}$

$9_{c_1} \times 2 + 9_{c_1} \times 1 + 9_{c_2} \times 2_{c_1} \times \frac{3!}{2!}$

$18 + 9 + 216$

$= 243$

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Hence the answer is 243

Q. 28 Let $f(x)=\left|(x-1)\left(x^2-2x-3\right)\right|+x-3, x \in \mathbb{R}$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval $(0,4)$, then $m+M$ is equal to _____.

Correct Answer:
3

Solution:

$$f(x)=\begin{cases}(x^2-1)(x-3)+(x-3), x \in (0,1] \cup [3,4) \\ -(x^2-1)(x-3)+(x-3), x \in [1,3]\end{cases}$$

$$f'(x)=\begin{cases}3x^2-6x, x \in (0,1) \cup (3,4) \\ -3x^2+6x+2; x \in (1,3)\end{cases}$$

$f(x)$ is non-derivable at $x=1$ and $x=3$

Also $f'(x)=0$, at $x=1+\sqrt{\frac{5}{3}}$

$\Rightarrow m+M=3$

Hence the correct answer is 3

Q. 29 Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x+\beta y=\lambda$, then $\lambda-\beta$ is equal to _____.

Correct Answer:
85

Solution:
Given

$$s: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1, e=\frac{5}{4}$$

$$\because b^2=a^2\left(e^2-1\right)$$

$$b^2=a^2\left(\frac{25}{26}-1\right)$$

$$b^2=a^2\left(\frac{9}{16}\right) \text{.....(1)}$$

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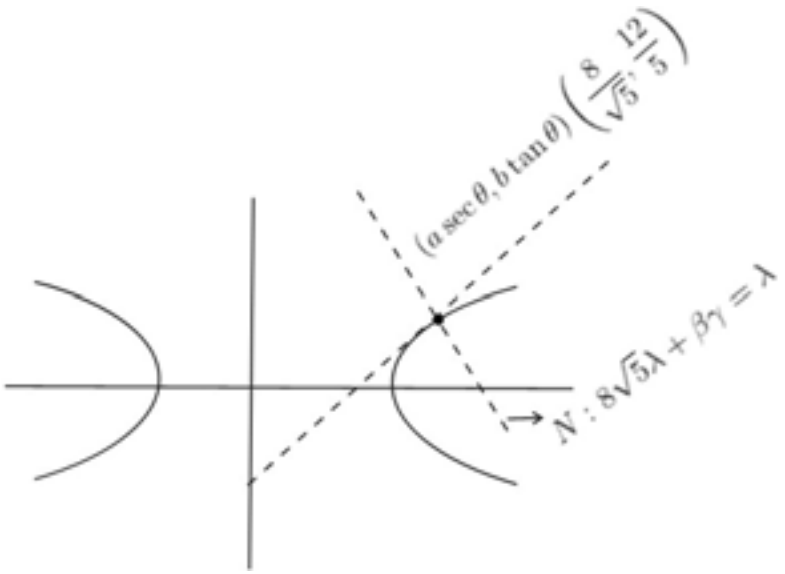

$$a\sec\theta = \frac{8}{\sqrt{5}}$$

$$\sec^2\theta = \frac{64}{5a^2}$$

$$b\tan\theta = \frac{12}{5} \qquad \ldots\ldots\because b^2 = a^2\left(\frac{9}{16}\right)$$

$$\tan^2\theta = \frac{144}{25b^2}$$

$$\tan^2\theta = \frac{144}{25} \times \frac{16}{9a^2}$$

$$\tan^2\theta = \frac{256}{25a^2}$$

$$\because \sec^2 a - \tan^2 a = 1$$

$$\frac{64}{5a^2} - \frac{25}{25a^2} = 1$$

$$\frac{320 - 256}{25a^2} = 1 \Rightarrow a^2 = \left(\frac{64}{25}\right)$$

$$\because b^2 = a^2\left(\frac{9}{16}\right) \Rightarrow b^2 = \frac{64}{25} \times \frac{9}{16} = \frac{36}{25}$$

The equation of the normal to the hyperola at any point $(x_1 y_1)$ lying on it is

$$\frac{x - x_1}{x_{1/a^2}} + \frac{y - y_1}{y_1/b^2} = 0$$

$$\frac{x - 8/\sqrt{5}}{\frac{8}{\sqrt{5}}/\frac{64}{25}} + \frac{y - 12/5}{\frac{12}{5}/\frac{36}{25}} = 0$$

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$$\frac{x-\frac{8}{\sqrt{5}}}{\frac{5\sqrt{5}}{8}}+\frac{y-12/5}{5/3}=0$$

$$\frac{8}{5\sqrt{5}}\left[x-\frac{8}{\sqrt{5}}\right]+\frac{3}{5}\left[y-\frac{12}{5}\right]=0$$

$$\frac{8}{5\sqrt{5}}x-\frac{64}{25}+\frac{3}{5}y-\frac{36}{25}=0$$

$$\frac{8\sqrt{5}}{25}x-\frac{64}{25}+\frac{3y}{5}-\frac{36}{25}=0$$

$$\frac{8\sqrt{5}x+15y-64-36}{25}=0$$

Ni :$8\sqrt{5}x+15y-100=0$ .....(1)

N : $8\sqrt{5}x+\beta y=\lambda$

so $\beta=15 \quad \lambda=100$

$\lambda-\beta=100-15=85$

Hence the answer is 85

Q. 30 Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____.

Correct Answer:
51

Solution:
$8x+4\sqrt{2}y=1, z=0$

$$\Rightarrow \frac{x-\frac{1}{8}}{1}=\frac{y-0}{-\sqrt{2}}=\frac{z-0}{0}=\lambda$$

$-8x-6\sqrt{3}z=1, \quad y=0$

$$\Rightarrow \frac{x+\frac{1}{8}}{3\sqrt{3}}=\frac{y-0}{0}=\frac{z-0}{-4}$$

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$$\begin{vmatrix} \frac{1}{4} & 0 & 0 \\ 1 & -\sqrt{2} & 0 \\ 3\sqrt{3} & 0 & -4 \end{vmatrix}$$

$$\Rightarrow \frac{1}{4}\left(4\sqrt{2}-0\right)=\sqrt{2}$$

$$\mathrm{d}=\frac{1}{\sqrt{51}}$$

$$\frac{1}{\mathrm{d}^2}=51$$

Hence the correct answer is 51

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Online 26th June Morning Shift

Physics

Q. 1 An expression for a dimensionless quantity P is given by $P = \frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be:

Option 1:
$[M^0\ L^{-1}\ T^0]$

Option 2:
$[ML^0\ T^{-2}]$

Option 3:
$[MLT^{-2}]$

Option 4:
$[ML^2\ T^{-2}]$

Correct Answer:
$[MLT^{-2}]$

Solution:

$$P = \left(\frac{\alpha}{\beta}\right)\log\left(\frac{kt}{\beta x}\right)$$

$$[P] = \left[\frac{\alpha}{\beta}\right]\left[\log_e\left(\frac{kt}{\beta x}\right)\right] = [M^0L^0T^0]$$

$$\because [\log_e a] = [M^0L^0T^0] = [a]$$

$$\therefore \left[\frac{\alpha}{\beta}\right] = \left[\log_e\left(\frac{kt}{\beta x}\right)\right] = [M^0L^0T^0]$$

$$[\alpha] = [\beta] \rightarrow (1)$$

$$\left[\frac{kt}{\beta x}\right] = [M^0L^0T^0]$$

$$[Kt] = [\text{ Energy }] = [M^1L^2T^{-2}]$$

$$[\beta] = \left[\frac{Kt}{x}\right] = [M^1L^1T^{-2}]$$

From eqn (1)

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$[\alpha] \doteq [\beta] = [\mathrm{M^1L^1T^{-2}}]$

The correct option is (3)

Q. 2 A person is standing in an elevator. In which situation, he experiences weight loss?

Option 1:
When the elevator moves upward with constant acceleration

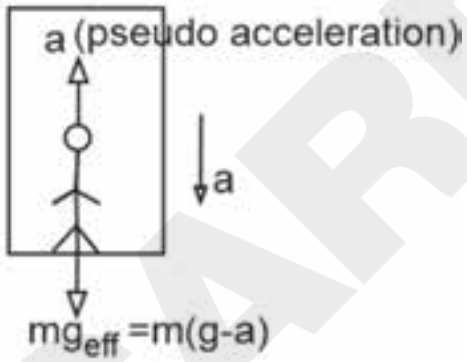
Option 2:
When the elevator moves downward with constant acceleration

Option 3:
When the elevator moves upward with uniform velocity

Option 4:
When the elevator moves downward with uniform velocity

Correct Answer:
When the elevator moves downward with constant acceleration

Solution:



Apparent Wt. $= \mathrm{mg_{eff}} = \mathrm{m\,(g - a)}$
Person will experience weight loss when the lift is accelerating downwards

The correct option is (2)

Q. 3 An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero ?

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Option 1:
Momentum

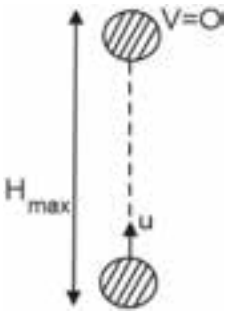
Option 2:
Potential Energy

Option 3:
Acceleration

Option 4:
Force

Correct Answer:
Momentum

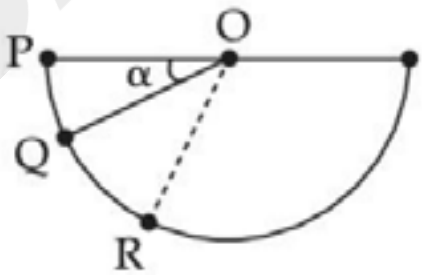
Solution:



→ At maximum height, its velocity becomes zero
→ Therefore , momentum of the object becomes zero

The correct option is (1)

Q. 4 A ball is released from rest from point P of a smooth semi-spherical vessel as shown in figure. The ratio of the centripetal force and normal reaction on the ball at point Q is A while angular position of point Q is α with respect to point P. Which of the following graphs represent the correct relation between A and α when ball goes from Q to R ?



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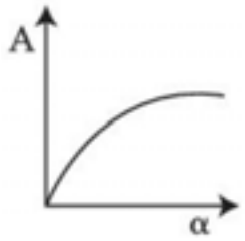
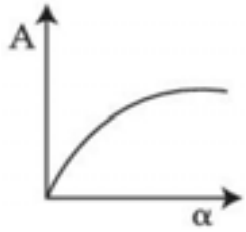
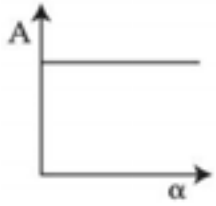
Option 1:



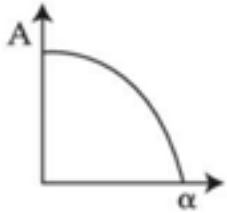
Option 2:



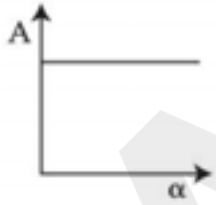
Option 3:



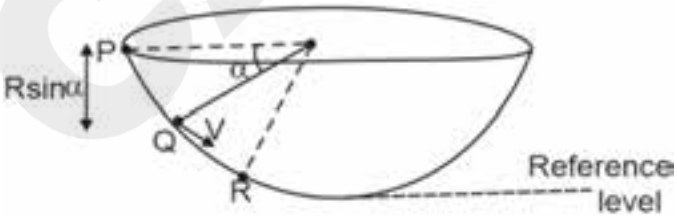
Option 4:



Correct Answer:



Solution:



Let the speed of ball at pt. Q be V

By energy conservation,

$$TE_p = TE_Q$$

$$mgR + 0 = \frac{1}{2}mv^2 + mg\,(R - R\sin\alpha)$$

$$\frac{1}{2}mv^2 = mgR\,\sin\alpha$$

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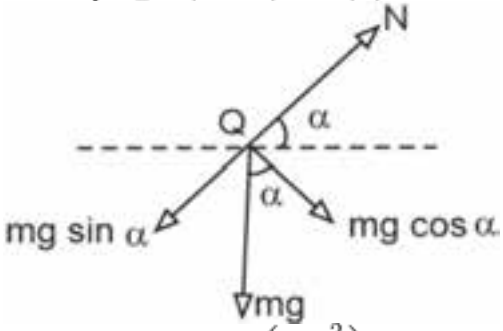
$$V = \sqrt{2gR\,(\sin\alpha)} \rightarrow (1)$$



$$N - mg\sin\alpha = \left(\frac{mv^2}{R}\right) = F_c$$

$$N - mg\sin\alpha = 2\,mg\sin\alpha \ \ (\text{from eqn } 1)$$

$$N = 3\,mg\sin\alpha \rightarrow (2)$$

$$\frac{F_c}{N} = A = \frac{2\,mg\sin\alpha}{3\,mg\sin\alpha} = \frac{2}{3} \rightarrow (3)$$

A is independent of α

$\therefore$ The correct option is (3)

Q. 5 A thin circular ring of mass M and radius R is rotating with a constant angular velocity $2\ rads^{-1}$ in a horizontal plane about an axis vertical to its plane and passing through the center of the ring. If two objects each of mass m be attached gently to the opposite ends of a diameter of ring, the ring will then rotate with an angular velocity (in $rads^{-1}$).

Option 1:

$\frac{M}{(M+m)}$

Option 2:

$\frac{(M+2m)}{2M}$

Option 3:

$\frac{2M}{(M+2m)}$

Option 4:

$\frac{2(M+2m)}{M}$

Correct Answer:

$\frac{2M}{(M+2m)}$

Solution:

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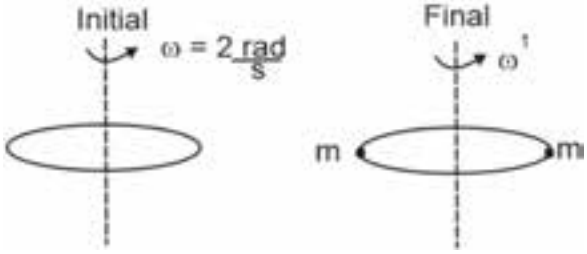


$L_i = I\omega$ $\qquad$ $L_f = (MR^2 + 2MR^2)\omega^1$

$= MR^2\omega$

By angular momentum conservation,

$L_i = L_f$

$MR^2\omega = \left(MR^2 + 2mR^2\right)\omega'$

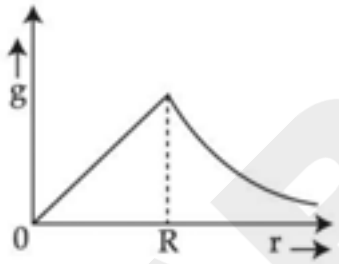
$\omega' = \dfrac{M\omega}{(M + 2m)}$

$\therefore \omega' = \dfrac{2M}{(M + 2m)}$

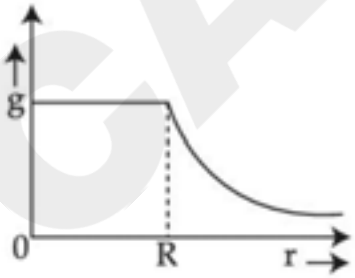
The correct option is (3)

Q. 6 The variation of acceleration due to gravity (g) with distance (r) from the center of the earth is correctly represented by :
(Given R= radius of earth)

Option 1:



Option 2:



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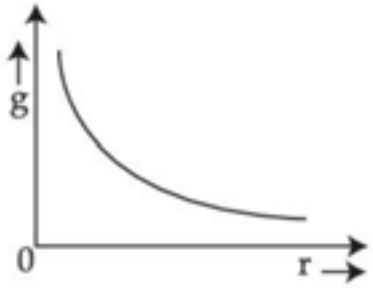
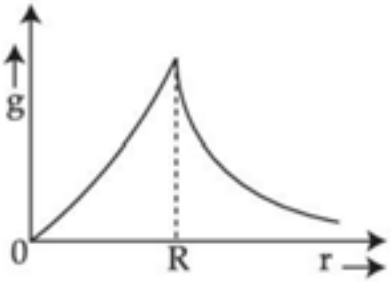
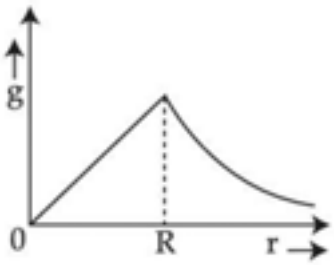
Option 3:



Option 4:



Correct Answer:

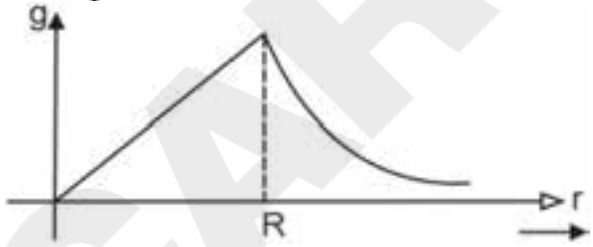


Solution:

Acceleration due to gravity from the centre of earth,

$$g = \frac{GM}{R^3}(r) \quad (r \leq R)$$

$$g = \frac{GM}{r^2} \quad (r > R)$$



The correct option is (1)

Q. 7 The efficiency of a Carnot's engine, working between steam point and ice point, will be :

Option 1:

26.81%

Option 2:
37.81%

Option 3:
47.81%

Option 4:
57.81%

Correct Answer:
26.81%

Solution:
For carnot engine ,

$$\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{273}{373}$$
$$= \frac{100}{373} = 0.2681$$
$$\eta\% = 26.81\%$$

The correct option is (1)

Q. 8 Time period of a simple pendulum in a stationary lift is $'T'$. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be :
(Where $g =$ acceleration due to gravity)

Option 1:
$\sqrt{\frac{6}{5}}\ T$

Option 2:
$\sqrt{\frac{5}{6}}\ T$

Option 3:
$\sqrt{\frac{6}{7}}\ T$

Option 4:
$\sqrt{\frac{7}{6}}\ T$

Correct Answer:
$\sqrt{\frac{6}{7}}\ T$

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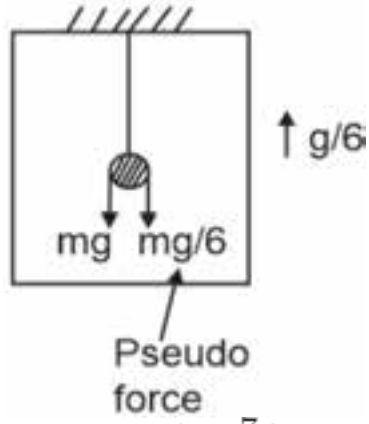
Solution:



$$g_{eff} = g + \frac{g}{6} = \frac{7g}{6}$$

$$T = 2\pi\sqrt{\frac{l}{g}} \quad (\text{Stationary Lift}) T' = 2\pi\sqrt{\frac{l}{g_{eff}}} \quad (\text{Moving Lift})$$

$$\Rightarrow T' = 2\pi\sqrt{\frac{6l}{7g}} = \sqrt{\frac{6}{7}}\,T$$

The correct option is (3)

Q. 9 A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel temperature of the gas increases by :
(R= universal gas constant)

Option 1:
$\frac{Mv^2}{7R}$

Option 2:
$\frac{Mv^2}{5R}$

Option 3:
$2\frac{Mv^2}{7R}$

Option 4:
$7\frac{Mv^2}{5R}$

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Correct Answer:

$$\frac{Mv^2}{5\mathrm{R}}$$

Solution:

$\vartheta = 1.4$

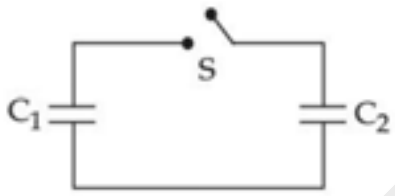
KE lost by vessel $=$ Increase in internal energy of gas

$$\frac{1}{2}\mathrm{mV}^2 = \frac{\mathrm{nR}\Delta\mathrm{T}}{\vartheta - 1} = \left(\frac{\mathrm{m}}{\mathrm{M}}\right)\frac{\mathrm{R}\,\Delta\mathrm{T}}{(0.4)}$$

$$\frac{\mathrm{MV}^2}{5\mathrm{R}} = \Delta\mathrm{T}$$

The correct option is (2)

Q. 10 Two capacitors having capacitance C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S. The amount of charge on the capacitor C_2, after equilibrium, is :



Option 1:

$$\frac{C_1C_2}{(C_1+C_2)}V$$

Option 2:

$$\frac{(C_1+C_2)}{C_1C_2}V$$

Option 3:

$$(C_1+C_2)\,V$$

Option 4:

$$(C_1-C_2)\,V$$

Correct Answer:

$$\frac{C_1C_2}{(C_1+C_2)}V$$

Solution:

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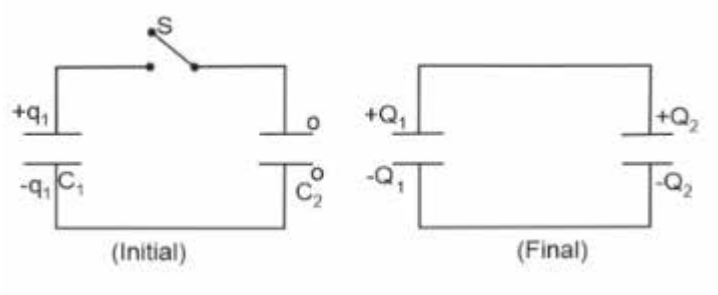

By charge conservation,

$$q_1 + 0 = Q_1 + Q_2$$
$$q_1 = Q_1 + Q_2 \rightarrow (1)$$

At steady state (equilibrium)

$$\text{Common Potential} = V_0 = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$
$$V_0 = \frac{C_1V}{C_1 + C_2}$$

$$Q_1 = C_1V_0 \quad \Big| \quad Q_2 = C_2V_0$$
$$Q_1 = \frac{C_1^2V}{C_1+C_2} \quad \Big| \quad Q_2 = \frac{C_1C_2V}{C_1+C_2}$$

The correct option is (1)

Q. 11 **Given below two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).**

Assertion (A) : Non-polar materials do not have any permanent dipole moment.

Reason (R): When a non-polar material is placed in an electric field, the centre of the positive charge distribution of it's individual atom or molecule coincides with the centre of the negative charge distribution.

In the light of above statements, choose the most appropriate answer from the options given below.

Option 1:
Both (A) and (R) are correct and (R) is the correct explanation of (A).

Option 2:
Both (A) and (R) are correct and (R) is not the correct explanation of (A).

Option 3:
(A) is correct but (R) is not correct.

Option 4:
(A) is not correct but (R) is correct.

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Correct Answer:
(A) is correct but (R) is not correct.

Solution:
In the absence of electric field ,the centre of the positive charge distribution of it's individual atom or molecule coincides with the centre of negative chatge distribution

$\rightarrow$ Therefore it has dipole moment only in the presence of electric field.

The correct option is (3)

Q. 12 The magnetic flux through a coil perpendicular to its plane is varying according to the relation $\phi = \left(5\mathrm{t}^3 + 4\mathrm{t}^2 + 2\mathrm{t} - 5\right)$ Weber. If the resistance of the coil is $5\mathrm{ohm}$, then the induced current through the coil at $t = 2s$ will be,

Option 1:
$15.6\ \mathrm{A}$

Option 2:
$16.6\ \mathrm{A}$

Option 3:
$17.6\ \mathrm{A}$

Option 4:
$18.6\ \mathrm{A}$

Correct Answer:
$15.6\ \mathrm{A}$

Solution:
$\phi = 5\mathrm{t}^3 + 4\mathrm{t}^2 + 2\mathrm{t} - 5$

$\mathrm{e} = \left|\frac{\mathrm{d}\phi}{\mathrm{dt}}\right| = 15\mathrm{t}^2 + 8\mathrm{t} + 2$

$\mathrm{at\ t} = 2\mathrm{s},$

$\mathrm{iR} = \left|\frac{\mathrm{d}\phi}{\mathrm{dt}}\right| = 60 + 16 + 2 = 78$

$\mathrm{i} = \frac{78}{5} = 15.6\,\mathrm{A}$

The correct option is (1)

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Q. 13 An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is :

Option 1:
0.4%

Option 2:
0.2%

Option 3:
0.8%

Option 4:
0.6%

Correct Answer:
0.8%

Solution:

$$\frac{\Delta R}{R} \times 100 = \frac{2\Delta l}{l} \times 100$$
$$= 0.8\,\%$$

The correct option is (3)

Q. 14 A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the alpha particle and proton is :

Option 1:
$1:4$

Option 2:
$4:1$

Option 3:
$2:1$

Option 4:
$1:2$

Correct Answer:
$2:1$

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Solution:

$\overline{\mathrm{V}} \perp \overline{\mathrm{B}} \quad (\text{given})$

$\mathrm{r} = \dfrac{\mathrm{mv}}{\mathrm{qB}}$

$\dfrac{\mathrm{r}_\alpha}{\mathrm{r_p}} = \dfrac{\mathrm{M}_\alpha}{\mathrm{M_p}} \times \dfrac{\mathrm{q_p}}{\mathrm{q}_\alpha}$

$= \dfrac{4}{1} \times \dfrac{1}{2} = \dfrac{2}{1}$

The correct option is (3)

Q. 15 If Electric field intensity of a uniform plane electro magnetic wave is given as

$\mathrm{E} = -301.6\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x + 452.4\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y \dfrac{\mathrm{V}}{\mathrm{m}}.$

Then, magnetic intensity $'\mathrm{H}'$ of this wave in Am^{-1} will be :

[Given : Speed of light in vacuum

$\mathrm{c} = 3 \times 10^8\ \mathrm{ms}^{-1}$, Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\mathrm{NA}^{-2}$]

Option 1:

$+0.8\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y + 0.8\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x$

Option 2:

$+1.0 \times 10^{-6}\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y + 1.5 \times 10^{-6}(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x$

Option 3:

$-0.8\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y - 1.2\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x$

Option 4:

$-1.0 \times 10^{-6}\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y - 1.5 \times 10^{-6}\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x$

Correct Answer:

$-1.0 \times 10^{-6}\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_y - 1.5 \times 10^{-6}\sin(\mathrm{k}z - \omega\mathrm{t})\hat{\mathrm{a}}_x$

Solution:

$(\mathrm{E}_0)_\mathrm{x} = 301.6$

$(\mathrm{B}_0)_\mathrm{x} = \dfrac{(\mathrm{E}_0)\,\mathrm{x}}{\mathrm{c}} = 10^{-6}$

$(\mathrm{E}_0)_\mathrm{y} = 452.4$

$(\mathrm{B}_0)_\mathrm{y} = \dfrac{(\mathrm{E}_0)_\mathrm{y}}{\mathrm{c}} = 1.5 \times 10^{-6}$

$\left(\left(\hat{E}_0\right) \times \left(\hat{B}_0\right)\right)$

Points in the direction of propagation which is along $+z^-$ axis

The correct option is (4)

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Q. 16 In free space, an electromagnetic wave of $3\mathrm{GHz}$ frequency strikes over the edge of an object of size $\frac{\lambda}{100}$, where λ is the wavelength of the wave in free space. The phenomenon, which happens there will be :

Option 1:
Reflection

Option 2:
Refraction

Option 3:
Diffraction

Option 4:
Scattering

Correct Answer:
Scattering

Solution:
$\rightarrow$ The electromagnetic wave bending around sharp edges and spreading out of waves past small aperture is due to phenomenon diffraction
(Size of aperture is comparable to wavelength)

$\rightarrow$ Scattering is the phenomenon in which wave or particle deviated tiny objects in the atmosphere

Therefore , The correct option is (4)

Q. 17 An electron with speed v and a photon with speed c have the same de-Broglie wavelength. If the kinetic energy and momentum of electron are $\mathrm{E_e}$ and $\mathrm{p_e}$ and that of photon are $\mathrm{E_{ph}}$ and $\mathrm{p_{ph}}$ respectively. Which of the following is correct?

Option 1:
$\frac{\mathrm{E_e}}{\mathrm{E_{ph}}}=\frac{2\mathrm{c}}{v}$

Option 2:
$\frac{\mathrm{E_e}}{\mathrm{E_{ph}}}=\frac{v}{2\mathrm{c}}$

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Option 3:

$$\frac{\mathrm{Pe_e}}{\mathrm{P_{ph}}} = \frac{2\mathrm{c}}{v}$$

Option 4:

$$\frac{\mathrm{pe_e}}{\mathrm{p_{ph}}} = \frac{v}{2\mathrm{c}}$$

Correct Answer:

$$\frac{\mathrm{E_e}}{\mathrm{E_{ph}}} = \frac{v}{2\mathrm{c}}$$

Solution:

$$\lambda_\mathrm{e} = \lambda_\mathrm{ph}$$

$$\frac{\mathrm{h}}{\mathrm{p_e}} = \frac{\mathrm{h}}{\mathrm{p_{ph}}}$$

$$\mathrm{p_e} = \mathrm{p_{ph}} \rightarrow (1)$$

$$\mathrm{E_e} = \frac{1}{2}\mathrm{m_e V^2}$$

$$\mathrm{E_{ph}} = \mathrm{m_{ph} c^2}$$

$$\frac{\mathrm{E_e}}{\mathrm{E_{ph}}} = \frac{(\mathrm{m_e V})\,\mathrm{V}}{2\,(\mathrm{m_{ph} c})\,\mathrm{c}}$$

$$\Rightarrow \frac{\mathrm{E_e}}{\mathrm{E_{ph}}} = \frac{\mathrm{V}}{2\mathrm{c}}$$

The correct option is (2)

Q. 18 How many alpha and beta particles are emitted when Uranium $_{92}\mathrm{U}^{238}$ decays to lead $_{82}\mathrm{Pb}^{206}$?

Option 1:

3 alpha particles and 5 beta particles

Option 2:

6 alpha particles and 4 beta particles

Option 3:

4 alpha particles and 5 beta particles

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Option 4:
8 alpha particles and 6 beta particles

Correct Answer:
8 alpha particles and 6 beta particles

Solution:
$_{92}\mathrm{U}^{238} \rightarrow {}_{82}\mathrm{Pb}^{206} + \mathrm{x}\ {}_{2}\alpha^{4} + \mathrm{y}\ {}_{-1}\beta^{0}$

Let $\mathrm{x}\ \&\ \mathrm{y}$ be the no.of alpha & beta particles emitted

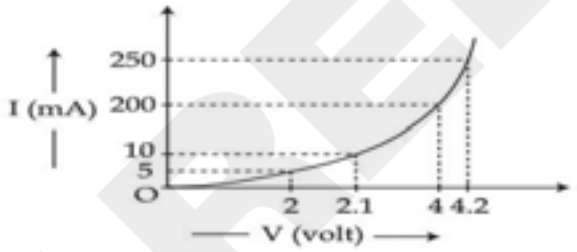
Comparing reactant & products ,

$$238 = 206 + 4\mathrm{x} + 0$$
$$32 = 4\mathrm{x}$$
$$\mathrm{x} = 8 \ \rightarrow (1)$$

$$92 = 82 + 2\mathrm{x} + \mathrm{y}\,(-1)$$
$$92 = 82 + 16 - \mathrm{y}$$
$$\mathrm{y} = 6 \ \rightarrow (2)$$

The correct option is (4)

Q. 19 The $\mathrm{I} - \mathrm{V}$ characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltage of $2\ \mathrm{V}$ and $4\ \mathrm{V}$ respectively, is :



Option 1:
$1 : 2$

Option 2:
$5 : 1$

Option 3:
$1 : 40$

Option 4:
$20 : 1$

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Correct Answer:
$5:1$

Solution:
Dynamic resistance at $\mathrm{V}=2\mathrm{V}$

$$\mathrm{R}_1=\frac{\Delta \mathrm{V}}{\Delta \mathrm{I}}=\frac{2.1-2}{10-5}$$
$$\mathrm{R}_1=2\times10^{-2}$$

Dynamic resistance at $\mathrm{V}=4\mathrm{V}$

$$\mathrm{R}_2=\frac{\Delta \mathrm{V}}{\Delta \mathrm{I}}=\frac{4.2-4}{250-200}$$
$$\mathrm{R}_2=0.4\times10^{-2}$$

$$\frac{\mathrm{R}_1}{\mathrm{R}_2}=5$$

The correct option is (2)

Q. 20 Choose the correct statement for amplitude modulation :

Option 1:
Amplitude of modulating signal is varied in accordance with the information signal.

Option 2:
Amplitude of modulated signal is varied in accordance with the information signal.

Option 3:
Amplitude of carrier signal is varied in accordance with the information signal.

Option 4:
Amplitude of modulated signal is varied in accordance with the modulating signal.

Correct Answer:
Amplitude of carrier signal is varied in accordance with the information signal.

Solution:
In amplitude modulation, the amplified of high frequency carrier wave is varied in accordance with message signal

The correct option is (3)

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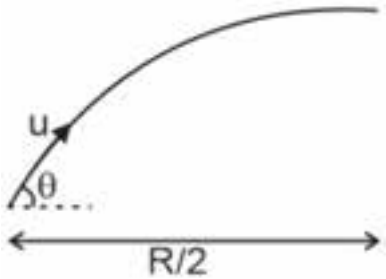
Q. 21 A fighter jet is flying horizontally at a certain altitude with a speed of $200\ \mathrm{ms}^{-1}$. When it passes directly overhead an anti-aircraft gun, a bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is $400\ \mathrm{m/s}$, the value of θ will be_____o.

Correct Answer:
60

Solution:



Bullet can strike jet only when it is horizontal
Horizontal distance = Horizontal distance travelled by jet

$$(\mathrm{u}\cos\theta)\,\mathrm{t} = \mathrm{vt}$$
$$(400)\cos\theta = 200$$
$$\cos\theta = \frac{1}{2}$$
$$\theta = 60^\circ$$

Q. 22 A ball of mass$0.5\ \mathrm{kg}$ is dropped from the height of $10\ \mathrm{m}$. The height, at which the magnitude of velocity becomes equal to the magnitude of acceleration due to gravity, is_____ m. [Use $g = 10\ \mathrm{m/s^2}$]

Correct Answer:
5

Solution:
$$\mathrm{m} = 0.5\,\mathrm{kg}$$
$$\mathrm{h} = 10\,\mathrm{m}$$
$$\mathrm{v} = 10\left(\frac{\mathrm{m}}{\mathrm{s}}\right)$$

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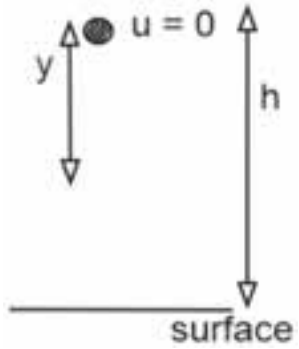
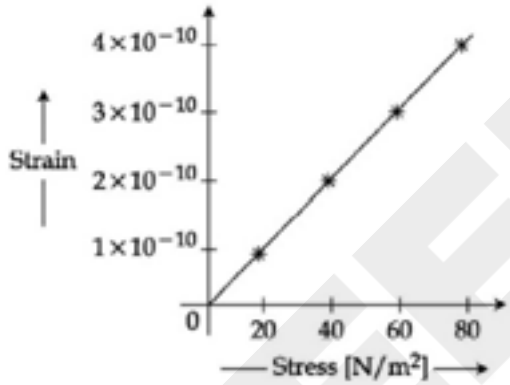


$\sqrt{2gy} = 10$
$2 \times 10 \times y = 100$
$y = 5\,m$

At heights 5 m from the surface or from the top the velocity becomes equal to the magnitude of acceleration due to gravity

Q. 23 The elastic behaviour of material for linear stress and linear strain, is shown in the figure. The energy density for a linear strain of 5×10^{-4} is_____kJ/m^3. Assume that material is elastic upto the linear strain of 5×10^{-4}.



Correct Answer:
25

Solution:

$$\text{Energy density} = \frac{1}{2} \times \text{stress} \times \text{strain}$$
$$= \frac{1}{2} (\text{strain})^2 \gamma$$
$$\because \text{slope of graph} = \frac{1}{\gamma} = \frac{10^{-10}}{20}$$
$$\therefore \ Y = 20 \times 10^{+10}$$

$$\text{Energy density} = \frac{1}{2} \times \left(5 \times 10^{-4}\right)^2 \times \left(20 \times 10^{10}\right)$$
$$= 250 \times 10^2$$
$$\text{Energy density} = 25 \frac{\text{kJ}}{\text{m}^3}$$

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Q. 24 The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5} m on another planet. The acceleration due to gravity on the planet will be_____ms^{-2}. (Take acceleration due to gravity on the surface of earth $= 10\ ms^{-2}$)

Correct Answer:
6

Solution:
For a given material , Y will be constant

$$\left.\begin{array}{l}\Delta l_E = 10^{-4}m \\ \Delta l_p = 6 \times 10^{-5}m\end{array}\right| g_P \rightarrow \text{acceleration due to gravity on planet}$$

$$Y = \frac{mg}{A} \times \frac{l}{\Delta l}$$

$$g\,\alpha\,\Delta l$$

$$\frac{g_E}{g_p} = \frac{\Delta l_E}{\Delta l_P}$$

$$\frac{10}{g_p} = \frac{10^{-4}}{6 \times 10^{-5}} \Rightarrow g_p = 6\,\frac{m}{s^2}$$

Q. 25 A $10\Omega, 20\ mH$ coil carrying constant current is connected to a battery of $20\ V$ through a switch. Now after switch is opened current becomes zero in $100\mu s$. The average e.m.f. induced in the coil is_____ V.

Correct Answer:
400

Solution:
$$R = 10\,\Omega$$
$$L = 20 \times 10^{-3}H$$

Current in the steady state ,
$$I_0 = \frac{v}{R} = 2A$$

$$\text{Avg.Emf induced} = e_{avg} = L\left|\frac{\Delta I}{\Delta t}\right|$$
$$= 200 \times 10^{-3}\left|\frac{2}{100 \times 10^{-6}}\right|$$
$$\Rightarrow e_{avg} = 400$$

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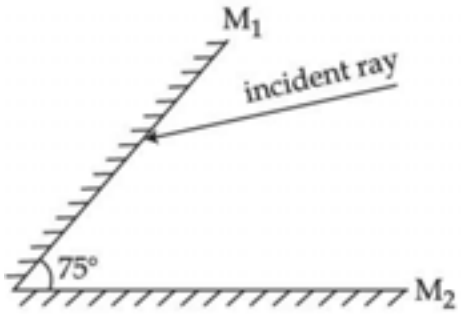
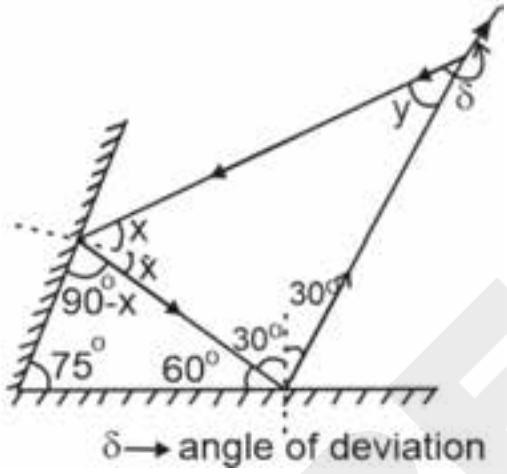
Q. 26 A light ray is incident, at an incident angle θ_1, on the system of two plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30°. The total deviation of the ray will be _____degree.



Correct Answer:
210

Solution:



$\delta \rightarrow$ angle of deviation.

$60^\circ + 75^\circ + 90^\circ - \mathrm{x} = 180$

$\mathrm{x} = 45^\circ$

$\text{Also }, 2\mathrm{x} + 60^\circ + \mathrm{y} = 180^\circ$

$\mathrm{y} = 30^\circ$

$\delta = 180^\circ + \mathrm{y} = 210^\circ$

The angle of deviation is 210°

Q. 27 In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with nineth main scale division. Then the value of vernier contant will be_____ $\times 10^{-2}$ mm.

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Correct Answer:
5

Solution:

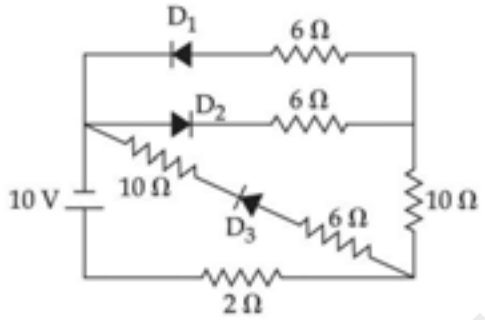
$$1\,\text{MSD} = \frac{1\text{cm}}{20} = 0.05\,\text{cm}$$

$$10\,\text{VSD} = 9\,\text{MSD}$$

$$\begin{aligned}\text{Vernier constant} = \text{Least counct} &= 1\text{MSD} - 1\,\text{VSD} \\ &= 0.1\,\text{MSD}\end{aligned}$$

$$\begin{aligned}\text{Vernier constant} &= 5 \times 10^{-3}\,\text{cm} \\ &= 5 \times 10^{-2}\,\text{mm}\end{aligned}$$

Q. 28 As per the given circuit, the value of current through the battery will be_____ A.

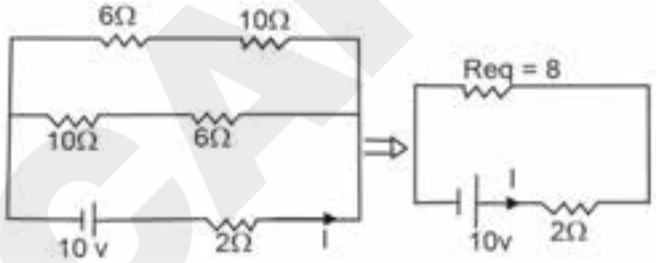


Correct Answer:
1

Solution:

$\rightarrow$ D_2 is reversed biased so it won't conduct current

Redrawing the circuit



$$I = \frac{10}{2 + R_{eq}} = \frac{10}{2+8}$$

$$I = 1A$$

The current through battery will be 1 A

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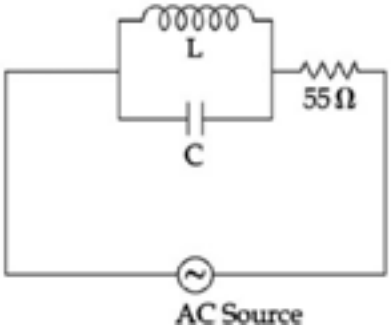
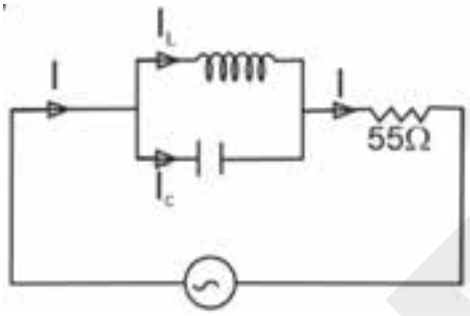
Q. 29 A $110\ \mathrm{V}, 50\ \mathrm{Hz}, \mathrm{AC}$ source is connected in the circuit (as shown in figure). The current through the resistance 55Ω, at resonance in the circuit, will be_____ A.



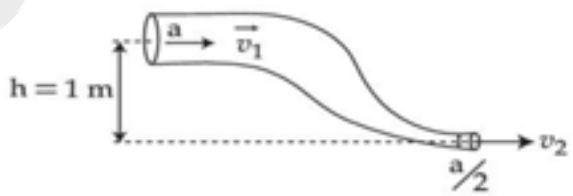
Correct Answer:
0

Solution:
At resonance , $\mathrm{I_L} - \mathrm{I_C} = 0$
i.e $\mathrm{I_L} = -\mathrm{I_C}$
Negative sign expresses, the $\mathrm{I_L} \& \mathrm{I_C}$ are of phase



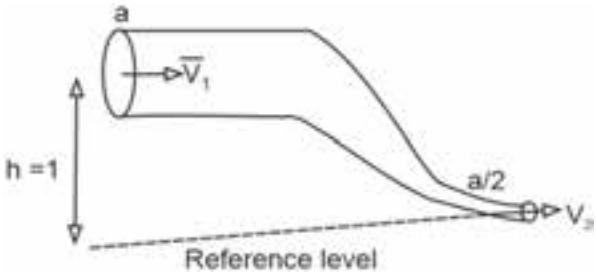
$\mathrm{I} = \mathrm{I_L} + \mathrm{I_C} = 0$
The current through the resistance 55Ω , at resonance in the circuit will be 0

Q. 30 An ideal fluid of density $800\mathrm{kgm}^{-3}$, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is $4100\ \mathrm{Pa}$. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6}\ \mathrm{ms}-1$ for x = _____.$(Given\ \ g = 10\ \mathrm{ms}^{-2})$



Correct Answer:
363

Solution:



By equation of continuity ,

$$av_1 = \left(\frac{a}{2}\right) v_2$$

$$\therefore v_2 = 2v_1 \ \rightarrow (1)$$

By Bernoulli's theorem ,

$$p_1 + \frac{1}{2}\delta v_1^2 + \delta gh = p_2 + \frac{1}{2}\delta v_2^2 + 0$$

$$(p_1 - p_2) = 4100\,pa$$

$$4100 + 800 \times 1 \times 10 = 400(3)\left(\frac{x}{36}\right)$$

$$12100 = (400)\left(\frac{x}{12}\right)$$

$$x = \frac{121 \times 12}{4}$$

$$x = 363$$

Chemistry

Q. 1 A commercially sold conc. HCl is $35\% HCl$ by mass. If the density of this commercial acid is $1.46\ g/mL,$ the molarity of this solution is :
(Atomic mass : $Cl = 35.5\ amu, H = 1 amu$)

Option 1:
$10.2\ M$

Option 2:
$12.5\ M$

Option 3:
$14.0\ M$

Option 4:
$18.2\ M$

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Correct Answer:
$14.0\ \mathrm{M}$

Solution:
Given,

Concentration of $\mathrm{HCl} = 35\%\ \mathrm{HCl}$ by mass.
$\text{density} = 1.46\ \mathrm{g/ml}$

Let us assume the total volume of Solution $= 1000\ \mathrm{ml}$

From density, total mass of Solution $= 1.46 \times 1000 = 1460\ \mathrm{g}$

$$\therefore\ \text{mass of HCl} = \frac{35}{100} \times 1460 = 511\ \mathrm{g}$$

$$\text{numeber of moles of HCl} = \frac{\text{mass of HCl in Solution}}{\text{molar mass of HCl}} = \frac{511}{36.5} = 14$$

$$\text{Molarity} = \text{no. of moles of HCl in 1000ml of solution} = 14\ \mathrm{M}$$

Hence, option (C) is correct.

Q. 2 An evacuated glass vessel weighs $40.0\ \mathrm{g}$ when empty, $135.0\ \mathrm{g}$ when filled with a liquid of density $0.95\ \mathrm{g\ mL^{-1}}$ and $40.5\ \mathrm{g}$ when filled with an ideal gas at $0.82\ \mathrm{atm\ at\ 250\ K}$. The molar mass of the gas in $\mathrm{g\ mol^{-1}}$ is :
(Given : $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$)

Option 1:
35

Option 2:
50

Option 3:
75

Option 4:
125

Correct Answer:
125

Solution:
Given, mass of empy glass vessel $= 40\,\mathrm{g}$
Total mass $= 135\,\mathrm{g}$.

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$\text{density of liquid } = 0.95 \text{ g ml}^{-1}$
$\text{R} = 0.082 \text{ L atm K}^{-} \text{ mol}^{-1}$

$\text{mass of Liquid} = 135 - 40 = 95 \text{ g.}$

$$\text{Volume of Liquid } = \frac{\text{mass}}{\text{density}} = \frac{95 \text{ g}}{0.95 \text{ g/ml}} = 100 \text{ ml}$$
$$= 0.1 \text{ L}$$

Now, for ideal gas-

$\text{mass of ideal gas} = 40.5 - 40 = 0.5 \text{ g}$

From ideal gas equation
$\text{PV} = \text{nRT}$

Putting values -
$0.82 \text{ atm} \times 0.1 \text{ L} = \left(\frac{0.5}{\text{M}}\right) \text{mol} \times 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 250 \text{ K}$

Solving equation, we get -
$\text{M} = 125 \text{ g mol}^{-1}$

Option (4) is correct

Q. 3 If the radius of the 3^{rd} Bohr's orbit of hydrogen atom is r_3 and the radius of 4^{th} Bohr's orbit is r_4. Then :

Option 1:
$\text{r}_4 = \frac{9}{16}\text{r}_3$

Option 2:
$\text{r}_4 = \frac{16}{9}\text{r}_3$

Option 3:
$r_4 = \frac{3}{4}r_3$

Option 4:
$r_4 = \frac{4}{3}r_3$

Correct Answer:
$\text{r}_4 = \frac{16}{9}\text{r}_3$

Solution:

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Bohr's radius is given as

$$r = 0.529 \times \frac{n^2}{Z}\text{Å}$$

where. n = prinicipal quantum number of orbit

Z = atomic number

$$r_3 = 0.529 \times \frac{3^2}{1}\text{Å}$$

$$r_4 = 0.529 \times \frac{4^2}{1}\text{Å}$$

$$\frac{r_4}{r_3} = \frac{4^3}{3^2} = \frac{16}{9} \Rightarrow r_4 = \frac{16}{9}r_3$$

Option (2) is correct

Q. 4 Consider the ions/molecule

$O_2^+, O_2, O_2^-, O_2^{2-}$

For increasing bond order the correct option is :

Option 1:

$O_2^{2-} < O_2^- < O_2 < O_2^+$

Option 2:

$O_2^- < O_2^{2-} < O_2 < O_2^+$

Option 3:

$O_2^- < O_2^{2-} < O_2^+ < O_2$

Option 4:

$O_2^- < O_2^+ < O_2^{2-} < O_2$

Correct Answer:

$O_2^{2-} < O_2^- < O_2 < O_2^+$

Solution:

From MOT i.e molecular orbital Theory,

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

N_b = number of bonding electrons .

N_a = number of antibonding electrons.

$$\text{for } O_2^+, N_b = 10,\ N_a = 5$$

$$\text{Bond order} = \frac{10-5}{2} = 2.5$$

$$\text{for } O_2,\ N_b = 10, N_a = 6$$

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$$\text{Bond order} = \frac{10-6}{2} = 2$$

for O_2^-, $N_b = 10$, $N_a = 7$

$$B \cdot O = \frac{N_b - N_a}{2} = \frac{10-7}{2} = 1.5$$

for O_2^{2-}, $N_b = 10$, $N_a = 8$

$$B \cdot O = \frac{N_b - N_a}{2} = \frac{10-8}{2} = \frac{2}{2} = 1$$

$\therefore$ Bond order : $O_2^{2-} < O_2^- < O_2 < O_2^+$

Option (1) is correct

Q. 5 The $\left(\frac{\partial E}{\partial T}\right)_P$ of different types of half cells are as follows :

A	B	C	D
1×10^{-4}	2×10^{-4}	0.1×10^{-4}	0.2×10^{-4}

(Where E is the electromotive force)

Which of the above half cells would be preferred to be used as reference electrode?

Option 1:
A

Option 2:
B

Option 3:
C

Option 4:
D

Correct Answer:
C

Solution:
$\left(\frac{\partial E}{\partial T}\right)_P$ of different half cells are given which represents the change in EMF of a cell with temperature.

A cell showing lesser variation of EMF with respect to temperature is preferred as it can be used over a wide range of temperature without much deviation from standard EMF of a cell.

Here, $\left(\frac{\partial E}{\partial T}\right)_p$ is lowest for Half cell "C" which is 0.1×10^{-4}

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Hence, the most preferrable half cell to be used as a reference electrode is C.

Option (3) is correct.

Q. 6 Choose the correct stability order of group 13 elements in their $+1$ oxidation state.

Option 1:
$\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

Option 2:
$\mathrm{Tl} < \mathrm{In} < \mathrm{Ga} < \mathrm{Al}$

Option 3:
$\mathrm{Al} < \mathrm{Ga} < \mathrm{Tl} < \mathrm{In}$

Option 4:
$\mathrm{Al} < \mathrm{Tl} < \mathrm{Ga} < \mathrm{In}$

Correct Answer:
$\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

Solution:
As we move form top to Bottom in group 13, stability of $+1$ oxidation state increases due to inert pair effect.

Hence correct order of stability:-
$\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

Option (1) is correct

Q. 7 Given below are two statements :

Statement I : According to the Ellingham diagram, any metal oxide with higher $\Delta \mathrm{G}^{\circ}$ is more stable than the one with lower $\Delta \mathrm{G}^{\circ}$.

Statement II : The metal involved in the formation of oxide placed lower in the Ellingham diagram can reduce the oxide of a metal placed higher in the diagram.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

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Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Statement I is incorrect but **Statement II** is correct.

Solution:
According to ellingham diagram, any metal oxide with lower ΔG° is more stable. Hence Statement I is false.

Statement II says that metal involved in the formation of oxide lower in ellingham diagram can reduce the oxide of a metal placed higher in the diagram, which is correct statement.

Thus, Statement I is correct and Statement II is incorrect.

Hence option (4) is correct.

Q. 8 Cosider the following reaction :

$$2HSO_4^-(aq) \xrightarrow[(2)\text{Hydrolysis}]{(1)\text{Electrolysis}} 2HSO_4^- + 2H^+ + \mathbf{A}$$

The dihedral angle in product $\mathbf{A}$ in its solid phase at $110\ K$ is :

Option 1:
104°

Option 2:
111.5°

Option 3:
90.2°

Option 4:
111.0°

Correct Answer:
90.2°

Solution:
$$2HSO_4^-(aq) \xrightarrow[(ii)\text{Hydrolysis}]{(i)\text{Electrolysis}} 2HSO_4^- + 2H^+ + H_2O_2$$

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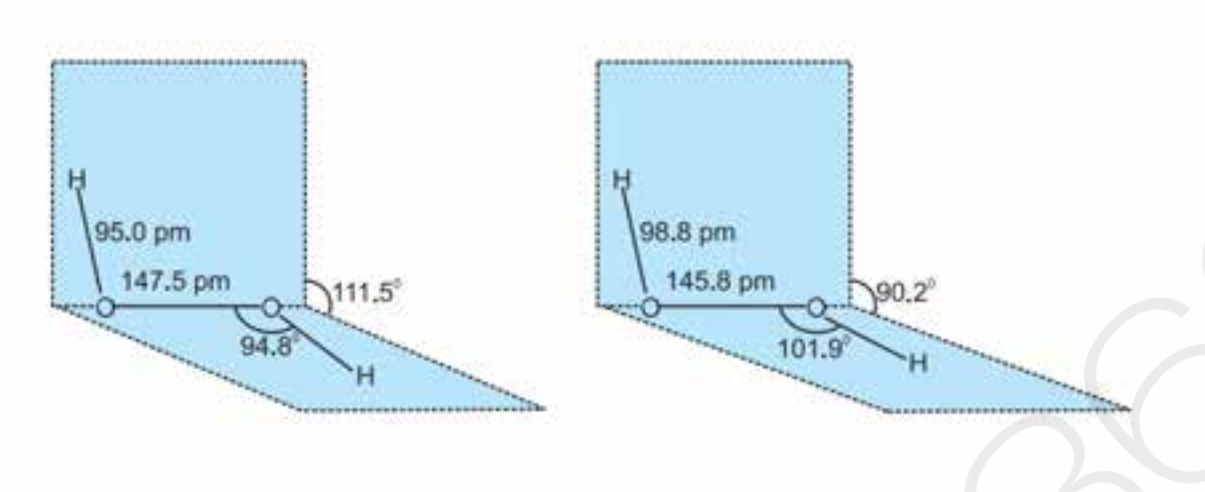
$A = H_2O_2$ in solid phase

from figure,



The dihedral angle of H_2O_2 in solid phase is 90.2°

Option (3) is correct.

Q. 9 The correct order of melting point is :

Option 1:
$Be > Mg > Ca > Sr$

Option 2:
$Sr > Ca > Mg > Be$

Option 3:
$Be > Ca > Mg > Sr$

Option 4:
$Be > Ca > Sr > Mg$

Correct Answer:
$Be > Ca > Sr > Mg$

Solution:
Fact based.

Melting point order of alkaline earth metals is

$Be > Ca > Sr > Mg$

Option (4) is correct.

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Q. 10 The correct order of melting points of hydrides of group 16 elements is :

Option 1:
$H_2\ S < H_2Se < H_2Te < H_2O$

Option 2:
$H_2O < H_2\ S < H_2Se < H_2Te$

Option 3:
$H_2\ S < H_2Te < H_2Se < H_2O$

Option 4:
$H_2Se < H_2\ S < H_2Te < H_2O$

Correct Answer:
$H_2\ S < H_2Se < H_2Te < H_2O$

Solution:
Melting point of hydrides of group 16 elements

H bonding and molar mass predominantly decides the melting poin of Hydrides in Group 16 elements.

In H_2O, due to excessive Hydrogen bonding , its melting point is larger than that of the other Group 16 hydrides.

$\therefore$ Order of melting point is

$H_2S < H_2Se < H_2Te < H_2O$

Option (1) is correct

Q. 11 Consider the following reaction :

$$A\ +\ \text{alkali}\ \rightarrow\ B\ (\text{Major Product})$$

If B is an oxoacid of phosphorus with no $P-H$ bond, then A is :

Option 1:
$\text{White}\ P_4$

Option 2:
$\text{Red}\ P_4$

Option 3:
P_2O_3

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Option 4:
H_3PO_3

Correct Answer:
$\text{Red } P_4$

Solution:
$\text{A} + \text{alkali} \rightarrow \text{B (major product)}$

Given , B is an oxoacid of phosphorus with no P-H bond, the reaction is ---

$\text{Red } P_4 + \text{alkali} \rightarrow H_4P_2O_6$

Hence , $\text{A} = \text{Red } P_4$

It is to be noted that $\text{White } P_4$ reacts with alkali to form H_3PO_2 which contains $\text{P} - \text{H}$ bond.

Option (2) is correct

Q. 12 Polar stratospheric clouds facilitate the formation of :

Option 1:
$ClONO_2$

Option 2:
$HOCl$

Option 3:
ClO

Option 4:
CH_4

Correct Answer:
$HOCl$

Solution:
Polar straospheric clouds facilitates the formation $HOCl$ by providing surface on which hydrolysis of $ClONO_2$ (Chlorine Nitrate) takes place.

Reaction given below demonstrates the formation of $HOCl$:

$$ClONO_2(g) + H_2O(g) \longrightarrow HOCl(g) + HNO_3(g)$$

Option (2) is correct

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Q. 13 Given below are two statements:

Statement I : In 'Lassaigne's Test', when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.

Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give $\mathrm{NaCN\ and\ Na_2S}$.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Both **Statement I** and **Statement II** are correct.

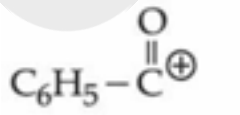
Solution:
Both the given statements are correct in the context of "Lassagne's Test" and decomposifion of Sodium Thiocyanate by excess sodium respectively.

Hence, option (1) is correct.

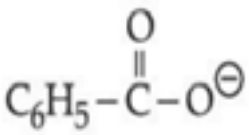
Q. 14 $(\mathrm{C_7H_5O_2})_2 \xrightarrow{h\nu} [\mathrm{X}] \rightarrow 2\dot{\mathrm{C}}_6\mathrm{H}_5 + 2\mathrm{CO}_2$

Consider the above reaction and indentify the intermidiate $'X'$

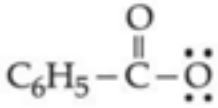
Option 1:



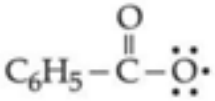
Option 2:



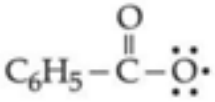
Option 3:



Option 4:

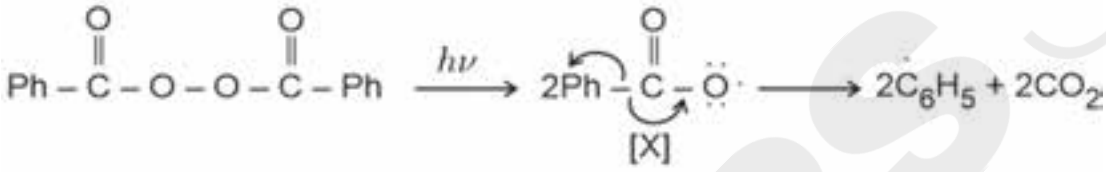


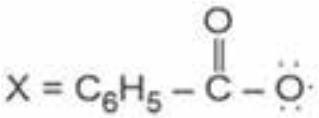
Correct Answer:



Solution:

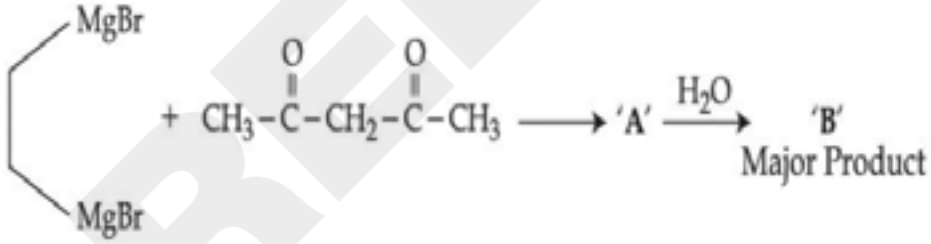
$(C_7H_5O_2)_2 \xrightarrow{h\nu} [X] \rightarrow 2\dot{C}_6H_5 + 2CO_2$





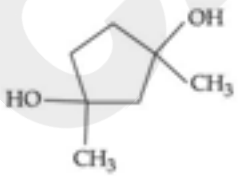
Option (4) is correct.

Q. 15



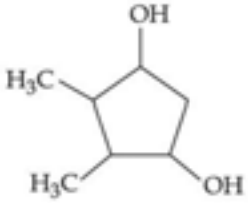
Consider the above reaction sequence and indentify the product B.

Option 1:

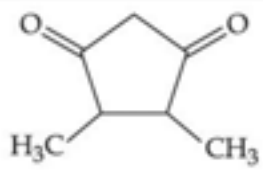


Option 2:

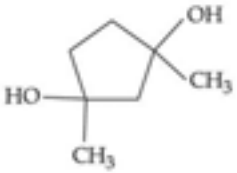
Option 3:



Option 4:

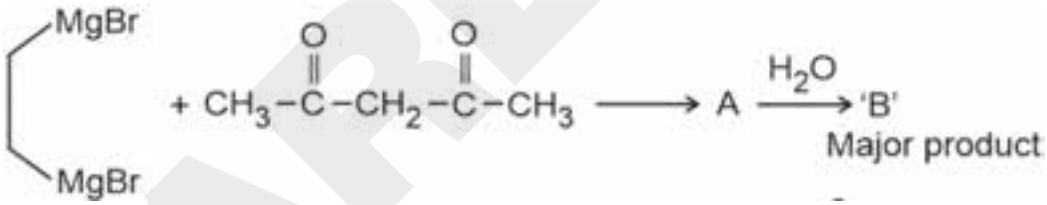


Correct Answer:



Solution:

Nucleophilic addition Reaction takes place in the first step - Given Reaction is -



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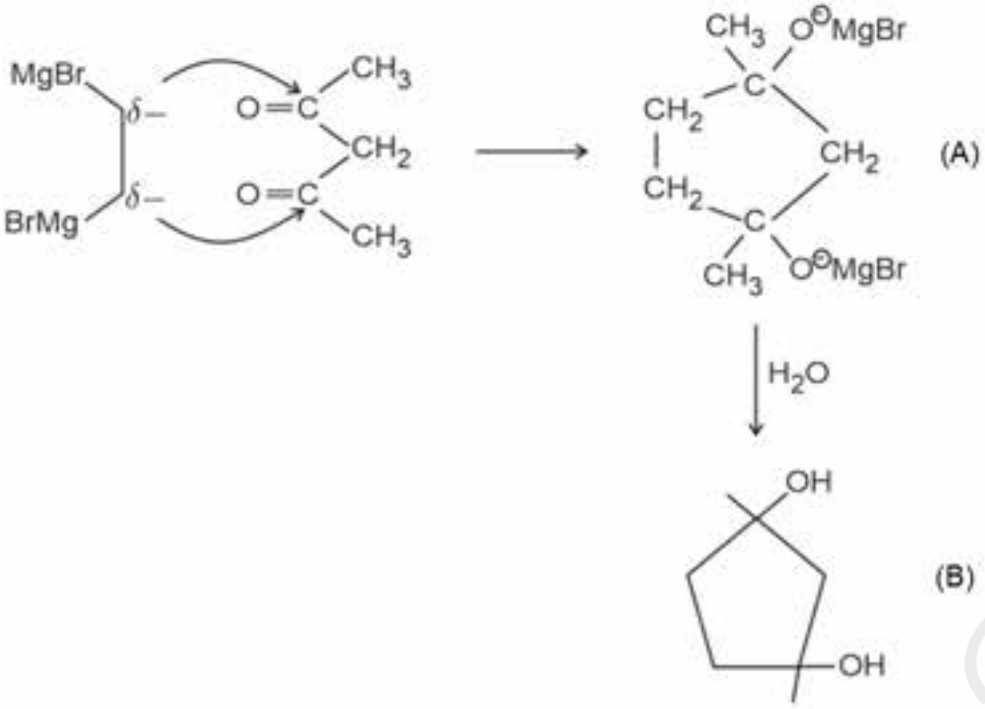
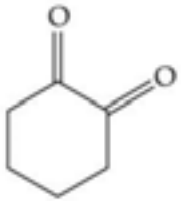


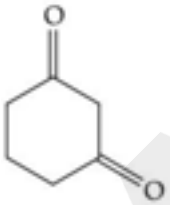
Hence option (1) is correct.

Q. 16 Which will have the highest enol content ?

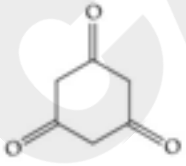
Option 1:



Option 2:



Option 3:



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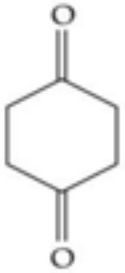
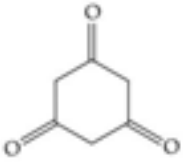
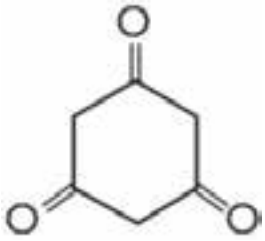
Option 4:

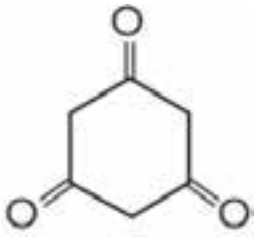
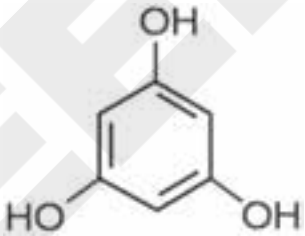


Correct Answer:



Solution:
The highest enol content is shown by



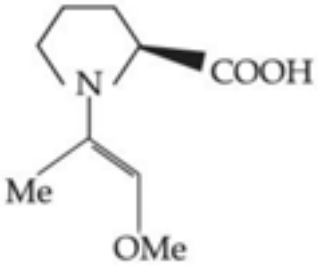
As  is a tautomer of 

Which is aromatic in nature and hence most stable in enol form.

Option (3) is correct.

Q. 17 Among the following structures, which will show the most stable enamine formation ?
(Where Me is $-CH_3$)

Option 1:



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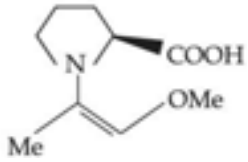
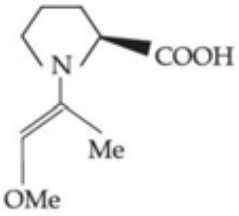
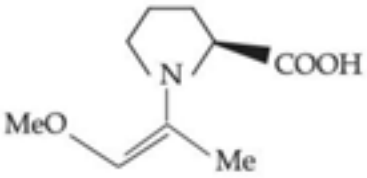
Option 2:



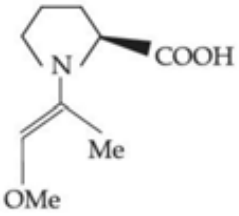
Option 3:



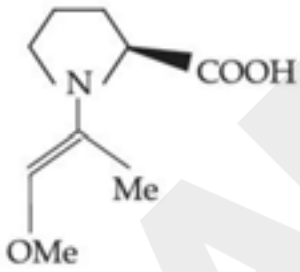
Option 4:



Correct Answer:



Solution:
The most stable enamine out of the given species is



It is most stable among the given ones due to steric factor.

Hence option (3) is correct.

Q. 18 Which of the following sets are **correct** regarding polymer?
(A) Copolymer : Buna-S
(B) Condensation polymer : Nylon-6,6
(C) Fibres : Nylon-6,6
(D) Thermosetting polymer : Terylene
(E) Homopolymer : Buna-N
Choose the **correct** answer from given options below :

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Option 1:
(A), (B) and (C) are correct

Option 2:
(B), (C) and (D) are correct

Option 3:
(A), (C) and (E) are correct

Option 4:
(A), (B) and (D) are correct

Correct Answer:
(A), (B) and (C) are correct

Solution:
As we have learnt,

Buna-S is a Copolymer of Butadiene and styrene.

Nylon-6,6 is Condensation polymer as well as fibre

Terylene is thermoplastic polymer.

Buna -N is a Copolymer of Butadiene and Acrylonitrile. It is not a homopolymer

Hence statements (A), (B) and (C) are correct.

Option (1) is correct.

Q. 19 A chemical which stimulates the secretion of pepsin is:

Option 1:
Anti histamine

Option 2:
Cimetidine

Option 3:
Histamine

Option 4:
Zantac

Correct Answer:
Histamine

Solution:
Fact based.

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Histamine is responsible for stimulation of secretion of pepsin.

Option (3) is correct.

Q. 20 Which statement is **not** true with respect to nitrate ion test?

Option 1:
A dark brown ring is formed at the junction of two solutions.

Option 2:
Ring is formed due to nitroferrous sulphate complex.

Option 3:
The brown complex is $[Fe\,(H_2O)_5\,(NO)]SO_4$

Option 4:
Heating the nitrate salt with conc. H_2SO_4, light brown fumes are evolved.

Correct Answer:
Ring is formed due to nitroferrous sulphate complex.

Solution:
In nitrate ion test, the Brown Ring is formed due to the formation of **nitrosoferrous** sulphate not nitroferrous sulphate . Hence statement given in Option (2) is incorrect.

Nitrosoferrous sulphate : $[Fe\,(H_2O)_5\,(NO)]SO_4$.
(Brown ring)

Option (2) is the answer.

Q. 21 For complete combustion of methanol

$$CH_3OH(l) + \frac{3}{2}O_2(\ g) \rightarrow CO_2(\ g) + 2H_2O(l)$$

the amount of heat produced as measured by bomb calorimeter is $726\ kJ\ mol^{-1}$ at $27^{\circ}C$. The enthalpy of combustion for the reaction is $-x\ kJ\ mol^{-1}$, where x is _____.(Nearest integer)
(Given : $R = 8.3JK^{-1}\ mol^{-1}$)

Correct Answer:
727

Solution:

$$CH_3OH(\ell) + \frac{3}{2}O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$$

$$\text{Given, } \Delta U = -726KJ/mol$$

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$$T = 27^{\circ}C = 300\ K$$
$$R = 8.3 JK^{-1}\ mol^{-1}$$

$$\Delta n_g = 1 - \frac{3}{2} = -\frac{1}{2}$$

Now, the relation between ΔU and ΔH is given as

$$\Delta H = \Delta U + (\Delta n_g) RT$$

$$\Rightarrow \Delta H = \left(-726 - \frac{1}{2} \times \frac{8.3 \times 300}{1000}\right) kJmol^{-1}$$
$$\Rightarrow \Delta H = -727.2\ kJ\ mol^{-1}$$

$$\text{Value of x is } 727.$$

Hence, the answer is 727

Q. 22 A 0.5 percent solution of potassium chloride was found to freeze at $-0.24^{\circ}C$. The percentage dissociation of potassium chloride is_____. (Nearest integer)

(Molal depression constant for water is $1.80\ K\ kg\ mol^{-1}$ and molar mass of KCl is $74.6\ g\ mol^{-1}$)

Correct Answer:
98

Solution:
Given , KCl soln is 0.5%

Assuming $100\ g$ of solution, the mass of KCl is $0.5\,g$

Given,

Depression of freezing point is $-0.24^{\circ}C$.
K_f for $KCl = 1.80\,k\ Kg\ mol^{-1}$
Molar mass of $KCl = 74.6\,g\ mol^{-1}$

Let us now calculate the molality of the solution

$$\text{molality of KCl} = \frac{\text{moles of KCl}}{\text{mass of solvent}} = \frac{0.5/74.6}{(99.5/1000)} mol\ kg^{-1}$$
$$\therefore \text{molality of KCl} = 0.06736\ mol\ kg^{-1}$$

$$\Delta T_f = i\ k_f\ m$$

$$\Rightarrow 0.24 = i \times 1.80 \times 0.06736$$

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$$\Rightarrow i = \frac{0.24}{1.80 \times 0.06736} = 1.98$$

Now, for the dissociation of KCl,

$$KCl \longrightarrow K^+ + Cl^-$$

The van't Hoff factor(i) is given as

$$i = 1 + \alpha$$

$$1.98 = 1 + \alpha \Rightarrow \alpha = 0.98$$

Thus, the % dissociation of KCl is 98%

Hence 98 is the answer.

Q. 23 $50\ mL$ of $0.1M CH_3COOH$ is being titrated against $0.1M NaOH$. When $25\ mL$ of $NaOH$ has been added, the pH of the solution will be _____ $\times 10^{-2}$. (Nearest integer)

(Given : $pK_a(CH_3COOH) = 4.76$)

$\log 2 = 0.30$
$\log 3 = 0.48$
$\log 5 = 0.69$
$\log 7 = 0.84$
$\log 11 = 1.04$

Correct Answer:
476

Solution:
$\text{Given}, pK_a(CH_3COOH) = 4.76$

$\text{moles of } CH_3COOH = 5 \text{ m mole}$
$\text{moles of } NaOH = 2.5 \text{ m mole}$

Chemical Reaction -

	$NaOH + CH_3COOH \longrightarrow$		$CH_3COONa + H_2O$
$t = 0$	$2.5\,mmol$	$5\,mmol$	0
$t = t$	0	$2.5\,mmol$	$2.5\,mmol$

So buffer is formed whose pH is given as

$$pH = pKa + \log\left(\frac{[Salt]}{[Acid]}\right)$$

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$$\Rightarrow \mathrm{pH} = \mathrm{pKa} + \log\left(\frac{2.5/75}{2.5/75}\right)$$
$$\Rightarrow \mathrm{pH} = \mathrm{pKa} + \log(1)$$
$$\Rightarrow \mathrm{pH} = \mathrm{pKa}$$
$$\Rightarrow \mathrm{pH} = 4.76 = 476 \times 10^{-2}$$

Hence 476 is the answer.

Q. 24 A flask is filled with equal moles of A and B. The half lives of A and B are $100\ \mathrm{s}$ and $50\ \mathrm{s}$ respectively and are independent of the initial concentration. The time required for the concentration of A to be four times that of B is _____ s.

(Given : $\ln 2 = 0.693$)

Correct Answer:
200

Solution:
Given , half Life of A, $\left(t_{\frac{1}{2}}\right)_{A} = 100\,\mathrm{sec}$
half life of B , $\left(t_{\frac{1}{2}}\right)_{B} = 50\,\mathrm{sec}$

Since, initial concentration is independent of half life signifies that Reaction is of first order.

$$\therefore \text{ for first order Reaction, } t_{\frac{1}{2}} = \frac{\ln 2}{k}.$$
$$K_A = \frac{\ln 2}{100},\ K_B = \frac{\ln 2}{50}$$

For I[st] order $R\alpha^{n} \rightarrow A = A_0\, e^{-kt}$

$$\text{For A} \rightarrow A_t = A_0\, e^{\left(-\frac{\ln 2}{100}t\right)}$$
$$\text{For B} \rightarrow B_t = B_0\, e^{\left(-\frac{\ln 2}{50}t\right)}$$

Since initial concentration of A and B are same.
$$\text{So } A_0 = B_0 \quad -(1)$$
and condition given is -
$$A_t = 4B_t \quad -(2)$$

Putting Both equation (1), (2) we get -

$$e^{-\frac{\ln 2}{100}t} = 4e^{-\frac{\ln 2}{50}t}$$
$$\Rightarrow e^{\left(\frac{\ln 2}{100}\right)t} = 4$$
$$\Rightarrow \frac{\ln 2}{100} \times t = \ln 4$$

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$$\Rightarrow \frac{\ln 2}{100} \times t = 2 \ln 2$$

$$\Rightarrow t = 200 \text{sec.}$$

Hence, answer is 200.

Q. 25 $2.0\ \text{g of } H_2$ gas is adsorbed on $2.5\ \text{g}$ of platinum powder at $300\ \text{K}$ and 1 bar pressure. The volume of the gas adsorbed per gram of the adsorbent is_____ mL.
(Given : $R = 0.083\ \text{L}^{-}\text{bar K}^{-1}\ \text{mol}^{-1}$)

Correct Answer:
9960

Solution:
$\text{Given, R} = 0.083\ \text{L bar K}^{-1}\,\text{mol}^{-1}$

$\text{mass of } H_2 \text{ adsorbed} = 2\ \text{g.}$
$\text{mass of platinum powder} = 2.5\ \text{g.}$

$\text{T} = 300\ \text{K}$
$\text{p} = 1\text{bar}$

$$\text{Volume of } H_2 = \frac{\text{nRT}}{\text{P}} = \left(\frac{2}{2} \times \frac{0.083 \times 300}{1}\right) \text{L}$$
$$= 24.9\,\text{L}$$
$$= 24900\,\text{mL}$$

$$2.5\ \text{g of Platinum absorbs} = 24900\ \text{ml of } H_2$$
$$\text{So, 1 g of Platinum absorbs} = \frac{24900}{2.5}\ \text{ml of } H_2$$
$$= 9960\,\text{ml of } H_2$$

9960 is the correct answer.

Q. 26 The spin-only magnetic moment value of the most basic oxide of vanadium among $V_2O_3,\ V_2O_4$ and V_2O_5 is_____ B.M. (Nearest integer)

Correct Answer:
3

Solution:
As we have learnt,

Oxides with lesser oxidation state of the metal ion is more basic.

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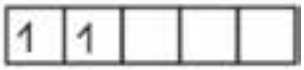
Thus, the most basic oxide among the given ones is V_2O_3 and the oxidation state of V is $+3$.

$\therefore V^{+3} \rightarrow [Ar]\, 3d^2$

$3d^2$: 

no.of unpaired electrons $= 2$.

$$\therefore \text{ magnetic moment}, \mu = \sqrt{n\,(n+2)}\ \text{B.M.}$$
$$= \sqrt{2\,(2+2)}\ \text{B.M}$$
$$= 2.84 \approx 3\ \text{B.M.}$$

Hence, 3 is the answer.

Q. 27 The spin-only magnetic moment value of an octahedral complex among $CoCl_3 \cdot 4NH_3$, $NiCl_2 \cdot 6H_2O$ and $PtCl_4 \cdot 2HCl$, which upon reaction with excess of $AgNO_3$ gives 2 moles of $AgCl$ is _____B.M. (Nearest Integer)

Correct Answer:
3

Solution:
The given complexes exist as

$CoCl_3 \cdot 4NH_3 : [Co(NH_3)_4Cl_2]Cl$

$NiCl_2 \cdot 6H_2O : [Ni\,(H_2O)_6]\,Cl_2$

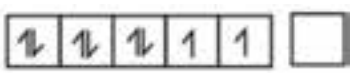
$PtCl_4 \cdot 2HCl : H_2\,[PtCl_6]$

Among the given complexes,

$NiCl_2 \cdot 6H_2O$ is an octahedral complex which gives 2 moles of $AgCl$ upon reaction with excess of $AgNO_3$.

$$[Ni\,(H_2O)_6]\,Cl_2 \xrightarrow{2AgNO_3} 2\,AgCl\downarrow + [Ni\,(H_2O)_6]\,(NO_3)_2$$

Ni is in +2 oxidation state.

Configuration $3d^8 4s^0$: 

No.of unpaired $e^- = 2$.

$$\text{magnetic moment} = \sqrt{n\,(n+2)}\ \text{B.M}$$
$$= \sqrt{2\,(2+4)}\ \text{BM}$$

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$$= 2.84 \approx 3 \text{ BM}$$

Hence 3 is answer

Q. 28 On complete combustion 0.30 g of an organic compound gave 0.20 g of carbon dioxide and 0.10 g of water. The percentage of carbon in the given organic compound is _____.(Nearest Integer)

Correct Answer:
18

Solution:
Combustion Reaction of given organic compound-

Balanced Reaction

$$C_xH_yO_z + \left(x + \frac{y}{4} - \frac{z}{2}\right) O_2 \rightarrow xCO_2 + \frac{y}{2} H_2O$$

$0.3\,g$ $0.2\,g$ $0.1\,g$

To find the percentage of carbon in given organic compound i.e. $C_xH_yO_z$, we need to find values of x, y, z.

$$\frac{^nCO_2}{^nH_2O} = \frac{x}{y/2} = \frac{0.2/44}{0.1/18} \Rightarrow \frac{2x}{y} = \frac{0.2 \times 18}{4.4}$$

$$\Rightarrow 2\frac{x}{y} = \frac{36}{44} \Rightarrow \frac{x}{y} = \frac{9}{22} \Rightarrow x = \frac{9}{22}y - (1)$$

Now,

$$\frac{^nC_xH_yO_z}{^nCO_2} = \frac{1}{x}$$

$$\Rightarrow \frac{0.3}{(12x + y + 16z)} \times \frac{(44)}{0.2} = \frac{1}{x}$$

$$\Rightarrow 132x = 24x + 2y + 32z$$

$$\Rightarrow \quad 54x = y + 16z$$

now putting the value of 'x' from equation (1) , we get

$$54\left(\frac{9}{22}y\right) = y + 16z$$

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$$\Rightarrow z = \frac{29}{22}y \quad -(2)$$

$\therefore$ We have -

$$x = \frac{9}{22}y,\ z = \frac{29}{22}y$$

Formula , $C_xH_yO_z$ can be written as $C_{\frac{9y}{22}}H_yO_{\frac{29y}{22}}$

which can be simplified into $C_9H_{22}O_{29}$

$$\begin{aligned}\%\text{ of carbon in } C_9H_{22}O_{29} &= \frac{\text{mass of C}}{\text{Total mass}} \times 100 \\ &= \frac{12 \times 9}{(12 \times 9 + 22 + 16 \times 29)} \times 100 \\ &= \frac{108}{594} \times 100 \\ &= 18.18\,\%\end{aligned}$$

Nearest integer is 18 which is the answer

Q. 29 Compound $'P'$ on nitration with $\text{dil. } HNO_3$ yields two isomers (A) and (B). These isomers can be separated by steam distillation. Isomers (A) and (B) show the intramolecular and intermolecular hydrogen bonding respectively. Compound (P) on reaction with $\text{conc. } HNO_3$ yields a yellow compound ' C ', a strong acid. The number of oxygen atoms is present in compound $'C'$_____.

Correct Answer:
7

Solution:
From the information given in question -

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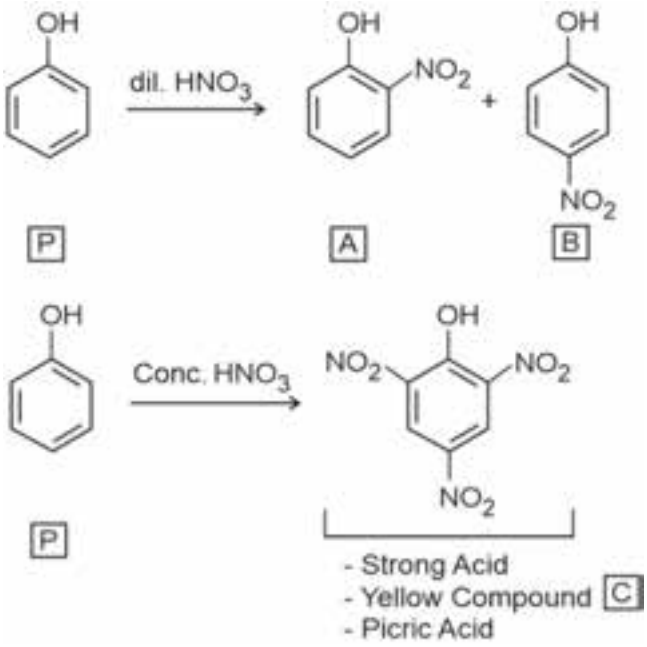


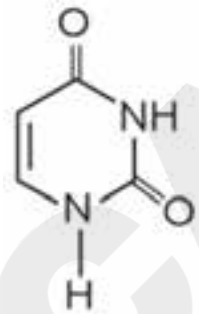
No.of oxygen in picric acid = 7
Hence 7 is the answer.

Q. 30 The number of oxygens present in a nucleotide formed from a base, that is present only in RNA is_____.

Correct Answer:
9

Solution:
Uracil is the base which is only present in RNA.

The structure of Uracil is -



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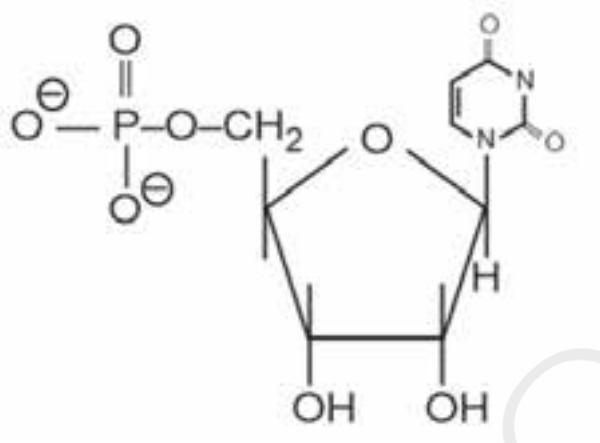
The structure of formed Nucleotide from the uracil is



The total number of oxygen atoms present = 9.

Maths

Q. 1 Let $f(x)=\frac{x-1}{x+1}, x \in \mathbf{R}-\{0,-1,1\}$. If $f^{n+1}(x)=f\left(f^{n}(x)\right)$ for all $n \in \mathbf{N}$, then $f^{6}(6)+f^{7}(7)$ is equal to:

Option 1:

$\frac{7}{6}$

Option 2:

$-\frac{3}{2}$

Option 3:

$\frac{7}{12}$

Option 4:

$-\frac{11}{12}$

Correct Answer:

$-\frac{3}{2}$

Solution:

$f(x)=\frac{x-1}{x+1}$

$\Rightarrow f^{2}(x)=f(f(x))=\frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1}=\frac{-1}{x}$

$\Rightarrow f^{3}(x)=f\left(f^{2}(x)\right) \doteq f\left(-\frac{1}{x}\right)=\frac{x+1}{1-x}$

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$$\Rightarrow f^4(x) = f\left(\frac{x+1}{1-x}\right) = -\frac{1}{x}$$

$$\Rightarrow f^6(x) = \frac{-1}{x} \Rightarrow f^6(6) = -\frac{1}{6}$$

$$f^7(x) = \frac{x+1}{1-x}$$

$$\Rightarrow f^7(7) = \frac{8}{-6} = -\frac{4}{3}$$

$$\therefore -\frac{1}{6} - \frac{4}{3} = \left(-\frac{3}{2}\right)$$

Hence the correct answer is option 2

Q. 2 $Let\ A = \left\{z \in \mathbf{C} : \left|\frac{z+1}{z-1}\right| < 1\right\}$ and $B = \left\{z \in \mathbf{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$.

$Then\ A \cap B\ is:$

Option 1:

a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

Option 2:

a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

Option 3:

an empty set

Option 4:

a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

Correct Answer:

a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

Solution:

Set A

$$\Rightarrow \left|\frac{z+1}{z-1}\right| < 1$$

$$\Rightarrow |z+1| < |z-1|$$

$$\Rightarrow (x+1)^2 + y^2 < (x-1)^2 + y^2$$

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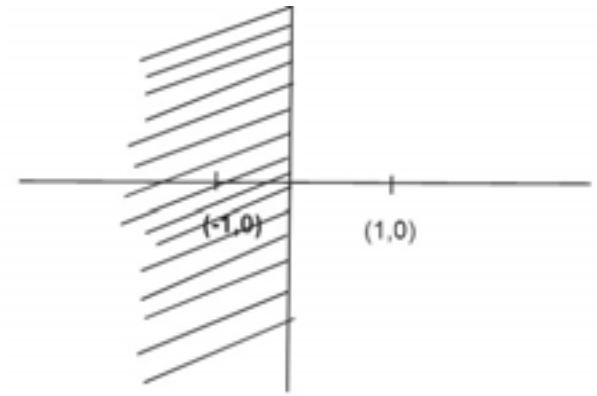
$\Rightarrow x < 0$



Set B

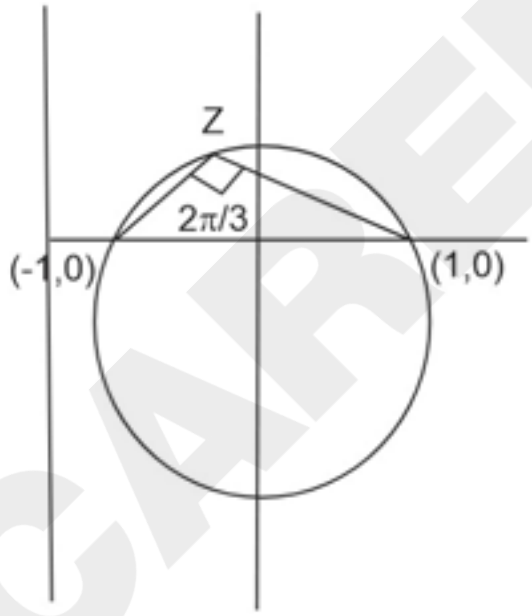
Using Rotation theorem and circular arc equation

$$\text{Centre}: \left(0, -\frac{1}{\sqrt{3}}\right)$$

Hence, $A \cap B$

$$\Rightarrow \text{Centre} \left(0, \frac{-1}{\sqrt{3}}\right)$$

Hence the correct answer is option 2.



Q. 3 Let

A be a 3×3 invertible matrix. If $|\operatorname{adj}(24\ A)| = |\operatorname{adj}(3\operatorname{adj}(2\ A))|$, then $|A|^2$ is equal to :

Option 1:

6^6

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Option 2:
2^{12}

Option 3:
2^{6}

Option 4:
1

Correct Answer:
2^{6}

Solution:
$|\operatorname{adj}(24 \cdot \mathrm{A})| = |\operatorname{adj} 3(\operatorname{adj} 2\mathrm{A})|$

$\Rightarrow |24A|^2 = |3\operatorname{adj}(2A)|^2$

$\Rightarrow \left(24^3|\mathrm{A}|\right)^2 = \left(3^3|\operatorname{adj}(2\mathrm{A})|\right)^2$

$= 3^6\left(|2\mathrm{A}|^2\right)^2$

$\Rightarrow 24^6|\mathrm{A}|^2 = 3^6 \times 2^{12} \cdot |\mathrm{A}|^4$

$\Rightarrow |\mathrm{A}|^2 = \dfrac{24^6}{3^6 \times 2^{12}} = 64$

Hence the correct answer is option 3

Q. 4 The ordered pair (a, b), for which the system of linear equations
$3x - 2y + z = b$
$5x - 8y + 9z = 3$
$2x + y + az = -1$
has no solution, is:

Option 1:
$\left(3, \dfrac{1}{3}\right)$

Option 2:
$\left(-3, \dfrac{1}{3}\right)$

Option 3:
$\left(-3, -\dfrac{1}{3}\right)$

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Option 4:

$\left(3, -\frac{1}{3}\right)$

Correct Answer:

$\left(-3, -\frac{1}{3}\right)$

Solution:

$$\begin{vmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{vmatrix} = 0$$

$3(-8a - 9) + 2(5a - 18) + 1(21) = 0$

$\Rightarrow a = -3$

$\text{Also } \Delta_2 = \begin{vmatrix} 3 & -2 & b \\ 5 & 8 & 3 \\ 2 & 1 & -1 \end{vmatrix}$

$\text{if } b = \frac{1}{3}, \Delta_2 = 0$

$\text{so b must be equal to } -\frac{1}{3}$

Hence the correct answer is option 3.

Q. 5 The remainder when $(2021)^{2023}$ is divided by 7 is:

Option 1:

1

Option 2:

2

Option 3:

5

Option 4:

6

Correct Answer:

5

Solution:

$(2021)^{2023} = (7\lambda - 2)^{2023}$

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$$= {}^{2023}C_0(7\lambda)^{2023} + \cdots - {}^{2023}C_{2023}2^{2023}$$

$$= 7t - 2^{2023}$$

$$\therefore -2^{2023} = -2 \times 2^{2022}$$

$$= -2 \times \left(2^3\right)^{674}$$

$$= -2(1+7\mu)^{674}$$

$$= -(7\alpha + 2)$$

$$\Rightarrow \text{ remainder } = -2 \text{ or } +5$$

Hence the correct answer is option 3

Q. 6 $\lim\limits_{x\to\frac{1}{\sqrt{2}}} \dfrac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$ is equal to:

Option 1:
$\sqrt{2}$

Option 2:
$-\sqrt{2}$

Option 3:
$\dfrac{1}{\sqrt{2}}$

Option 4:
$-\dfrac{1}{\sqrt{2}}$

Correct Answer:
$-\dfrac{1}{\sqrt{2}}$

Solution:

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sin\left(\sin^{-1}\sqrt{1-x^2}\right)-x}{1-\tan\left(\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sqrt{1-x^2}-x}{1-\left(\frac{\sqrt{1-x^2}}{x}\right)}$$

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$$\lim_{x\to\frac{1}{\sqrt{2}}}(-x) = -\frac{1}{\sqrt{2}}$$

Hence the correct answer is option 4

Q. 7 Let $f, g : \mathbf{R} \to \mathbf{R}$ be two real valued functions defined as $f(x) = \begin{cases} -|x+3|, & x < 0 \\ e^x & , x \geqslant 0 \end{cases}$ and $g(x) = \begin{cases} x^2 + k_1 x, & x < 0 \\ 4x + k_2, & x \geqslant 0 \end{cases}$, where k_1 and k_2 are real constants. If $(g \circ f)$ is differentiable at $x = 0$, then $(g \circ f)(-4) + (g \circ f)(4)$ is equal to :

Option 1:
$4\left(e^4 + 1\right)$

Option 2:
$2\left(2e^4 + 1\right)$

Option 3:
$4e^4$

Option 4:
$2\left(2e^4 - 1\right)$

Correct Answer:
$2\left(2e^4 - 1\right)$

Solution:

$$f(x) = \left\{ \begin{array}{ll} x+3; & x < -3 \\ -(x+3); & -3 \leq x < 0 \\ e^x & ; x \geq 0 \end{array} \right\}$$

$$g(x) = \left\{ \begin{array}{ll} x^2 + k_1 x & ; x < 0 \\ 4x + k_2 & ; x \geq 0 \end{array} \right\}$$

$$g(f(x)) = \left\{ \begin{array}{ll} f(x)^2 + k_1 f(x); & f(x) < 0 \\ 4f(x) + k_2; & f(x) \geq 0 \end{array} \right\}$$

$$g(f(x)) = \left\{ \begin{array}{ll} (x+3)^2 + k_1(x+3); & x < -3 \\ (x+3)^2 - k_1(x+3); & -3 \leq x < 0 \\ 4e^x + k_2 & ; \quad x > 0 \end{array} \right\}$$

Check Continuity at $x = 0$

$$gof(0) = g\left(f\left(0^-\right)\right) = g\left(f\left(0^+\right)\right)$$

$$4 + k_2 = 9 - 3k_1 = 4 + k_2$$

$$3k_1 + k_2 = 5 \qquad \text{.....(a)}$$

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Differentiate

$$(g(f(x))) = \left\{ \begin{array}{cc} 2(x+3)+k_1; & x<-3 \\ 2(x+3)-k_1 & ;-3 \le x<0 \\ 4e^x & ;x \ge 0 \end{array} \right\}$$

$6 - k_1 = 4$

$k_1 = 2$ .....(b)

$\therefore k_1 = 2 \quad , \quad k_2 = -1$

$$gof(x) = \left\{ \begin{array}{cc} (x+3)^2+2\cdot(x+3)\cdot; & x<-3 \\ (x+3)^2-2(x+3), & ;-3 \le x<0 \\ 4e^x-1 & ;x \ge 0 \end{array} \right\}$$

$gof(-4) + gof(4) = 4e^4 - 2$

$\Rightarrow 2\left(2e^4-1\right)$

Hence the correct answer is option 4.

Q. 8 The sum of the absolute minimum and the absolute maximum values of the function $f(x) = \left|3x - x^2 + 2\right| - x$ in the interval $[-1, 2]$ is :

Option 1:
$\frac{\sqrt{17}+3}{2}$

Option 2:
$\frac{\sqrt{17}+5}{2}$

Option 3:
5

Option 4:
$\frac{9-\sqrt{17}}{2}$

Correct Answer:
$\frac{\sqrt{17}+3}{2}$

Solution:

$$f(x) = \begin{cases} x^2-4x-2, & \forall x \in \left(-1, \frac{3-\sqrt{17}}{2}\right) \\ -x^2+2x+2, & \forall x \in \left(\frac{3-\sqrt{17}}{2}, 2\right) \end{cases}$$

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$f'(x)$ when $x \in \left(-1, \frac{3-\sqrt{17}}{2}\right)$

$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$

$f'(x) = 2(x-2)$

$\Rightarrow f'(x)$ is always $\downarrow$

$f(2) = 2$

$f(-1) = 3$

$\dot{f}\left(\frac{3-\sqrt{17}}{2}\right) = \frac{\sqrt{17}-3}{2}$

$f'(x)$ when $x \in \left(\frac{3-\sqrt{17}}{2}, 2\right)$

$f'(x) = -2x + 2$

$f'(x) = -2(x-1)$

$f'(x) = 0 \qquad$ when $x = 1$

$f(1) = 3$

Absolute minimum value $= \frac{\sqrt{17}-3}{2}$

Absolute maximum value $= 3$

Sum $= \frac{\sqrt{17}-3}{2} + 3 = \frac{\sqrt{17}+3}{2}$

Q. 9 Let S be the set of all the natural numbers, for which the line $\frac{x}{a} + \frac{y}{b} = 2$ is a tangent to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ at the point $(a, b), ab \neq 0$. Then:

Option 1:
$S = \phi$

Option 2:
$n(S) = 1$

Option 3:
$S = \{2k : k \in \mathbf{N}\}$

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Option 4:

$S = \mathbf{N}$

Correct Answer:

$S = \mathbf{N}$

Solution:

$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$

Slope of tangent at(a, b)

$n \cdot \left(\frac{x}{a}\right)^{n-1} \cdot \frac{1}{a} + n \left(\frac{y}{b}\right)^{n-1} \frac{1}{b} \frac{dy}{dx} = 0$

$\left.\frac{dy}{dx}\right|_{(a,b)} = -\frac{b}{a}$

$\therefore$ Equation of tangent

$(y - b) = -\frac{b}{a}(x - a)$

$\frac{x}{a} + \frac{y}{b} = 2 \quad : \forall \quad n \in N$

Hence the correct answer is option 4

Q. 10 The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is:

Option 1:

$4(2\sqrt{3} + \sqrt{6} - 4)$

Option 2:

$4(4\sqrt{3} + \sqrt{6} - 4)$

Option 3:

$8(4\sqrt{3} + 3\sqrt{6} - 9)$

Option 4:

$8(4\sqrt{3} + 2\sqrt{6} - 9)$

Correct Answer:

$8(4\sqrt{3} + 2\sqrt{6} - 9)$

Solution:

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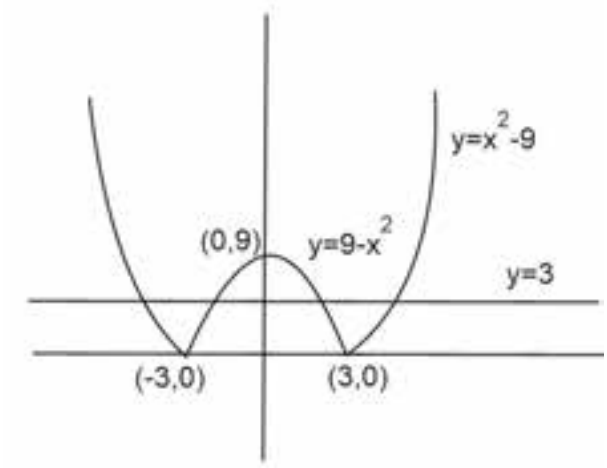

$$= 2\int_0^3 (\sqrt{9+y} - \sqrt{9-y})dy + 2\int_3^9 (\sqrt{9-y})dy$$

$$= 2\left[\int_0^3 (9+y)^{1/2}dy - \int_0^3 (9-y)^{1/2}dy + \int_3^9 (9-y)^{1/2}dy\right]$$

$$= 2\left[\frac{2}{3}\left[(9+y)^{3/2}\right]_0^3 + \frac{2}{3}\left[(9-y)^{3/2}\right]_0^3 - \frac{2}{3}\left[(9-y)^{3/2}\right]_3^9\right]$$

$$= \frac{4}{3}[12\sqrt{12} - 27 + 6\sqrt{6} - 27 - (0 - 6\sqrt{6})]$$

$$= \frac{4}{3}[24\sqrt{3} + 12\sqrt{6} - 54]$$

$$= 8[4\sqrt{3} + 2\sqrt{6} - 9]$$

Hence the correct answer is option 4

Q. 11 Let R be the point $(3, 7)$ and let P and Q be two points on the line $x + y = 5$ such that PQR is an equilateral triangle. Then the area of $\triangle\mathrm{PQR}$ is:

Option 1:

$\frac{25}{4\sqrt{3}}$

Option 2:

$\frac{25\sqrt{3}}{2}$

Option 3:

$\frac{25}{\sqrt{3}}$

Option 4:

$\frac{25}{2\sqrt{3}}$

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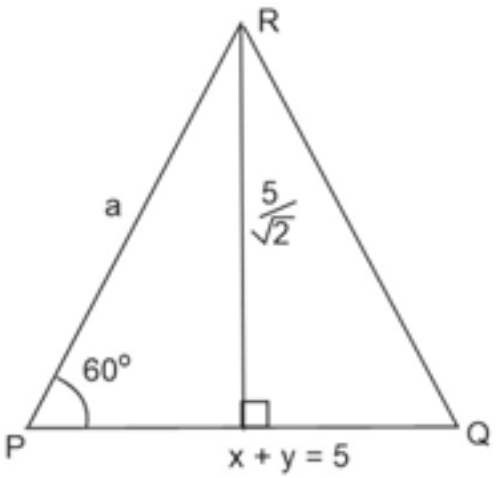
Correct Answer:

$\frac{25}{2\sqrt{3}}$

Solution:



$$\sin 60^\circ = \frac{5/\sqrt{2}}{a}$$

$$a = \frac{5\sqrt{2}}{3}$$

$$\text{Area of } \triangle PQR = \frac{\sqrt{3}}{4}a^2 = \frac{25}{2\sqrt{3}}$$

Hece the correct answer is option 4.

Q. 12 Let C be a circle passing through the points $A(2, -1)$ and $B(3, 4)$. The line segment AB is not a diameter of C. If r is the radius of C and its centre lies on the circle $(x-5)^2 + (y-1)^2 = \frac{13}{2}$, then r^2 is equal to:

Option 1:

32

Option 2:

$\frac{65}{2}$

Option 3:

$\frac{61}{2}$

Option 4:

30

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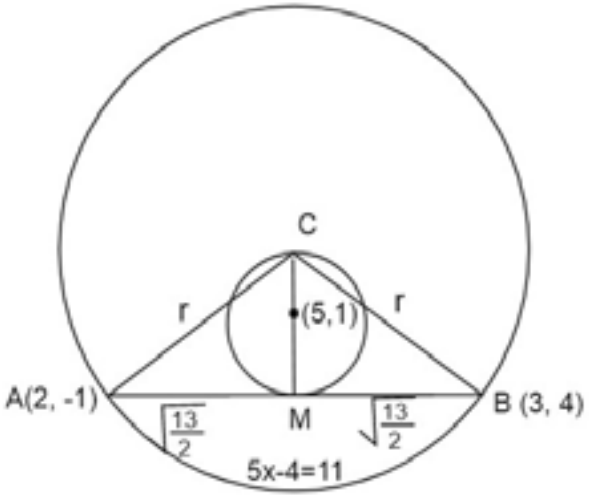
Correct Answer:

$\frac{65}{2}$

Solution:



$AB = \sqrt{26}$

$r^2 = CM^2 + AM^2$

$= \left(2.\sqrt{\frac{13}{2}}\right)^2 + \left(\sqrt{\frac{13}{2}}\right)^2$

$r^2 = \frac{65}{2}$

Q. 13 Let the normal at the point P on the parabola $y^2 = 6x$ pass through the point $(5, -8)$. If the tangent at P to the parabola intersects its directrix at the point Q, then the ordinate of the point Q is:

Option 1:

-3

Option 2:

$-\frac{9}{4}$

Option 3:

$-\frac{5}{2}$

Option 4:

-2

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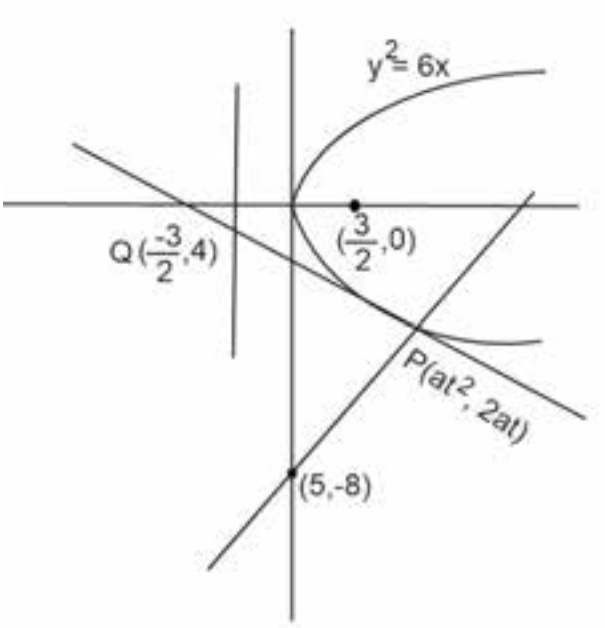
Correct Answer:

$-\frac{9}{4}$

Solution:



Equation of normal : $\mathrm{y} = -\mathrm{tx} + 2\mathrm{at} + \mathrm{at}^3 \qquad \because \left(\mathrm{a} = \frac{3}{2}\right)$

Since passing through $(5, -8)$, we get $\mathrm{t} = -2$

Co-ordinate of $\mathrm{Q} : (6, -6)$

Equation of tangent at $\mathrm{Q} : \mathrm{x} + 2\mathrm{y} + 6 = 0$

Put $\mathrm{x} = \frac{-3}{2}$ to get $\mathrm{R} = \left(\frac{-3}{2}, \frac{-9}{4}\right)$

Hence the correct answer is option 2.

Q. 14 If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is:

Option 1:

$\cos^{-1}\left(\frac{29}{4}\right)$

Option 2:

$\sec^{-1}\left(\frac{29}{4}\right)$

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Option 3:

$\cos^{-1}\left(\frac{2}{29}\right)$

Option 4:

$\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Correct Answer:

$\sec^{-1}\left(\frac{29}{4}\right)$

Solution:

$l_1: \frac{x-2}{3}=\frac{y+1}{-2}=\frac{z-2}{0}$

$l_2: \frac{x-1}{1}=\frac{y+3/2}{\alpha/2}=\frac{z+5}{2}$

$l_3: \frac{x-1}{-3}=\frac{y-\frac{1}{2}}{-2}=\frac{z-0}{4}$

$l_1 \text{ perpendicular } l_2 :\Rightarrow \frac{|3-\alpha+0|}{\sqrt{13}\sqrt{1+\frac{\alpha^2}{4}}+4}=0\Rightarrow \alpha=3$

Angle between l_2 and l_3

$\cos\theta=\frac{|1\times(-3)+(-2)(\alpha/2)+2\times 4\,|}{\sqrt{1+4+\frac{\alpha^2}{4}}\sqrt{9+16+4}}$

$\cos\theta=\frac{|\,-3-\alpha+8\,|}{\sqrt{5+\frac{\alpha^2}{4}}\sqrt{29}} \quad \text{put } \alpha=3$

$\cos\theta=\frac{2}{\sqrt{\frac{29}{4}}\sqrt{29}}=\frac{4}{29}$

$\theta=\cos^{-1}\left(\frac{4}{29}\right)\Rightarrow\theta=\sec^{-1}\left(\frac{29}{4}\right)$

Hence the correct asnwer is option 2

Q. 15 Let the plane $2x+3y+z+20=0$ be rotated through a right angle about its line of intersection with the plane $x-3y+5z=8$. If the mirror image of the point $\left(2,-\frac{1}{2},2\right)$ in the rotated plane is $B(a,b,c)$, then:

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Option 1:

$$\frac{a}{8}=\frac{b}{5}=\frac{c}{-4}$$

Option 2:

$$\frac{a}{4}=\frac{b}{5}=\frac{c}{-2}$$

Option 3:

$$\frac{a}{8}=\frac{b}{-5}=\frac{c}{4}$$

Option 4:

$$\frac{a}{4}=\frac{b}{5}=\frac{c}{2}$$

Correct Answer:

$$\frac{a}{8}=\frac{b}{5}=\frac{c}{-4}$$

Solution:

Let equation of rotated plane be :

$$(2x+3y+z+20)+\lambda(x-3y+5z-8)=0$$

$$(2+\lambda)x+(3-3\lambda)y+1(1+5\lambda)z+20-8\lambda=0$$

Above plane is perpendicular to $2x+3y+z+20=0$

So, $(2+\lambda)\cdot 2+(3-3\lambda)\cdot 3+(1+5\lambda)\cdot 1=0 \Rightarrow \lambda=7$

$\Rightarrow$ Equation of rotated plane:

$$x-2y+4z-4=0$$

Mirror image of $A\left(2,\frac{-1}{2},2\right)$ in rotated plane is $B(a,b,c)$

Equation of $AB: \frac{x-2}{1}=\frac{y+1/2}{-2}=\frac{z-2}{4}=k$

Let coordinate of B be $\left(2+k,-\frac{1}{2}-2k,2+4k\right)$

Midpoint of AB is $\left(2+\frac{k}{2},-\frac{1}{2}-k,2+2k\right)$ which will lie on the plane $x-2y+4z-4=0$

$$\text{Hence } k=\frac{-2}{3}$$

$$\text{Therefore B is } \left(\frac{4}{3},\frac{5}{6},-\frac{2}{3}\right)\equiv\left(\frac{8}{6},\frac{5}{6},\frac{-4}{6}\right)$$

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So, $\frac{a}{8}=\frac{b}{5}=\frac{c}{-4}$

Hence the correct answer is option 1.

Q. 16 If $\vec{a}\cdot\vec{b}=1,\vec{b}\cdot\vec{c}=2$ and $\vec{c}\cdot\vec{a}=3$, then the value of $[\vec{a}\times(\vec{b}\times\vec{c}),\vec{b}\times(\vec{c}\times\vec{a}),\vec{c}\times(\vec{b}\times\vec{a})]$ is:

Option 1:
0

Option 2:
$-6\vec{a}\cdot(\vec{b}\times\vec{c})$

Option 3:
$12\vec{c}\cdot(\vec{a}\times\vec{b})$

Option 4:
$-12\vec{b}\cdot(\vec{c}\times\vec{a})$

Correct Answer:
0

Solution:

$\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=3\vec{b}-\vec{c}$

$\vec{b}\times(\vec{c}\times\vec{a})=(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}=\vec{c}-2\vec{a}$

$\vec{c}\times(\vec{b}\times\vec{a})=(\vec{c}\cdot\vec{a})\vec{b}-(\vec{c}\cdot\vec{b})\vec{a}=3\vec{b}-2\vec{a}$

So the required expression equals:

$[3\vec{b}-\vec{c},\vec{c}-2\vec{a},3\vec{b}-2\vec{a}]$

$(3\vec{b}-\vec{c})\cdot[(\vec{c}-2\vec{a})\times(3\vec{b}-2\vec{a})]$

$(3\vec{b}-\vec{c})\cdot\left[3(\vec{c}\times\vec{b})-2(\vec{c}\times\vec{a})-6(\vec{a}\times\vec{b})\right]$

$-6[\vec{b}\ \vec{c}\ \vec{a}]+6[\vec{c}\ \vec{a}\ \vec{b}]$

$=0$

Q. 17 Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 5 heads, then the probability of getting atmost two heads is:

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Option 1:
$\frac{275}{6^5}$

Option 2:
$\frac{36}{5^4}$

Option 3:
$\frac{181}{5^5}$

Option 4:
$\frac{46}{6^4}$

Correct Answer:
$\frac{46}{6^4}$

Solution:
Let $\mathrm{P(H)=x} \quad \mathrm{P(T)=1-x}$

$\mathrm{P(4H,1T)=P(5H)}$

$^5\mathrm{C}_4\mathrm{x}^4(1-\mathrm{x})^1 = {}^5\mathrm{C}_5\mathrm{x}^5$

$5(1-\mathrm{x})=\mathrm{x}$

$6\mathrm{x}=5=0 \quad :\mathrm{x}=\frac{5}{6}$

$\mathrm{P(atmost\ 2H)}$

$=\mathrm{P(0H,5T)+P(1H,4T)+P(2H,3T)}$

$={}^5\mathrm{C}_0\left(\frac{1}{6}\right)^5+{}^5\mathrm{C}_1\frac{5}{6}\left(\frac{1}{6}\right)^4+{}^5\mathrm{C}_2\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)^3$

$=\frac{1}{6^5}(1+25+250)=\frac{276}{6^5}$

$=\frac{46}{6^4}$

Q. 18 The mean of the numbers $a, b, 8, 5, 10$ is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then $25\mathrm{M}$ is equal to :

Option 1:
60

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Option 2:

55

Option 3:

50

Option 4:

45

Correct Answer:

60

Solution:

$\text{mean } = 6$

$$\frac{a+b+8+5+10}{5} = 6$$

$$a+b=7$$

$$b=7-a$$

Variance:

$$\sigma^2 = \frac{\sum_{i=1}^{5}(x_i-\bar{x})^2}{n}$$

$$6.8 = \frac{(a-6)^2+(b-6)^2+(8-6)^2+(5-6)^2+(10-6)^2}{5}$$

$$34 = (a-6)^2+(7-a-6)^2+4+1+18$$

$$a^2-7a+12=0 \Rightarrow a=4 \text{ or } a=3$$

$$a=4 \quad a=3$$

$$b=3 \quad b=4$$

Mean deviation about mean

$$M = \frac{\sum_{i=1}^{5}|x_i-x|}{n}$$

$$M = \frac{|a-6|+|b-6|+|8-6|+|5-6|+|10-6|}{5}$$

$\text{when } a=3, b=4$ and $\text{when } a=4, b=3$

$$M = \frac{3+2+2+1+4}{5} \quad \text{and} \quad M = \frac{2+3+2+1+7}{5}$$

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$M = \frac{12}{5}$ and $M = \frac{12}{5}$

$25M = 25 \times \frac{12}{5} = 60$

Hece the correct answer is option 1.

Q. 19 Let $f(x) = 2\cos^{-1} x + 4\cot^{-1} x - 3x^2 - 2x + 10, x \in [-1, 1]$. If $[a, b]$ is the range of the function f, then $4a - b$ is equal to:

Option 1:
11

Option 2:
$11 - \pi$

Option 3:
$11 + \pi$

Option 4:
$15 - \pi$

Correct Answer:
$11 - \pi$

Solution:

$$f'(x) = \frac{-2}{\sqrt{1-x^2}} - \frac{4}{1+x^2} - 6x - 2$$

$$= -2\left[\frac{1}{\sqrt{1-x^2}} + \frac{2}{1+x^2} + 3x + 1\right]$$

$f'(x) < 0 \Rightarrow f(x)$ is a decreasing function

$f(1) = \pi + 5$

$f(-1) = 5\pi + 5$

Range: $[a, b] \equiv [\pi + 5, 5\pi + 5]$

$a = \pi + 5, b = 5\pi + 5 \Rightarrow 4a - b = 11 - \pi$

Hence the correct answer is option 2

Q. 20 Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $p \nabla q \Rightarrow ((p \Delta q) \nabla r)$ is a tautology. Then $(p \nabla q) \Delta r$ is logically equivalent to:

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Option 1:
$(p\Delta r) \vee q$

Option 2:
$(\mathrm{p}\Delta \mathrm{r}) \wedge \mathrm{q}$

Option 3:
$(\mathrm{p} \wedge \mathrm{r})\Delta \mathrm{q}$

Option 4:
$(p\nabla r) \wedge q$

Correct Answer:
$(p\Delta r) \vee q$

Solution:
$\text{Case-I if } \Delta \equiv \nabla \equiv \wedge$

$(\mathrm{p} \wedge \mathrm{q}) \rightarrow ((\mathrm{p} \wedge \mathrm{q}) \wedge \mathrm{r})$

It can be false if r is false

So not a tautology

$\text{Case-II if } \Delta \equiv \nabla \equiv \mathrm{V}$

$(\mathrm{p} \vee \mathrm{q}) \rightarrow ((\mathrm{p} \vee \mathrm{q}) \vee \mathrm{r}) \equiv \text{ tautology}$

$\text{then } (\mathrm{p} \vee \mathrm{q}) \vee \mathrm{r} \equiv (\mathrm{p}\Delta \mathrm{r}) \vee \mathrm{q}$

$\text{Case} - \text{III if } \Delta = \mathrm{V}, \nabla = \Lambda$

$\text{then}(\mathrm{p} \wedge \mathrm{q}) \rightarrow \{(\mathrm{p} \vee \mathrm{q}) \wedge \mathrm{r}\} : \text{Not a tautology}$

$(\text{ check } \mathrm{p} \rightarrow \mathrm{T}, \mathrm{q} \rightarrow \mathrm{T}, \mathrm{r} \rightarrow \mathrm{F})$

$\text{Case} - \text{IV if } \Delta = \wedge, \quad \nabla = \mathrm{v}$

$(\mathrm{p} \vee \mathrm{q}) \rightarrow \{(\mathrm{p} \wedge \mathrm{q}) \vee \mathrm{r}\}$

Not a tautology

Q. 21 The sum of the cubes of all the roots of the equation $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ *is* _____.

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Correct Answer:
36

Solution:
$x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$

$x = 0$ is ot the root of this eqn, so divide it by x^2

$$x^2 - 3x - 2 + \frac{3}{x} + \frac{1}{x^2} = 0$$

$$x^2 + \frac{1}{x^2} - 2 + 2 - 3\left(x - \frac{1}{x}\right) - 2 = 0$$

$$\left(x - \frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right) = 0$$

$$x - \frac{1}{x} = 0 \qquad x - \frac{1}{x} = 3$$

$$x^2 - 1 = 0 \qquad x^2 - 3x - 1 = 0$$

$$x = \pm 1 \qquad \gamma + \delta = 3.$$

$$\alpha = 1, \beta = -1, \gamma\delta = -1$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3$$

$$1 - 1 + (\gamma + \delta)((\gamma + \delta)^2 - 3\gamma\delta)$$

$$0 + 3(9 - 3(-1))$$

$$3(12) = 36$$

Hence answer is 36

Q. 22 There are ten boys $B_1, B_2, \ldots, B_{10}$ and five girls $G_1, G_2, \ldots, G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of a group, is_____.

Correct Answer:
1120

Solution:
$n(B) = 10, \quad n(a) = 5$

The number of ways of forming a group of 3 girls of 3 boys

$= {}^{10}C_3 \times {}^{5}C_3$

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$$= \frac{10 \times 9 \times 8}{3 \times 2} \times \frac{5 \times 4}{2} = 1200$$

The number of ways whe two particular boys $\mathrm{B}_1 \text{ and } \mathrm{B}_2$ be the member of group together

$$= {}^{8}\mathrm{C}_1 \times {}^{5}\mathrm{C}_3 = 8 \times 10 = 80$$

Number of ways boys $\mathrm{B}_1 \text{ and } \mathrm{B}_2$ are not in the same group together

Required no.of selections = $1200 - 80 = 1120$

Hence answer is 1120

Q. 23 Let the common tangents to the curves $4\left(x^2+y^2\right)=9 \text{ and } y^2=4x$ intersect at the point Q. Let an ellipse, centered at the origin O, has lengths of semi-minor and semi-major axes equal to $\mathrm{O}Q$ and 6 , respectively. If e and l respectively denote the eccentricity and the length of the latus rectum of this ellipse, then $\frac{l}{\mathrm{e}^2}$ is equal to_____.

Correct Answer:
4

Solution:
$x^2+y^2=\frac{9}{4} \quad and \quad y^2=4x$

Eqn tangent in slope form

$$\mathrm{y}=\mathrm{mx} \pm \frac{3}{2}\sqrt{\left(1+\mathrm{m}^2\right)} \quad \text{.....(1)}$$

$$\mathrm{y}=\mathrm{mx}+\frac{1}{\mathrm{m}} \quad \text{.....(2)}$$

Compare (1) and (2)

$$\pm\frac{3}{2}\sqrt{\left(1+\mathrm{m}^2\right)}=\frac{1}{\mathrm{m}}$$

$$9\mathrm{m}^2\left(1+\mathrm{m}^2\right)=4$$

$$9\mathrm{m}^4+9\mathrm{m}^2-4=0$$

$$9\mathrm{m}^4+12\mathrm{m}^2-3\mathrm{m}^2-4=0$$

$$3\mathrm{m}^2\left(3\mathrm{m}^2+4\right)-1\left(3\mathrm{m}^2+4\right)=0$$

$$\mathrm{m}^2=-\frac{4}{3}(\text{ rejected })$$

$$\mathrm{m}^2=\frac{1}{3} \Rightarrow \mathrm{m}=\pm\frac{1}{\sqrt{3}}$$

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Eqn of common tangent

$$y = \frac{1}{\sqrt{3}}x + \sqrt{3} \quad \text{on x axis } y = 0$$

$$OQ = -3 \quad b = |OQ| = 3 \quad a = 6$$

$$b^2 = a^2\left(1 - e^2\right) \Rightarrow e^2 = 1 - \frac{9}{36} \Rightarrow e = \left(\frac{3}{4}\right)$$

$$l = \frac{2b^2}{a} = \frac{2 \times 9}{6} = 3$$

$$\frac{l}{e^2} = \frac{3}{3/4} = 4$$

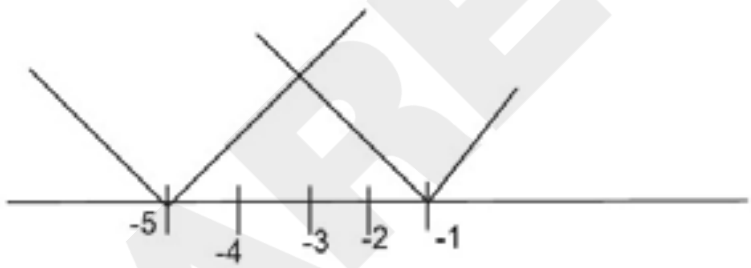
Hence answer is 4.

Q. 24

Let $f(x) = \max\{|x+1|, |x+2|, \ldots, |x+5|\}$. Then $\int_{-6}^{0} f(x)dx$ is equal to_____.

Correct Answer:
21

Solution:
$f(x) = \max\{|x+1|, |x+2|, |x+3|, |x+4|, |x+5|\}$



$$\int_{-6}^{0} f(x)dx = \int_{-6}^{-3} |x+1|dx + \int_{-3}^{0} |x+5|dx$$

$$= -\int_{-6}^{-3} (x+1)dx + \int_{-3}^{0} (x+5)dx$$

$$= -\left[\frac{x^2}{2} + x\right]_{-6}^{-3} + \left[\frac{x^2}{2} + 5x\right]_{-3}^{0}$$

$$= -\left[\left(\frac{9}{2} - 3\right) - (18 - 6)\right] + \left[0 - \left(\frac{9}{2} - 15\right)\right]$$

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$$= -\left[\frac{3}{2} - 12\right] + \frac{21}{2} = \frac{21}{2} + \frac{21}{2} = 21$$

Hence answer is 21

Q. 25 Let the solution curve $y = y(x)$ of the differential equation $\left(4 + x^2\right) dy - 2x\left(x^2 + 3y + 4\right) dx = 0$ pass through the origin. Then $y(2)$ is equal to_____.

Correct Answer:
12

Solution:

$\left(4 + x^2\right) dy - 2x\left(x^2 + 3y + 4\right) dx = 0$

$\left(x^2 + 4\right) \frac{dy}{dx} = 2x^3 + 6xy + 8x$

$\left(x^2 + 4\right) \frac{dy}{dx} - 6xy = 2x^3 + 8x$

$\frac{dy}{dx} - \frac{6x}{x^2 + 4} y = \frac{2x^3 + 8x}{x^2 + 4}$

L.I. $\frac{dy}{dx} + Py = \phi \quad P = \frac{-6x}{x^2 + 4} \quad \phi = \frac{2x^3 + 8x}{x^2 + 4}$

$I.F = e^{-\int \frac{6x}{x^2+4} dx} = e^{-3\log_e\left(x^2+4\right)}$

$= e^{\log_e\left(x^2+4\right)^{-3}} = \frac{1}{\left(x^2 + 4\right)^3}$

hence, its solution is

$y \cdot \frac{1}{\left(x^2 + 4\right)^3} = \int \frac{2x^3 + 8x}{\left(x^2 + 4\right)^3 \left(x^2 + 4\right)} dx$

$\frac{y}{\left(x^2 + 4\right)^3} = \int \frac{2x\left(x^2 + 4\right)}{\left(x^2 + 4\right)^3 \left(x^2 + 4\right)} dx \qquad \text{let } x^2 + 4 = t \text{ so } 2xdx = dt$

$\frac{y}{\left(x^2 + 4\right)^3} = \int \frac{dt}{t^3} \Rightarrow \frac{y}{\left(x^2 + 4\right)^3} = \frac{-1}{2\left(x^2 + 4\right)^2} + c$

Passes through origin $(0, 0)$

$0 = \frac{-1}{2 \times 18} + c$

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$$\frac{y}{(x^2+4)^3} = \frac{-1}{2(x^2+4)^2} + \frac{1}{32} \Rightarrow y = \frac{-(x^2+4)}{2} + \frac{(x^2+4)^3}{32}$$

$$y(2) = -\frac{8}{2} + \frac{8 \times 8 \times 8}{32} = 12$$

Hence answer is 12

Q. 26 If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to_____.

Correct Answer:
80

Solution:

$$\sin 10^\circ \left(\frac{1}{2} \cdot 2\sin 20^\circ \cdot \sin 40^\circ\right) \cdot \sin 10^\circ \cdot \sin(60^\circ - 10^\circ) \cdot \sin(60^\circ + 10^\circ)$$

$$\sin 10^\circ . \frac{1}{2} \cdot (\cos 20^\circ - \cos 60^\circ) \cdot \frac{1}{4}\sin 30^\circ$$

$$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{2}\sin 10^\circ \left(\cos 20^\circ - \frac{1}{2}\right)$$

$$= \frac{1}{32}(2\sin 10^\circ \cdot \cos 20^\circ - \sin 10^\circ)$$

$$= \frac{1}{32}(\sin 30^\circ - \sin 10^\circ - \sin 10^\circ)$$

$$= \frac{1}{32}\left(\frac{1}{2} - 2\sin 10^\circ\right)$$

$$= \frac{1}{64}(1 - 4\sin 10^\circ)$$

$$= \frac{1}{64} - \frac{1}{16}\sin 10^\circ$$

$$\text{Hence } \alpha = \frac{1}{64}$$

$$16 + \alpha^{-1} = 80$$

Hence answer 80

Q. 27 Let $A = \{n \in \mathbf{N} : \text{H.C.F. } (n, 45) = 1\}$ and
Let $B = \{2k : k \in \{1, 2, \ldots\ldots, 100\}\}$. Then the sum of all the elements of $A \cap B$ is_____.

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Correct Answer:
5264

Solution:
Sum of elements in $\mathrm{A} \cap \mathrm{B}$

$$= \underbrace{(2+4+6+\cdots+200))}_{\text{multiple of 2}} - \underbrace{(6+12+\cdots+198)}_{\text{mutiple of 2\&3}}$$

$$- \underbrace{(10+20+\cdots+200)}_{\text{mutiple of 5\&2 ie 10}} + \underbrace{(30+60+\cdots\cdot+180}_{\text{multipleof2,5\&3 ie 30}}$$

$$= 5264$$

Hence answer is 5264

Q. 28 The value of the integral $\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2}-x^3\right)\frac{\sin x}{1+\cos^2 x}\,dx$ is equal to_____.

Correct Answer:
6

Solution:
$$\mathrm{I} = \frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi \mathrm{x}^2}{2}-\mathrm{x}^3\right)\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\mathrm{dx}$$

$$\mathrm{I} = \frac{48}{\pi^4}\int_0^{\pi}\mathrm{x}^2\left(\frac{3\pi}{2}-\mathrm{x}\right)\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\mathrm{dx} \quad \text{.....(1)}$$

Apply King Property

$$\mathrm{I} = \frac{48}{\pi^4}\int_0^{\pi}(\pi-\mathrm{x})^2\left(\frac{\pi}{2}+\mathrm{x}\right)\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\mathrm{dx} \quad \text{.....(2)}$$

Eqn (1) + (2)

$$\mathrm{I} = \frac{12}{\pi^3}\int_0^{\pi}\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\left[\pi^2+(\pi-2)\mathrm{x}(\pi-2\mathrm{x})\right]\mathrm{dx} \text{.....(3)}$$

Apply Kings's Property again

$$\mathrm{I} = \frac{12}{\pi^3}\int_0^{\pi}\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\left[\pi^2+(\pi-2)(\pi-\mathrm{x})(2\mathrm{x}-\pi)\right]\mathrm{dx} \text{.....(4)}$$

Adding eqn (3) and (4)

$$\mathrm{I} = \frac{6}{\pi^2}\int_0^{\pi}\frac{\sin \mathrm{x}}{1+\cos^2 \mathrm{x}}\left[2\pi+(\pi-2)(\pi-2\mathrm{x})\right]\mathrm{dx} \text{.....(5)}$$

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Apply Kings Property again

$$I = \frac{6}{\pi^2}\int_0^{\pi} \frac{\sin x}{(1+\cos^2 x)}[2\pi + (\pi-2)(2x-\pi)]dx \quad \text{.....(6)}$$

Eqn (5)+(6)

$$I = \frac{12}{\pi}\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$$

$$\text{Let } \cos x = t \Rightarrow \sin x dx = -dt$$

$$I = \frac{12}{\pi}\int_1^{-1} \frac{-dt}{1+t^2} = \frac{-12}{\pi}\left[\tan^{-1}-1-\tan^{-1}1\right] = \frac{-12}{\pi}\times\frac{-2\pi}{4}$$

$$= 6$$

Hence answer is 6

Q. 29 Let $A = \sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ and $B = \sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$. Then $A+B$ is equal to _____.

Correct Answer:
1100

Solution:

$$A = \sum_{i=1}^{10}\sum_{J=1}^{10}\min\{i,j\}$$

$$B = \sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$$

$$A = \sum_{i=1}^{10}\min(i,1)+\min(i,2)+\cdots+\min(i,10)$$

$$= \underbrace{(1+1++\cdots+1)}_{19 \text{ time}} + \underbrace{((2+2+2+\cdots+2)}_{17 \text{ times}} + \underbrace{(3+3+3+\cdots+3)}_{15 \text{ time}}$$

$$+\ldots\cdot(10)1 \text{ times}$$

$$B = \sum_{i=1}^{10}\max(i,1)+\max(i,2)+\cdots+\max(i,10)$$

$$= \underbrace{10+10+\cdots+10)}_{19 \text{ times}} + \underbrace{(9+9+\cdots+9)}_{17 \text{ times}} + \cdots + (1)\text{times}$$

$$A+B = 20(1+2+3+\cdots+10)$$

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$$= 20 \times \frac{10 \times 11}{2} = 10 \times 110 = 1100$$

Hence answer 1100

Q. 30 Let $S = (0, 2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$. Let $y = y(x), x \in S$, be the solution curve of the differential equation $\frac{dy}{dx} = \frac{1}{1+\sin 2x}, y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve $y = y(x)$ with the curve $y = \sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to _____.

Correct Answer:
42

Solution:

$$\frac{dy}{dx} = \frac{1}{1+\sin 2x}$$

$$\int dy = \int \frac{dx}{(\sin x + \cos x)^2}$$

$$\int dy = \int \frac{\sec^2 x}{(1+\tan x)^2}$$

$$y(x) = -\frac{1}{1+\tan x} + c$$

$$y\left(\frac{\pi}{4}\right) = \frac{1}{2} = \frac{-1}{2} + c$$

$$c = 1$$

$$y(x) = \frac{-1}{1+\tan x} + 1$$

$$y(x) = \frac{-1+1+\tan x}{1+\tan x}$$

$$y(x) = \frac{\tan x}{1+\tan x}$$

Solving with $y = \sqrt{2}\sin x$

$$\frac{\tan x}{1+\tan x} = \sqrt{2}\sin x$$

$$\sin x = 0 \quad , \frac{1}{\sqrt{2}} = \sin x + \cos x$$

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$x = \pi$ $\quad \frac{1}{2} = \sin\left(x + \frac{\pi}{4}\right)$

$$\sin\frac{\pi}{6} = \sin\left(x + \frac{\pi}{4}\right)$$

$$x + \frac{\pi}{4} = \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}$$

$$x = \frac{5\pi}{6} - \frac{\pi}{4}$$

$$x = \frac{7\pi}{12}$$

$$x = \frac{13\pi}{6} - \frac{\pi}{4}$$

$$x = \frac{23\pi}{12}$$

Sum of solution

$$= \pi + \frac{7\pi}{12} + \frac{23\pi}{12}$$

$$= \frac{42\pi}{12} = \frac{k\pi}{12}$$

$$\Rightarrow k = 42$$

Hence answer is 42

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Online 26th June Evening Shift

Physics

Q. 1 The dimension of mutual inductance is :

Option 1:
$[\mathrm{ML^2T^{-2}A^{-1}}]$

Option 2:
$[\mathrm{ML^2T^{-3}A^{-1}}]$

Option 3:
$[\mathrm{ML^2T^{-2}A^{-2}}]$

Option 4:
$[\mathrm{ML^2T^{-3}A^{-2}}]$

Correct Answer:
$[\mathrm{ML^2T^{-2}A^{-2}}]$

Solution:
The dimension of mutual inductance is

$$[\mathrm{M}] = \frac{[\mathrm{e}]}{\left[\frac{\mathrm{di}}{\mathrm{dt}}\right]} = \frac{[\mathrm{W/Q}]}{\left[\frac{\mathrm{I}}{\mathrm{t}}\right]} = \left[\frac{\mathrm{W}}{\mathrm{I^2}}\right]$$
$$[\mathrm{M}] = [\mathrm{M^1L^2T^{-2}A^{-2}}]$$

The correct option is (3)

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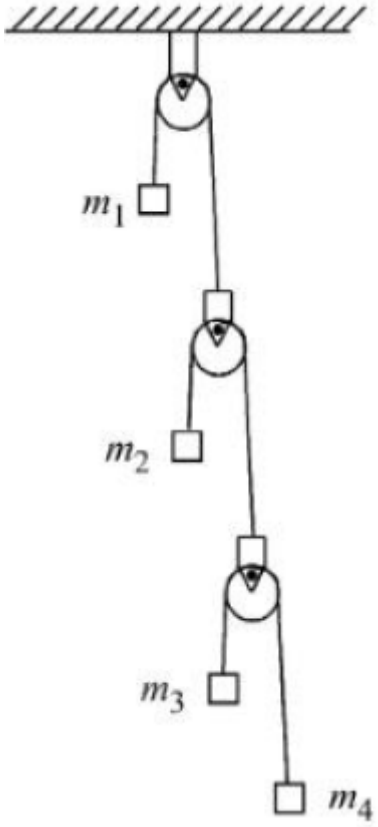
Q. 2 In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement ?



Option 1:
$4a_1 + 2a_2 + a_3 + a_4 = 0$

Option 2:
$a_1 + 4a_2 + 3a_3 + a_4 = 0$

Option 3:
$a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

Option 4:
$2a_1 + 2a_2 + 3a_3 + a_4 = 0$

Correct Answer:
$4a_1 + 2a_2 + a_3 + a_4 = 0$

Solution:

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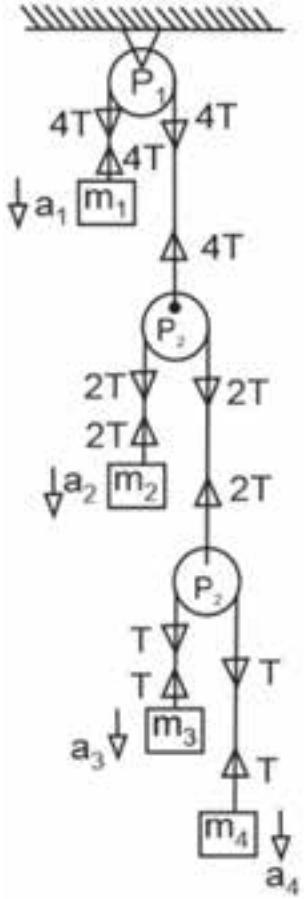

By virtual work method ,

$$\Sigma \overline{T} . \overline{a} = 0$$

$$-4\mathrm{T}\,\mathrm{a}_1 - 2\mathrm{T}\mathrm{a}_2 - \mathrm{T}\mathrm{a}_3 - \mathrm{T}\mathrm{a}_4 = 0$$

$$4\mathrm{a}_1 + 2\mathrm{a}_2 + \mathrm{a}_3 - \mathrm{a}_4 = 0$$

The correct option is (1)

Q. 3 Arrange the four graphs in descending order of total work done; where $\mathrm{W}_1, \mathrm{W}_2, \mathrm{W}_3$ and W_4 are the work done corresponding to figure $\mathrm{a}, \mathrm{b}, \mathrm{c}$ and d respectively.

Figure-a

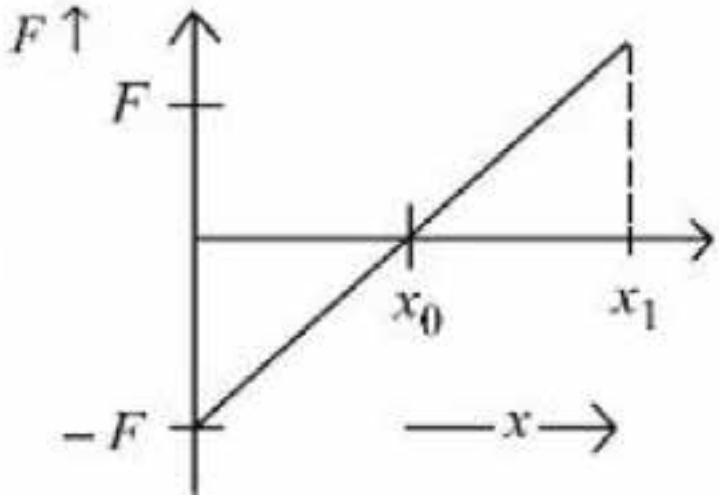

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Figure-b

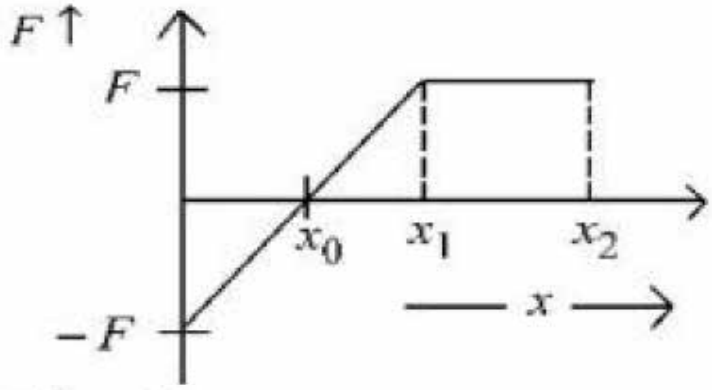


Figure-c

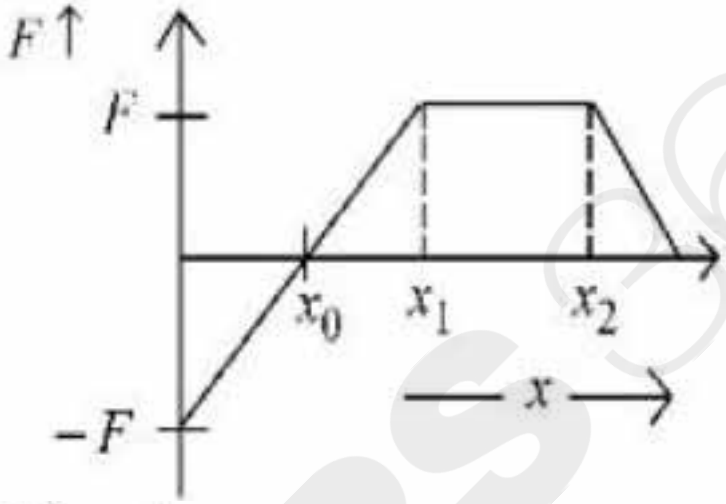
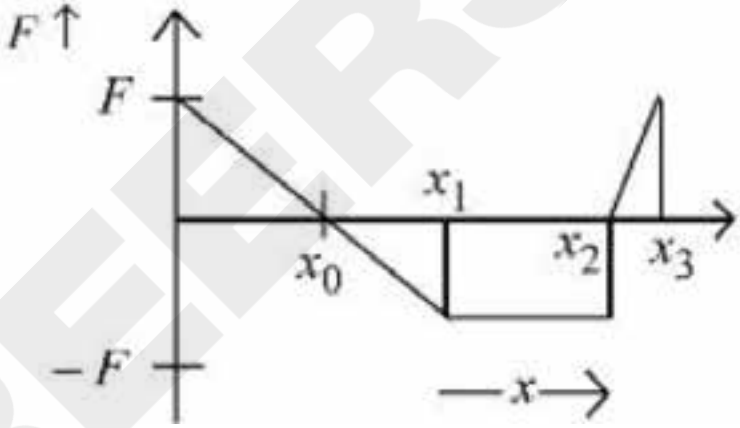


Figure-d



Option 1:

$W_3 > W_2 > W_1 > W_4$

Option 2:

$W_3 > W_2 > W_4 > W_1$

Option 3:

$W_2 > W_3 > W_4 > W_1$

Option 4:

$W_2 > W_3 > W_1 > W_4$

Correct Answer:

$W_3 > W_2 > W_1 > W_4$

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Solution:

$W_1 = 0$

(By symmetry , $x_1 = 2x_0$)

$W_2 = F(x_2 - x_1)$

$W_3 = F(x_2 - x_1) + \frac{1}{2} \times (x_3 - x_2) F$

$W_4 \Rightarrow \text{negative}$

The correct option is (1)

Q. 4 A solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is -

Option 1:

$\frac{2}{5}$

Option 2:

$\frac{2}{7}$

Option 3:

$\frac{1}{5}$

Option 4:

$\frac{7}{10}$

Correct Answer:

$\frac{2}{7}$

Solution:

For solid spherical ball,

$$I = \frac{2}{5} mr^2$$

$$KE_{Rotationd} = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times \left(\frac{2}{5} mr^2\right) \frac{v^2}{r^2}$$

$$= \frac{2}{5} \left(\frac{1}{2} mv^2\right)$$

$$KE_{Rotation} = \frac{2}{5} KE_{transalalion}$$

$$KE_{total} = KE_R + KE_T$$

$$= \frac{7}{5} KE_T$$

$$\frac{KE_R}{KE_{total}} = \frac{2}{7}$$

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The correct option is (2)

Q. 5 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.
Reason R: At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false

Option 4:
A is false but R is true

Correct Answer:
A is false but R is true

Solution:
$\rightarrow \mathrm{Radius\ at\ equator} = \mathrm{Radius\ at\ pole} + 21\ \mathrm{km}$
$\rightarrow$ As we move from pole to equator, acceleration due to gravity will decrease
$\rightarrow$ The direction of acceleration due to gravity is towards the earth

The correct option is (4)

Q. 6 If ρ is the density and η is coefficient of viscosity of fluid which flows with a speed ν in the pipe of diameter d, the correct formula for Reynolds number $\mathrm{R_e}$ is :

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Option 1:
$\mathrm{R_e} = \frac{\eta \mathrm{d}}{\mathrm{p}\upsilon}$

Option 2:
$\mathrm{R_e} = \frac{\mathrm{p}\upsilon}{\eta \mathrm{d}}$

Option 3:
$\frac{\rho \nu \mathrm{d}}{\eta}$

Option 4:
$\frac{\eta}{\rho \nu \mathrm{d}}$

Correct Answer:
$\frac{\rho \nu \mathrm{d}}{\eta}$

Solution:
$$\text{Raynolds number} = \frac{\text{Inertial force}}{\text{Viscous force}}$$
$$= \frac{\delta \mathrm{a} \nu^2}{\eta \mathrm{a} \left(\frac{\nu}{\mathrm{d}}\right)}$$
$$\mathrm{R_e} = \frac{\delta \nu \mathrm{d}}{\eta}$$
The correct option is (3)

Q. 7 A flask contains argon and oxygen in the ratio of $3:2$ in mass and the mixture is kept at $27\mathrm{C}^{\circ}$. The ratio of their average kinetic energy per molecule respectively will be :

Option 1:
$3:2$

Option 2:
$9:4$

Option 3:
$2:3$

Option 4:
$1:1$

Correct Answer:
$1:1$

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Solution:

$\mathrm{T} = 300\ \mathrm{K}$

$\frac{\mathrm{m_{Ar}}}{\mathrm{m_{O_2}}} = \frac{3}{2}$

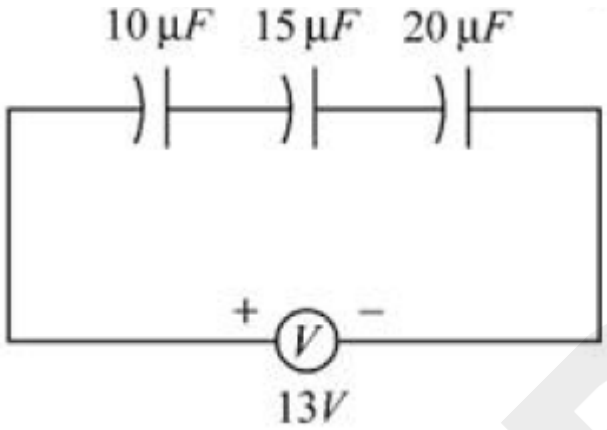
$\mathrm{Avg.KE\ per\ molecule} = \frac{3}{2}\mathrm{K_B T}$

The avg. KE per molecule is & independent of mass

$\therefore$ At a given temperáture,it will be same for Argon and Oxygen.

The correct option is (4)

Q. 8 The charge on capacitor of capacitance $15\mu\mathrm{F}$ in the figure given below is :



Option 1:

$60\mu c$

Option 2:

$130\mu c$

Option 3:

$260\mu c$

Option 4:

$585\mu c$

Correct Answer:

$60\mu c$

Solution:

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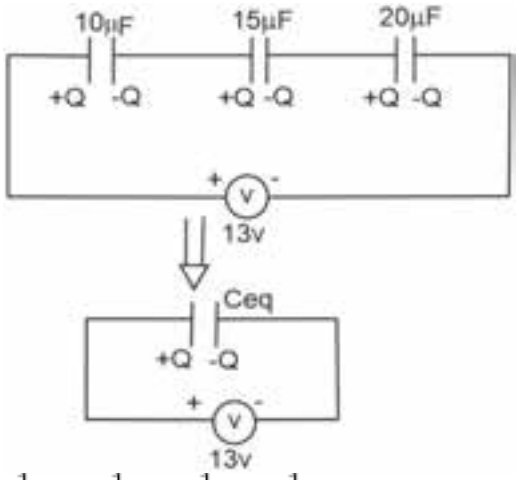


$$\frac{1}{c_{eq}} = \frac{1}{10} + \frac{1}{15} + \frac{1}{20}$$

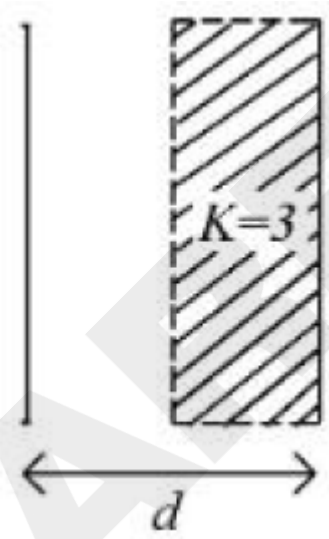
$$\frac{1}{c_{eq}} = \frac{13}{60}$$

$$c_{eq} = \frac{60}{13}$$

$$Q = (c_{eq})\, v = 60\mu c$$

The correct option is (1)

Q. 9 A parallel plate capacitor with plate area A and plate seperation $d = 2$ m has a capacitance of $4\,\mu F$. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant $K = 3$ (as shown in figure) will be :



Option 1:

$2\mu F$

Option 2:

$32\mu F$

Option 3:

$6\mu F$

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Option 4:
$8\mu\mathrm{F}$

Correct Answer:
$6\mu\mathrm{F}$

Solution:
$$\mathrm{C}=\frac{\mathrm{A}\varepsilon_0}{\mathrm{d}}=4\mu\mathrm{F}$$
Let the new capacitance be C'
$$\mathrm{C}'=\frac{\mathrm{A}\varepsilon_0}{\frac{\mathrm{d}/2}{1}+\frac{(\mathrm{d}/2)}{3}}=\frac{\mathrm{A}\varepsilon_0}{\left(\frac{4}{3}\right)\left(\frac{\mathrm{d}}{2}\right)}=\frac{3\mathrm{A}\varepsilon_0}{2\mathrm{d}}$$
$$=\frac{3}{2}\mathrm{C}=6\mu\mathrm{F}$$

The correct option is (3)

Q. 10 Sixty four conducting drops each of radius $0.02\,\mathrm{m}$ and each carrying a charge of $5\,\mu\mathrm{C}$ are combined to form a bigger drop. The ratio of surface density of bigger drop to the smaller drop will be :

Option 1:
$1:4$

Option 2:
$4:1$

Option 3:
$1:8$

Option 4:
$8:1$

Correct Answer:
$1:4$

Solution:
Let the radius of small drop and bigger drop be $\mathrm{r}\ \&\ \mathrm{R}$ respectively
By volume conservation,

$$64\times\left(\frac{4}{3}\pi\mathrm{r}^3\right)=\frac{4}{3}\pi\mathrm{R}^3$$
$$\mathrm{R}=4\mathrm{r}\rightarrow(1)$$

Let the charge on small drop and bigger drop be $\mathrm{q}\ \&\ \mathrm{Q}$ respectively
By charge conservation,

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$64q = Q$

$64 \times 5 \times 10^{-6} = Q$

$Q = 320 \times 10^{-6} C$

Surface charge density of small drop $= \sigma_1 = \frac{q}{4\pi r^2}$

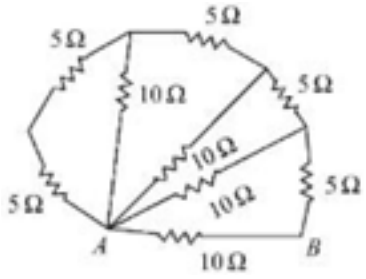
Surface charge density of bigger drop $= \sigma_2 = \frac{Q}{4\pi R^2}$

$$\frac{\sigma_1}{\sigma_2} = \frac{q}{r^2} \times \frac{R^2}{Q}$$
$$= \frac{1}{64} \times \frac{16}{1}$$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{1}{4}$$

The correct option is (1)

Q. 11 The equivalent resistance between points A and B in the given network is :



Option 1:
65Ω

Option 2:
20Ω

Option 3:
5Ω

Option 4:
2Ω

Correct Answer:
5Ω

Solution:

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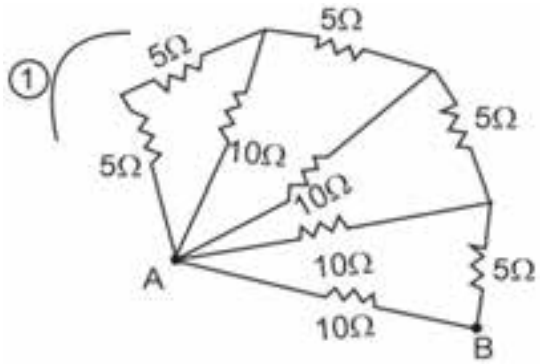

After combining $10\Omega \,\&\, (5+5)\,\Omega$ in parallel we get 5Ω then further combining with 5Ω. We again get 10Ω and this pattern continues

$R_{AB} = 5\Omega$

The correct option is (3)

Q. 12 A bar magnet having a magnetic moment of $2.0 \times 10^5\ JT^{-1}$, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5}T$. The work done in rotating the magnet slowly through 60° from the direction of field is :

Option 1:
$14\ J$

Option 2:
$8.4\ J$

Option 3:
$4\ J$

Option 4:
$1.4\ J$

Correct Answer:
$14\ J$

Solution:

$M = 2 \times 10^5$
$B = 14 \times 10^{-5}T$
$\theta_1 = 0^\circ \quad (\bar{M} \,||\, \bar{B}$ direction of field $)$
$\theta_2 = 60^\circ$

$$\begin{aligned} w_{ext} &= \Delta U = U_f - U_i \\ &= (-MB\cos\theta_2) - (-MB\cos\theta_1) \\ &= (-MB\cos 60^\circ) - (-MB\cos 0^\circ) \\ &= \frac{-MB}{2} + MB \\ &= \frac{MB}{2} \end{aligned}$$

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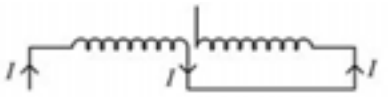
$$W_{ext} = \frac{2 \times 10^5 \times 14 \times 10^{-5}}{2} = 14 \text{ J}$$

Q. 13 Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :



Option 1:

$$\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{M}$$

Option 2:

$$L_1 + L_2 + M$$

Option 3:

$$L_1 + L_2 + 2M$$

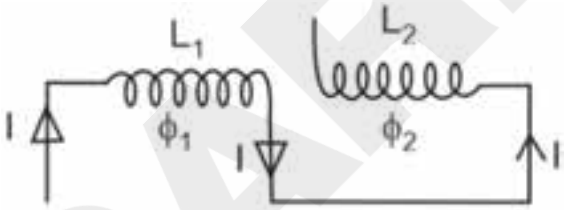
Option 4:

$$L_1 + L_2 - 2M$$

Correct Answer:

$$L_1 + L_2 - 2M$$

Solution:



$\rightarrow$ Flux due to coil 1 and coil 2 are in opposite direction

$\rightarrow$ The current is same through both the coil

$$\therefore L_{eq} = L_1 + L_2 - 2M$$

The correct option is (4)

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Q. 14 A metaliic conductor of length $1\mathrm{m}$ rotates in a vertical plane parallel to east-west direction about one of its end with angular velocity $5\,\mathrm{rad\,s^{-1}}$. If the horizontal component of earth's magnetic field is $0.2\times10^{-4}\mathrm{T}$, then emf induced between the two end of the conductor is :

Option 1:
$5\mu\mathrm{v}$

Option 2:
$50\mu\mathrm{v}$

Option 3:
$5\mathrm{mv}$

Option 4:
$50\mathrm{mv}$

Correct Answer:
$50\mu\mathrm{v}$

Solution:

emf induced between the two ends of the conductors is

$$\mathrm{e}=\frac{1}{2}\mathrm{B}\omega\mathrm{l}^2$$
$$=\frac{2\times10^{-5}\times5\times1}{2}$$
$$=5\times10^{-5}$$
$$\Rightarrow \mathrm{e}=50\mu\mathrm{v}$$

The correct option is (2)

Q. 15 Which is the correct ascending order of wavelengths ?

Option 1:
$\lambda_{\mathrm{visible}}<\lambda_{\mathrm{X-ray}}<\lambda_{\mathrm{gamma-ray}}<\lambda_{\mathrm{microwave}}$

Option 2:
$\lambda_{\mathrm{gamma-ray}}<\lambda_{\mathrm{X-ray}}<\lambda_{\mathrm{visible}}<\lambda_{\mathrm{microwave}}$

Option 3:
$\lambda_{\mathrm{X-ray}}<\lambda_{\mathrm{gamma-ray}}<\lambda_{\mathrm{visible}}<\lambda_{\mathrm{microwave}}$

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Option 4:
$\lambda_{\text{microwave}} < \lambda_{\text{visible}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}}$

Correct Answer:
$\lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

Solution:
Energy of electromagnetic waves are in the order

$$E_{\gamma-\text{ray}} > E_{\text{X-ray}} > E_{\text{visibleray}} > E_{\text{microwave}}$$
$$E = \frac{hc}{\lambda}$$

$$\therefore \lambda_{\gamma-\text{ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visibleray}} < \lambda_{\text{microwave}}$$

The correct option is (2)

Q. 16 For a specific wavelength $670\ \text{nm}$ of light coming from a galaxy moving with velocity v, the observed wavelength is 670.7nm.
The value of v is :

Option 1:
$3 \times 10^8\ \text{ms}^{-1}$

Option 2:
$3 \times 10^{10}\ \text{ms}^{-1}$

Option 3:
$3.13 \times 10^5\ \text{ms}^{-1}$

Option 4:
$4.48 \times 10^5\ \text{ms}^{-1}$

Correct Answer:
$3.13 \times 10^5\ \text{ms}^{-1}$

Solution:
We know that,
From doppler effect in Light that ,

$$\frac{\Delta\lambda}{\lambda} = \frac{-V}{c}$$
$$\frac{0.7}{670} = \frac{V}{3 \times 10^8}$$

$$V = \frac{7}{6700} \times 3 \times 10^8$$

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$$= \frac{21}{67} \times 10^6$$

$$\Rightarrow \mathrm{V} = 3.13 \times 10^5 \ \mathrm{m/s}$$

The correct option is (3)

Q. 17 A metal surface is illuminated by a radation of wavelength 4500 A. The ejected photo-electron enters a constant magnetic field of $2\mathrm{mT}$ making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius $2\ \mathrm{mm}$. the work function of the metal is approximately :

Option 1:
$1.36\,\mathrm{eV}$

Option 2:
$1.69\,\mathrm{eV}$

Option 3:
$2.78\,\mathrm{eV}$

Option 4:
$2.23\,\mathrm{eV}$

Correct Answer:
$1.36\,\mathrm{eV}$

Solution:

$\lambda = 4500\text{Å}$
$\mathrm{B} = 2 \times 10^{-3}\ \mathrm{T}$
$\bar{V} \perp \bar{B}$
$\mathrm{r} = 2\ \mathrm{mm} = 2 \times 10^{-3}\ \mathrm{m}$

We know that,

$$\mathrm{r} = \frac{\mathrm{mv}}{\mathrm{qB}}$$
$$2 \times 10^{-3} = \frac{\mathrm{mv}}{1.6 \times 10^{-19} \times 2 \times 10^{-3}}$$

$$\rightarrow \mathrm{mv} = 6.4 \times 10^{-25}\ \mathrm{kg}\left(\frac{\mathrm{m}}{\mathrm{s}}\right) \rightarrow (1)$$

By Einstein's photoelectric equation

$$\mathrm{hf} = \phi_0 + \frac{\mathrm{p}^2}{2\mathrm{m}} \rightarrow (2)$$

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$$\left(\because \mathrm{KE}=\frac{\mathrm{p}^2}{2\mathrm{m}}\right)$$

$$\frac{\mathrm{p}^2}{2\mathrm{m}}=\frac{6.4\times 6.4\times 10^{-50}}{2\times 9.1\times 10^{-31}}\ \mathrm{J}$$
$$=\frac{6.4\times 4}{2\times 9.1}=\mathrm{eV}$$
$$\frac{\mathrm{p}^2}{2\mathrm{m}}=\frac{25.6}{18.2}\mathrm{eV}=1.41\mathrm{eV}\rightarrow(3)$$

$$\text{Energy in ev}=\mathrm{hf(ev)}=\frac{12400}{4500}$$
$$=2.755\,\mathrm{eV}\ \rightarrow(4)$$

From eqn (2),(3) & (4)

$$\mathrm{hf}=\phi_0+\frac{\mathrm{p}^2}{2\mathrm{m}}$$
$$2.755=\phi_0+1.41\,\mathrm{ev}$$
$$\phi_0=1.345\,\mathrm{ev}$$
$$\phi_0\simeq 1.36\,\mathrm{ev}$$

The correct option is (1)

Q. 18 A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process. The effective half-life of the nucleus will be :

Option 1:
3.75 hours

Option 2:
0.56 hours

Option 3:
0.26 hours

Option 4:
1.80 hours

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Correct Answer:
$1.80\ \text{hours}$

Solution:
Let the half life for 1st, 2nd and effective be $\mathrm{T}_1, \mathrm{T}_2 \text{ and } \mathrm{T}$ respectively

$$\because\ \lambda_{\text{eff}} = \lambda_1 + \lambda_2$$
$$\therefore\ \frac{1}{\mathrm{T}_{\text{eff}}} = \frac{1}{\mathrm{T}_1} + \frac{1}{\mathrm{T}_2}$$
$$= \frac{1}{3} + \frac{1}{4.5}$$

$$\frac{1}{\mathrm{T}_{\text{eff}}} = \frac{3+2}{9}$$

$$\mathrm{T}_{\text{eff}} = 1.8\,\text{hrs}$$

The correct option is (4)

Q. 19 The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

Option 1:
External input is necessary to sustain ac signal in output.

Option 2:
A portion of the output power is returned back to the input.

Option 3:
Feedback can be achicved by LR network.

Option 4:
The basc-collector junction must be forward biased.

Correct Answer:
A portion of the output power is returned back to the input.

Solution:
$\rightarrow$ The feedback here mean a portion of the output power is retuned back to the input

$\rightarrow$ The positive feedback is the feedback which is in phase with the input

$\rightarrow$ The negative feedback is the feedback which is out of phase with resped to input

$\rightarrow$ The correct option is (2)

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Q. 20 A sinusoidal wave $y(t) = 40\sin\left(10 \times 10^6 \pi t\right)$ is amplitude modulated by anothet sinusoidal wave $x(t) = 20\sin(1000\pi t)$. The amplitude of minimum frequency component of modulated signal is :

Option 1:
0.5

Option 2:
0.25

Option 3:
20

Option 4:
10

Correct Answer:
10

Solution:
$y(t) = 40\sin\left(10 \times 10^7 \pi t\right)$
$x(t) = 20\sin(1000\,\pi t)$
$\omega_c = 10^7\pi$
$\omega_m = 10^3\pi$
$A_c = 40$
$A_m = 20$

Equation of modulated $= (A_c + A_m \sin\omega_m t)(\sin\omega_c t)$

$$= A_c\left(1 + \frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t$$
$$= A_c\left(1 + \mu\sin\omega_m t\right)\sin\omega_c t$$
$$= A_c\sin\omega_c t + \frac{uA_c}{2}\left[\cos\left(\omega_c - \omega_m\right)t - \cos\left(\omega_c + \omega_m\right)t\right]$$

Amplitude of minimum frequency
$= \frac{\mu A_c}{2} = \left(\frac{1}{2}\right) \times \frac{40}{2} = 10$
The correct option is (4)

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Q. 21 A ball is projected vertically upward with an initial velocity of $50\ \text{ms}^{-1}$ at $\text{t} = 0\ \text{s}$. At $\text{t} = 2\ \text{s}$, another ball is projected vertically upward with same $\text{At t} =$ _____ s, second ball will meet the first ball $\left(\text{g} = 10\ \text{ms}^{-2}\right)$.

Correct Answer:
6

Solution:
Ball 1
$\text{u}_1 = 50\ \text{m/s} \quad (\text{t} = 0)$

Ball 2
$\text{u}_2 = 50\ \text{m/s} \quad (\text{t} = 25)$

They will meet when the height from the ground is same for both the ball at a given inatant

$$\text{S}_1 = \text{u}_1\text{t} + \frac{1}{2}\text{a}_1\text{t}^2$$
$$\text{S}_2 = \text{u}_2(\text{t} - 2) + \frac{1}{2}\text{a}_2\,(\text{t} - 2)^2$$

$$\text{For } \text{S}_1 = \text{S}_2$$

$$\text{u}_1\text{t} + \frac{1}{2}\,(-9)\,\text{t}^2 = \text{u}_2\,(\text{t} - 2) + \frac{1}{2} + (-9)\,(\text{t} - 2)^2$$
$$50\text{t} - \frac{1}{2}\text{gt}^2 = 50\text{t} - 100 - \frac{1}{2}\text{g}\left(\text{t}^2 - 4\text{t} + 4\right)$$
$$0 = -100 + 2\text{gt} - 2\text{g}$$
$$120 = 20\text{t}$$
$$\text{t} = 6\text{s}$$

Q. 22 A batsman hits back a ball of mass $0.4\ \text{kg}$ straight in the direction of the bowler without changing its initial speed of $15\ \text{ms}^{-1}$. The impulse imparted to the ball is _____Ns.

Correct Answer:
12

Solution:
Let the ball is initially & along the $+\text{x}$-axis

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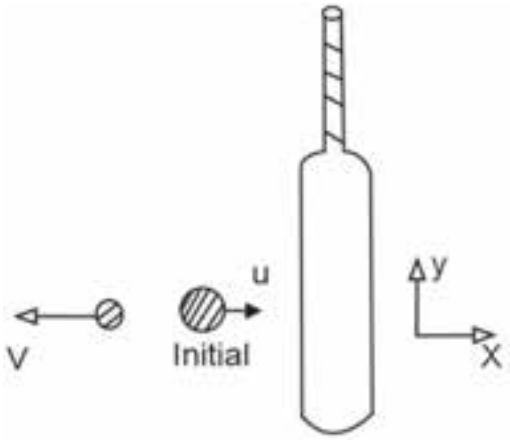
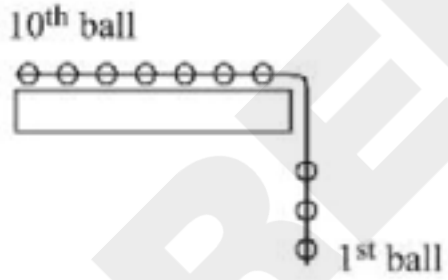


$$\bar{u} = 15\left(\hat{i}\right)$$
$$\bar{v} = 15\left(\hat{-i}\right)$$
$$\text{Impulse} = \Delta\bar{p} = m\left(\bar{v} - \bar{u}\right)$$
$$= 0.4\left(15\left(-\hat{i}\right) - 15\hat{i}\right)$$

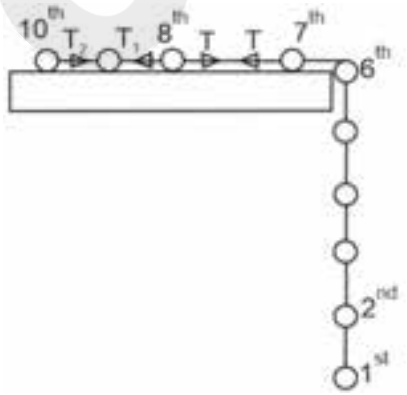
$$\text{Magnitude of Impulse} = |\Delta\bar{p}| = \left|-12\hat{i}\right| = 12\,\text{Ns}$$

Q. 23 A system to 10 balls each of mass $2\ \text{kg}$ are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7^{th} and 8^{th}ball is _____N when 6^{th} ball just leaves the table.



Correct Answer:
36

Solution:



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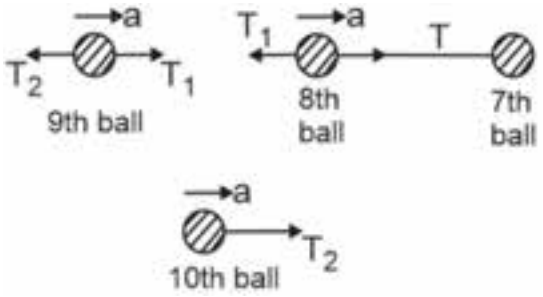
Net downward force $= 6\,\mathrm{mg}$

$= 120\,\mathrm{N}$

$$a = \frac{\text{Total force applied}}{\text{Total mass}}$$

$$= \frac{120}{(10 \times 2)} = 6\mathrm{m/s^2}$$



On 10th ball

$T_2 = ma = 12\,\mathrm{N}$

On 9th ball

$T_1 - T_2 = ma$

$T_1 = 24\,\mathrm{N}$

On 8th ball

$T - T_1 = ma$

$T = 36\,\mathrm{N}$

Tension on the string between the 7th & 8th ball is 36 N.

Q. 24 A geyser heats water flowing at a rate of $2.0\ \mathrm{kg}$ per minute from $30^\circ\mathrm{C}$ to $70^\circ\mathrm{C}$. If geyser operates on a gas burner, the rate of combustion of fuel will be_____$\mathrm{g\,min^{-1}}$

[Heat of combustion $= 8 \times 10^3\,\mathrm{Jg^{-1}}$,

Specific heat of water $= 4.2\,\mathrm{Jg^{-1}\,^0C^{-1}}$]

Correct Answer:

42

Solution:

Energy required for heating water in one minute $(\Delta t = 1\ \mathrm{min})$

$$\frac{E}{\Delta t} = \frac{mc\Delta T}{\Delta t}$$

$$= \frac{2000 \times 4.2 \times 40}{1\ \mathrm{min}}$$

$$\frac{E}{\Delta t} = 336 \times 10^3 \left(\frac{\mathrm{J}}{\mathrm{min}}\right)$$

Rate of combustion of fuel is let say R

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$$\therefore R = \frac{(E/\Delta t)}{\text{Heat of combustion}}$$
$$= \frac{336 \times 10^3\ \text{J/min}}{8 \times 10^3\ \text{Jg}^{-1}}$$
$$R = 42\left(\frac{g}{min}\right)$$

Q. 25 A heat engine operates with the cold reservoir at temperature $324\ K$.The minimum temperature of the hot reservoir, if the heat engine takes $300\ J$ heat from the hot reservoir and delivers $180\ J$heat to the cold reservoir per cycle, is_____K.

Correct Answer:
540

Solution:
$T_c = 324\ K$
$Q_H = 300\ J$
$Q_c = 180\ J$

By carnot's theorem,
$$\frac{Q_H}{Q_c} = \frac{T_H}{T_c}$$
$$\frac{300}{180} = \frac{T_H}{324}$$
$$T_H = 540\ K$$

Q. 26 A set of 20 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is _____Hz.

Correct Answer:
152

Solution:
$f, f+4, \& f+8 \ldots, 2f$
For n^{th} term in arithmetic progression,

$$t_n = a + (n-1)d$$
$$2f = f + (20-1)4$$
$$2f = f + 76$$
$$f = 76\ Hz$$

The frequency of the last fork is
$$2f = 152\,Hz$$

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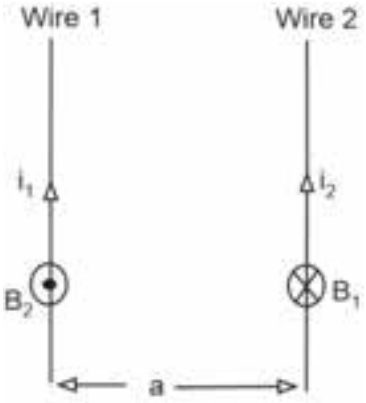
Q. 27 Two $10\ \text{cm}$ long, straight wires, each carrying a current of $5\ \text{A}$ are kept parallel to each other. If each wire experienced a force of $10^{-5}\ \text{N}$, then separation between the wires is _____ cm.

Correct Answer:
5

Solution:



$$F_{2/1} = \frac{\mu_0 i_1 i_2}{2\pi a} l_2$$

$$10^{-5} = \frac{2 \times 10^{-7} \times 5 \times 5 \times 10^{-1}}{a}$$

$$a = 50 \times 10^{-3}\ \text{m}$$

$$a = 5\ \text{cm}$$

Q. 28 A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}\ \text{m}$. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi\ \text{m}^2$. The value of x is _____.

Correct Answer:
9

Solution:
Circle of illumination

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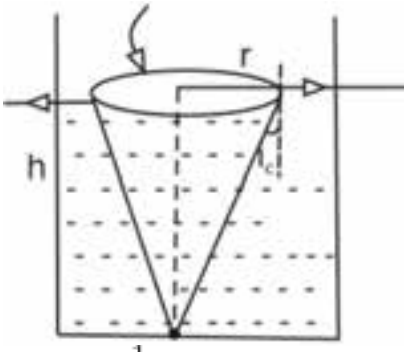
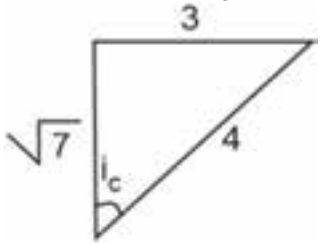


$$\sin i_c = \frac{1}{u}$$

$$\sin i_c = \frac{3}{4}$$

$$\tan i_c = \frac{r}{n} = \frac{3}{\sqrt{7}}$$



$r = 3\ \text{m}$

$\text{Area } = \pi r^2 = 9\pi = x\pi$

$\therefore x = 9$

Q. 29 A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in $1\ \text{cm}$ on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is_____ $\times 10^{-6}\ \text{m}$.

Correct Answer:
5

Solution:

$$1\text{MSD} = \frac{1}{40}\ \text{cm} = \frac{1}{40}\ \text{cm}$$

$$50\ \text{VSD } = 49\,\text{MSD}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD}$$
$$= 1\text{MSD} - \frac{49}{50}\text{MSD}$$

$$\text{LC} = \frac{1}{50}\text{MSD} = \frac{1}{2000}\ \text{cm}$$

$$\text{LC} = 5 \times 10^{-6}\ \text{m}$$

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Q. 30 The stopping potential for photoelectrons emitted from a surface illuminated by light of wavelength $6630\ \text{Å}$ is $0.42\ \text{V}$. If the threshold frequency is $\text{x} \times 10^{13}/\text{s}$, where x is_____ (nearest integer).
(Given, speed light $= 3 \times 10^8\ \text{m/s}$, Planck's constant $= 6.63 \times 10^{-34}\text{Js}$)

Correct Answer:
35

Solution:
$\lambda = 6630\text{Å}$
$\text{V}_0 = 0.42\ \text{V}$
$\text{f}_0 = \text{ threshold frequency } = \text{x} \times 10^{13}$
$\text{Energy of incident photon} = \frac{12400*}{6630}(\text{eV})$
$\text{hf} = 1.87\text{eV}$

By Einstein's photodectric equation,
$\text{hf} = \phi_0 + \text{eV}_0$
$\phi_0 = \text{hf}_0 = 1.87\text{eV} - 0.42\text{eV}$
$\text{f}_0 = \frac{1.45\text{eV}}{\text{h}}$

$\text{f}_0 = \frac{1.45 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$
$= 0.35 \times 10^{15}$
$\text{f}_0 = 35 \times 10^{13}\ \text{Hz}$

$\therefore \text{x} = 35$

Chemistry

Q. 1 The number of radial and angular nodes in 4d orbital are, respectively

Option 1:
1 and 2

Option 2:
3 and 2

Option 3:
1 and 0

Option 4:
2 and 1

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Correct Answer:
1 and 2

Solution:
As we have learnt,

$$\text{For 4 d orbital, n} = 4, \text{l} = 2$$

$$\therefore \text{Radial nodes} = \text{n} - \ell - 1 = 1$$

$$\text{Angular nodes} = \ell = 2$$

Hence, the answer is **Option (1)**

Q. 2 Match **List I** with **List II.**

List I **Enzyme**	**List II** **Conversion of**
A. Invertase	I. Starch into maltose
B. Zymase	II. Maltose into glucose
C. Diastase	III.Glucose into ethanol
D. Maltase	IV. Cane sugar into glucose

Choose the most appropriate answer from the options given below :

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-III, B-II, C-I, D-IV

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-IV, B-III, C-I, D-II

Solution:
The correct match of Enzyme ard the conversion carried out by it is given below:

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(A) Invertase: Cane sugar into Glucose (IV)

(B) Zymase: Glucose into ethanol (III)

(C) Diastase: Starch into maltose (I)

(D) Maltose: Maltose into Glucose (II)

Hence, the answer is **Option (3)**

Q. 3 Which of the following elements is considered as a metalloid?

Option 1:
Sc

Option 2:
Pb

Option 3:
Bi

Option 4:
Te

Correct Answer:
Te

Solution:
Fact Based.

$\mathrm{Sc, Pb \text{ and } Bi}$ are metals while Te is a metalloid.

Hence, the answer is Option (4)

Q. 4 The role of depressants in 'Froth Floation method' is to

Option 1:
selectively prevent one component of the ore from coming to the froth.

Option 2:
reduce the consumption of oil for froth formation.

Option 3:
stabilize the froth.

Option 4:
enhance non-wettability of the mineral particles.

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Correct Answer:
selectively prevent one component of the ore from coming to the froth.

Solution:
The role of depressant is to selectively prevent one component of the ore from coming to the froth.

For example, NaCN is used as a depressant in the separation of ZnS from PbS. It selectively forms a complex with ZnS forming the tetracyanozincate(II) complex while PbS is unaffected.

$$\mathrm{ZnS + 4\ NaCN \longrightarrow Na_2[Zn(CN)_4] + Na_2S}$$

Hence, answer is **Option(1)**

Q. 5 Boiling of hard water is helpful in removing the temporary hardness by converting calcium hydrogen carbonate and magnesium hydrogen carbonate to

Option 1:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Option 2:
$\mathrm{CaCO_3}$ and $\mathrm{MgCO_3}$

Option 3:
$\mathrm{Ca(OH)_2}$ and $\mathrm{MgCO_3}$

Option 4:
$\mathrm{Ca(OH)_2}$ and $\mathrm{Mg(OH)_2}$

Correct Answer:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Solution:
During removal of temporary hardness by boiling, $\mathrm{Ca^{2+}}$ and $\mathrm{Mg^{2+}}$ are removed as their carbonate and hydroxide respectively

$$\mathrm{Ca\,(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2}$$

$$\mathrm{Mg\,(HCO_3)_2 \xrightarrow{\Delta} Mg(OH)_2 + 2CO_2}$$

Herce, **Option (1)** is correct.

Q. 6 s-block element which **cannot** be qualitatively confirmed by the flame test is

Option 1:
Li

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Option 2:
Na

Option 3:
Rb

Option 4:
Be

Correct Answer:
Be

Solution:
Be does not respond to the flame test.

The colour imparted by other elements are:

Li - Crimson Red

Na - Yellow

Rb - Red violet

Hence, **Option (4)** is correct.

Q. 7 The oxide which contains an odd electron at the nitrogen atom is

Option 1:
N_2O

Option 2:
NO_2

Option 3:
N_2O_3

Option 4:
N_2O_5

Correct Answer:
NO_2

Solution:
NO_2 contains an odd electron at Nitrogen atom.

$O \leftarrow \dot{N} = O$

Hence, **Option (2)** is correct.

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Q. 8 Which one of the following is an example of disproportionation reaction?

Option 1:
$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

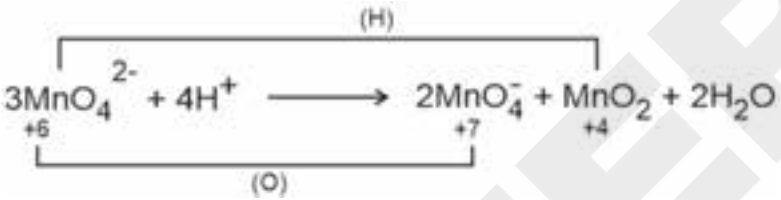
Option 2:
$MnO_4^- + 4H^+ + 4e^- \rightarrow MnO_2 + 2H_2O$

Option 3:
$10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^+ + 8H_2O + 5I_2$

Option 4:
$8MnO_4^- + 3\ S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Correct Answer:
$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

Solution:
In a disproportionation reaction, the same species is oxidised as well as reduced.



Hence, **Option (1)** is correct.

Q. 9 The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

Option 1:
Ce (At. No. 58)

Option 2:
La (At. No. 57)

Option 3:
Lu (At. No. 71)

Option 4:
Gd (At. No. 64)

Correct Answer:
Ce (At. No. 58)

Solution:

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Electronic configuration of Ce is $[\mathrm{Xe}]\ 4\mathrm{f}^1 5\mathrm{d}^1 6\mathrm{s}^2$

$\therefore\ \mathrm{Ce}^{4+}$ has a noble gas configuration

Thus, Ce^{3+} is likely to deviate to its (+4) oxidation state.

Hence, **Option(1)** is correct.

Q. 10 The measured BOD values for four different water samples (A-D) are as follows: A= 3 ppm; B=18 ppm; C=21 ppm D=4 ppm. The water samples which can be called as highly polluted with organic wastes, are

Option 1:
A and B

Option 2:
A and D

Option 3:
B and C

Option 4:
B and D

Correct Answer:
B and C

Solution:
As we have learnt,

Highly polluted water has a BOD of more than 17 ppm.

Thus, water samples B (18 ppm) and C (21 ppm) are highly polluted.

Hence, the correct answer is **Option (3)**

Q. 11 The correct order of nucleophilicity is

Option 1:
$\mathrm{F}^- > \mathrm{OH}^-$

Option 2:
$\mathrm{H_2\ddot{O}} > \mathrm{OH}^-$

Option 3:
$\mathrm{R\ddot{O}H} > \mathrm{RO}^-$

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Option 4:
$NH_2^- > NH_3$

Correct Answer:
$NH_2^- > NH_3$

Solution:
While comparing nucleophiles of same period, the nucleophilic strength follows basic strength.

Thus, $OH^- > F^-$

Next, neutral species are poorer nucleophiles than their anionic form

$\therefore \quad OH^- > H_2O$

$RO^- > ROH$

$NH_2^- > NH_3$

Hence, **Option (4)** is correct.

Q. 12 Oxidation of toluene to benzaldehyde can be easily carried out with which of the following reagents?

Option 1:
CrO_3/ acetic acid, H_3O^+

Option 2:
CrO_3 /acetic anhydride, H_3O^+

Option 3:
$KMnO_4/HCl, H_3O^+$

Option 4:
CO/HCl, anhydrous $AlCl_3$

Correct Answer:
CrO_3 /acetic anhydride, H_3O^+

Solution:
Benzaldehyde can be obtained from Toluene upon reaction with CrO_3 in the presence of acetic anhydride followed by acidic hydrolysis.

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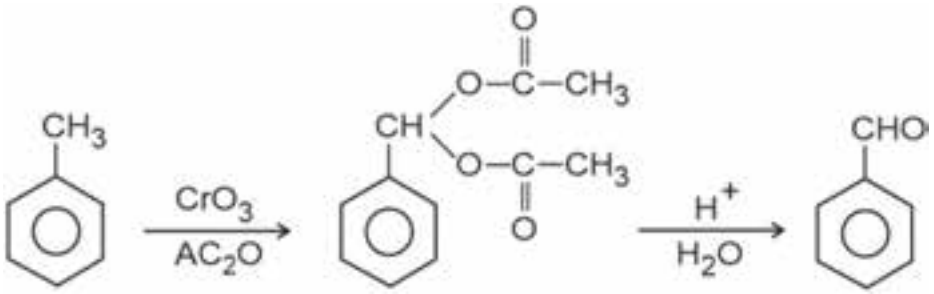
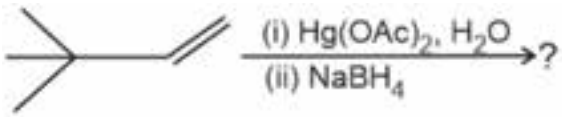
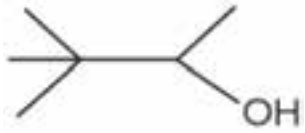


Hence, **Option (2)** is correct.

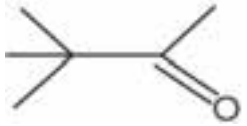
Q. 13 The major product in the following reaction



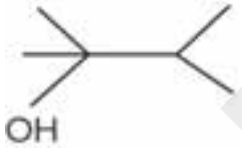
Option 1:



Option 2:



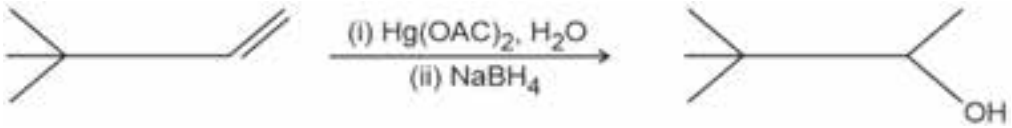
Option 3:



Option 4:

Correct Answer:

Solution:
The reaction is given as,



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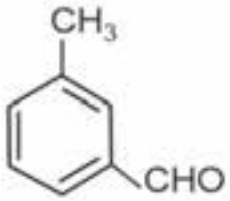
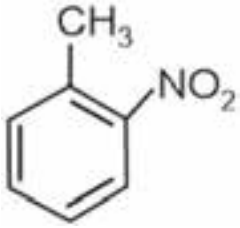
Hence, **Option (1)** is correct.

Q. 14 Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product?

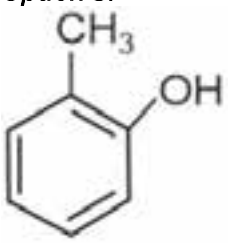
Option 1:



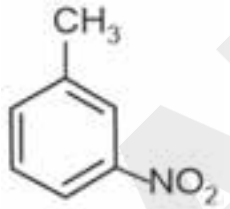
Option 2:



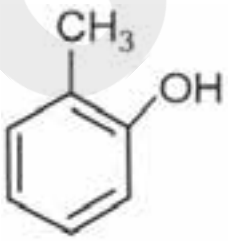
Option 3:



Option 4:



Correct Answer:



Solution:

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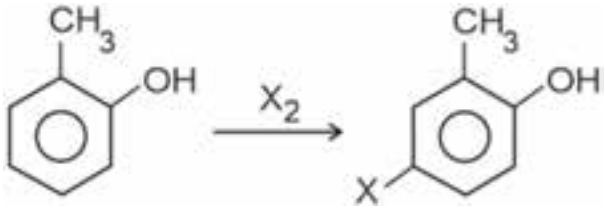
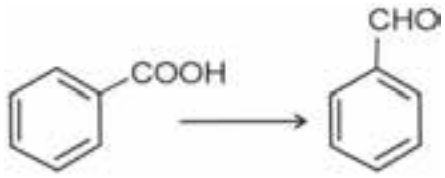


Electrophilic substitution occurs with respect to the more activating group. OH is a more activating than CH_3 and as a result, the electrophile is added at the para position of OH and meta position of CH_3 group.

Hence, Option (3) is correct.

Q. 15 The reagent, from the following, which converts benzoic acid to benzaldehyde in one step is



Option 1:
$LiAlH_4$

Option 2:
$KMnO_4$

Option 3:
MnO

Option 4:
$NaBH_4$

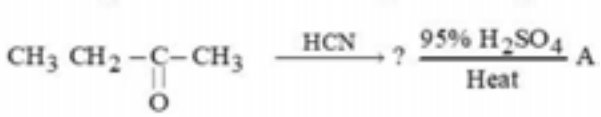
Correct Answer:
MnO

Solution:
MnO on reaction with $PhCOOH$ in the presence of formic acid will form $Ph\ CHO$

$$Ph\ COOH + HCOOH \xrightarrow[\Delta]{MnO} PhCHO + CO_2 + H_2O$$

Hence, **Option (3)** is correct.

Q. 16 The final product 'A' in the following reaction sequence



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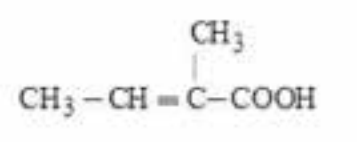
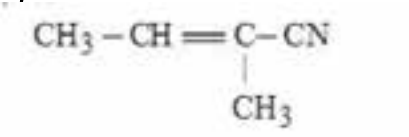
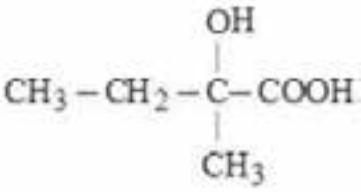
Option 1:



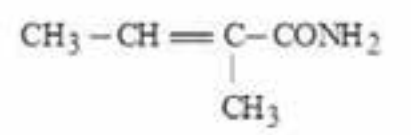
Option 2:



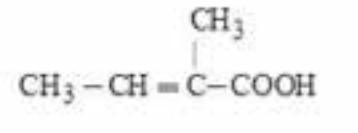
Option 3:



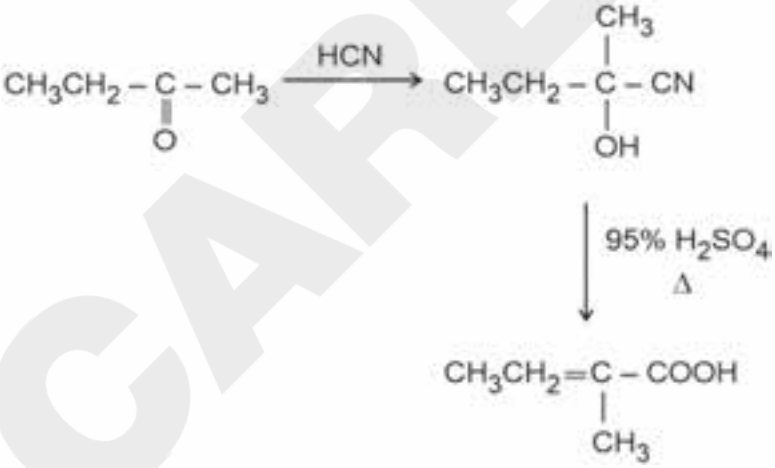
Option 4:



Correct Answer:



Solution:



Hence, **Option (1)** is correct.

Q. 17 Which statement is NOT correct for p-toluenesulphonyl chloride?

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Option 1:
It is known as Hinsberg's reagent.

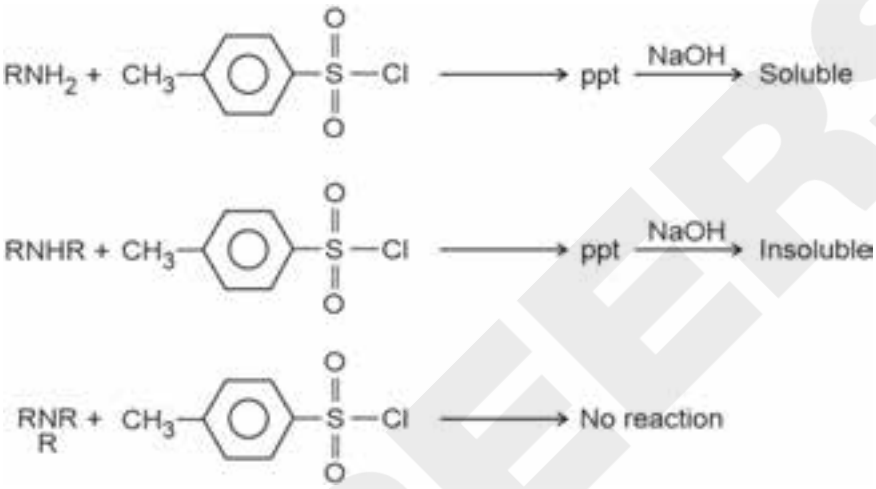
Option 2:
It is used to distinguish primary and secondary amines.

Option 3:
On treatment with secondary amine, it leads to a product, that is soluble in alkali.

Option 4:
It doesn't react with tertiary amines.

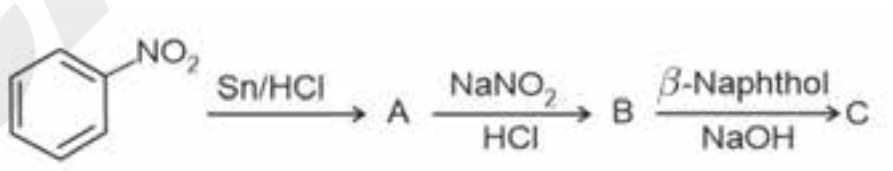
Correct Answer:
On treatment with secondary amine, it leads to a product, that is soluble in alkali.

Solution:
p-Toluene sulphonyl chloride is also known as Hinsberg's reagent and is used to distinguish between primary, secondary and tertiary amines.

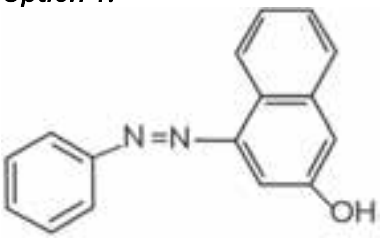


Hence, Statement in Option (3) is incorrect.
Answer is **Option (3)**

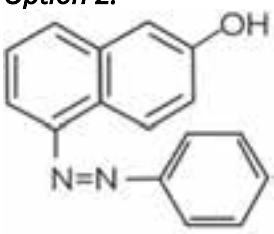
Q. 18 The final product 'C' in the following series of reactions



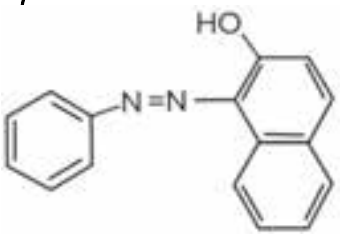
Option 1:



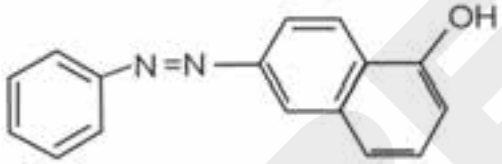
Option 2:



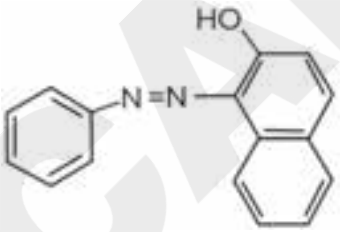
Option 3:



Option 4:



Correct Answer:



Solution:
The reaction sequence occurs as

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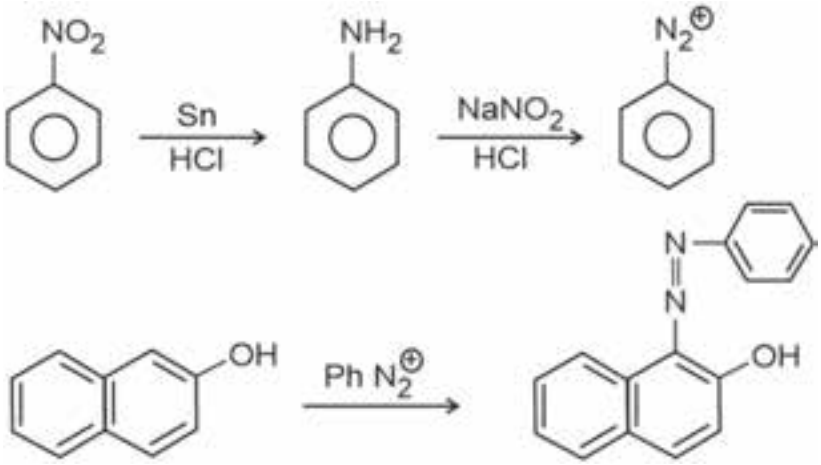
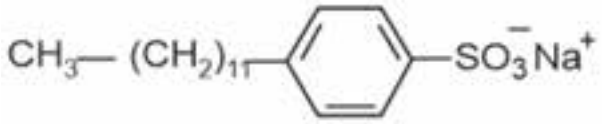


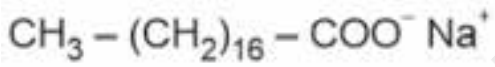
Hence, **Option (3)** is correct.

Q. 19 Which of the following is NOT an example of synthetic detergent?

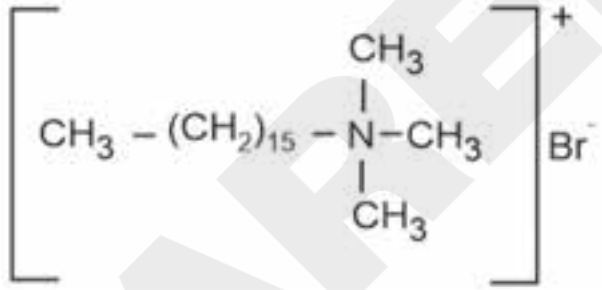
Option 1:



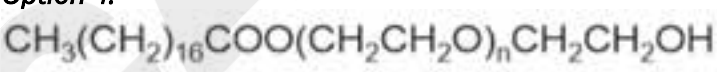
Option 2:



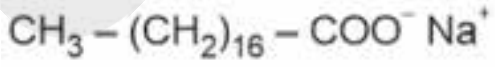
Option 3:



Option 4:



Correct Answer:



Solution:
Sodium stearate is a soap and not a detergent.

$$CH_3 - (CH_2)_{16} - COO^{\ominus}\, Na^{\oplus}$$

All other given species are examples of synthetic detergent.

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Hence, answer is **Option (2)**

Q. 20 Which one of the following is a water soluble vitamin, that is not excreted easily?

Option 1:
Vitamin B_2

Option 2:
Vitamin B_1

Option 3:
Vitamin B_6

Option 4:
Vitamin B_{12}

Correct Answer:
Vitamin B_{12}

Solution:
Fact Based.

Vitamin B_{12} is a water soluble vitamin which is not excreted easily.

Herce, **Option (4)** is correct.

Q. 21 CNG is an important transportation fuel. When $100\ \text{g}$ CNG is mixed with $208\ \text{g}$ oxygen in vehicles, it leads to the formation of CO_2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbone dioxide, produced in grams is ___. [nearest interger]

Correct Answer:
143

Solution:
Combustion reaction of Methane is given as

$$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$$

$$\text{moles of methane } = \frac{100}{16} = 6.25$$

$$\text{moles of oxygen } = \frac{208}{32} = 6.5$$

Limiting reagent is O_2

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$\therefore$ moles of $CO_2 = \frac{6.5}{2} = 3.25$

$\therefore$ mass of $CO_2 = 3.25 \times 44 = 143$ g

Hence, the answer is 143

Q. 22 In a solid AB, A atoms are in ccp arrangement and B atoms occupy all the octaheral sites. If two atoms from the opposite faces are removed, then the resultant stoichiometry of the compund is A_x B_y, The value of x is _____[nearest integer]

Correct Answer:
3

Solution:
Given,

A atoms form a CCP while B atoms occupy all the Octahedral sites.

Now, A atoms from the opposite faces are removed

So, effective no.of A atoms $= 4 - 1$

$= 3$

effective no.of B atoms $= 4$

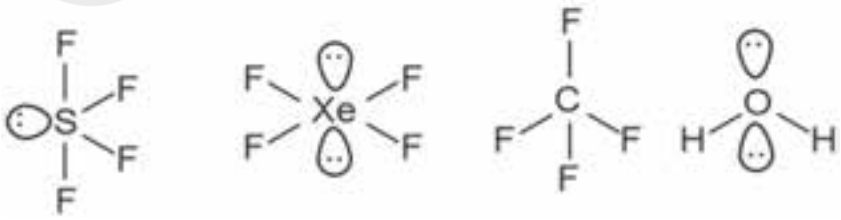
$\therefore$ x = 3

Thus, answer is 3

Q. 23 Amongst SF_4, XeF_4, CF_4 and H_2O, the number of species with two lone pairs of electrons on central atom is _____.

Correct Answer:
2

Solution:
The structures of the compounds are given below:



Thus, two compounds XeF_4 and H_2O have two lone pairs on the central atom.

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Hence, the answer is 2.

Q. 24 A fish swimming in water body when taken out from the water body is covered with a film of water of weight $36\ \mathrm{g}$. When it is subjected to cooking at $100^{\circ}\mathrm{C}$, then the internal energy for vaporization in $\mathrm{kJmol^{-1}}$ is _____. [nearest integer]

[Assume steam to be an ideal gas. Given $\Delta_{\mathrm{vap}}\ \mathrm{H}^{\Theta}$ for water at $373\ \mathrm{K}$ and 1 bar is $41.1\ \mathrm{kJ\ mol^{-1}}; \mathrm{R} = 8.31\ \mathrm{J\ K^{-1}\ mol^{-1}}$]

Correct Answer:
38

Solution:
$\mathrm{H_2O}(l) \rightleftharpoons \mathrm{H_2O(g)}$

$\Delta \mathrm{H} = \Delta \mathrm{U} + (\Delta \mathrm{n})_{\mathrm{g}} \mathrm{RT}$

$\Rightarrow \Delta \mathrm{U} = \Delta \mathrm{H} - (\Delta \mathrm{n})_{\mathrm{g}} \mathrm{RT}, (\Delta \mathrm{n_g} = 1)$

$$= \left(41.1 - \frac{1 \times 8.31 \times 373}{1000}\right)\ \mathrm{kJmol^{-1}}$$
$$= (41.1 - 3.1)\ \mathrm{kJmol^{-1}}$$
$$= 38\ \mathrm{kJmol^{-1}}$$

Hence, the answer is 38.

Q. 25 The osmotic pressure exerted by a solution prepared by dissolving $2.0\ \mathrm{g}$ of protein of molar mass $60\ \mathrm{kg\ mol^{-1}}$ in $200\ \mathrm{mL}$ of water at $27^{\circ}\mathrm{C}$ is _____Pa.[integer value]

$(\text{use } \mathrm{R} = 0.083\ \mathrm{L\ bar\ mol^{-1}\ K^{-1}})$

Correct Answer:
415

Solution:
$$\mathrm{n} = \frac{2}{60 \times 10^3} = 3.33 \times 10^{-5}$$

$$\therefore [\mathrm{Protein}] = \mathrm{C} = 1.67 \times 10^{-5}$$

$$\therefore \pi = \mathrm{CRT} = 1.67 \times 10^{-5} \times 0.083 \times 300$$
$$= 4.15 \times 10^{-3}\ \mathrm{bar}$$
$$= 415\ \mathrm{Pa}\ (\because 1\ \mathrm{bar} = 10^5 \mathrm{Pa})$$

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Hence, the answer is 415.

Q. 26 40% of HI undergoes deomposition to H_2 and I_2 $at\ 300\ K \cdot \Delta G^{\ominus}$ for this decompostion reaction at one atmopsphere pressure is_____ $Jmol^{-1}$.[nearest integer]

The decomposition reaction is given as

$$HI \rightleftharpoons \frac{1}{2}H_2 + \frac{1}{2}I_2$$

$(\text{Use } R = 8.31\ J\ K^{-1}\ mol^{-1}; \log 2 = 0.3010, \ln 10 = 2.3, \log 3 = 0.477\ )$

Correct Answer:
2735

Solution:

$$HI \rightleftharpoons \frac{1}{2}H_2 + \frac{1}{2}I_2$$

P_0

$0.6P_0 \quad 0.2P_0 \quad 0.2P_0$

Given,

$$P_T = 1 = (0.6 + 0.2 + 0.2)P_0$$

$$\Rightarrow P_0 = 1$$

$$\therefore K_P = \frac{P_{H_2}^{1/2} P_{I_2}^{1/2}}{P_{HI}} = \frac{(0.2)}{(0.6)} = \frac{1}{3}$$

$$\therefore \Delta G^{\circ} = -RT \ln K_P = +8.31 \times 300 \times 2.3 \times \log 3$$

$$= 2735\ J\ mole^{-1}$$

Hence, answer is 2735

Q. 27 $Cu(s) + Sn^{2+}(0.001M) \rightarrow Cu^{2+}(0.01M) + Sn(s)$

The Gibbs free energy change for the above reaction at $298\ K$ is $x \times 10^{-1}\ kJ\ mol^{-1}$.

The value of x is _____.[nearest integer]

Given: $E^{\ominus}_{Cu^{2+}/Cu} = 0.34\ V; E^{\ominus}_{Sn^{2+}/Sn} = -0.14\ V; F = 96500 C mol^{-1}$]

Correct Answer:
983

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Solution:

$\mathrm{E^o}$ for the given reection $= -0.14 - 0.34 = -0.48$

Applying Nernst equation.

$$\mathrm{E} = \mathrm{E}^0 - \frac{0.059}{2} \log \frac{[\mathrm{Cu}^{2+}]}{[\mathrm{Sn}^{2+}]}$$

$$= -0.48 - 0.0295 \ \log 10$$

$$= -0.5095$$

$$\therefore \Delta \mathrm{G}=\text{-n F E} = 2 \times 96500 \times 0.5095 \ \mathrm{J}$$

$$= 98335 \ \mathrm{J}$$

$$= 98.34 \ \mathrm{kJ}$$

Hence, the answer is 983.

Q. 28 Catalyst A reduces the activation energy for a reaction by $10 \ \mathrm{kJ \ mol^{-1}}$ at $300 \ \mathrm{K}$.

The ratio of rate constants, $\frac{\mathrm{k_T, \ Catalysed}}{\mathrm{k_T, \ Uncatalysed}}$ is e^x. The value of x is____.[nearest integer]

[Assume that the pre- exponential factor is same in both the cases.

Given Given $\mathrm{R} = 8.31 \ \mathrm{J \ K^{-1} \ mol^{-1}}$]

Correct Answer:

4

Solution:

From Arhenius equation ,

$$\mathrm{K} = \mathrm{Ae}^{-\mathrm{Ea/RT}}$$

$$\therefore \ \mathrm{K_{uncat}} = \mathrm{A} \exp(\frac{-\mathrm{Ea}}{\mathrm{RT}})$$

$$\mathrm{K_{cat}} = \mathrm{A} \ \exp(\frac{-\mathrm{Ea} + 10}{\mathrm{RT}})$$

$$\therefore \frac{\mathrm{K_{Cat}}}{\mathrm{K_{uncat}}} = \ \exp(\frac{-\mathrm{Ea} + 10 - \mathrm{Ea}}{\mathrm{RT}})$$

$$= \ \exp(\frac{10}{\mathrm{RT}})$$

$$= \exp\left(\frac{10 \times 10^3}{8.31 \times 300}\right)$$

$$= \exp(4)$$

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$\therefore x = 4$

Hence, the answer is 4

Q. 29 Reaction of $[Co(H_2O)_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g} - orbitals of the product is_____.

Correct Answer:
6

Solution:
The given reaction occurs as

$$[Co(H_2O)_6]^{2+} + 6NH_3 \xrightarrow{O_2} [Co(NH_3)_6]^{3+} + 6H_2O$$

It is to be noted that d^7 configuration in a strong ligand field undergoes oxidation readily to form a d^6 complex.

Thus, configuration of Co^{3+} in strong ligand field is $t_{2g}^6 e_g^0$.

Hence, the arswer is 6

Q. 30 The moles of methane required to produce $81\ g$ of water after complete combustion is _____$\times 10^{-2}$ mol.[nearest integer]

Correct Answer:
225

Solution:
$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

$$\text{moles of water} = \frac{81}{18}$$

$$\therefore \text{ moles of methane} = \frac{1}{2} \times \frac{81}{18}$$

$$= 2.25$$

$$= 225 \times 10^{-2}$$

Hence, the answer is 225

Maths

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Q. 1 Let $\mathrm{f}:\mathbb{R}\to\mathbb{R}$ be defined as $\mathrm{f(x)}=\mathrm{x}-1$ and $\mathrm{g}:\mathbb{R}-\{1,-1\}\to\mathbb{R}$ be defined as $\mathrm{g(x)}=\dfrac{\mathrm{x}^2}{\mathrm{x}^2-1}$ Then the function $f\circ g$ is

Option 1:
one-one but not onto

Option 2:
onto but not one-one

Option 3:
both one-one and onto

Option 4:
neither one-one nor onto

Correct Answer:
neither one-one nor onto

Solution:
$f(x)=x-1$

$g(x)=\dfrac{x^2}{x^2-1}$

$f\circ g=f(g(x))=\dfrac{x^2}{x^2-1}-1=\dfrac{1}{x^2-1}$

since $f\circ g(x)$ is even function hance many one function
and also Range $\in(-\infty,-1]\cup(0,\infty)$ which is nat equal ts co-domain.

hence into function.

Q. 2 If the system of equations
$\alpha\mathrm{x}+\mathrm{y}+\mathrm{z}=5,\mathrm{x}+2\mathrm{y}+3\mathrm{z}=4,\mathrm{x}+3\mathrm{y}+5\mathrm{z}=\beta$
: has infinitely many solutions, then the ordered pair (α,β) is equal to :

Option 1:
$(1,-3)$

Option 2:
$(-1,3)$

Option 3:
$(1,3)$

Option 4:
$(-1,-3)$

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Correct Answer:
$(1,3)$

Solution:
$\alpha x + y + z = 5$

$x + 2y + 3z = 4$

$x + 3y + 5z = \beta$

$$\Delta = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{vmatrix} = (\alpha - 1)$$

$$\Delta_1 = \begin{vmatrix} 5 & 1 & 1 \\ 4 & 2 & 3 \\ \beta & 3 & 5 \end{vmatrix} = (\beta - 3)$$

$$\Delta_2 = \begin{vmatrix} \alpha & 5 & 1 \\ 1 & 4 & 3 \\ 1 & \beta & 5 \end{vmatrix} = 20\alpha - 3\alpha\beta - 14 + \beta$$

$$\Delta_3 = \begin{vmatrix} \alpha & 1 & 5 \\ 1 & 2 & 4 \\ 1 & 3 & \beta \end{vmatrix} = 2\alpha\beta - 12\alpha + 9 - \beta$$

$$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$$

for infinitaly many solutions.hence $\alpha = 1, \beta = 3$

$$\therefore (\alpha, \beta) \equiv (1,3)$$

Q. 3 If $A = \sum_{n=1}^{\infty} \frac{1}{(3+(-1)^n)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{(3+(-1)^n)^n}$, then $\frac{A}{B}$ is equal to :

Option 1:
$\frac{11}{9}$

Option 2:
1

Option 3:
$-\frac{11}{9}$

Option 4:
$-\frac{11}{3}$

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Correct Answer:

$$-\frac{11}{9}$$

Solution:

$$A=\sum_{n=1}^{\infty}\frac{1}{(3+(-1)^n)^n}$$

$$=\frac{1}{2}+\frac{1}{4^2}+\frac{1}{2^3}+\frac{1}{4^4}+\frac{1}{2^5}+\frac{1}{4^6}+\cdots$$

$$=\frac{\frac{1}{2}}{1-\frac{1}{4}}+\frac{\frac{1}{16}}{1-\frac{1}{16}}=\frac{2}{3}+\frac{1}{15}=\frac{11}{15}$$

$$B=\sum_{n=1}^{\infty}\frac{(-1)^n}{(3+(-1)^n)^n}$$

$$=\frac{-1}{2}+\frac{1}{4^2}-\frac{1}{2^3}+\frac{1}{4^4}\cdots\cdots-$$

$$=\frac{-\frac{1}{2}}{1-\frac{1}{4}}+\frac{\frac{1}{16}}{1-\frac{1}{16}}=-\frac{2}{3}+\frac{1}{15}=\frac{-9}{15}$$

$$\therefore \quad \frac{A}{B}=\frac{-11}{9}$$

Q. 4 $\lim\limits_{x\to 0}\dfrac{\cos(\sin x)-\cos x}{x^4}$ is equal to :

Option 1:

$$\frac{1}{3}$$

Option 2:

$$\frac{1}{4}$$

Option 3:

$$\frac{1}{6}$$

Option 4:

$$\frac{1}{12}$$

Correct Answer:

$$\frac{1}{6}$$

Solution:

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$$\lim_{x\to 0}\frac{\text{cas}(\sin x)-\cos x}{x^4}$$

$$\lim_{x\to 0}\frac{2\sin\left(\frac{x+\sin x}{2}\right)\left(\sin\left(\frac{x-\sin x}{2}\right)\left(\frac{x^2-\sin^2 x}{4}\right)\right.}{x^4\left(\frac{x+\sin x}{2}\right)\left(\frac{x-\sin x}{2}\right)}$$

$$\lim_{x\to 0}\frac{x^2-\sin^2 x}{2x^4}=\lim_{x\to 0}\frac{x^2-\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots\right)^2}{2x^4}$$

$$\lim_{x\to 0}\frac{x^4\left(\frac{2}{3!}+\cdots-\cdots\right)}{2x^4}$$

$$=\frac{1}{6}$$

Q. 5 Let $f(x)=\min\{1,1+x\sin x\}, 0\le x\le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ardered pair (m,n) is equal to

Option 1:
(2,0)

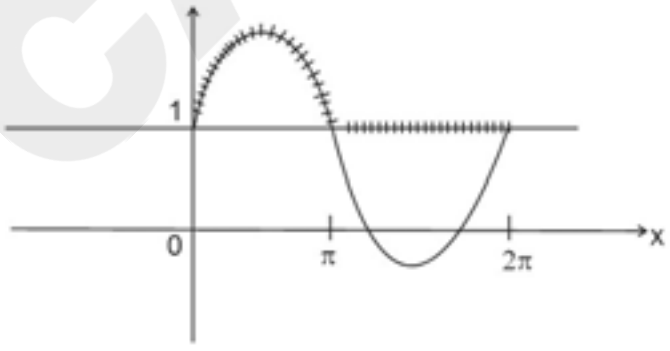
Option 2:
(1,0)

Option 3:
(1,1)

Option 4:
(2,1)

Correct Answer:
(1,0)

Solution:
$\text{f(x)}=\min\{1,1+\text{x}\sin\text{x}\};0\le\text{x}\le 2\pi$



Non differentiable at $\text{x}=\pi$

$\therefore\quad \text{m}=1,\text{n}=0\quad ;(\text{m},\text{n})\equiv(1,0)$

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Q. 6 Cosider a cuboid of sides $2x, 4x \text{ and } 5x$ and a closed hemisphere of radius r. If the sum of their surface areas is a constant k, then the ratio $x : r$, for which the sum of their volumes is maximum, is:

Option 1:
2:5

Option 2:
19:45

Option 3:
3:8

Option 4:
19:15

Correct Answer:
19:45

Solution:

$$S = 2\left(8x^2 + 20x^2 + 10x^2\right) + \left(2\pi r^2 + \pi r^2\right)$$

$$K = 76x^2 + 3\pi r^2$$

$$\text{Volume } (V) = 40x^3 + \frac{2}{3}\pi r^3$$

$$V(r) = 40\left(\frac{k - 3\pi r^2}{76}\right)^{3/2} + \frac{2}{3}\pi r^3$$

$$V'(r) = 40\left(\frac{3}{2}\right)\left(\frac{k - 3\pi r^2}{76}\right)^{1/2}\left(-\frac{3\pi}{30}r\right) + 2\pi r^2$$

$$v'(r) = 0$$

$$60\left(\frac{3\pi r}{38}\right)\left(\frac{k - 3\pi r^2}{76}\right)^{1/2} = 2\pi r^2$$

$$\frac{90}{19} \quad x = 2n$$

$$\frac{x}{x} = \frac{19}{45}$$

Q. 7 The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to:

Option 1:
$\frac{32}{3}$

Option 2:
$\frac{40}{3}$

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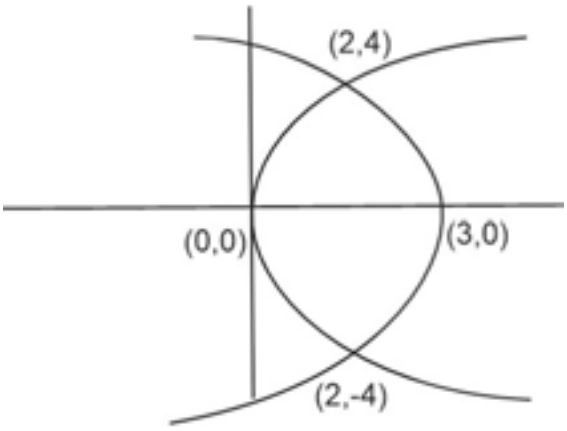
Option 3:
16

Option 4:
19

Correct Answer:
16

Solution:



$$\begin{aligned}\text{Area} &= 2\int_0^4 \left(3 - \frac{y^2}{16} - \frac{y^2}{8}\right) dy \\ &= 2\left[3y - \frac{3y^3}{48}\right]_0^4 \\ &= 2\left[12 - \frac{64}{16}\right] = 16\end{aligned}$$

Q. 8 If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx = g(x) + c, g(1) = 0,$ then $g\left(\frac{1}{2}\right)$ is equal to :

Option 1:
$\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) + \frac{\pi}{3}$

Option 2:
$\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) + \frac{\pi}{3}$

Option 3:
$\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) - \frac{\pi}{3}$

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Option 4:

$$\frac{1}{2}\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)-\frac{\pi}{6}$$

Correct Answer:

$$\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)+\frac{\pi}{3}$$

Solution:

$$\int\frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx$$

$$\int\frac{\sqrt{1-x^2}}{x+x^2}dx$$

$$\text{let} x=\sin\theta$$

$$\int\frac{\cos^2\theta d\theta}{\sin\theta(1+\sin\theta)}$$

$$\int\frac{1-\sin\theta}{\sin\theta}d\theta=\int(\operatorname{cosec}\theta-1)d\theta$$

$$=\ln|\operatorname{cosec}\theta-\cot\theta|-\theta+c$$

$$=\ln\left|\frac{1-\sqrt{1-x^2}}{x}\right|-\sin^{-1}x+c$$

$$\therefore g(x)=\ln\left|\frac{1-\sqrt{1-x^2}}{2}\right|-\sin^{-1}x+\mu$$

$$0=0-\frac{\pi}{2}+\mu\Rightarrow\mu=\frac{\pi}{2}$$

$$g(x)=\ln\left|\frac{1-\sqrt{1-x^2}}{x}\right|-\sin^{-1}x+\frac{\pi}{2}$$

$$\therefore g\left(\frac{1}{2}\right)=\ln\left|\frac{2-\sqrt{3}}{1}\right|-\frac{\pi}{6}+\frac{\pi}{2}$$

$$g\left(\frac{1}{2}\right)=\ln\left|\frac{\sqrt{3}-1}{\sqrt{3}+1}\right|+\frac{\pi}{3}$$

Q. 9 If $y=y(x)$ is the solution of the differential equation $x\frac{dy}{dx}+2y=xe^x, y(1)=0$ then the local maximum value of the function $z(x)=x^2y(x)-e^x, x\in\mathbb{R}$ is :

Option 1:

$1-e$

Option 2:

0

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Option 3:

$\frac{1}{2}$

Option 4:

$\frac{4}{e} - e$

Correct Answer:

$\frac{4}{e} - e$

Solution:

$x\frac{dy}{dx} + 2y = xe^x$

$\frac{dy}{dx} + \left(\frac{2}{x}\right)y = e^x$

$If = e^{\int \frac{2}{x}dx} = x^2$

$\therefore \left(x^2\right) = \int x^2 e^x dx$

$y\left(x^2\right) = e^x\left(x^2 - 2x + 2\right) + c$

$\text{at} x = 1, y = 0$

$0 = e + c \Rightarrow c = -e$

$\therefore y(x) = \frac{e^x\left(x^2 - 2x + 2\right) - e}{x^2}$

$z(x) = e^x\left(x^2 - 2x + 2\right) - e - e^x$

$z(x) = e^x\left(x^2 - 2x + 1\right) - e$

$z'(x) = 2e^x(x-1) + e^x(x-1)^2$

$= e^x(x-1)[x+1]$

local maximum of x=-1

$z(-1) = e^{-1}(4) - e = \frac{4}{e} - e$

Q. 10 If the solution of the differential equation $\frac{dy}{dx} + e^x\left(x^2 - 2\right)y = \left(x^2 - 2x\right)\left(x^2 - 2\right)e^{2x}$ satisfies $y(0) = 0$, then the value of $y(2)$ is_____

Option 1:
-1

Option 2:
10

Option 3:
0

Option 4:
e

Correct Answer:
0

Solution:

$$\frac{dy}{dx}+e^{x}\left(x^{2}-2\right)y=\left(x^{2}-2x\right)\left(x^{2}-2\right)e^{2x}$$

$$I\cdot F\cdot=e^{\int e^{x}\left(x^{2}-2\right)dx}$$

$$=e^{\left[\left(x^{2}-2\right)e^{x}-\int 2xe^{x}dx\right]}$$

$$=e^{\left(x^{2}-2x\right)e^{x}}$$

$$y\cdot e^{\left(x^{2}-2x\right)e^{x}}=\int e^{\left(x^{2}-2x\right)e^{x}}\left(x^{2}-2x\right)\left(x^{2}-2\right)e^{2x}dx$$

$$e^{x}\left(x^{2}-2x\right)=t$$

$$=e^{\left[\left(x^{2}-2\right)e^{x}-2e^{x}(x)+2e^{x}\right]}\left\{2(x-1)e^{x}+e^{x}\left(x^{2}-2x\right)\right\}dx=dx$$

$$y\cdot e^{t}=\int e^{t}\cdot tdt$$

$$ye^{t}=te^{t}-e^{t}+c$$

$$ye^{\left(x^{2}-2x\right)e^{x}}=\left(x^{2}-2x\right)e^{x}e^{\left(x^{2}-2x\right)e^{x}}-e^{\left(x^{2}-2x\right)e^{x}}+c$$

$$\text{at } x=0, y=0$$

$$0=0-1+c\Rightarrow c=\frac{1}{=}$$

$$\therefore \text{ at } x=2$$

$$y(1)=0-1+1$$

$$y=0$$

Q. 11 If m is the slope of a common tangent to the curves $\frac{\mathrm{x}^{2}}{16}+\frac{\mathrm{y}^{2}}{9}=1$ and $\mathrm{x}^{2}+\mathrm{y}^{2}=12$, then $12\mathrm{m}^{2}$ is equal to :

Option 1:
6

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Option 2:
9

Option 3:
10

Option 4:
12

Correct Answer:
9

Solution:

$\frac{x^2}{16}+\frac{y^2}{9}=1 \qquad \mathrm{y}=\mathrm{m}x\pm\sqrt{16\mathrm{m}^2+9}$

$\therefore \quad 16m^2+9=\mathrm{m}x\pm\sqrt{12\mathrm{m}^2+12}$

$\mathrm{m}^2=\frac{3}{4}$

$\therefore 12\mathrm{m}^2=9$

$\because \qquad x^2+y^2=12$

$\therefore \qquad y=mx\pm\sqrt{2m^2+12}$

Q. 12 The locus of the mid point of the line segment joining the point $(4,3)$ and the points on the ellipse $\mathrm{x}^2+2\mathrm{y}^2=4$ is an ellipse with eccentricity :

Option 1:
$\frac{\sqrt{3}}{2}$

Option 2:
$\frac{1}{2\sqrt{2}}$

Option 3:
$\frac{1}{\sqrt{2}}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{1}{\sqrt{2}}$

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Solution:

$$\frac{x^2}{4}+\frac{y^2}{2}=1$$

$$p:(2\cos\theta,\sqrt{2}\sin\theta)$$

$$\therefore \quad \hbar=\frac{4+2\cos\theta}{2}, \quad k=\frac{3+\sqrt{2}\sin\theta}{2}$$

$$(h-2)^2+\left(\frac{2k-3}{\sqrt{2}}\right)^2=1$$

$$\therefore \quad 2\left(1-e^2\right)=1$$

$$1-e^2=\frac{1}{2}$$

$$e=\frac{1}{\sqrt{2}}$$

Q. 13 The normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ at the point $(8,3\sqrt{3})$ on it passe the point :

Option 1:

$\left(15,-2\sqrt{3}\right)$

Option 2:

$\left(9,2\sqrt{3}\right)$

Option 3:

$\left(-1,9\sqrt{3}\right)$

Option 4:

$\left(-1,6\sqrt{3}\right)$

Correct Answer:

$\left(-1,9\sqrt{3}\right)$

Solution:

$$\frac{x^2}{a^2}-\frac{y^2}{9}=1$$

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$p : (8, 3\sqrt{3})$ lies on Hyperbola

$a^2 = 16$

$\frac{2x}{16} - \frac{2y}{9}y' = 0$

Equation of Normal:

$m_N = -\frac{1}{y'}\Big)_p = \frac{-16y}{9x}\Big)_p = \frac{-48\sqrt{3}}{72} = -\frac{2\sqrt{3}}{3}$

$y - 3\sqrt{3} = \frac{-2\sqrt{3}}{3}(x - 8)$

$\sqrt{3}y - 9 = -2x + 16$

$2x + \sqrt{3}y = 25$

Q. 14 If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point :

Option 1:
(2,-2,0)

Option 2:
(-2,2,0)

Option 3:
(1,0,2)

Option 4:
(-1,0,-2)

Correct Answer:
(1,0,2)

Solution:

$(2x + y - 5z) + \lambda(3x - y + 4z - 7) = 0$

$(2 + 3\lambda)x + (1 - \lambda)y + (-5 + 4\lambda)z - 7\lambda = 0$

For this plane to be perpendicular to 2x+y-5z=0

$2(2 + 3\lambda) + (1 - \lambda) - 5(-5 + 4\lambda) = 0$

$-15\lambda = -30$

$\lambda = 2$

$\therefore$ Required plane : $8x - y + 3z = 14$

Q. 15 If the lines $\vec{r} = (\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{j} - \hat{k})$ and $\vec{r} = (\alpha\hat{i} - \hat{j}) + \mu(2\hat{i} - 3\hat{k})$ are coplanar, then the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is:

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Option 1:
$\frac{2}{9}$

Option 2:
$\frac{2}{11}$

Option 3:
$\frac{4}{11}$

Option 4:
2

Correct Answer:
$\frac{2}{11}$

Solution:
lines ort co plonar.

$$\begin{vmatrix} 1-2 & 0 & 1 \\ 0 & 3 & -1 \\ 2 & 0 & -3 \end{vmatrix} = 0$$

$(1-\alpha)(-5)-6=0$

$\alpha - 1 = \frac{2}{3} \quad \Rightarrow \quad \alpha = \frac{5}{3}$

Nermal vector of plans: $(-9,-2,-6)$

$\therefore$ Equation of plans: $-9(x-1)-2(y+1)-6(2-1)=0$

$9x+2y+6z-13=0$

$\therefore$ Distance from $\left(\frac{5}{3},0,0\right) = \frac{15-13}{\sqrt{81+4+36}} = \frac{2}{11}$

Q. 16 Let $\tilde{a}=\hat{i}+\hat{j}+2\hat{k}, \tilde{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\tilde{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\tilde{v}$ be a vector in the plane of $\tilde{a}$ and $\tilde{b}$ whose projection on $\tilde{c}$ is $\frac{2}{\sqrt{3}}$. If $\tilde{v}\cdot\hat{j}=7$, then $\tilde{v}\cdot(\hat{i}+\hat{k})$ is equal to :

Option 1:
6

Option 2:
7

Option 3:
8

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Option 4:
9

Correct Answer:
9

Solution:
$\bar{v} = \lambda\bar{a} + \mu\bar{b}$

$\bar{v} = (\lambda + 2k)\hat{i} + (\lambda - 3\mu)\hat{j} + (2\lambda + k)\hat{k}$

v. $\frac{\vec{c}}{|\vec{c}|} = \frac{2}{\sqrt{3}}$

$\text{als } 0\bar{5}, j = 7$

$\geq \lambda - 2\mu = 7$

$\lambda + 2\mu - \lambda + 3\mu + 2\lambda + \mu = 2$

$2\lambda + 6\mu = 2$

$x + 3\mu = 1$

$\therefore \lambda = 4, \mu = -1$

$\therefore \bar{j} \cdot (\hat{i} + \hat{k}) = (2\vec{i} + \vec{j} + 7\hat{k}) \cdot (\hat{i} + \hat{k})$

$= 9$

Q. 17 The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70 . If the correct mean is 16 , then the correct variance is equal to:

Option 1:
10

Option 2:
36

Option 3:
43

Option 4:
60

Correct Answer:
43

Solution:
$\bar{x} = 15$

$r = 2$

$n = 50$

let incorrect observation =x_1

correct obsenvation =x_1^*

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$\therefore \quad x_1 + x_2 + \ldots - x_{50} = 750$

$x_1^* + x_2 + \ldots x_{50} = 800$

$x_1^* - x_1 = 50$

also $x_1^* + x_1 = 70$

$\therefore \quad x_1 = 10, \quad x_1^* = 60$

$\therefore \dfrac{x_1^2 + x_2^2 + - - x_{50}^2}{50} - 225 = 4$

$x_1^2 + x_2^2 + \cdots - x_{50}^2 = 11450$

$(x_1^*)^2 + x_2^2 + \cdots - x_{50}^2 = 11450 + 3500$

$= 14950$

$= \dfrac{14950}{50} - 256$

$= 299 - 256$

$= 43$

Q. 18 $16 \sin(20°) \sin(40°) \sin(80°)$ is equal to :

Option 1:

$\sqrt{3}$

Option 2:

$2\sqrt{3}$

Option 3:

3

Option 4:

$4\sqrt{3}$

Correct Answer:

$2\sqrt{3}$

Solution:

$16 \sin 20° \sin 40° \sin 80°$

$16 \dfrac{\sin 60°}{4} = 4\left(\dfrac{\sqrt{3}}{2}\right) = 2\sqrt{3}$

Q. 19 If the inverse trignometric functions take principal values, then $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right) + \dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)$ is equal to:

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Option 1:
0

Option 2:
$\frac{\pi}{4}$

Option 3:
$\frac{\pi}{3}$

Option 4:
$\frac{\pi}{6}$

Correct Answer:
$\frac{\pi}{3}$

Solution:
$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$

$\cos^{-1}\left(\left(\frac{3}{10}\times\frac{3}{5}\right)+\frac{2}{5}\times\frac{4}{5}\right)$

$\cot^{-1}\left(\frac{9+16}{50}\right)=\frac{\pi}{3}$

Q. 20 Let $r\in\{p, q, \sim p, \sim q\}$ be such that the logical statement $r\vee(\sim p)\Rightarrow(p\wedge q)\vee r$ is a tautology. Then r is equal to :

Option 1:
p

Option 2:
q

Option 3:
$\sim p$

Option 4:
$\sim q$

Correct Answer:
p

Solution:
$r\vee(\sim p)\rightarrow(p\wedge q)\vee r$
when $r\equiv P$

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Then
$\mathrm{rv}(\sim \mathrm{p}) \rightarrow (\mathrm{p} \wedge \mathrm{q}) \vee \mathrm{r}$ is tautology

Q. 21 Let $\mathrm{f} : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\mathrm{f}(\mathrm{x}+\mathrm{y}) = 2^{\mathrm{x}}\mathrm{f}(\mathrm{y}) + 4^{\mathrm{y}}\mathrm{f}(\mathrm{x}), \forall \mathrm{x}, \mathrm{y} \in \mathbb{R}$. If $\mathrm{f}(2) = 3$, then $14 \cdot \dfrac{\mathrm{f}'(4)}{\mathrm{f}'(2)}$ is equal to

Correct Answer:
248

Solution:
$\mathrm{f}(\mathrm{x}+\mathrm{y}) = 2^{\mathrm{x}}\mathrm{f}(\mathrm{y}) + 4^{\mathrm{y}}\mathrm{f}(\mathrm{x})$
$\mathrm{f}(2) = 3$

if $\mathrm{y} = 2$

put $\mathrm{x} = 2$

$\therefore \mathrm{f}'(\mathrm{x}+2) = 3 \cdot 2^{\mathrm{x}} \ln 2 + 16\mathrm{f}'(\mathrm{x})$
$\mathrm{f}'(4) = 12 \ln 2 + 16\mathrm{f}'(2)$
$\mathrm{f}(2+\mathrm{y}) = 4\mathrm{f}(\mathrm{y}) + 4\mathrm{y}(3)$
$\mathrm{f}'(2+\mathrm{y}) = 4\mathrm{f}'(\mathrm{y}) + 4\mathrm{y} \ln 4(3)$
$\mathrm{Put}\ \mathrm{y} = 2$
$\mathrm{f}'(4) = 4\mathrm{f}'(2) + 96 \ln 2 \ldots - (\mathrm{ii})$
$\mathrm{from}(i)\&(ii)$
$\mathrm{f}'(2) = 7 \ln 2$
$\mathrm{f}'(4) = 124 \ln 2$

$$\therefore \quad 14 \cdot \frac{f'(4)}{f'(2)} = \frac{14(124 \ln 2)}{7 \ln 2}$$
$$= 248 \ \mathrm{Ans.}$$

Q. 22 Let p and q be two real numbers such that $\mathrm{p}+\mathrm{q} = 3$ and $\mathrm{p}^4 + \mathrm{q}^4 = 369$. Then $\left(\dfrac{1}{\mathrm{p}} + \dfrac{1}{\mathrm{q}}\right)^{-2}$ is equal to

Correct Answer:
4

Solution:

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$p+q=3$

$p^4+q^4=369$

$9=p^2+q^2+2pq$

$81+4(pq)^2-36pq=369+2p^2q^2$

$2(pq)^2-36pq-288=0$

$pq=\dfrac{24,}{\downarrow \text{rejectad}}-6$

$\therefore \quad pq=-6$

Now $\left(\dfrac{1}{p}+\dfrac{1}{q}\right)^{-2}=\dfrac{p^2q^2}{(p+q)^2}=\dfrac{36}{9}=4$

Q. 23

If $z^2+z+1=0, z\in\mathbb{C}$, then $\left|\sum_{n=1}^{15}\left(z^n+(-1)^n\dfrac{1}{z^n}\right)^2\right|$ is equal to _____.

Correct Answer:

2

Solution:

$z^2+z+1=0$

$z=\omega,\omega^2$

$\left|\sum_{n=1}^{15}\left(z^n+(-1)^n\dfrac{1}{z^n}\right)^2\right|$

$\left|\sum_{n=1}^{15}\left(\omega^{2n}+\dfrac{1}{\omega^n}+2(-1)^n\right)\right|$

$\left|\dfrac{\omega^2\left(1-\omega^{30}\right)}{1-\omega^2}+\dfrac{1}{\omega^2}\dfrac{\left(1-\frac{1}{\omega^{30}}\right)}{\left(1-\frac{1}{\omega^2}\right)}-2\right|$

$|0+0-2|$

2

Q. 24

Let $X=\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}, Y=\alpha I+\beta X+\gamma X^2$ and

$Z=\alpha^2 I-\alpha\beta X+\left(\beta^2-\alpha\gamma\right)X^2, \alpha,\beta,\gamma\in\mathbb{R}$. If $Y^{-1}=\begin{bmatrix}1/5&-2/5&1/5\\0&1/5&-2/5\\0&0&1/5\end{bmatrix}$,

then $\mathbf{n}: (\alpha-\beta+\gamma)^2$ is equal to

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Correct Answer:
100

Solution:

$$x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore y = \alpha I + \beta x + \gamma x^2$$

$$y = \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix}$$

&

$$z = \begin{bmatrix} \alpha^2 & -\alpha\beta & \beta^2 - \alpha r \\ 0 & \alpha^2 & -\alpha\beta \\ 0 & 0 & \alpha^2 \end{bmatrix}$$

$$\text{As } yy^{-1} = I$$

then

$$\alpha = 5, \quad \beta = 10, \gamma = 15$$

$$\therefore (\alpha - \beta + \gamma)^2 = 100$$

Q. 25 The total number of 3 -digit numbers, whose greatest common divisor with 36 is 2 , is

Correct Answer:
150

Solution:

$$36 = 2^2 3^2$$

Number should have prime factor 2 anly once \& it should not have 3 as factor

$$102, 106, 110, \ldots 98 \Rightarrow 225 \text{ numbers}$$

$$\text{Required numbers } = 225 - 75 \text{ (multiple of 3) } = 150$$

Q. 26 If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots + \left({}^{60}C_{20}\right) = \frac{m}{n}{}^{60}C_{20}$ m and n are coprime, then $\text{m} + \text{n}$ is equal to

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Correct Answer:
102

Solution:
$40c_0 + 41c_1 + 42c_2 + \cdots c_{20}$

$41c_0 + 4_{c_1} + 42c_2 + - \cdots + 59c_{19} + 60_{c_{20}}$

$\{$ using $n_{c_r} + {}^{n}c_{r-1} = \mathrm{n} + 1c_r\}$

$= \dfrac{61}{41} \quad 60_{c_{20}}$

$\therefore \quad \mathrm{m} = 61, \mathrm{n} = 41$

$\therefore \mathrm{m} + \mathrm{n} = 102$

Q. 27 If $\mathrm{a}_1(>0), \mathrm{a}_2, \mathrm{a}_3, \mathrm{a}_4, \mathrm{a}_5$ are in a G.P., $\mathrm{a}_2 + \mathrm{a}_4 = 2\mathrm{a}_3 + 1$ and $3\mathrm{a}_2 + \mathrm{a}_3 = 2\mathrm{a}_4$, then $\mathrm{a}_2 + \mathrm{a}_4 + 2\mathrm{a}_5$ is equal to____

Correct Answer:
40

Solution:
$\mathrm{a}_1, \mathrm{a}_2, \ldots, \mathrm{a}_5 \longrightarrow \mathrm{G} \cdot \mathrm{p}.$

$\mathrm{a}_2 + \mathrm{a}_4 = 2\mathrm{a}_3 + 1; \quad 3\mathrm{a}_2 + \mathrm{a}_3 = 2\mathrm{a}_4$

$\mathrm{ar} + \mathrm{ar}^3 = 2\mathrm{a}^2 + 1; 3\mathrm{ar} + \mathrm{ar}^2 = 2\,\mathrm{ar}^3$

$2\mathrm{r}^2 - \mathrm{r} - 3 = 0$

$\mathrm{r} = -1 \cdot \dfrac{3}{2}$

Now

$\dfrac{3\mathrm{a}}{2} + \dfrac{27\mathrm{a}}{8} = \dfrac{3\mathrm{a}}{2} + 1$

$\therefore \quad \mathrm{n} = \dfrac{3}{2}$

$\mathrm{a} = \dfrac{8}{3}$

$\mathrm{a}_2 + \mathrm{a}_4 + 2\mathrm{a}_5 = \mathrm{ar} + \mathrm{ax}^3 + 2\mathrm{ax}^4$

$= 4 - 0 \text{ ins}$

Q. 28 The integral $\dfrac{24}{\pi} \displaystyle\int_0^{\sqrt{2}} \dfrac{(2 - \mathrm{x}^2)\,\mathrm{dx}}{(2 + \mathrm{x}^2)\sqrt{4 + \mathrm{x}^4}}$ is equal to

Correct Answer:
3

Solution:

$$\frac{24}{\pi}\int_0^{\sqrt{2}}\frac{(2-x^2)\,dx}{(2+x^2)\sqrt{4+x^4}}$$

$$\frac{24}{\pi}\int_0^{\sqrt{2}}\frac{\left(\frac{2}{x^2}-1\right)}{\left(\frac{2}{x}+x\right)\sqrt{\frac{4}{x^2}+x^2}}dx$$

$$\text{let}\frac{2}{x}+x=5$$

$$\frac{24}{\pi}\int_{\infty}^{2\sqrt{2}}-\frac{dt}{t\sqrt{t^2-4}}$$

$$\frac{24}{\pi}\times\frac{1}{2}\left[\sec^{-1}\left(\frac{t}{2}\right)\right]_{2\sqrt{2}}^{\infty}$$

$$\frac{12}{\pi}\left[\frac{\pi}{2}-\frac{\pi}{4}\right]$$

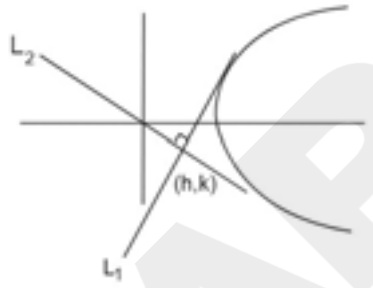
$$=3$$

Q. 29

Let a line L_1 be tangent to the hyperbola $\frac{x^2}{16}-\frac{y^2}{4}=1$ and let L_2 be the line passing through the origin and perpendicular to L_1. If the locus of the point of :intersection of L_1 andL_2 is $\left(x^2+y^2\right)^2=\alpha x^2+\beta y^2$, then $\alpha+\beta$ is equal to

Correct Answer:
12

Solution:



$$\frac{x^2}{16}-\frac{y^2}{4}=1$$

$$L_1:\quad\frac{x\sec\theta}{4}-\frac{y\tan\theta}{2}=1$$

$$m_1=\frac{1}{2\sin\theta}$$

$$L_2:\quad y=-2\sin\theta x$$

passes through (h,k)

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$$k = -2(\sin\theta)h^h$$

$$\sin\theta = \frac{-k}{2h}$$

$$\text{from} L_1 :$$

$$\frac{h}{4}\frac{2h}{\sqrt{4k^2+k^2}} - \frac{k}{2}\left(\frac{-k}{\sqrt{4k^2-k^2}}\right) = 1$$

$$\left(x^2+y^2\right)^2 = 16x^2 - 4y^2$$

$$\therefore \quad \alpha = 16, \quad \beta = -4$$

$$\therefore \alpha + \beta = 12$$

Q. 30 If the probability that a randomly chosen 6 -digit number formed by using digits 1 and 8 only is a multiple of 21 is p, then $96p$ is equal to____

Solution:

divisible by 21 when -

Case-1 All digits ore 1=1

case-2 All ligits are 8=1

Case-3 three digisare, three digits are $8 = \frac{6!}{3!3!}$

$= 20$

$\therefore$ Requirad Probability = $\frac{22}{2^6} = \frac{11}{32} = p$

$$\therefore \quad 96P = 33$$

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27 June Morning Shift

Physics

Q. 1 A projectile is laughed at angle $'\alpha'$ with the horizontal with a velocity 20ms^{-1}. After 10s, its incliation with horizontal is $'\beta'$. The value of $\tan\beta$ will be : $\left(\text{g} = 10\text{ms}^{-2}\right)$

Option 1:
$\tan\alpha + 5\sec\alpha$

Option 2:
$\tan\alpha - 5\sec\alpha$

Option 3:
$2\tan\alpha - 5\sec\alpha$

Option 4:
$2\tan\alpha - 5\sec\alpha$

Correct Answer:
$\tan\alpha - 5\sec\alpha$

Solution:
At t=10s,

$$V_x = u\cos\alpha = constant$$

$$V_y = u\sin\alpha + (-g)t$$

$$\tan\beta = \frac{\text{V}_\text{y}}{\text{V}_\text{x}} = \frac{20\sin\alpha - 10\times 10}{20\cos\alpha} = \tan\alpha - 5\sec\alpha$$

Q. 2 A girl standing on a road hold her umbrella at 45° with velocity to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}\text{kmh}^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is :

Option 1:
30kmh^{-1}

Option 2:
$\frac{25}{\sqrt{2}}\text{kmh}^{-1}$

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Option 3:

$\frac{30}{\sqrt{2}}\mathrm{kmh}^{-1}$

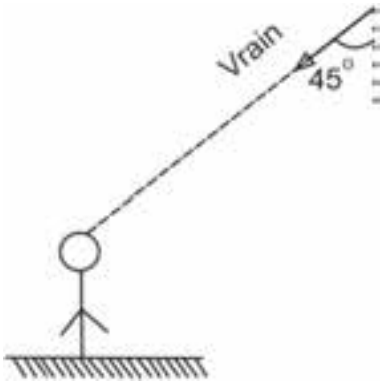
Option 4:

$25\mathrm{kmh}^{-1}$

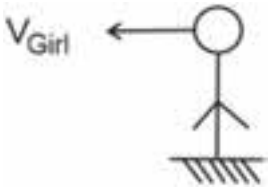
Correct Answer:

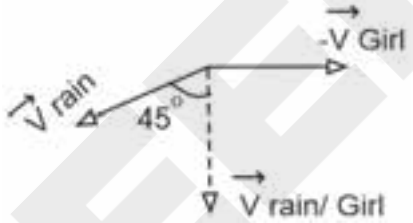
$\frac{30}{\sqrt{2}}\mathrm{kmh}^{-1}$

Solution:



When girl starts running $\vec{V}_{\text{rain / Girl}} = \vec{V}_{\text{rain}} - \vec{V}_{\text{Girl}}$





$\mathrm{V}_{\text{Girl}} = \mathrm{V}_{\text{rain}} \sin 45^\circ$

$15\sqrt{2} = \mathrm{Vrain} \times \frac{1}{\sqrt{2}}$

$\mathrm{V}_{\text{rain}} = 30 \text{ km/hr}$

$\mathrm{V}_{\text{rain/Girl}} = \mathrm{V}_{\text{rain}} \cos 45^\circ = 30 \cos 45^\circ = \frac{30}{\sqrt{2}}\mathrm{km/hr}$

Hence the correct answer is option 3.

Q. 3 A silver wire has a mass $(0.6 \pm 0.006)\,\mathrm{g}$, radius $(0.5 \pm 0.005)\,\mathrm{mm}$ and length $(4 \pm 0.04)\,\mathrm{cm}$. The maximum percentage error in the measurement of its density will be :

Option 1:

4%

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Option 2:
3%

Option 3:
6%

Option 4:
7%

Correct Answer:
4%

Solution:

$$\text{density } (\rho) = \frac{\text{mass}}{\text{volume}} = \frac{\text{m}}{\pi \text{r}^2 \text{l}}$$

$$\frac{\Delta\rho}{\text{p}} \times 100 = \left(\frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta l}{l}\right) \times 100$$

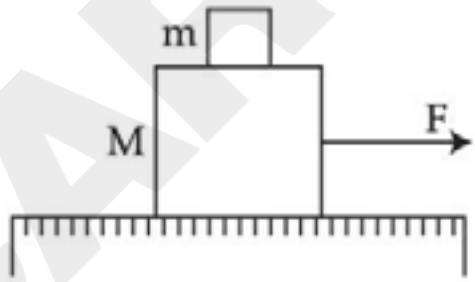
$$= \left(\frac{0.006}{0.6} + \frac{2 \times 0.005}{0.5} + \frac{0.04}{4}\right) \times 100$$

$$= 1 + 2 + 1$$

$$= 4\%$$

Hence the correct answer is option 1.

Q. 4 A system of two blocks of masses $\text{m} = 2\text{kg}$ and $\text{M} = 8\text{kg}$ is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the block move together will be :



Option 1:
9.8N

Option 2:
39.2N

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Option 3:

49N

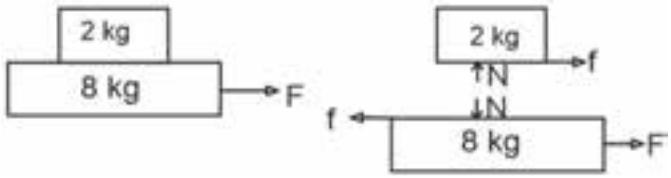
Option 4:

78.4N

Correct Answer:

49N

Solution:



$$f_l = \mu \text{ N}$$

$$= 0.5 \times 20 = 10 \text{ N}$$

Let both blocks are moving without slipping then

$$\text{F} = (2 + 8) \times \text{a}$$

$$\text{a} = \frac{\text{F}}{10}$$

For the upper block,

$$\text{f} = \text{ma} \leq \text{f}_\text{l}$$

$$2 \times \frac{\text{F}}{10} \leqslant 10$$

$$\text{F} \leqslant 50 \text{ N}$$

Hence the correct answer is option 3.

Q. 5 Two blocks of masses 10kg and 30kg are placed on the same straight line with coordinates $(0, 0)\ \text{cm}$ and $(\text{x}, 0)\ \text{cm}$ respectively. The block of 10kg is moved on the same line through a distance of 6cm towards the other block. The distance through which the block of 30kg must be moved to keep the position of centre of mass of the system unchanged is :

Option 1:

4cm towards the 10kg block

Option 2:

2cm away from the 10kg block

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Option 3:
2cm towards the 10kg block

Option 4:
4cm away from the 10kg block

Correct Answer:
2cm towards the 10kg block

Solution:

If the centre of mass does not change then

$$\Delta x_{cm} = \frac{m_1\Delta x_1 + m_2\Delta x_2}{m_1 + m_2} = 0$$

$$m_1\Delta x_1 = -m_2\Delta x_2$$

A negative sign implies both blocks will move in the opposite direction

$$10 \times 6 = 30 \times \Delta x_2$$

$\Delta x_2 = 2\ \text{cm}$ towards left

Hence the correct answer is option 3.

Q. 6 A 72Ω galvanomter is shunted by resistance of 8Ω . The percentage of the total current which passes through the galvanometer is :

Option 1:
0.1%

Option 2:
10%

Option 3:
25%

Option 4:
0.25%

Correct Answer:
10%

Solution:

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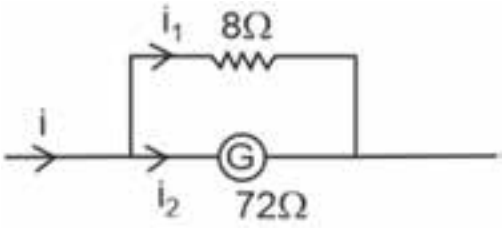

$$i_2 = \frac{i \times 8}{72 + 8} = \frac{i}{10}$$

Percentage of current passing through galvanometer is

$$= \frac{\left(\frac{i}{10}\right)}{i} \times 100 = 10\%$$

Hence the correct answer is option 2.

Q. 7 Given below are two statements :

Statements I : The law of gravitation hold goods for any pair of bodies in the universe.

Statement II : The weight of any person become zero when the person is at the center of the earth.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both **Statments I** and **Statements II** are true

Option 2:
Both **Statments I** and **Statements II** are false

Option 3:
Statments I is true and **Statements II** is false.

Option 4:
Statments I is false and **Statements II** is true.

Correct Answer:
Both **Statments I** and **Statements II** are true

Solution:
At center of earth acceleration due to gravity is zero,therefore weight is zero and law of gravitation holds for any two bodies which are having masse .

Hence the correct answer is option 1.

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Q. 8 What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass ?

(Assume the collision to be head-on elastic collison)

Option 1:

50.0%

Option 2:

66.6%

Option 3:

55.6%

Option 4:

33.3%

Correct Answer:

55.6%

Solution:

By linear momentum conservation

$$\mathrm{m(u)+5m(0)=m(v_1)+5m(v_2)}$$

$$\mathrm{u=v_1+5v_2} \quad \text{.....(1)}$$

Coefficient of restitution equation

$$\mathrm{e=\frac{v_2-v_1}{u-0}=1}$$

$$\mathrm{v_2-v_1=u} \quad \text{.....(2)}$$

Add (1) and (2)

$$\mathrm{6v_2=2u}$$

$$\mathrm{v_2=\frac{u}{3}}$$

Percentage energy of the bigger block $= \frac{\frac{1}{2}\times 5\mathrm{m}\times\frac{\mathrm{u}^2}{9}}{\frac{1}{2}\mathrm{mu}^2}\times 100 = 55.5\%$

Hence the correct answer is option 3.

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Q. 9 The velocity of a small ball of mass $'m'$ and density d_1, when dropped in a container filled with glycerine, becomes constant after some time. If the density of glycerine is d_2, then the viscous force acting on the ball, will be :

Option 1:

$mg\left(1-\frac{d_1}{d_2}\right)$

Option 2:

$mg\left(1-\frac{d_2}{d_1}\right)$

Option 3:

$mg\left(\frac{d_1}{d_2}-1\right)$

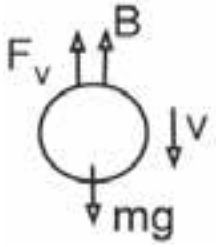
Option 4:

$mg\left(\frac{d_2}{d_1}-1\right)$

Correct Answer:

$mg\left(1-\frac{d_2}{d_1}\right)$

Solution:



$B+F_V=mg$

$F_v=mg-B$

$\because B=d_2vg=d_2\times\frac{m}{d_1}g$

$\therefore F_v=mg\left(1-\frac{d_2}{d_1}\right)$

Hence the correct answer is option 2.

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Q. 10 The susceptibility of a paramagnetic material is 99. The permeability of the material in $\mathrm{Wb/A-m}$, is :

(Permeability of free space $\mu_0 = 4\pi \times 10^{-7}\mathrm{Wb/A-m}$)

Option 1:
$4\pi \times 10^{-7}$

Option 2:
$4\pi \times 10^{-4}$

Option 3:
$4\pi \times 10^{-5}$

Option 4:
$4\pi \times 10^{-6}$

Correct Answer:
$4\pi \times 10^{-5}$

Solution:

$\mu_r = 1 + \lambda \qquad \lambda \rightarrow$ susceptibility

$\frac{\mu}{\mu_0} = 1 + 99 \qquad \mu_r \rightarrow$ relatine permeability

$\mu = 100 \times \mu_o = 100 \times 4\pi \times 10^{7}$

$= 4\pi \times 10^{-5}$

Hence the correct answer is option 3

Q. 11 The current flowig through an ac circuit is given by $\mathrm{I} = 5\sin(120\pi \mathrm{t})\ \mathrm{A}$.

How long will the current take to reach the peak value starting from zero?

Option 1:
$\frac{1}{60}\mathrm{s}$

Option 2:
$60\mathrm{s}$

Option 3:
$\frac{1}{120}\mathrm{s}$

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Option 4:

$\frac{1}{240}\text{s}$

Correct Answer:

$\frac{1}{240}\text{s}$

Solution:

$\text{I} = 5\sin(120\pi\text{t})$

$\text{t} = 0, \text{I} = 0$

$5 = 5\sin(120\pi\text{t})$

$\sin(120\pi\text{t}) = 1$

$120\pi\text{t} = \frac{\pi}{2}$

$\text{t} = \frac{1}{240}\ \text{second}$

Hence the correct answer is option 4.

Q. 12 Match **List I** with **List II** :

	List I		**List II**
a	Ultraviolet rays	i	Study crystal structure
b	Microwaves	ii	Greenhouse effect
c	Infrared waves	iii	Sterilizing surgical instrument
d	X-rays	iv	Radar system

Choose the correct answer from the options given below:

Option 1:

$(\text{a}) - (\text{iii}), (\text{b}) - (\text{iv}), \text{c} - (\text{ii}), (\text{d}) - (\text{i})$

Option 2:

$(\text{a}) - (\text{iii}), (\text{b}) - (\text{i}), \text{c} - (\text{ii}), (\text{d}) - (\text{iv})$

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Option 3:
$(a)-(iv),(b)-(iii),c-(ii),(d)-(i)$

Option 4:
$(a)-(iii),(b)-(iv),c-(i),(d)-(ii)$

Correct Answer:
$(a)-(iii),(b)-(iv),c-(ii),(d)-(i)$

Solution:
- X-rays are used for the study of crystal structures.
- Ultraviolet rays are used for sterilising the surgical instrument
- Infrared waves emitted from the earth's surface are responsible for the greenhouse effect.
- Microwaves are used in the Radar system

Hence the correct answer is option 1.

Q. 13 An α article and a carbon 12 atom has kinetic energy K. The ratio of their de-Broglie wavelengths $(\lambda_\alpha : \lambda_{C12})$ is :

Option 1:
$1:\sqrt{3}$

Option 2:
$\sqrt{3}:1$

Option 3:
$3:1$

Option 4:
$2:\sqrt{3}$

Correct Answer:
$\sqrt{3}:1$

Solution:
$$\lambda=\frac{h}{\sqrt{2mk}}$$

$$\frac{\lambda_\alpha}{\lambda_{c12}}=\frac{\sqrt{m_{c12}}}{\sqrt{m_\alpha}}=\sqrt{\frac{12}{4}}=\sqrt{3}$$

Hence the correct answer is option 2

Q. 14 A force of $10N$ acts on a charged particle placed between two plates of a charged capacitors. If one plate of capacitor is removed, the the force acting o that particle will be:

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Option 1:
5N

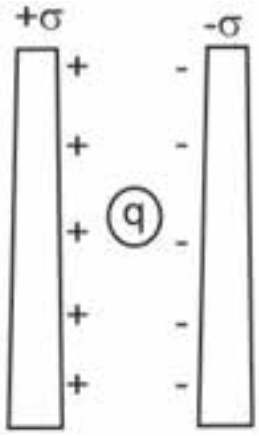
Option 2:
10N

Option 3:
20N

Option 4:
Zero

Correct Answer:
5N

Solution:

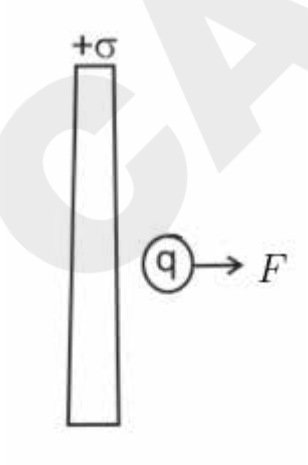


$$F_{inital} = q \times E$$

$$= q \times \frac{\sigma}{\varepsilon_0} = 10 \text{ N}$$

$\sigma \rightarrow$ Surface charge density

If one plate is removed



$$F = qE'$$

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$= q_1 \times \frac{\sigma}{2\varepsilon_0} = 5\ \text{N}$

Hence the correct answer is option 1.

Q. 15 The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

Option 1:
6s

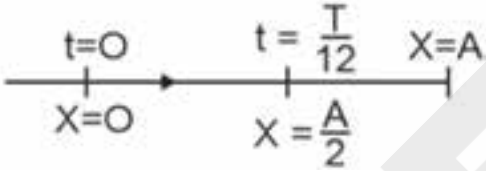
Option 2:
8s

Option 3:
12s

Option 4:
36s

Correct Answer:
36s

Solution:



$\text{T} \rightarrow \text{time period},$

$\frac{\text{T}}{12} = 3$

$\text{T} = 36\ \text{second}$

Hence the correct answer is option 4.

Q. 16 An observer moves towards a stationary source of sound with a velocity equal to one-fifth of the velocity of sound. The percentage change in the frequency will be :

Option 1:
20%

Option 2:
10%

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Option 3:
5%

Option 4:
0%

Correct Answer:
20%

Solution:
Doppler's Effect

$$f' = f\left[\frac{V + V_0}{V}\right]$$

$$f' = f\left(\frac{V + \frac{V}{5}}{V}\right) = f \times \frac{\frac{6V}{5}}{V}$$

$$f' = \frac{6f}{5}$$

Percentage change in frequency

$$= \frac{f' - f}{f} \times 100$$

$$= \frac{\frac{6f}{5} - f}{f} \times 100$$

$$= 20\%$$

Hence the correct answer is option 1.

Q. 17 Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be :

Option 1:
$\sin^{-1}\left(\sqrt{n}\right)$

Option 2:
$\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Option 3:
$\sin^{-1}\left(\sqrt{2n}\right)$

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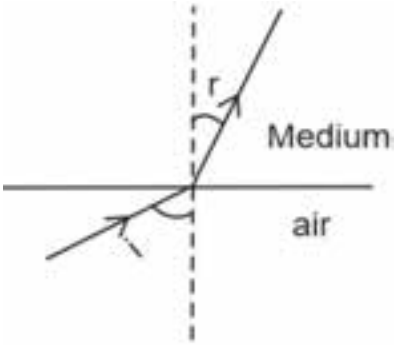
Option 4:

$2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Correct Answer:

$2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Solution:



$i = 2r$

By snell's law

$1x\sin i = \sqrt{2n}\sin r$

$\sin(2r) = \sqrt{2n}\sin r$

$2\sin r\cos r = \sqrt{2n}\sin r$

$\cos r = \frac{\sqrt{n}}{\sqrt{2}}$

$r = \cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

$i = 2r = 2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Q. 18 A hydrogen atom in its ground state absorbs 10.2eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of :

(Given, Planck's constant $= 6.6 \times 10^{-34}\text{Js}$).

Option 1:

$2.10 \times 10^{-34}\text{Js}$

Option 2:

$1.05 \times 10^{-34}\text{Js}$

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Option 3:

$3.15 \times 10^{-34}\text{Js}$

Option 4:

$4.2 \times 10^{-34}\text{Js}$

Correct Answer:

$1.05 \times 10^{-34}\text{Js}$

Solution:

$$E_f = E_i + \Delta E$$

$$-\frac{13.6}{n_f^2} = \frac{-13.6}{1^2} + 10.2\text{eV}$$

$$n_f^2 = \frac{-13.6}{-3.4} = 4$$

$$n_f = 2$$

$\therefore$ The electron will jump from the ground state to the first excited state orbit.

According to Bohr's model

$$L = \frac{nh}{2\pi}, \qquad L \rightarrow \text{ angular momentum}$$

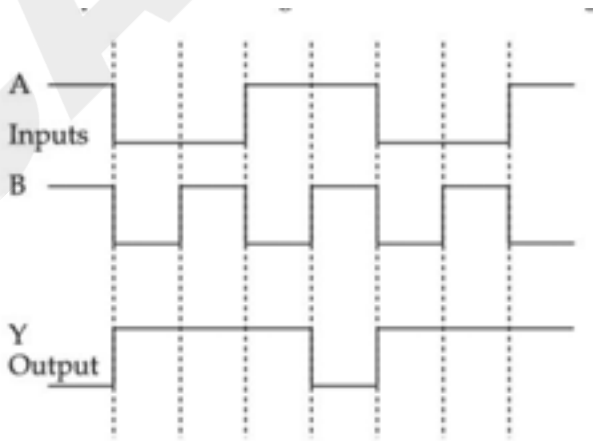
$$L_i = \frac{h}{2\pi}, \quad L_f = \frac{2h}{2\pi} = \frac{h}{\pi}$$

$$\Delta L = L_f - L_i = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34}}{2 \times 3.14}$$

$$\Delta L = 1.05 \times 10^{-34}\ \text{J} - \text{s}$$

Hence the correct answer is option 2.

Q. 19 Identify the correct Logic gate for the following output (Y) of two inputs A and B.



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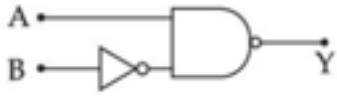
Option 1:

Option 2:

Option 3:

Option 4:



Correct Answer:

Solution:
From the input and output diagrams,

$\mathrm{Y}=\overline{\mathrm{A}\cdot\mathrm{B}}$ this expression

AND gate + NOT gate = NAND gate

Hence the correct answer is option 2.

Q. 20 A mixture of hydrogen and oxygen has volume $2000\mathrm{cm}^3$, temperature $300\mathrm{K}$, pressure $100\mathrm{kPa}$ and mass $0.76\mathrm{g}$. The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be :

[Take gas constant $\mathrm{R}=8.3\mathrm{JK}^{-1}\mathrm{mol}^{-1}$]

Option 1:

$\frac{1}{3}$

Option 2:

$\frac{3}{1}$

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Option 3:

$\frac{1}{16}$

Option 4:

$\frac{16}{1}$

Correct Answer:

$\frac{3}{1}$

Solution:

$\mathrm{PV} = \mathrm{nRT}$

$100 \times 10^3 \times 2000 \times 10^{-6} = \mathrm{n} \times 8.3 \times 300$

$\mathrm{n} = \frac{2}{25}$

Let n_1 moles of hydrogen

n_2 moles of oxygen

$\mathrm{n}_1 + \mathrm{n}_2 = \frac{2}{25}$ .....(1)

$\mathrm{n}_1 \times 2 + \mathrm{n}_2 \times 32 = 0.76 = \frac{19}{25}$ .....(2)

The solving (1) and (2)

$\mathrm{n}_1 = 0.06$

$\mathrm{n}_2 = 0.02$

$\frac{\mathrm{n}_1}{\mathrm{n}_2} = \frac{0.06}{0.02} = 3$

Hence the correct answer is option 2.

Q. 21 In a carnot engine, the temperature of reservoir is $527^\circ\mathrm{C}$ and that of sink is $200\mathrm{K}$. If the worked by the engine when it tranfers heat from reservoir to sink is $12000\mathrm{kJ}$, the quantity of heat absorbed by the engine from reservoir is _____$\times 10^6\,\mathrm{J}$.

Correct Answer:

16

Solution:

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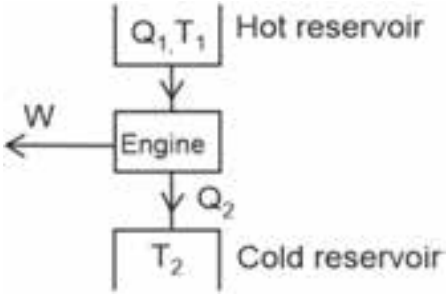


$$Q_1 = W + Q_2$$

$$Q_1 = 12 \times 10^6 + Q_2 \quad \text{.....(1)}$$

$$\frac{Q_2}{Q_2} = \frac{T_2}{T_1} = \frac{200}{800} = \frac{1}{4}$$

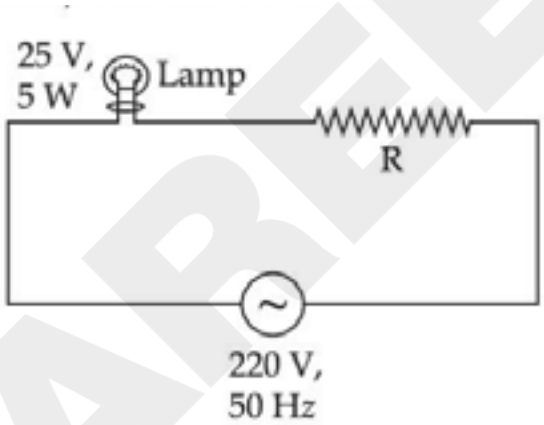
$$Q_2 = \frac{Q_1}{4} \quad \text{.....(2)}$$

Solving (1) and (2)

$$Q_1 = 16 \times 10^6 \text{ J}$$

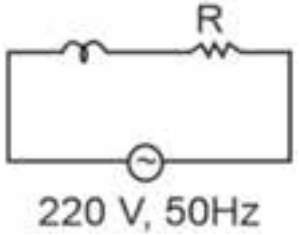
Hence the answer is 16×10^6 J

Q. 22 A $200\text{V}, 50\text{Hz}\,\text{AC}$ source is connected to a $25\text{V}, 5\text{W}$ lamp and an addition resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be _____.



Correct Answer:
975

Solution:



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For its peak brightness, the actual voltage lamp has to be $25\mathrm{V}$.

Resistance of lamp $\mathrm{R_1} = \dfrac{\mathrm{V^2}}{\mathrm{P}} = \dfrac{25 \times 25}{5} = 125\Omega$

Voltage across lamp $\mathrm{V_1} = \dfrac{\mathrm{V} \times \mathrm{R_1}}{\mathrm{R_1} + \mathrm{R_2}}$

$$25 = \frac{220 \times 125}{125 + \mathrm{R}}$$

$$125 + \mathrm{R} = 260 \times 5 = 1100$$

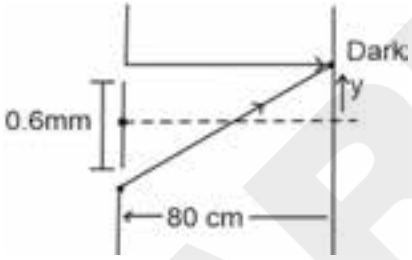
$$\mathrm{R} = 1100 - 125$$

$$\mathrm{R} = 975\Omega$$

Q. 23 In Young,s double slit experiment the two slits are $0.6\mathrm{mm}$ distance apart. Interference pattern is observed on a screen at a distance $80\mathrm{cm}$ from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____nm.

Correct Answer:
450

Solution:



$\mathrm{d} = 0.6\ \mathrm{mm}, \quad \mathrm{D} = 80\ \mathrm{cm}.$

For dark fringe

$$\mathrm{y} = (2\mathrm{n} - 1)\frac{\mathrm{D}\lambda}{2\mathrm{d}}$$

For 1st dark $\mathrm{n} = 1$

$$\mathrm{y} = \frac{\mathrm{D}\lambda}{2\mathrm{d}}$$

$$0.3\ \mathrm{mm} = \frac{80\ \mathrm{cm} \times \lambda}{2 \times 0.6\ \mathrm{mm}}$$

$$\lambda = 450\ \mathrm{nm}$$

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Hence the answer is $450\ \text{nm}$

Q. 24 A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $\text{x} \times 10^{-10}\text{m}$. The value of x is _____.

(Given $\text{hc} = 1242\text{eVnm}$)

Correct Answer:
114

Solution:

$$\text{E} = -13.6\frac{\text{z}^2}{\text{n}^2}\text{eV}$$

When electron jumps, excited state energy difference will be provided by light $(\text{z} = 3)$

$$\text{E}_\text{f} - \text{E}_\text{i} = \frac{\text{hc}}{\lambda}$$

$$\left(-13.6 \times \frac{9}{9}\right) - \left(-13.6 \times \frac{9}{1^2}\right) = \frac{\text{hc}}{\lambda}$$

$$9 \times 13.6 - 13.6 = \frac{\text{hc}}{\lambda}$$

$$8 \times 13.6\ \text{eV} = \frac{1242}{\lambda}\text{eVnm}$$

$$\lambda = 114 \times 10^{-10}\ \text{m}$$

Hence the answer is $114 \times 10^{-10}\ \text{m}$

Q. 25 A cell, shunted by a 8Ω resistance, is balanced across a potentiometer wire of length 3m. The balancing length is 2m when the cell is shunted by 4Ω resistance. The value of internal resistance of the cell will be _____Ω.

Correct Answer:
8

Solution:

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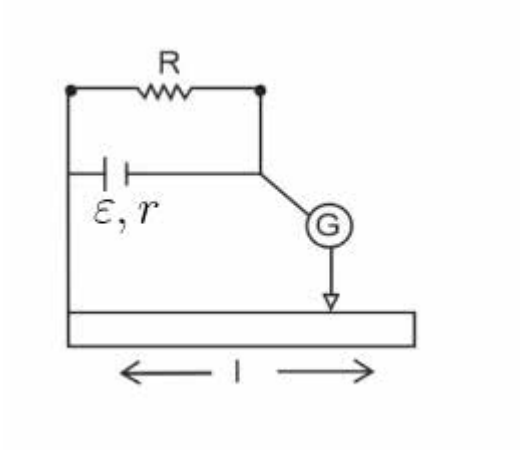


$$\frac{\varepsilon}{\mathrm{R}+\mathrm{r}} \times \mathrm{R}=\mathrm{K} l$$

In the first case

$$\frac{\varepsilon}{8+\mathrm{r}} \times 8=\mathrm{K} \times 3 \quad \text{.....(1)}$$

In the second case

$$\frac{\varepsilon}{4+\mathrm{r}} \times 4=\mathrm{K} \times 2 \quad \text{.....(2)}$$

Solve (1) and (2)

$$\mathrm{r}=8 \Omega$$

Hence the answer is 8Ω

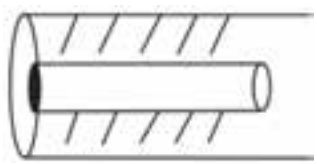
Q. 26 The current density in a cylindrical wire of radius $4 \mathrm{mm}$ is $4 \times 10^{6} \mathrm{Am}^{-2}$. The current through the outer portion of the wire between radial distances $\frac{\mathrm{R}}{2}$ and R is _____ $\pi \mathrm{A}$.

Correct Answer:
48

Solution:
Current Density

$$\mathrm{J}=\frac{\mathrm{I}}{\mathrm{A}}$$

$$\mathrm{I}=\mathrm{JA}$$



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$$I = J \times \left[\pi R^2 - \pi\left(\frac{R^2}{2}\right)^2\right]$$

$$= J \times \left(\pi R^2 - \frac{\pi R^2}{4}\right)$$

$$= J \times \frac{3\pi R^2}{4}$$

$$= 4 \times 10^6 \times \frac{3\pi}{4} \times 16 \times 10^{-6}$$

$$= 48\pi A$$

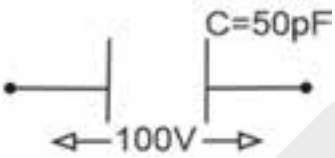
Hence the answer is $48\pi A$

Q. 27 A capacitor of capacitance $50pF$ is charged by $100V$ source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____nJ.

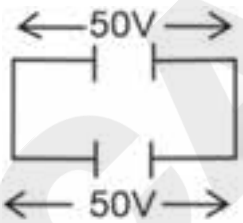
Correct Answer:
125

Solution:

Initial :



Finally



$$\text{Initial energy} = \frac{1}{2}CV^2 = U_1$$

$$\text{Final energy } = \frac{1}{2}C\left(\frac{V}{2}\right)^2 + \frac{1}{2} \times C \times \left(\frac{V}{2}\right)^2$$

$$= \frac{CV^2}{4} = U_2$$

$$\text{Loss in energy } = U_1 - U_2$$

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$$= \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{CV^2}{4}$$

$$= \frac{1}{4} \times 50 \times 10^{-12} \times 100 \times 100$$

$$= \frac{50 \times 10^4 \times 10^{-3} \times 10^{-9}}{4}$$

$$= 125 \text{ nJ}$$

Hence answer is 125 nJ.

Q. 28 The height of a transmitting antenna at the top of a tower is 25m and that of receiving antenna is 49m. The maximum distance between them, for satisfactory communication in $\text{LOS}(\text{Line} - \text{of} - \text{Sight})$ is $\text{K}\sqrt{5} \times 10^2\text{m}$. The value of K is _____.

(Assume radius of Earth is $64 \times 10^{+5}\text{m}$) [Calculate upto nearest integer value]

Correct Answer:
192

Solution:
Here height of transmitting antenna $\text{h}_\text{T} = 25\text{m}$

Height of receiving antenna $\text{h}_\text{R} = 49\text{m}$

Maximum distance for LOS

$$\text{d} = \sqrt{2\text{Rh}_\text{R}} + \sqrt{2\text{Rh}_\text{T}}$$

$$= \sqrt{2\text{R}}\left(\sqrt{\text{h}_\text{R}} + \sqrt{\text{h}_\text{T}}\right)$$

$$= \sqrt{2 \times 64 \times 10^5} \times (\sqrt{49} + \sqrt{25})$$

$$= 8 \times 10^2 \times \sqrt{20} \times (7 + 5)$$

$$= 96 \times 10^2 \times 2\sqrt{5}$$

$$= 192\sqrt{5} \times 10^2$$

$$\text{K} = 192$$

Hence the answer is 192

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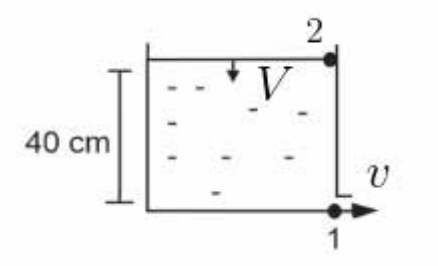
Q. 29 Tha area of cross-section of a large tank is $0.5\mathrm{m}^2$. It has a narrow opening near the bottom having area of cross-section $1\mathrm{cm}^2$. A load of $25\mathrm{kg}$ is applied on the water at the top in the tank. Neglecting the speed of water in the tank, the velocity of the water, coming out of the opening at the time when the height of water level in the tank is $40\mathrm{cm}$ above the bottom, will be _____cms^{-1}.

[Take $\mathrm{g} = 10\mathrm{ms}^{-2}$]

Correct Answer:
300

Solution:



Applying Bernoulli's equation at points 1 & 2 we get,

$$P_0 + 0 + \frac{1}{2}\rho V^2 = \left(P_0 + \frac{Mg}{A}\right) + \rho g h + \frac{1}{2}\rho v^2$$

Speed of free surface (V) is negligible compared to the speed at the orifice (v)

$$\frac{1}{2}\rho v^2 = \frac{Mg}{A} + \rho g h$$

$$\frac{10^3}{2} \times v^2 = \frac{25 \times 10}{0.5} + 10^3 \times 10 \times 0.4$$

$$\frac{10^3}{2} v^2 = 500 + 4000$$

$$v^2 = \frac{4500 \times 2}{10^3} = 9 \times \frac{10^3}{10^3}$$

$$v = 3 \text{ m/s}$$

$$v = 300 \text{ cm/s}$$

Hence answer is 300 cm/s

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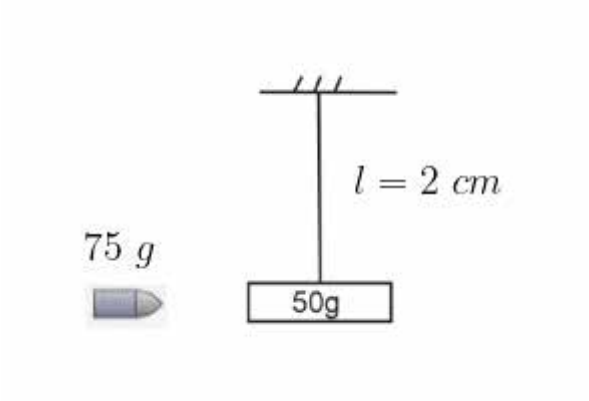
Q. 30 A pendulum of length $2\mathrm{m}$ consists of a wooden bob of mass $50\mathrm{g}$. A bullet of mass $75\mathrm{g}$ is fired towards the stationary Bob with a speed v. The bullet emerges out of the Bob with a speed $\frac{\mathrm{v}}{3}$ and the Bob just completes the vertical circle. The value of v is _____ms^{-1}.

(if $\mathrm{g} = 10\mathrm{m/s}^2$).

Correct Answer:
10

Solution:



Linear momentum conservation

$$75 \times \mathrm{V} = 50 \times \mathrm{V}' + 75 \times \frac{\mathrm{V}}{3}$$

$$\mathrm{V}' \times 50 = 2 \times \frac{75}{3} \times \mathrm{V}$$

$$\mathrm{V}' = \frac{2 \times 75}{50 \times 3} \times \mathrm{V}$$

$$= \frac{2 \times 3}{2 \times 3}\mathrm{V} = \mathrm{V}$$

The bob just completes the vertical circular motion

$$\therefore \mathrm{V} = \sqrt{5\mathrm{gl}}$$

$$\mathrm{V} = \sqrt{5 \times 10 \times 2}$$

$$\mathrm{V} = 10\ \mathrm{m/s}$$

Hence answer is 10.

Chemistry

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Q. 1 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : At $10^{\circ}\mathrm{C}$, the density of a 5M solution of KCl [atomic masses of K&Cl are $39\&35.5\ \mathrm{g\ mol^{-1}}$ respectively], is 'x' $\mathrm{gml^{-1}}$. The solution is cooled to $-21^{\circ}C$. The molality of the solution will remain unchanged.

Reason (R): The molality of a solution does not change with temperature as mass remains unaffected with temperature.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A).**

Option 2:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

Option 3:
(A) is true but **(R)** is false.

Option 4:
(A) is false but **(R)** is true.

Correct Answer:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A).**

Solution:
The temperature does not change the mass of the solute or the solvent and hence, the molality of solution remains the same.

Hence both Assertion and Reason are true and the Reason correctly explains the Assertion.

Hence , the correct answer is **Option (1).**

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Q. 2 Based upon VSEPR theory, match the shape (geometry) of the molecules in **List-I** with the molecules in **List - II** and select the **most appropriate** option.

List-I	List-II
(Shape)	**(Molecules)**
(A) T-Shaped	(I)$\mathrm{XeF_4}$
(B) Trigonal planar	(II)$\mathrm{SF_4}$
(C) Square planar	(III)$\mathrm{ClF_3}$
(D) See-Saw	(IV)$\mathrm{BF_3}$

Option 1:
$(\mathrm{A})-(\mathrm{I}),(\mathrm{B})-(\mathrm{II}),(\mathrm{C})-(\mathrm{III}),(\mathrm{D})-(\mathrm{IV})$

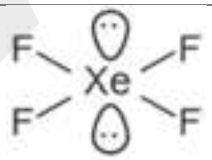
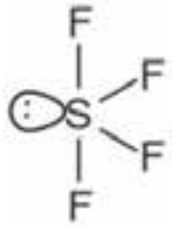
Option 2:
$(\mathrm{A})-(\mathrm{III}),(\mathrm{B})-(\mathrm{IV}),(\mathrm{C})-(\mathrm{I}),(\mathrm{D})-(\mathrm{II})$

Option 3:
$(\mathrm{A})-(\mathrm{III}),(\mathrm{B})-(\mathrm{IV}),(\mathrm{C})-(\mathrm{II}),(\mathrm{D})-(\mathrm{I})$

Option 4:
$(\mathrm{A})-(\mathrm{IV}),(\mathrm{B})-(\mathrm{III}),(\mathrm{C})-(\mathrm{I}),(\mathrm{D})-(\mathrm{II})$

Correct Answer:
$(\mathrm{A})-(\mathrm{III}),(\mathrm{B})-(\mathrm{IV}),(\mathrm{C})-(\mathrm{I}),(\mathrm{D})-(\mathrm{II})$

Solution:

Molecules	Structure	Hybridisation	Shape
(I) $\mathrm{XeF_4}$		$\mathrm{sp^3d^2}$	Square Planar
(II) $\mathrm{SF_4}$		$\mathrm{sp^3d}$	See Saw

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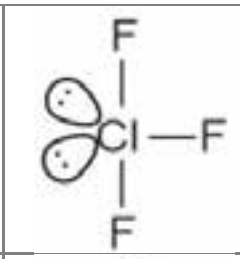
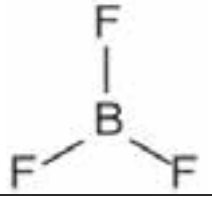
(III) ClF_3		sp^3d	T-Shaped
(IV) BF_3		sp^2	Trigonal Planar

Hence, the correct answer is **Option (2).**

Q. 3 Match **List - I** with **List - II.**

List - I	List - II
(A) Spontaneous process	(I) $\Delta H < 0$
(B) Process with $\Delta P = 0, \Delta T = 0$	(II) $\Delta G_{T,P} < 0$
(C) $\Delta H_reaction$	(III) Isothermal and isobaric process
(D) Exothermic Process	(IV) [Bond energies of molecules in reactants] -[Bond energies of product molecules]

Choose the correct answer from the options given below :

Option 1:

$(A) - (III), (B) - (II), (C) - (IV), (D) - (I)$

Option 2:

$(A) - (II), (B) - (III), (C) - (IV), (D) - (I)$

Option 3:

$(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$

Option 4:

$(A) - (II), (B) - (I), (C) - (III), (D) - (IV)$

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Correct Answer:

$(A)-(II),(B)-(III),(C)-(IV),(D)-(I)$

Solution:

(A) For spontoneous process $\Rightarrow \Delta G_{T,p} < 0 (II)$

(B) Process with $\Delta P = 0 \rightarrow$ Isobaric process

$\Delta T = 0 \rightarrow$ Isothermal process (III)

$(C)\ \Delta H_{reaction} = [\text{Bond energies of molecules in reactents}] - [\text{Bond energies of product molecules}]\ (IV)$

(D) Exothermic process , $\Delta H < 0$ (I)

Hence, the correct answer is **Option (2)**

Q. 4 Match **List - I** with **List - II.**

List-I	List-II
(A) Lyophilic colloid	(I) Liquid-liquid colloid
(B) Emulsion	(II) Protective colloid
(C) Positively charged colloid	(III) $FeCl_3 + NaOH$
(D) Negatively charged colloid	(IV) $FeCl_3$ + hot water

Choose the correct answer from the options given below :

Option 1:

$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

Option 2:

$(A)-(III),(B)-(I),(C)-(IV),(D)-(II)$

Option 3:

$(A)-(II),(B)-(I),(C)-(III),(D)-(IV)$

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Option 4:
$(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

Correct Answer:
$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

Solution:
(A) Lyophillic colloid $\rightarrow$ Protective colloid (II)
(B) Emulsion $\rightarrow$ liquid in liquid colloidal solutions (I)
(C) Positively charged colloid $\rightarrow$ If $FeCl_3$ is added to the excess of hot water, a positively changed sol of ferric oxide is formed due to ad of Fe^{3+} ions. (IV)

$$Fe_2O_3 \cdot xH_2O|Fe^{3+}$$

(D) Negatively charged colloid $\rightarrow$ when ferric chloride is added to $NaOH$ solution a negatively charged sol is obtained with adsorption of $\overline{O}H$ ions. (III)

$$Fe_2O_3 \cdot xH_2O|\overline{O}H$$

Hence, the correct answer is **Option (1)**

Q. 5 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.
Reason (R): Both O^{2-} and Mg^{2+} are isoelectronic species.
In the light of the above statements, choose the **correct** answer from the options given below.

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A).

Option 2:
Both (A) and (R) are true but (R) is not the correct explanation of (A).

Option 3:
(A) is true but (R) is false.

Option 4:
(A) is false but (R) is true.

Correct Answer:
(A) is false but (R) is true.

Solution:
As we have learnt,

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$\rightarrow$ Ionic radius order $\Rightarrow Mg^{+2} < O^{2-}$

$\rightarrow$ Both Mg^{2+} and O^{2-} are isoelectronic with 10 electrons.

Thus, Assertion (A) is false while Reason (R) is true.

Hence, the correct answer is Option (4).

Q. 6 Match **List - I** with **List - II.**

List - I	List - II
(A) Concentration of gold ore	(I) Aniline
(B) Leaching of alumina	(II) $NaOH$
(C) Froth stabiliser	(III) SO_2
(D) Blister copper	(IV) $NaCN$

Choose the **correct** answer from the options given below.

Option 1:
$(A)-(IV), (B)-(III), (C)-(II), (D)-(I)$

Option 2:
$(A)-(IV), (B)-(II), (C)-(I), (D)-(III)$

Option 3:
$(A)-(III), (B)-(II), (C)-(I), (D)-(IV)$

Option 4:
$(A)-(II), (B)-(IV), (C)-(III), (D)-(I)$

Correct Answer:
$(A)-(IV), (B)-(II), (C)-(I), (D)-(III)$

Solution:
(A) NaCN is used for concentration of Gold ore. (IV)
(B) Leaching of alumina is done by $NaOH$ (II)
(C) Foth stabiliser $\rightarrow$ Aniline (I)
(D) Blister copper $\rightarrow$ due to the evolution of SO_2 (III)

Hence, the correct answer is **Option (2)**

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Q. 7 Addition of H_2SO_4 to BaO_2 produces :

Option 1:
BaO, SO_2 and H_2O

Option 2:
$BaHSO_4$ and O_2

Option 3:
$BaSO_4, H_2$ and O_2

Option 4:
$BaSO_4$ and H_2O_2

Correct Answer:
$BaSO_4$ and H_2O_2

Solution:
As we have learnt,

The given reaction is used for the preparation of H_2O_2

$$BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$$

Hence, the correct answer is **Option (4).**

Q. 8 $BeCl_2$ reacts with $LiAlH_4$ to give :

Option 1:
$Be + Li\,[AlCl_4] + H_2$

Option 2:
$Be + AlH_3 + LiCl + HCl$

Option 3:
$BeH_2 + LiCl + AlCl_3$

Option 4:
$BeH_2 + Li\,[AlCl_4]$

Correct Answer:
$BeH_2 + LiCl + AlCl_3$

Solution:
The given reactios is used for the preparation of BeH_2

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$$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$$

Hence, the correct answer is **Option (3)**

Q. 9 Match **List - I** with **List - II.**

List - I	**List - II**
(Si-Compounds)	**(Si-Polymeric/Other Products)**
(A) $(CH_3)_4 Si$	(I) Chain Silicone
(B) $(CH_3) Si(OH)_3$	(II) Dimeric Silicone
(C) $(CH_3)_2 Si(OH)_2$	(III) Silane
(D) $(CH_3)_3 Si(OH)$	(IV) 2D-Silicone

Choose the correct answer from the options given below :

Option 1:
$(A) - (III), (B) - (II), (C) - (I), (D) - (IV)$

Option 2:
$(A) - (IV), (B) - (I), (C) - (II), (D) - (III)$

Option 3:
$(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

Option 4:
$(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

Correct Answer:
$(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

Solution:
As we have learnt,

(A) $(CH_3)_4 Si$ is called Silane (III)

(B) $(CH_3) Si(OH)_3$ Polymerise to form 2D Silicone (IV)

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(C) $(CH_3)_2 Si(OH)_2$ Polymerise to form chain Silicone (I)

(D) $(CH_3)_3 Si(OH)$ Form dimeric silicone (II)

Hence, the correct answer is **Option (4)**

Q. 10 Heating white phosphorus with conc. $NaOH$ solution gives mainly :

Option 1:
Na_3P and H_2O

Option 2:
H_3PO and NaH

Option 3:
$P(OH)_3$ and NaH_2PO_4

Option 4:
PH_3 and NaH_2PO_2

Correct Answer:
PH_3 and NaH_2PO_2

Solution:
As we have learnt,

White Phosphorus disproportionates in alkali to form Phosphine and Hypophosphite.

The reaction is given as:

$$P_4 + 3NaOH + H_2O \longrightarrow PH_3 + 3NaH_2PO_2$$

Hence, the correct answer is **Option (4)**

Q. 11 Which of the following will have maximum stabilization due to crystal field?

Option 1:
$[Ti(H_2O)_6]^{3+}$

Option 2:
$[Co(H_2O)_6]^{2+}$

Option 3:
$[Co(CN)_6]^{3-}$

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Option 4:
$[Cu\,(NH_3)_4]^{2+}$

Correct Answer:
$[Co(CN)_6]^{3-}$

Solution:
Stability $\propto$ CFSE

$[Co(CN)_6]^{3-}$ Oxidation number of $Co = +3$

CN^- is strong field ligand

$[Co(CN)_6]^{3-} \Rightarrow d^6(\text{low spin}) : \; t_{2g}^6\; e_g^0$

$$CFSE = \left[-(0.4 \times n_{t_{2g}}) + (0.6 \times n_{eg})\right] \Delta_0$$
$$= -(0.4 \times 6) + (0.6 \times 0)\; \Delta_0$$
$$= -2.4\; \Delta_0$$

$[Co(CN)_6]^{3-}$ has the maximum $CFSE$ among given complexes.

Hence, the correct answer is **Option (3).**

Q. 12 Given below are two Statements :

Statement I : Classical smog occurs in cool humid climate. It is a reducing mixture of smoke, fog and sulphur dioxide.

Statement II : Photochemical smog has components, ozone, nitric oxide, acrolein, formaldehyde, PAN etc.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

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Solution:
As we have learnt,

Classical smog occurs in cool humid climate. It is a reducing mixture of smoke, fog and sulphurdioxide.

Photochemical smog has components Ozone, nitric oxide, acrolein, formaldehyde, PAN etc.

Both statement I and II are correct.

Hence, the correct answer is **Option (1)**

Q. 13 Which of the following is structure of a separating funnel?

Option 1:

Option 2:

Option 3:

Option 4:

Correct Answer:

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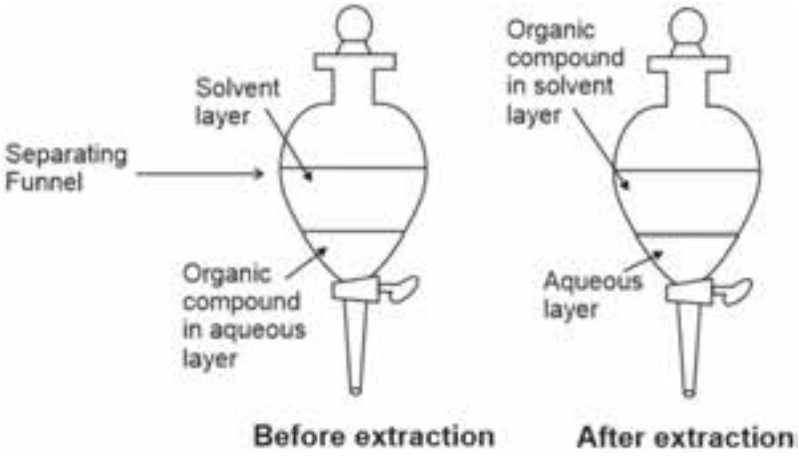
Solution:
Fact based.

The structure of a Separating funnel is given below



Hence, the correct answer is **Option (1)**

Q. 14 $'\mathbf{A}'$ and $'\mathbf{B}'$ respectively are :

$$\mathrm{A} \xrightarrow[(2)\ \mathrm{Zn-H_2O}]{(1)\ \mathrm{O_3}} \text{Ethane-1,2-dicarbaldehyde} + \text{Glyoxal/Oxaldehyde}$$

$$\mathrm{B} \xrightarrow[(2)\ \mathrm{Zn-H_2O}]{(1)\ \mathrm{O_3}} \text{5-oxohexanal}$$

Option 1:
$1-\text{methylcyclohex}-1,3-\text{diene}\ \&\ \text{cyclopentene.}$

Option 2:
$\text{Cyclohex}-1,3-\text{diene}\ \&\ \text{cyclopentene}$

Option 3:
$1-\text{methylcyclohex}-1,4-\text{diene}\ \&\ 1-\text{methylcyclopent}-1-\text{ene}$

Option 4:
$\text{Cyclohex}-1,3-\text{diene}\ \ \&\ 1-\text{methylcyclopent}-1-\text{ene}$

Correct Answer:
$\text{Cyclohex}-1,3-\text{diene}\ \ \&\ 1-\text{methylcyclopent}-1-\text{ene}$

Solution:

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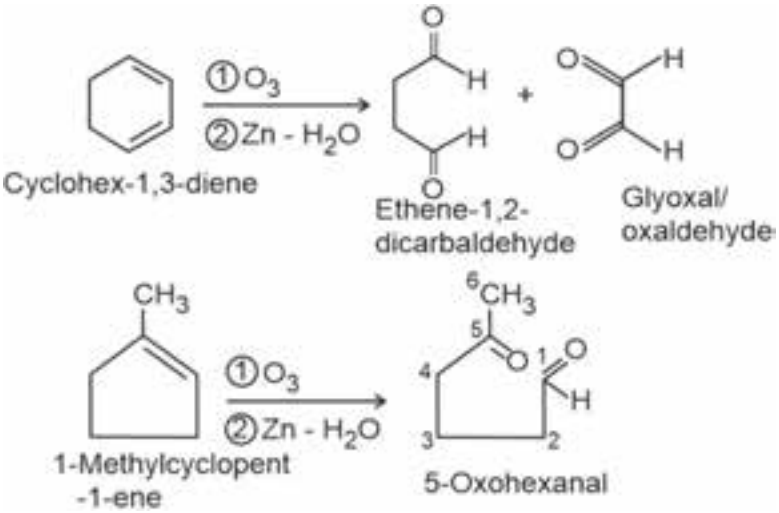
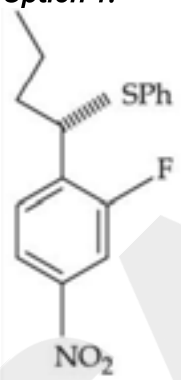


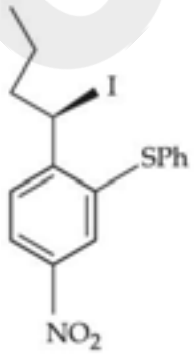
Hence, the correct answer is **Option (4)**

Q. 15 The major product of the following reaction is :

Option 1:



Option 2:



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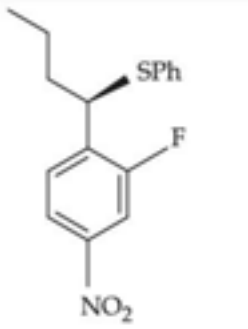
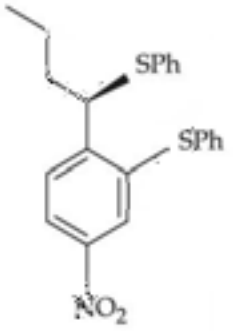
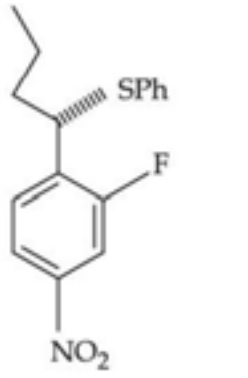
Option 3:



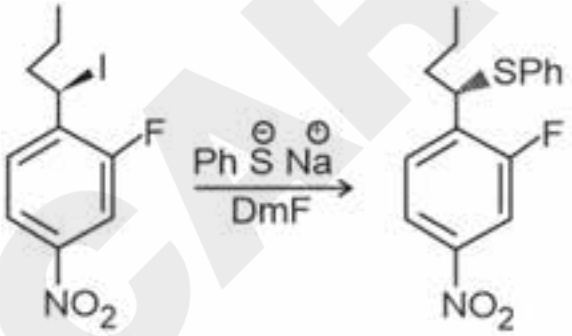
Option 4:



Correct Answer:



Solution:



$\rightarrow$ It is a SN^2 reaction

$\rightarrow$ alkyl halide is more reactive than aryl halide towards nucleophilic Substitution reactions.

Hence, the correct answer is **Option (1).**

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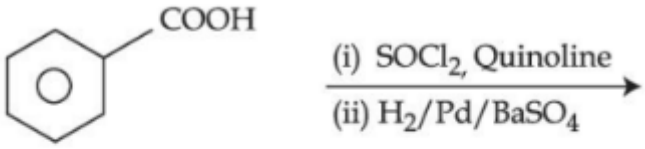
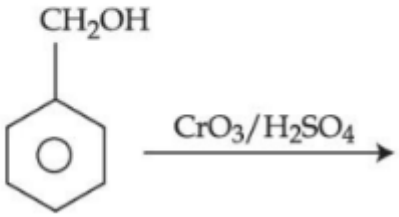
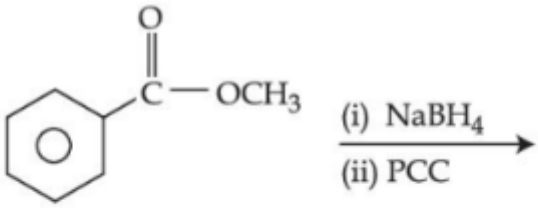
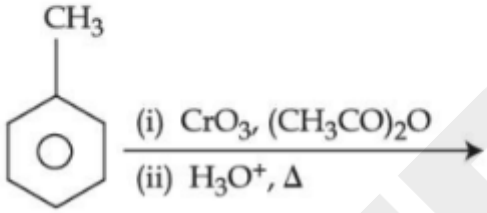
Q. 16 Which of the following reactions will yield benzaldehyde as a product?

(A) 

(B) 

(C) 

(D) 

Option 1:
(B) and (C)

Option 2:
(C) and (D)

Option 3:
(A) and (D)

Option 4:
(A) and (C)

Correct Answer:
(A) and (D)

Solution:

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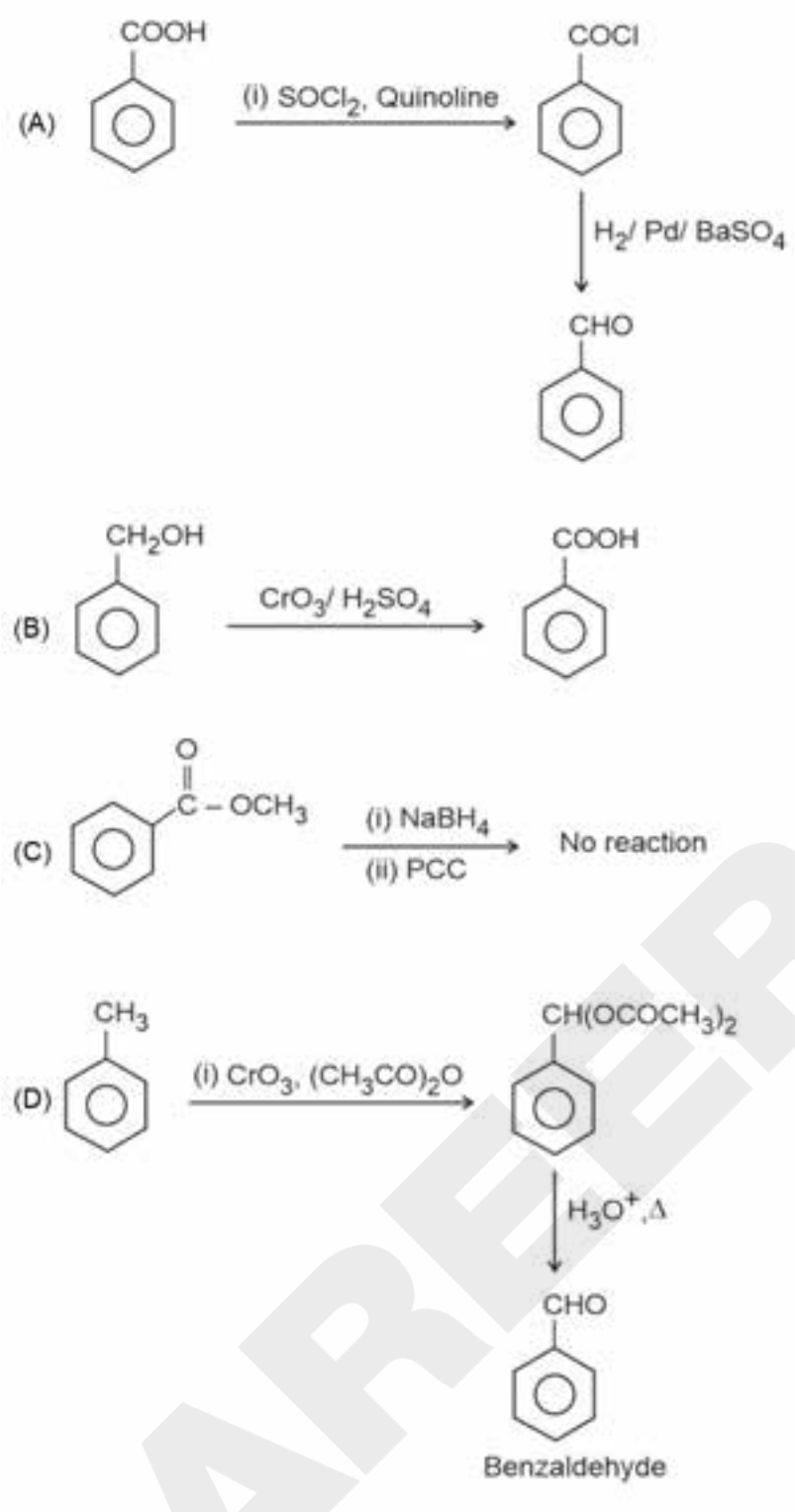

Hence, the correct answer is **Option (3)**

Q. 17 Given below are two statements :

Statement - I : In Hofmann degradation reaction, the migration of only an alkyl group takes place from carbonyl carbon of the amide to the nitrogen atom.

Statement - II : The group is migrated in Hofmann degradation reaction to electron deficient atom.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

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Option 1:
Both Statement I and Statement II are correct.

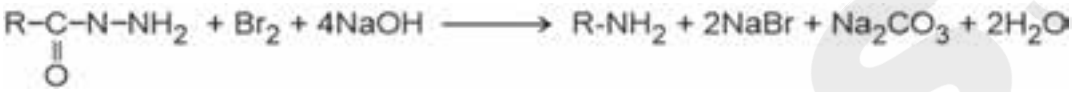
Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

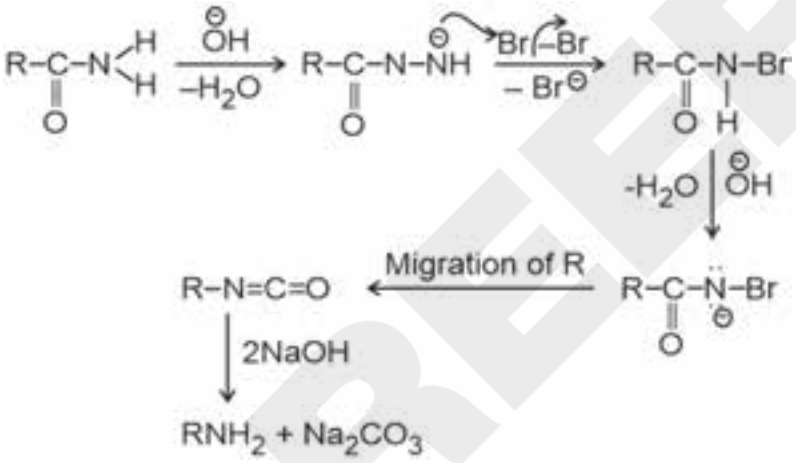
Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Statement I is incorrect but Statement II is correct.

Solution:
Hofmann degradation reaction:



Mech:



* R may be alkyl or aryl group.
* R group is migrated to the electron-deficient nitrogen atom.

Hence, the correct answer is **Option (4)**

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Q. 18 Match **List - I** with **List - II.**

List - I	**List - II**
(Polymer)	(**Used in)**
(A) Bakelite	(I) Radio and television cabinets
(B) Glyptal	(II) Electrical switches
(C) PVC	(III) Paints and Lacquers
(D) Polystyrene	(IV) Water pipes

Choose the correct answer from the options given below :

Option 1:
$(A)-(II),(B)-(III),(C)-(IV),(D)-(I)$

Option 2:
$(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

Option 3:
$(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

Option 4:
$(A)-(II),(B)-(III),(C)-(I),(D)-(IV)$

Correct Answer:
$(A)-(II),(B)-(III),(C)-(IV),(D)-(I)$

Solution:

Polymer **Used in**

(A) Bakelite $\longrightarrow$ making electrical switches (II)

(B) Glyptal $\longrightarrow$ paints and lacquers (III)

(C) PVC $\longrightarrow$ water pipes (IV)

(D) Polystyrene $\longrightarrow$ manufacture of Radio and television Cabinets (I)

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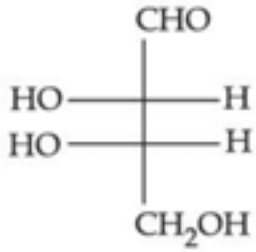
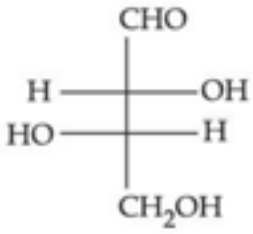
Hence, the correct answer is **Option (1)**

Q. 19 L-isomer of a compound $'A'\ (C_4H_8O_4)$ gives a positive test with $[Ag\ (NH_3)_2]^+$. Treatment 'A' with acetic anhydride yields triacetate derivative. Compound ' A' produces an optically active compound (B) and an optically inactive compound (C) on treatment with bromine water and HNO_3 respectively. Compound (A) is :

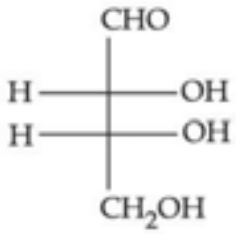
Option 1:



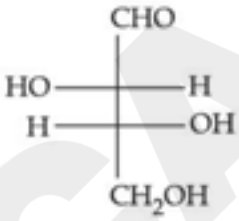
Option 2:



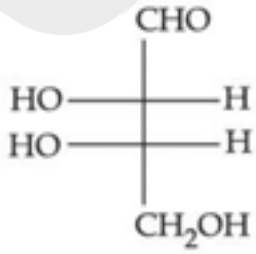
Option 3:



Option 4:



Correct Answer:



Solution:

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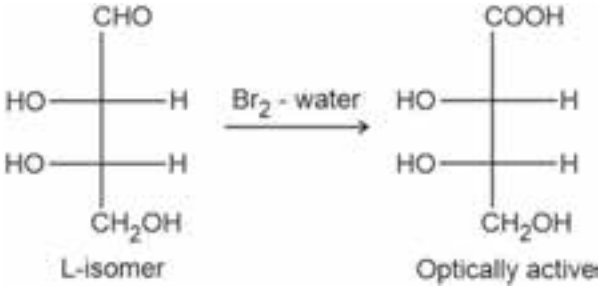
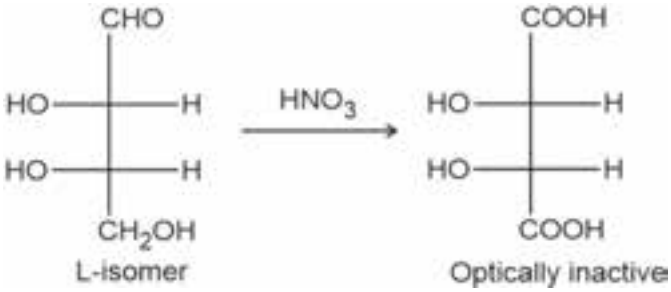


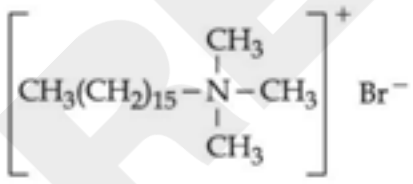
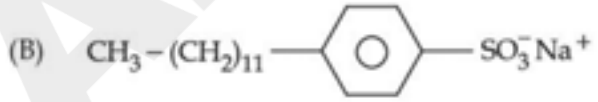


L- isomer(option 1 compound) gives positive test with $[Ag(NH_3)]_2]^+$ due to presence of $-CHO$ group.

L- isomer(option 1 compound) Yield triacetate derivative with acetic anhydride; due to presence of $3\ -OH$ group.

Hence, the correct answer is **Option (1)**

Q. 20 Match **List -I** with **List-II**

List-I	List-II
(A) 	(I) Dishwashing powder
(B) 	(II) Toothpaste
(C) $C_{17}H_{35}COO^-Na^+ + Na_2CO_3 +$ Rosinate	(III) Laundry soap
(D) $CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$	(IV) Hair conditioner

Choose the **correct** answer from the options given below:

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Option 1:
(A) - (III), (B) - (II), (C) - (IV), (D) - (I)

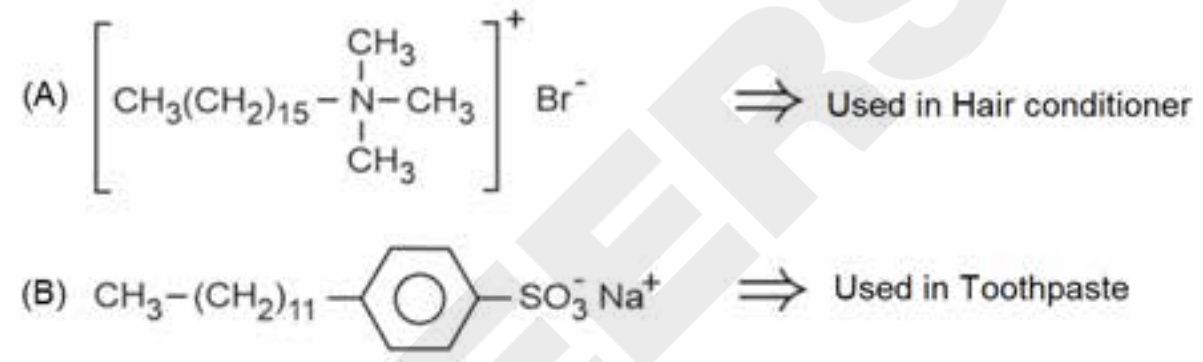
Option 2:
(A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Option 3:
(A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Option 4:
(A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Correct Answer:
(A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Solution:



$(C)\ C_{17}H_{35}COO^{-}Na^{+} + Na_2CO_3 + \text{Rosinate} \Rightarrow \text{Used in Laundry Soap}$

$(D) CH_3(CH_2)_{16}COO(CH_2CH_2O)_H CH_2CH_2OH \Rightarrow \text{Used in Dishwashing Powder}$

Hence, the correct answer is **Option (2).**

Q. 21 Metal deficiency defect is shown by $Fe_{0.93}O$. In the crystal, some Fe^{2+} cations are missing and loss of positive charge is compensated by the presence of Fe^{3+} ions. The percentage of Fe^{2+} ions in the $Fe_{0.93}O$ crystals is_____ . (Nearest integer)

Correct Answer:
85

Solution:
In $Fe_{0.93}O$, Iron present as Fe^{+2} & Fe^{+3}

Suppose : $Fe^{2+} = x, Fe^{3+} = 0.93 - x$

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$Fe_{0.93}O$ is electrically neutral, thus charge balance on the compound gives us

$$x \times 2 + (0.93 - x) \times 3 - 2 \times 1 = 0$$
$$2x + 2.79 - 3x - 2 = 0$$
$$x = .79$$
$$Fe^{+2} = 0.79, Fe^{+3} = 0.14$$

Percentage of Fe^{2+} ion present

$$= \frac{0.79}{0.93} \times 100 = 84.94\%$$
$$= 85 \text{ (Nearest integer)}$$

Hence, answer is 85.

Q. 22 If the uncertainty in velocity and position of a minute particle in space are, $2.4 \times 10^{-26}\ (ms^{-1})$ and $10^{-7}(\ m)$ respectively. The mass of the particle in g is_____.(Nearest integer)
(Given : $h = 6.626 \times 10^{-34} Js$)

Correct Answer:
22

Solution:
$$\text{Uncentainty in velocity } (\Delta V) = 2.4 \times 10^{-26} \text{ m/s}$$

$$\text{Uncertainty in position } (\Delta x) = 10^{-7} \text{ m}$$

Mass of particle in g = ?

$$\Delta P \cdot \Delta x = \frac{h}{4\pi}$$
$$m\Delta V \cdot \Delta x = \frac{h}{4\pi}$$

$$m = \frac{h}{\Delta v \cdot \Delta x \cdot 4\pi} = \frac{6.626 \times 10^{-34}}{2.4 \times 10^{-26} \times 10^{-7} \times 4 \times 3.14}$$
$$m = 0.02198 \text{ kg}$$
$$m = 21.98 \text{ g}$$
$$m \simeq 22 \text{ g (Nearest integer)}$$

Hence, the answer is 22

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Q. 23 $2\ \mathrm{g}$ of a non-volatile non-electrolyte solute is disolved in $200\ \mathrm{g}$ of two different solvents A and B whose ebullioscopic constants are in the ratio of $1:8$. The elevation in boiling points of $A \text{ and } \mathrm{B}$ are in the ratio $\frac{x}{y}(x:y)$. The value of y is_____. (Nearest Integer)

Correct Answer:
8

Solution:
Given

$$\frac{(\mathrm{K_b})_\mathrm{A}}{(\mathrm{K_b})_\mathrm{B}} = \frac{1}{8}$$

We know,

$$\Delta \mathrm{T_b} = \mathrm{K_b\ m}$$

$$\frac{(\Delta \mathrm{T_b})_\mathrm{A}}{(\Delta \mathrm{T_b})_\mathrm{B}} = \frac{(\mathrm{K_b})_\mathrm{A}\mathrm{m}}{(\mathrm{K_b})_\mathrm{B}\mathrm{m}} = \frac{1}{8} = \frac{\mathrm{x}}{\mathrm{y}}$$

$$\frac{\mathrm{x}}{\mathrm{y}} = \frac{1}{8}, \quad \mathrm{y} = 8$$

Hence, the answer is 8

Q. 24 $2\mathrm{NOCl(g)} \rightleftharpoons 2\mathrm{NO(g)} + \mathrm{Cl_2(\ g)}$
In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be $0.4\ \mathrm{mol/L}$. The equilibrium constant at $30^\circ\mathrm{C}$ is_____$\times 10^{-4}$.

Correct Answer:
125

Solution:
The equlibrium can be solved as

$$\begin{array}{lccccc} & 2\mathrm{NOCl(g)} & \rightleftharpoons & 2\mathrm{NO(g)} & + & \mathrm{Cl_2(g)} \\ \mathrm{t}=0 & 2\mathrm{M} & & 0 & & 0 \\ \mathrm{t}=\mathrm{t_{eq}} & (2-\mathrm{x})\mathrm{M} & & \mathrm{xM} & & \frac{\mathrm{x}}{2}\mathrm{M} \\ \mathrm{t}=\mathrm{t_{eq}} & 1.6\mathrm{M} & & 0.4\mathrm{M} & & 0.2\mathrm{M}\ (\because \mathrm{x}=0.4\mathrm{M\ at\ eq^m}) \end{array}$$

$$\mathrm{K_c} = \mathrm{K_{eq}} = \frac{(0.4)^2 \times 0.2}{(1.6)^2} = \frac{0.4 \times 0.4 \times 0.2}{1.6 \times 1.6}$$

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$$= \frac{2}{16} \times 10^{-1} = \frac{1}{8} \times 10^{-1}$$
$$= 0.125 \times 10^{-1}$$
$$= 125 \times 10^{-4}$$

Hence, the answer is 125

Q. 25 The limiting molar conductivities of $\mathrm{NaI}, \mathrm{NaNO_3}$ and $\mathrm{AgNO_3}$ are $12.7, 12.0$ and $13.3 \mathrm{mSm^2\ mol^{-1}}$, respectively (all at $25^{\circ}\mathrm{C}$). The limiting molar conductivity of AgI at this temperature is_____$\mathrm{mSm^2\ mol^{-1}}$.

Correct Answer:
14

Solution:
$\Lambda_m^{\infty}(\mathrm{Ag\ I}) = ?$

From Kohlrausch's law of infinite dilution, we have

$$\Lambda_m^{\infty}(\mathrm{Ag\ I}) = \Lambda_m^{\infty}(\mathrm{AgNO_3}) + \Lambda_m^{\infty}(\mathrm{NaI}) - \Lambda_m^{\infty}(\mathrm{NaNO_3})$$
$$= 13.3 + 12.7 - 12.0$$
$$= 26.0 - 12.0$$
$$= 14\ \mathrm{mSm^2\ mol^{-1}}$$

Hence, the answer is 14.

Q. 26 The rate constant for a first order reaction is given by the following equation :

$$\ln k = 33.24 - \frac{2.0 \times 10^4\ \mathrm{K}}{T}$$

The Activation energy for the reaction is given by_____ $\mathrm{kJ mol^{-1}}$. In Nearest integer)

(Given : $\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$)

Correct Answer:
166

Solution:
$$\ln k = \ln A - \frac{E_a}{RT} \ldots\ldots\ldots (i)$$

Given:

$$\ln k = 33.24 - \frac{2.0 \times 10^4 \mathrm{K}}{T} \ldots\ldots (ii)$$

On comparing Eq.(i) & (ii), we have

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$$\frac{Ea}{R} = 2.0 \times 10^4$$

$$Ea = 2.0 \times 10^4 \times 8.3 \text{ J/mol}$$

$$= 16.6 \times 10^4 \text{ J/mol}$$

$$= 166 \times 10^3 \text{ J/mol}$$

$$= 166 \text{ KJ/mol}$$

Hence,the answer is 166.

Q. 27 The number of statement(s) correct from the following for Copper (at. no. 29) is/are_____.

(A) $Cu(II)$ complexes are always paramagnetic
(B) $Cu(I)$ complexes are generally colourless
(C) $Cu(I)$ is easily oxidized
(D) In Fehling solution, the active reagent has $Cu(I)$

Correct Answer:
3

Solution:
Correct statements:

(A) Cu (II) Complexes are always paramagnetic because of a d^9 configuration
(B) Cu (I) complexes are generally colourless because of a d^{10} configuration
(C) Cu (I) is easily oxidized
(D) Given statement is incorect because Fehling solution, the active reagent has Cu(II).

Hence, the correct answer is 3.

Q. 28 Acidified potassium permanganate solution oxidises oxalic acid. The spin-only magnetic moment of the mangenese product formed from the above reaction is_____ B.M. (Nearest Integer)

Correct Answer:
6

Solution:
$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \rightarrow 2MnSO_4 + K_2SO_4 + 10CO_2 + 8H_2O$$

Manganese gets reduced to Mn^{2+}

Mn^{2+} has 5 unpaired electron

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$$\mu = \sqrt{n(n+2)} \quad B \cdot M.$$
$$= \sqrt{5(5+2)} \quad B \cdot M$$
$$= \sqrt{35} \quad B \cdot M.$$
$$= 5.91 \quad B \cdot M.$$

Thus, the magnetic moment when rounded off to the nearest integer gives a value equal to 6.

Hence, answer is 6.

Q. 29 Two elements A and B which form 0.15 moles of A_2 B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is_____ times of atomic weight of B.

Correct Answer:
2

Solution:
Given in Q:

$$\text{Molar mass of } A_2B = \text{ Molar mass of } AB_3$$

$$2A + B = A + 3B$$

A=Atomic weight of A
B= Atomic weight of B

$$A = 2B$$

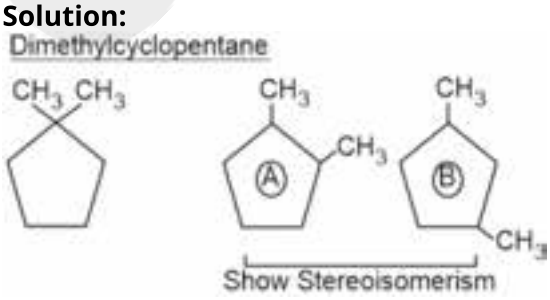
Thus, atomic weight of A is 2 times of atomic weight of B

Hence, answer is 2.

Q. 30 Total number of possible stereoisomers of dimethyl cyclopentane is _____.

Correct Answer:
6

Solution:



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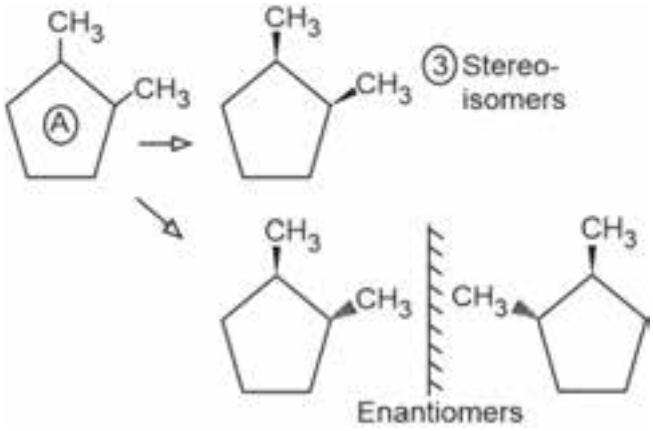

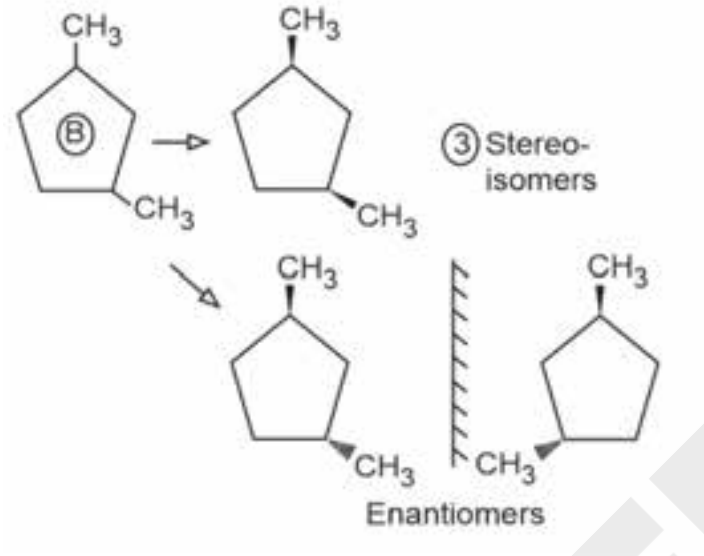

Hence, the answer is 6

Maths

Q. 1 The area of the polygon, whose vertices are the non-real roots of the equation $\bar{z} = iz^2 \, is :$

Option 1:

$\frac{3\sqrt{3}}{4}$

Option 2:

$\frac{3\sqrt{3}}{2}$

Option 3:

$\frac{3}{2}$

Option 4:

$\frac{3}{4}$

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Correct Answer:

$\frac{3\sqrt{3}}{4}$

Solution:

$\text{Let } z = x + iy$

$\text{so } \bar{z} = x - iy$

$\bar{z} = iz^2 \Rightarrow x - iy = i\left(x^2 - y^2 + 2xyi\right)$

$\Rightarrow x - iy = -2xy + \left(x^2 - y^2\right) i$

Comparing: $x = -2xy \quad \& \quad -y = x^2 - y^2$

$\Rightarrow x(1 + 2y) = 0 \quad \& \; x^2 - y^2 + y = 0$

So either $x = 0 \quad \text{or} \quad y = -1/2$

$\text{If } x = 0; -y^2 + y = 0 \Rightarrow y = 0 \text{ or } 1 \Rightarrow (0,0), (0,1)$

$\text{If } y = -1/2, x^2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \Rightarrow x = \pm\frac{\sqrt{3}}{2}$

$\Rightarrow \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

So the non-real solutions are $z = i, \frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}$

It is a triangle whose area $= \frac{1}{2} \times \frac{2\sqrt{3}}{2} \times \frac{3}{2} = \frac{3\sqrt{3}}{4}$

Hence the correct answer is option 1

Q. 2 Let the system of linear equations $x + 2y + z = 2, \alpha x + 3y - z = \alpha, -\alpha x + y + 2z = -\alpha$ be inconsistent. Then α is equal to :

Option 1:

$\frac{5}{2}$

Option 2:

$-\frac{5}{2}$

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Option 3:

$\frac{7}{2}$

Option 4:

$-\frac{7}{2}$

Correct Answer:

$-\frac{7}{2}$

Solution:

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ \alpha & 3 & -1 \\ -\alpha & 1 & 2 \end{vmatrix}$$

$= 0 \Rightarrow 1 \times 7 - 2 \times (\alpha) + 1 \times 4\alpha = 0$

$\Rightarrow \alpha = -7/2$

$$\Delta_x = \begin{vmatrix} 2 & 2 & 1 \\ \alpha & 3 & -1 \\ -\alpha & 1 & 2 \end{vmatrix}$$

$= 2 \times 7 - 2 \times \alpha + 1 \times 4\alpha$

at $\alpha = -7/2$,

$\Delta_x = 14 + 2 \times \frac{7}{2} - 4 \times \frac{7}{2} = 7 \neq 0$

So the system of equations is inconsistent for $\alpha = \frac{-7}{2}$

Hence the correct answer is option 4

Q. 3 If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$, where a, b, c are in A.P. and $|a| < 1, |b| < 1, |c| < 1, abc \neq 0$,

then :

Option 1:

x, y, z are in A.P.

Option 2:

x, y, z are in G.P.

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Option 3:

$\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

Option 4:

$\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1-(a+b+c)$

Correct Answer:

$\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

Solution:

$x=\frac{1}{1-a}, y=\frac{1}{1-b}, z=\frac{1}{1-c}$

$\Rightarrow a=1-\frac{1}{x}, b=1-\frac{1}{y}, c=1-\frac{1}{z}$

Since a, b, c are in A.P

$\Rightarrow 1-\frac{1}{x}, 1-\frac{1}{y}, 1-\frac{1}{z}$ are in A.P

$\Rightarrow \quad -\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z}$ are in A.P.

$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

Hence the correct answer is option 3.

Q. 4 Let $\frac{dy}{dx}=\frac{ax-by+a}{bx+cy+a}$, where a, b, c are constants, represent a circle passing through the point $(2,5)$. Then the shortest distance of the point $(11,6)$ from this circle is :

Option 1:

10

Option 2:

8

Option 3:

7

Option 4:

5

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Correct Answer:

8

Solution:

Let the equation of the circle be

$$x^2+y^2+2gx+2fy+k=0$$

Passes through $(2,5)$

$$\Rightarrow 4+25+4g+10f+k=0 \Rightarrow 4g+10f+k=-2g \quad \text{.....(1)}$$

Differentiating with respect to x,

$$2x+2y\frac{dy}{dx}+2g+2f\frac{dy}{dx}=0$$

$$\Rightarrow \frac{dy}{dx}=\frac{-x-g}{y+f}=\frac{ax-by+a}{bx+cy+a}$$

Comparing $b=0, g=1, f=-1, c=1$

$$K=-29-4+10=-23$$

So equation of circle is

$$x^2+y^2+2x-2y-23=0$$

$$\text{Centre: } (-1,1), \quad \text{Radius } =5$$

Shortest distance of $(11,6)$ from the circle

$$13-5=8$$

Hence the correct answer is option 2.

Q. 5 Let a be an integer such that $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists, where $[t]$ is greatest integer $\leqslant t$.

Then a is equal to:

Option 1:

-6

Option 2:

-2

Option 3:

2

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Option 4:

6

Correct Answer:

-6

Solution:

$$\lim_{x\to 7^+} = \frac{18-1-[-x]}{[2]-3a}$$

$$= \lim_{x\to 7^+} \frac{17-(-x-1)}{x-3a} = \lim_{x\to 7^+} \frac{18+x}{x-3a} = \frac{25}{7-3a}$$

$$\lim_{x\to 7^-} = \frac{18-1-[-x]}{[x]-3a} = \lim_{x\to 7^-} \frac{17-(-x)}{x-1-3a} = \frac{24}{6-3a}$$

For limit to exist $\text{LHL} = \text{RHL} \Rightarrow \frac{25}{7-3a} = \frac{24}{6-3a}$

$\Rightarrow 150-75a = 168-72a \Rightarrow 3a = -18 \Rightarrow a = -6$

Hence the correct answer is option 1.

Q. 6 The number of distinct real roots of $x^4 - 4x + 1 = 0$ is :

Option 1:

4

Option 2:

2

Option 3:

1

Option 4:

0

Correct Answer:

2

Solution:

$f(x) = x^4 - 4x + 1$

$f'(x) = 4x^3 - 4 = 0$

$\Rightarrow x = 1$

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$f''(x) = 12x^2, \quad f''(1) : 12 > 0$

$\Rightarrow x = 1$ is a poit of minima

$f(1) : 1 - 4 + 1 = -2$

For $x < 1, \quad f(n)$ is decreasing and for $x > 1, \quad f(x)$ is in increasing

$f(-\infty) = \infty, \quad f(1) = -2, \quad f(\infty) = \infty$

From intermediate value theorem , $f(x)$ will have 2 real roots one less than 1 and other greater than 1.

Hence the correct answer is option 2

Q. 7 The lengths of the sides of a triangle are $10 + x^2, 10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k^2$ is equal to:

Option 1:
5

Option 2:
8

Option 3:
10

Option 4:
12

Correct Answer:
10

Solution:
Semiperimeter s of the triangle

$= \frac{10 + x^2 + 10 + x^2 + 20 - 2x^2}{2} = 20$

$\text{Area of } \Delta = \sqrt{S(S-a)(S-b)(S-c)}$

$\Rightarrow \Delta^2 = 20\left(10 - x^2\right)\left(10 - x^2\right)\left(2x^2\right)$

$\Rightarrow \Delta = 2\sqrt{10}x\left(10 - x^2\right) = 2\sqrt{10}\left(10x - x^3\right)$

$\Delta' = 2\sqrt{10}\left(10 - 3x^2\right) = 0 \Rightarrow x = \pm\sqrt{\frac{10}{3}}, x = \sqrt{\frac{10}{3}}$ will be point of maxima so,

$3x^2 = 3 \times \frac{10}{3} = 10$

Hence the correct answer is option 3

Q. 8 If $\cos^{-1}\left(\frac{y}{2}\right)=\log_e\left(\frac{x}{5}\right)^5, |y|<2$, then :

Option 1:
$x^2y''+xy'-25y=0$

Option 2:
$x^2y''-xy'-25y=0$

Option 3:
$x^2y''-xy'+25y=0$

Option 4:
$x^2y''+xy'+25y=0$

Correct Answer:
$x^2y''+xy'+25y=0$

Solution:

$\cos^{-1}\frac{y}{2}=5\log_e\frac{x}{5}$

$\Rightarrow \frac{y}{2}=\cos\left(5\log_e\frac{x}{5}\right)$

$\Rightarrow y'=-2\sin\left(5\log\frac{x}{5}\right)\times\frac{5}{x/5}\times\frac{1}{5}$

$=\frac{-10\sin\left(5\log_e\frac{x}{5}\right)}{x}$

$\Rightarrow y''=\frac{-x\times10\cos\left(5\log_e\frac{x}{5}\right)\times\frac{5}{x/5}\times\frac{1}{5}+10\sin\left(5\log_e\frac{x}{5}\right)}{x^2}$

$\Rightarrow x^2y''=-50\cos\left(5\log_e\frac{x}{5}\right)+10\sin\left(5\log_e\frac{x}{5}\right)$

$\Rightarrow x^2y''=-25y-xy'$

$\Rightarrow x^2y''+xy'+25y=0$

Hence the correct answer is option 4

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Q. 9 If $\int \frac{(x^2+1)\,e^x}{(x+1)^2}\,dx = f(x)e^x + C$, where C is a constant, then $\frac{d^3f}{dx^3}\ at\ x=1$ is equal to :

Option 1:
$-\frac{3}{4}$

Option 2:
$\frac{3}{4}$

Option 3:
$-\frac{3}{2}$

Option 4:
$\frac{3}{2}$

Correct Answer:
$\frac{3}{4}$

Solution:

We have to convert this integral into the form $\int e^x\,(f(x)+f'(x))\,dx$ whose integral is $e^x f(1)+c$.

So let us assume $f(x)$ to be

$$f(x)=\frac{ax+b}{x+1} \Rightarrow f'(x)=\frac{a(x+1)-(ax+b)}{(x+1)^2}$$

$$=\frac{a-b}{(x+1)^2}$$

$$f(x)+f'(x)=\frac{(ax+b)(x+1)+a-b}{(x+1)^2}=x^2+1$$

$$\Rightarrow a=1, a+b=0 \Rightarrow b=-1$$

So $f(x)=\frac{x-1}{x+1}$

$$f'(x)=\frac{2}{(x+1)^2}, f''(x)=\frac{-4}{(x+1)^3}, f'''(x): \frac{12}{(x+1)^4}$$

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$$f'''(1) : \frac{12}{(2)^4} : \frac{12}{16} = \frac{3}{4}$$

Hence the correct answer is option 2

Q. 10 The value of the integral $\int_{-2}^{2} \frac{|x^3+x|}{(e^{x|x|}+1)}dx$ is equal to :

Option 1:

$5e^2$

Option 2:

$3e^{-2}$

Option 3:

4

Option 4:

6

Correct Answer:

6

Solution:

$$I = \int_{-2}^{2} \frac{|x^3+x|}{e^{x|x|}+1}dx \quad \text{.....(1)}$$

$$I(x) = I(a+b-x)$$

$$\Rightarrow I = \int_{-2}^{2} \frac{|-x^3-x|}{e^{-x|-x|}+1}dx$$

$$= \int_{-2}^{2} \frac{|x^3+x|}{e^{-x|x|}+1} = \int_{-2}^{2} \frac{e^{x|x|}|x^3+x|}{e^{x|x|}+1}dx \quad \text{.....(2)}$$

Add eqn (1) and (2)

$$\Rightarrow 2I = \int_{-2}^{2} |x^3+x|\,dx = 2\int_{0}^{2} |x^3+x|\,dx$$

$$\Rightarrow I = \int_{0}^{2} (x^3+x)\,dx = 6$$

Hence the correct aswer is option 4.

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Q. 11 If $\frac{dy}{dx}+\frac{2^{x-y}\left(2^{y}-1\right)}{2^{x}-1}=0, x, y>0, y(1)=1$, then $y(2)$ is equal to :

Option 1:
$2+\log_2 3$

Option 2:
$2+\log_3 2$

Option 3:
$2-\log_3 2$

Option 4:
$2-\log_2 3$

Correct Answer:
$2-\log_3 2$

Solution:
$$\frac{dy}{dx}=-\frac{2^{x}}{2^{x}-1}\times\frac{2^{y}-1}{2^{y}}$$

$$\Rightarrow\int\frac{2^{y}\ln 2}{2^{y}-1}dy=-\int\frac{2^{x}\ln 2}{2^{x}-1}dx$$

$$\Rightarrow \ln|2^{y}-1|=-\ln|2^{x}-1|+C$$

$$\Rightarrow \text{ at } x=1, y=1\Rightarrow c=0$$

$$\Rightarrow \ln\left(2^{y-1}\right)=\ln\left(\frac{1}{2^{x}-1}\right)$$

$$\Rightarrow 2^{y}-1=\frac{1}{2^{x}-1}\Rightarrow 2^{y}=1+\frac{1}{2^{x}-1}$$

$$\text{at } x=2, 2^{y}=1+\frac{1}{3}\&=\frac{4}{3}$$

$$\Rightarrow \quad y=\log_2\frac{4}{3}=2-\log_2 3$$

Hence the correct aswer is option 3

Q. 12 In an isosceles triangle ABC, the vertex A is $(6,1)$ and the equation of the base BC is $2x+y=4$. Let the point B lie on the line $x+3y=7$. If (α,β) is the centroid of ΔABC, then $15(\alpha+\beta)$ is equal to :

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Option 1:
39

Option 2:
41

Option 3:
51

Option 4:
63

Correct Answer:
51

Solution:
Foot of perpendicular from $\mathrm{A}(6,1)$ to BC

$$\frac{\mathrm{x}-6}{2}=\frac{\mathrm{y}-1}{1}=\frac{-1\left(6\mathrm{x}^2+1-4\right)}{2^2+1^2}=-\frac{9}{5}$$

$$\Rightarrow \mathrm{x}=\frac{12}{5}, \mathrm{y}=\frac{-4}{5}$$

Centroid divide medium in $2:1$

$$\frac{\mathrm{A}\,(6,1)\,\mathrm{G}\,(\alpha,\beta)}{2:1}\mathrm{D}\left(\frac{12}{5},\frac{-4}{5}\right)$$

$$\text{So } \alpha=\frac{6\times 1+\frac{12}{5}\times 2}{2+1}, \quad \beta=\frac{1\times 1+2\times\left(-\frac{4}{5}\right)}{2+1}$$

$$\alpha+\beta=\frac{30+24+5-8}{15}=\frac{51}{15}\Rightarrow 15(\alpha+\beta)=51$$

Hence the correct answer is option 3.

Q. 13 Let the eccentricity of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b,$ be $\frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$, then $\mathrm{a}^2+\mathrm{b}^2$ is equal to:

Option 1:
29

Option 2:
31

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Option 3:
32

Option 4:
34

Correct Answer:
31

Solution:
$e^2 = 1 - \frac{b^2}{a^2} = \frac{1}{16} \Rightarrow \frac{b^2}{a^2} = \frac{15}{16}$ .....(1)

Also ellipse passes through $\left(-4\sqrt{\frac{2}{5}}, 3\right)$

So $\frac{32}{5a^2} + \frac{9}{b^2} = 1 \quad \text{Put } b^2 = \frac{15a^2}{16}$

$\Rightarrow \frac{32}{5a^2} + \frac{9}{15a^2} \times 16 = 1 \Rightarrow \frac{80}{5a^2} = 1 \Rightarrow a^2 = 16, b^2 = 15$

$a^2 + b^2 = 31$

Hence the correct answer is option 2

Q. 14 If two straight lines whose direction cosines are given by the relations $l + m - n = 0, 3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is :

Option 1:
6

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
6

Solution:
$l + m = n$ .....(1) $\quad 3l^2 + m^2 + cnl = 0$ .....(2)

Substituting $n = 1 + m$ in eqn(2)

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$\Rightarrow \quad 3l^2 + m^2 + c(l+m)l = 0$

$\Rightarrow \quad (3+c)l^2 + cml + m^2 = 0$

$\Rightarrow \quad (3+c)\left(\frac{l}{m}\right)^2 + c\left(\frac{l}{m}\right) + 1 = 0$

Since lines are parallel $\Rightarrow$ this quadratic will have repeated roots $\Rightarrow D = 0$

$\Rightarrow c^2 - 4(3+c) = 0$

$\Rightarrow c^2 - 4c - 12 = 0 \Rightarrow c = 6 \text{ or } c = -2$

So positive value of $c = 6$

Hence the correct answer is option 1.

Q. 15 Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{b}| \in \{1, 2, \ldots, 10\}$ is :

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
0

Solution:
Let $\vec{b} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{x} \\ \hat{x} & y & z \\ 2 & -3 & 2 \end{vmatrix} = \vec{a}$$

$\Rightarrow (2y + 3z)\hat{i} - (2x - 2z)\hat{j} + (-3x - 2y)\hat{k} = \hat{i} + \hat{j} - \hat{k}$

$2y + 3z = 1$....(1) $\qquad 2y - 3z = 1$.....(2)

$3x - 2y = -1 \qquad \Rightarrow 3x + 2y = 1$.....(3)

This system of equation is inconsistent So no such $\vec{b}$ exist.

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Hence the correct answer is option 1.

Q. 16 Five numbers x_1, x_2, x_3, x_4, x_5 are randomly selected from the numbers $1, 2, 3, \ldots, 18$ and are arranged in the increasing order $(x_1 < x_2 < x_3 < x_4 < x_5)$. The probability that $x_2 = 7$ and $x_4 = 11$ is :

Option 1:
$\frac{1}{136}$

Option 2:
$\frac{1}{72}$

Option 3:
$\frac{1}{68}$

Option 4:
$\frac{1}{34}$

Correct Answer:
$\frac{1}{68}$

Solution:
$\text{If} \quad x_2 = 7, \quad x_1 < 7 \Rightarrow x, \in \{1, 2, 3, 4, 5, 6\}$

$x_3 \in \{8, 9, 10\}, x_5 \in \{12, 13, 14, 15, 16, 17, 18\}$

$\text{So Probability} = \frac{{}^{6}C_1 \times {}^{3}C_1 \times {}^{7}_{7}C_1}{{}^{18}C_5}$

$= \frac{6 \times 3 \times 7 \times 8 \times 4 \times 8 \times 2 \times 1}{18 \times 17 \times +6 \times 15 \times 14} = \frac{1}{68}$

Hence the correct answer is option 3.

Q. 17 Let X be a random variable having binomial distribution $B(7, p)$. If $P(X = 3) = 5P(X = 4)$, then the sum of the mean and the variance of X is :

Option 1:
$\frac{105}{16}$

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Option 2:

$\frac{7}{16}$

Option 3:

$\frac{77}{36}$

Option 4:

$\frac{49}{16}$

Correct Answer:

$\frac{77}{36}$

Solution:

$\mathrm{P(X=3)} = {}^{7}\mathrm{C}_3 \times \mathrm{p}^3 \times (1-\mathrm{p})^4$

$\mathrm{P(X=4)} = {}^{7}\mathrm{C}_4 \times \mathrm{p}^4 \times (1-\mathrm{p})^3$

$\text{Given } \mathrm{P(x=3)} = 5\mathrm{p(x=4)}$

$\Rightarrow 7\mathrm{c}_3 \times \mathrm{p}^3(1-\mathrm{p})^4 = 5 \times 7\mathrm{c}_4 \times \mathrm{p}^4 \times (1-\mathrm{p})^3$

$\Rightarrow 1-\mathrm{p} = 5\mathrm{p} \Rightarrow \mathrm{p} = \frac{1}{6}$

$\text{Mean} = \mathrm{np} = 7 \times \frac{1}{6} = \frac{7}{6}$

$\text{Variance} = \mathrm{np}(1-\mathrm{p}) = 7 \times \frac{1}{6} \times \frac{5}{6} = \frac{35}{36}$

$\text{Mean} + \text{Variance} = \frac{7}{6} + \frac{35}{36} = \frac{77}{36}$

Hence the correct answer is option 3.

Q. 18 The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to :

Option 1:

-1

Option 2:

$-\frac{1}{2}$

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Option 3:

$-\frac{1}{3}$

Option 4:

$-\frac{1}{4}$

Correct Answer:

$-\frac{1}{2}$

Solution:

Using Summation of cosine series

$$\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=\frac{\sin 3\times\frac{2\pi}{2\times 7}}{\sin\frac{2\pi}{2\times 7}}\times\cos\left(\frac{\frac{2\pi}{7}+\frac{6\pi}{7}}{2}\right)$$

$$=\frac{\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}}\times\cos\frac{4\pi}{7}$$

$$=\frac{\sin\left(\pi-\frac{4\pi}{7}\right)\cos\frac{4\pi}{7}}{\sin\frac{\pi}{7}}$$

$$=\frac{2\sin\frac{4\pi}{7}\cos\frac{4\pi}{7}}{2\sin\frac{\pi}{7}}$$

$$=\frac{\sin\frac{8\pi}{7}}{2\sin\frac{\pi}{7}}=-\frac{1}{2}$$

Hence the correct answer is option 2

Q. 19 $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to :

Option 1:

$\frac{11\pi}{12}$

Option 2:

$\frac{17\pi}{12}$

Option 3:

$\frac{31\pi}{12}$

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Option 4:

$-\frac{3\pi}{4}$

Correct Answer:

$\frac{11\pi}{12}$

Solution:

$$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)=\sin^{-1}\left(\sin\frac{\pi}{3}\right)=\frac{\pi}{3}$$

$$\cos^{-1}\left(\cos\frac{7\pi}{6}\right)=\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)=\frac{5\pi}{6}$$

$$\tan^{-1}\left(\tan\frac{3\pi}{4}\right)=\tan^{-1}(-1)=\frac{-\pi}{4}$$

$$\text{Sum }=\frac{\pi}{3}+\frac{5\pi}{6}-\frac{\pi}{4}$$

$$=\frac{(4+10-3)\pi}{12}=\frac{11\pi}{12}$$

Hence the correct answer is option 1

Q. 20 The boolean expression $(\sim(p\wedge q))\vee q$ is equivalent to :

Option 1:

$q\rightarrow(p\wedge q)$

Option 2:

$\mathrm{p}\rightarrow\mathrm{q}$

Option 3:

$\mathrm{p}\rightarrow(\mathrm{p}\rightarrow\mathrm{q})$

Option 4:

$\mathrm{p}\rightarrow(\mathrm{p}\vee\mathrm{q})$

Correct Answer:

$\mathrm{p}\rightarrow(\mathrm{p}\vee\mathrm{q})$

Solution:

			Question		A	C	D		
p	q	$p\wedge q$	$\sim(p\wedge q)$	$(\sim(p\wedge q)\vee q)$	$q\rightarrow(p\wedge q)$	$b\rightarrow q$	$b\rightarrow(p\rightarrow q)$	$p\vee q$	$b\rightarrow(p\vee q)$
T	T	T	F	T	T	T	T	T	T
T	F	F	T	T	T	F	F	T	T

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F	T	F	T	T	F	T	T	T	T
F	F	F	T	t	T	T	T	F	T

Hence the correct answer is option 4.

Q. 21 Let $f : \mathbf{R} \to \mathbf{R}$ be a function defined by $f(x) = \frac{2e^{2x}}{e^{2x} + e}$. Then $f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \ldots + f\left(\frac{99}{100}\right)$ is equal to_____.

Correct Answer:
99

Solution:

$$f(x) = \frac{2e^{2x}}{e^{2x} + e}, f(1-x) = \frac{2e^{2-2x}}{e^{2-2x} + e} = \frac{2e}{e^{2x} + e}$$

$$\Rightarrow f(x) + f(1-x) = 2$$

$$s = f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + \cdots f\left(\frac{50}{100}\right) + \cdots - f\left(\frac{93}{100}\right) + f\left(\frac{99}{100}\right)$$

$$s = f\left(\frac{99}{100}\right) + f\left(\frac{98}{100}\right) + \cdots f\left(\frac{50}{100}\right) + \cdots f\left(\frac{2}{100}\right) + f\left(\frac{1}{100}\right)$$

Adding

$$2s = 99 \times (f(r) + f(1-r))$$

$$\Rightarrow s = \frac{99}{2} \times 2 = 99.$$

Hence answer is 99.

Q. 22 If the sum of all the roots of the equation $e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to_____.

Correct Answer:
45

Solution:
The equation can be rewritten as

$$2e^{3x} - 22e^{2x} + 81e^{x} - 90 = 0$$

Let the roots are α, β, γ

$$\text{Let } e^{x} = t, e^{\alpha} = t_1, e^{\beta} = t_2, e^{\gamma} = t_3$$

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$2t^3 - 22t^2 + 81t - 90 = 0$

$\text{Product of roots } = t, t_2t_3 = e^{\alpha} \cdot e^{\beta} \cdot e^{\gamma} = \frac{90}{2} = 45$

$\Rightarrow e^{\alpha+\beta+\gamma} = 45$

$\Rightarrow \alpha + \beta + \gamma = \log_e 45$

$\text{So } p = 45$

Hence the answer is 45

Q. 23 The positive value of the determinant of the matrix A, whose $\text{Adj}(\text{Adj}(A)) = \begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix}$ is_____.

Correct Answer:
14

Solution:
$|\text{Ad}_j(\text{Adj}(A))| = |A|^{(n-1)^2} = |A|^{2^2} = |A|^4.$

$|\text{ Adj}(\text{Adj } A)| = 14 \times 14 \times 14 \begin{vmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}$

$\Rightarrow |A|^4 = 14 \times 14 \times 14 \times [1 \times 3 - 2 \times (-5) - 1 \times (-1)]$

$\Rightarrow |A|^4 = 14^4 \Rightarrow |A| = \pm 14$

$+\text{ve value} = 14.$

Hence answer is $14.$

Q. 24 The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes there should be at least 2 blue cubes, is_____.

Correct Answer:
28

Solution:
Let $n_1, n_2, n_3, n_4, n_5 \text{ and } n_6$ are number of blue balls in the gaps of red balls

Here $n_1 \geqslant 0, n_2 \geqslant 2, n_3 \geqslant 2, n_4 \geqslant 2, n5 \geqslant 2, n_6 \geqslant 0$

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Take $n_1 = n_1^1, n_2 - 2 = n_2^1, n_3 - 2 = n_3^1, n_4 - 2 - n_4^1, n_5 - 2 = n_5^1, n_6 = n_6^1$

$n_1 + n_2 + n_3 + n_4 + n_5 + n_6 = 11$

$\Rightarrow n_1 + (n_2 - 2) + (n_3 - 2) + (n_4 - 2) + (n_5 - 2) + n_6 = 11 - 8$

$\Rightarrow n_1' + n_2' + n_3' + n_4' + n_5' + n_6' = 3$

no of non-negative integral solutions

$= {}^{6+3-1}C_{3-1} = {}^8C_2 = 28$

Hence answer is 28

Q. 25

If the coefficient of x^{10} in the binomial expansion of $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}} + \frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ is $5^k \cdot l$, where $l, k \in \mathbf{N}$ and l is co-prime to 5 , then k is equal to_____.

Correct Answer:
5

Solution:

$T_{r+1} = {}^{60}Cr \frac{x^{\frac{60-r}{2}}}{5^{\frac{60-r}{4}}} \times \frac{5^{\frac{r}{2}}}{x^{\frac{r}{3}}}$

$= 6C_r x^{30-\frac{r}{2}-\frac{r}{3}} \cdot 5^{\frac{r}{2}+\frac{r}{4}-15}$

$30 - \frac{r}{2} - \frac{r}{3} = 10$

$\Rightarrow \frac{5r}{6} = 20 \Rightarrow r = 24$

so $T_{2.5}\ {}^{60}C_{24} \times x^{10} \times 5^3$

$= \frac{60!}{24!36!} \times 5^3 \times x^{10}$

No. of $5's$ in $60! = \left[\frac{60}{5}\right] + \left[\frac{60}{25}\right] + \cdots$

$= 12 + 2 = 14$

No. of $5'$ in $24! = \left[\frac{24}{5}\right] + \left[\frac{24}{5^2}\right] + \cdots$

$= 4$

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$$\text{No. of } 5' \text{ in } 36! = \left[\frac{36}{5}\right] + \left[\frac{36}{5^2}\right] + \cdots$$

$$= 7 + 1$$

$$= 8$$

$$\text{So No. of } 5' \text{ in} \frac{60!}{24!36!} : 14 - 4 - 8 = 2.\text{so overall no of factors of 5 in } T_{25}$$

$$= 3 + 2 = 5.$$

Hence answer is 5

Q. 26 Let $A_1 = \{(x, y) : |x| \leqslant y^2, |x| + 2y \leqslant 8\}$ *and*

$A_2 = \{(x, y) : |x| + |y| \leqslant k\}$. If $27(\text{ Area } A_1) = 5 (\text{Area } A_2)$, then k is equal to :

Correct Answer:
6

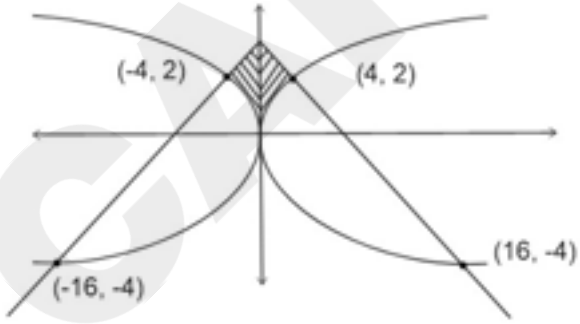
Solution:
$A_1 = \{(x, y) : |x| \leq y^2, |x| + 2y \leq 8\}$

Point of intersection of

$|x| = y^2 \& \quad |x| = 8 - 2y$ are

$y^2 = 8 - 2y \Rightarrow y^2 + 2y - 8 = 0 \Rightarrow y = -4 \text{ or } y = 2$

$y = -4 \Rightarrow x \pm 16, \quad y = 2 \Rightarrow x = \pm 4$



$$A_1 = 2\int_0^4 \left[\left(4 - \frac{x}{2}\right) - (\sqrt{x})\right] dx$$

$$= \frac{2}{2}\left[\ 4x - \tfrac{x^2}{4} - \tfrac{2}{3}x^{3/2}\ \right]_0^4$$

$$= 2\left[16 - 4 - \frac{16}{3}\right] = \frac{40}{3}$$

$$A_2 = \{(x, y); |x| + |y| < 1\}$$

$$\text{Area } A_2 = 4 \times \frac{1}{2} \times k \times k = 2k^2$$

$$27A_1 = 5A_2$$

$$\Rightarrow 27 \times \frac{40}{3} = 5 \times 2k^2$$

$$\Rightarrow k^2 = 36 = k = 6.$$

Hence answer is 6.

Q. 27 If the sum of the first ten terms of the series
$\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + \cdots$
is $\frac{m}{n}$, where m and n are co-prime numbers, then $m + n$ is equal to_____.

Correct Answer:
276

Solution:

$$S = \frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + \cdots$$

$$= \frac{1}{4 \times 1^4 + 1} + \frac{2}{4 \times 2^4 + 1} + \frac{3}{4 \times 3^4 + 1} + \frac{4}{4 \times 5^4} + \frac{5}{4 \times 5^4 + 1} + \cdots$$

$$T_r = \frac{r}{4r^4 + 1}$$

$$S_{10} = \sum_{r=1}^{10} \frac{r}{4r^4 + 1} = \sum_{r=1}^{10} \frac{r}{4r^4 + 4r^2 + 1 - 4r^2}$$

$$= \sum_{r=1}^{10} \frac{r}{(2r^2 + 1)^2 - (2r)^2}$$

$$= \sum_{r=1}^{10} \frac{r}{(2r^2 + 2r + 1)(2r^2 - 2r + 1)} = \frac{1}{4} \sum_{r=1}^{10} \left(\frac{1}{2r^2 - 2r + 1} - \frac{1}{2r^2 + 2\lambda + 1}\right)$$

$$= \frac{1}{4}\left[\left(1 - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{13}\right) + \left(\frac{1}{13} - \frac{1}{25}\right) + \cdots \left(\frac{1}{2 \times 10^2 - 2 \times 10 + 1} - \frac{1}{2 \times 10^2 + 2 \times 10 + 1}\right)\right]$$

$$= \frac{1}{4} + \left[1 - \frac{1}{221}\right] = \frac{1}{4} \times \frac{220}{221} = \frac{55}{221} = \frac{m}{n}$$

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$\mathrm{m}+\mathrm{n}: 55+221=276$

Q. 28 A rectangle R with end points of one of its sides as $(1,2)$ and $(3,6)$ is inscribed in a circle. If the equation of a diameter of the circle is $2x-y+4=0$, then the area of R is_____.

Correct Answer:
16

Solution:
Let the cetre of the circle is $C(\alpha, 4+2\alpha)$

It will be equidistant from $(1,2)(3,6)$

$\Rightarrow (\alpha-1)^2+(4+2\alpha-2)^2=(\alpha-3)^2+(4+2\alpha-6)^2$

$\Rightarrow \alpha^2-2\alpha+1+4\alpha^2+8\alpha+4=\alpha^2-6\alpha+3+4\alpha^2-8\alpha+4$

$\Rightarrow 20\alpha=8 \Rightarrow \alpha=\frac{4}{5}; \quad \mathrm{c}: \left(\frac{4}{5}, \frac{28}{5}\right).$

$\text{distance b/w } (1,2) \quad \&(3,6)=2\sqrt{5}=\text{length.}$

$\text{Equation of line joining}(1,2)\text{and}(3,6)\text{is}$

$\mathrm{y}-2=\frac{6-2}{3-1}(\mathrm{x}-1)$

$2\mathrm{x}-\mathrm{y}=0$

$\text{distance of centre from } 2\mathrm{x}-\mathrm{y}=0 \text{ is}$

$\mathrm{d}=\frac{\left|2\times\frac{4}{5}-\frac{28}{5}\right|}{\sqrt{2^2+1^2}}=\frac{4}{\sqrt{5}} \text{ units}$

$\text{So Breadth } 2\times\frac{4}{\sqrt{5}}=\frac{8}{\sqrt{5}}\text{units}$

$\text{Area of R}=\mathrm{l}\times\mathrm{b}=2\sqrt{5}\times\frac{8}{\sqrt{5}}=16\text{sq.units.}$

So answer is 16sq.units.

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Q. 29 A circle of radius 2 unit passes through the vertex and the focus of the parabola $y^2 = 2x$ and touches the parabola $y = \left(x - \frac{1}{4}\right)^2 + \alpha,$ where $\alpha > 0$. Then $(4\alpha - 8)^2$ is equal to_____.

Correct Answer:
63

Solution:

$\text{Vertex} : (0,0), \text{Focus} : \left(\frac{1}{2}, 0\right) \quad \text{r} = 2$

$\text{Let the centre of circle is } \left(\frac{1}{4}, \text{k}\right),$

$\text{r} = \sqrt{\left(\frac{1}{4}\right)^2 + \text{x}^2} = 2 \Rightarrow \text{k}^2 = 4 - \frac{1}{16} = \frac{63}{16}$

$\Rightarrow \text{k} = \frac{+3}{4}\sqrt{7}.$

$\text{so centre} : \left(\frac{1}{4}, \pm\frac{3\sqrt{7}}{4}\right) \qquad \text{r} = 2$

$\text{If centre is } \left(\frac{1}{4}, \frac{3\sqrt{7}}{4}\right), \quad \alpha = 2 + \frac{3\sqrt{7}}{4}$

$\text{If centre is } \left(\frac{1}{4}, -\frac{3\sqrt{7}}{4}\right), \alpha = 2 - \frac{3\sqrt{7}}{4}$

$\text{In either case; } (4\alpha - 8)^2 = (\pm 3\sqrt{7})^2 = 63$

So answer is 63.

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Q. 30 Let the mirror image of the point (a, b, c) with respect to the plane $3x - 4y + 12z + 19 = 0$ be $(a - 6, \beta, \gamma)$. If $a + b + c = 5$, then $7\beta - 9\gamma$ is equal to_____.

Correct Answer:
137

Solution:

$$\frac{x - a}{3} = \frac{y - b}{-4} = \frac{3 - c}{12} = -\frac{2(3a - 4b + 12c + 19)}{3^2 + 4^2 + 12^2}$$

$\text{Here } x = a - 6, y = \beta, z = \gamma.$

$$\Rightarrow \frac{a - 6 - a}{3} = \frac{\beta - b}{-4} = \frac{\gamma - c}{12} = \frac{-2(3a - 4b + 12c + 19)}{169}$$

$\Rightarrow 3a - 4b + 12c = 150$ .....(1)

$a + b + c = 5$

$\Rightarrow 3a + 3b + 3c = 15$ .....(2)

Subtracting Eqn (1)-(2)

$\Rightarrow 7b - 9c = -135$ .....(3)

$$\frac{\beta - b}{-4} = -2 \Rightarrow b = \beta - 8,$$

$$\frac{\gamma - c}{12} = -2 \Rightarrow c = \gamma + 24,$$

Substituting in (3)

$7(\beta - 8) - 9(\gamma + 24) = -135$

$\Rightarrow 7\beta - 9\gamma = 56 + 216 - 135$

$= 272 - 135 = 137$

So answer is 137

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27 June Evening Shift

Physics

Q. 1 The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantify will be :

Option 1:
$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

Option 2:
$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]$

Option 3:
$\left[\mathrm{ML}^{2}\ \mathrm{T}^{-1}\right]$

Option 4:
$\left[\mathrm{M}^{-1}\mathrm{L}^{3}\ \mathrm{T}^{0}\right]$

Correct Answer:
$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

Solution:

$$\begin{aligned}[\mathrm{A}] &= [\mathrm{pascal}-\mathrm{second}] \\ &= \frac{\mathrm{Force}}{\mathrm{area}} \times \mathrm{second} \\ &= \left[\frac{\mathrm{MLT}^{-2}}{\mathrm{L}^{2}} \times \mathrm{T}\right] \\ &= \left[\mathrm{ML}^{-1}\mathrm{T}^{-1}\right]\end{aligned}$$

Correct ans is (1).

Q. 2 The distance of the Sun from earth is $1.5 \times 10^{11}\mathrm{m}$ and its angular diameter is $(2000)s$ when observed from the earth. The diameter of the Sun will be :

Option 1:
$2.45 \times 10^{10}\ \mathrm{m}$

Option 2:
$1.45 \times 10^{10}\ \mathrm{m}$

Option 3:
$1.45 \times 10^{9}\ \mathrm{m}$

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Option 4:

0.14×10^9 m

Correct Answer:

1.45×10^9 m

Solution:

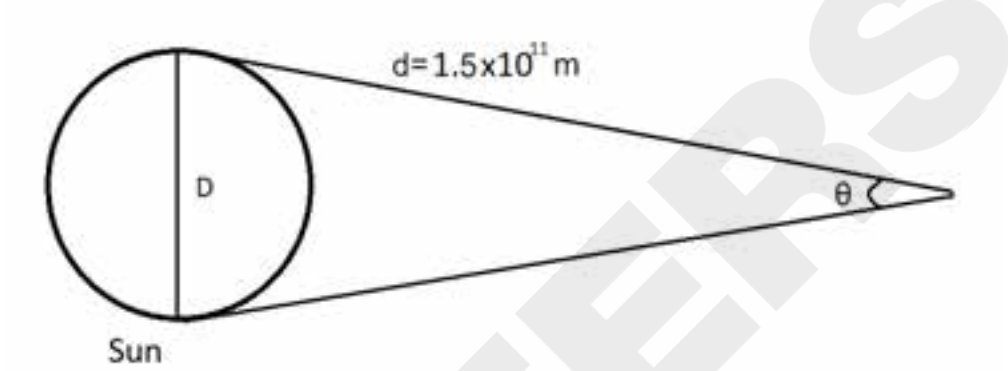
1 degree = 60 minute

1 minute = 60 second

1 degree = 3600 second

$$1\,\text{second} = \frac{1}{3600}\,\text{degree}$$
$$= \frac{1}{3600} \times \frac{\pi}{180}\,\text{rad.}$$

$$2000\,\text{second} = \frac{2000}{3600} \times \frac{\pi}{180} = \frac{2\pi}{36 \times 18}$$



$$\theta = \frac{\text{D}}{\text{d}}$$
$$\text{diameter} = \frac{2\pi}{36 \times 18} \times 1.5 \times 10^{11}$$
$$= 1.45 \times 10^9 \text{m}$$

Correct ans is (3)

Q. 3 When a ball is dropped anto a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is :

Option 1:

19.6 m

Option 2:

29.4 m

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Option 3:

$39.2\mathrm{m}$

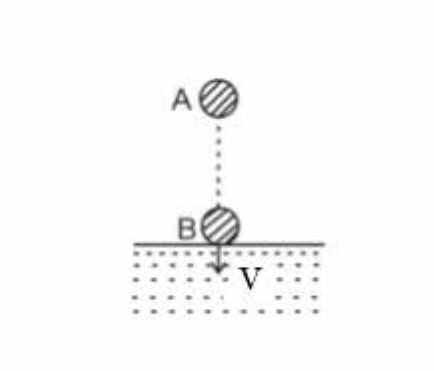
Option 4:

$73.5\mathrm{m}$

Correct Answer:

$29.4\ \mathrm{m}$

Solution:



$\mathrm{v} = \sqrt{2\mathrm{gh}} = \sqrt{2 \times 9.8 \times 4.9} = 9.8\,\mathrm{m/s}$

$\text{time for A to B} = \sqrt{\dfrac{2\mathrm{h}}{\mathrm{g}}} = \sqrt{\dfrac{2 \times 4.9}{9.8}} = 1\,\mathrm{sec}$

distance travelled by ball in 3 sec in lake $= 9.8 \times 3$

$\therefore$ depth of lake $= 29.4\,\mathrm{m}$

Correct answer is (2)

Q. 4 One end of a massless spring of spring constant k and natural length l_0 is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing trough fixed end, then the elongation of the spring will be :

Option 1:

$\dfrac{\mathrm{k} - \mathrm{m}\omega^2\mathrm{l}_0}{\mathrm{m}\omega^2}$

Option 2:

$\dfrac{\mathrm{m}\omega^2\mathrm{l}_0}{\mathrm{k} + \mathrm{m}\omega^2}$

Option 3:

$\dfrac{\mathrm{m}\omega^2\mathrm{l}_0}{\mathrm{k} - \mathrm{m}\omega^2}$

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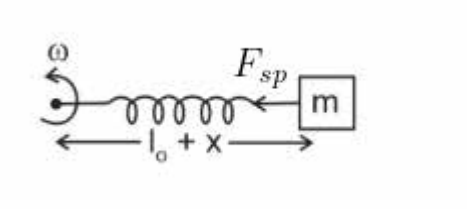
Option 4:

$\frac{k+m\omega^2 l_0}{m\omega^2}$

Correct Answer:

$\frac{m\omega^2 l_0}{k-m\omega^2}$

Solution:



Let elongation in spring is x ,
Using the centripetal force equation

$$F_{spring} = m\omega^2 (l_0 + x)$$
$$kx = m\omega^2 (l_0 + x)$$
$$kx = m\omega^2 l_0 + m\omega^2 x$$
$$x\left(k - m\omega^2\right) = m\omega^2 l_0$$
$$x = \frac{m\omega^2 l_0}{k - m\omega^2}$$

The correct answer is (3)

Q. 5 A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x\left(u^2 - gL\right)}$. The value of x is-

Option 1:
3

Option 2:
2

Option 3:
1

Option 4:
5

Correct Answer:
2

Solution:

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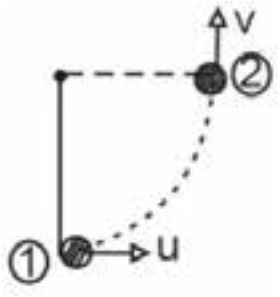


$$\overrightarrow{\Delta v} = \mathrm{v}\hat{\mathrm{j}} - \mathrm{u}\hat{\mathrm{i}}$$
$$\left|\overrightarrow{\Delta v}\right| = \sqrt{\mathrm{v}^2 + \mathrm{u}^2}$$

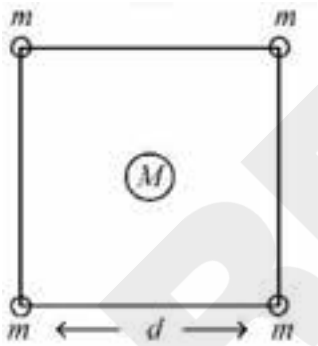
Applying mechanical energy conservation. at 1 & 2.

$$\frac{1}{2}\mathrm{mu}^2 + 0 = \frac{1}{2}\mathrm{mv}^2 + \mathrm{mgl}$$
$$\mathrm{v}^2 = \mathrm{u}^2 - 2\mathrm{gl}$$

$$\therefore \Delta \mathrm{v} = \sqrt{\mathrm{u}^2 - 2\mathrm{gl} + \mathrm{u}^2} = \sqrt{2\left(\mathrm{u}^2 - 2\mathrm{gl}\right)}$$

The correct answer is (2)

Q. 6 Four spheres each of mass m form a square of side d(as shown in figutre). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is :



Option 1:

$$-\frac{\mathrm{Gm}}{\mathrm{d}}\left[\left(4+\sqrt{2}\right)\mathrm{m} + 4\sqrt{2}\mathrm{M}\right]$$

Option 2:

$$-\frac{\mathrm{Gm}}{\mathrm{d}}\left[\left(4+\sqrt{2}\right)\mathrm{M} + 4\sqrt{2}\mathrm{m}\right]$$

Option 3:

$$-\frac{\mathrm{Gm}}{\mathrm{d}}\left[3\mathrm{m}^2 + 4\sqrt{2}\mathrm{M}\right]$$

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Option 4:

$$-\frac{Gm}{d}\left[6m^2+4\sqrt{2}M\right]$$

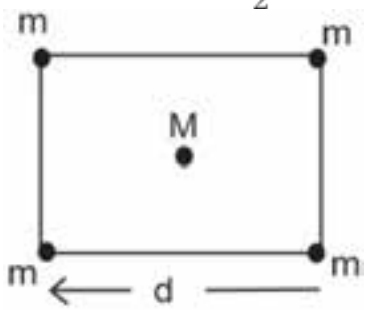
Correct Answer:

$$-\frac{Gm}{d}\left[\left(4+\sqrt{2}\right)m+4\sqrt{2}M\right]$$

Solution:

For gravitational potential energy of the system, total no.of pair to write energy equation is

$${}^{n}C_2={}^{5}C_2=\frac{5\times 9}{2}=10$$

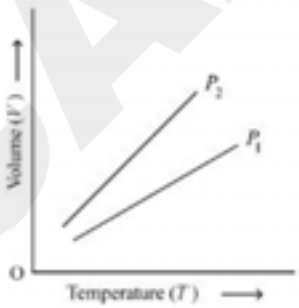


$$U_{system}=\left[4\times\left(\frac{-Gm^2}{d}\right)\right]+\left[2\times\left(\frac{-Gm^2}{\sqrt{2}d}\right]+\left[4\times\left(\frac{-GMm}{\left(\frac{d}{\sqrt{2}}\right)}\right]\right.\right.$$

$$=\frac{-Gm}{d}\left[4m+\sqrt{2}m+4\sqrt{2}M\right]$$

The correct answer is (1)

Q. 7 For a perfect gas, two pressures P_1 and P_2 are shown in figure. The graph shows :



Option 1:

$P_1 > P_2$

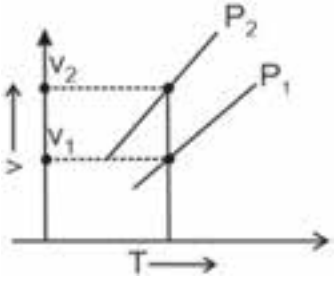
Option 2:
$P_1 < P_2$

Option 3:
$P_1 = P_2$

Option 4:
Insufficient data to draw any conclusion

Correct Answer:
$P_1 > P_2$

Solution:



Ideal gas equation

$$PV = nRT$$

For the same temperature as can be seen from the graph,
$V_2 > V_1$
$\therefore P_2 > P_1$

The correct answer is (1)

Q. 8 According to kinetic theory of gases,
A. The motion of the gas molecules freezes at $0^\circ C$.
B. The mean free path of gas molecules decreases if the density of molecules is increased.
C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.
D. Average kinctic energy per molecule per degree of freedom is $\frac{3}{2}k_B T$ (for monoatomic gases).
Choose the most appropriate answer from the options given below:

Option 1:
A and C only

Option 2:
B and C only

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Option 3:
A and B only

Option 4:
C and D only

Correct Answer:
B and C only

Solution:

Mean free path, $\lambda = \dfrac{\mathrm{kT}}{\sqrt{2}\pi \mathrm{d}^2 \rho}$

Mean free path is directly proportional to temperature (T) and inversely proportional to the density(ρ)

The correct answer is (2)

Q. 9 A lead bullet penetrates into a solid object and melts. Assuming that 40% of its kinetic energy is used to heat it, the initial speed of bullet is :
(Given, initial temperature of the bullet $= 127^{\circ}\mathrm{C}$,
Melting point of the bullet $= 327^{\circ}\mathrm{C}$,
Latent heat of fusion of lead $= 2.5 \times 10^4\ \mathrm{J\ kg^{-1}}$,
Specific heat capacity of lead $= 125\ \mathrm{J/kgK}$)

Option 1:
$125\ \mathrm{ms^{-1}}$

Option 2:
$500\ \mathrm{ms^{-1}}$

Option 3:
$250\ \mathrm{ms^{-1}}$

Option 4:
$600\ \mathrm{ms^{-1}}$

Correct Answer:
$500\ \mathrm{ms^{-1}}$

Solution:

Let initial speed $= \mathrm{u}$

$$\underset{127^{\circ}C}{\text{bullet}} \xrightarrow{Q_1} \underset{327^{\circ}C}{\text{bullet}} \xrightarrow{Q_2} \underset{327^{\circ}C}{\text{bullet melts}}$$

total heat required.

$$\begin{aligned}\mathrm{Q} &= \mathrm{Q_1} + \mathrm{Q_2} \\ &= \mathrm{ms\Delta T} + \mathrm{mL_f} \\ &= \mathrm{m}\left[125 \times 200 + 2.5 \times 10^4\right]\end{aligned}$$

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Given,

$40\%\,\mathrm{KE} = \mathrm{Q}$

$\frac{40}{100} \times \frac{1}{2}\mathrm{mu}^2 = \mathrm{m} \times \left[250 \times 100 + 2.5 \times 10^4\right]$

$\frac{2}{5} \times \frac{\mathrm{u}^2}{2} = 5 \times 10^4$

$\mathrm{u}^2 = 25 \times 10^4$

$\mathrm{u} = 5 \times 10^2 \mathrm{m/s}$

The correct answer is (2)

Q. 10 The equation of a particle executing simple harmonic motion is given by $\mathrm{x} = \sin \pi \left(\mathrm{t} + \frac{1}{3}\right) \mathrm{m}.$ At $\mathrm{t} = 1\mathrm{s}$, the speed of particle will be

(Given : $\pi = 3.14$)-

Option 1:

$0\ \mathrm{cm\,s^{-1}}$

Option 2:

$157\ \mathrm{cm\,s^{-1}}$

Option 3:

$272\ \mathrm{cm\,s^{-1}}$

Option 4:

$314\ \mathrm{cm\,s^{-1}}$

Correct Answer:

$157\ \mathrm{cm\,s^{-1}}$

Solution:

$\mathrm{x} = \sin\left(\pi \mathrm{t} + \frac{\pi}{3}\right)$

Speed of particle.

$\mathrm{v} = \frac{\mathrm{dx}}{\mathrm{dt}} = \left[\cos\left(\pi \mathrm{t} + \frac{\pi}{3}\right)\right] \times \pi$

at $\mathrm{t} = 1$ sec.

$\mathrm{v} = \pi \times \cos\left(\pi + \frac{\pi}{3}\right)$

$= \pi \times \left(-\cos\frac{\pi}{3}\right)$

$= -\pi \times \frac{1}{2} = \frac{-3.14}{2}\,\mathrm{m/s}$

$\mathrm{speed} = 157\,\mathrm{cm/s}$

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The correct answer is (2)

Q. 11 If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be :

Option 1:

$\frac{q}{\epsilon_0}$

Option 2:

$\frac{q}{2\ \epsilon_0}$

Option 3:

$\frac{q}{4\ \epsilon_0}$

Option 4:

$\frac{q}{2\pi\ \epsilon_0}$

Correct Answer:

$\frac{q}{2\ \epsilon_0}$

Solution:

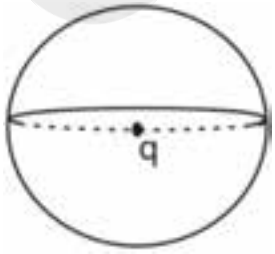
Using Gauss's law:

$$\text{Total flux } (\phi) = \frac{q_{enc}}{\epsilon_0}$$

$$\phi_{\text{hemisphere}} = \phi_{\text{curve}} + \phi_{\text{flat}}$$

$$\frac{q}{\epsilon_0} = \phi_{\text{curve}} + \phi_{\text{flat}} \ \ldots(1)$$

Consider complete sphere charge kept at centre flux through hemispherical part will be $\frac{q}{2\ \epsilon_0}$



$\therefore$ from equation (1)

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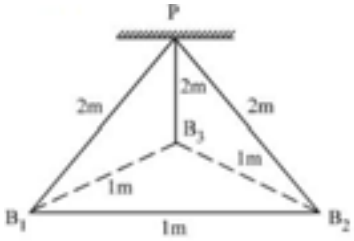
$$\phi_{\text{flat}} = \frac{q}{\epsilon_0} - \frac{q}{2\,\epsilon_0} = \frac{q}{2\,\epsilon_0}$$

The correct answer is (2)

Q. 12 Three identical charged balls each of charge $2\,\text{C}$ are suspended from a common point P by silk threads of $2\,\text{m}$ each (as shown in figure). They form an equilateral triangle of side $1\,\text{m}$. The ratio of net force on a charged ball to the force between any two charged balls will be :



Option 1:

$1 : 1$

Option 2:

$1 : 4$

Option 3:

$\sqrt{3} : 2$

Option 4:

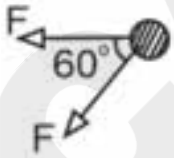
$\sqrt{3} : 1$

Correct Answer:

$\sqrt{3} : 1$

Solution:

The electric force on charge kept at B_1



$$\begin{aligned}\text{F}_{\text{net}} &= \sqrt{\text{F}^2 + \text{F}^2 + 2\text{F} \times \text{F} \times \cos 60^\circ} \\ &= \sqrt{3}\text{F}\end{aligned}$$

Where F is the electrostatic force between any two charge

$$\frac{\text{F}_{\text{net}}}{\text{F}} = \frac{\sqrt{3}\text{F}}{\text{F}} = \sqrt{3} : 1$$

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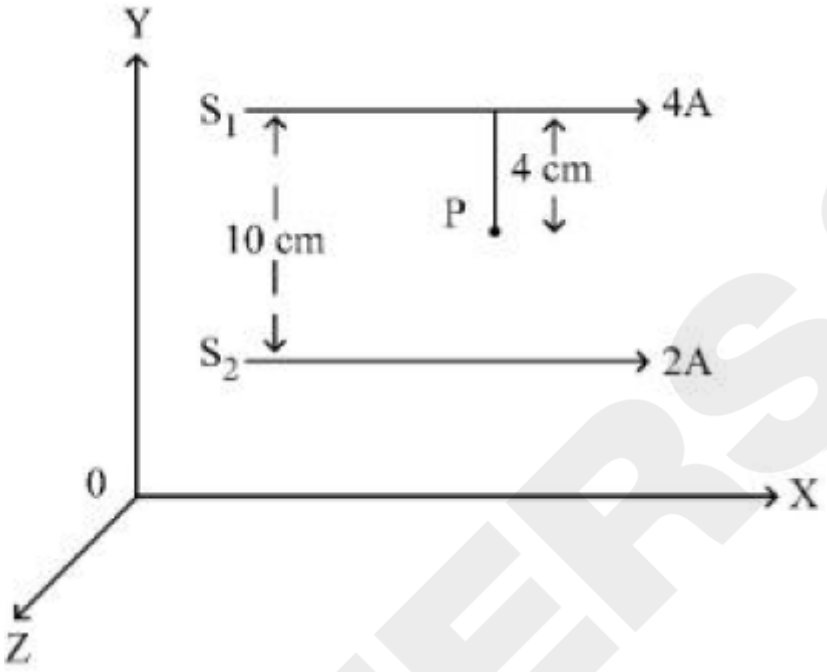
The correct answer is (4)

Q. 13 Two long parallel conductors S_1 and S_2 are separated by a distance $10\ cm$ and carrying currents of $4\ A$ and $2\ A$ respectively. The conductors are placed along $x-$ axis in $X-Y$ plane. There is a point P located between the conductors (as shown in figure).
A charge particle of 3π coulomb is passing through the point P with velocity $\vec{v}=(2\hat{i}+3\hat{j})m/s$; where $\hat{i}_i \& \hat{j}$ represents unit vector along $x \& y$ axis respectively.
The force acting on the charge particle is $4\pi \times 10^{-5}(-x\hat{i}+2\hat{j})N$. The value of x is :



Option 1:
2

Option 2:
1

Option 3:
3

Option 4:
-3

Correct Answer:
3

Solution:

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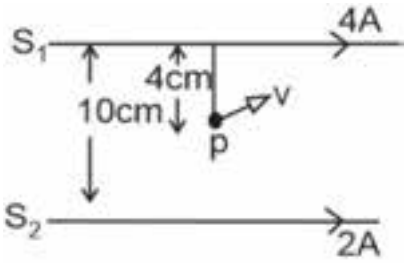

Charge particle will experience a force due to magnetic field produced by current-carrying wired using Lorentz force equation

$$\vec{F} = q\left(\vec{v} \times \vec{B}\right)$$

To calculate the magnetic field at 'P'

$$\vec{B} = \mathrm{B}_1\left(-\hat{\mathrm{k}}\right) + \mathrm{B}_2\hat{\mathrm{k}}$$

$$= \frac{\mu_0 \times 4}{2\pi \times 4 \times 10^{-2}}\left(-\hat{\mathrm{k}}\right) + \frac{\mu_0 \times 2}{2\pi \times 6 \times 10^{-2}}\left(\hat{\mathrm{k}}\right)$$

$$= \frac{\mu_0}{\pi} \times 10^2\left[\frac{-1}{2} + \frac{1}{6}\right]\hat{\mathrm{k}}$$

$$\vec{B} = \frac{\mu_0}{3\pi} \times 10^2\left(-\hat{\mathrm{k}}\right)$$

$$\vec{F} = 3\pi\left[\left(2\hat{\mathrm{i}} + 3\hat{\mathrm{j}}\right) \times \left(\frac{\mu_0}{3\pi} \times 10^{+2}\right)\left(-\hat{\mathrm{k}}\right)\right]$$

$$= 10^{+2} \times \mu_0\left[2\hat{\mathrm{j}} - 3\hat{\mathrm{i}}\right]$$

$$= 4\pi \times 10^{-7} \times 10^2\left[2\hat{\mathrm{j}} - 3\hat{\mathrm{i}}\right]$$

$$= 4\pi \times 10^{-5} \times \left[-3\hat{\mathrm{i}} + 2\hat{\mathrm{j}}\right]$$

$$\mathrm{x} = 3$$

The correct answer is (3)

Q. 14 If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

Option 1:

RC

Option 2:

$\frac{\mathrm{L}}{\mathrm{R}}$

Option 3:

$\sqrt{\mathrm{LC}}$

Option 4:

$\frac{\mathrm{L}}{\mathrm{C}}$

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Correct Answer:
$\frac{L}{C}$

Solution:
$RC \rightarrow$ time constant for R-C circuit

$\frac{L}{R} \rightarrow$ time constant for L-R circuit

$\sqrt{LC} \rightarrow$ angular frequency for L-C oscillation

only $\frac{L}{C}$ does not have the dimension of time

The correct answer is (4)

Q. 15 Given below are two statements :
Statement I: A time-varying electric field is a source of changing magnetic fields and vice-versa. Thus a disturbance in an electric or magnetic field creates EM waves.
Statement II: In a material medium, the EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 \in_0}}$.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both statement I and statement II are true

Option 2:
Both statement I and statement II are false

Option 3:
Statement I is correct but statement II is false

Option 4:
Statement I is incorrect but statement II is true

Correct Answer:
Statement I is correct but statement II is false

Solution:
Statement I is correct as a time-varying electric field is a source of changing magnetic fields and vice-versa as per the Maxwell's equations.

In the material medium speed of light wave changes.
$v = \frac{c}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$

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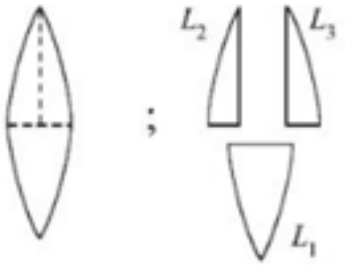
For vacuum, the speed of light wave is

$$v = \frac{1}{\sqrt{\mu_o \epsilon_o}}$$

The correct answer is (3)

Q. 16 A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figures). Choose the incorrect option for the reported pieces.



Option 1:

$$\text{Power of } L_1 = \frac{P}{2}$$

Option 2:

$$\text{Power of } L_2 = \frac{P}{2}$$

Option 3:

$$\text{Power of } L_3 = \frac{P}{2}$$

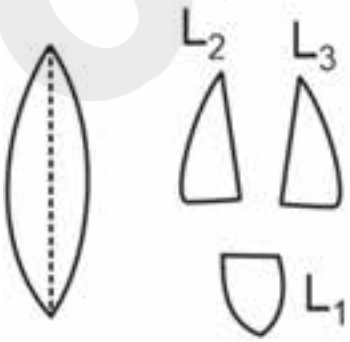
Option 4:

$$\text{Power of } L_1 = P$$

Correct Answer:

$$\text{Power of } L_1 = \frac{P}{2}$$

Solution:



Len's maker's formula.

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$$P = \frac{1}{f} = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For convex Lens

$R_1 = +R,\ R_2 = -R$

$$P = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{2}{R}\right).$$

for $L_1, R_1 \& R_2$ remain the same as the original Lens

$\therefore$ even after cutting,

L_1 , power will be the same as P. (original Lens). & for L_2 & L_3 power become half.

The correct answer is (1)

Q. 17 If wave gets refracted into a denser medium, then which of the following is true?

Option 1:

Wavelength, speed and frequency decreases.

Option 2:

Wavelength increases, speed decreases and frequency remains constant..

Option 3:

Wavelength and speed decreases but frequency remains constant.

Option 4:

Wavelength,speed and frequency increases.

Correct Answer:

Wavelength and speed decreases but frequency remains constant.

Solution:

$V = \lambda f \qquad V = \frac{C}{\mu}$

When the wave goes from rarer to denser medium speed decreases, the frequency remains the same, and therefore wavelength decreases.

As the number of waves per unit from one medium to another medium remains constant to maintain continuity.

$\therefore$The correct answer is (3)

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Q. 18 Given below are two statements:

Statement I: In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_1 - E_2$

Statement II: The jumping of electron from higher energy orbit (E_2) to lower cnergy orbit (E_1) is associated with frequency of radiation given as $f = (E_2 - E_1)/h$

This condition is Bohr's frequency condition.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I and statement II are true.

Option 2:
Both statement I and statement II are false.

Option 3:
Statement I is correct but Statement II is false.

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Statement I is incorrect but Statement II is true.

Solution:
When an electron jumps from a lower energy state to a higher energy state, then energy will be required radiation will be absorbed.
hence Statement - I is wrong
Statement II is correct

The correct answer is (4)

Q. 19 For a transistor to act as a switch, it must be operated in

Option 1:
Active region.

Option 2:
Saturation state only.

Option 3:
Cut-off state only.

Option 4:
Saturation and cult-off state.

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Correct Answer:
Saturation and cult-off state.

Solution:
For a transistor as a switch, it operates in saturation and cut-off region.

For amplifier it perates in an active state

The correct answer is (4)

Q. 20 We do not transmit low frequency signal to long distances because-
(a) The size of the antenna should be comparable to signal wavelength which is unreal solution for a signal of longer wavelength.
(b) Effective power radiated by a long wavelength baseband signal would be high.
(c) We want to avoid mixing up signals transmitted by different transmitter simultaneously.
(d) Low frequency signal can be sent to long distances by superimposing with a high frequency wave as well.
Therefore, the most suitable option will be :

Option 1:
All statements are true

Option 2:
$(\mathrm{a}), (\mathrm{b})$ and (c) are true only

Option 3:
$(\mathrm{a}), (\mathrm{c})$ and (d) are true only

Option 4:
$(\mathrm{b}), (\mathrm{c})$ and (d) are true only

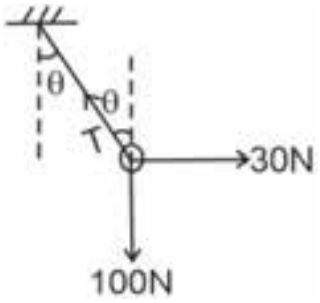
Correct Answer:
$(\mathrm{a}), (\mathrm{c})$ and (d) are true only

Solution:
- The size of the antenna should be comparable with the wavelength of the wave.
- If the frequency is very small, the wavelength will be large (in km)
- So antenna size will be in Km which is not feasible
- The correct answer is (3)

Q. 21 A mass of $10\ \mathrm{kg}$ is suspended vertically by a rope of length $5\ \mathrm{m}$ from the roof. A force of $30\ \mathrm{N}$ is applied at the middle point of rope in horizontal direction. The angle made by upper half of the rope with vertical is $\theta = \tan^{-1}\left(\mathrm{x} \times 10^{-1}\right)$. The value of x is_____.
(Given, $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Correct Answer:
3

Solution:

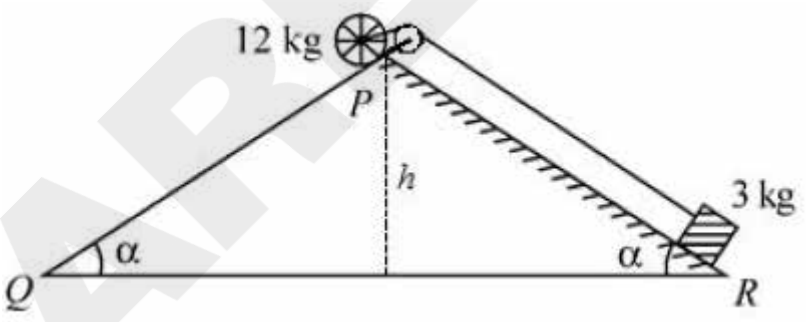


$$\mathrm{T}\cos\theta = 100 \ ---(1)$$
$$\mathrm{T}\sin\theta = 30 \ ---(2)$$

eqn (2) / eqn (1)

$$\tan\theta = \frac{30}{100} = \frac{3}{10}$$
$$\theta = \tan^{-1}\left[3 \times 10^{-1}\right]$$
$$\mathrm{x} = 3$$

Q. 22 A rolling wheel of $12\ \mathrm{kg}$ is on an inclined plane at position P and connected to a mass of $3\ \mathrm{kg}$ through a string of fixed length and pulley as shown in figure.
Consider PR as friction free surface.
The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{\mathrm{xgh}}\ \mathrm{m/s}$. The value of x is _____.



Correct Answer:
3

Solution:
By mechanical energy conservation,

$$12\mathrm{gh} = 3\mathrm{gh} + \frac{1}{2} \times 3 \times \mathrm{v}^2 + \frac{1}{2} \times 12 \times \mathrm{v}^2 + \frac{1}{2} \times \mathrm{I}\frac{\mathrm{v}^2}{\mathrm{R}^2}$$
$$\mathrm{I} = 12 \times \mathrm{R}^2$$

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$$9gh = \frac{3v^2}{2} + \frac{12v^2}{2} + \frac{12v^2}{2}$$
$$9gh = \frac{27}{2}v^2$$

$$v^2 = \frac{2}{3}gh$$
$$v = \sqrt{\frac{2}{3}gh} = \frac{1}{2} \times \sqrt{\frac{8}{3}gh}$$
$$x = 2.67$$

The correct answer is (3)

Q. 23 A diatomic gas $(\gamma = 1.4)$ does $400\,J$ of work when it is expanded isobarically. The heat given to the gas in the process is _____ J.

Correct Answer:
1400

Solution:
$Q = \Delta U + W$, $W = 400\,J$
For Isobaric process,

$$Q = nC_p\Delta T$$
$$\Delta U = nC_v\Delta T$$

$$\frac{Q}{\Delta U} = \gamma = 1.4$$
$$Q = \frac{Q}{1.4} + 400$$
$$Q\left(1 - \frac{1}{1.4}\right) = 400, Q = \frac{400 \times 1.4}{0.4}$$
$$Q = 1400J$$

Q. 24 A particle executes simple harmonic motion. Its amplitude is $8\ cm$ and time period is $6\ s$. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is _____ s .

Correct Answer:
1

Solution:

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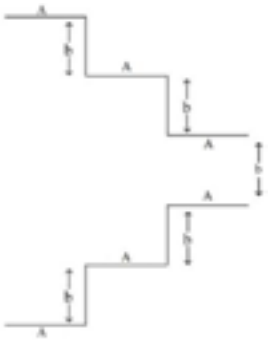
Time taken from the extreme position to $X = \frac{A}{2}$ is $\frac{T}{6}$

$t = \frac{T}{6} = \frac{6}{6} = 1\,\text{second}$

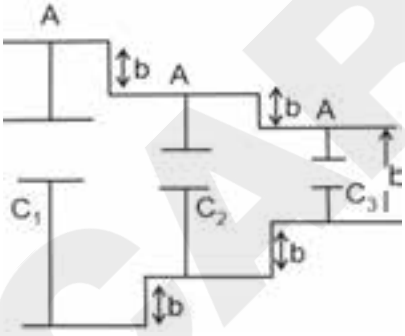
The correct answer is (1)

Q. 25 A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b, as shown in the figure. The capacitance of the arrangement is $\frac{x}{15}\frac{\epsilon_0 A}{b}$. The value of x is _____.



Correct Answer:
23

Solution:



$C_1 = \frac{\epsilon_0 A}{5b}$

$C_2 = \frac{\epsilon_0 A}{3b}$

$C_3 = \frac{\epsilon_0 A}{b}$

$C_1, C_2 \,\&\, C_3$ are in parallel

$\because\ C_{eq} = C_1 + C_2 + C_3$

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$$= \frac{\epsilon_0 A}{5b} + \frac{\epsilon_0 A}{3b} + \frac{\epsilon_0 A}{b}$$
$$= \frac{23}{15}\frac{\epsilon_0 A}{b}$$

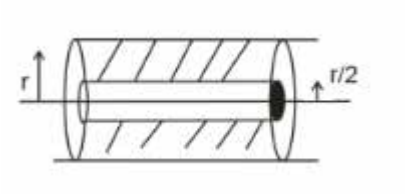
The answer is 23

Q. 26 The current density in a cylindrical wire of radius $r = 4.0\ mm$ is $1.0 \times 10^6\ A/m^2$. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi A$; where x is _____.

Correct Answer:
12

Solution:

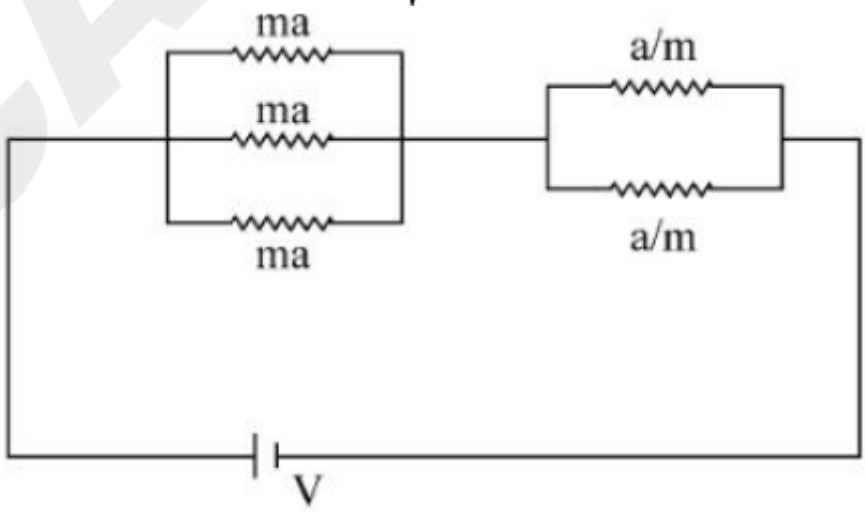
$$J = \frac{I}{A}$$



$$I = \left[J \times \left[\pi r^2 - \frac{\pi r^2}{4}\right]\right] = J \times \frac{3\pi r^2}{4}$$
$$= 1 \times 10^6 \times \frac{3}{4} \times \pi \times 16 \times 10^{-6}$$
$$= 12\pi$$

The answer is 12

Q. 27 In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____.



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Correct Answer:
3

Solution:

$$\begin{aligned}\text{Req} &= \frac{\text{ma}}{3} + \frac{(\text{a/m})}{\text{a}^2} \\ &= \frac{\text{ma}}{3} + \frac{\text{a}^2}{2\text{m}} \\ &= \frac{2\text{m}^2\text{a} + 3\text{a}}{6\text{m}} \\ &= \frac{\text{ma}}{3} + \frac{\text{a}}{2\text{m}}\end{aligned}$$

For R_{eq} to be minimum,

$$\frac{\text{dReq}}{\text{dm}} = 0$$

$$\frac{\text{a}}{3} - \frac{\text{a}}{2\text{m}^2} = 0$$

$$\frac{\text{a}}{3} = \frac{\text{a}}{2\text{m}^2} \Rightarrow \text{m}^2 = \frac{3}{2}$$

$$\text{m} = \sqrt{\frac{3}{2}}$$

The answer is (3)

Q. 28 A deuteron and a proton moving with equal kinetic energy enter into to a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{\text{r}_\text{d}}{\text{r}_\text{p}}$ will be $\sqrt{\text{x}} : 1$ where x is _____.

Correct Answer:
2

Solution:

$$\text{r} = \frac{\text{mV}}{\text{qB}} = \sqrt{\frac{2\text{mK}}{\text{qB}}}$$

If kinetic energy is the same

$$\text{r} \propto \sqrt{\frac{\text{m}}{\text{q}}}$$

$$\frac{\text{r}_\text{d}}{\text{r}_\text{p}} = \frac{\sqrt{\frac{\text{m}_\text{d}}{\text{q}_\text{d}}}}{\sqrt{\frac{\text{m}_\text{p}}{\text{q}_\text{p}}}} = \sqrt{\frac{\text{m}_\text{d}}{\text{m}_\text{p}}} \times \sqrt{\frac{\text{q}_\text{p}}{\text{q}_\text{d}}}$$

$$= \sqrt{\frac{2}{1}} \times \sqrt{\frac{1}{1}} = \sqrt{2} : 1$$

Ans : (2)

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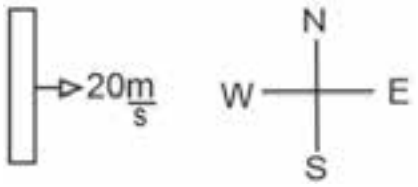
Q. 29 A metallic rod of length $20\ \text{cm}$ is placed in North-South direction and is moved at a constant speed of $20\ \text{m/s}$ towards East. The horizontal component of the Earth's magnetic field at that place is $4 \times 10^{-3}\ \text{T}$ and the angle of dip is $45°$. The emf induced in the rod is _____mV.

Correct Answer:
16

Solution:

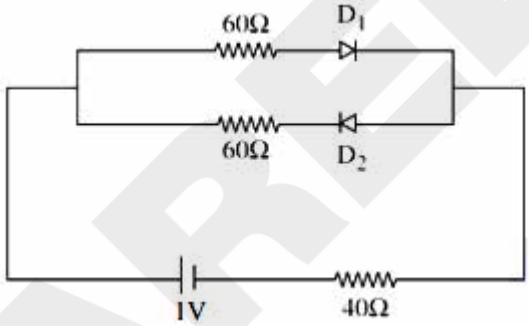


$\text{B}_\text{V} = \text{B}_\text{H}$
Vertical component (B_V) will produced motional emf.

$$\begin{aligned}\text{E} &= \text{B}_\text{V} \times \text{V} \times l \\ &= 4 \times 10^{-3} \times 20 \times 20 \times 10^{-2} \\ &= 16 \times 10^{-3} = 16\,\text{mV}\end{aligned}$$

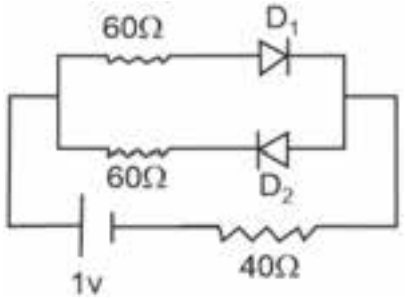
The answer is 16

Q. 30 The cut-off voltage of the diodes (shown in the figure) in forward bias is $0.6\ \text{V}$. The current through the resister of $40\ \Omega$ is_____ mA.



Correct Answer:
4

Solution:



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Since D_2 in reverse
therefore, there will be no current in D_2.
Voltage in $D_1 = 0 \cdot 6\,V$.
Remaining $1 - 0.6 = 0.4\,V$ will cause current in $(60 + 40)\,\Omega$:

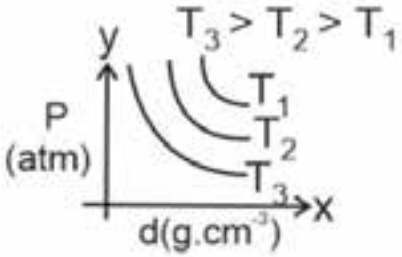
$$\therefore\ i = \frac{0.4}{100} = 4\,mA$$

Ans : (4)

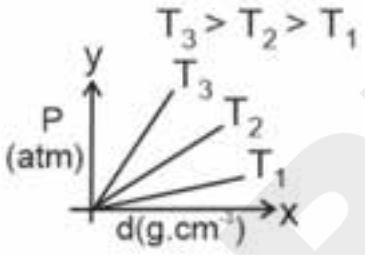
Chemistry

Q. 1 Which amongst the given plot is the correct plot for pressure (p) vs density (d) for an ideal gas?

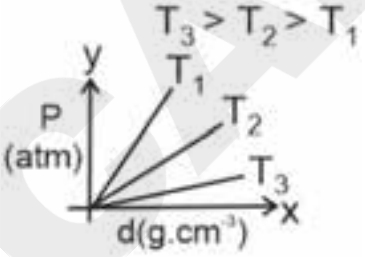
Option 1:



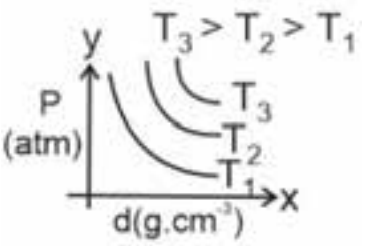
Option 2:



Option 3:



Option 4:



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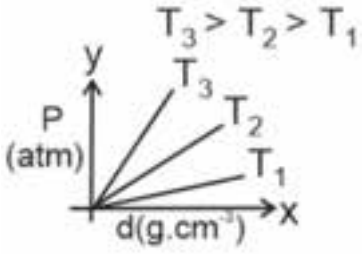
Correct Answer:



Solution:
From ideal gas equation,

$$PV = nRT$$

$$\Rightarrow P = \frac{dRT}{M_0}$$

$$\Rightarrow P = \left(\frac{RT}{M_0}\right) d$$

$\therefore$ P vs d will be a straight line passing through origin.

Also, $\text{Slope} \propto T$

So, greater the temperature, greater is the slope of the plot.

Hence, the answer is **Option (2)**

Q. 2 Identify the **incorrect** statement for PCl_5 from the following.

Option 1:
In this molecule, orbitals of phosphorous are assumed to undergo $sp^3\ d$ hybridization.

Option 2:
The geometry of PCl_5 is trigonal bipyramidal.

Option 3:
PCl_5 has two axial bonds stronger than three equatorial bonds.

Option 4:
The three equatorial bonds of PCl_5 lie in a plane.

Correct Answer:
PCl_5 has two axial bonds stronger than three equatorial bonds.

Solution:
The structure of PCl_5 is

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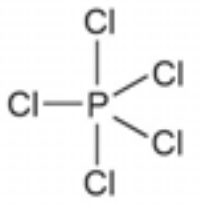

P is $\mathrm{sp^3d}$ hybridised.

The two axial bonds are longer and weaker than the equatorial bonds.

Hence, statement in option (3) is in correct.

Thus, answer is **Option (3)**

Q. 3 **Statement I:** Leaching of gold with cyanide ion in absence of air $/\mathrm{O_2}$ leads to cyano complex of Au(III).

Statement II: Zine is oxidized during the displacement reaction carried out for gold extraction.

In the light of the above statement, choose the **correct** answer from the options given below.

Option 1:
Both statement I and staement II are correct.

Option 2:
Both statement I and statement II are incorrect.

Option 3:
Statement I is correct but statement II is incorrect.

Option 4:
Statement I is incorrect but statement II is correct.

Correct Answer:
Statement I is incorrect but statement II is correct.

Solution:
In leaching of Gold, the complex formed is $\mathrm{[Au(CN)_2]^-}$which contains Au (I) and the complex is obtained in presence of air.

Statement I is not correct.

Reduction of the dicyonoaurum(I) complex is carried out by Zn wherein Zn is oxidised to $\mathrm{Zn^{+2}}$.

Statement II is correct

Hence, the answer is **Option (4)**

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Q. 4 The correct order of increasing intermolecular hydrogen bond strength is

Option 1:
$HCN < H_2O < NH_3$

Option 2:
$HCN < CH_4 < NH_3$

Option 3:
$CH_4 < HCN < NH_3$

Option 4:
$CH_4 < NH_3 < HCN$

Correct Answer:
$CH_4 < HCN < NH_3$

Solution:
The correct order of intermolecular forces is

$$CH_4 < HCN < NH_3 < H_2O$$

Hence, the correct answer is **Option (3)**

Q. 5 The correct order of increasing ionic radii is

Option 1:
$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Option 2:
$N^{3-} < O^{2-} < F^{-} < Na^{+} < Mg^{2+}$

Option 3:
$F^{-} < Na^{+} < O^{2-} < Mg^{2+} < N^{3-}$

Option 4:
$Na^{+} < F^{-} < Mg^{2+} < O^{2-} < N^{3-}$

Correct Answer:
$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Solution:
In isoelectronic species, the ionic radii decreases with the increase in positive change on the ion

Thus, the order of ionic radii is

$$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$$

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Hence, answer is **Option (1)**

Q. 6 The gas produced by treating an aqueous solution of ammoninum chloride with sodium nitrite is

Option 1:
$\mathrm{NH_3}$

Option 2:
$\mathrm{N_2}$

Option 3:
$\mathrm{N_2O}$

Option 4:
$\mathrm{Cl_2}$

Correct Answer:
$\mathrm{N_2}$

Solution:
Ammonium Chloride reacts with Sodium nitrite and forms Ammonium nitrite which gives $\mathrm{N_2}$ gas.

$$\mathrm{NH_4Cl + NaNO_2 \rightarrow NH_4NO_2 \longrightarrow N_2 + H_2O}$$

Hence, the answer is **Option (2)**

Q. 7 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Flourine forms one oxoacid.

Reason R: Flourine has smallest size amongst all halogens and is highly electronegative.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

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Option 4:
A is not correct but **R** is correct.

Correct Answer:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Solution:
Fluorine forms only one oxoacid, HOF.

Higher oxoacids are not formed because F is smaller in size and also highly electronegative.

Thus, both assertion (A) and reason (R) are correct and (R) correctly explains (A).

Thus, **Option (1)** is correct.

Q. 8 In 3d series, the metal having the highest $M^{2+}|M$ standard electrode potential is

Option 1:
Cr

Option 2:
Fe

Option 3:
Cu

Option 4:
Zn

Correct Answer:
Cu

Solution:
As we have learnt,

Among the elements of the 3d series, only Copper has a positive value of $E^0_{M^{2+}|M}$.

Other metals have negative values of Reduction potential for the bivalent cation.

Hence, **Option (3)** is correct.

Q. 9 The 'f' orbitals are half and completely filled, respectively in lanthanide ions

$[\text{Given : Atomic no. Eu, 63; Sm, 62; Tm, 69; Tb, 65; Yb, 70; Dy, 66}]$

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Option 1:
Eu^{2+} and Tm^{2+}

Option 2:
Sm^{2+} and Tm^{3+}

Option 3:
Tb^{4+} and Yb^{2+}

Option 4:
Dy^{3+} and Yb^{3+}

Correct Answer:
Tb^{4+} and Yb^{2+}

Solution:
The ions having fully filled or half filled ' f ' orbital configuration are

Tb^{4+} : $[Xe]\ 4f^{7}$
Yb^{2+} : $[Xe]4f^{14}$

Hence, **Option (3)** is correct.

Q. 10 Arrange the following coordination compounds in the increasing order of magnetic moments.$(\text{Atomic numbers: } Mn = 25; Fe = 26\ )$

A. $[FeF_6]^{3-}$

B. $[Fe(CN)_6]^{3-}$

C. $[MnCl_6]^{3-}$ (high spin)

D. $[Mn(CN)_6]^{3-}$

Choose the correct answer from the options given below:

Option 1:
$A < B < D < C$

Option 2:
$B < D < C < A$

Option 3:
$A < C < D < B$

Option 4:
$B < D < A < C$

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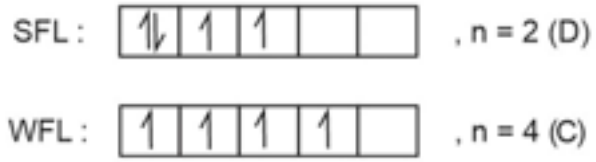
Correct Answer:
$B < D < C < A$

Solution:
Fe^{3+} has d^5 configuration

Its configuration is given below:

Mn^{+3} has d^4 configuration

Its configuration is given below:



Hence, the order of magnetic moment is

$(A) > (C) > (D) > (B)$

Hence, **Option (2)** is correct.

Q. 11 On the surface of polar stratospheric clouds, hydrolysis of chlorine nitrate gives A and B while its reaction with HCl produces B and C.

A,B and Care, respectively

Option 1:
$HOCl, HNO_3, Cl_2$

Option 2:
$Cl_2, HNO_3, HOCl$

Option 3:
$HClO_2, HNO_2, HOCl$

Option 4:
$HOCl, HNO_2, Cl_2O$

Correct Answer:
$HOCl, HNO_3, Cl_2$

Solution:

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Chlorine Nitrate on hydrolysis gives Nitric Acid and Hydochloric Acid.

$$ClONO_2 + H_2O \longrightarrow HNO_3 + HOCl$$

Also, upon rection with HCl, Chlorine Nitrate forms molecular Chlorine and Nitric Acid.

$$ClONO_2 + HCl \longrightarrow Cl_2 + HNO_3$$

$$\therefore A = HOCl, B = HNO_3, C = Cl_2$$

Hence, **Option (1)** is correct.

Q. 12 Which of the following is most stable?

Option 1:

Option 2:

Option 3:

Option 4:

Correct Answer:

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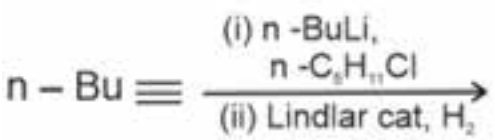
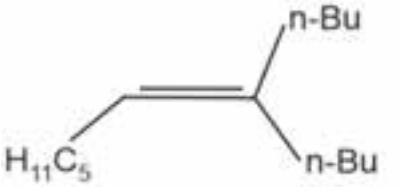
Solution:
Among the given species, ion in Option (1) is Aromatic. Hence, it is most stable.

Thus, answer is **Option (1).**

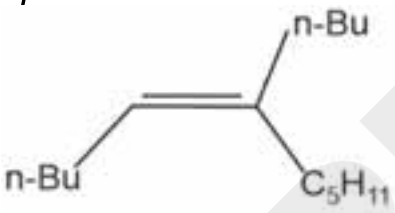
Q. 13 What will be the major product of following sequence of reactions?



Option 1:

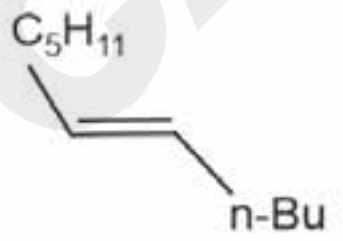


Option 2:



Option 3:

Option 4:



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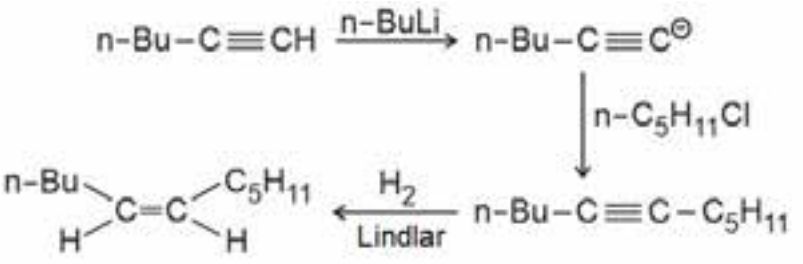
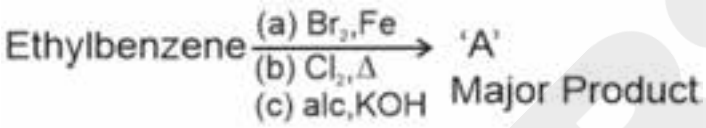
Correct Answer:

Solution:
The reaction sequence occurs as

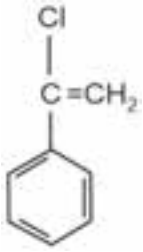


Hence, **Option (3)** is correct.

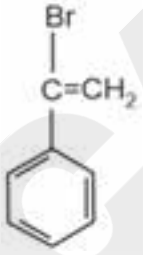
Q. 14 Product 'A' of following sequence of reactions is



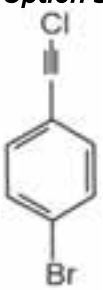
Option 1:



Option 2:



Option 3:



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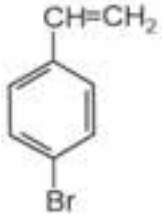
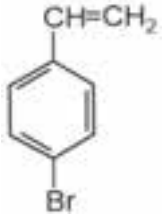
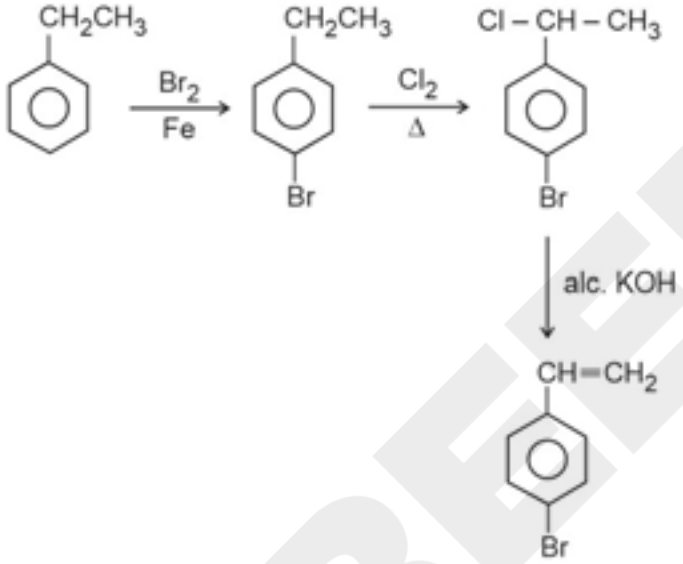
Option 4:



Correct Answer:



Solution:
The reaction sequence is given below



Hence, **Option (4)** is correct

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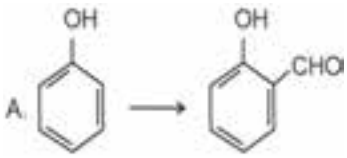
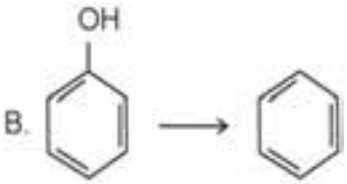
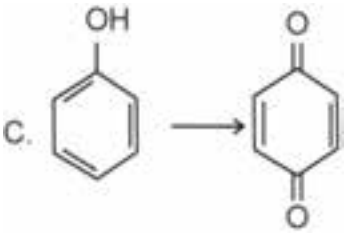
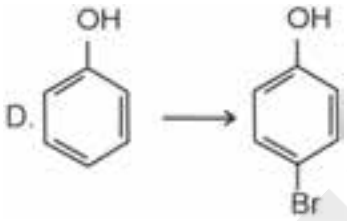
Q. 15 Match List I with List II.

List I	List II
A. 	I. Br_2 in CS_2
B. 	II. $Na_2Cr_2O_7/H_2SO_4$
C. 	III. Zn
D. 	IV. $CHCl_3/NaOH$

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-IV, B-III, C-I, D-II

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:
The correct match of the reaction and reagent is given below

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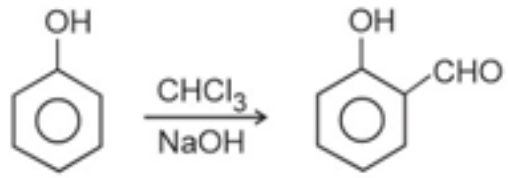
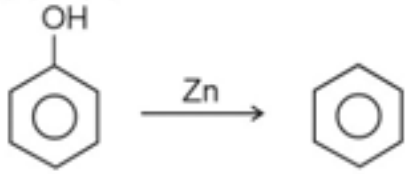
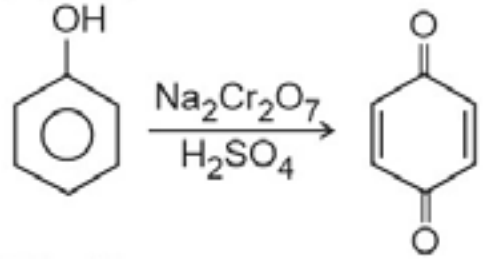
(A) – (iv)



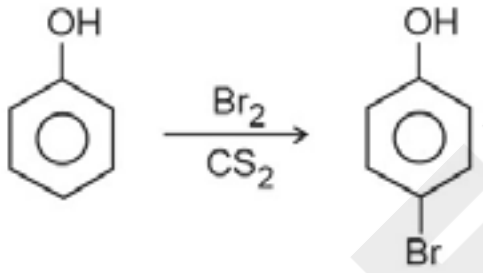
(B) – (iii)



(C) – (ii)



(D) – (i)



Hence, the **option(1)** is correct.

Q. 16 Decarboxylation of all six possible forms of diaminobenzoic acids $C_6H_3\ (NH_2)_2\ COOH$ yields three products A, B and C. Three acids give a product 'A', two acids gives a product 'B' and one acid give a product 'C' The melting point of product 'C' is

Option 1:

$63^{\circ}C$

Option 2:

$90^{\circ}C$

Option 3:

$104^{\circ}C$

Option 4:

$142^{\circ}C$

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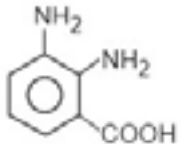
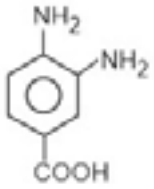
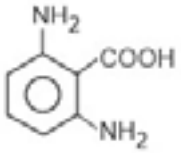
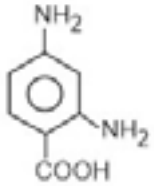
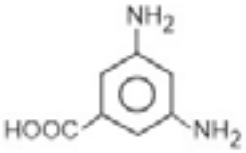
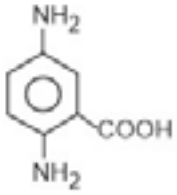
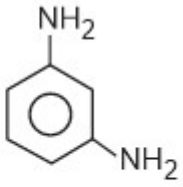
Correct Answer:

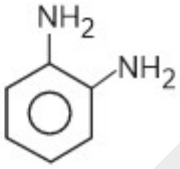
$142^{\circ}C$

Solution:

6 forms of diaminobenzoic acid are

$\therefore$ A is 

B is 

C is

C has a melting point of $142^{\circ}C$

Hence, **Option(4)** is correct.

Q. 17 Which is true about Buna-N?

Option 1:

It is a linear polymer of $1,3-$ butadiene.

Option 2:

It is obtained by copolymerization of $1,3-$butadiene and styrene.

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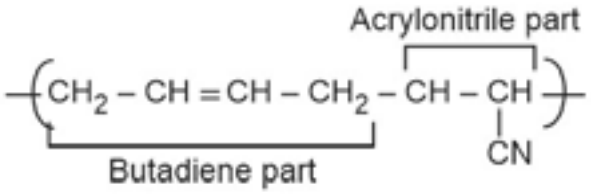
Option 3:
It is obtained by copolymerization of $1,3$—butadiene and acrylonitrile.

Option 4:
The suffix N in Buna-N stands for its natural occurrence.

Correct Answer:
It is obtained by copolymerization of $1,3$—butadiene and acrylonitrile.

Solution:
Buna - N is a copolymer of 1,3- Butadiene and Acrylonitrile.



Hence, option (3) is correct.

Q. 18 Given below are two statements.

Statement I: Maltose has two $\alpha-\mathrm{D}$—glucose units linked at C_1 and C_4 and is a reducing sugar.

Statement II: Maltose has two monosaccharides:
$\alpha-\mathrm{D}-\text{glucose}$ and $\beta-\mathrm{D}-\text{glucose}$ linked at C_1 and C_6 and it is a non-reducing sugar.

In the light of the above statements, choose the **correct** answer from the options given below.

Option 1:
Both Statment I and Statement II are true.

Option 2:
Both Statment I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Maltose is a disaccharide of $\alpha-\mathrm{D}-\text{Glucose}$ with $\mathrm{C}_1-\mathrm{C}_4$ linkage.

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Thus, statement I is true and II is false.

Hence, **Option (3)** is correct.

Q. 19 Mach List I with List II.

List I	List II
A. Antipyretic	I. Reduces pain
B. Analgesic	II.Reduces stress
C. Tranquilizer	III. Reduces fever
D. Antacid	IV. Reduces acidity(stomach)

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-I, B-IV, C-II, D-III

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-III, B-I, C-II, D-IV

Solution:
The correct matching of the drug and its use are given below:

(A) Antipyretic - (III) Reduces fever
(B) Analgesic - (I) Reduces Pain
(C) Tranquiliser - (II) Reduces Stress
(D) Antacid - (IV) Reduces Acidity

Hence, **Option (1)** is correct.

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Q. 20 Match **List I** with **List II.**

List I (Anion)	**List II** (gas evolved on reaction with dil. H_2SO_4)
A. CO_3^{2-}	I. Colourless gas which turns lead acetate acetate paper black.
B. S^{2-}	II. Colourless gas which turns acidified potassium dichromate solution green.
C. SO_3^{2-}	III.Brown fumes which turns acidified KI solution containing starch blue.
D. NO_2^-	IV. Colourless gas evolved with brisk effervescence, which turns lime water milky.

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-II, B-I, C-IV, D-III

Option 3:
A-IV, B-I, C-III, D-II

Option 4:
A-IV, B-I, C-II, D-III

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
The correct matching of the Anion and the gas evolved on reaction with dil. H_2SO_4 is

$(A)CO_3{}^{2-}$ — (IV) Colourless gas evolved with brisk effervescence, which turns lime water milky.

$(B)S^{2-}$ — (I) Colourless gas which turns lead acetate acetate paper black

$(C)SO_3{}^{2-}$ — (II) Colourless gas which turns acidified potassium dichromate solution green.

$(D)NO_2^-$ — (IV) Colourless gas evolved with brisk effervescence, which turns lime water milky.

Hence, **Option (4)** is correct.

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Q. 21 116g of a substance upon dissociation reaction, yields 7.5g of hydrogen, 60g of oxygen and 48.5g of carbon. Given that the atomic masses of H,O and C are 1,16 and 12, respectively. The data agrees with how many formulae of the following?

_____.

$\mathrm{A.CH_3COOH}$ $\mathrm{B.HCHO}$ $\mathrm{C.CH_3OOCH_3}$ $\mathrm{D.CH_3CHO}$

Correct Answer:
2

Solution:
In 116 g of the compound ,

$$\text{moles of H} = 7.5$$

$$\text{moles of O} = \frac{60}{16} = 3.75$$

$$\text{moles of C} = \frac{48.5}{12} = 4.04$$

$$\therefore \text{Approximately, C:H:O} = 1:2:1$$

$$\therefore \text{Empirical formula} = \mathrm{CH_2O}$$

Thus,molecules (A) and (B) have this empirical formula.

Answer is 2

Q. 22 Consider the following set of quantum numbers.

	n	l	m_l
A.	3	3	-3
B.	3	2	-2
C.	2	1	$+1$
D.	2	2	$+2$

The number of correct sets of quantum numbers is _____.

Correct Answer:
2

Solution:
As we have learnt,

ℓ can take values from 0 to $(\mathrm{n}-1)$

m can take values from $(-\ell)$ to $(+\ell)$

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Therefore, the permissible set of Quantum numbers are

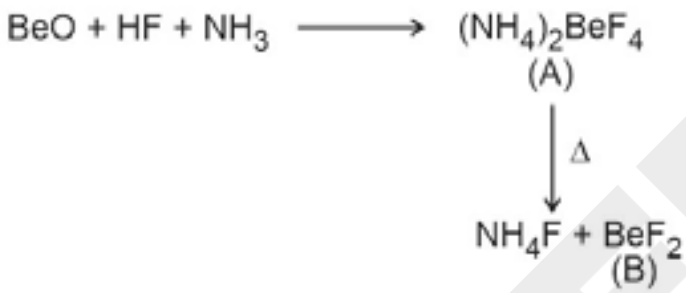
	n	l	m
(B)	3	2	−2
(C)	2	1	+1

Answer is 2

Q. 23 BeO reacts with HF in presence of ammonia to give [A] which on thermal decomposition produces [B] and ammonium fluoride. Oxidation state of Be in [A] is_____.

Correct Answer:
2

Solution:
The reaction is given as



Throughout the course of reaction, Be has an oxidation state of +2.

Answer is 2

Q. 24 When 5 moles of He gas expand isothermally and reversibly at 300 K from 10 liter to 20 liter, the magnitude of the maximum work obtained is _____J. [nearest integer]

$$(\text{Given : } \mathrm{R} = 8.3 \text{ J K}^{-1} \text{ mol}^{-1} \text{ and } \log 2 = 0.3010 \text{ })$$

Correct Answer:
8630

Solution:
Given,
$\mathrm{n}=5$
$\mathrm{T}=300 \text{ K}$
$\mathrm{V}_1 = 10$
$\mathrm{V}_2 = 20$

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Isothermal work $= -\mathrm{nRT}\ln\left(\frac{\mathrm{V_2}}{\mathrm{V_1}}\right)$

$= -5 \times 8.3 \times 300 \times \ln 2$

$= -5 \times 8.3 \times 300 \times 2.303 \times 0.3010$

$= -8630.37$

Thus, magnitude of work is 8630 J when rounded off to the nearest integer.

Answer is 8630

Q. 25 A solution containing 2.5×10^{-3} kg of a solute dissolved in 75×10^{-3} kg of water boils at 373.535 K. The molar mass of the solute is _____ $\mathrm{g\ mol^{-1}}$.[nearest integer]
$(\text{Given : } \mathrm{K_b\,(H_2O) = 0.52\ K\ kg\ mol^{-1}}$ and
boiling point of water $= 373.15$ K)

Correct Answer:
45

Solution:
Given,

$\Delta \mathrm{T_b} = 0.385\ \mathrm{K}$

$\Rightarrow \quad \mathrm{K_b \cdot m} = 0.385$

$\Rightarrow \quad \mathrm{m} = \frac{0.385}{\mathrm{kb}} = \frac{0.300}{0.52} = 0.74$

Now, using expression for Molality, we have

$\mathrm{m} = 0.74 = \frac{\mathrm{n}}{75 \times 10^{-3}} = \frac{2.5}{\mathrm{M_0} \times 75 \times 10^{-3}}$

$\Rightarrow 0.74 = \frac{10^2}{3\mathrm{M_0}}$

$\Rightarrow \mathrm{M_0} = \frac{100}{3 \times 0.74} = 45.04$

Hence, the answer is 45

Q. 26 pH value of 0.001 M NaOH solution is_____.

Correct Answer:
11

Solution:
Given

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$$\left[OH^{-}\right] = [NaOH\,] = 10^{-3}$$

$$\therefore pOH = 3.$$

$$\therefore pH = 11$$

Hence, answer is 11

Q. 27 For the reaction taking place in the cell:

$$Pt(s)\left|H_2(g)|H^{+}(aq)||Ag^{+}(aq)\right|Ag(s)$$

$$E^{\circ}_{cell} = +0.5332\ V.$$

The value of $\Delta_f G^{\Theta}$ for the reaction given below is _____$kJ\ mol^{-1}$.(in nearest integer)

Reaction: $\frac{1}{2}H_2 + Ag^{+} \longrightarrow H^{+} + Ag$

Correct Answer:
51

Solution:

$$\Delta G^{\circ} = -n\ f\ E^{\circ}$$

$$= -1 \times 96500 \times 0.5332$$

$$= -51453\ Jmol^{-1}$$

$$= -51.453\ kJ\ mol^{-1}$$

Answer is 51

Q. 28 It has been found that for a chemical reaction with rise in temperature by 9 K the rate constant gets doubled. Assuming a reaction to be occurring at 300 K, the value of activation enegy is found to be _____$kJ\ mol^{-1}$.[nearest integer]

$$(Given\ \ \ln 10 = 2.3, R = 8.3\ J\ K^{-1}\ mol^{-1}, \log 2 = 0.30)$$

Correct Answer:
59

Solution:

$$K_{300} = A\exp\left(\frac{-Ea}{300R}\right)$$

$$K_{309} = A\exp\left(\frac{-Ea}{309R}\right)$$

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Given,

$$K_{309} = 2\,K_{300}$$

$$\exp\left(\frac{-Ea}{309R}\right) = 2\exp\left(\frac{-Ea}{300R}\right)$$

$$\frac{-Ea}{309R} = \ln 2 - \frac{Ea}{300R}$$

$$\Rightarrow \frac{Ea}{R}\left(\frac{1}{300} - \frac{1}{309}\right) = \ln 2$$

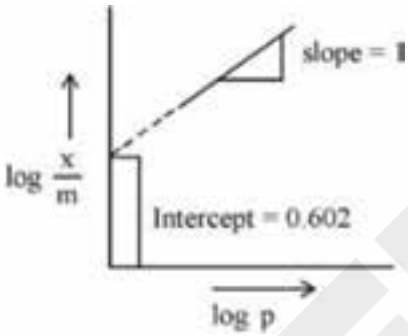
$$\Rightarrow \quad Ea = \frac{300 \times 309 \times \ln 2 \times R}{9}$$

$$\Rightarrow \quad Ea = 58988.1\ J$$

$$= 58.9\ kJ$$

Hence, answer is 59

Q. 29



If the initial pressure of a gas is 0.03 atm, the mass of the gas adsorbed per gram of the adsorbent is _____ $\times 10^{-2}$ g

---

Correct Answer:
12

Solution:
From Freundlich adsorption isotherm

$$\frac{x}{m} = kp^{1/n}$$

$$\Rightarrow \log\left(\frac{x}{m}\right) = \log k + \frac{1}{n}\log p$$

$$\therefore \log k = 0.602 = \log 4$$

$$\Rightarrow k = 4$$

$$\text{and } \frac{1}{n} = 1 \Rightarrow n = 1$$

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$$\therefore \left(\frac{x}{m}\right) = 4p = 4 \times 0.03$$
$$= 0.12 = 12 \times 10^{-2}$$

Hence,answer is 12

Q. 30 0.25 g of an organic compound containing chlorine gave 0.40 g of silver chloride in Carius estimation, The percentage of chlorine present in the compound is _____.[in nearest integer]

(Given: Molar mass of Ag is 108 g mol^{-1} and that of Cl is 35.5 g mol^{-1})

Correct Answer:
40

Solution:

$$\text{mass of AgCl} = 0.4$$
$$\text{moles of AgCl} = \frac{0.4}{143.5} = \text{moles of Cl}$$
$$\therefore \text{ mass of Cl} = \frac{0.4}{143.5} \times 35.5$$
$$\therefore \quad \%\text{by mass of Cl} = \frac{0.4 \times 35.5}{143.5 \times 0.25} \times 100$$
$$= 39.58\%$$

Hence,the answer is 40 when rounded off to the nearest integer

Maths

Q. 1 The number of points of intersection of $|z-(4+3i)| = 2$ and $|z| + |z-4| = 6, z \in \mathbb{C}$, is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
2

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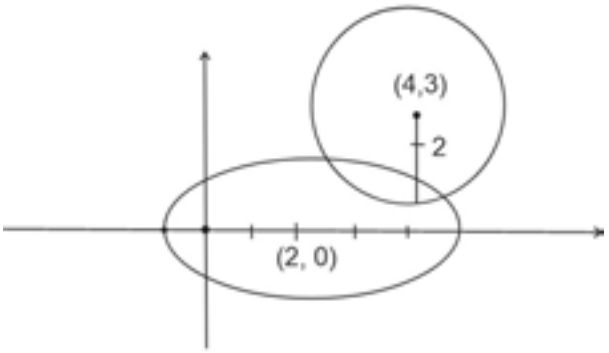
Solution:

$|z-(4+3i)|=2$ is a circle with centre: (4,3) and radius =2.

$|z|+|z-4|=6$ is an ellipse with foci:

$$(0,0)(4,0), \quad 2a=6 \Rightarrow a=3, \quad 2ae=4 \Rightarrow ae=2 \Rightarrow e=\frac{2}{3},$$

$$b^2=a^2\left(1-e^2\right)=9\left(1-\frac{4}{9}\right)=5 \Rightarrow b=\sqrt{5}.$$



So there are 2 points of intersection

Q. 2 Let $f(x)=\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}, a \in \mathbb{R}$. Then the sum of the squares of all the values of A, for which $2f'(10)-f'(5)+100=0$, is

Option 1:
117

Option 2:
106

Option 3:
125

Option 4:
136

Correct Answer:
125

Solution:

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$$f(x) = a\begin{vmatrix} 1 & -1 & 0 \\ x & a & -1 \\ x^2 & ax & a \end{vmatrix} = a\left(1\left(a^2+ax\right)+1\left(ax+x^2\right)\right]$$

$$= ax^2 + 2a^2x + a^3 = a(x+a)^2.$$

$$f'(x) = 2a(x+a)$$

$$f'(10) = 2a(10+a) = 2a^2 + 20a$$

$$f'(5) = 2a(5+a) = 2a^2 + 10a.$$

$$\Rightarrow f'(10) - f'(5) + 100 = 0$$

$$\Rightarrow \left(2a^2+20a\right) - 2a^2 - 10a + 100 = 0$$

$$\Rightarrow 2a^2 + 30a + 100 = 0$$

$$\Rightarrow a^2 + 15a + 50 = 0$$

$$\Rightarrow a = -5, -10.$$

$$\text{sum of squares } = 5^2 + 10^2 = 125$$

Q. 3 Let for some real numbers α & β, $\mathrm{a} = \alpha - \mathrm{i}\beta$. If the system of eqution $4\mathrm{ix} + (1+\mathrm{i})\mathrm{y} = 0$ and $8\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)\mathrm{x} + \bar{\mathrm{a}}\mathrm{y} = 0$ has more tha one solution, then $\frac{\alpha}{\beta}$ is equal to

Option 1:

$-2+\sqrt{3}$

Option 2:

$2-\sqrt{3}$

Option 3:

$2+\sqrt{3}$

Option 4:

$-2-\sqrt{3}$

Correct Answer:

$2-\sqrt{3}$

Solution:

This is a homogeneous system of linear equations

$$(4\mathrm{i})\mathrm{x} + (1+\mathrm{i})\mathrm{y} = 0$$

$$\left(8\mathrm{e}^{\mathrm{i}2\pi/3}\right)\mathrm{x} + (\bar{\mathrm{a}})\mathrm{y} = 0$$

If it has more than one solution

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$$\Rightarrow \quad \Delta = 0 \Rightarrow \begin{vmatrix} 4i & 1+i \\ 8e^{i2\pi/3} & \bar{a} \end{vmatrix} = 0$$

$$\Rightarrow \quad 4\mathrm{i}(\alpha + \mathrm{i}\beta) - 8(1+\mathrm{i})\left(-\frac{1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}\right) = 0$$

$$\Rightarrow \quad \mathrm{i}\alpha - \beta - [-1 + \sqrt{3}\mathrm{i} - \mathrm{i} - \sqrt{3}] = 0$$

$$\Rightarrow \quad -\beta + \sqrt{3} + 1 + \mathrm{i}(\alpha - \sqrt{3} + 1) = 0$$

$$\Rightarrow \quad \beta = \sqrt{3} + 1 \text{ and } \alpha = \sqrt{3} - 1$$

$$\frac{\alpha}{\beta} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{3+1-2\sqrt{3}}{3-1} = 2 - \sqrt{3}$$

Hence the correct answer is option 2.

Q. 4 Let A & B be two 3×3 matrices such that $\mathrm{AB} = \mathrm{I}$ and $|\mathrm{A}| = \frac{1}{8}$. Then $|\operatorname{adj}(\mathrm{B}\ \operatorname{adj}(2\mathrm{A}))|$ is equal to

Option 1:
16

Option 2:
32

Option 3:
64

Option 4:
128

Correct Answer:
64

Solution:

$$|\mathrm{AB}| = 1 \Rightarrow |\mathrm{A}||\mathrm{B}| = 1 \Rightarrow |\mathrm{B}| = 8$$

$$|\operatorname{adj}(\mathrm{B}\ \operatorname{adj}(2\mathrm{A}))| = |\mathrm{B}\ \operatorname{adj}(2\mathrm{A})|^2$$

$$= |\mathrm{B}|^2 |\operatorname{adj}(2\mathrm{A})|^2$$

$$= |\mathrm{B}|^2 \left(|2\mathrm{A}|^2\right)^2 = |\mathrm{B}|^2 \left(\left(2^3|\mathrm{A}|\right)^2\right)^2$$

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$$= 8 \times 8 \times \left(8 \times \frac{1}{8}\right)^4$$

$$= 64$$

Hence the correct answer is option 3.

Q. 5 Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \ldots$. Then $4S$ is equal to

Option 1:

$\left(\frac{7}{3}\right)^2$

Option 2:

$\frac{7^3}{3^2}$

Option 3:

$\left(\frac{7}{3}\right)^3$

Option 4:

$\frac{7^2}{3^3}$

Correct Answer:

$\left(\frac{7}{3}\right)^3$

Solution:

$$S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \cdots$$

$$\frac{S}{7} = \frac{2}{7} + \frac{6}{7^2} + \frac{12}{7^3} + \frac{20}{7^4} + \cdots$$

Substract $\frac{6S}{7} = 2 + \frac{4}{7} + \frac{6}{7^2} + \frac{8}{7^3} + \frac{10}{7^4} + \cdots$

$$\Rightarrow \frac{6}{7^2}S = \frac{2}{7} + \frac{4}{7^2} + \frac{6}{7^3} + \frac{8}{7^4} + \cdots$$

Substract $\Rightarrow \frac{6}{7}\left(1 - \frac{1}{7}\right)S = 2 + \frac{2}{7} + \frac{2}{7^2} + \frac{2}{7^3} + \frac{2}{7^4} + \cdots$

$$\Rightarrow \frac{36}{49}S = 2\left(\frac{1}{1 - 1/7}\right)$$

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$$\Rightarrow \frac{36}{49}S = \frac{7}{3} \Rightarrow S = \frac{7^3}{2^2 \cdot 3^3}$$

$$\Rightarrow 4S = \left(\frac{7}{3}\right)^3$$

Hence the correct answer is option 3

Q. 6 If $a_1, a_2, a_3 \ldots\ldots$ and $b_1, b_2, b_3 \ldots\ldots$ are A.P., and $a_1 = 2, a_{10} = 3, a_1b_1 = 1 = a_{10}b_{10}$ then a_4b_4 is equal to

Option 1:
$\frac{35}{27}$

Option 2:
1

Option 3:
$\frac{27}{28}$

Option 4:
$\frac{28}{27}$

Correct Answer:
$\frac{28}{27}$

Solution:

$a_1 = 2, \quad a_{10} = 2 + (10-1)d = 3 \Rightarrow d = \frac{1}{9},$

$a_4 = 2 + (4-1) \times \frac{1}{9} = \frac{7}{3}$

$a_1b_1 = 1 \Rightarrow b_1 = \frac{1}{2}, \quad a_{10}b_{10} = 1 \Rightarrow b_{10} = \frac{1}{3}$

$b_{10} = \frac{1}{2} + (10-1) \times d = \frac{1}{3} \Rightarrow 9d = -\frac{1}{6} \Rightarrow d = -\frac{1}{5^4}$

$b_4 = \frac{1}{2} + (4-1) \times \left(-\frac{1}{54}\right) = \frac{1}{2} - \frac{1}{18} = \frac{4}{9}$

$a_4b_4 = \frac{7}{3} \times \frac{4}{9} = \frac{28}{27}$

Hence the correct answer is option 4.

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Q. 7 If $m \text{ and } n$ respectively are the number of local maximum and local minimum points of the function $f(x)=\int_0^{x^2} \frac{t^2-5t+4}{2+e^t} dt$, then the ordered pair (m, n) is

Option 1:
$(3,2)$

Option 2:
$(2,3)$

Option 3:
$(2,2)$

Option 4:
$(3,4)$

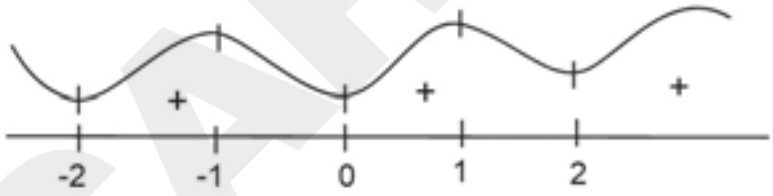
Correct Answer:
$(2,3)$

Solution:

$$f'(a)=\frac{2x-\left((x^2)^2-5x^2+4\right)}{2+e^{x^2}}$$

$$=\frac{2x\left[(x^2-1)(x^2-4)\right]}{2+e^{x^2}}$$

$$=\frac{2(x+2)(x+1)x(x-1)(x-2)}{2+e^{x^2}}$$



So $\{-2,0,2\}$ are poits of Minima $\Rightarrow n=3$

and $\{-1,1\}$ are poits is Maxixma $\Rightarrow m=2$

$(m, n)=(2,3)$

Hence the correct answer is option 2

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Q. 8 Let f be a differentiable function in $\left(0, \frac{\pi}{2}\right)$ If $\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x$, then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to

Option 1:
$6 - 9\sqrt{2}$

Option 2:
$6 - \frac{9}{\sqrt{2}}$

Option 3:
$\frac{9}{2} - 6\sqrt{2}$

Option 4:
$\frac{9}{\sqrt{2}} - 6$

Correct Answer:
$6 - \frac{9}{\sqrt{2}}$

Solution:

$$\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x. \quad , t(0, \pi/2)$$

Differentiating both sides,

$$\sin x \cdot \cos^2 x \, f(\cos x) = 3\sin^2 x \cos x - \sin x$$

$$\Rightarrow \quad f(\cos x) = \frac{3\sin x}{\cos x} - \frac{1}{\cos^2 x} = 3\tan x - \sec^2 x$$

If $\cos x = \frac{1}{\sqrt{3}}, \tan x = \sqrt{2}, \quad \sec x = \sqrt{3}$

$$\Rightarrow f'(\cos x) \cdot (-\sin x) = 3\sec^2 x - 2\sec x \tan x \sec x$$

$$\Rightarrow f'\left(\frac{1}{\sqrt{3}}\right) \cdot \left(\frac{-\sqrt{2}}{\sqrt{3}}\right) = 3 \times 3 - 2 \times 3 \times \sqrt{2}$$

$$\Rightarrow \frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right) = 6 - \frac{9}{\sqrt{2}}$$

Hence the correct answer is option 2

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Q. 9 The integral $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}dx$, where $[\cdot]$ denotes the greater integer function, is equal to

Option 1:

$1+6\log_e\left(\frac{6}{7}\right)$

Option 2:

$1-6\log_e\left(\frac{6}{7}\right)$

Option 3:

$\log_e\left(\frac{7}{6}\right)$

Option 4:

$1-7\log_e\left(\frac{6}{7}\right)$

Correct Answer:

$1+6\log_e\left(\frac{6}{7}\right)$

Solution:

$$I=\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}dx$$

$$=\int_{1/2}^1 \frac{1}{7}dx+\int_{1/3}^{1/2}\frac{1}{7^2}dx+\int_{1/4}^{1/3}\frac{1}{7^3}dx+\int_{1/5}^{7^4}\frac{1}{7^4}dx+\cdots$$

$$=\left(1-\frac{1}{2}\right)\frac{1}{7}+\left(\frac{1}{2}-\frac{1}{3}\right)\frac{1}{7^2}+\left(\frac{1}{3}-\frac{1}{4}\right)\frac{1}{7^3}+\left(\frac{1}{4}-\frac{1}{5}\right)\frac{1}{7^4}+\cdots$$

$$=\left(\frac{1}{7}+\frac{1}{2}\cdot\frac{1}{7^2}+\frac{1}{3}\cdot\frac{1}{7^3}+\frac{1}{4}\cdot\frac{1}{7^4}+\cdots\right)$$

$$-\left(\frac{1}{2}\cdot\frac{1}{7}+\frac{1}{3}\cdot\frac{1}{7^2}+\frac{1}{4}\cdot\frac{1}{7^3}+\frac{1}{5}\cdot\frac{1}{7^4}+\cdots\right)$$

$$=\log_e\left(1-\frac{1}{7}\right)-7\left(\frac{-1}{7}-\log_e\left(1-\frac{1}{7}\right)\right)$$

$$=1+6\log_e\frac{6}{7}$$

Hence the correct answer is option 1.

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Q. 10 If the solution curve of the differential equation $\left(\left(\tan^{-1} y\right)-x\right) dy=\left(1+y^2\right) dx$ passes through the point $(1,0)$, then the abscissa of the point on the curve whose ordinate is $\tan(1)$, is

Option 1:
$2e$

Option 2:
$\frac{2}{e}$

Option 3:
2

Option 4:
$\frac{1}{e}$

Correct Answer:
$\frac{2}{e}$

Solution:
$$\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^1 y}{1+y^2}$$

Linear differential equation

Integral factor

$$\text{If } =e^{\int \frac{1}{1+y^2}dy}=e^{\tan^{-1} y}$$

$$\Rightarrow \quad x\times e^{\tan^{-1} y}=\int \frac{\tan^{-1} y e^{\tan^{-1} y}}{1+y^2}dy$$

Let $\tan^{-1} y=t \rightarrow x\times e^{\tan^{-1} y}=\int \underset{I}{t}\underset{II}{e^t}dt$

$$\Rightarrow x\times e^{\tan^{-1} y}=te^t-\int e^t dt$$

$$\Rightarrow x\times e^{\tan^{-1} y}=\tan^{-1} y e^{\tan^{-1} y}-e^{\tan^{-1} y}+c$$

$$x=1, y=0 \Rightarrow 1\times e^0=0-e^0+c \Rightarrow c=2$$

$$y=\tan 1 \Rightarrow x\times e^1=1\times e^1-e^1+2 \Rightarrow x=2/e$$

Hence the correct answer is option 2

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Q. 11 If the equation of the parabola, whose vertex is at $(5, 4)$ and the directrix is $3x + y - 29 = 0,$ is $x^2 + ay^2 + bxy + cx + dy + k = 0$, then $a + b + c + d + k$ is equal to

Option 1:
575

Option 2:
-575

Option 3:
576

Option 4:
-576

Correct Answer:
-576

Solution:
Calculate foot of directrix

$$\frac{x-5}{3} = \frac{y-4}{1} = -1\frac{(3\times 5+4-29)}{3^2+1^2}$$

$$\Rightarrow x = 8, y = 5.$$

Vertex is mid-point of focus and foot at directrix so coordinates of focus $= (2, 3)$

Using $PS = PM$, equation of parabola is

$$\sqrt{(x-2)^2+(y-3)^2} = \left|\frac{3x+y-29}{\sqrt{10}}\right|$$

$$\Rightarrow 10\left[x^2-4x+4+y^2-6y+9\right] = 9x^2+y^2+841+6xy-58y-174x$$

$$\Rightarrow x^2+9y^2-6xy+134x-2y-711 = 0.$$

Compare with $x^2 + ay^2 + bxy + cx + dy + x = 0$

$$a+b+c+d+k : 9-6+134-2-711 = 143-719$$

$$= -576$$

Hence the correct answer is option 4.

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Q. 12 The set of value of k, for which the circle $C: 4x^2+4y^2-12x+8y+k=0$ lies inside the fourth quadrant and the point $\left(1,-\frac{1}{3}\right)$ lies on or inside the circle C, is

Option 1:
an empty set

Option 2:
$\left(6,\frac{65}{9}\right]$

Option 3:
$\left[\frac{80}{9},10\right)$

Option 4:
$\left(9,\frac{92}{9}\right]$

Correct Answer:
$\left(9,\frac{92}{9}\right]$

Solution:

$\text{Cirde}: x^2+y^2-3x+2y+\frac{k}{4}=0$

$\text{Centre}: \left(\frac{3}{2},-1\right) \text{ radius}: \sqrt{\left(\frac{3}{2}\right)^2+(1)^2-\frac{k}{4}}$

Radius must be lies than 1 for the circle to lie in the fourth quadrant.

$\Rightarrow \quad \frac{9}{4}+1-\frac{k}{4}<1 \Rightarrow k>9$ .....(1)

Point $\left(1,-\frac{1}{3}\right)$ lies on or inside the circle

$\Rightarrow s_1 \leqslant 0 \Rightarrow 1+\frac{1}{9}-3-\frac{2}{3}+\frac{k}{4} \leqslant 0 \Rightarrow \frac{k}{4} \leqslant \frac{8}{3}-\frac{1}{9}=\frac{23}{9}$

$\Rightarrow k \leqslant \frac{92}{9}$ .....(2)

From (1) and (2)

$k \in \left(9,\frac{92}{9}\right]$

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Hence the correct answer is option 4.

Q. 13 Let the foot of the perpendicular from the point $1, 2, 4$ on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P. Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is

Option 1:
5

Option 2:
$\frac{50}{13}$

Option 3:
4

Option 4:
$\frac{63}{13}$

Correct Answer:
5

Solution:
Let the foot of perpendicular be

$P(4\lambda - 2, 2\lambda + 1, 3\lambda - 1)$

$A : (1, 2, 4) \quad AP \perp L$

$\Rightarrow (4\lambda - 2 - 1, 2\lambda + 1 - 2, 3\lambda - 1 - 4) - (4, 2, 3) = 0$

$\Rightarrow 4(4\lambda - 3) + 2(2\lambda - 1) + 3(3\lambda - 5) = 0$

$\Rightarrow 29\lambda = 29 \Rightarrow \lambda = 1$

$\Rightarrow P(2, 3, 2)$

Distance of P from $3x + 4y + 12z + 23 = 0$ is

$$d = \frac{3 \times 2 + 4 \times 3 + 2 \times 12 + 23}{\sqrt{3^2 + 4^2 + 12^2}}$$

$$= \frac{65}{13} = 5$$

Hence the correct answer is option 1

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Q. 14 The shortest distance between the lines $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ and $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$ is

Option 1:
$\frac{18}{\sqrt{5}}$

Option 2:
$\frac{22}{3\sqrt{5}}$

Option 3:
$\frac{46}{3\sqrt{5}}$

Option 4:
$6\sqrt{3}$

Correct Answer:
$\frac{18}{\sqrt{5}}$

Solution:

$L_1=\vec{a_1}+\lambda\vec{b_1}=(3,2,1)+\lambda(2,3,-1)$

$L_2=\vec{a_2}+\mu\vec{b_2}=(-3,6,5)+\mu(2,1,3)$

Shortest Distance $=\left|\frac{(\vec{a_2}-\vec{a_1})\cdot\left(\vec{b_1}\times\vec{b_2}\right)}{\left|\vec{b_1}\times\vec{b_2}\right|}\right|$

$\tilde{b}_1\times\vec{b_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-1\\2&1&3\end{vmatrix}=10\hat{i}-8\hat{j}-4\hat{k}$

$\text{S.D.}=\left|\frac{(-6,4,4)\cdot(10,-8,-4)}{\sqrt{10^2+8^2+4^2}}\right|$

$=\frac{60+32+16}{\sqrt{180}}=\frac{18}{\sqrt{5}}$

Hence the correct answer is option 1

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Q. 15 Let $\overrightarrow{a}$ and $\overrightarrow{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\overrightarrow{a}$ and $\overrightarrow{b}$ be cute, $|\vec{a}| = 1$ and $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$. If $\vec{c} = 2\sqrt{2}(\vec{a} \times \vec{b}) - 2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is

Option 1:

$\frac{\pi}{4}$

Option 2:

$-\frac{\pi}{4}$

Option 3:

$\frac{5\pi}{6}$

Option 4:

$\frac{3\pi}{4}$

Correct Answer:

$\frac{3\pi}{4}$

Solution:

Area of parallelogram $= \frac{1}{2}\left|\overrightarrow{d_1} \times \overrightarrow{d_2}\right| = 2\sqrt{2}$

$\Rightarrow |\vec{a} \times \vec{b}| = 4\sqrt{2}$

$|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$

$|\vec{a}||\vec{b}| \cos\theta = |\vec{a}||\vec{b}| \sin\theta$

$\Rightarrow \tan\theta = 1 \Rightarrow \theta = \pi/4$

$|\vec{a} \times \vec{b}| = 4\sqrt{2} \Rightarrow |\vec{a}||\vec{b}| \sin\pi/4 = 4\sqrt{2}$

$= |\vec{b}| = 8$

$\vec{c} = 2\sqrt{2}(\vec{a} \times \vec{b}) - 2\vec{b}$

$|\vec{c}|^2 = (2\sqrt{2}(\vec{a} \times \vec{b}))^2 + 4|\vec{b}|^2 + 4\sqrt{2}(\vec{a} \times \vec{b}) - \vec{b}$

$= 256 + 256 = 512 \Rightarrow |\vec{c}| = 16\sqrt{2}$

$\vec{c} + 2\vec{b} = 2\sqrt{2}(\vec{a} \times \vec{b})$

$\Rightarrow |\vec{c}|^2 + 4|\vec{b}|^2 + 4|\vec{b}||\vec{c}| \cos\phi = (2\sqrt{2}|\vec{a} \times \vec{b}|)^2$

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$\Rightarrow 512+256+4\times 8\times 16\sqrt{2}\cos\phi=256$

$\Rightarrow \cos\phi=-\frac{1}{\sqrt{2}}\Rightarrow \phi=\frac{3\pi}{4}$

Hence the correct answer is option 4

Q. 16 The mean and variance of the data $4,5,6,6,7,8,x,y$, where $x<y$, are 6 and $\frac{9}{4}$ respectively. Then x^4+y^2 is equal to

Option 1:
162

Option 2:
320

Option 3:
674

Option 4:
420

Correct Answer:
320

Solution:

$\text{Mean}=\frac{4+5+6+6+7+8+x+y}{8}=6\Rightarrow x+y=12$.....(1)

$\text{Varciance}=\frac{\sum x_i^2}{x}-(\bar{x})^2\Rightarrow \frac{4^2+5^2+6^2+6^2+7^2+8^2+x^2+y^2-(6)^2}{8}=\frac{9}{4}$

$\Rightarrow 16+25+36+36+4y+64+x^2+y^2=18+288$

$\Rightarrow x^2+y^2=306-226=80$ .....(2)

$(1)^2\Rightarrow x^2+y^2+2xy=144\Rightarrow 2xy=144-80=64\Rightarrow xy=32$.....(3)

From eqn (1) and (2) $x=4,\quad y=8$

Hence the correct answer is option 2.

$x^4+y^2=256+64=320$

Q. 17 If a point $A(x,y)$ yes in the region bounded by the y-axis, straight lines $2y+x=6$ and $5x-6y=30$ then the probability that $y<1$ is

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Option 1:

$\frac{1}{6}$

Option 2:

$\frac{5}{6}$

Option 3:

$\frac{2}{3}$

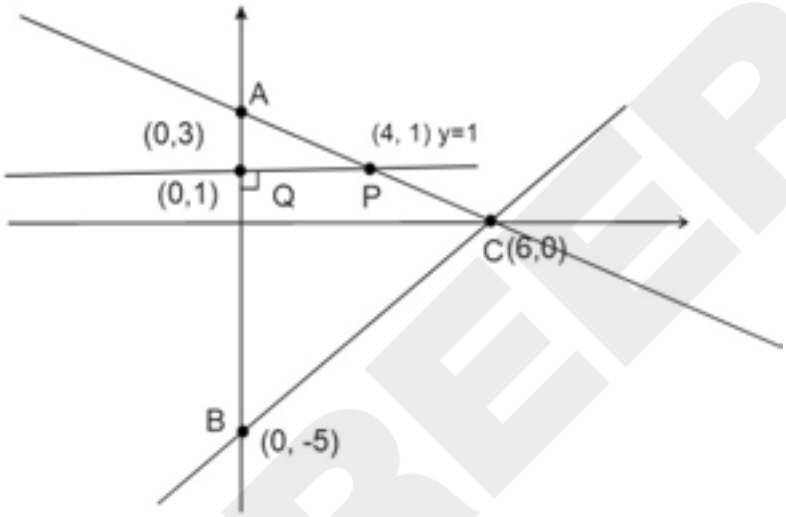
Option 4:

$\frac{666}{777}$

Correct Answer:

$\frac{5}{6}$

Solution:



$$\text{Area of } \triangle ABC = \frac{1}{2} \times 8 \times 6 = 24$$

$$\text{Area of} \triangle APQ = \frac{1}{2} \times 2 \times 4 = 4$$

$$\text{So } P(y < 1) = \frac{24 - 4}{24} = \frac{20}{24} = \frac{5}{6}$$

Hence the correct answer is option 2

Q. 18 The value of $\cot\left(\sum_{n=1}^{50} \tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is

Option 1:
$\frac{26}{25}$

Option 2:
$\frac{25}{26}$

Option 3:
$\frac{50}{51}$

Option 4:
$\frac{52}{51}$

Correct Answer:
$\frac{26}{25}$

Solution:

$$\sum_{n=1}^{50} \tan^{-1}\left(\frac{(n+1)-(n)}{1+(n+1)(n)}\right) = \sum_{n=1}^{50} \left[\tan^{-1}(n+1) - \tan^{-1}(n)\right]$$

$$= \left(\tan^{-1} 2 - \tan^{-1} 1\right) + \left(\tan^{-1} 3 - \tan^{-1} 2\right) + \ldots - \left(\tan^{-1} 51 - \tan^{-1} 50\right)$$

$$= \tan^{-1}(51) - \tan^{-1}(1)$$

$$= \tan^{-1} \frac{51-1}{1+51\times 1} = \tan^{-1} \frac{50}{52} = \tan^{-1} \frac{25}{26}$$

$$\text{so } \cot\left(\tan^{-1}\left(\frac{25}{26}\right)\right) = \frac{26}{25}$$

Hence the correct answer is option 1

Q. 19 $\alpha = \sin 36^\circ$ is a root of which of the following equation?

Option 1:
$16x^4 - 10x^2 - 5 = 0$

Option 2:
$16x^4 + 20x^2 - 5 = 0$

Option 3:
$16x^4 - 20x^2 - 5 = 0$

Option 4:
$16x^4 - 10x^2 + 5 = 0$

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Correct Answer:

$16x^4 - 20x^2 - 5 = 0$

Solution:

$\cos 72^\circ = 1 - 2\sin^2 36^\circ$

$\Rightarrow \frac{\sqrt{5}-1}{4} = 1 - 2\alpha^2 \Rightarrow \sqrt{5} = 5 - 8\alpha^2$

$\Rightarrow 5 = 25 + 64\alpha^4 - 80\alpha^2$

$\Rightarrow 64\alpha^4 - 80\alpha^2 + 20 = 0$

$\Rightarrow 16\alpha^4 - 20\alpha^2 + 5 = 0$

$\Rightarrow 16x^4 - 20\alpha^2 + 5 = 0$

Hence the correct answer is option 3

Q. 20 Which of the following statemet is a tautology?

Option 1:

$((\sim q) \wedge p) \wedge q$

Option 2:

$((\sim q) \wedge p) \wedge (p \wedge (\sim p))$

Option 3:

$((\sim q) \wedge p) \vee (p \vee (\sim p))$

Option 4:

$(p \wedge q) \wedge (\sim (p \wedge q))$

Correct Answer:

$((\sim q) \wedge p) \vee (p \vee (\sim p))$

Solution:

$t = \text{tautology}, c = \text{contradiction}$

$(A)((\sim q) \wedge p) \wedge q \equiv ((\sim q) \wedge q) \wedge p \equiv c \wedge p \equiv c$

$(B) \quad ((\sim q) \wedge p) \wedge (b \wedge (\sim p)) \equiv ((\sim q) \wedge p) \wedge c \equiv c$

$(C)((\sim q) \wedge p) \vee (p \vee (\sim p)) \equiv ((\sim q) \wedge p) \vee t \equiv t$

$(D)(p \wedge q) \wedge (\sim (p \wedge q)) \equiv c$

Hence the correct answer is option 3

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Q. 21 Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define $f : S \to S$ as

$$f(n) = \begin{cases} 2n, & \text{if } n = 1, 2, 3, 4, 5 \\ 2n - 11, & \text{if } n = 6, 7, 8, 9, 10 \end{cases}$$

Let $g : S \to S$ be a function such that $f \circ g(n) = \begin{cases} n + 1 & , \text{ if } n \text{ is odd} \\ n - 1, & \text{if } n \text{ is even} \end{cases}$, Then $g(10)(g(1) + g(2) + g(3) + g(4) + g(5))$ is equal to _____.

Correct Answer:
190

Solution:

$$f(g(n)) = \begin{cases} 2g(n), & \text{if } g(n) = 1, 2, 3, 4, 5 \\ 2g(n) - 11, & if(g(n) = 6, 7, 8, 9, 10 \end{cases}$$

$$f(1) = 2, f(2) = 4, f(3) = 6, f(4) = 8, f(5) = 10$$

$$f(6) = 1, f(7) = 3, f(8) = 5, f(9) = 7, f(10) = 9$$

$$f(g(n)) = \begin{cases} n + 1, & \text{if} \quad n = 1, 3, 5, 7, 9 \\ n - 1, & \text{if} \quad n = 2, 4, 0, 8, 10 \end{cases}$$

$$\Rightarrow f(g(1)) = 2, fog(3) = 4, fog(5) = 6, fog(7) = 8, fog(9) = 10$$

$$\Rightarrow fog(2) = 1, fog(4) = 3, fog(6) = 5, fog(8) = 7, fog(10) = 9$$

$$\Rightarrow g(1) = f^{-1}(2) = 1, \quad g(2) = f^{-1}(1) = 6, \quad g(3) = f - 1(4) = 2,$$

$$g(4) = f^{-1}(3) = 7, \quad g(5) = f^{-1}(6) = 3, g(10) = f^{-1}(g) = 10$$

$$g(10)(g(1) + g(2) + g(3) + g(4) + g(5)) = 10(1 + 6 + 2 + 7 + 3) = 190$$

Hence answer is 190

Q. 22 Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0, \lambda > 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to ____.

Correct Answer:
98

Solution:

$$\alpha + \beta = 4\lambda, \qquad \alpha\beta = 5$$

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$\alpha+\gamma=3\sqrt{2}+2\sqrt{3}, \alpha\gamma=3\sqrt{3}\lambda$

$\beta+\gamma=3\sqrt{2}$

$\Rightarrow \alpha+\beta+\alpha+\gamma=4\lambda+3\sqrt{2}+2\sqrt{3}$

$\Rightarrow 2\alpha+3\sqrt{2}=4\lambda+3\sqrt{2}+2\sqrt{3}$

$\Rightarrow \alpha=2\lambda+\sqrt{3}, \beta=2\lambda-\sqrt{3},$

$\alpha\beta=5 \Rightarrow 4\lambda^2-3=5 \Rightarrow \lambda=\pm\sqrt{2}, \lambda>0 \text{ so } \lambda=\sqrt{2}.$

$(\alpha+2\beta+\gamma)^2=(\alpha+\beta+\beta+\gamma)^2$

$=(4\lambda+3\sqrt{2})^2=(4\sqrt{2}+3\sqrt{2})^2=98.$

Hence answer is 98.

Q. 23 Let A be a matrix of order 2×2 whose entries are from the set $\{0,1,2,3,4,5\}$.If the sum of all the entries of A is a prime number $\mathrm{p}, 2<\mathrm{p}<8$, then the number of such matrices A is _____.

Correct Answer:
180

Solution:

$\mathrm{A}=\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \mathrm{b}=3,5,7$

$\mathrm{b}=3 \Rightarrow \mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}=3, \quad \mathrm{a},\mathrm{b},\mathrm{c},\mathrm{d}\in\{0,1,2,3\}$

$\text{So} \quad \mathrm{n}_1={}^{3+4-1}\mathrm{c}_{4-1}={}^{6}\mathrm{c}_3=20$

$\mathrm{p}=5 \Rightarrow \mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}=5, \quad \mathrm{a},\mathrm{b},\mathrm{c},\mathrm{d}\in\{0,1,2,3,4,5\}$

$\Rightarrow \mathrm{n}_2={}^{5+4-1}\mathrm{c}_{4-1}={}^{8}\mathrm{C}_3=56.$

$\mathrm{p}=7 \Rightarrow \mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}=7, \quad \mathrm{a},\mathrm{b},\mathrm{c},\mathrm{d}\in\{0,1,2,3,4,5\}$

$\mathrm{n}_3={}^{7+4-1}\mathrm{C}_{4-1}-(6,1,0,0)-(7,0,0,0)$

$={}^{10}\mathrm{c}_3-\frac{4!}{2!}-\frac{4!}{3}!$

$=120-12-4=104$

$\text{Total}=\mathrm{n}_1+\mathrm{n}_2+\mathrm{n}_3=20+56+104=180$

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Q. 24 If the sum of the coefficient of all the positive powers of x, in the Binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral values of n is _____.

Correct Answer:
57

Solution:

$$\left(a^n + \frac{2}{x^5}\right)^7 = {}^7c_0x^{7n} + {}^7c_1 \cdot x^{6n-5} \cdot 2 + {}^7c_2 \cdot x^{5n-10} \cdot 2^2 + {}^7/3 \cdot x^{4n-15} \cdot 2^3$$

$$+{}^7c_4x^{3n-20}2^4 + {}^7c_5x^{2n-25}2^5 + {}^7\,c_6x^{n-30}2^6 + {}^7c_7x^{-35}2^7$$

Sum of coeffficiet from beginning

$s_1 = 1, s_2 = 1 + 14 = 15, S_3 = 1 + 14 + 84 = 99$

$s_4 = 1 + 14 + 84 + 280 = 379, S_5 = 1 + 14 + 24 + 280 + 560 = 939$

$\text{So} \quad 3n - 20 > 0 \quad \& \quad 2n - 25 \leqslant 0$

$\Rightarrow n > \frac{20}{3} \quad \& \quad n \leqslant \frac{25}{2}$

$\Rightarrow n = 7, 8, 9, 10, 11, 12$

$\text{sum} \ = 7 + 8 + 9 + 10 + 11 + 12 = 57$

Hence answer is 57

Q. 25 Let $[t]$ denote the greatest integer $\leq t$ and t denote the fractional part of t. The integral value of α for which the left hand limit of the function $f(x) = [1 + x] + \frac{\alpha^{2[x]+\{x\}} + [x] - 1}{2[x] + \{x\}}$ at $x = 0$ is equal to $\alpha - \frac{4}{3}$, is _____

Correct Answer:
3

Solution:

$\lim\limits_{x\to0^-}[x] = -1, \ \lim\limits_{x\to0^-}\{x\} = 1$

$\text{So} \ \lim\limits_{x\to0^-} f(x)$

$= \lim\limits_{x\to-}\left(1 + [x] + \frac{\alpha^{2[x]+\{x\}} + [x] - 1}{2[x] + \{x\}}\right) = \alpha - \frac{4}{3}$

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$$\Rightarrow \quad 1-1+\frac{\alpha^{2(-1)+1}+(-1)-1}{2(-1)+1}=\alpha-\frac{4}{3}$$

$$\Rightarrow \quad \frac{\frac{1}{\alpha}-2}{-1}=\alpha-\frac{4}{3} \Rightarrow -\frac{1}{2}+2=\alpha-\frac{4}{3}$$

$$\Rightarrow \alpha+\frac{1}{\alpha}=\frac{10}{3} \Rightarrow 3\alpha^2-10\alpha+3=0$$

$$\Rightarrow \alpha=3 \text{ OR } \alpha=1/3$$

$$\text{Since } \alpha=\text{integer},$$

$$\alpha=3$$

Hence answer is 3

Q. 26 If $y(x)=(x^x)^x, x>0$ then $\frac{d^2x}{dy^2}+20$ at $x=1$ is equal to _____.

Correct Answer:
16

Solution:

$$y=(x^x)^x=x^{x\times x}=x^{x^2}$$

$$\Rightarrow \ln y=x^2\ln x$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx}=\frac{x^2}{x}+2x\ln x$$

$$\text{at } x=1, y=1, \quad y'=1$$

$$y''=y'(1+2,\ln x)+y(1+2(1+\ln x))$$

$$\Rightarrow y''(1)=1(1+0)+1(1+2(1+0))=4.$$

$$\frac{d^2x}{dy^2}=-\frac{\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^3}=\frac{-y}{(1)^3}-4$$

$$\text{So}\frac{d^2x}{dy^2}=-4+20=16$$

Hence answer is 16

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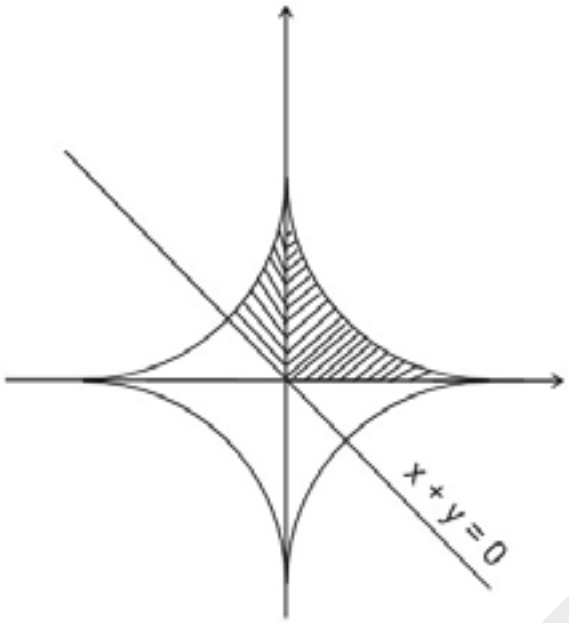
Q. 27 If the area of the region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \leq 1, x + y \geq 0, y \geq 0\right\}$ is A, then $\frac{256A}{\pi}$ is equal to _____.

Correct Answer:
36

Solution:



Desired area in second quadrant $= \frac{1}{2}$ Area in first quadrant

So total Area $\frac{3}{2}\int_0^1 \left(1 - x^{2/3}\right)^{3/2} dx$

$\text{Let } x^{1/3} = \sin\theta \Rightarrow 1 - x^{2/3} = \cos^2\theta,$

$\Rightarrow x \cdot \sin^3\theta \Rightarrow dx = 3\sin^2\theta\cos\theta d\theta.$

$\text{Area} = \frac{3}{2}\int_0^{\pi/2} \left(\cos^2\theta\right)^{3/2} \times 3\sin^2\theta\cos\theta d\theta.$

$= \frac{9}{2}\int_0^{\pi/2} \sin^2\theta 4s^4\theta d\theta = \frac{9}{2} \times \frac{1 \times 3 \times 1}{6 \times 4 \times 2} \times \frac{\pi}{2} = \frac{9\pi}{64}$

$\Rightarrow \frac{256\ A}{\pi} = 36$

Hence the answer is 36

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Q. 28 Let $y = y(x)$ be the solution of the differential equation $(1-x^2)\,dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx, -1 < x < 1$ and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}y(x)dx = k$, then k^{-1} is equal to _____.

Correct Answer:
320

Solution:

$$\frac{dy}{dx} - \frac{x}{1-x^2}y = \frac{x^3+2}{\sqrt{1-x^2}},$$

$$\text{I.F.} = e^{\int \frac{-x}{1-x^2}dx} = \sqrt{1-x^2}$$

$$\Rightarrow y\sqrt{1-x^2} = \int (x^3+2)\,dx \Rightarrow y\sqrt{1-x^2} = \frac{x^4}{4} + 2x + c$$

$$y(0) = 0 \Rightarrow c = 0 \Rightarrow \quad y\sqrt{1-x^2} = \frac{x^4}{4} + 2x$$

$$\Rightarrow \int_{-1/2}^{1/2} y\sqrt{1-x^2}dx = \int_{-1/2}^{1/2} \left(\frac{x^4}{4} + 2x\right)dx = 2\int_0^{1/2} \frac{x^4}{4}dx = \frac{1}{2} \times \left(\frac{x^5}{5}\right)_0^{1/2}$$

$$= \frac{1}{320} = k \Rightarrow k^{-1} = 320$$

Hence the correct is 320

Q. 29 Let a circle C of radius 5 lie below the x-axis. The line $L_1 : 4x + 3y + 2 = 0$ passes through the center P of the circle C and intersects the line $L_2 : 3x - 4y - 11 = 0$ at Q. The line L_2 touches C at the point Q. Then the distance of P from the line $5x - 12y + 51 = 0$ is _____.

Correct Answer:
11

Solution:
Point of intersection of $L_1 : 4x + 3y + 2 = 0$ and $L_2 : 3x - 4y - 11 = 0$ is

$Q\,(1, -2)$.

Centre P of the circle lies on L_1

$$\text{Slope of } L_1 : \tan\theta = -\frac{4}{3} \Rightarrow \cos\theta : \frac{-3}{5}, \sin\theta = \frac{4}{5}$$

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$$\text{Coordinates of P} : (1-5\cos\theta, -2-5\sin\theta)$$

$$= (4, -6)$$

distance of $\mathrm{P}(4,-6)$ from $5\mathrm{x}-12\mathrm{y}+51$ is

$$\mathrm{d} = \left|\frac{5\times 4-12\times(-6)+51}{\sqrt{5^2+122}}\right| = \frac{20+72+51}{13} = \frac{143}{13} = 11$$

Hence the answer is 11

Q. 30 Let $\mathrm{S} = \{\mathrm{E}_1, \mathrm{E}_2, \ldots\ldots\ldots, \mathrm{E}_8\}$ be a ample space of a random experiment such that $\mathrm{P}(\mathrm{E}_n) = \frac{n}{36}$ for every $n = 1, 2, \ldots., 8$. Then the number of elements in the set $\left\{\mathrm{A} \subseteq \mathrm{S} : \mathrm{P}(\mathrm{A}) \geq \frac{4}{5}\right\}$ is _____.

Correct Answer:
19

Solution:

$$\mathrm{P}(1) = \frac{1}{36}, \mathrm{P}(2) = \frac{2}{36}, \cdots \mathrm{P}(8) = \frac{8}{36}$$

$$\mathrm{P}(\mathrm{A}) \geqslant \frac{4}{5} \Rightarrow \mathrm{P}(\mathrm{A}) \geqslant \frac{144}{180}$$

Total number of subsets of $\mathrm{S} = 2^8 = 256$

Number of subsets such that $5(\text{ sum of elements }) \geqslant 144$

$$\Rightarrow 5\ (\text{Sum of removed items}) < 180 - 144 = 36$$

If 1 item is removed : $5\mathrm{x} < 36$

$$\Rightarrow \mathrm{n} = \{1,2,3,4,5,6,7\}, \mathrm{n}_1 = 7$$

If 2 items are removed : $5(\mathrm{x}+\mathrm{y}) < 36$

$$\Rightarrow \mathrm{x}+\mathrm{y} < 72$$

$$\Rightarrow (x, y) = (1,2), (1,3)(1,4), (1,5), (1,6), (2,3), (2,4), (2,5), (3,4)$$

$$\mathrm{n}_2 = 9.$$

If 3 items are removed $\mathrm{x}+\mathrm{y}+\mathrm{z} < 7.2$

$$\Rightarrow (\mathrm{x}, \mathrm{y}, \mathrm{z}) = (1,2,3), (1,2,4), \mathrm{n}_3 = 2$$

If no item is removed, $\mathrm{n}_0 = 1$

$$\text{Total} = \mathrm{n}_0 + \mathrm{n}_1 + \mathrm{n}_2 + \mathrm{n}_3$$

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$$= 1 + 7 + 9 + 2 = 19$$

Hence the answer is 19

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28 June Morning Shift

Physics

Q. 1 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coeffcient of viscosity.

Reason R : Coefficient of viscosity $= \dfrac{\text{Force}}{\text{Velocity gradient}}$

Choose the correct answer from the options given below :

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ true, and $\mathbf{R}$ is correct explanation of $\mathbf{A}$.

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

Option 3:
$\mathbf{A}$ is true but $\mathbf{R}$ is false.

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is true.

Correct Answer:
$\mathbf{A}$ is true but $\mathbf{R}$ is false.

Solution:

viscous force

$$\mathrm{F} = -\eta \mathrm{A}\frac{\mathrm{dv}}{\mathrm{dy}}$$

$$\eta = \frac{\mathrm{F}}{\mathrm{A}} \times \frac{\mathrm{dy}}{\mathrm{dv}}$$

$$[\eta] = \ [\text{Pressure}] \ \text{x} \ [\text{time}]$$

Therefore Assertio A is correct, Reason R is incorrect

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Hence the correct option is 3

Q. 2 A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $\mathrm{a}=\mathrm{k}^2\mathrm{rt}^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

Option 1:
$Zero$

Option 2:
$\mathrm{mk}^2\mathrm{r}^2\mathrm{t}^2$

Option 3:
$\mathrm{mk}^2\mathrm{r}^2\mathrm{t}$

Option 4:
$\mathrm{mk}^2\mathrm{rt}$

Correct Answer:
$\mathrm{mk}^2\mathrm{r}^2\mathrm{t}$

Solution:
$\mathrm{a}=\mathrm{k}^2\mathrm{rt}^2$

$\because$ since centripetal acceleration

$$\mathrm{a_c}=\frac{\mathrm{v}^2}{\mathrm{r}}=\mathrm{k}^2\mathrm{rt}^2$$

$$\mathrm{v}^2=\mathrm{k}^2\mathrm{r}^2\mathrm{t}^2$$

$$\mathrm{v}=\mathrm{krt}$$

Here speed is variable, therefore particle is performing non-uniform circular motion

$$\text{Power }(\mathrm{P})=\mathrm{F_{tangential}}\times\text{ speed}$$

$$=\mathrm{ma_t}\times\mathrm{v}$$

$$\mathrm{a_t}=\frac{\mathrm{dv}}{\mathrm{dt}}=\frac{\mathrm{d}}{\mathrm{dt}}(\mathrm{krt})=\mathrm{kr}$$

$$\mathrm{p}=\mathrm{mkr}\times\mathrm{krt}$$

$$\mathrm{p}=\mathrm{mk}^2\mathrm{r}^2\mathrm{t}$$

Hence the correct option is 3

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Q. 3 Motion of a particle in $\mathrm{x-y}$ plane is described by a set of following equations $x=4\sin\left(\frac{\pi}{2}-\omega t\right)\mathrm{m}$ and $y=4\sin(\omega t)\mathrm{m}$. The path of the particle will be :

Option 1:
circular

Option 2:
helical

Option 3:
parabolic

Option 4:
elliptical

Correct Answer:
circular

Solution:

$$\mathrm{x}=4\sin\left(\frac{\pi}{2}-\omega\mathrm{t}\right)$$

$$\mathrm{x}=4\cos\omega\mathrm{t} \quad \text{.....(1)}$$

$$\mathrm{y}=4\sin\omega\mathrm{t} \quad \text{.....(2)}$$

Square both equation and add

$$\mathrm{x}^2+\mathrm{y}^2=4\left[\cos^2\omega\mathrm{t}+\sin^2\omega\mathrm{t}\right]$$

$$\mathrm{x}^2+\mathrm{y}^2=4$$

this is equation of circle of radius 2m and centre at origin.

Hence correct option is 1

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Q. 4 Match List-I with List-II

	List-I		List-II
A	Moment of inertia of solid sphere of radius R about any tangent.	I.	$\frac{5}{3}MR^2$
B	Moment of inertia of hollow sphere of radius (R) about any tangent.	II.	$\frac{7}{5}MR^2$
C	Moment of inertia of circular ring of radius (R) about its diameter.	III.	$\frac{1}{4}MR^2$
D	Moment of inertia of circular disc of radius (R)about any diameter.	IV	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-IV, D-III

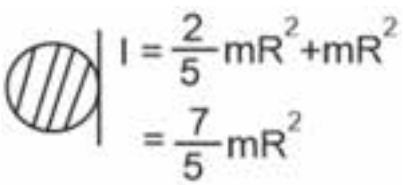
Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-II, B-I, C-III, D-IV

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
For solid Sphere



For hollow sphere

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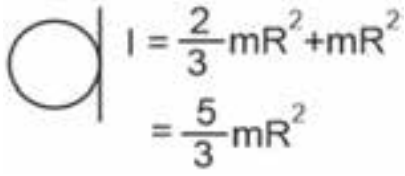
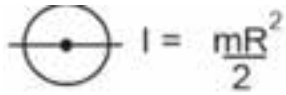


For ring



Hence the correct option is 1

Q. 5 Two planets $\mathrm{A\ and\ B}$ of equal mass are having their period of revolutions $\mathrm{T_A\ and\ T_B\ such\ that\ T_A = 2T_B}$. These planets are revolving in the circular orbits of radii $\mathrm{r_A\ and\ r_B}$ respectively. Which out of the following would be the correct relationship of their orbits?

Option 1:
$2r_A^2 = r_B^3$

Option 2:
$r_A^3 = 2r_B^3$

Option 3:
$r_A^3 = 4r_B^3$

Option 4:
$T_A^2 - T_B^2 = \frac{\pi^2}{GM}\left(r_B^3 - 4r_A^3\right)$

Correct Answer:
$r_A^3 = 4r_B^3$

Solution:
According to Kepler's law

$T^2 \propto r^3$

$T \rightarrow \text{time period}$

$r \rightarrow \text{radius of orbit}$

$\frac{T_A^2}{T_B^2} = \frac{r_A^3}{r_B^3}, \qquad \frac{T_A}{T_B} = 2$

$2^2 = \left(\frac{r_A}{r_B}\right)^3$

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$$4=\frac{r_A^3}{r_B^3}$$

$$4r_B^3=r_A^3$$

Hence correct option is 3

Q. 6 A water drop of diameter $2\ \mathrm{cm}$ is broken into 64 equal droplets. The surface tension of water is $0.075\ \mathrm{N/m}$. In this process the gain in surface energy will be :

Option 1:
$2.8\times10^{-4}\ \mathrm{J}$

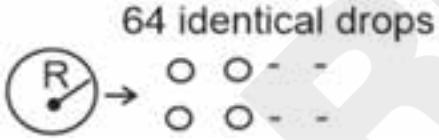
Option 2:
$1.5\times10^{-3}\ \mathrm{J}$

Option 3:
$1.9\times10^{-4}\ \mathrm{J}$

Option 4:
$9.4\times10^{-5}\ \mathrm{J}$

Correct Answer:
$2.8\times10^{-4}\ \mathrm{J}$

Solution:



Let radius of small drops = r

Volume conservation of water

$$\frac{4}{3}\pi R^3=64\times\frac{4}{3}\pi r^3$$

$$r^3=\frac{R^3}{64},r=\frac{R}{4}$$

Initial surface energy

$$U_1=S\times4\pi R^2$$

Final surface energy is system

$$U_2=64\times5\times4\pi r^2$$

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$$= 64 \times 5 \times 4\pi \times \frac{R^2}{16}$$

$$= 4 \times 5 \times 4\pi R^2$$

$$= 4U_1$$

Gain in energy

$$\Delta U = U_2 - U_1$$

$$= 4U_1 - U_1$$

$$= 3U_1$$

$$= 3 \times 5 \times 4\pi R^2$$

$$= 3 \times 0.075 \times 4\pi \times 1 \times 10^{-4}$$

$$= 0.225 \times 4\pi \times 10^{-4}$$

$$= 2.8 \times 10^{-4}\ J$$

Hence correct option is 1

Q. 7 Given below are two statements :

Statement - I : When μ amount of an ideal gas undergoes adiabatic change from state $(P_1,\ V_1,\ T_1)$ to state $(P_2,\ V_2,\ T_2)$, then work done is $W = \frac{\mu R\,(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_p}{C_V}$ and $R =$ universal gas constant.

Statement - II : In the above case, when work is done on the gas, the temperature of the gas would rise.

Choose the correct answer from the options given below:

Option 1:
Both statement-I and statement-II are true.

Option 2:
Both statement-I and statement-II are false.

Option 3:
Statement-I is true but statement-II is false.

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Option 4:
Statement-I is false but statement-II is true.

Correct Answer:
Both statement-I and statement-II are true.

Solution:

$$W = \frac{\mu R\left(T_2 - T_1\right)}{1-\gamma}$$

When work is done on the gas. W is taken as -ve.

Here $(1-\gamma)$ will be negative

$\therefore \mathrm{T}_2 > \mathrm{T}_1$

Temperature will rise

Hence the correct option is 1

Q. 8 Given below are two statements :
Statement-I: A point charge is brought into an electric field. The value of the electric field at a point near the charge may increase if the charge is positive.

Statement II: An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.
Choose the correct answer from the options given below :

Option 1:
Both statement-I and statement-II are true.

Option 2:
Both statement-I and statement-II are false.

Option 3:
Statement-I is true but statement-II is false.

Option 4:
Statement-I is false but statement-II is true.

Correct Answer:
Both statement-I and statement-II are true.

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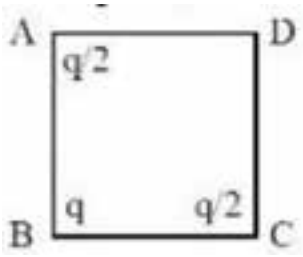
Solution:
For both positive/negative charges, near the charge, the electric field lines density increases therefore field near the charges increases

In a non-uniform electric field, the force on positive and negative charges will be different so the net force is not equal to zero.

Hence the correct option is 1

Q. 9 The three charges $q/2, q \text{ and } q/2$ are placed at the corners $A, B \text{ and } C$ of a square of side $'a'$ as shown in figure. The magnitude of electric field (E) at the corner D of the square, is :



Option 1:
$\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

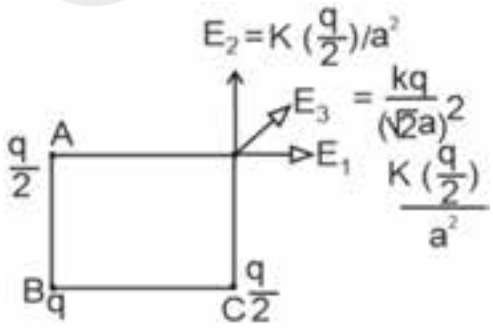
Option 2:
$\frac{q}{4\pi\ \epsilon_0\ a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

Option 3:
$\frac{q}{4\pi\epsilon_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

Option 4:
$\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

Correct Answer:
$\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

Solution:



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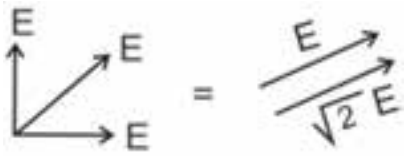
$$\text{Let } \frac{kq}{2a^2} = E$$

$$E_1 = E$$

$$E_2 = E$$

$$E_3 = \frac{kq^2}{2a^2} = E$$



$$E\text{net } = E + \sqrt{2}E$$

$$= (\sqrt{2}+1)\frac{kq}{2a^2}$$

$$= \frac{q}{4\pi\varepsilon_0 a^2}\left[\frac{\sqrt{2}}{2} + \frac{1}{2}\right]$$

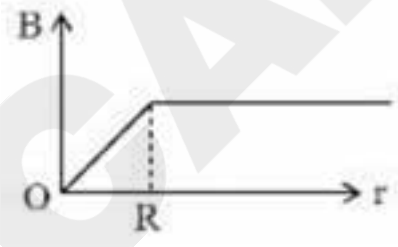
$$= \frac{q}{4\pi\varepsilon_0 a^2}\left[\frac{1}{\sqrt{2}} + \frac{1}{2}\right]$$

Hence correct option is 1

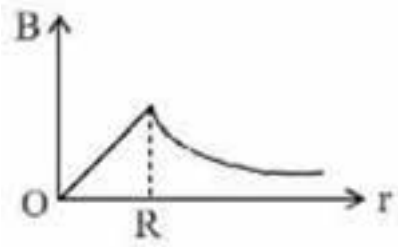
Q. 10 An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.

Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

Option 1:



Option 2:



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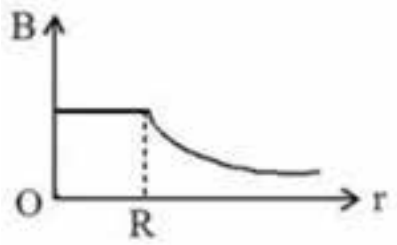
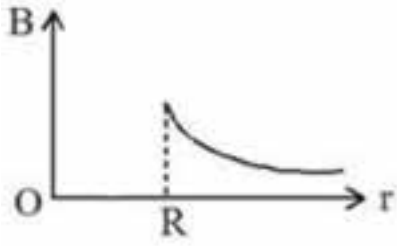
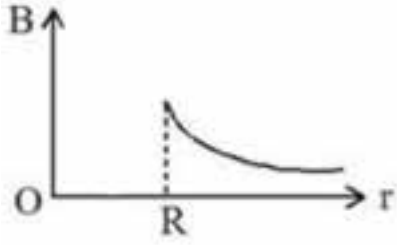
Option 3:



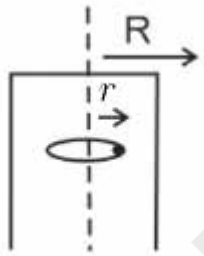
Option 4:



Correct Answer:



Solution:



For $r \leq R$

$$\oint \vec{B} \cdot d\vec{l} = \mu_o I_{enc}$$

$$\mathrm{B} \times 2\pi \mathrm{r} = \mu_o \times 0$$

$$\mathrm{B} = 0$$

For $\mathrm{r} \geq \mathrm{R}$,

$$\oint \vec{B} \cdot d\vec{l} = \mu_o I_{enc}$$

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$$B \times 2\pi r = \mu_o i$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B \propto \frac{1}{r}$$

Hence the correct option is 4

Q. 11 A radar sends an electromagnetic signal of electric field $(E_o) = 2.25\ V/m$ and magnetic field $(B_o) = 1.5 \times 10^{-8}\ T$ which strikes a target on line of sight at a distance of $3\ km$ in a medium. After that, a part of signal (echo) reflects back towards the radar with same velocity and by same path. If the signal was transmitted at time $t = 0$ from radar, then after how much time echo will reach to the radar?

Option 1:

$2.0 \times 10^{-5}\ s$

Option 2:

$4.0 \times 10^{-5}\ s$

Option 3:

$1.0 \times 10^{-5}\ s$

Option 4:

$8.0 \times 10^{-5}\ s$

Correct Answer:

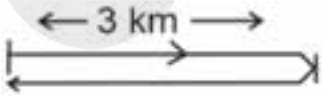
$4.0 \times 10^{-5}\ s$

Solution:

Speed of wave will be

$$v = \frac{E_0}{B_0}$$

$$= \frac{2.25}{1.5 \times 10^{-8}} = \frac{2.25}{1.5} \times 10^8 \frac{m}{s}$$



$$\text{time} = \frac{6\ km}{\left(\frac{2.25}{1.5}\right) \times 10^8}$$

$$= \frac{6 \times 1.5}{2.25} \times \frac{10^3}{10^8}$$

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$= 4 \times 10^{-5} \text{sec}$

Hence the correct option is 2

Q. 12 The refracting angle of a prism is A and refractive index of the material of the prism is $\cot(\mathrm{A}/2)$. Then the angle of minimum deviation will be -

Option 1:
$180 - 2\ \mathrm{A}$

Option 2:
$90 - \mathrm{A}$

Option 3:
$180 + 2\ \mathrm{A}$

Option 4:
$180 - 3\mathrm{A}$

Correct Answer:
$180 - 2\ \mathrm{A}$

Solution:
For Prism

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin(A/2)}$$

$\mu \rightarrow$ refractive index

$\mathrm{A} \rightarrow$ refracting angle of prism

$\delta_{\mathrm{m}} \rightarrow$ minimum deviation angle

$$\mu = \cot\frac{A}{2}$$

$$\cot\left(\frac{\mathrm{A}}{2}\right) = \frac{\sin\left(\frac{\mathrm{A}+\delta_{\mathrm{m}}}{2}\right)}{\sin(\mathrm{A}/2)}$$

$$\frac{\cos\left(\frac{\mathrm{A}}{2}\right)}{\sin(\mathrm{A}/2)} = \frac{\sin\left(\mathrm{A} + \frac{\delta_{\mathrm{m}}}{2}\right)}{\sin(\mathrm{A}/2)}$$

$$\cos\left(\frac{\mathrm{A}}{2}\right) = \sin\left(\frac{\mathrm{A} + \delta_{\mathrm{m}}}{2}\right)$$

$$90^{\circ} - \frac{\mathrm{A}}{2} = \frac{\mathrm{A} + \delta_{\mathrm{m}}}{2}$$

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$$90^\circ - \mathrm{A} = \frac{\delta_\mathrm{m}}{2}$$

$$\delta_\mathrm{m} = 180^\circ - 2\mathrm{A}$$

Hence the correct option is 1

Q. 13 The aperture of the objective is $24.4\ \mathrm{cm}$. The resolving power of this telescope, if a light of wavelength 2440Å is used to see the object will be :

Option 1:
8.1×10^6

Option 2:
10.0×10^7

Option 3:
8.2×10^5

Option 4:
1.0×10^{-8}

Correct Answer:
8.2×10^5

Solution:
$\text{Resolving power (R.P.)}$

$$\text{R.P.} = \frac{1}{1.22\frac{\lambda}{\mathrm{a}}} = \frac{\mathrm{a}}{1.22\lambda}$$

$$= \frac{24.4 \times 10^{-2}}{1.22 \times 2440 \times 10^{-10}}$$

$$= 8.2 \times 10^5$$

Hence the correct option is 3

Q. 14 The de Broglie wavelengths for an electron and a photon are $\lambda_e \text{ and } \lambda_p$ respectively. For the same kinetic energy of electron and photon, which of the following presents the correct relation between the de Broglie wavelengths of two?

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Option 1:
$\lambda_{\mathrm{p}} \propto \lambda_e^2$

Option 2:
$\lambda_{\mathrm{p}} \propto \lambda_e$

Option 3:
$\lambda_{\mathrm{p}} \propto \sqrt{\lambda_e}$

Option 4:
$\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

Correct Answer:
$\lambda_{\mathrm{p}} \propto \lambda_e^2$

Solution:
For Photon

$$\lambda_p = \frac{h}{p} = \frac{hc}{E} \quad \text{.....(1)}$$

For electron

$$\lambda_e = \frac{h}{\sqrt{2mk}} \quad \text{.....(2)}$$

From (1) and (2)

$\lambda_p \alpha \lambda_e^2$

Hence correct answer is option 1

Q. 15 The Q-value of a nuclear reaction and kinetic energy of the projectile particle, $\mathrm{K_p}$ are related as:

Option 1:
$\mathrm{Q} = \mathrm{K_p}$

Option 2:
$(\mathrm{K_p} + \mathrm{Q}) < 0$

Option 3:
$\mathrm{Q} < \mathrm{K_p}$

Option 4:
$(\mathrm{K_p} + \mathrm{Q}) > 0$

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Correct Answer:
$(K_p + Q) > 0$

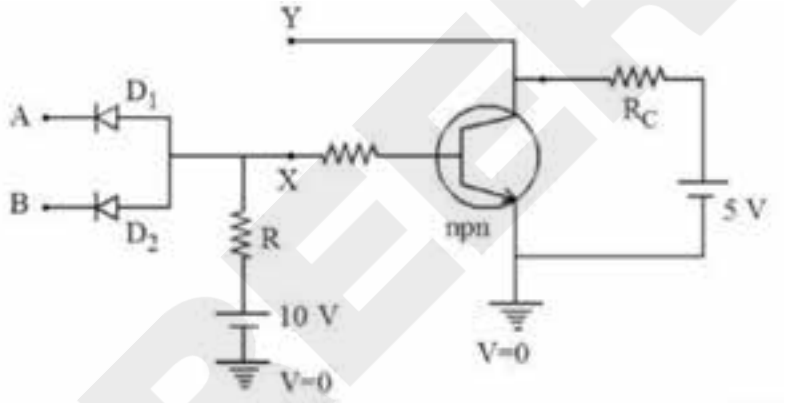
Solution:
Let a, c and d are the other entities involved in the reaction

p is the projectile

$a + p \rightarrow c + d$
$Q = k_c + k_d - k_p$
$Q + k_p = k_c + k_d$
$Q + k_p > 0$

Hence the correct option is 4

Q. 16 In the following circuit, the correct relation between output (Y) and inputs A and B will be :



Option 1:
$Y = AB$

Option 2:
$Y = A + B$

Option 3:
$Y = \overline{AB}$

Option 4:
$Y = \overline{A + B}$

Correct Answer:
$Y = \overline{AB}$

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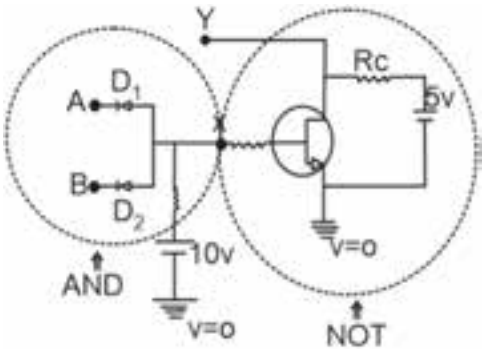
Solution:



Here the circuit can be said to be composed of two parts. The first part is AND gate and the second part is NOT gate realisation. So, overall it is the NAND gate.

$$Y = \overline{AB}$$

Hence the correct option is 3

Q. 17 For using a multimeter to identify a diode from electrical components, choose the correct statement out of the following about the diode :

Option 1:
It is two terminal device which conducts current in both directions.

Option 2:
It is two terminal device which conducts current in one direction only

Option 3:
It does not conduct current gives an initial deflection which decays to zero.

Option 4:
It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

Correct Answer:
It is two terminal device which conducts current in one direction only

Solution:
A diode is two terminal devices which conducts current preferentially in forward biased condition only.

Hence the correct option is 2

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Q. 18 Given below are two statements : One is labelled as **Assertion** A and the other is labelled as **Reason** R.
Assertion A : n-p-n transistor permits more current than a p-n-p transistor.
Reason R : Electrons have greater mobility as a charge carrier.
Choose the correct answer from the options given below :

Option 1:
Both A and R true, and R is correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R true, and R is correct explanation of A.

Solution:
n-p-n transistor permits more current than p-n-p transistor as electrons which are majority charge carriers in n-p-n have higher mobility, than holes. Which are majority carriers in p-n-p transistor

Q. 19 Match List-I with List-II

	List -I		List-II
A.	Television signal	I.	$03\mathrm{KHz}$
B.	Radio signal	II.	$20\mathrm{KHz}$
C.	High Quality Music	III.	$02\mathrm{MHz}$
D.	Human speech	IV	$06\mathrm{MHz}$

Choose the correct answer from the options given below :

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Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-IV, B-III, C-I, D-II

Option 3:
A-IV, B-III, C-II, D-I

Option 4:
A-I, B-II, C-IV, D-III

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:

$$\text{Television signal} = 6\text{MHz}$$

$$\text{Radio signal} = 2\text{MHz}$$

$$\text{High Quality music} = 20\text{KHz}$$

$$\text{Human speech} = 3\text{kHz}$$

Hence correct option is 3

Q. 20 The velocity of sound in a gas, in which two wavelengths $4.08\ \text{m}$ and $4.16\ \text{m}$ produce 40 beats in $12\ \text{s}$, will be :

Option 1:
$282.8\ \text{ms}^{-1}$

Option 2:
$175.5\ \text{ms}^{-1}$

Option 3:
$353.6\ \text{ms}^{-1}$

Option 4:
$707.2\ \text{ms}^{-1}$

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Correct Answer:

$707.2\ \mathrm{ms}^{-1}$

Solution:

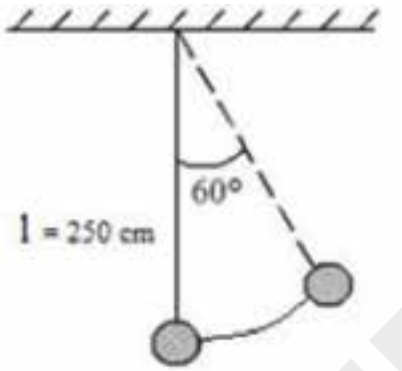
$\text{beat frequeny}\,(\mathrm{f_b}) = |\mathrm{f_1} - \mathrm{f_2}|$

$$\frac{40}{12} = \frac{\mathrm{V}}{4.08} - \frac{\mathrm{V}}{4.16}$$

$$\mathrm{V} = 707.2\ \mathrm{m/s}$$

Hence correct option is 4

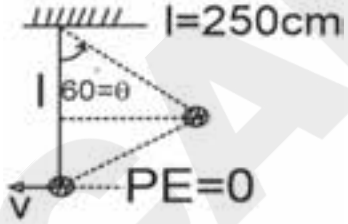
Q. 21 A pendulum is suspended by a string of length $250\ \mathrm{cm}$. The mass of the bob of the pendulum is $200\ \mathrm{g}$. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob, the maximum velocity attained by the bob will be _____ms^{-1}. (if $\mathrm{g} = 10\ \mathrm{m/s^2}$)



Correct Answer:

5

Solution:



Maximum velocity will be when Bob is in a vertical position.

By mechanical energy conversation

$$\mathrm{mgl}\ (1 - \cos\theta) = \frac{1}{2}\mathrm{mv}^2$$

$$10 \times 2.5 \times \left(1 - \frac{1}{2}\right) = \frac{\mathrm{v}^2}{2}$$

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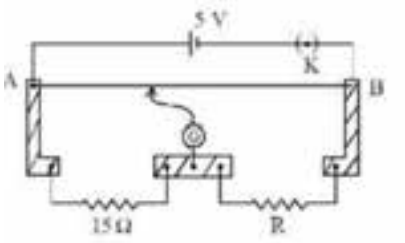
$$25 \times \frac{1}{2} = \frac{v^2}{2}$$

$$v = 5 \text{ m/s}.$$

Hence answer is 5

Q. 22 A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω. The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A is 2 cm, then the determined value of R will be _____Ω.



Correct Answer:
19

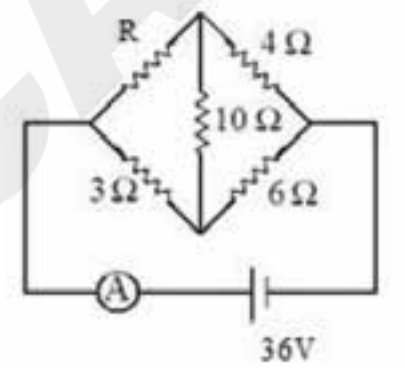
Solution:
Using the balanced wheat stone bridge condition,

$$\frac{43+2}{15} = \frac{57}{R}$$

$$R = \frac{57 \times 15}{45} = 19\Omega$$

Hence answer is 19

Q. 23 Current measured by the ammeter (A) in the reported circuit when no current flows through 10Ω resistance, will be _____.A.



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Correct Answer:
10

Solution:
If no current flows in 10Ω resistor.

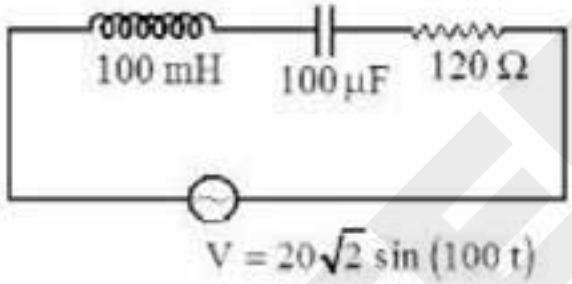
$$\frac{R}{3}=\frac{4}{6}$$

$$R=\frac{12}{6}=2\Omega$$

$$I_A=\frac{36\times(6+9)}{6\times 9}=\frac{36\times 15}{54}=10A$$

Hence answer is 10

Q. 24 An AC source is connected to an inductance of 100mH, a capacitance of $100\mu\text{F}$ and a resistance of 120Ω as shown in figure. The time in which the resistance having a thermal capacity $2\ \text{J}/^\circ\text{C}$ will get heated by 16°C is_____s.



Correct Answer:
15

Solution:
$$X_L=\omega L=100\times 100\times 10^{-3}=10\Omega$$

$$X_C=\frac{1}{\omega C}=\frac{1}{100\times 10^2\times 10^{-6}}=100\Omega$$

$$X_C-X_L=90\Omega$$

$$z=\sqrt{(90)^2+(120)^2}=150\Omega$$

$$I_{rms}=\frac{20}{150}=\frac{2}{15}\ A$$

To increase the temperature of the resistor by 16°C, heat required $=32\text{J}$

$$I^2Rt=32$$

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$$\left(\frac{2}{15}\right)^2 \times 120 \times t = 32$$

$$t = \frac{32 \times 15 \times 15}{4 \times 120}$$

$$t = 15$$

Hence answer is 15

Q. 25 The position vector of $1\ \text{kg object is } \vec{r} = (3\hat{i} - \hat{j})\text{m}$ and its velocity $\vec{v} = (3\hat{j} + \hat{k})\text{ms}^{-1}$. The magnitude of its angular momentum is $\sqrt{x}\text{Nm}$ where x is_____.

Correct Answer:
91

Solution:
$$|\vec{L}| = |\vec{r} \times (m\vec{v})|$$

$$= |(3i - j) \times 1 \times (3j + \hat{k})|$$

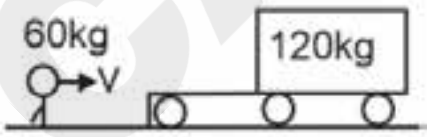
$$= |-i - 3j + 9\hat{k}|$$

$$= \sqrt{91}$$

Hence answer is 91

Q. 26 A man of $60\ \text{kg}$ is running on the road and suddenly jumps into a stationary trolly car of mass $120\ \text{kg}$. Then, the trolly car starts moving with velocity $2\ \text{ms}^{-1}$. The velocity of the running man was_____ ms^{-1}, when he jumps into the car.

Correct Answer:
6

Solution:


$$60 \times v = (120 + 60) \times 2$$

$$60v = 360$$

$$v = 6\ \text{m/s}$$

Hence answer is 6

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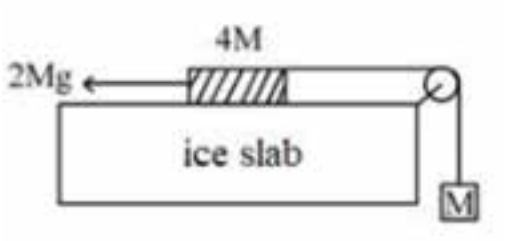
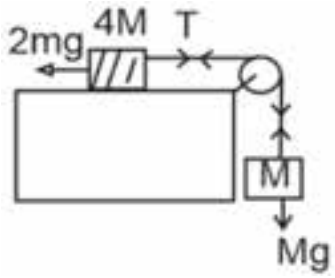
Q. 27 A hanging mass m is connected to a four times bigger mass by using a stringpulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by $2\mathrm{Mg}$ force. In this situation, tension in the string is $\frac{x}{5}\mathrm{Mg}$ for $\mathrm{x} =$ _____ . Neglect mass of the string and friction of the block (bigger mass) with ice slab. (Given $g =$ acceleration due to gravity)



Correct Answer:
6

Solution:



$$\mathrm{a} = \frac{2\mathrm{Mg} - \mathrm{Mg}}{4\mathrm{M} + \mathrm{M}} = \frac{\mathrm{g}}{5}$$

$$\mathrm{T} - \mathrm{Mg} = \mathrm{M} \times \frac{\mathrm{g}}{5}$$

$$\mathrm{T} = \frac{6\mathrm{Mg}}{5}$$

Hence answer is 6

Q. 28 The total internal energy of two mole monoatomic ideal gas at temperature $\mathrm{T} = 300\ \mathrm{K}$ will be_____ J. (Given $\mathrm{R} = 8.31\ \mathrm{J/mol.K}$)

Correct Answer:
7479

Solution:

$$\mathrm{U} = \mathrm{nC_V T} \qquad \mathrm{C_V} = \frac{\mathrm{fR}}{2} = \frac{3\mathrm{R}}{2}$$

$$= 2 \times \frac{3\mathrm{R}}{2} \times 300$$

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$= 3 \times 8.31 \times 300$

$= 7479 \text{ J}$

Hence the correct answer is 7479J

Q. 29 A singly ionized magnesium atom $(\text{A} = 24)$ ion is accelerated to kinetic energy 5keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5T. The radius of path formed will be _____cm.

Correct Answer:
10

Solution:
$\text{R} = \dfrac{\text{mv}}{\text{qB}}$

$= \dfrac{\sqrt{2\text{mKE}}}{\text{qB}}$

$= \sqrt{2 \times 24 \times 1.67 \times 10^{-27} \times 5 \times 1.6 \times 10^{-19}}$

$= 1.6 \times 10^{-19} \times 0.5$

$= 10\text{cm}$

Hence correct answer is 10

Q. 30 A telegraph line of length 100 km has a capacity of $0.01\mu\text{F/km}$ and it carries an alternating current at 0.5kilo cycle per second. If minimum impedance is required, then the value of the inductance that needs to be introduced in series is_____mH. $(\text{if } \pi = \sqrt{10})$

Correct Answer:
100

Solution:
Total Capacitance $= 0.01 \times 100 = 1\mu\text{F}$

$\omega = 500 \times 2\pi = 1000\pi \text{ rad/s}$

$\omega\text{L} = \dfrac{1}{\omega \text{ c}}$

$\text{L} = \dfrac{1}{\omega^2\text{c}} = \dfrac{1}{10^6\pi^2 \times 10^{-6}} = \dfrac{1}{10}\text{H}$

$= 100\text{mH}$

Hence answer is 100

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Chemistry

Q. 1 The **incorrect** statement about the imperfections in solids is:

Option 1:
Schottky defect decreases the density of the substance.

Option 2:
Interstitial defect increases the density of the substance.

Option 3:
Frenkel defect does not alter the density of the substance.

Option 4:
Vacancy defect increases the density of the substance.

Correct Answer:
Vacancy defect increases the density of the substance.

Solution:
As we have learnt,

Schottky defect and Vacancy defect both decrease the density of the solid.

Interstitial defect increases the density of the solid.

Frenkel defect does not change the density of the solld.

So, statements given in Option (1), (2) and (3) are correct but that in Option (4) is incorrect

Hence, **Option(4)** is the answer.

Q. 2 The Zeta potential is related to which property of colloids?

Option 1:
Colour

Option 2:
Tyndall effect

Option 3:
Charge on the surface of colloidal particles

Option 4:
Brownian movement

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Correct Answer:
Charge on the surface of colloidal particles

Solution:
The Zeta potential is related to charge on the surface of colloidal particles.

Hence, **Option (3)** is correct.

Q. 3 Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is

Option 1:
$3s^2, 3p^4$

Option 2:
$3d^{10}, 4s^2, 4p^4$

Option 3:
$4d^{10}, 5s^2, 5p^4$

Option 4:
$2s^2, 2p^4$

Correct Answer:
$3s^2, 3p^4$

Solution:
The element "E" will be Se (Selenium), it belongs to the period 4 and group 16 of the periodic table.

Just above Se, S (sulfur) is present. The valence shell electronic configuration of Sulfur is $3s^2, 3p^4$

Hence, **Option (1)** is correct

Q. 4 Given are two statements one is labelled as **Assertion A** and other is labelled as **Reason R.**

Assertion A: Magnesium can reduce Al_2O_3 at a temperature below $1350^\circ C$, while above $1350^\circ C$ aluminium can reduce MgO.

Reason R: The melting and boiling points of magnesium are lower than those of aluminium.

In light of the above statements, choose most appropriate answer from the options given below:

Option 1:
Both **A** and **R** are correct, and **R** is correct explanation of **A.**

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Option 2:
Both **A** and **R** are correct, but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct **R** is not correct.

Option 4:
A is not correct, **R** is correct.

Correct Answer:
Both **A** and **R** are correct, but **R** is NOT the correct explanation of **A.**

Solution:
As we have learnt,

In Ellingham diagram, Mg can reduce Al_2O_3 at a temperature below $1350^{\circ}C$, while above $1350^{\circ}C$ Al can reduce MgO. The melting and boiling points of magnesium are lower than those of Aluminium.

$$\text{M.P. of Mg} = 924\ \text{K}, \text{B.P.of Mg} = 2745\ \text{K}$$
$$\text{M.P. of Al=933 K}, \text{B.P. of Al} = 2740\ \text{K}$$

Both Assertion A and Reason R are correct. But R is not the correct explanation of A, because A can be explained by Gibb's free energy.

Hence, **Option (2)** is correct.

Q. 5 Dihydrogen reacts with CuO to give

Option 1:
CuH_2

Option 2:
Cu

Option 3:
Cu_2O

Option 4:
$Cu(OH)_2$

Correct Answer:
Cu

Solution:
Reaction will be

$$H_2 + CuO \longrightarrow Cu + H_2O$$

Hence, **Option (2)** will be correct.

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Q. 6 Nitrogen gas is obtained by thermal decomposition of

Option 1:
$\mathrm{Ba\,(NO_3)_2}$

Option 2:
$\mathrm{Ba\,(N_3)_2}$

Option 3:
$\mathrm{NaNO_2}$

Option 4:
$\mathrm{NaNO_3}$

Correct Answer:
$\mathrm{Ba\,(N_3)_2}$

Solution:
Very pure Nitrogen can be obtained by the thermal decomposition of Sodium or Barium azide.

$$\mathrm{Ba\,(N_3)_2 \rightarrow Ba + 3N_2}$$

Hence, **Option (2)** is correct.

Q. 7 Given below are two statements:

Statement **I:** The pentavalent oxide of group-15 element, $\mathrm{E_2O_5}$, is less acidic than trivalent oxide, $\mathrm{E_2O_3}$, of the same element.

Statement **II:** The acidic character of trivalent oxide of group 15 elements, $\mathrm{E_2O_3}$, decreases down the group.

In light of the above statement, choose **most appropriate** answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I true, but statement II is false.

Option 4:
Statement I is false but statement II is true.

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Correct Answer:
Statement I is false but statement II is true.

Solution:
E_2O_5 will be more acidic than E_2O_3 because in E_2O_5 , E has +5 oxidation state and in E_2O_3, E has +3 oxidation state. Cation in higher oxidation state more will be acidic.

Statement (1) is wrong

Acidic character of Oxides decreases down the group because metallic nature increases down the group.

So, **Option (4)** is correct.

Q. 8 Which one of the lanthanoids given below is the most stable in divalent from?

Option 1:
Ce (Atomic Number 58)

Option 2:
Sm (Atomic Number 62)

Option 3:
Eu (Atomic Number 63)

Option 4:
Yb (Atomic Number 70)

Correct Answer:
Eu (Atomic Number 63)

Solution:
Fact based.

Eu is most stable in divalent form among them.

$E^0_{M^{3+}|M^{2+}} : Eu(-0.35\ V), Yb(-1.05\ V)$

Hence, **Option (3)** is correct.

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Q. 9 Given below are two statements:

Statement **I:** $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[Ni(CO)_4]$ is tetrahedral, paramagnetic and with sp^3- hybridication for Ni.

Statement **II:** $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ both have same d-electron configuration, have same geometry and are paramagnetic.

In light the above statements, choose the **correct** answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but statement II is false.

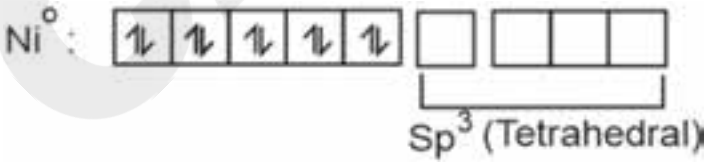
Option 4:
Statement I is incorrect but statement II is true.

Correct Answer:
Both Statement I and Statement II are false.

Solution:
$[Ni(CN)_4]^{2-}$: d^8 configuration due to SFL (CN^-)

Zero unpaired $e^{\ominus}$: (diamagnetic)

$[Ni(CO)_4]$: d^{10} configuration due to SFL (CO) and excitation of $e^{\ominus}$



Zero unpaired $e^{\ominus}$: diamagnetic

$[NiCl_4]^{2-}$: d^8 configuration due to WFL $(Cl^{\ominus})$

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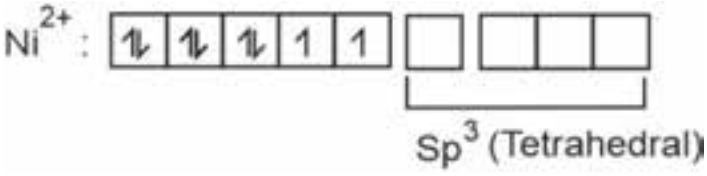

2 unpaired $e^{\ominus}$: Paramagnetic

Both statements I and II are false.

Hence, **Option (2)** is correct

Q. 10 Which amongst the following is **not** a pesticide?

Option 1:
DDT

Option 2:
Organophosphates

Option 3:
Dieldrin

Option 4:
Sodium arsenite

Correct Answer:
Sodium arsenite

Solution:
Among the given compounds, Pesticides are DDT, Organo-phosphates and Dieldrin.

Sodium arsenite is not a pesticide.

Hence,**Option (4)** is correct.

Q. 11 Which one of the following techniques is **not** used to spot components of a mixture separated on thin layer chromatographic plate?

Option 1:
I_2 (Solid)

Option 2:
U.V. Light

Option 3:
Visualisation agent as a component of mobile phase

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Option 4:
Spraying of an appropriate reagent

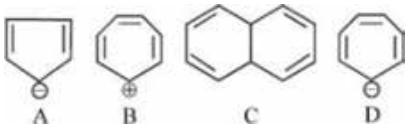
Correct Answer:
Visualisation agent as a component of mobile phase

Solution:
Visualisation agent as a component of mobile phase is not used to spot components of a mixture separated on thin layer chromatographic plate.

$I_2(solid)$, U.V.right and spraying of ninhydrin solution used to spot the components of a mixture.

Hence, **Option (3)** is correct.

Q. 12 Which of the following structures are aromatic in nature?



Option 1:
A,B,C and D

Option 2:
Only A and B

Option 3:
Only A and C

Option 4:
Only B,C and D

Correct Answer:
Only A and B

Solution:
As we have learnt,

Cyclic, planar, completely conjugated systems having $(4n+2)\pi$ electrons in complete conjugation are aromatic in nature.

Among the given species, A and B satify all the above conditions and hence, they are aromatic.

C is not completely conjugated and is Non Aromatic while D has 8π electrons in conjugation and is Anti Aromatic.

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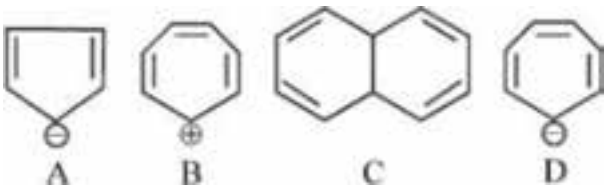
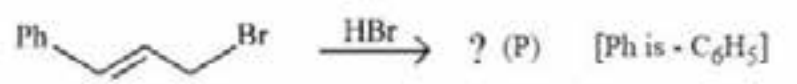


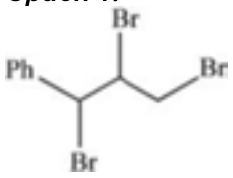
So, A and B are aromatic.

Hence, **Option (2)** is correct.

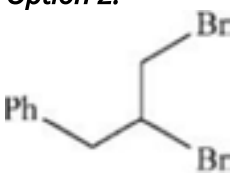
Q. 13 The major product (p) in the reaction



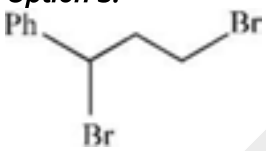
Option 1:



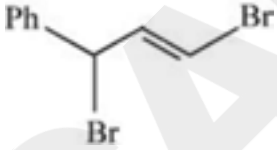
Option 2:



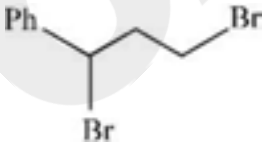
Option 3:



Option 4:



Correct Answer:



Solution:

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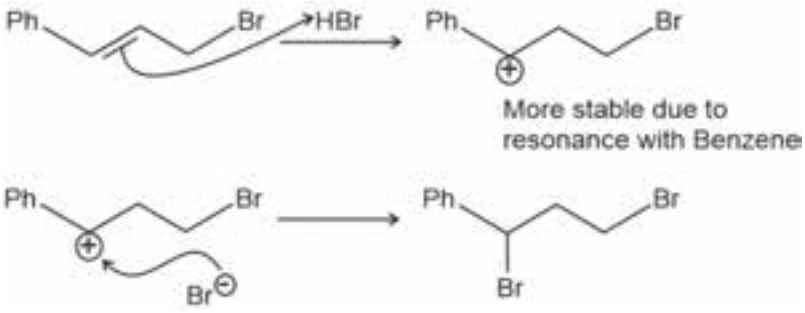
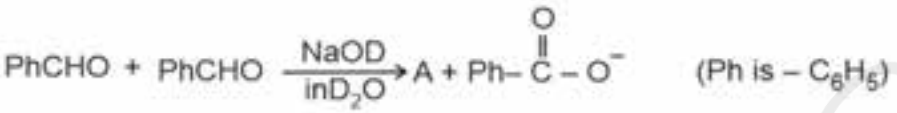
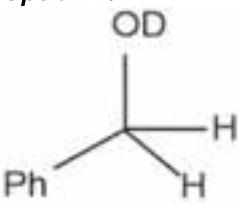


Hence, **Option (3)** is correct.

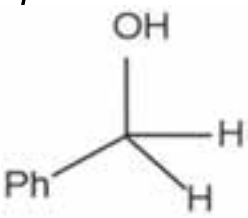
Q. 14 The correct structure of product 'A' formed in the following reaction,



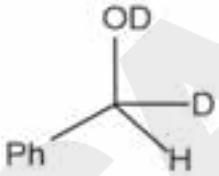
Option 1:



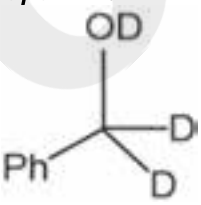
Option 2:



Option 3:



Option 4:



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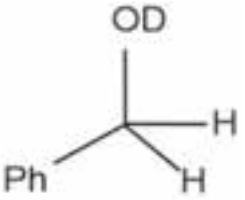
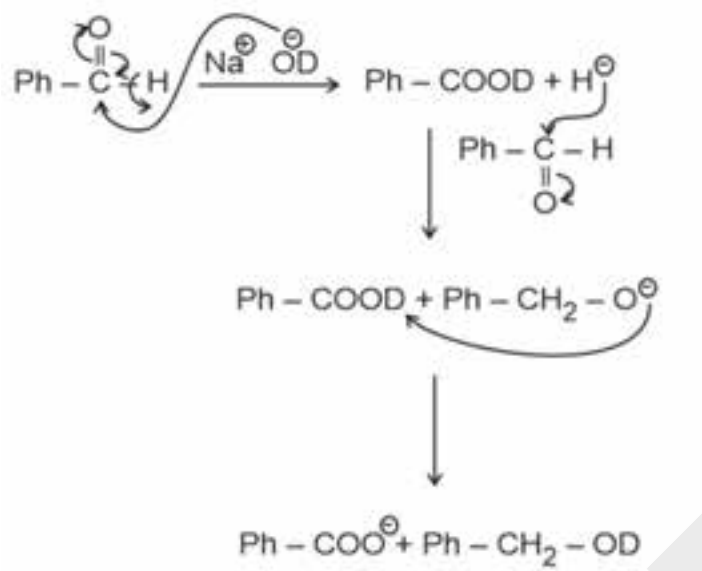
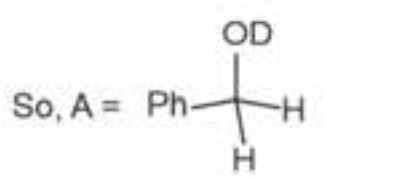
Correct Answer:



Solution:

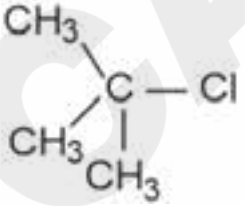




Hence, **Option (1)** is correct.

Q. 15 Which one of the following compounds is inactive towards S_N1 reaction?

Option 1:



Option 2:

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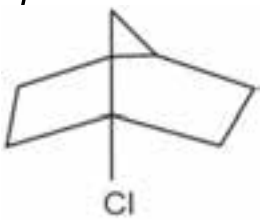
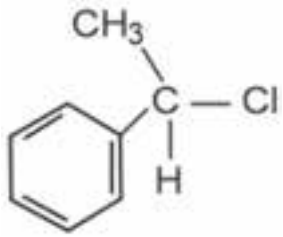
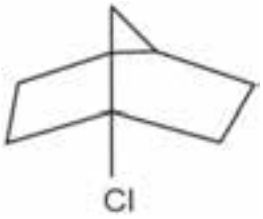
Option 3:



Option 4:



Correct Answer:



Solution:

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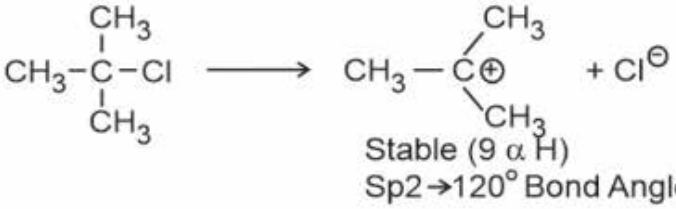
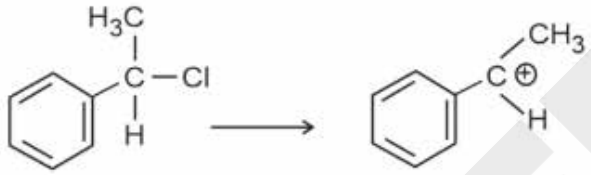
1) 

Stable (9 α H)
Sp2 → 120° Bond Angle (planar)

2)

Resonance
(stable)

3)
Very unstable
due to $Sp^2 \rightarrow 120^\circ$ bond angle
not possible
due to structure.
Not favour making carbocation
so, not favor for Sn^1 reaction.

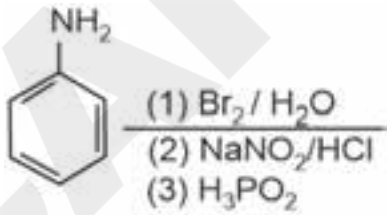
4) 

Stable (resonance)
inactive towards SN^1 reaction

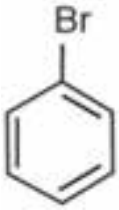
Compound (3) will inactive towards S_N1 reaction

Hence, **Option (3)** is correct.

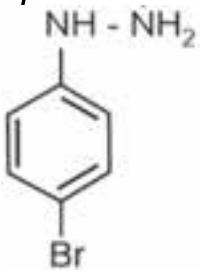
Q. 16 Identify the major product formed in the following sequence of reactions:



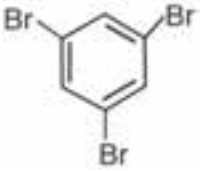
Option 1:



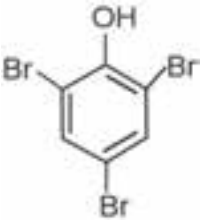
Option 2:



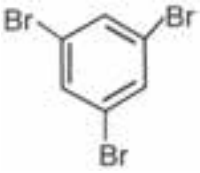
Option 3:



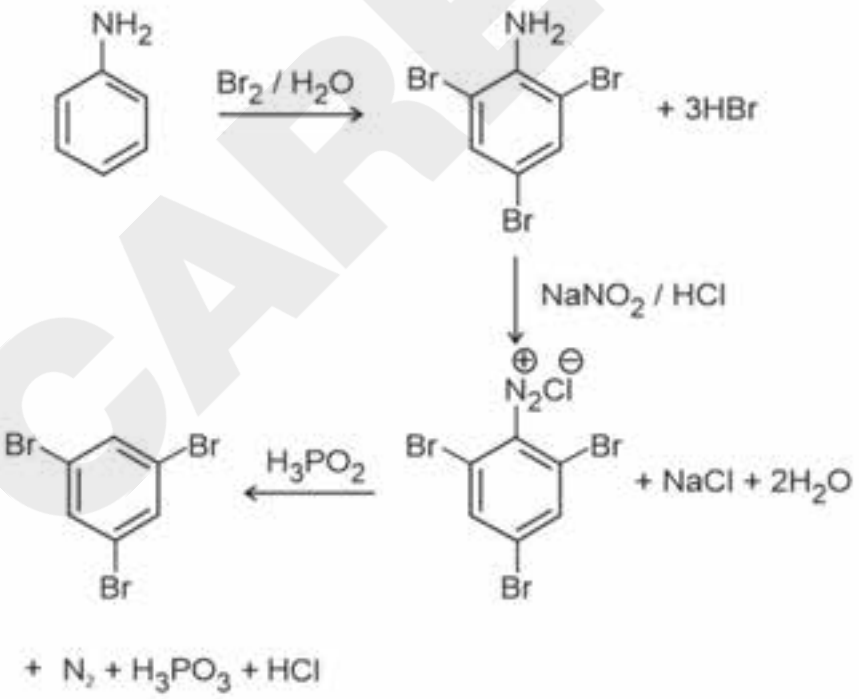
Option 4:



Correct Answer:



Solution:



Hence,**Option (3)** is correct.

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Q. 17 A primary aliphatic amine on reaction with nitrous acid in cold $(273\ \mathrm{K})$ and there after raising temperature of reaction mixture to room temperature $(298\ \mathrm{K})$, gives a/an

Option 1:
nitrile

Option 2:
alcohol

Option 3:
diazonium salt

Option 4:
secondary amine

Correct Answer:
alcohol

Solution:

$$\mathrm{R-NH_2 + HNO_2 \xrightarrow[(273\ K)]{NaNO_2+HCl} R-N_2^{\oplus}Cl^{\ominus} \xrightarrow[(298\ K)]{H_2O} \underset{(alcohol)}{ROH} + N_2 + HCl}$$

Hence,**Option (2)** is correct.

Q. 18 Which one of the following is **NOT** a copolymer?

Option 1:
Buna-S

Option 2:
Neoprene

Option 3:
PHBV

Option 4:
Butadiene-styrene

Correct Answer:
Neoprene

Solution:
The polymers made from two or more than two monomers, is called copolymers.

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Neoprene is polychoroprene and it is not a copolymer. Neoprene is made of one monomer chloroprene.

Hence,**Option (2)** is correct.

Q. 19 Stability of α - Helix structure of proteins depends upon

Option 1:
dipolar interaction

Option 2:
H-bonding interaction

Option 3:
van der Waals forces

Option 4:
π—stacking interaction

Correct Answer:
H-bonding interaction

Solution:
As we have learnt,

The $\alpha - \text{Helix structure}$ of proteins is stabilised by H-Bonding interaction.

Hence, **Option (2)** is correct.

Q. 20 The formula of the purple colour formed in Laissaigne's test for sulphur using sodium nitroprusside is

Option 1:
$\text{NaFe}\left[\text{Fe(CN)}_6\right]$

Option 2:
$\text{Na}\left[\text{Cr}\left(\text{NH}_3\right)_2\left(\text{NCS}\right)_4\right]$

Option 3:
$\text{Na}_2\left[\text{Fe(CN)}_5\text{(NO)}\right]$

Option 4:
$\text{Na}_4\left[\text{Fe(CN)}_5\text{(NOS)}\right]$

Correct Answer:
$\text{Na}_4\left[\text{Fe(CN)}_5\text{(NOS)}\right]$

Solution:

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The purple or violet colouration due to the formation of the complex compound $Na_4[Fe(CN)_5NOS]$ confirms the presence of sulphide ion in the salt.

$$\underset{\text{Sodium nitroprusside}}{Na_2[Fe(CN)_5NO]} + Na_2S \rightarrow \underset{\text{Purple complex}}{Na_4[Fe(CN)_5NOS]}$$

Hence,**Option (4)** is correct.

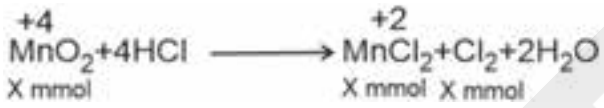
Q. 21 A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2. The Cl_2 gas is passed into a solution of KI and $60.0\ mL$ of $0.1M Na_2S_2O_3$ is required to titrate the liberated iodine, The percentage of MnO_2 in the sample is _____. (Nearest integer)

$[\text{Atomic masses (in u)} Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23,\ K = 39, S = 32]$

Correct Answer:
13

Solution:
We have to find the mass of MnO_2 for that we need to find the moles of MnO_2.

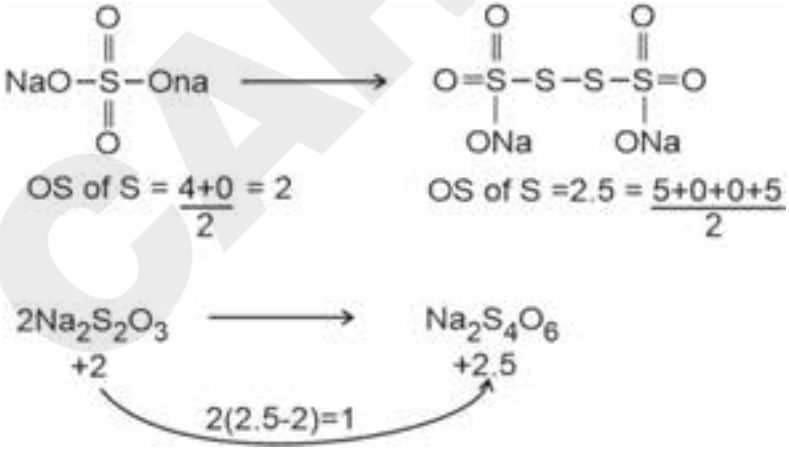
(1)



(2) $Cl_2 + 2kI \longrightarrow 2KC1 + I_2$
x mmol x mmol

(3) $I_2 + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2NaI$
x mmol x mmol

Now, $Na_2S_2O_3$ n-factor will be



So, $x\ \text{mmol} = 60\text{ml} \times 0.1\text{M} \times \text{n factor}$

$$x = 60\text{ml} \times 0.1\text{M} \times L$$

$$x = 6\ \text{mmol}$$

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$$x = 6$$

So, from Reaction $(1), (2) \text{and} (3),$ used mmol of MnO_2 will 6 mmol,
mole of $MnO_2 \times n - \text{factor}^{a} = 6\text{mmol}$

$$\text{mole} \times (4-2) = 6 \times 10^{-3}$$
$$\text{mole of } MnO_2 = 3 \times 10^{-3}$$

$$\Rightarrow \text{ Mass of } MnO_2 = \text{ mole } \times \text{ molar mass}$$
$$= 3 \times 10^{-3} \times (55 + 16 \times 2)$$
$$= 3 \times 10^{-3} \times 87$$
$$= 261 \times 10^{-3} \text{ gram}$$

$$\Rightarrow \% \text{ of } MnO_2 = \frac{\text{mass of } MnO_2}{\text{Mass of Sample}} \times 100$$
$$= \frac{261 \times 10^{-3}}{2} \times 100$$
$$= 13.05$$
$$\approx 13$$

Ans=13

Q. 22 If the work function of a metal is 6.63×10^{-19} J, the maximum wavelength of the photon required to remove a photoelectron from the metal is _____ nm. (Nearest integer)

$$[\text{ Given } : h = 6.63 \times 10^{-34} \text{ J s, and } c = 3 \times 10^{8} \text{ m s}^{-1} \text{ }]$$

Correct Answer:
300

Solution:
We know,

$$\phi = \frac{hc}{\lambda}$$
$$\lambda = \frac{hc}{\phi} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{6.63 \times 10^{-19}} \text{ m}$$
$$\lambda = 3 \times 10^{-7} \text{ m}$$
$$\lambda = 300 \times 10^{-9} \text{ m}$$
$$\lambda = 300 \text{ nm}$$

Answer is 300

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Q. 23 The hybridization of P exhibited in PF_5 is sp^xd^y.

The value of y is _____

Correct Answer:
1

Solution:
The hybridisation of P in PF_5 is sp^3d

So, $y = 1, x = 3$

Answer is 1

Q. 24 4.0 L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 20 L. The amount of heat absorbed in the expansion is _____L atm.

Correct Answer:
0

Solution:
In vaccum, for Free expansion.

$$\text{Pext } = 0, w = 0, q = 0, \Delta U = 0, \Delta T = 0$$

So, the amount of heat aborbed in this expansion is **Zero L atm.**

Hence, the answer is 0

Q. 25 The vapour pressures of two volatile liquids A and B at $25^\circ C$ are 50 Torr and 100 Torr., respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is _____.

Correct Answer:
14

Solution:

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From Raoult's Law

$$P_A = P_A^0 X_A$$

$$P_B = P_B^0 X_B$$

Thus,

$P_A = 50 \times 0.3 = 15 \text{ Torr}$

$P_B = 100 \times 0.7 = 70 \text{ Torr}$

Now, mole fraction of B in vapor phase, y_B is given as

$$y_B = \frac{P_B}{P_A + P_B} = \frac{70}{85} = \frac{14}{17}$$

Answer is 14

Q. 26 The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23}. If specific conductance of he solution is $3 \times 10^{-5} Sm^{-1}$, the limiting molar conducivity of the solution is $x \times 10^{-3}$ S m^2 mol^{-1}. The value of x is _____.

Correct Answer:
3

Solution:
Let the solubility of A_2X_3 be s molL^{-1}.

We have,

$$A_2X_3 \longrightarrow 2A^{3+} + 3X^{2-}$$

$$2s \quad 3s$$

k_{sp} of $A_2X_3 = (2s)^2(3s)^3$

$1.1 \times 10^{-23} = 108s^5$

$s = 1 \times 10^{-5}$ mol/L

$$s = \frac{10^{-5}}{10^{-3}} \text{ mol/m}^3 \quad (1 \text{ L} = 10^{-3} \text{ m}^3)$$

$s = 10^{-2}$ mol/m^3

Now for a sparingly soluble salt,

$$\Lambda_m^\infty = \frac{K}{C} \text{ Sm}^2\text{mol}^{-1} \qquad [\text{all terms in SI units}]$$

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$$\Lambda_m^\infty = \frac{3 \times 10^{-5}}{10^{-2}} \text{ Sm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^\infty = 3 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

$$x = 3$$

Answer is 3

Q. 27 The quantity of elctricity in Faraday needed to reduce $1 \text{ mol of } Cr_2O_7^{-2} \text{ to } Cr^{3+}$ is _____.

Correct Answer:
6

Solution:
given,

$$Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$$

$$Cr = +6 \quad Cr = +3$$

$$\text{n-factor } = 2(6-3) = 6$$

So, 6 moles of electrons are required to reduce 1 mole of $Cr_2O_7^{2-}$

$$1 \text{ mol of electron} = 1 \text{ Faraday.}$$

$$6 \text{ mol of } e^{\ominus} = 6 \text{ Faraday.}$$

Answer is 6

Q. 28 For a first order reaction $A \rightarrow B$, the rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. The time required for 67% completion of reaction is $x \times 10^{-1}$ times the half life of reaction. The value of x is _____ (Nearest integer)

$$(\text{Given : } \log 3 = 0.4771 \text{ })$$

Correct Answer:
16

Solution:
For the first order reaction,

$$kt = \ln\left(\frac{a_0}{a_t}\right) \text{ and } t_{1/2} = \frac{\ln 2}{k}$$

Given, $t_{67\%} = x \times 10^{-1} \times t_{1/2}$

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So, we can write

$$\Rightarrow \frac{\ln\left(\frac{a_0}{a_0-0.67a_0}\right)}{k} = x \times 10^{-1} \times \frac{\ln 2}{k}$$

$$\Rightarrow \frac{\ln\left(\frac{a_0}{0.33a_0}\right)}{k} = x \times 10^{-1} \times \frac{\ln 2}{k}$$

$$\Rightarrow \frac{\ln 3}{\ln 2} = x \times 10^{-1}$$

$$\Rightarrow \frac{\log 3}{\log 2} = x \times 10^{-1}$$

$$\Rightarrow \frac{0.4771}{0.3010} = x \times 10^{-1}$$

$$\Rightarrow \quad x \times 10^{-1} = 1.585$$

$$\Rightarrow \quad x = 15.85$$

$$\Rightarrow \quad x = 16 \text{ (Nearest Integer)}$$

Answer is 16

Q. 29 Number of complexes which will exhibit synergic bonding amongst, $[Cr(CO)_6]$, $[Mn(CO)_5]$ and $[Mn_2(CO)_{10}]$ is _____.

Correct Answer:
3

Solution:
As we have learnt,

The metal-carbon bond in metal carbonyls possess both σ and π character. The M-C σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of the metal.

The M-C π bond is formed by the donation of a pair of electrons from a filled d orbital of Metal into the vacant antibonding π^* orbital of carbon monoxide.

The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal. This bonding in metal carbonyls is called synergic bonding.

Since, all three complexes are metal carbonyl complexes, all of them would exhibit synergic bonding.

Hence answer is 3

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Q. 30 In the estimation of bromine, $0.5\ \text{g}$ of an organic compound gave $0.40\ \text{g}$ of silver bromide. The percentage of bromine in the given compound is ____% (nearest integer)

(Relative atomic masses of Ag nad Br are 108u and 80u, respectively).

Correct Answer:
34

Solution:
$0.5\ \text{g}$ of an organic compound gave $0.40\ \text{g}$ of AgBr.

$$\text{mole of Br} = \text{mole of AgBr} = \frac{0.4}{108+80} = \frac{0.4}{188}$$

$$\Rightarrow \text{mass of Br} = \frac{0.4}{188} \times 80 = 0.17$$

So, in 0.5 gram of an organic compound,0.17 gram of Br is present

$$\% \text{ Br in compound} = \frac{0.17}{0.5} \times 100\%$$

$$= 34\%$$

Answer is 34 (nearest integer)

Maths

Q. 1 If $\sum_{k=1}^{31} \left({}^{31}\text{C}_k\right)\left({}^{31}\text{C}_{k-1}\right) - \sum_{k=1}^{30} \left({}^{30}\text{C}_k\right)\left({}^{30}\text{C}_{k-1}\right) = \frac{\alpha\,(60!)}{(30!)\,(31!)}$, where $\alpha \in \text{R}$, then the value of 16α is equal to

Option 1:
1411

Option 2:
1320

Option 3:
1615

Option 4:
1855

Correct Answer:
1411

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Solution:

$$\sum_{k=1}^{31} {}^{31}C_k \, {}^{31}C_{k-1} - \sum_{k=1}^{30} {}^{30}C_k \, {}^{30}C_{k-1}$$

$$= \left({}^{31}C_1 \, {}^{31}C_0 + {}^{31}C_2 \, {}^{31}C_1 + \cdots {}^{31}C_{31} \, {}^{31}C_{30}\right) - \left({}^{30}C_1 \, {}^{30}C_0 + {}^{30}C_2 \, {}^{30}C_1 + \cdots {}^{30}C_{30} \, {}^{30}C_{20}\right)$$
$$= \left({}^{31}C_1 \, {}^{31}C_{31} + {}^{31}C_2 \, {}^{31}C_{30} + \cdots {}^{31}C_{31} \, {}^{31}C_1\right) - \left({}^{30}C_1 \, {}^{30}C_{30} + {}^{30}C_2 \, {}^{30}C_{29} + \cdots {}^{30}C_{30} \, {}^{30}C_1\right)$$
$$= {}^{62}C_{32} - {}^{60}C_{31}$$

$$= \frac{62!}{32!30!} - \frac{60!}{31!29!} = \frac{60!\left[62 \times 61 - 32 \times 30\right]}{32!30!}$$
$$= \frac{60!}{30!31!}\left(\frac{62 \times 61 - 32 \times 30}{32}\right)$$

$$\Rightarrow \alpha = \frac{31 \times 61 - 16 \times 30}{16}$$
$$\Rightarrow 16\alpha = 1891 - 480 = 1411$$

Q. 2 Let a function $f : N \to N$ be defined by

$$f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \ldots \\ n - 1, & n = 3, 7, 11, 15, \ldots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \ldots . \end{cases}$$

then ,f is

Option 1:
one-one but not onto

Option 2:
onto but not one-one

Option 3:
neither one-one nor onto

Option 4:
one-one and onto

Correct Answer:
one-one and onto

Solution:

n	1	2	3	4	5	6	7	8	9
$f(n)$	1	4	2	8	3	12	6	16	5

As can be seen , All the odd numbers are given by the third sequence

All the $4n$ type numbers are given by the first sequence
All the $4n+2$ type numbers are given by the second sequence

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So the function is one-one and onto.

Option (D)

Q. 3 If the system of linear equations

$2x + 3y - z = -2$
$x + y + z = 4$
$x - y + |\lambda|z = 4\lambda - 4$
where $\lambda \in \mathbb{R}$ has no solution, then

Option 1:
$\lambda = 7$

Option 2:
$\lambda = -7$

Option 3:
$\lambda = 8$

Option 4:
$\lambda^2 = 1$

Correct Answer:
$\lambda = -7$

Solution:
For No solution,
$\Delta = 0 \quad \text{But} \quad \Delta_x \text{ or } \Delta_y \text{ or } \Delta_z$ is non zero

$$\Rightarrow \Delta = \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & |\lambda| \end{vmatrix} = 0$$

$$\Rightarrow 2|\lambda| + 3 + 1 + 1 + 2 - 3|\lambda| = 0 \Rightarrow |\lambda| = 7 \Rightarrow \lambda = \pm 7$$

$$\Delta_z = \begin{vmatrix} 2 & 3 & -2 \\ 1 & 1 & 4 \\ 1 & -1 & 4\lambda - 4 \end{vmatrix} = 8\lambda - 8 + 12 + 2 + 2 + 8 - 12\lambda + 12$$
$$= -4\lambda + 28$$

$\text{For } \lambda = 7, \Delta_z = 0, \quad \text{For } \lambda = -7, \Delta_z \neq 0$
So $\lambda = -7$, No solution

Option (B)

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Q. 4 Let A be a matrix of order 3×3 and $\det(A) = 2$. Then $\det\left(\det(A)\ \text{adj}\left(5\,\text{adj}\left(A^3\right)\right)\right)$ is equal to _____.

Option 1:
512×10^6

Option 2:
256×10^6

Option 3:
1024×10^6

Option 4:
256×10^{11}

Correct Answer:
512×10^6

Solution:
$\det\left(2\,\text{adj}\left(5\,\text{adj}\left(A^3\right)\right)\right)$

$= 2^3\left|\text{adj}\left(5\,\text{adj}\left(A^3\right)\right)\right|$
$= 2^3\left|5\,\text{adj}\left(A^3\right)\right|^2$
$= 2^3\left(5^3\left|\text{adj}\left(A^3\right)\right|\right)^2$
$= 2^3\,5^6\left(\left|A^3\right|^2\right)^2 = 2^3 5^6\ |A|^{12} = 2^3\,5^6\,2^{12}$
$= 2^{15}\,5^6 = 2^9\,10^6 = 512 \times 10^6$

Option (A)

Q. 5 The total number of 5-digit numbers, formed by using the digits 1,2,3,5,6,7 without repetition, which are multiple of 6, is

Option 1:
36

Option 2:
48

Option 3:
60

Option 4:
72

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Correct Answer:
72

Solution:
The number must be multiple of both 2 & 3

For 3, sum of digits must be divisible by 3

$1+2+3+5+6+7=24$, so can remove 3 or 6

$(1,2,5,6,7) \rightarrow \quad -----[2] = 2 \times 4! = 2 \times 24 = 48$
$(1,2,3,5,7) \rightarrow \quad -----[2] = 4! = 24$

Total $= 48 + 24 = 72$

Option (D)

Q. 6 Let $A_1, A_2, A_3, \cdots$ be an increasing geometric progresssion of positive real numbers. If $A_1A_3A_5A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then the value of $A_6 + A_8 + A_{10}$ is equal to

Option 1:
33

Option 2:
37

Option 3:
43

Option 4:
47

Correct Answer:
43

Solution:
Let $A_1 = a$ & common ratio $= r$.

$$a \cdot ar^2 \cdot ar^4 \cdot ar^6 = \frac{1}{1296} \Rightarrow a^4r^{12} = \frac{1}{1296} \Rightarrow a^2r^6 = \frac{1}{36} \Rightarrow ar^2 = \frac{1}{6}$$

$$\Rightarrow A_4 = \frac{1}{6}, \quad A_2 + A_4 = \frac{7}{36}$$

$$\Rightarrow A_2 = \frac{7}{36} - \frac{1}{6} = \frac{1}{36} = ar \Rightarrow r^2 = 6 \Rightarrow r = \sqrt{6}, a = \frac{1}{36\sqrt{6}}$$

$$A_6 + A_8 + A_{10} = ar^5 + ar^7 + ar^9 = \frac{1}{36\sqrt{6}}\left(6^2\sqrt{6} + 6^3\sqrt{6} + 6^4\sqrt{6}\right)$$

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$$= 1 + 6 + 6^2 = 43$$

Option (C)

Q. 7 Let $[t]$ denote the greatest integer less than or equal to t. Then, the value of the integral $\int_0^1 \left[-8x^2 + 6x - 1\right] dx$ is equal to

Option 1:
-1

Option 2:
$\frac{-5}{4}$

Option 3:
$\frac{\sqrt{17} - 13}{8}$

Option 4:
$\frac{\sqrt{17} - 16}{8}$

Correct Answer:
$\frac{\sqrt{17} - 13}{8}$

Solution:
Let $f(x) = -8x^2 + 6x - 1, \quad f(0) = -1, f(1) = -3,$

$$f\left(\frac{3}{8}\right) = -8 \times \frac{9}{64} + 6 \times \frac{3}{8} - 1 = \frac{1}{8} \Rightarrow f(x) \in [-3, 1/8] \text{in } x \in [0, 1].$$

So when
$-8x^2 + 6x - 1 = -3;\ 8x^2 - 6x - 2 = 0 \Rightarrow 4x^2 - 3x - 1 = 0 \Rightarrow 4x^2 - 4x + x - 1 = 0$
$\Rightarrow x = 1 \text{ OR } x = -1/4.$

$$\text{When } -8x^2 + 6x - 1 = -2; \Rightarrow 8x^2 - 6x - 1 = 0 \Rightarrow x = \frac{6 \pm \sqrt{36 + 32}}{16} = \frac{3 \pm \sqrt{17}}{8}$$
$\text{When } -8x^2 + 6x - 1 = -1 \Rightarrow -8x^2 + 6x = 0 \Rightarrow x = 0 \quad \text{oR} \quad x = 3/4.$
$\text{When } -8x^2 + 6x - 1 = 0 \Rightarrow x = 1/4 \text{ or } 1/2.$

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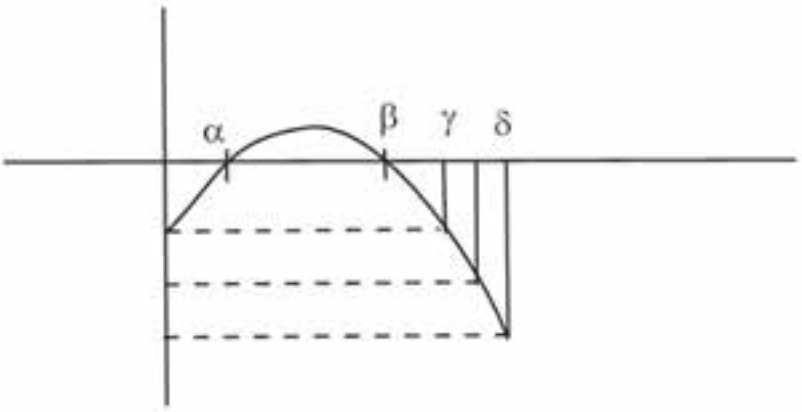

$$\text{So } \int_0^1 \left[-8x^2+6x-1\right] dx$$
$$= \int_0^{1/4} (-1)dx + \int_{1/4}^{1/2} 0dx + \int_{1/2}^{3/4} (-1)dx + \int_{3/4}^{\frac{3+\sqrt{7}}{2}} (-2)dx$$
$$+ \int_{\frac{3+\sqrt{17}}{2}}^{1} (-3)dx$$

$$= \frac{-1}{4} + 0 - \frac{1}{4} - 2\left(\frac{3+\sqrt{17}}{8} - \frac{3}{4}\right) - 3\left(1 - \frac{3+\sqrt{17}}{8}\right)$$
$$= \frac{-2-2-6-2\sqrt{17}+12-24+9+3\sqrt{17}}{8} = \frac{\sqrt{17}-13}{8}$$

Option (C).

Q. 8 Let $f : \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$$

where $a, b, c \in \mathbb{R}$ and $[t]$ denotes greatest integer less than or equal to t.Then, which of the following statements is true?

Option 1:
There exists $a, b, c \in \mathbb{R}$ such that f is continuous on $\mathbb{R}$.

Option 2:
If f is discontinuous at exactly one point, then $a + b + c = 1$

Option 3:
If f is discontinuous at exactly one point, then $a + b + c \neq 1$

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Option 4:

f is discontinuous at atleast two points, for any values of a, b and c

Correct Answer:

If f is discontinuous at exactly one point, then $a+b+c \neq 1$

Solution:

$f\left(0^{-}\right)=0$

$f\left(0^{+}\right)=a-1=f(0)$

$f\left(1^{-}\right)=a/e-1,\ f\left(1^{+}\right)=b-1,\ f(1)=b.$

$f\left(2^{-}\right)=b-1,\ f\left(2^{+}\right)=-c,\ f(2).$

So f has a discontinuity at $x=1.$

If f is continuous at $x=0 \Rightarrow a-1=0 \Rightarrow a=1.$

f is continuous at $x=2 \Rightarrow b-1=-c \Rightarrow b+c=1$

So $a+b+c=2 \neq 1.$

Option (C)

Q. 9 The area of the region $S=\left\{(x, y): y^2 \leq 8x, y \geq \sqrt{2}x, x \geq 1\right\}$ is

Option 1:

$\frac{13\sqrt{2}}{6}$

Option 2:

$\frac{11\sqrt{2}}{6}$

Option 3:

$\frac{5\sqrt{2}}{6}$

Option 4:

$\frac{19\sqrt{2}}{6}$

Correct Answer:

$\frac{11\sqrt{2}}{6}$

Solution:

Points of intersection of these curves are

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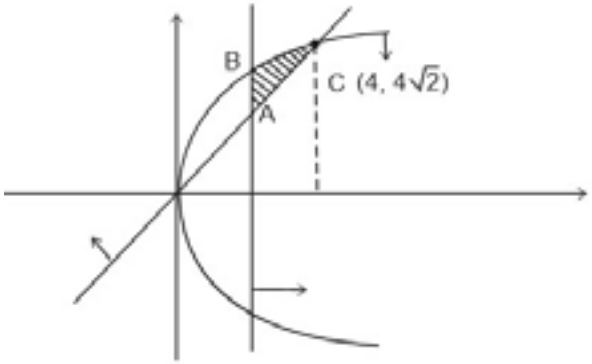
$\left(1, 2\sqrt{2}\right), \left(1, -2\sqrt{2}\right), (0, 0), \left(4, 4\sqrt{2}\right), \left(1, \sqrt{2}\right)$.



$$
\begin{aligned}
I &= \int_1^4 \left(2\sqrt{2}\sqrt{x} - \sqrt{2}x\right)\, dx \\
&= \left[2\sqrt{2} \times \frac{2}{3}x^{3/2} - \sqrt{2}\frac{x^2}{2}\right]_1^4 \\
&= \frac{32\sqrt{2}}{3} - 8\sqrt{2} - \frac{4\sqrt{2}}{3} + \frac{\sqrt{2}}{2} = \frac{64\sqrt{2} - 48\sqrt{2} - 8\sqrt{2} + 3\sqrt{2}}{6} \\
&= \frac{11\sqrt{2}}{6}
\end{aligned}
$$

Option (B)

Q. 10 Let the solution curve $y = y(x)$ of the differential equation

$$\left[\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}}\right] x\frac{dy}{dx} = x + \left[\frac{x}{\sqrt{x^2 - y^2}} + e^{\frac{y}{x}}\right] y$$

pass through the points $(1, 0)$ and $(2\alpha, \alpha), \alpha > 0$. Then α is equal to

Option 1:

$\frac{1}{2}\exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$

Option 2:

$\frac{1}{2}\exp\left(\frac{\pi}{3} + e - 1\right)$

Option 3:

$\exp\left(\frac{\pi}{6} + \sqrt{e} + 1\right)$

Option 4:

$2\exp\left(\frac{\pi}{3} + \sqrt{e} - 1\right)$

Correct Answer:

$\frac{1}{2}\exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$

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Solution:

$$\text{Let } \frac{y}{x} = t \Rightarrow \frac{dy}{dx} = t + \frac{x\,dt}{dx}.$$

$$\left[\frac{1}{\sqrt{1-t^2}} + e^t\right] x\left(t + \frac{x\,dt}{dx}\right) = x + \left[\frac{1}{\sqrt{1-t^2}} + e^t\right] t^x$$

$$\Rightarrow \frac{t}{\sqrt{1-t^2}} + \frac{x}{\sqrt{1-t^2}}\frac{dt}{dx} + t\,e^t + x\,e^t\frac{dt}{dx} = 1 + \frac{t}{\sqrt{1-t^2}} + t\,e^t$$

$$\Rightarrow \frac{x}{\sqrt{1-t^2}}\frac{dt}{dx} + x\,e^t\frac{dt}{dx} = 1$$

$$\Rightarrow \int \frac{dt}{\sqrt{1-t^2}} + \int e^t\,dt = \int \frac{dx}{x}$$

$$\Rightarrow \sin^{-1} t + e^t = \ln x + c$$
$$\Rightarrow \sin^{-1}\left(\frac{y}{x}\right) + e^{y/x} = \ln x + c, \text{ at } x = 1, y = 0$$
$$\Rightarrow c = 1, \text{ Also at } x = 2\alpha, y = \alpha$$
$$\Rightarrow \sin^{-1}\left(\frac{\alpha}{2\alpha}\right) + e^{\alpha/2\alpha} = \ln 2\alpha + 1$$
$$\Rightarrow \frac{\pi}{6} + e^{1/2} = 1 + \ln 2\alpha \Rightarrow \ln 2\alpha = \frac{\pi}{6} + \sqrt{e} - 1$$
$$\Rightarrow \alpha = \frac{1}{2}\exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$$

Option (A)

Q. 11 Let $y = y(x)$ be the solution of the differential equation $x\left(1-x^2\right)\frac{dy}{dx} + \left(3x^2y - y - 4x^3\right) = 0, x > 1, \text{with } y(2) = -2. \text{Then } y(3)$ is equal to

Option 1:
-18

Option 2:
-12

Option 3:
-6

Option 4:
-3

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Correct Answer:
-18

Solution:

$$\frac{dy}{dx}+\frac{(3x^2-1)}{x(1-x^2)}y=\frac{4x^3}{x(1-x^2)}$$

$$I.F=e^{\int\left(\frac{x^2-1}{x(1-x^2)}+\frac{2x^2}{x(1-x^2)}\right)dx}=e^{-\ln x-\ln\left(1-x^2\right)}=\frac{1}{x(1-x^2)}$$

$$\Rightarrow y\times\frac{1}{x(1-x^2)}=\int\frac{4x^3}{x^2(1-x^2)^2}dx=\frac{+2}{1-x^2}+c$$

$$\text{at } x=2, y=-2 \Rightarrow c=1$$
$$\Rightarrow y=2x+x\left(1-x^2\right)$$

$$\text{at } x=3;\ y=6+3(1-9)=-18$$

Option (A)

Q. 12 The number of real solutions of $x^7+5x^3+3x+1=0$ is equal to _____.

Option 1:
0

Option 2:
1

Option 3:
3

Option 4:
5

Correct Answer:
1

Solution:

$$f(x)=x^7+5x^3+3x+1$$
$$f'(x)=7x^6+15x^2+3>0\,\forall x\in R$$

$\Rightarrow f(x)$ is stictly increasing
$\Rightarrow f(x)$ will have exactly 1 real root .

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Option (B)

Q. 13 Let the eccentricity of the hyperbola $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\sqrt{\frac{5}{2}}$ and length of its latus rectum be $6\sqrt{2}$, If $y=2x+c$ is a tangent to the hyperbola H, then the value of c^2 is equal to

Option 1:
18

Option 2:
20

Option 3:
24

Option 4:
32

Correct Answer:
20

Solution:

$$e^2=1+\frac{b^2}{a^2}=\frac{5}{2} \Rightarrow \frac{b^2}{a^2}=\frac{3}{2} \quad --(1)$$

$$\frac{2b^2}{a^2}=6\sqrt{2} \Rightarrow 2b^2=6\sqrt{2}\,a \Rightarrow 3a^2=6\sqrt{2}\,a \Rightarrow a=2\sqrt{2}, b=2\sqrt{3}.$$

Equation of tangent of slope $m=2$ is

$$y=2x+\sqrt{a^2\times 2^2-b^2}$$
$$y=2x+\sqrt{32-12} \quad c=\sqrt{20} \Rightarrow c^2=20$$

Option (B)

Q. 14 If the tangents drawn at the points $O(0,0)$ and $P\left(1+\sqrt{5},2\right)$ on the circle $x^2+y^2-2x-4y=0$ intersect at the point Q. then the area of the triangle OPQ is equal to

Option 1:
$\frac{3+\sqrt{5}}{2}$

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Option 2:
$\frac{4+2\sqrt{5}}{2}$

Option 3:
$\frac{5+3\sqrt{5}}{2}$

Option 4:
$\frac{7+3\sqrt{5}}{2}$

Correct Answer:
$\frac{5+3\sqrt{5}}{2}$

Solution:
$\text{Tangent at } (0,0) \text{ is } \mathrm{T}=0$
$\Rightarrow \mathrm{x}\cdot 0+\mathrm{y}\cdot 0-(\mathrm{x}+0)-2(\mathrm{y}+0)=0$
$\Rightarrow \mathrm{x}+2\mathrm{y}=0 \ ---(1)$

$\text{Tangent at}(1+\sqrt{5},2) \text{ is}$
$\mathrm{x}(1+\sqrt{5})+\mathrm{y}\cdot 2-(\mathrm{x}+1+\sqrt{5})-2(\mathrm{y}+2)=0$
$\Rightarrow \sqrt{5}\mathrm{x}=\sqrt{5}+5 \Rightarrow \mathrm{x}=\sqrt{5}+1 \ ---(2)$

Point of intersection of (1) & (2) is $\left(\sqrt{5}+1,\frac{-(\sqrt{5}+1)}{2}\right)$

$$\text{Area of } \Delta\mathrm{OPQ}=\left|\frac{1}{2}\right|\begin{vmatrix}1 & 0 & 0\\ 1 & 1+\sqrt{5} & 2\\ 1 & 1+\sqrt{5} & \frac{-\sqrt{5}+1}{2}\end{vmatrix}|$$

$$=\frac{1}{2}(\sqrt{5}+1)\left(\frac{\sqrt{5}+1}{2}+2\right)=\frac{1}{2}\left[\frac{(\sqrt{5}+1)(\sqrt{5}+5)}{2}\right]$$
$$=\frac{5+5\sqrt{5}+\sqrt{5}+5}{4}=\frac{10+6\sqrt{5}}{4}=\frac{5+3\sqrt{5}}{2}$$
Option (C)

Q. 15 If two distinct points Q,R lie on the line of intersection of the planes $-\mathrm{x}+2\mathrm{y}-\mathrm{z}=0 \text{ and } 3\mathrm{x}-5\mathrm{y}+2\mathrm{z}=0 \text{ and } \mathrm{PQ}=\mathrm{PR}=\sqrt{18}$ where the point P is $(1,-2,3)$, then the area of the triangle PQR is equla to

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Option 1:
$\frac{2}{3}\sqrt{38}$

Option 2:
$\frac{4}{3}\sqrt{38}$

Option 3:
$\frac{8}{3}\sqrt{38}$

Option 4:
$\sqrt{\frac{152}{3}}$

Correct Answer:
$\frac{4}{3}\sqrt{38}$

Solution:

$\text{Let z} = \text{k}$

$$-x + 2y = k \ ---(1)$$
$$3x - 5y = -2k \ ---(2)$$
$$(1) \times 3 \Rightarrow -3\text{x} + 6\text{y} = 3\text{k} \ ---(3)$$
$$(2) + (3) \Rightarrow \text{y} = \text{k}; \quad \text{x} = \text{k}$$

Also both the lines pass through $(0, 0, 0)$ So theie line of intersection is
$\frac{\text{x}}{1} = \frac{\text{y}}{1} = \frac{\text{z}}{1} = \text{k}.$

Let the Foot of Perpendicular from P to Line QR is

$\text{S}(\text{k}, \text{k}, \text{k}). \quad \text{P}(1, -2, 3)$

$\overrightarrow{PS} = (\text{k} - 1, \text{k} + 2, \text{k} - 3). \quad \overrightarrow{PS} \cdot \vec{b} = 0$
$\Rightarrow \text{k} - 1 + \text{k} + 2 + \text{k} - 3 = 0 \Rightarrow \text{k} = 2/3.$

$$\text{S}\left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right), \quad |\overrightarrow{PS}| = \sqrt{\frac{1}{9} + \frac{64}{9} + \frac{49}{9}} = \frac{\sqrt{114}}{3}.$$
$$\text{RS} = \text{QS} = \sqrt{18 - \frac{114}{9}} = \sqrt{\frac{48}{9}} = \frac{4}{\sqrt{3}}.$$

$$\text{Area of } \Delta\text{PQR} = 2 \times \frac{1}{2} \times \text{RS} \times \text{PS} = \frac{\sqrt{114}}{3} \times \frac{4}{\sqrt{3}} = \frac{4}{3}\sqrt{38}$$

Option (B)

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Q. 16 The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points $(2, 1, 3)$ and $(0, 1, 2)$, respectively, is

Option 1:

$\frac{\pi}{3}$

Option 2:

$\frac{\pi}{4}$

Option 3:

$\frac{\pi}{6}$

Option 4:

$\frac{\pi}{12}$

Correct Answer:

$\frac{\pi}{3}$

Solution:

$\text{Let } P_1 : (5x + 2y + 13z - 29) + \lambda(8x - 7y + z - 20) = 0$

Passes through $(2, 1, 3)$

$\Rightarrow (10 + 8 + 39 - 29) + \lambda(16 - 7 + 3 - 20) = 0 \Rightarrow \lambda = \frac{7}{2}$

$\Rightarrow P_1 : (10x + 16y + 26z - 58) + (56x - 49y + 73 - 140) = 0$

$\Rightarrow 66x - 33y + 33z - 198 = 0 \Rightarrow 2x - y + z - 6 = 0 \; --- (1)$

$\text{Let } P_2 : (5x + 8y + 13z - 29) + \mu(8x - 7y + z - 20) = 0$

Passes through $(0, 1, 2)$

$\Rightarrow (0 + 8 + 26 - 29) + \mu(0 - 7 + 2 - 20) = 0 \Rightarrow \mu = 1/5$

$\Rightarrow P_2 : (25x + 40y + 65z - 145) + 8x - 7y + z - 20 = 0 \Rightarrow 33x + 33y + 663 - 165 = 0$

$\Rightarrow x + y + 2z - 5 = 0 \; --- (2)$

$\cos\theta = \left| \frac{(2, -1, 1) \cdot (1, 1, 2)}{\sqrt{2^2 + 1^2 + 1^2}\sqrt{1^2 + 1^2 + 2^2}} \right| = \frac{3}{6} = \frac{1}{2} \Rightarrow \theta = \pi/3$

Option (A)

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Q. 17 Let the plane $P : \vec{r} \cdot \vec{a} = d$ contain the line of intersection of two planes $\vec{r} \cdot \left(\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - \hat{\mathrm{k}}\right) = 6 \; and \; \vec{r} \cdot \left(-6\hat{\mathrm{i}} + 5\hat{\mathrm{j}} - \hat{\mathrm{k}}\right) = 7.$ If the plane P passes through the point $\left(2, 3, \frac{1}{2}\right)$, then the value of $\frac{|13\,\vec{a}|^2}{d^2}$ is equal to

Option 1:
90

Option 2:
93

Option 3:
95

Option 4:
97

Correct Answer:
93

Solution:
Let the equation of Plane P is

$$(\mathrm{x} + 3\mathrm{y} - 3 - 6) + \lambda(-6\mathrm{x} + 5\mathrm{y} - 3 - 7) = 0$$

Passes through $(2, 3, 1/2)$

$$\Rightarrow \left(2 + 9 - \frac{1}{2} - 6\right) + \lambda\left(-12 + 15 - \frac{1}{2} - 7\right) = 0$$
$$\Rightarrow \quad \lambda = 1 \Rightarrow \quad \mathrm{P} : -5\mathrm{x} + 8\mathrm{y} - 23 - 13 = 0$$
$$\Rightarrow \vec{r} \cdot \left(-5\hat{i} + 8\hat{j} - 2\hat{k}\right) = 13.$$
$$\Rightarrow \frac{|13\vec{a}|}{d^2} = \frac{13^2\,(5^2 + 8^2 + 2^2)}{13^2} = 93$$

Option (B)

Q. 18 The probability, that in a randomly selected 3-digit number at least two digits are odd, is

Option 1:
$\frac{19}{36}$

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Option 2:
$\frac{15}{36}$

Option 3:
$\frac{13}{36}$

Option 4:
$\frac{23}{36}$

Correct Answer:
$\frac{19}{36}$

Solution:
Case 1 3 odd digits $\underline{5} \times \underline{5} \times \underline{5} = 125.$
Case 2 2 odd- 1 even digit

(i) if 0 is selected $_\ \underline{0}\ \underline{0}\ \ 2 \times 5 \times 5 = 50$
(ii) if 0 is not selected $__\ _\ \ 3 \times 4 \times 5 \times 5 = 300$

Total = 475,
No.of 3 digit numbers = 900

$$\mathrm{P} = \frac{475}{900} = \frac{19}{36}$$
Option (A)

Q. 19 Let AB and PQ be two vertical poles, $160\ \mathrm{m}$ apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to

Option 1:
$\frac{3-2\sqrt{2}}{2}$

Option 2:
$\frac{3+\sqrt{2}}{2}$

Option 3:
$\frac{3-2\sqrt{2}}{4}$

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Option 4:

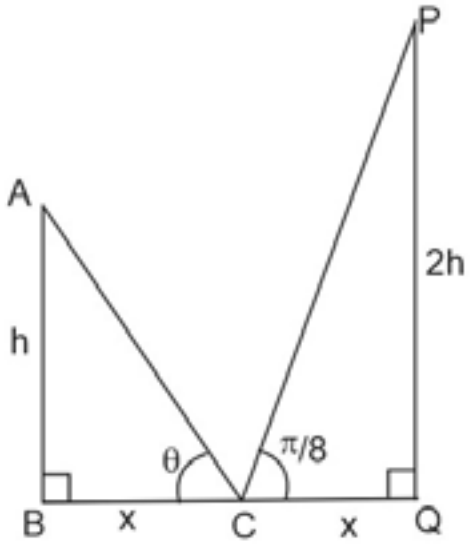
$\frac{3-\sqrt{2}}{4}$

Correct Answer:

$\frac{3-2\sqrt{2}}{4}$

Solution:

$\tan\theta = \frac{h}{x}$, $\tan\frac{\pi}{8} = \frac{2h}{x}$



$\Rightarrow \frac{\tan\theta}{\tan\frac{\pi}{8}} = \frac{1}{2}$

$\Rightarrow \tan\theta = \frac{1}{4}\tan\frac{\pi}{8}$

$\Rightarrow \tan^2\theta = \frac{1}{4}\tan^2\frac{\pi}{8} = \frac{1}{4}\left(\frac{1-\cos\frac{\pi}{4}}{1+\cos\frac{\pi}{4}}\right) = \frac{1}{4}\left(\frac{1-1/\sqrt{2}}{1+1/\sqrt{2}}\right)$

$= \frac{1}{4}\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right) = \frac{1}{4}\frac{(\sqrt{2}-1)^2}{(2-1)} = \frac{3-2\sqrt{2}}{4}$

Option (C)

Q. 20 Let p, q, r be three logical statements. Consider the compound statements
$S_1 : ((-p) \vee q) \vee ((-p) \vee r)$ and
$S_2 : p \rightarrow (q \vee r))$
Then, which of the following is **NOT** true?

Option 1:

If S_2 is True, then S_1 is True

Option 2:

If S_2 is False, then S_1 is False

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Option 3:
If S_2 is False, then S_1 is True

Option 4:
If S_1 is False, then S_2 is False

Correct Answer:
If S_2 is False, then S_1 is True

Solution:
$$S_1 : ((\sim p) \vee q) \vee ((\sim p) \vee r)$$
$$\equiv (\sim p) \vee (q \vee r).$$

$$S_2 : p \rightarrow (q \vee r) \equiv \sim p \vee (q \vee r)$$

$$\Rightarrow S_1 \equiv S_2$$

$\Rightarrow$ Option (C) is not True

Q. 21 Let R_1 and R_2 be relations on the set $\{1, 2, \ldots, 50\}$ such that
$R_1 = \{(p, p^n) : p$ is a prime and $n \geq 0$ is an integer $n \geq 0\}$ and
$R_2 = \{(p, p^n) : p$ is a prime and $n = 0$ or $1\}$.
Then, the number of elements in $R_1 - R_2$ is _____.

Correct Answer:
8

Solution:
$$R_1 - R_2 = \{(p, p^n) : p \text{ is a prime and } n \geq 2 \text{ is an integer}\}$$
$$= \left\{(2, 2^2), (2, 2^3), (2, 2^4), (2, 2^5), (3, 3^2), (3, 3^3), (5, 5^2), (7, 7^2)\right\}.$$

No.of elements in $R_1 - R_2 = 8$

Q. 22 The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5 . Then, the correct variance is equal to_____.

Correct Answer:
17

Solution:
$$\frac{\sum_{i=1}^{14} x_i + 5}{15} = 8$$

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$$\Rightarrow \sum_{i=1}^{14} xi = 115 \quad \Rightarrow \quad \sum_{i=1}^{14} xi + 20 = 123$$

$$\Rightarrow \text{Real Mean} = \frac{\sum_{i=1}^{14} xi + 20}{15} = \frac{13}{15}$$

$$\frac{\sum_{i=1}^{14} x\,i^2 + 5^2}{15} - (8)^2 = 9 \Rightarrow \quad \sum_{i=1}^{14} x\,i^2 = 1070.$$

So Real variance $= \frac{\sum_{i=1}^{14} x\,i^2 + 20^2}{15} - (9)^2$

$$= \frac{1070 + 400}{15} - (9)^2$$

$$= \frac{1470}{15} - 81$$

$$= 98 - 81 = 17.$$

Q. 23 The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is

Correct Answer:
2

Solution:

$\text{Let } e^x = t; \quad t > 0$

$t^4 + 4t^3 - 58t^2 + 4t + 1 = 0$

$$\Rightarrow \quad t^2 + \frac{1}{t^2} + 4\left(t + \frac{1}{t}\right) - 58 = 0$$

$$\text{Let } t + \frac{1}{t} = p \Rightarrow p^2 - 2 + 4p - 58 = 0$$

$\Rightarrow p^2 + 4p - 60 = 0 \Rightarrow p^2 + 10p - 6p - 60 = 0$
$\Rightarrow p = -10 \text{ or } p = 6 \text{ But since } t > 0; p > 0$

$$\text{So } p = 6 \Rightarrow t + \frac{1}{t} = 6 \Rightarrow t^2 - 6t + 1 = 0$$

$$\Rightarrow t = \frac{6 \pm \sqrt{36 - 4}}{2} = 3 \pm 2\sqrt{2}$$

Both are + ve , So 2 as responding values of x will be $\ln\left(3 + 2\sqrt{2}\right)$ & $\ln\left(3 - 2\sqrt{2}\right)$

$\Rightarrow$ 2 Real solutions

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Q. 24 If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}, \quad \vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ *and* $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} \cdot \vec{c} = 5, \vec{b} \perp \vec{c}$, *then* $122\,(c_1 + c_2 + c_3)$ is equal to_____.

Correct Answer:
150

Solution:

$\vec{a} \cdot \vec{c} = 5 \Rightarrow 2C_1 + C_2 + 3C_3 = 5 \; --- (1)$
$\vec{b} \perp \vec{c} = 3C_1 + 3C_2 + C_3 = 0 \; --- (2)$

$$\vec{a}, \vec{b}, \vec{c} \text{ are coplanar} \Rightarrow \begin{vmatrix} C_1 & C_2 & C_3 \\ 2 & 1 & 3 \\ 3 & 3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow -8C_1 + 7C_2 + 3C_3 = 0 \; --- (3)$$

Eliminating C_3 from (1) & (3) $\Rightarrow 10C_1 - 6C_1 = 5 \; --- (4)$
from (2) & (3) $\Rightarrow 17C_1 + 2C_2 = 0 \; --- (5)$
$\Rightarrow 51C_1 + 6C_2 = 0 \; --- (6)$

$$\Rightarrow 61C_1 = 5 \Rightarrow C_1 = \frac{5}{61}, \; C_2 = \frac{-1}{2} \times 17C_1 = \frac{-85}{122}.$$

$C_3 = -3\,(C_1 + C_2)$

$$\text{So } C_1 + C_2 + C_3 = -2\,(C_1 + C_2) = \left(\frac{85}{61} - \frac{10}{61}\right)$$

$$\Rightarrow 122\,(C_1 + C_2 + C_3) = 75 \times 2 = 150$$

Q. 25 A ray of light passing through the point $P(2, 3)$ reflects on the x-axis at point A and the reflected ray passes through the point $Q(5, 4)$. Let R be the point that divides the line segment AQ internally into the ratio $2 : 1$. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β). Then, the value of $7\alpha + 3\beta$ is equal to ____.

Correct Answer:
31

Solution:

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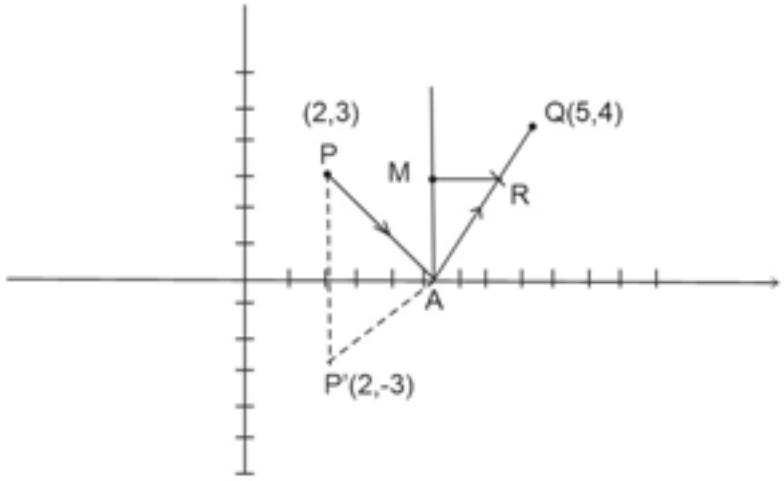

p' be reflection of p about x-axis

$\text{Slope of } p'Q \quad {}^{m}p'_{Q} = {}^{m}A_{Q} = \dfrac{4+3}{5-2} = \dfrac{7}{3}.$

$\Rightarrow AQ : y - 4 = \dfrac{7}{3}(x-5) \quad y = 0 \Rightarrow x = \dfrac{-12}{7} + 5 = \dfrac{23}{7}.$

So Coordinates of $M : \left(\dfrac{23}{7}, \dfrac{8}{3}\right) = (\alpha, \beta)$

$\Rightarrow 7\alpha + 3\beta = 23 + 8 = 31.$

Q. 26 Let l be a line which is normal to the curve $y = 2x^2 + x + 2$ at a point P on the curve. If the point $Q\,(6, 4)$ lies on the line l and O is origin. then the area of the triangle OPQ is equla to_____.

Correct Answer:
13

Solution:
$y = 2x^2 + x + 2$

$\text{Let P is } (h, k).$

$\dfrac{dy}{dx}\Big|_{h,k} = 4x + 1\Big|_{h,k} = 4h + 1 = m_l$

$\Rightarrow m_N = \dfrac{-1}{4h+1}$

$l : y - k = \dfrac{-1}{4h+1}(x - h)$

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$$\text{Passes through}(6,4) = 4 - k = \frac{-1}{4h+1}(6-h)$$
$$\Rightarrow (4h+1)(4-k) = h-6$$
$$\Rightarrow \text{ Also } k = 2h^2 + h + 2$$
$$\Rightarrow (4h+1)\left(4 - 2h^2 - h - 2\right) = h-6$$
$$\Rightarrow (4h+1)\left(2h^2 + h - 2\right) + h - 6 = 0$$
$$\Rightarrow 8h^3 + 4h^2 - 8h + 2h^2 + h - 2 + h - 6 = 0 \Rightarrow 8h^3 + 6h^2 - 6h - 8 = 0$$
$$\Rightarrow 4h^3 + 3h^2 - 3h - 4 = 0 \Rightarrow h = 1 \text{ satisfy this eqn}$$

$$\text{For } h = 1, k = 5 \Rightarrow Q(1,5)$$

$$\triangle OPQ = |\frac{1}{2}\begin{vmatrix} 1 & 0 & 0 \\ 1 & 6 & 4 \\ 1 & 1 & 5 \end{vmatrix}| = \frac{1}{2}(30-4) = 13$$

Q. 27 Let $A = \{1, a_1, a_2 \ldots\ldots a_{18}, 77\}$ be a set of integers with $1 < a_1 < a_2 < \ldots\ldots a_{18} < 77$. Let the set $A + A = \{x + y : x, y \in A\}$ contain exactly 39 elements. Then, the value of $a_1 + a_2 + \ldots + a_{18}$ is equal to_____.

Correct Answer:
702

Solution:
The minimum no. of different sums of 2 elements (not neccessarly distinct) at a set containg 20 distinct elements is 39, and it occurs ; if f the elements are in A. P

For e.g $\text{Let } B = \{1, 2, 3, \cdots 20\}$

$$\text{Min sum} = 1 + 1 = 2, \text{ Max sum} = 20 + 20 = 40$$
So No.of diff sums $= \{2, 3, 4, \cdots 40\} = 39 \text{ sums}$

So $a_1, a_2, \cdots a_{12} = 5, 9, 13, \cdots 73$
So $a_1, a_2, \cdots a_{12} = \frac{18}{2}[5 + 73] = 702$

Q. 28 The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^3 + \frac{3}{x^k}\right)^{12}, x \neq 0$ is $2^8 \cdot \ell$, where ℓ is an odd integer, is _____.

Correct Answer:
2

Solution:
$$T_{r+1} = {}^{12}C_r \left(2x^3\right)^{12-r} \left(\frac{3}{x^k}\right)^r$$

$= {}^{12}C_r \cdot 2^{12-r} \cdot 3^r \cdot x^{36-3r-kr}$

For constant term, $36 - r(3+k) = 0$
$\Rightarrow r = \dfrac{36}{3+k}; k \in N.$

So $3 + k = \{4, 6, 9, 12, 18, 36\}.$
$\Rightarrow k = \{1, 3, 6, 9, 15, 33\}.$

Corresponding $r = \{9, 6, 4, 3, 2, 1\}.$

(i) $T_{10} = {}^{12}C_9 \cdot 2^3 \cdot 3^9$
No. of 25 in $\dfrac{12!}{9!3!}$ are $\left(\left[\dfrac{12}{2}\right] + \left[\dfrac{12}{4}\right] + \left[\dfrac{12}{2}\right] + \cdots\right)$
$-\left(\left[\dfrac{9}{2}\right] + \left[\dfrac{9}{4}\right] + \left[\dfrac{9}{2}\right] + \cdots\right) - \left(\left[\dfrac{3}{2}\right] + \cdots\right)$

$= 10 - 7 - 1 = 2$

So No. of $2s = 3 + 2 = 5x$

(ii) $T_7 = {}^{12}C_6 \cdot 2^6 \cdot 3^6$
No. of 2s in $\dfrac{12!}{6!6!} = 10 - 4 - 4 = 2.$
So No. of $2s = 6 + 2 = 8$

(iii) $T_5 = {}^{12}C_4 \times 2^8 \times 3^4$
No. of 2s in $\dfrac{12!}{8!4!} = 10 - (4 + 2 + 1) - (2 + 1) = 0$
No. of $2s = 8 + 0 = 8$

(iv) $r = 3, 2, 1$ are rejected as No.of $2s > 8.$

So No.of positive integers $k = 2.$
Which are 3 & 6

Q. 29 The number of elements in the set
$\{z = a + ib \in \mathbb{C} : a, b \in \mathbb{Z} \text{ and } 1 < |z - 3 + 2i| < 4\}$ is

Correct Answer:
40

Solution:

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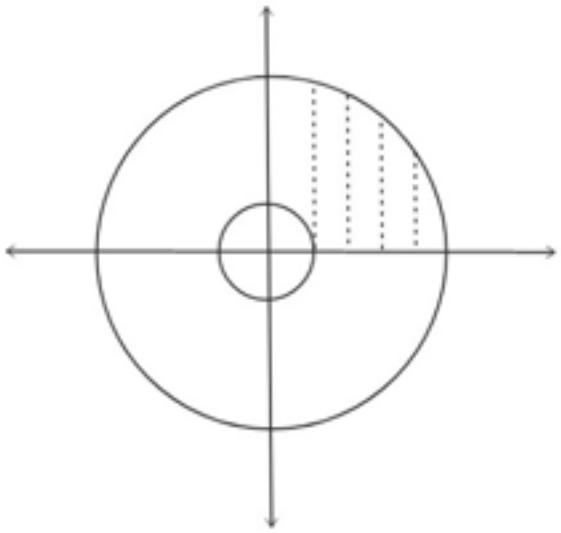
Let us shift the origin to $(3, -2)$

So $1 < |\mathbb{Z}| < 4$

No. of integral coordinates in the given region will be the same as in $1 < |\mathbb{Z} - 3 + 2i| < 4$



No.of such point in the first quadrat are

$(1,1), (2,1), (3,1), (1,2)\,(2,2), (3,2), (1,3), (2,3) = 8$

So in II, III & IV quadrant = 8,8,8.

On + x-axis : $(2,0)\ \&\ (3,0) = 2$

On -ve x-axis = 2

On +ve y-axis = 2

On -ve y-axis = 2

Total = $8 + 8 + 8 + 8 + 2 + 2 + 2 + 2 = 40.$

Q. 30 Let the lines $y + 2x = \sqrt{11} + 7\sqrt{7}$ and $2y + x = 2\sqrt{11} + 6\sqrt{7}$ be normal to a circle $C : (x-h)^2 + (y-k)^2 = r^2$. If the line $\sqrt{11}y - 3x = \frac{5\sqrt{77}}{3} + 11$ is tangent to the circle C, then the value of $(5h - 8k)^2 + 5r^2$ is equal to_____.

Correct Answer:

816

Solution:

Point of intersection will be centre

$2x + y = \sqrt{11} + 7\sqrt{7}$

$\Rightarrow 4x + 2y = 2\sqrt{11} + 14\sqrt{7} - - - -1$

$x + 2y = 2\sqrt{11} + 6\sqrt{7} - - - -2$

$(1) - (2) \Rightarrow 3x = 8\sqrt{7} \Rightarrow x = \frac{8\sqrt{7}}{3},$

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$$y = \sqrt{11} + 7\sqrt{7} - \frac{16\sqrt{7}}{3} = \sqrt{11} + \frac{5\sqrt{7}}{3}$$

$r =$ distance of centre from tangent

$$= \left| \frac{3\left(\frac{8\sqrt{7}}{3}\right) - \sqrt{11}\left(\sqrt{11} + \frac{5\sqrt{7}}{3}\right) + \frac{5\sqrt{77}}{3} + 11}{\sqrt{9+11}} \right|$$

$$= \frac{8\sqrt{7}}{2\sqrt{5}} = 4\sqrt{\frac{7}{5}}$$

$$h = \frac{8\sqrt{7}}{3}, k = \sqrt{11} + \frac{5\sqrt{7}}{3}, r = 4\sqrt{\frac{7}{5}}$$

$$(5h - 8k)^2 + 5r^2$$

$$= \left(\frac{40\sqrt{7}}{3} - 8\sqrt{11} - 40\frac{\sqrt{7}}{3} \right)^2 + 5 \times 16 \times \frac{7}{5}$$

$$= 64 \times 11 + 112 = 816$$

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28 June Evening Shift

Physics

Q. 1 Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as $v_2 = \frac{n}{m^2}v_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two systems respectively are:

Option 1:

$\frac{n^3}{m^3} L_1 = L_2$ and $\frac{n^2}{m} T_1 = T_2$

Option 2:

$L_1 = \frac{n^4}{m^2} L_2$ and $T_1 = \frac{n^2}{m} T_2$

Option 3:

$L_1 = \frac{n^2}{m} L_2$ and $T_1 = \frac{n^4}{m^2} T_2$

Option 4:

$\frac{n^2}{m} L_1 = L_2$ and $\frac{n^4}{m^2} T_1 = T_2$

Correct Answer:

$\frac{n^3}{m^3} L_1 = L_2$ and $\frac{n^2}{m} T_1 = T_2$

Solution:

Dimension of length (L)

$$L = \frac{v^2}{a}$$

$$\frac{v_2^2}{a_2} = \frac{\left[\frac{n}{m^2}v_1\right]^2}{\left[\frac{a_1}{mn}\right]}$$

$$\frac{[v_2]^2}{[a_2]} = \frac{n^3}{m^3}\frac{[v_1]^2}{[a_1]}$$

Or

$$[L_2] = \left[\frac{n^3}{m^3}\right][L_1]$$

Similarly

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$$[T] = \left[\frac{V}{a}\right]$$

$$[T_2] = \frac{n^2}{m}[T_1]$$

Hence, the correct answer is (1)

Q. 2 A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$ where t is in second and α is in rad s^{-2}. $\text{At } t = 0,$ the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad. The most appropriate expression for the angular position of the ball is :

Option 1:
$\frac{3}{2}t^4 - t^2 + 10t$

Option 2:
$\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

Option 3:
$\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$

Option 4:
$2t^4 - \frac{t^3}{2} + 5t + 4$

Correct Answer:
$\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

Solution:

$\alpha = 6t^2 - 2t$

$\frac{d\omega}{dt} = 6t^2 - 2t$

$\int_{10}^{\omega} d\omega = \int_{t=0}^{t} (6t^2 - 2t)\, dt$

$\omega - 10 = \left[\frac{6t^3}{3} \frac{-2t^2}{2}\right]_0^t$

$\omega = 10 + 2t^3 - t^2$ --------(1)

$\frac{d\theta}{dt} = 10 + 2t^3 - t^2$

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$$\int_{\theta=4}^{\theta} d\theta = \int_{t=0}^{t} \left(10 + 2t^3 - t^2\right) dt$$

$$\theta - 4 = \left[10t + \frac{2t^4}{4} - \frac{t^3}{3}\right]_0^t$$

$$\theta = \frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$$

Hence 2 is correct answer.

Q. 3 A block of mass 2 kg moving on a horizontal surface with speed of $4\ ms^{-1}$ enters a rough surface ranging from $x = 0.5\ m \text{ to } x = 1.5\ m$. The retarding force in this range of rough surface is related to distance by $F = -kx \text{ where } k = 12 Nm^{-1}$. The speed of the block as it just crosses the rough surface will be :

Option 1:
zero

Option 2:
$1.5\ ms^{-1}$

Option 3:
$2.0\ ms^{-1}$

Option 4:
$2.5\ ms^{-1}$

Correct Answer:
$2.0\ ms^{-1}$

Solution:
$$F = -12x$$

$$mv\frac{dv}{dx} = -12x$$

$$vdv = -\frac{12}{m}xdx$$

$$\int_{v=4}^{v} vdv = -6\int_{x=0.5}^{x=1.5} xdx$$

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$$\left[\frac{v^2}{2}\right]_4^v = -6\left[\frac{x^2}{2}\right]_{0.5}^{1.5}$$

$$\frac{v^2-16}{2} = -\frac{6}{2}\left[1.5^2 - 0.5^2\right]$$

$$\frac{v^2}{2} - 8 = -3 \times 2$$

$$\frac{v^2}{2} = 8 - 6 = 2$$

$$\frac{v^2}{2} = 8 - 6 = 2$$

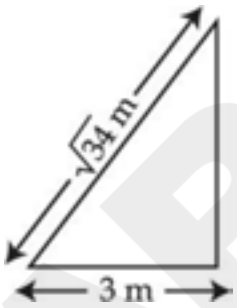
$$v^2 = 4$$

$$v = 2\text{m/s}$$

Hence 3 is the correct answer.

Q. 4 A $\sqrt{34}$ m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 3 m away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be:

$(\text{ Use } g = 10 \text{ m/s}^2.)$



Option 1:

$$\frac{6}{\sqrt{110}}$$

Option 2:

$$\frac{3}{\sqrt{113}}$$

Option 3:

$$\frac{3}{\sqrt{109}}$$

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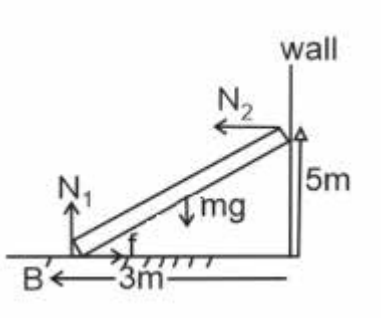
Option 4:

$\frac{2}{\sqrt{109}}$

Correct Answer:

$\frac{3}{\sqrt{109}}$

Solution:



Torque about B

$N_2 \times 5 = mg \times \frac{3}{2}$

$N_2 = \frac{10 \times 10 \times 3}{2 \times 5} = 30\ N$

$N_1 = mg = 100\ N$

$N_2 = f = 30\ N$

$E_f = \sqrt{100^2 + 30^2} = 10\sqrt{109}$

$F_w = 30N = N_2.$

$\frac{F_w}{F_f} = \frac{30}{10\sqrt{109}} = \frac{3}{\sqrt{109}}$

Hence option 3 is the correct answer.

Q. 5 Water falls from a 40 m high dam at the rate of 9×10^4 kg per hour. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydro electric energy number of 100 W lamps, that can be lit, is :

$\left(\text{Take } g = 10\ ms^{-2}\right)$

Option 1:

25

Option 2:

50

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Option 3:
100

Option 4:
18

Correct Answer:
50

Solution:

$$\frac{\text{PE}}{\text{time}} = \frac{9 \times 10^4 \times 40}{60 \times 60}$$

$$= \frac{36 \times 10^6}{3600} = 10^4 \frac{\text{J}}{\text{s}}$$

Electrical energy per sec

$$= 50\% \times 10^4$$
$$= \frac{10^4}{2}\ \text{J/s} = \frac{10^4}{2} W$$

Let N no.of bulbs lit

$$\text{N} \times 100 = \frac{10^4}{2}$$

$$\text{N} = \frac{100}{2} = 50$$

Hence correct answer is 2

Q. 6 Two objects of equal masses placed at certain distance from each other attracts each other with a force of F. If one-third mass of one object is transferred to the other object, then the new force will be :

Option 1:
$\frac{2}{9}\ \text{F}$

Option 2:
$\frac{16}{9}\ \text{F}$

Option 3:
$\frac{8}{9}\ \text{F}$

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Option 4:

F

Correct Answer:

$\frac{8}{9}\ \mathrm{F}$

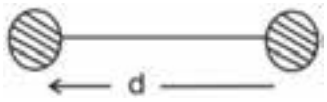
Solution:

Case-1

$$\mathrm{F}=\frac{\mathrm{Gm}^2}{\mathrm{d}^2}$$

Case-2

$$\mathrm{m}+\mathrm{m}/3=\frac{4\mathrm{m}}{3} \qquad \mathrm{m}-\mathrm{m}/3=2\mathrm{m}/3$$



$$\mathrm{F}'=\frac{\mathrm{G}\times\frac{4\mathrm{m}}{3}\times\frac{2\mathrm{m}}{3}}{\mathrm{d}^2}=\frac{8}{9}\frac{\mathrm{Gm}^2}{\mathrm{d}^2}=\frac{8}{9}\mathrm{F}$$

Hence the correct option is 3

Q. 7 A water drop of radius $1\mu\mathrm{m}$ falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is $1.8\times10^{-5}\mathrm{Nsm}^{-2}$ and its density is negligible as compared to that of water $10^6\mathrm{gm}^{-3}$. Terminal velocity of the water drop is :

$(\text{Take acceleration due to gravity } = 10\ \mathrm{ms}^{-2}\ )$

Option 1:

$145.4\times10^{-6}\ \mathrm{ms}^{-1}$

Option 2:

$118.0\times10^{-6}\ \mathrm{ms}^{-1}$

Option 3:

$132.6\times10^{-6}\ \mathrm{ms}^{-1}$

Option 4:

$123.4\times10^{-6}\ \mathrm{ms}^{-1}$

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Correct Answer:

$123.4 \times 10^{-6}\ \mathrm{ms}^{-1}$

Solution:

$$V_T = \frac{2}{9} r^2 \frac{(\sigma - \rho)g}{\eta}$$

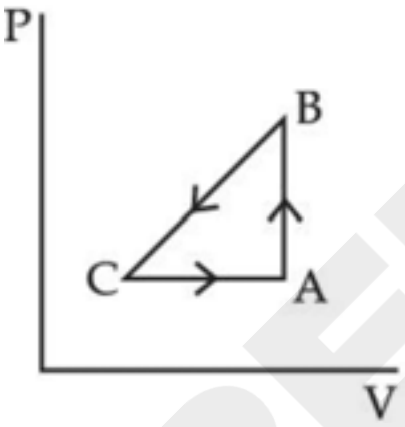
$$\rho = 0$$

$$V_T = \frac{2}{9} \times \frac{(1 \times 10^{-6})^2 \times (10^3) \times 10}{1.8 \times 10^{-5}}$$

$$= 123.4 \times 10^{-6}\ \mathrm{m/s}$$

Hence option 4 is correct.

Q. 8 A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60 J of heat during CA. A work of 50 J is done on the gas during the part BC. The internal energy of the gas at A is 1560 J. The workdone by the gas duringh the part CA is :



Option 1:

$20\ \mathrm{J}$

Option 2:

$30\ \mathrm{J}$

Option 3:

$-30\ \mathrm{J}$

Option 4:

$-60\ \mathrm{J}$

Correct Answer:

$30\ \mathrm{J}$

Solution:

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$\Delta U_{AB} = 40$ J as proces is Isochoric

$\mathrm{Q_{BC}} = \Delta \mathrm{U} + \mathrm{W_{BC}}$
$\Delta \mathrm{U} + \mathrm{W_{Bc}} = 0$

$\Delta \mathrm{U_{BC}} = -\mathrm{W_{BC}}$
$= -[-50] = 50$ J
$\mathrm{U_C} = \mathrm{U_A} + \Delta \mathrm{U_{AB}} + \Delta \mathrm{U_{BC}} = 1650$
For CA process
$Q_{CA} = -60J$
$\Delta \mathrm{U_{CA}} + \mathrm{W_{CA}} = -60$
$-90 + \mathrm{WCA} = -60$
$\mathrm{W_{CA}} = +30\mathrm{J}$
Hence option 2 is correct.

Q. 9 What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

Option 1:
The velocity of atomic oxygen remains same

Option 2:
The velocity of atomic oxygen doubles

Option 3:
The velocity of atomic oxygen becomes half

Option 4:
The velocity of atomic oxygen becomes four times

Correct Answer:
The velocity of atomic oxygen doubles

Solution:
$\mathrm{V_{rms}} = \sqrt{\frac{3\mathrm{RT}}{\mathrm{M_0}}}$
T is double and the Oxygen molecule is dissociated into atomic oxygen. Hence the molar mase is halved.
$V_{rms} = \sqrt{\frac{3R \times 2T}{\frac{M_0}{2}}} = 2V_{\mathrm{rms}}$
Hence option 2 is correct.

Q. 10 Two point charges A and B of magnitude $+8 \times 10^{-6}\mathrm{C}$ and $-8 \times 10^{-6}\mathrm{C}$ respectively are placed at a distance d apart. The electric field at the middle point O between the charges is $6.4 \times 10^{4}\mathrm{NC}^{-1}$. The distance 'd' between the point charges A and B is :

Option 1:
$2.0\ \mathrm{m}$

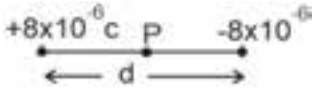
Option 2:
$3.0\ \mathrm{m}$

Option 3:
$1.0\ \mathrm{m}$

Option 4:
$4.0\ \mathrm{m}$

Correct Answer:
$3.0\ \mathrm{m}$

Solution:



Electric field at $'P'$

$$\mathrm{E}=\frac{\mathrm{Kq}}{\left(\frac{\mathrm{d}}{2}\right)^2}\times 2=8\frac{Kq}{d^2}$$

So,

$$\mathrm{E}=\frac{8\times 9\times 10^9\times 8\times 10^{-6}}{\mathrm{d}^2}=6.4\times 10^4$$

$$\mathrm{d}^2=9$$

$$\mathrm{d}=3\mathrm{m}$$

Hence option 2 is correct.

Q. 11 Resistance of the wire is measured as 2Ω and 3Ω at $10^{\circ}\mathrm{C}$ and $30^{\circ}\mathrm{C}$ respectively. Temperature co-efficient of resistance of the material of the wire is :

Option 1:
$0.033^{\circ}\mathrm{C}^{-1}$

Option 2:
$-0.033^{\circ}\mathrm{C}^{-1}$

Option 3:
$0.011^{\circ}\mathrm{C}^{-1}$

Option 4:
$0.055^{\circ}\mathrm{C}^{-1}$

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Correct Answer:

$0.033^{\circ}\mathrm{C}^{-1}$

Solution:

$\mathrm{R}_1 = \mathrm{R}_0(1 + \alpha \times 10) = 2\Omega$

$R_2 = \mathrm{R}_0(1 + \alpha + 30) = 3\Omega$

taking the ratio of both the equation we get

$$\frac{1 + 10\alpha}{1 + 30\alpha} = \frac{2}{3}$$

$$3 + 30\alpha = 2 + 60\alpha$$

$$\alpha = \frac{1}{30}$$

on solving,

$\alpha = 0.033/^{\circ}\mathrm{C}$

Hence option 1 is correct.

Q. 12 The space inside a straight current carrying solenoid is filled with a magnetic material having magnetic susceptibility equal to 1.2×10^{-5}. What is fractional increase in the magnetic field inside solenoid with respect to air as medium inside the solenoid?

Option 1:

1.2×10^{-5}

Option 2:

1.2×10^{-3}

Option 3:

1.8×10^{-3}

Option 4:

2.4×10^{-5}

Correct Answer:

1.2×10^{-5}

Solution:

$\vec{B}_1 = \mu_0 \eta i \ [without\ materiel]$

$\vec{B}_2 = \mu n i \ [with\ material]$

and $\mu = \mu_0 (1 + \lambda)$

$\lambda \rightarrow \text{susceptibility}$

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So fraction increase

$$= \frac{B_2 - B_1}{B_1} = \frac{\mu_0(1+\lambda)ni - \mu_0 ni}{\mu_0 ni}$$

$$= \lambda = 1.2 \times 10^{-5}$$

Hence option 1 correct.

Q. 13 Two parallel, long wires are kept 0.20 m apart in a vacuum, each carrying a current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately:

Option 1:

1

Option 2:

2.4

Option 3:

1.4

Option 4:

2

Correct Answer:

1.4

Solution:

$$\frac{dF}{dl} = \frac{\mu \mathrm{o} I_1 I_2}{2\pi d}$$

$$= \frac{4\pi \times 10^{-7} \times x \times x}{2\pi \times 0.2} = 2 \times 10^{-6}$$

$$x = \sqrt{2} = 1.4$$

Hence option 3 is correct

Q. 14 A coil is placed in a time-varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :

(Assume the coil to be short circuited.)

Option 1:

Halved

Option 2:

Quadrupled

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Option 3:
The same

Option 4:
Doubled

Correct Answer:
Doubled

Solution:
using Faradays law

$$\epsilon = \left| -\frac{d\phi}{dt} \right| = N\frac{dB}{dt}A$$

$$As\ N \text{ is half} \Rightarrow \epsilon' = \frac{\epsilon}{2}$$

Now wire of length l was forming N loops

$$As\ N \text{ is half} \Rightarrow l' = \frac{l}{2}$$

So overall for wire its length is halved & radius is doubled

$$R = \frac{\rho l}{\pi r^2}$$

$$R' = R/8$$

$$P_I = \epsilon^2/R$$

$$P_f = \frac{(\epsilon/2)^2}{(R/8)} = \frac{2\varepsilon^2}{R} = 2P_I$$

Hence option 4 (D) is correct

Q. 15 An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of $60\ \mathrm{Vm}^{-1}$. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

Option 1:

$$\mathrm{E}_y = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8\mathrm{t}\right)\right]\hat{j}\mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 2\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8\mathrm{t}\right)\right]\hat{k}\ \mathrm{T}$$

Option 2:

$$\mathrm{E}_y = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8\mathrm{t}\right)\right]\hat{j}\mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^3\left(x - 3\times 10^8\mathrm{t}\right)\right]\hat{k}\ \mathrm{T}$$

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Option 3:

$$\mathrm{E}_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 60 \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Option 4:

$$\mathrm{E}_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^4 \left(x - 4 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 60 \sin\left[\frac{\pi}{4} \times 10^4 \left(x - 4 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Correct Answer:

$$\mathrm{E}_y = 60 \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Solution:

$\text{speed of wave in vaccum} = 3 \times 10^8\ \text{m/s}$

$E_0 = CB_0$

$$B_0 = \frac{60}{3 \times 10^8} = 2 \times 10^{-7}\ \mathrm{T}$$

Hence option 2 is correct.

Q. 16 In young's double slit experiment performed using monochromatic light of wavelength λ, when a glass plate $(\mu = 1.5)$ of thickness $\mathrm{x}\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :

Option 1:

3

Option 2:

2

Option 3:

1.5

Option 4:

0.5

Correct Answer:

2

Solution:

For the intensity to remain the same the position must be of maxima.

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'so path difference must be $n\lambda$

$(1.5-1)x\lambda = n\lambda$

$x = 2n, n = 0, 1, 2 \ldots$

$x = 0, 2, 4, 6 \ldots$

Hence option B is correct.

Q. 17 Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2, respectively are incident on a metallic surface.

$\text{If } \lambda_1 = 3\lambda_2 \text{ then :}$

Option 1:

$\mathrm{K}_1 > \frac{\mathrm{K}_2}{3}$

Option 2:

$\mathrm{K}_1 < \frac{\mathrm{K}_2}{3}$

Option 3:

$\mathrm{K}_1 = \frac{\mathrm{K}_2}{3}$

Option 4:

$\mathrm{K}_2 = \frac{\mathrm{K}_1}{3}$

Correct Answer:

$\mathrm{K}_1 < \frac{\mathrm{K}_2}{3}$

Solution:

Using Einstein's photoelectric effect eqn

$E = \phi + k_{\mathrm{Emx}}$

$\frac{hC}{\lambda_1} = \Phi + K_1 - (1)$

$\frac{hC}{\lambda_2} = \phi + K_2 - (2)$

taking the ratio of Eqn (1) with Eqn (2)

And using $\lambda_1 = 3\lambda_2$

$\frac{\lambda_2}{\lambda_1} = \frac{\phi + K_1}{\phi + K_2} = \frac{1}{3}$

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$\Rightarrow 3K_1 = K_2 - 2\phi$

$\Rightarrow K_1 < K_2/3$

Hence Option B is correct.

Q. 18 Following statements related to radioactivity are given below:

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of $\log_e$ (no. of undecayed nuclei) Vs. time represents the reciprocal of the mean lifetime (τ).

(D) Product of decay constant (λ) and half-life time $\left(T_{1/2}\right)$ is not constant.

Choose the **most appropriate** answer from the options given below :

Option 1:
(A) and (B) only

Option 2:
(B) and (D) only

Option 3:
(B) and (C) only

Option 4:
(C) and (D) only

Correct Answer:
(B) and (C) only

Solution:
A) Statement A is False

Radioactivity is a random and spontaneous process and is dependent on physical conditions like temperature, the presence of fast-moving particles etc.

Radioactivity is a nuclear process, it is not concerned in any manner with the extranuclear part of the atom. It is independent of any chemical property of the element for example it does not effect the electronic configuration of the element.

B) Statement B is True

As $N = Noe^{-\lambda t}$

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where N=The number of un-decayed nuclei in the radioactive sample

(C) **Statement C is True**

$$N = Noe^{-\lambda t}$$

$$\ln(N) = -\lambda t + \ln N_0$$

so slope$= -\lambda$

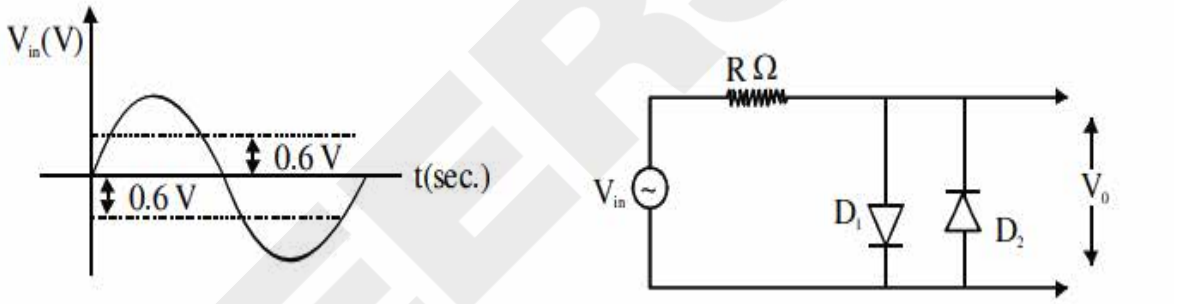
(D) **Statement D is False**

$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

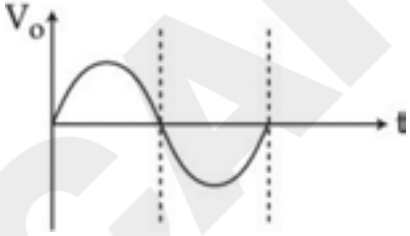
so, $t_{\frac{1}{2}} \times \lambda = \ln 2 = \text{ constant}$

Hence option C is correct.

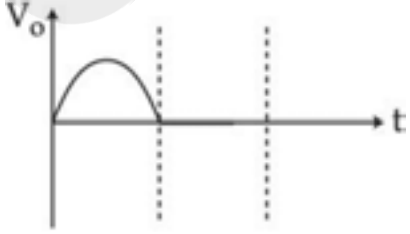
Q. 19 In the given circuit the input voltage V in is shown in figure. The cut- in voltage of p-n junction diode $(D_1 \text{ or } D_2)$ is $0.6\ V$. Which of the following output voltage (V_o) waveform across the diode is **correct** ?



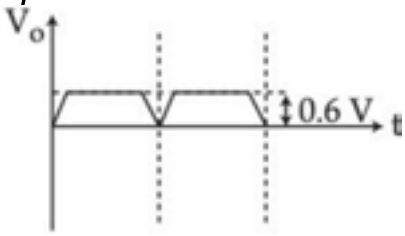
Option 1:



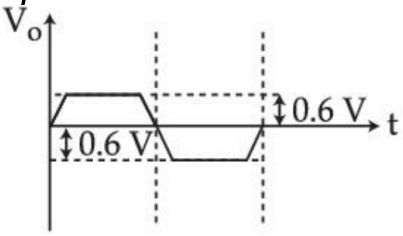
Option 2:



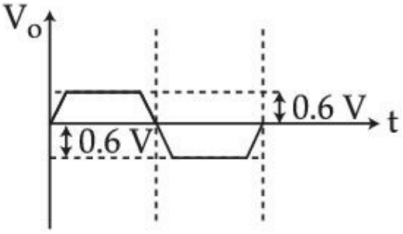
Option 3:



Option 4:



Correct Answer:



Solution:

In +ve half cycle

$\mathrm{D}_1 \rightarrow F.B.; \quad \mathrm{D}_2 \rightarrow R.B.$

In -ve half cycle

$\mathrm{D}_2 \rightarrow F.B.; \quad \mathrm{D}_1 \rightarrow R.B.$

So in both the cycles one of the diode will be in forward biased.

$For \quad |\mathrm{V}_{in}| \leq 0.6 \text{ V}, |\mathrm{Vo}| = \mathrm{V}_{in}$

$For \quad |\mathrm{V}_{in}| > 0.6 \text{ V}, |\mathrm{Vo}| = 0.6\ V$

so the correct graph will be

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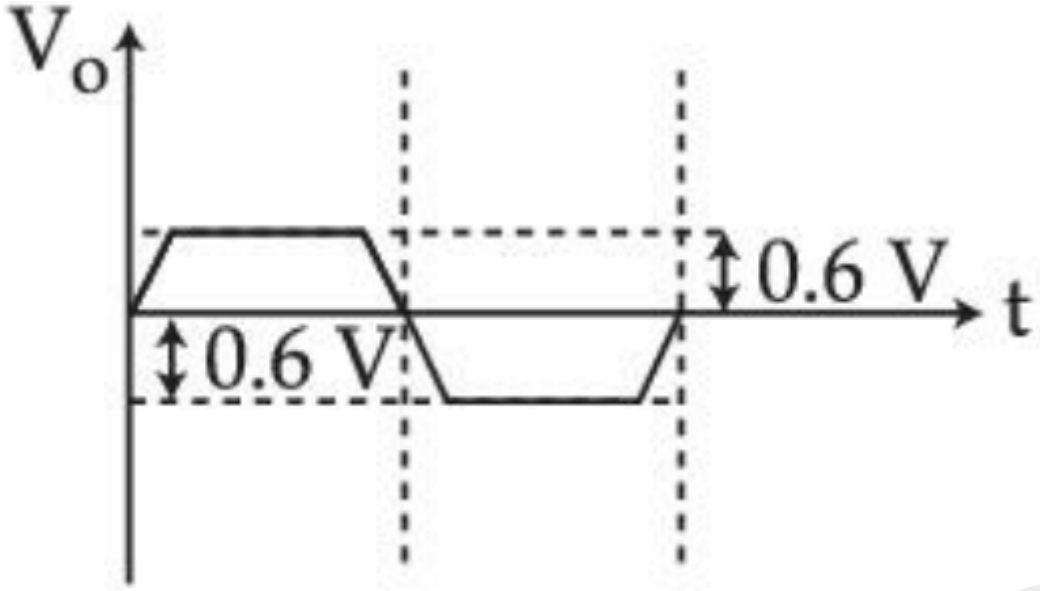

Hence D is correct option.

Q. 20 Amplitude modulated wave is represented by

$V_{AM} = 10\left[1 + 0.4\cos\left(2\pi \times 10^4 t\right)\right]\cos\left(2\pi \times 10^7 t\right)$. The total bandwidth of the amplitude modulated wave is:

Option 1:
$10\ \text{kHz}$

Option 2:
$20\ \text{MHz}$

Option 3:
$20\ \text{kHz}$

Option 4:
$10\ \text{MHz}$

Correct Answer:
$20\ \text{kHz}$

Solution:

$$V_{AM} = 10\left[1 + 0.4\cos\left(2\pi \times 10^4 t\right)\right] \times \cos\left(2\pi \times 10^7 t\right]$$

$\omega = 2\pi f_m = 2\pi \times 10^4$

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Bandwidth = $2 \times f_m$

$= 2 \times 10^4 = 20\text{kHz}$

Hence option 3 is correct.

Q. 21 A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.99 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is _____.

Correct Answer:
150

Solution:
\begin{aligned}

$$x_{\text{mean}} = \frac{1.22 + 1.23 + 1.19 + 1.20}{4}$$

$$= 1.21$$

$$\Delta x_1 = |x_1 - x_{\text{mean}}|$$

simil|arly $\Delta x_2, \Delta x_3, \Delta x_4$

$$\Delta x_{\text{mean}} = \frac{0.01 + 0.02 + 0.02 + 001}{4}$$

$$= 0.015$$

So, the percentage error

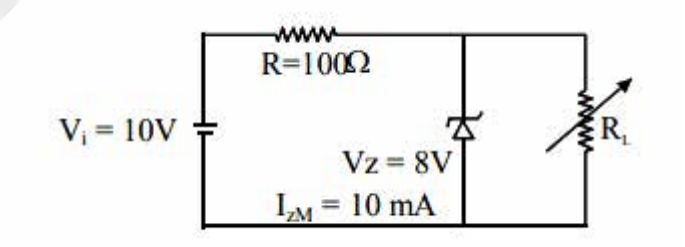
$$= \frac{\Delta x_{\text{mean}}}{x_{\text{mean}}} \times 100$$

$$= \frac{0.015}{1121} \times 100$$

$$= \frac{150}{121}\%$$

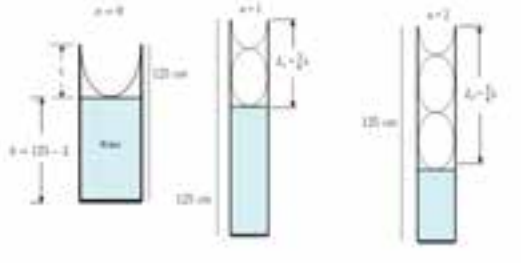
Hence 150 is the correct answer

Q. 22 A zener of breakdown voltage $\text{V}_\text{Z} = 8\ \text{V}$ and maximum zener current, $\text{I}_\text{ZM} = 10\ \text{mA}$ is subjected to an input voltage $\text{V}_\text{i} = 10\ \text{V}$ with series resistance $\text{R} = 100\ \Omega$. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is _____.



Correct Answer:
2

Solution:



The current through R

$$I = \frac{2}{100} = 20 \text{ mA}$$

Case I- For R_L to max current through it should be minimum

So $I_Z = 10mA \; and \;\; I_L = 10mA$

$$\mathrm{V_L} = \mathrm{I_L R_L}$$
$$8 = 10 \times 10^{-3} \times \mathrm{R_{L_{max}}}$$
$$\frac{4}{5} \times 10^3 = \mathrm{R_{L_{max}}}$$
$$800 = \mathrm{R_{L_{max}}}$$

Case II- For R_L to min current through it should be maximum

So $I_Z = 0 \; mA \; and \;\; I_L = 20mA$

$$\mathrm{V_L} = \mathrm{I_{L_{max}}} \times \mathrm{R_{L_{min}}}$$
$$\frac{8}{20} \times 10^3 = \mathrm{R_{L_{min}}}$$
$$400 = \mathrm{R_{L_{min}}}$$
$$\frac{\mathrm{R_{L_{max}}}}{\mathrm{R_{L_{min}}}} = \frac{800}{400} = 2$$

Q. 23 In a Young's double slit experiment, an angular width of the fring is 0.35° on a screen placed at 2m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index $7/5, \text{ is } \frac{1}{\alpha}$. The value of α is _____.

Correct Answer:
4

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Solution:
Angular fringe width

$$\Theta = \frac{\lambda}{D}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\mu_2}{\mu_1} = \frac{7/5}{1} = \frac{7}{5}$$

$$\lambda_2 = \frac{5}{7} \times \lambda_1 = \frac{5}{7} \times 450$$

$$\frac{\theta_1}{\lambda_1} = \frac{\theta_2}{\lambda_2}$$

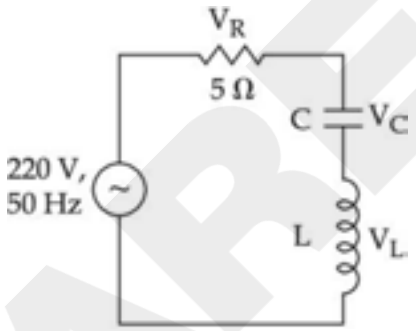
$$\theta_2 = \frac{\lambda_2}{\lambda_1} \times \theta_1$$

$$= \frac{5}{7} \times \frac{450}{450} \times 0.35$$

$$= \frac{5}{7} \times 0.35 = \frac{1}{4}$$

Hence 4 is correct answer.

Q. 24 In the given circuit, the magnitude of $\mathrm{V_L}$ and $\mathrm{V_C}$ are twice that of $\mathrm{V_R}$. Given that $f = 50$ Hz, the inductance of the coil is . $\frac{1}{\mathrm{K}\pi}\mathrm{mH}$. The value of K is _____.



Correct Answer:
0

Solution:
$V_L = 2V_R$

$i \times \omega L = 2 \times i \times R$
$L = \frac{2R}{\omega} = \frac{2\times5}{2\pi\times50} = \frac{1}{10\pi}$
$= \frac{100}{\pi}\mathrm{mH_2}$

$$K = \frac{1}{100} \approx 0$$

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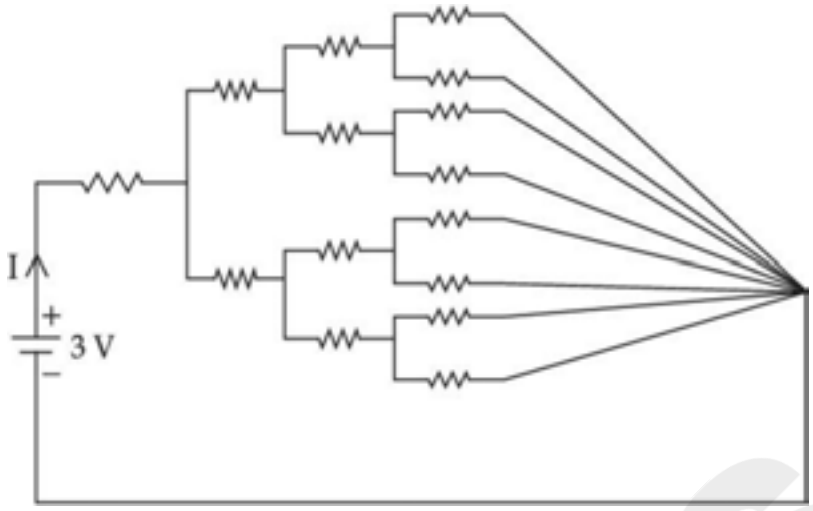
Hence 0 is correct answer.

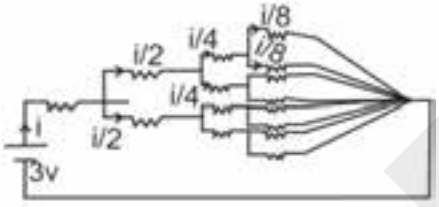
Q. 25 All resistances in the figure are $1\ \Omega$ each. The value of current 'I' is $\frac{a}{5}\ A$.

The value of a is _____.



Correct Answer:
8

Solution:



Using Kirchoff's Law

$$-3+i\times 1+\frac{i}{2}\times 1+\frac{i}{4}\times 1+\frac{i}{8}\times 1=0$$

$$3=i+\frac{i}{2}+\frac{i}{4}+\frac{i}{8}$$

$$=\frac{(8+4+2+1)\,i}{8}=\frac{15}{8}i$$

$$i=\frac{24}{15}=\frac{8}{5}$$

Hence 8 is correct answer

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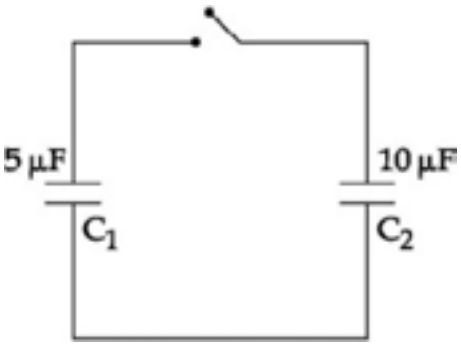
Q. 26 A capacitor C_1 of capcitance $5\ \mu F$ is charged to a potential of 30 V using a battery. The battery is then removed and the charged capacitor is connected to an uncharged capacitor C_2 of capcitance $10\ \mu F$ as shown in figure. When the switch is closed charge flows between the capacitors. At equilibrium, the charge on the capacitor C_2 is _____μC.



Correct Answer:
100

Solution:
A capacitor C_1 is charged to a potential of 30 V using a battery

then initial charge on it will be = $Q = C_1V = 5 \times 10^{-6} \times 30 = 150\mu\ C$

Let charge q flown in circuit

The potential of both will be the same under an equilibrium state

$$\frac{q}{10} = \frac{150-q}{5}, \quad q = 100\mu\ C$$

Hence 100 is correct answer

Q. 27 A tuning fork of frequency 340 Hz resonates in the m^{th} mode with an air column of length 125 cm in a cylindrical tube closed at one end. When water is slowly poured into it, the minimum height of water required for observing resonance once again is _____cm.

(Velocity of sound in air is $340\ ms^{-1}$)

Correct Answer:
50

Solution:
For resonance to occur

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$$f = (2n-1)\frac{V}{4l}$$

for $f = 340Hz \ and \ V = 340 \ m/s$

$$l = \frac{2n-1}{4}$$

So Height of water required = $h = 1.25 - l$

For n=0

$$l = 0.25m \ \ and \ h = 1m$$

For n=1

$$l = 0.75m \ \ and \ h = 0.5m$$

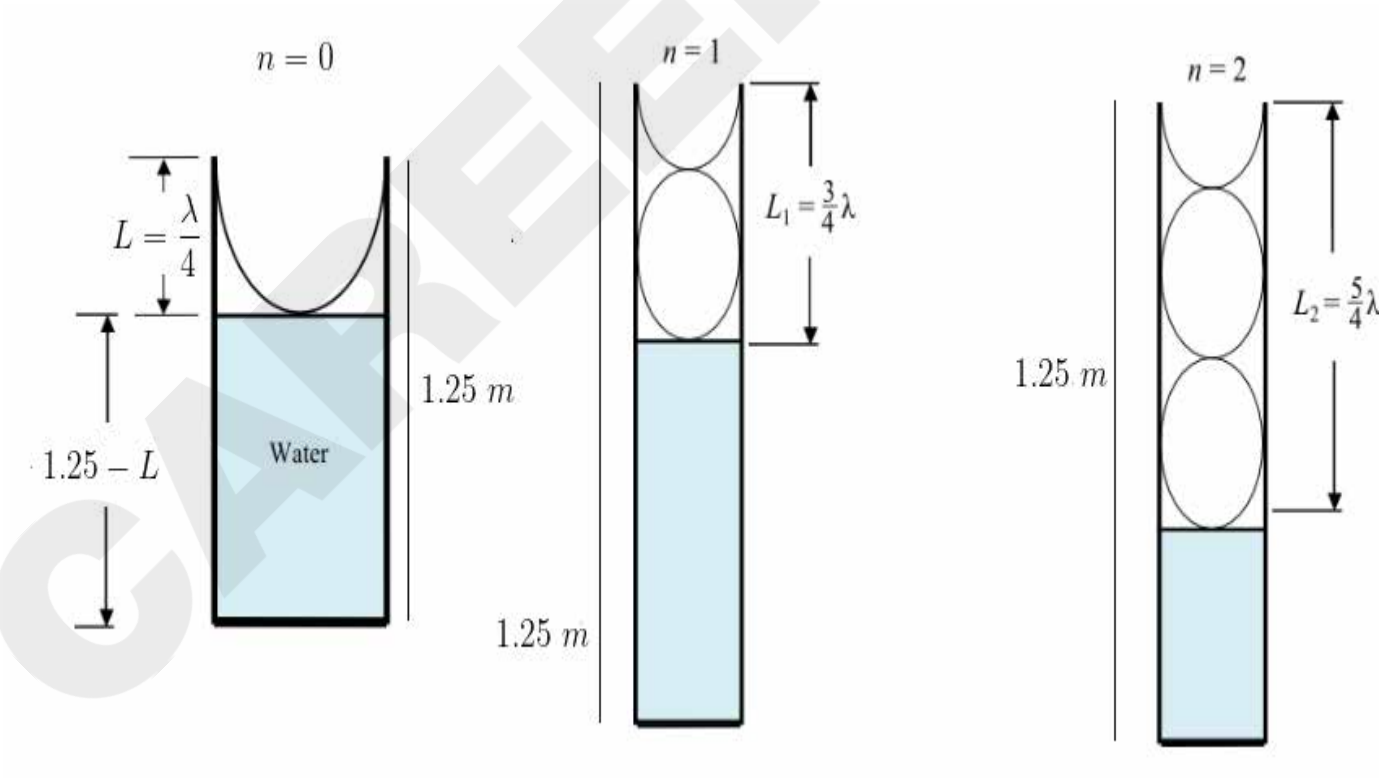
For n=2

$$l = 1.25m \ \ and \ h = 0m$$

h=0 means no water is there in the tube (which is already given in the question)

So the minimum value of h will be 0.5 m (i.e for n=2 case)

(see the below figure for n=0, n=1 and n=2 for better understanding)



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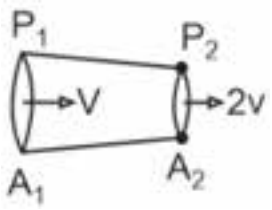
Q. 28 A liquid of density $750\ \mathrm{kgm}^{-3}$ flows smoothly through a horizontal pipe that tapers in cross-sectional area from $\mathrm{A}_1 = 1.2 \times 10^{-2}\ \mathrm{m}^2$ to $\mathrm{A}_2 = \dfrac{\mathrm{A}_1}{2}$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____$\times 10^{-3}\ \mathrm{m}^3\ \mathrm{s}^{-1}$.

Correct Answer:
24

Solution:



$$A_1V_1 = A_2V_2$$
$$2AV = AV_2 \quad \text{_____(1)}$$
$$V_2 = 2V$$

Bernoulli's equation

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$
$$\frac{3}{2}\rho v^2 = P_1 - P_2$$
$$V = \sqrt{\frac{2\,(P_1 - P_2)}{3\rho}}$$
$$= \sqrt{\frac{2\pi \times 4500}{3 \times 750}} = 2\ \mathrm{m/s}$$

So,

$$Q = A_1 v = 24 \times 10^{-3}\ \mathrm{m}^3/\mathrm{sec}$$

Hence 24 is the correct answer

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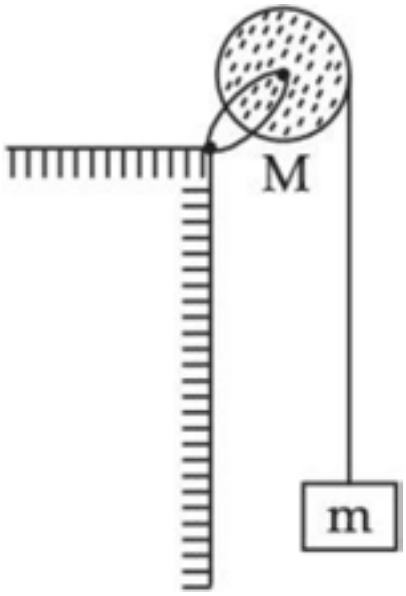
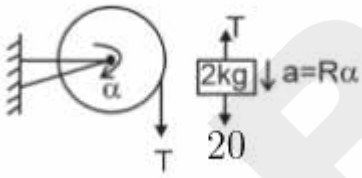
Q. 29 A uniform disc with mass $\mathrm{M} = 4\ \mathrm{kg}$ and radius $\mathrm{R} = 10\ \mathrm{cm}$ is mounted on a fixed horizontal axle as shown in figure. A block with mass $\mathrm{m} = 2\ \mathrm{kg}$ hangs from a massless cord that is wrapped around the rim of the disc. During the fall of the block, the cord does not slip and there is no friction at the axle. The tension in the cord is _____ N.

$\left(\text{ Take } \mathrm{g} = 10\ \mathrm{ms}^{-2}\right)$



Correct Answer:
10

Solution:



$R = 0.1\ m$

$I = \dfrac{MR^2}{2} = 0.02$

$20 - T = 2a - (1)$

$T \times 0.1 = 0.02 \times \alpha = 0.02 \times \dfrac{a}{0.1} - (2)$

$\Rightarrow T = 2a, a = 5\ \mathrm{m/s^2}$

$\mathrm{So},\ T = 10\ \mathrm{N}$

Hence 10 is the correct answer

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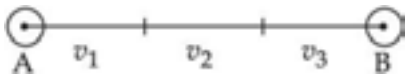
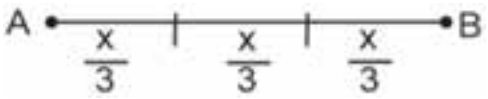
Q. 30 A car covers AB distance with first one-third at velocity $v_1\ ms^{-1}$, second one-third at $v_2\ ms^{-1}$ and last one-third at $v_3\ ms^{-1}$. If $v_3 = 3v_1, v_2 = 2v_1$ and $v_1 = 11\ ms^{-1}$ then the average velocity of the car is _____ms^{-1}.



Correct Answer:
18

Solution:



$$\text{Avg speed} = \frac{x}{\left(\frac{x}{3v_1} + \frac{x}{3v_2} + \frac{x}{3v_3}\right)}$$
$$= \frac{3}{\left(\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}\right)}$$
$$= \frac{3}{\frac{1}{11} + \frac{1}{22} + \frac{1}{33})}$$
$$= 18\ \text{m/s}$$

Hence 18 is correct option.

Chemistry

Q. 1 Compound A contains 8.7% Hydrogen 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is,

Given: Atomic masses of C,H and N are 12, 1 and 14 amu respectively.

The molar mass of the compound A is $162\ g\ mol^{-1}$.

Option 1:
$C_4H_6\ N_2$

Option 2:
$C_2H_3\ N$

Option 3:
$C_5H_7\ N$

Option 4:
$C_{10}H_{14}\ N_2$

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Correct Answer:
$C_{10}H_{14}\ N_2$

Solution:
Given,

$H \rightarrow 8.7\ \%$
$C \rightarrow 74\ \%$
$N \rightarrow 17.3\ \%$

Assume 100 g of compound

$$H \rightarrow 8.7\ g \rightarrow \frac{8.7}{1} = 8.7\ moles$$

$$C \rightarrow 74\ g \rightarrow \frac{74}{12} = 6.16\ moles$$

$$N \rightarrow 17.3\ g \frac{17.3}{14} = 1.23\ moles$$

Thus, the simplest ratio

$H{=}7\ C{=}5\ N{=}1$

Empirical formula will be C_5NH_7 having weight of $81\ g\ mol^{-1}$

But given compound 'A' has mass of $162\ g\ mol^{-1}$

So the formula will be $C_{10}N_2H_{14}$

Hence, **Option (4)** is correct.

Q. 2 Consider the following statements:

(A) The principle quantum number 'n' is a positive integer with values of $'n' = 1, 2, 3, \ldots$

(B) The azimuthal quantum number 'l' for a given 'n' (principal quantum number) can have values as $'l' = 0, 1, 2, \ldots .n$

(C) Magnetic orbital quantum number 'm' for a particular 'l' (azimuthal quantum number) has $(2l + 1)$ value.

(D) $\pm 1/2$ are the two possible orientations of electron spin.

(E) $\text{For } l = 5,$ there will be a total of 9 orbital

Which of the above statement are **correct** ?

Option 1:
$(A), (B) \text{ and } (C)$

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Option 2:
$(A), (C), (D)$ and (E)

Option 3:
$(A), (C)$ and (D)

Option 4:
$(A), (B), (C)$ and (D)

Correct Answer:
$(A), (C)$ and (D)

Solution:
- Statement (A) is Correct as ' n ' is a positive integer with values of $'\ n\ ' = 1, 2, 3 \ldots\ldots$

- Statement (B) is incorrect as 'ℓ' Can have values from 0 to $(n-1)$

- Statement (C) is Correct as m can take values from $-\ell$ to $+\ell$

$\therefore$ Total values is $2\ell + 1$

Statement (D) is correct as spin value $= \pm\frac{1}{2}$

Statement (E) is incorrect as for $l = 5$, no. of orbitals $= 2l + 1 = 11$

Hence, **Option (3)** is correct.

Q. 3 In the structure of SF_4, the lone pair of electrons on S is in.

Option 1:
equatorial position and there are two lone pair- bond pair repulsions at $90°$.

Option 2:
equatorial position and there are three lone pair- bond pair repulsions at $90°$.

Option 3:
axial position and there are three lone pair - bond pair repulsion at $90°$.

Option 4:
axial position and there are two lone pair - bond pair repulsion at $90°$.

Correct Answer:
axial position and there are two lone pair - bond pair repulsion at $90°$.

Solution:
SF_4 structure-

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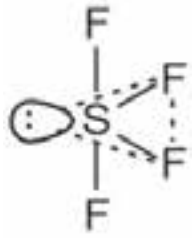

See-saw Shape $(\mathrm{sp^3d})$

As the lone pair of electrons on ' S ' is in equatiorial position and there are two lone pair-bond pair repulsion at 90°.

Hence, **Option (4)** is correct.

Q. 4 A student needs to prepare a buffer solution of propanoic acid and its sodium salt with $\mathrm{pH} = 4$.

The ratio of $\frac{\left[\mathrm{CH_3CH_2COO^-}\right]}{\left[\mathrm{CH_3CH_2COOH}\right]}$ required to make buffer is_____.

$\text{Given : } \mathrm{K_a\,(CH_3CH_2COOH)} = 1.3 \times 10^{-5}$

Option 1:
0.03

Option 2:
0.13

Option 3:
0.23

Option 4:
0.33

Correct Answer:
0.13

Solution:
Given,

$\mathrm{pH} = 4, \mathrm{Ka\,(CH_3CH_2COOH)} = 1.3 \times 10^{-5}$

We need to find out the ratio of $\frac{\left[\mathrm{CH_3CH_2COO^-}\right]}{\left[\mathrm{CH_3CH_2COOH}\right]}$

As we know

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$$\mathrm{pH} = \mathrm{pKa} + \log \frac{[\ \mathrm{Salt}\ ]}{[\ \mathrm{Acid}\ ]}$$

$$4 = 5 - \log 1.3 + \log \frac{\left[\mathrm{CH_3CH_2COO^-}\right]}{\left[\mathrm{CH_3CH_2COOH}\right]}$$

$$\log \frac{\left[\mathrm{CH_3CH_2COO^-}\right]}{\left[\mathrm{CH_3CH_2COOH}\right]} = \log 1.3 - 1 = \log \frac{1.3}{10}$$

comparing Both sides, we have

$$\frac{\left[\mathrm{CH_3CH_2COO^-}\right]}{\left[\mathrm{CH_3CH_2COOH}\right]} = \frac{1.3}{10} = 0.13$$

Hence, **Option(2)** is correct.

Q. 5 Match **List - I** with **List - II**

List - I	List - II
(A) negatively charged sol	(I) $\mathrm{Fe_2O_3 \cdot} x\mathrm{H_2O}$
(B) macromolecular colloid	(II) CdS sol
(C) positively charged sol	(III) Starch
(D) Cheese	(IV) a gel

Choose the **correct** answer from the options given below:

Option 1:
$(\mathrm{A}) - (\mathrm{II}), (\mathrm{B}) - (\mathrm{III}), (\mathrm{C}) - (\mathrm{IV}), (\mathrm{D}) - (\mathrm{I})$

Option 2:
$(\mathrm{A}) - (\mathrm{II}), (\mathrm{B}) - (\mathrm{I}), (\mathrm{C}) - (\mathrm{III}), (\mathrm{D}) - (\mathrm{IV})$

Option 3:
$(\mathrm{A}) - (\mathrm{II}), (\mathrm{B}) - (\mathrm{III}), (\mathrm{C}) - (\mathrm{I}), (\mathrm{D}) - (\mathrm{IV})$

Option 4:
$(\mathrm{A}) - (\mathrm{I}), (\mathrm{B}) - (\mathrm{III}), (\mathrm{C}) - (\mathrm{II}), (\mathrm{D}) - (\mathrm{IV})$

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Correct Answer:
$(A)-(II),(B)-(III),(C)-(I),(D)-(IV)$

Solution:
The correct combination of the given colloids are

(A) Negatively charged sol = CdS (II)

(B) Macromolecular colloid = Starch (III)

(C) Positively charged sol = $Fe_2O_3 \cdot xH_2O$ (I)

(D) Cheese = a gel (IV)

Hence, **Option(3)** is correct.

Q. 6 Match **List - I** with **List - II:**

List - I (Oxide)	**List -II (Nature)**
(A) Cl_2O_7	(I) Amphoteric
(B) Na_2O	(II) Basic
(C) Al_2O_3	(III) Neutral
(D) N_2O	(IV) Acidic

Choose the **correct** answer from the options given below:

Option 1:
$(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

Option 2:
$(A)-(IV),(B)-(II),(C)-(I),(D)-(III)$

Option 3:
$(A)-(II),(B)-(IV),(C)-(III),(D)-(I)$

Option 4:
$(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

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Correct Answer:
$(A)-(IV),(B)-(II),(C)-(I),(D)-(III)$

Solution:
The correct combination of the oxides and their nature is given below :

$Cl_2O_7 \longrightarrow$ Acidic

$Na_2O \longrightarrow$ Basic

$Al_2O_3 \longrightarrow$ Amphoteric

$N_2O \longrightarrow$ neutral

Thus, the correct combination is

$(A)-(IV),(B)-(II),(C)-(I),(D)-(III)$

Hence, **Option (2)** is correct.

Q. 7 In the metallurgical extraction of copper, following reaction is used:

$FeO + SiO_2 \rightarrow FeSiO_3$

FeO and $FeSiO_3$ respectively are

Option 1:
gangue and flux.

Option 2:
flux and slag.

Option 3:
slag and flux.

Option 4:
gangue and slag.

Correct Answer:
gangue and slag.

Solution:
In the extraction of Copper, SiO_2 is added as a flux to remove the impurity of FeO (gangue) in the form of $FeSiO_3$ (slag)

$FeO + SiO_2 \rightarrow FeSiO_3$

$FeO =$ Gangue

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$FeSiO_3 = Slag$

Hence, **Option (4)** is correct

Q. 8 Hydrogen has three isotopes : protium $\left({}^1H\right)$, deuterium $\left({}^2H \text{ or } D\right)$ and tritium $\left({}^3H \text{ or } T\right)$. They have nearly same chemical properties but different physical properties. They differ in

Option 1:
number of protons.

Option 2:
atomic number.

Option 3:
electronic configuration.

Option 4:
atomic mass.

Correct Answer:
atomic mass.

Solution:
Three isotopes of. $H: {}^1H, {}^2H, {}^3H$ have different neutrons hence different mass number.

Hence, **Option (4)** is correct.

Q. 9 Among the following, basic oxide is:

Option 1:
SO_3

Option 2:
SiO_2

Option 3:
CaO

Option 4:
Al_2O_3

Correct Answer:
CaO

Solution:

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As we have learnt,

SO_3 and SiO_2 are acidic oxides.

Al_2O_3 is amphoteric oxide.

CaO is Basic in nature.

Hence, **Option (3)** is correct.

Q. 10 Among the given oxides of nitrogen ; $N_2O, N_2O_3,\ N_2O_4$ and N_2O_5, the number of compound/(s) having N - N bond is:

Option 1:
1

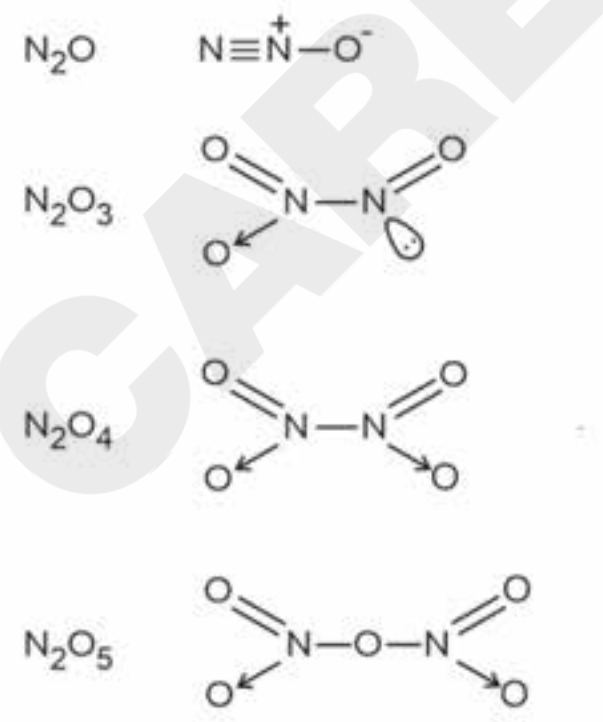
Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
3

Solution:



So 3 compounds are having N-N bonds.

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Hence, **Option (3)** is correct

Q. 11 Which of the following oxoacids of sulphur contains "S" in two different oxidation states?

Option 1:
$H_2S_2O_3$

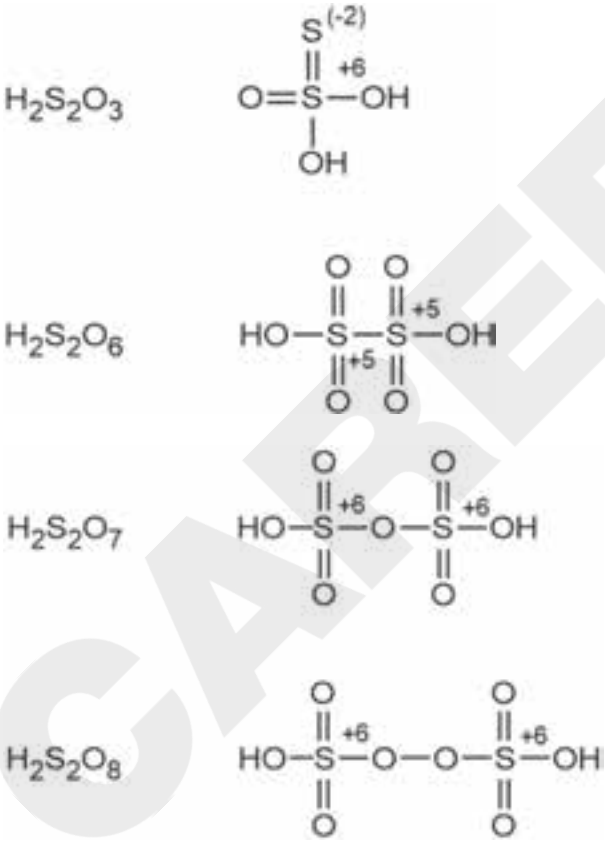
Option 2:
$H_2S_2O_6$

Option 3:
$H_2S_2O_7$

Option 4:
$H_2S_2O_8$

Correct Answer:
$H_2S_2O_3$

Solution:



So in $H_2S_2O_3$, sulfur Contains "S" in two different oxidation States.

Hence, **Option (1)** is correct.

Q. 12 Correct statement about photo-chemical smog is:

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Option 1:
It occures in humid climate.

Option 2:
It is a mixture of smoke, fog and SO_2.

Option 3:
It is reducing smog.

Option 4:
It results from reaction of unsaturated hydrocarbons.

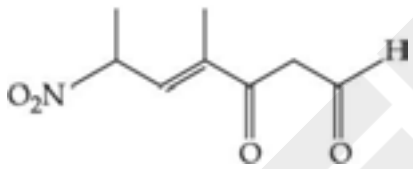
Correct Answer:
It results from reaction of unsaturated hydrocarbons.

Solution:
Photochemical smog is the result of reaction of unsaturated hydrocarbons and NO_x in presence of sunlight.

It occurs in warm, dry and sunny climate.

Hence,**Option (4)** is Correct.

Q. 13 The correct IUPAC name of the following compound is:



Option 1:
4-methyl-2-nitro-5-oxohept-3-enal

Option 2:
4-methyl-5-oxo-2-nitrohept-3-enal

Option 3:
4-methyl-6-nitro-3-oxohept-4-enal

Option 4:
6-formyl-4-methyl-2-nitrohex-3-enal

Correct Answer:
4-methyl-6-nitro-3-oxohept-4-enal

Solution:

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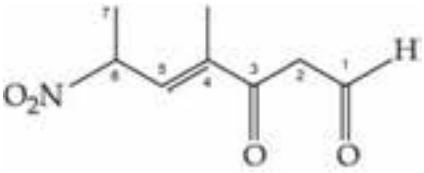
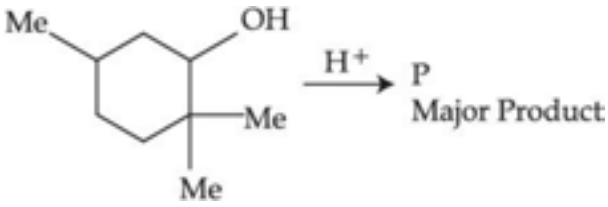


IUPAC name: 4 methyl -6-nitro-3-oxohept-4-enal.

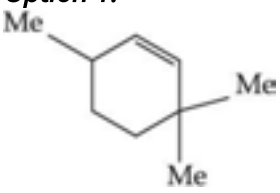
Hence, **Option (3)** is correct.

Q. 14 The major product (P) of the given reaction is

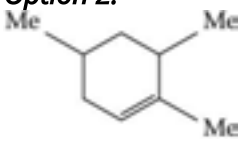
(where, Me is $-CH_3$)



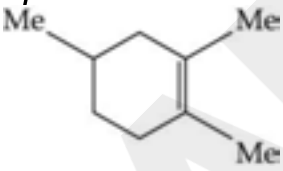
Option 1:



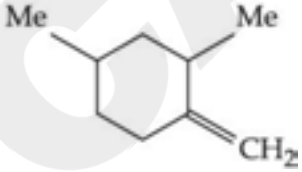
Option 2:



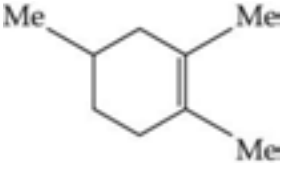
Option 3:



Option 4:



Correct Answer:



Solution:

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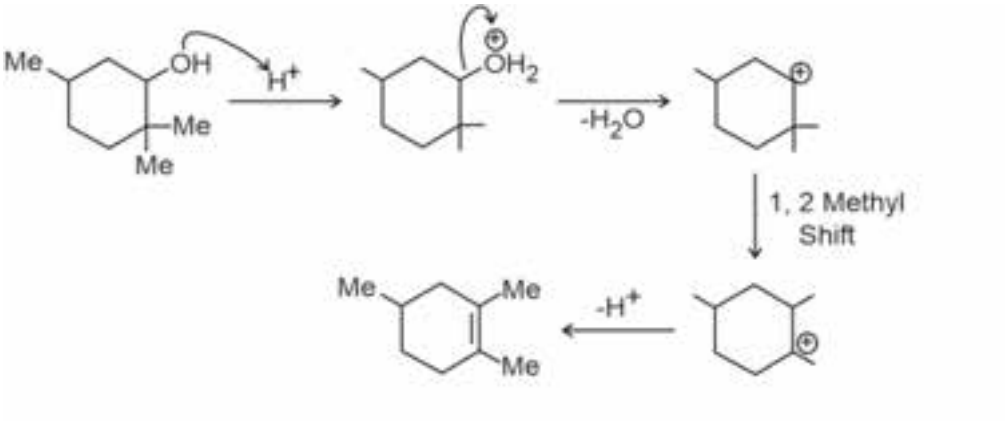
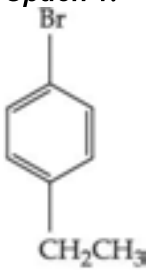


Hence, **Option (3)** is correct.

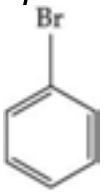
Q. 15 $A \xrightarrow[(ii)\ CN^-]{(i)\ Cl_2, \Delta} \underset{(iii)\ H_2O/H^+}{} 4\text{-Bromophenyl acetic acid.}$

In the above reaction 'A' is

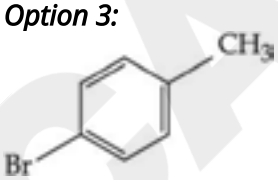
Option 1:



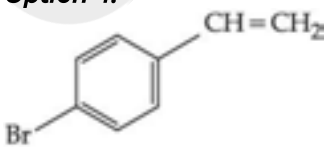
Option 2:



Option 3:



Option 4:



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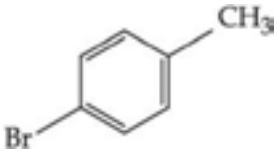
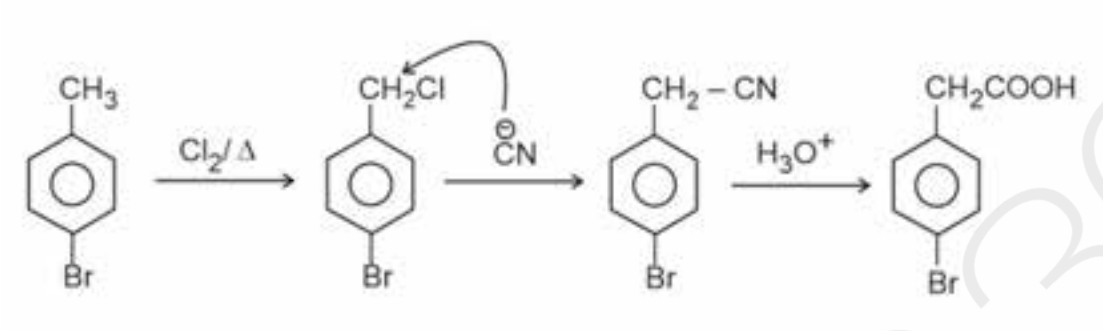
Correct Answer:



Solution:

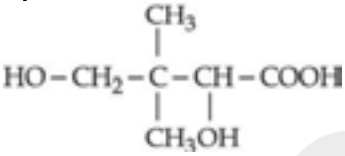
$$\text{A} \xrightarrow[\text{(ii)CN}^- \ \text{(iii)H}_2\text{O/H}^+]{\text{(i)Cl}_2,\Delta} 4-\text{Bromophenyl acetic acid}$$



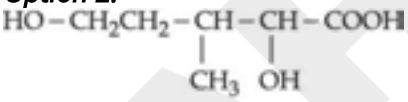
Hence, **Option (3)** is correct.

Q. 16 Isobutyraldehyde on reaction with formaldehyde and K_2CO_3 gives compound 'A'. Compound 'A' reacts with KCN and yields compound 'B', which on hydrolysis gives a stable compound 'C'. The compound 'C' is

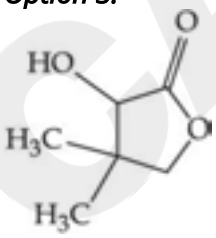
Option 1:



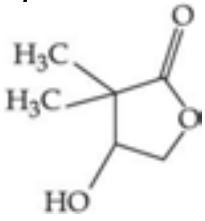
Option 2:



Option 3:



Option 4:



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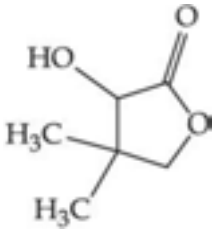
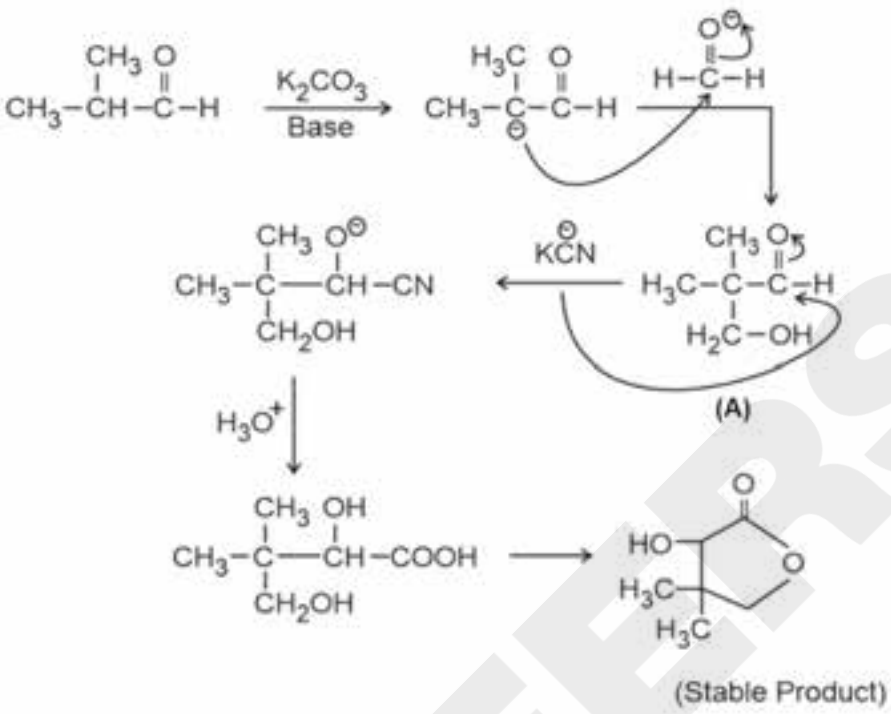
Correct Answer:



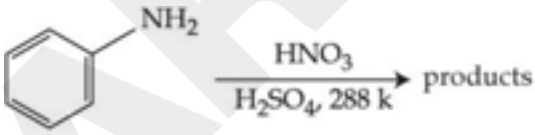
Solution:

According to the question, the reaction will be-



Hence, **Option (3)** is correct

Q. 17 With respect to the following reaction, consider the given statement:



(A) o-Nitroaniline and p-nitroaniline are the predominant products.

(B) p-Nitroaniline and m-nitroaniline are the predominant products.

(C) HNO_3 acts as an acid.

(D) H_2SO_4 acts as an acid.

Option 1:

(A) and (C) are correct statements.

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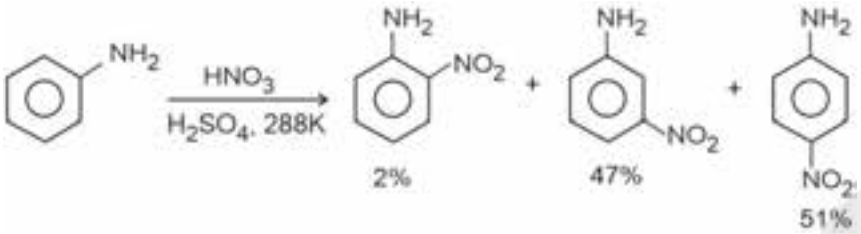
Option 2:
(A) and (D) are correct statements.

Option 3:
(B) and (D) are correct statements.

Option 4:
(B) and (C) are correct statements.

Correct Answer:
(B) and (D) are correct statements.

Solution:



$$\underset{\text{Base}}{HNO_3} + \underset{\text{Acid}}{H_2SO_4} \longrightarrow NO_2^+$$

Hence $\mathrm{p-Nitroaniline\ and\ m-nitroaniline}$ are predominant products and H_2SO_4 acts as an acid.

Hence, **Option (3)** is correct.

Q. 18 Given below are two statements, one is **Assertion (A)** and other is **Reason (R)**.

Assertion (A) : Natural rubber is a linear polymer of isoprene called cis-polyisoprene with elastic properties.

Reason (R) : The cis- polyisoprene molecules consist of various chains held together by strong polar interactions with coiled structure.

In the light of the above statement, choose the **correct** one from the options given below:

Option 1:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A).**

Option 2:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A).**

Option 3:
(A) is true but **(R)** is false.

Option 4:
(A) is false but **(R)** is true.

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Correct Answer:
(A) is true but **(R)** is false.

Solution:
Assertion is correct but Reason says that cis-polyisoprene molecules consists of various chains held together by strong polar interactions with coiled structure, which is incorrect.

The chains are held together by weak Vander Waal's interactions.

Hence ,**Option (3)** is Correct.

Q. 19 When sugar 'X' is boiled with dilute H_2SO_4 in alcoholic solution,two isomers 'A' and 'B' are formed 'A' on oxidation with HNO_3 yields saccharic acid where as 'B' is laevorotatory, The compound 'X' is:

Option 1:
Maltose

Option 2:
Sucrose

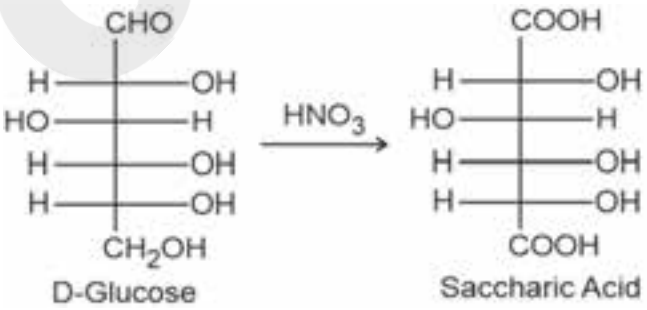
Option 3:
Lactose

Option 4:
Strach

Correct Answer:
Sucrose

Solution:

$$\underset{[\alpha]=66.6^\circ}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\substack{D-Glucose \\ [\alpha]=+52.7^\circ \\ (A)}}{C_6H_{12}O_6} + \underset{\substack{D-Fructose \\ [\alpha]=-92.2^\circ \\ (B)}}{C_6H_{12}O_6}$$



Hence, **Option (2)** is correct.

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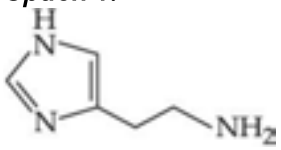
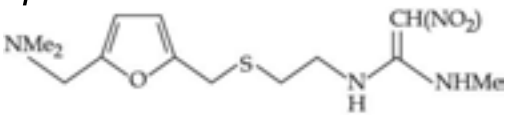
Q. 20 The drug tegamet is:

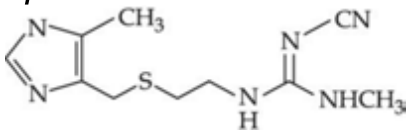
Option 1:



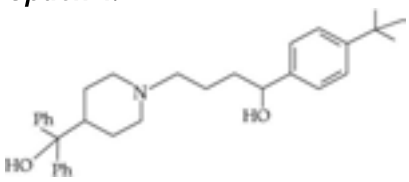
Option 2:



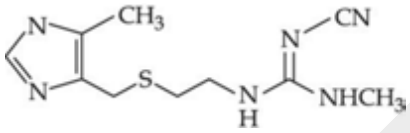
Option 3:



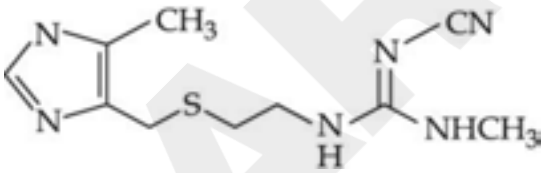
Option 4:



Correct Answer:



Solution:
Tegamet is name of cimetidine which is



Hence **Option (3)** is correct.

Q. 21 100 g of an ideal gas is kept in a cylinder of 416 L volume at $27^{\circ}\mathrm{C}$ under 1.5 bar pressure. The molar mass of the gas is _____ $\mathrm{gmol}-1$ (Nearest integer)

$$(\text{Given : R} = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1})$$

Correct Answer:
4

Solution:

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Given,

$$\mathrm{m} = 100\ \mathrm{g}$$
$$\mathrm{V} = 416\ \mathrm{L}$$
$$\mathrm{T} = 27^{\circ}\mathrm{C} = 300\ \mathrm{K}$$
$$\mathrm{P} = 1.5\ \mathrm{bar}$$
$$\mathrm{M} = ?$$

From Ideal gas equation, we have

$$\mathrm{P\ V = n\ R\ T}$$
$$(1.5\ \mathrm{bar}\ )(416) = \frac{100}{\mathrm{M}} \times 0.083 \times 300$$

On Calculating-

$$\mathrm{M} = 3.99 \approx 4$$

Hence, (4) is correct answer

Q. 22 For combustion of one mole of magnesium in an open container at 300 K and 1 bar pressure, $\Delta_{\mathrm{C}}\mathrm{H}^{\ominus} = -601.70\ \mathrm{kJ\ mol^{-1}}$, the magnitude of change in internal energy for the reaction is _____kJ.(Nearest integer)

$(\mathrm{Given:\ R = 8.3\ J\ K^{-1}\ mol^{-1}}\ )$

Correct Answer:
600

Solution:
Given,

$$\mathrm{T} = 300\ \mathrm{K}$$

$$\mathrm{P} = 1\ \mathrm{bar}$$

$$\Delta_{\mathrm{C}}\mathrm{H}^{0} = -601.7\ \mathrm{kJ\ mol^{-1}}$$

$$\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$$

Reaction of Combustion-

$$\mathrm{Mg(s)} + \frac{1}{2}\mathrm{O_2(g)} \longrightarrow \mathrm{MgO(s)}$$

$$\Delta\mathrm{H} = \Delta\mathrm{U} + \Delta\mathrm{ngRT}$$

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$$-601.7 = \Delta U(\text{kJ mol}^{-1}) - \frac{1 \times 8.3 \times 300}{2 \times 1000}$$

On solving-

$$\Delta U = -600.455 \text{ kJ} \approx -600 \text{ kJ}$$

Hence, 600 is the answer

Q. 23 2.5 g of protein containing only glycine $(C_2H_5NO_2)$ is dissolved in water to make 500 mL of solution. The osmotic pressure of this solution at 300 K is found to be 5.03×10^{-3} bar. The total number of glycine units present in the protein is_____.

$(\text{Given : R} = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1})$

Correct Answer:
330

Solution:
Given, $\text{T}=300\text{K}$

$$\text{P} = 5.03 \times 10^{-3} \text{ bar}$$

$$\text{R} = 0.083 \text{ L bar K}^{-1}\text{mol}^{-1}$$

Osmotic pressure -

$$\pi = \text{CRT}$$

$$5.03 \times 10^{-3} = \text{C}(0.083)(300)$$

$$\text{C} = 0.202 \times 10^{-3}\text{M}$$

$$\text{moles of protein} = 0.202 \times 10^{-3} \times 0.5$$

$$= 10^{-4} \times 1.01$$

We have-

$$\frac{\text{given mass}}{\text{molar mass}} = \text{no.of moles}$$

$$\therefore \frac{2.5}{\text{M}} = 1.01 \times 10^{-4}$$

$$\text{M(molar mass of protein)} = 24752$$

$$\text{no of glycerine units} = \frac{24752}{75} = 330.03$$

Hence, 330 is correct answer

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Q. 24 For the given reactions

$$Sn^{2+} + 2e^- \rightarrow Sn$$

$$Sn^{4+} + 4e^- \rightarrow Sn$$

the electrode potentials are; $E^{O}_{Sn^{2+}/Sn} = -0.140\ V$ and $E^{S}_{Sn^{4+}/Sn} = 0.010\ V$. The magnitude of standared electrode potential for $Sn^{4+}|Sn^{2+}$ i.e. $E^{0}_{Sn^{4+}|Sn^{2+}}$ is _____ $\times 10^{-2}$ V.

(Nearest integer)

Correct Answer:
16

Solution:

$$Sn^{2+} + 2e^- \rightarrow Sn,\ E^{0}_{Sn^{2+}|Sn} = -0.14V,\ \Delta G^{0}_{1} = -2F(-0.14) \quad -(i)$$
$$Sn^{4+} + 4e^- \rightarrow Sn,\ E^{0}_{Sn^{4+}|Sn} = 0.01V,\ \Delta G^{0}_{2} = -4F(0.01) \quad -(ii)$$

Operation $(ii) - (i)$ gives us

$$Sn^{4+} + 2e^- \rightarrow Sn^{2+}, E^{0}_{Sn^{4+}|Sn^{2+}}, \Delta G^{0}_{3} = -2FE^{0}$$

Thus, we can write

$$\Delta G^{0}_{3} = \Delta G^{0}_{2} - \Delta G^{0}_{1}$$

$$-2E^{0} \times F = -(0.04 + 0.28) \times F$$

$$E^{\circ} = 0.16\ Volt\ = 16 \times 10^{-2}\ V$$

Hence, 16 is correct answer

Q. 25 A radioactive element has a half life of 200 days. The percentage of original activity remaining after 83 days is _____.(Nearest integer)

(Given : antilog 0.125=1.333,antilog 0.693=4.93)

Correct Answer:
75

Solution:

$$\text{Half life } = 200 \text{ days}$$

$$t_{1/2} = \frac{\ln 2}{k}$$

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$$k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{200} \quad -(1)$$

Now, integrated rate equation for a first order reaction is given as

$$A = A_0 e^{-kt}$$

$$kt = \ln \frac{A_0}{A_t} \quad -(2)$$

Using equations (1) and (2), we have

$$\frac{\ln 2}{200}(83) = \ln\left(\frac{A_0}{A_{83}}\right)$$

$$\log\left(\frac{A_0}{A_{83}}\right) = \frac{\log 2}{200}(83) = \frac{0.3 \times 83}{200} \simeq 0.125$$

$$\left(\frac{A_0}{A_{83}}\right) = \text{antilog}(0.125) = 1.333 = \frac{4}{3}$$

In percentage -

$$\frac{A_{83}}{A_0} = \frac{3}{4} \times 100\% = 75\%$$

Hence, (75) is answer

Q. 26 $[Fe(CN)_6]^{4-}$
$[Fe(CN)_6]^{3-}$
$[Ti(CN)_6]^{3-}$
$[Ni(CN)_4]^{2-}$
$[Co(CN)_6]^{3-}$

Among the given complexes, number of paramagnetic complexes is_____.

Correct Answer:
2

Solution:
$[Fe(CN)_6]^{4-}$ - is diamagnetic

$[Fe\,(CN)_6]^{3-}$ - is paramagnetic with 1 unpaired e^-

$[Ti(CN)_6]^{-3}$ - is paramagnetic with 1 unpaired e^-

$[Ni(CN)_4]^{2-}$ - is diamagnetic

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$[Co(CN)_6]^{3-}$ - is diamagnetic

Thus, out of the given complexes, only 2 are paramagnetic.

Hence, 2 is the answer

Q. 27 (a) $CoCl_3.4NH_3$, (b) $CoCl_3 \cdot 5NH_3$ (c) $CoCl_3 \cdot 6NH_3$ and (d) $CoCl(NO_3)_2 \cdot 5NH_3$.

Number of complex(es) which will exist in cis-trans form is/are _____.

Correct Answer:
1

Solution:
Out of given complexes, only $CoCl_3 \cdot 4NH_3$ in $[Co(NH_3)_4Cl_2]Cl$ form can exhibit cis-trans isomers.

Other complexes are

$CoCl_3 \cdot 5NH_3 = [Co(NH_3)_5Cl]Cl_2$

$CoCl_3 \cdot 6NH_3 = [Co(NH_3)_6]Cl_3$

$CoCl(NO_3)_2 \cdot 5NH_3 = [Co(NH_3)_5Cl](NO_3)_2$
or $[Co(NH_3)_5(NO_3)]Cl(NO_3)$

All these other complexes cannot show cis-trans isomerism.

Hence, (1) is answer

Q. 28 The complete combustion of 0.492 g of an organic compound containing 'C', 'H' and 'O' gives 0.793g of CO_2 and 0.442 g of H_2O. The percentage of oxygen composition in the organic compound is _____.(nearest integer)

Correct Answer:
46

Solution:
Given, Organic compound $= 0.492$ g

$CO_2 = 0.793$ g.

$H_2O = 0.442$ g.

moles of CO_2 = moles of C in product formed

$= \frac{0.793}{44}$

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$$\therefore \text{ Weight of C} = \frac{0.793}{44} \times 12 = 0.216 \text{ g}$$

$$\text{moles of 'H'} = \frac{0.442}{18} \times 2$$

$$\text{Weight of 'H'} = \frac{0.442}{18} \times 2 \times 1 = 0.049 \text{ g}$$

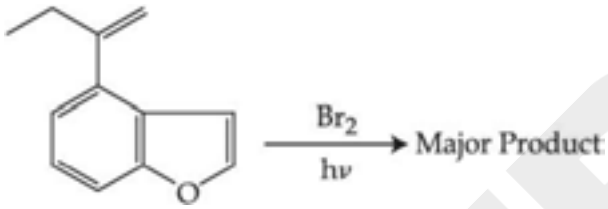
$$\therefore \text{ Weight of O} = \text{ Total wt. - wt.of H} - \text{wt} \cdot \text{of C}$$

$$= 0.492 - 0.216 - 0.049 = 0.227 \text{ g}$$

$$\% \text{ O} = \frac{0.227}{0.492} \times 100 = 46.13 \%.$$

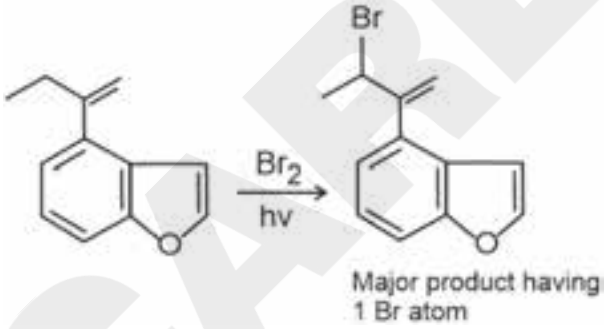
Hence, 46 is correct answer

Q. 29 The major product of the following reaction contains _____ bromine atom(s).



Correct Answer:
1

Solution:



Hence, 1 is the correct option.

Q. 30 $0.01 \text{ M } KMnO_4$ solution was added to $20.0 \text{ mL of } 0.05\text{M Mohr}' \text{s}$ salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of $KMnO_4$ solution left in the burette after the end point is _____mL. (nearest integer)

Correct Answer:
30

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Solution:
The formula for Mohr's Salt is $(NH_4)_2SO_4 \cdot Fe(SO_4) \cdot 6H_2O$

When oxidised with $KMnO_4$, only Fe^{2+}will get oxidised to Fe^{3+}

Thus, n-factor of Mohr's salt is 1.

Now, from law of equivalence, we have,

$N_1V_1 = N_2V_2$

Putting values ___

$0.01 \times 5 \times v_1 = 0.05 \times 1 \times 20$

$v_1 = 20\ ml$

Thus 20 ml of $KMnO_4$ will be used in the titration.

Volume left $= 50 - 20 = 30\ ml$

Hence, 30 is answer

Maths

Q. 1 Let $R_1 = \{(a, b) \in \mathbf{N} \times \mathbf{N} : |a - b| \leqslant 13\}$ and $R_2 = \{(a, b) \in \mathbf{N} \times \mathbf{N} : |a - b| \neq 13\}$. Then on $\mathbf{N}$:

Option 1:
Both $R_1 and\ R_2$ are equivalence relations

Option 2:
Neither $R_1\ nor\ R_2$ is an equivalence relation

Option 3:
R_1 is an equivalence relation but R_2 is not

Option 4:
R_2 is an equivalence relation but R_1 is not

Correct Answer:
Neither $R_1\ nor\ R_2$ is an equivalence relation

Solution:
R_1 is not transitive as $(1, 12) \in R_1$ and $(12, 24) \in R_1$ but $(1, 24) \notin R_1$

R_2 is not transitive as $(1, 10) \in R_2$ and $(10, 14) \in R_2$ but $(1, 14) \notin R_2$

Hence the correct option is 2

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Q. 2 Let $f(x)$ be a quadratic polynomial such that $f(-2)+f(3)=0$. If one of the roots of $f(x)=0$ is -1, then the sum of the roots of $f(x)=0$ is equal to:

Option 1:
$\frac{11}{3}$

Option 2:
$\frac{7}{3}$

Option 3:
$\frac{13}{3}$

Option 4:
$\frac{14}{3}$

Correct Answer:
$\frac{11}{3}$

Solution:
Let roots be $-1, \alpha$

$\therefore \quad f(x)=a(x+1)(x-\alpha)$

Using $f(-2)+f(3)=0$

$a(-2+1)(-2-\alpha)+a(3+1)(3-\alpha)=0$

$\Rightarrow \alpha+2+4(3-\alpha)=0$

$\Rightarrow \alpha+2+12-4\alpha=0$

$\Rightarrow \alpha=\frac{14}{3}$

$\text{Sum of roots}=-1+\frac{14}{3}=\frac{11}{3}$

Hence the correct answer is option 1

Q. 3 The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4 so that C_2 receives atleast 4 and atmost 7 candies, C_3 receives atleast 2 and atmost 6 candies, is equal to:

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Option 1:
205

Option 2:
615

Option 3:
510

Option 4:
430

Correct Answer:
430

Solution:
$4 \leq c_2 \leq 7$

$2 \leq c_3 \leq 6$

Now for solution of $C_1 + C_2 + C_3 + C_4 = 30$ with given conditions, the number of solutions equal

$\text{coefficient of } x^{30} \text{ in } \left(1 + x + x^2 + \cdots\right)\left(x^4 + x^5 + x^6 + x^7\right)$

$\left(x^2 + x^3 + x^4 + x^5 + x^6\right)\left(1 + x + x^2 + \ldots\right)$

$= \text{ coefficient of } x^{30} \text{ in } x^6\left(1 + x + x^2 + \cdots\right)^2$

$\left(1 + x + x^2 + x^3\right)\left(1 + x + x^2 + \ldots + x^4\right)$

$= \text{ coofficient of } x^{24} \text{ in } \frac{1}{(1-x)^2} \cdot \frac{(x^4-1)}{(x-1)} \cdot \frac{(x^5-1)}{(x-1)}$

$= \text{ coefficient of } x^{24} \text{ in } \left(x^4 - 1\right)\left(x^5 - 1\right)(1-x)^{-4}$

$= \text{coefficient of } x^{24} \text{ in } \left(x^9 - x^5 - x^4 + 1\right)\left(1 + 4x + {}^5C_2x^2 + \ldots...\right)$

$= {}^{18}C_{15} - {}^{22}C_{19} - {}^{23}C_{20} + {}^{27}C_{24}$

$= {}^{18}C_3 - {}^{22}C_3 - {}^{23}C_3 + {}^{27}C_3$

$= 816 - 1540 - 1771 + 2925$

$= 430$

Hence correct option is 4

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Q. 4 The term independent of x in the expansion of $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}, x \neq 0$ is:

Option 1:
$\frac{7}{40}$

Option 2:
$\frac{33}{200}$

Option 3:
$\frac{39}{200}$

Option 4:
$\frac{11}{50}$

Correct Answer:
$\frac{33}{200}$

Solution:

$\text{General term in } \left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$

$= {}^{11}C_r\left(\frac{5}{2}\right)^{11-r}\left(x^3\right)^{11-r}\left(-\frac{1}{5}\right)^r\left(x^{-2}\right)^r$

$= {}^{11}C_r\frac{5^{11-2r}(-1)^r}{2^{11-r}}x^{33-3r-2r}$

Constant term in $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$

$= 1 \cdot \text{constant term in second bracket} - 1.\text{coefficient of } x^{-2} \text{in second bracket}$

$+3 \text{ coefficient of } x^{-3} \text{in second bracket}$ .....(i)

For constant term in second bracket $33-5r=0$

$\Rightarrow r=\frac{33}{5} \Rightarrow \text{ no such term}$

$\text{For } x^{-2} \text{ term } \Rightarrow 33-5r=-2 \Rightarrow r=7$

$\text{For } x^{-3} \text{ term } \Rightarrow 33-5r=-3 \Rightarrow r=\frac{36}{5}$

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$\Rightarrow$ no such term

$\therefore$ Required answer $= -1 \times {}^{11}C_7 \frac{5^{-3}}{2^4}(-1)$

$$= \frac{{}^{11}C_7}{5^3 \cdot 2^4}$$

$$= \frac{33}{200}$$

Hence answer is option 2

Q. 5 If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is $1:7$ and $a+n=33$, then the value of n is :

Option 1:
21

Option 2:
22

Option 3:
23

Option 4:
24

Correct Answer:
23

Solution:
Let common difference of the AP obtained be d.

$\therefore$ First AM $= a+d$

Last mean $= 100-d$

Their ratio $= \frac{a+d}{100-d} = \frac{1}{7} \Rightarrow 7a+8d=100$ .....(i)

Also $100 = a+(n+2-1)d$

$\Rightarrow 100 = a+(n+1)d$

$\Rightarrow 100 = 33-n+(n+1)d \quad (\text{ given } a+n=33)$

$\Rightarrow d = \frac{n+67}{n+1}$

Using (i)

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$$7(33-n)+\frac{8(n+67)}{n+1}=100$$

$$\Rightarrow 7n^2-132n-667=0$$

$$\Rightarrow n=23$$

Hence the correct option is 3.

Q. 6 Let $f, g: R \to R$ be functions defined by

$$f(x)=\begin{cases}[x], & x<0\\ |1-x|, & x\geqslant 0\end{cases} \text{ and } g(x)=\begin{cases}e^x-x, & x<0\\ (x-1)^2-1, & x\geqslant 0\end{cases}$$

where $[x]$ denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly :

Option 1:
One point

Option 2:
Two points

Option 3:
Three points

Option 4:
Four points

Correct Answer:
Two points

Solution:

$$g\left(0^-\right)=e^0-0=1$$

$$g(0)=g\left(0^+\right)=1-1=0$$

$$\therefore g(x) \text{ is discontinuous at } x=0$$

$$f\left(0^-\right)=-1$$

$$f(0)=f\left(0^+\right)=1$$

$$\therefore f(x) \text{ is discontinuous at } x=0$$

$$\therefore \text{ we need to check fog continuity at } x=0, \text{at } x \text{ where } g(x)=0$$

$$\text{At } x=0$$

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$\text{fog}(0) = \text{f}(0) = 1$

$\text{fog}\left(0^{+}\right) = \text{f}\left(0^{-}\right) = -1$

$\therefore$ Discontinuous at $\text{x} = 0$

At $\text{x} = 2$

$fog\left(2^{+}\right) = f\left(0^{+}\right) = 1$

$\text{fog}\left(2^{-}\right) = \text{f}\left(0^{-}\right) = -1$

$\therefore$ Discontinuous at $\text{x} = 2$

Hence the correct answer is option 2.

Q. 7 Let $\text{f} : \mathbf{R} \to \mathbf{R}$ be a differentiable function such that $\text{f}\left(\frac{\pi}{4}\right) = \sqrt{2}, \text{f}\left(\frac{\pi}{2}\right) = 0$ and $\text{f}'\left(\frac{\pi}{2}\right) = 1$ and let $\text{g}(\text{x}) = \int_{\text{x}}^{\pi/4} (\text{f}'(\text{t}) \sec\text{t} + \tan\text{t} \sec\text{t}\, \text{f}(\text{t}))\, \text{dt}$ for $\text{x} \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^{-}} g(x)$ is equal to:

Option 1:

2

Option 2:

3

Option 3:

4

Option 4:

-3

Correct Answer:

3

Solution:

$\text{Since } \frac{\text{d}}{\text{dx}}(\text{f}(\text{x}) \sec\text{x}) = \text{f}'(\text{x}) \sec\text{x} + \text{f}(\text{x}) \sec\text{x} \tan\text{x}$

$\therefore \text{g}(\text{x}) = \int_{\text{x}}^{\pi/4} (\text{f}'(\text{t}) \sec\text{t} + \tan\text{t} \cdot \sec\text{t} \cdot \text{f}(\text{t}))\, \text{dt}$

$= \text{f}(\text{t}) \cdot \sec(\text{t})|_{\text{x}}^{\pi/4}$

$= \text{f}\left(\frac{\pi}{4}\right) \cdot \sec\left(\frac{\pi}{4}\right) - \text{f}(\text{x}) \sec(\text{x})$

$= \sqrt{2} \cdot \sqrt{2} - \text{f}(\text{x}) \cdot \sec\text{x}$

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$$= 2 - f(x) \cdot \sec x$$

Now $\lim\limits_{x \to \frac{\pi}{2}^-} g(x) = 2 - \lim\limits_{x \to \frac{\pi}{2}^-} f(x) \cdot \sec x$

$$= 2 - \lim_{x \to \frac{\pi}{2}^-} \frac{f(x)}{\cos x}$$

Using L'H Rule

$$= 2 - \lim_{x \to \frac{\pi}{2}^-} \frac{f'(x)}{-\sin x}$$

$$= 2 - \frac{f'(\pi/2)}{-1}$$

$$= 2 + 1$$

$$= 3$$

Hence the correct answer is option 2

Q. 8 Let $f : \mathbf{R} \to \mathbf{R}$ be a continuous function satisfying $f(x) + f(x + k) = n$, for all $x \in \mathbf{R}$ where $k > 0$ and n is a positive integer. If $I_1 = \int_0^{4nk} f(x)dx$ and $I_2 = \int_{-k}^{3k} f(x)dx$, then :

Option 1:
$I_1 + 2I_2 = 4nk$

Option 2:
$I_1 + 2I_2 = 2nk$

Option 3:
$I_1 + nI_2 = 4n^2k$

Option 4:
$I_1 + nI_2 = 6n^2k$

Correct Answer:
$I_1 + nI_2 = 4n^2k$

Solution:
$f(x) + f(x + k) = n$

Put $x \to x + k$

$f(x + k) + f(x + 2k) = n$

Subtract these equations

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$f(x) - f(x+2k) = 0$

$\Rightarrow f(x+2k) = f(x)$

$f(x)$ is periodic with period $2k$

$$\text{Now } I_1 = \int_0^{4nk} f(x)dx = \int_0^{2n(2k)} f(x)dx = 2n\int_0^{2k} f(x)dx$$

$$\text{and } I_2 = \int_{-k}^{3k} f(x)dx = \int_{-k}^{4k-k} f(x)dx = \int_0^{4k} f(x)dx$$

$$= \int_0^{2\cdot(2k)} f(x)dx = 2\int_0^{2k} f(x)dx$$

$$\text{Now } \int_0^{2k} f(x)dx = \int_0^{k} f(x)dx + \int_k^{2k} f(x)dx$$

$\text{Put } x = t + k$ in second integral

$$= \int_0^k f(x)dx + \int_0^k f(t+k)dt$$

$$= \int_0^k (f(x) + f(x+k))dx$$

$$= \int_0^k ndx$$

$$= nk$$

$$\therefore I_1 = 2n^2k, \quad I_2 = 2nk$$

Checking options,

$$I_1 + nI_2 = 2n^2k + 2n^2k = 4n^2k$$

Hence correct answer is option 3

Q. 9 The area of the bounded region enclosed by the curve $y = 3 - \left|x - \frac{1}{2}\right| - |x+1|$ and the $x-$axis is:

Option 1:

$\frac{9}{4}$

Option 2:

$\frac{45}{16}$

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Option 3:

$\frac{27}{8}$

Option 4:

$\frac{63}{16}$

Correct Answer:

$\frac{27}{8}$

Solution:

$y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$

For $x < -1$

$y = 3 - \left(\frac{1}{2} - x\right) - (-1 - x)$

$= 3 + x - \frac{1}{2} + x + 1$

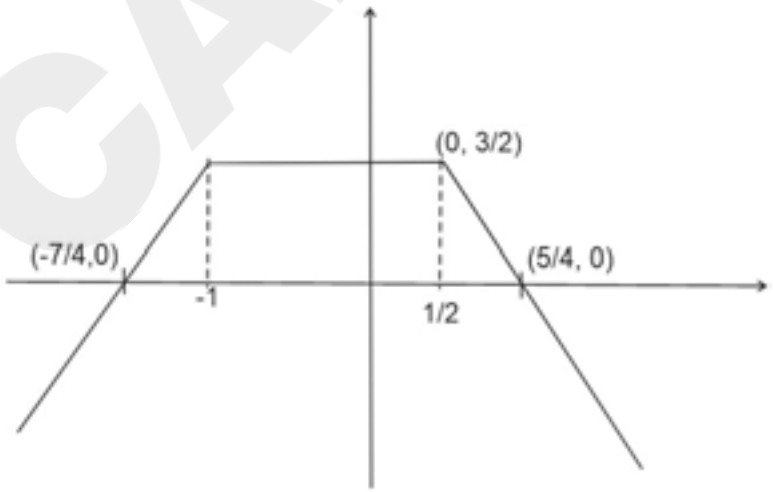
$= 2x + \frac{7}{2}$

For $-1 \leq x < \frac{1}{2}$

$y = 3 + x - \frac{1}{2} - x - 1 = \frac{3}{2}$

For $x \geqslant \frac{1}{2}$

$y = 3 - x + \frac{1}{2} - x - 1 = -2x + \frac{5}{2}$



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$$\text{Area } = \frac{1}{2}\left(\frac{3}{2}\times\frac{3}{4}\right)+\frac{3}{2}\times\frac{3}{2}+\frac{1}{2}\left(\frac{3}{2}\times\frac{3}{4}\right)$$

$$=\frac{9}{16}+\frac{9}{4}+\frac{9}{16}$$

$$=\frac{9}{8}+\frac{9}{4}$$

$$=\frac{27}{8}$$

Hence the correct answer is option 3.

Q. 10 Let $x = x(y)$ be the solution of the differential equation $2ye^{x/y^2}\,dx+\left(y^2-4xe^{x/y^2}\right)dy=0$ such that $x(1)=0$. Then, $x(e)$ is equal to :

Option 1:
$e\log_e(2)$

Option 2:
$-e\log_e(2)$

Option 3:
$e^2\log_e(2)$

Option 4:
$-e^2\log_e(2)$

Correct Answer:
$-e^2\log_e(2)$

Solution:

$$2ye^{x/y^2}dx+\left(y^2-4xe^{x/y^2}\right)dy=0$$

$$2ye^{x/y^2}dx-4xe^{x/y^2}dy+y^2dy=0$$

$$2e^{x/y^2}(ydx-2xdy)+y^2dy=0$$

$$2e^{x/y^2}\left(\frac{y^2dx-2xydy}{y}\right)+y^2dy=0$$

$$2e^{x/y^2}\left(\frac{y^2dx-2xydy}{y^4}\right)+\frac{dy}{y}=0$$

$$2e^{x/y^2}d\left(\frac{x}{y^2}\right)+\frac{dy}{y}=0$$

Integrating

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$2e^{x/y^2} + \ln y + c = 0$

As $x(1) = 0 \Rightarrow (0, 1)$ satisfies it

$2 \cdot 1 + 0 + c = 0 \Rightarrow c = -2$

Putting $y = e$

$2e^{x/e^2} + \ln e - 2 = 0$

$\Rightarrow 2e^{x/e^2} = 1$

$\Rightarrow e^{x/e^2} = \frac{1}{2}$

$\Rightarrow x = e^2 \ln\left(\frac{1}{2}\right) = -e^2 \ln(2)$

Hence the correct answer is option 4.

Q. 11 Let the slope of the tangent to a curve $y = f(x)$ at (x, y) be given by $2 \tan x(\cos x - y)$. If the curve passes through the point $(\pi/4, 0)$, then the value of $\int_0^{\pi/2} y dx$ is equal to :

Option 1:
$(2 - \sqrt{2}) + \frac{\pi}{\sqrt{2}}$

Option 2:
$2 - \frac{\pi}{\sqrt{2}}$

Option 3:
$(2 + \sqrt{2}) + \frac{\pi}{\sqrt{2}}$

Option 4:
$2 + \frac{\pi}{\sqrt{2}}$

Correct Answer:
$2 - \frac{\pi}{\sqrt{2}}$

Solution:
$\frac{dy}{dx} = 2 \tan x(\cos x - y)$

$\Rightarrow \frac{dy}{dx} = 2 \sin x - 2 \tan x \cdot y$

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$$\Rightarrow \frac{dy}{dx} + 2\tan x \cdot y = 2\sin x$$

$$\text{I.F.} = e^{\int 2\tan x dx} = e^{\ln \sec^2 x} = \sec^2 x$$

$$\Rightarrow y \cdot \sec^2 x = 2\int \sin x . \sec^2 x dx$$

$$\Rightarrow y \cdot \sec^2 x = 2\int \tan x \cdot \sec x dx$$

$$\Rightarrow y \cdot \sec^2 x = 2\sec x + c$$

$$\text{Putting } \left(\frac{\pi}{4}, 0\right)$$

$$0 = 2\sqrt{2} + c \Rightarrow c = -2\sqrt{2}$$

$$\Rightarrow y \cdot \sec^2 x = 2\sec x - 2\sqrt{2}$$

$$\Rightarrow y = 2\cos x - 2\sqrt{2}\cos^2 x$$

$$\Rightarrow y = 2\cos x - \sqrt{2}(1 + \cos 2x)$$

$$\therefore \quad \int_0^{\pi/2} y dx = \int_0^{\pi/2} 2\cos x dx - \sqrt{2}\int_0^{\pi/2} (1 + \cos 2x) dx$$

$$= 2\sin x|_0^{\pi/2} - \sqrt{2}\left(x + \frac{\sin 2x}{x}\right)\Big|_0^{\pi/2}$$

Hence the correct answer is option 2.

Q. 12 Let a triangle be bounded by the lines $L_1 : 2x + 5y = 10; L_2 : -4x + 3y = 12$ and the line L_3, which passes through the point $P(2, 3)$, intersects L_2 at A and L_1 at B. If the point P divides the line-segment AB, internally in the ratio $1 : 3$, then the area of the triangle is equal to:

Option 1:

$$\frac{110}{13}$$

Option 2:

$$\frac{132}{13}$$

Option 3:

$$\frac{142}{13}$$

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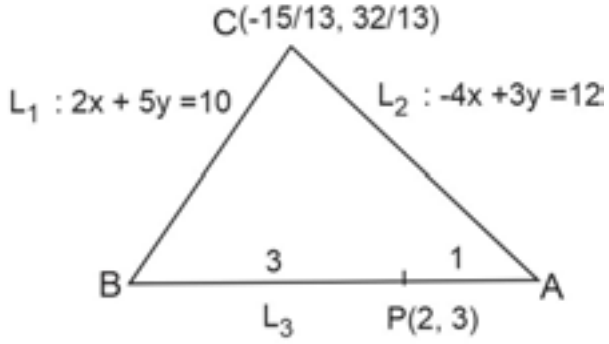
Option 4:

$\frac{151}{13}$

Correct Answer:

$\frac{132}{13}$

Solution:



$\text{Let A be } \left(a, 4+\frac{4a}{3}\right) \quad (\because \text{A lies on } L_2)$

$\text{and B be } \left(b, 2-\frac{2}{5}b\right) \quad (\because \text{B lies on } L_1)$

Using section formula for point P

$2=\frac{3a+b}{4} \text{ and } 3=\frac{3\left(4+\frac{4a}{3}\right)+\left(2-\frac{2b}{5}\right)}{4}$

$\Rightarrow 3a+b=8 \text{ and } 4a-\frac{2b}{5}=-2$

$\Rightarrow \quad a=\frac{3}{13}, b=\frac{95}{13}$

$\therefore \quad \text{A is } \left(\frac{3}{13}, \frac{56}{13}\right), \text{B is } \left(\frac{95}{13}, \frac{-12}{13}\right)$

$\therefore \text{ Area of } \triangle ABC=\frac{1}{2}\begin{vmatrix} \frac{3}{13} & \frac{56}{13} & 1 \\ \frac{95}{13} & \frac{-12}{13} & 1 \\ \frac{-15}{13} & \frac{32}{13} & 1 \end{vmatrix} = \frac{1}{2}\cdot\frac{1}{13^3}\begin{vmatrix} 3 & 56 & 1 \\ 95 & -12 & 1 \\ -15 & 32 & 1 \end{vmatrix}$

$=\frac{132}{13}$

Hence the correct answer is option 2

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Q. 13 Let $a > 0,\ b > 0$. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}l$ and $(e')^2 = \frac{11}{8}l'$, then the value of $77a + 44\ b$ is equal to :

Option 1:
100

Option 2:
110

Option 3:
120

Option 4:
130

Correct Answer:
130

Solution:

Given $e^2 = \frac{11}{14}l$

$\Rightarrow 1 + \frac{b^2}{a^2} = \frac{11}{14} \cdot \left(\frac{2b^2}{a}\right)$

$\Rightarrow \frac{a^2 + b^2}{a^2} = \frac{22b^2}{14a}$

$\Rightarrow 7\left(a^2 + b^2\right) = 11b^2a$ .....(i)

Also $e'^2 = \frac{11}{8}l'$

$\Rightarrow 1 + \frac{a^2}{b^2} = \frac{11}{8} \cdot \frac{2a^2}{b}$

$\Rightarrow b^2 + a^2 = b^2\left(\frac{11a^2}{4b}\right)$

$\Rightarrow 4\left(a^2 + b^2\right) = 11a^2b$ .....(ii)

Divide eqn (i) and (ii)

$\frac{7}{4} = \frac{b}{a}$

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$\text{Also } 4\left(a^2+b^2\right)=11a^2b$

$\Rightarrow 4\left(1+\left(\frac{b}{a}\right)^2\right)=11b$

$\Rightarrow 11b=4\left(1+\frac{49}{16}\right)$

$\Rightarrow 11b=4\left(\frac{65}{16}\right)$

$\Rightarrow b=\frac{65}{44}$

$\Rightarrow a=\frac{4b}{7}=\frac{4}{7}\cdot\frac{65}{44}=\frac{65}{77}$

$\therefore 77a+44b=65+65=130$

Hence the correct answer is option 4.

Q. 14 Let $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$, where $\alpha\in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15\left(\alpha^2+4\right)}$, then the value of $2|\vec{a}|^2+(\vec{a}\cdot\vec{b})|\vec{b}|^2$ is equal to :

Option 1:
10

Option 2:
7

Option 3:
9

Option 4:
14

Correct Answer:
14

Solution:

$\text{Area of parallelogram}=|\vec{a}\times\vec{b}|$

$=\left|\begin{vmatrix} i & j & k \\ \alpha & 2 & -1 \\ -2 & \alpha & 1 \end{vmatrix}\right|$

$=\left|(2+\alpha)i-(\alpha-2)j+\left(\alpha^2+4\right)k\right|$

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$= \sqrt{(2+\alpha)^2 + (\alpha-2)^2 + (\alpha^2+4)^2}$

Given that this area also equals $\sqrt{15\,(\alpha^2+4)}$

$\Rightarrow (2+\alpha)^2 + (\alpha-2)^2 + \left(\alpha^2+4\right)^2 = 15\left(\alpha^2+4\right)$

$\Rightarrow 2\left(\alpha^2+4\right) + \left(\alpha^2+4\right)^2 = 15\left(\alpha^2+4\right)$

$\Rightarrow \left(\alpha^2+4\right) = 13$

$\Rightarrow \alpha^2 = 9$

Now for $2|\vec{a}|^2 + (\vec{a}\cdot\vec{b})|\vec{b}|^2$

$|\vec{a}|^2 = \alpha^2 + 4 + 1 = 9 + 5 = 14$

$\vec{a}\cdot\vec{b} = -2\alpha + 2\alpha - 1 = -1$

$|\vec{b}|^2 = 4 + \alpha^2 + 1 = 14$

$\therefore 2|\vec{a}|^2 + (\vec{a}\cdot\vec{b})|\vec{b}|^2$

$= 2(14) + (-1)(14)$

$= 14$

Hence the correct answer is option 4

Q. 15 If vertex of a parabola is $(2,-1)$ and the equation of its directrix is $4x - 3y = 21$, then the length of its latus rectum is :

Option 1:
2

Option 2:
8

Option 3:
12

Option 4:
16

Correct Answer:
8

Solution:

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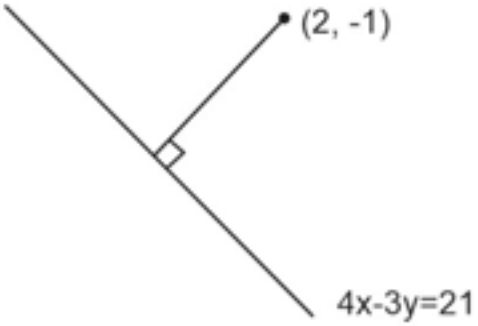

$a =$ distance between vertex and directrix

$$= \frac{|4(2) - 3(-1) - 21|}{\sqrt{4^2 + 3^2}}$$

$$= \frac{10}{5}$$

$$= 2$$

$$\text{Length of L . R.} = 4a = 8$$

Hence the correct option is 2

Q. 16 Let the plane $ax + by + cz = d$ pass through $(2, 3, -5)$ and is perpendicular to the planes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$.

If a, b, c, d are integers $d > 0$ and $\gcd(|a|, |b|, |c|, d) = 1$,

then the value of $a + 7\ b + c + 20\ d$ is equal to :

Option 1:
18

Option 2:
20

Option 3:
24

Option 4:
22

Correct Answer:
22

Solution:
DR's of the required plane are given by $\vec{n_1} \times \vec{n_2}$

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$$= \begin{vmatrix} i & j & k \\ 2 & 1 & -5 \\ 3 & 5 & -7 \end{vmatrix}$$

$= (-7+25)\mathrm{i} - (-14+15)\mathrm{j} + (10-3)\mathrm{k}$

$= 18\mathrm{i} - \mathrm{j} + 7\mathrm{k}$

So, the equation of required plane is

$18\mathrm{x} - \mathrm{y} + 7\mathrm{z} = 2$

$\text{Using}(2, 3, -5)$

$36 - 3 - 35 = \lambda \Rightarrow \lambda = -2$

$\text{Plane} : 18\mathrm{x} - \mathrm{y} + 7\mathrm{z} = -2$

$\text{But } \mathrm{d} > 0$

$-18\mathrm{x} + \mathrm{y} - 7\mathrm{z} = 2$

$\therefore \mathrm{a} = -18, \mathrm{b} = 1, \mathrm{c} = -7, \mathrm{d} = 2$

$\therefore \mathrm{a} + 7\mathrm{b} + \mathrm{c} + 20\mathrm{d}$

$= -18 + 7 - 7 + 40$

$= 22$

Hence the correct answer is option 4

Q. 17 The probability that a randomly chosen one-one function from the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$ to the set $\{1, 2, 3, 4, 5\}$ satisfies $\mathrm{f(a)} + 2\mathrm{f(b)} - \mathrm{f(c)} = \mathrm{f(\ d)}$ is :

Option 1:

$\frac{1}{24}$

Option 2:

$\frac{1}{40}$

Option 3:

$\frac{1}{30}$

Option 4:

$\frac{1}{20}$

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Correct Answer:

$\frac{1}{20}$

Solution:

Total one-one functions $= 5 \times 4 \times 3 \times 2 = 120$

For $f(a) + 2f(b) - f(c) = f(d)$,

following are possible values of

$(f(a), f(b), f(c), f(d))$:

$(5,1,3,4), (4,2,3,5), (1,3,2,5)$

$(5,1,4,3), (4,2,5,3), (1,3,5,2)$

$\therefore$ Probabelity $= \frac{6}{120} = \frac{1}{20}$

Hence the correct answer is option 4

Q. 18 The value of $\lim\limits_{n\to\infty} 6\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to:

Option 1:

1

Option 2:

2

Option 3:

3

Option 4:

6

Correct Answer:

3

Solution:

$\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)$

$= \tan^{-1}\left(\frac{1}{1+(r^2+3r+2)}\right)$

$= \tan^{-1}\left(\frac{1}{1+(r+1)(r+2)}\right)$

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$$= \tan^{-1}\left(\frac{(r+2)-(r+1)}{1+(r+1)(r+2)}\right)$$

$$= \tan^{-1}(r+2) - \tan^{-1}(r+1)$$

$$\therefore \quad \sum_{1}^{n} \tan^{-1}\left(\frac{1}{r^2+3r+3}\right) = \left(\tan^{-1}3 - \tan^{-1}2\right)$$
$$+ \left(\tan^{-1}4 - \tan^{-1}3\right) + \left(\tan^{-1}5 - \tan^{-1}4\right)$$
$$+ - - \cdots + \left(\tan^{-1}(n+2) - \tan^{-1}(n+1)\right)$$

$$= \tan^{-1}(n+2) - \tan^{-1}2$$

$$\therefore 6\tan\left(\tan^{-1}(n+2) - \tan^{-1}2\right)$$

$$= 6 \cdot \frac{\tan\left(\tan^{-1}(n+2)\right) - \tan\left(\tan^{-1}2\right)}{1+\tan\left(\tan^{-1}(n+2)\right)\cdot\tan\left(\tan^{-1}2\right)}$$

$$= 6 \cdot \frac{n+2-2}{1+2(n+2)}$$

$$= \frac{6n}{2n+5}$$

$$\therefore \lim_{n\to\infty} \frac{6n}{2n+5} = \frac{6}{2+\frac{5}{n}} = 3$$

Hence answer is option 3

Q. 19 Let $\tilde{a}$ be a vector which is perpendicular to the vector $3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}$. If $\tilde{a} \times (2\hat{i} + \hat{k}) = 2\hat{i} - 13\hat{j} - 4\hat{k}$, then the projection of the vector $\tilde{a}$ on the vector $2\hat{i} + 2\hat{j} + \hat{k}$ is :

Option 1:
$\frac{1}{3}$

Option 2:
1

Option 3:
$\frac{5}{3}$

Option 4:
$\frac{7}{3}$

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Correct Answer:

$\frac{5}{3}$

Solution:

Let $3\mathrm{i}+\frac{1}{2}\mathrm{j}+2\mathrm{k}=\vec{c}$

and $2\mathrm{i}+\mathrm{k}=\vec{b}$

Given $\quad \vec{a}\perp\vec{c}\Rightarrow\vec{a}\cdot\vec{c}=0$

and $\quad \vec{a}\times\vec{b}=2\mathrm{i}-13\mathrm{j}-4\mathrm{k}.$

$\Rightarrow \quad (\vec{a}\times\vec{b})\times\vec{c}=(2\mathrm{i}-13\mathrm{j}-4\mathrm{k})\times\vec{c}$

$\Rightarrow \quad (\vec{a}\cdot\vec{c})\vec{b}-(\vec{b}\cdot\vec{c})\vec{a}=(2\mathrm{i}-13\mathrm{j}-4\mathrm{k})\times\left(3\mathrm{i}+\frac{1}{2}\mathrm{j}+2\mathrm{k}\right)$

$$\Rightarrow 0-(6+2)\vec{a}=\begin{vmatrix} i & j & k \\ 2 & -13 & -4 \\ 3 & \frac{1}{2} & 2 \end{vmatrix}$$

$\Rightarrow -8\vec{a}=-24\mathrm{i}-16\mathrm{j}+40\mathrm{k}$

$\Rightarrow \vec{a}=3\mathrm{i}+8\mathrm{j}-5\mathrm{k}$

Projection of $\vec{a}$ *on* $2\mathrm{i}+2\mathrm{j}+\mathrm{k}$

$$=\frac{(3\mathrm{i}+8\mathrm{j}-5\mathrm{k})\cdot(2\mathrm{i}+2\mathrm{j}+\mathrm{k})}{\sqrt{4+4+1}}$$

$$=\frac{5}{3}$$

Hence correct option is 3

Q. 20 If $\cot\alpha=1$ and $\sec\beta=-\frac{5}{3}$, where $\pi<\alpha<\frac{3\pi}{2}$ and $\frac{\pi}{2}<\beta<\pi$, then the value of $\tan(\alpha+\beta)$ and the quadrant in which $\alpha+\beta$ lies, respectively are :

Option 1:

$-\frac{1}{7}$ and $\mathrm{IV}^{\mathrm{th}}$ quadrant

Option 2:

7 and I^{st} quadrant

Option 3:

-7 and $\mathrm{IV}^{\mathrm{th}}$ quadrant

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Option 4:

$\frac{1}{7}$ and I^{st} quadrant

Correct Answer:

$-\frac{1}{7}$ and $\mathrm{IV}^{\mathrm{th}}$ quadrant

Solution:

$\cot\alpha = 1$

$\Rightarrow \tan\alpha = 1$

$\text{and}\sec\beta = -\frac{5}{3}$

$\Rightarrow \tan\beta = -\frac{4}{3}(\text{As } \beta \in \mathbb{III} \text{ quadrant})$

$\therefore \tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \cdot \tan\beta}$

$= \frac{1 - \frac{4}{3}}{1 + \frac{4}{3}\cdot 1} = -\frac{1}{7}$

$\text{As } \pi < \alpha < \frac{3\pi}{2} \text{ and } \frac{\pi}{2} < \beta < \pi$

$\Rightarrow \quad \frac{3\pi}{2} < \alpha+\beta < \frac{5\pi}{2} \Rightarrow \text{IV or I quadrant}$

$\text{Also } \tan(\alpha+\beta) < 0 \Rightarrow \text{ IV th quadrant}$

Hence correct option is 1

Q. 21 Let the image of the point $\mathrm{P}(1,2,3)$ in the line $\mathrm{L}: \frac{x-6}{3} = \frac{y-1}{2} = \frac{z-2}{3}$ be Q. Let $\mathrm{R}(\alpha,\beta,\gamma)$ be a point that divides internally the line segment PQ in the ratio $1:3$. Then the value of $22(\alpha+\beta+\gamma)$ is equal to_____

Correct Answer:

125

Solution:

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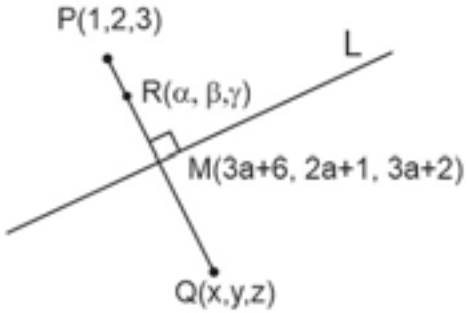

Let $\mathrm{M}(3\mathrm{a}+6, 2\mathrm{a}+1, 3\mathrm{a}+2)$ be the mid point of PQ.

$\therefore \mathrm{PM} \perp \mathrm{L}$

$\Rightarrow \overrightarrow{PM} \cdot \vec{b} = 0$

$\Rightarrow ((3\mathrm{a}+5)\mathrm{i} + (2\mathrm{a}-1)\mathrm{j} + (3\mathrm{a}-1)\mathrm{k}) \cdot (3\mathrm{i} + 2\mathrm{j} + 3\mathrm{k}) = 0$

$\Rightarrow \mathrm{a} = -\dfrac{5}{11}$

$\therefore \mathrm{M}$ is $\left(\dfrac{51}{11}, \dfrac{1}{11}, \dfrac{7}{11}\right)$

As R divides PQ in ratio 1:3, hence R is the mid-point of PM

$\mathrm{R} = \left(\dfrac{62}{22}, \dfrac{23}{22}, \dfrac{40}{22}\right)$

$\therefore 22(\alpha + \beta + \gamma) = 62 + 23 + 40 = 125$

Hence answer is 125

Q. 22 Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62 , and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is_____.

Correct Answer:
0

Solution:

$\mathrm{Variance} = 20$

$\Rightarrow \dfrac{\sum (x_i - \bar{x})^2}{n} = 20$

$\Rightarrow \dfrac{\sum (\mathrm{x_i} - 62)^2}{7} = 20$

$\Rightarrow \sum (\mathrm{x_i} - 62)^2 = 140$

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$\Rightarrow (62-x_1)^2+(62-x_2)^2+\ldots.+(62-x_7)^2=140$

Now if one student (say x_1) gets less than 50 marks then

$\text{LHS} \ > 12^2+(62-x_2)^2+\cdots+(62-x_7)^2$

$\Rightarrow \ \text{LHS} \ > 144$

In such a case LHS cannot equal $\text{RHS}(=140)$

So, no student can fail

Hence answer is 0

Q. 23 If one of the diameters of the circle $x^2+y^2-2\sqrt{2}x-6\sqrt{2}y+14=0$ is a chord of the circle $(x-2\sqrt{2})^2+(y-2\sqrt{2})^2=r^2$, then the value of r^2 is equal to_____.

Correct Answer:
10

Solution:
$C_1: x^2+y^2-2\sqrt{2}x-6\sqrt{2}y+14=0$

$C_2: x^2+y^2-4\sqrt{2}x-4\sqrt{2}y+(16-r^2)=0$

Common Chord is $C_1-C_2=0$

$2\sqrt{2}x-2\sqrt{2}y-2+r^2=0$

It is a diameter of C_1, so it passes through its centre $(\sqrt{2},3\sqrt{2})$

$2\sqrt{2}(\sqrt{2})-2\sqrt{2}(3\sqrt{2})-2+r^2=0$

$\Rightarrow r^2=2-4+12$

$\Rightarrow r^2=10$

Hence answer is 10

Q. 24 If $\lim\limits_{x\to 1}\dfrac{\sin(3x^2-4x+1)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of (a - b) is equal to _____.

Correct Answer:
11

Solution:
For a finite limit, as numerator $\to 0$, so denominator $\to 0$

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$\therefore 2(1)^3 - 7(1)^2 + a(1) + b = 0$

$\Rightarrow a + b = 5 \quad \ldots\ldots \text{ (i)}$

Using L'H Rule given limit equals

$$\lim_{x \to 1} \frac{(6x - 4)\cos(3x^2 - 4x + 1) - 2x}{6x^2 - 14x + a}$$

$\text{As numerator } \to 0, \text{so denominalor } \to 0$

$6(1)^2 - 14(1) + a = 0$

$\Rightarrow a = 8$

$\Rightarrow b = -3 \text{ (using } (i))$

$\text{So } a - b = 11$

Hence answer is 11

Q. 25 Let for $n = 1, 2, \ldots\ldots, 50,\ S_n$ be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of

$$\frac{1}{26} + \sum_{n=1}^{50} \left(S_n + \frac{2}{n+1} - n - 1 \right)$$ is equal to_____

Correct Answer:
41651

Solution:

$a = n^2, \quad r = \frac{1}{(n+1)^2}$

$\therefore S_n = \frac{n^2}{1 - \frac{1}{(n+1)^2}} = \frac{n^2(n+1)^2}{n^2 + 2n}$

$= \frac{n(n+1)^2}{n+2} = \frac{n\,(n^2 + 2n + 1)}{n+2}$

Using long division

$S_n = n^2 + 1 - \frac{2}{n+2}$

$\therefore S_n + \frac{2}{n+1} - n - 1$

$= n^2 + 1 - \frac{2}{n+2} - n - 1 + \frac{2}{n+1}$

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$$= n^2 - n + 2\left[\frac{1}{n+1} - \frac{1}{n+2}\right]$$

Apply summation

$$\sum_{1}^{50}\left(S_n + \frac{2}{n+1} - n - 1\right)$$

$$= \sum n^2 - \sum n + 2\sum\left(\frac{1}{n+i} - \frac{1}{n+2}\right)$$

$$= \frac{50 \times 51 \times 101}{6} - \frac{50 \times 51}{2} - 2\left(\frac{1}{52} - \frac{1}{2}\right)$$

$$\therefore \frac{1}{26} + \sum\left(S_n + \frac{2}{n+1} - n - 1\right)$$

$$= \frac{1}{26} + \frac{50 \times 51 \times 101}{6} - \frac{50 \times 51}{2} - 2\left(\frac{1}{52} - \frac{1}{2}\right)$$

$$= 41651$$

Hence answer is 41651

Q. 26 If the system of linear equations
$2x - 3y = \gamma + 5$,
$\alpha x + 5y = \beta + 1$, where $\alpha, \beta, \gamma \in \mathbf{R}$ has infinitely many solutions, then the value of $|9\alpha + 3\beta + 5\gamma|$ is equal to_____.

Correct Answer:
58

Solution:
$2x - 3y = \gamma + 5$

$\alpha x + 5y = \beta + 1$

For inifinite solutions, these two lines should co-incide

$$\frac{2}{\alpha} = \frac{-3}{5} = \frac{\gamma + 5}{\beta + 1}$$

$$\Rightarrow \frac{2}{\alpha} = -\frac{3}{5} \Rightarrow \alpha = -\frac{10}{3} \quad \text{and}$$

$$\frac{\gamma + 5}{\beta + 1} = \frac{-3}{5}$$

$$\Rightarrow 5\gamma + 25 = -3\beta - 3$$

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$\Rightarrow 3\beta + 5\gamma = -28$

$\therefore |9\alpha + 3\beta + 5\gamma| = |-30 - 28| = 58$

Hence answer is 58.

Q. 27 Let $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where $i = \sqrt{-1}$. Then, the number of elements in the set $\{n \in \{1, 2, \ldots, 100\} : A^n = A\}$ is_____.

Correct Answer:
25

Solution:

$$A^2 = \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (1+i)^2 - i & 1+i \\ -i+1 & -i \end{bmatrix}$$

$$= \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix}$$

$A^4 = A^2 \cdot A^2$

$$= \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix} \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$\therefore A^4 = I.$

$\Rightarrow A^{4p+1} = (A^4)^p \cdot A = I \cdot A = A$

$\therefore n$ should be of type $4p + 1$

$n = 1, 5, 9, \ldots, 97$

Total of 25 elements

Hence answer is 25

Q. 28 Sum of squares of modulus of all the complex numbers z satisfying $\bar{z} = iz^2 + z^2 - z$ is equal to _____.

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Correct Answer:
2

Solution:

$\text{Let } z = x + iy$

$\therefore x - iy = i(x + iy)^2 + (x + iy)^2 - x - iy.$

$\Rightarrow 2x = i\left(x^2 - y^2 + i(2xy)\right) + \left(x^2 - y^2 + 2ixy\right)$

$\Rightarrow 2x = \left(x^2 - y^2 - 2xy\right) + i\left(x^2 - y^2 + 2xy\right)$

$\therefore \quad x^2 - y^2 - 2xy = 2x \text{ and}$

$x^2 - y^2 + 2xy = 0$

Subtracting these

$-4xy = 2x$

$\Rightarrow 2x + 4xy = 0$

$\Rightarrow 2x(1 + 2y) = 0$

$\Rightarrow x = 0 \text{ or } y = -\frac{1}{2}$

$\text{When } x = 0 \Rightarrow 0 - y^2 + 0 = 0 \Rightarrow y = 0$

$When\ y = -\frac{1}{2}$

$x^2 - \frac{1}{4} - x = 0$

$4x^2 - 4x - 1 = 0$

$x = \frac{4 \pm \sqrt{16 + 16}}{8} = \frac{1 \pm \sqrt{2}}{2}$

$\therefore \quad z = 0 \text{ or } \frac{1 + \sqrt{2}}{2} - \frac{1}{2}i \text{ or } \frac{1 - \sqrt{2}}{2} - \frac{1}{2}i$

$|z_1|^2 + |z_2|^2 + |z_3|^2$

$= 0 + \frac{(1 + \sqrt{2})^2}{4} + \frac{1}{4} + \frac{(1 - \sqrt{2})^2}{4} + \frac{1}{4}$

$= \frac{1}{2} + \frac{6}{4} = 2$

Hence answer is 2

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Q. 29 Let $S = \{1, 2, 3, 4\}$. Then the number of elements in the set
$\{f : S \times S \to S : f \text{ is onto and } f(a, b) = f(b, a) \geqslant a \forall (a, b) \in S \times S\}$ is_____.

Correct Answer:
37

Solution:
As $f(a, b) = f(b, a) \geqslant a$

If a or b equals 4, then image has to be 4

$\therefore \quad (1,4),(4,1),(2,4),(4,2),(3,4),(4,3),(4,4)$must have 4 as their image.

And$(1,3),(3,1),(2,3),(3,2),(3,3)$can have 3 or 4 as their image

And$(1,2),(2,1),(2,2)$ can have 2 or 3 or 4 as their image

And$(1,1)$ can have only 1 as image (as function has to be onto)

$\therefore$ Total such functions $= 2^3 \cdot 3^2 = 72$ (2^3 because we have to choose image

for $(1,3),(2,3), and\ (3,3)$ only and not for $(3,1)$ and $(3,2)$ as their image equal the image of $(1,3) and (2,3)$ respectively)

Now out of these we need to remove cases which are not onto. 1 and 4 are always the image of same element. So three cases remain

1. When 3 is not an image (A)

Such functions $= 2^2 = 4$

2. When 2 is not an image (B)

Such functions $= 2^3 \cdot 2^2 = 32$

3. When both 2 and 3 are not the images $(A \cap B)$

Such functions =1

$\therefore$ Answer $=$ Total- $(A \cup B)$

$= 72 - (4 + 32 - 1)$

$= 72 - 35$

$= 37$

Hence answer is 37

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Q. 30 The maximum number of compound propositions, out of $\mathrm{p} \vee \mathrm{r} \vee \mathrm{s}, \mathrm{p} \vee \mathrm{r} \vee \sim \mathrm{s}, \mathrm{p} \vee \sim \mathrm{q} \vee \mathrm{s}, \sim \mathrm{p} \vee \sim \mathrm{r} \vee \mathrm{s}, \sim \mathrm{p} \vee \sim \mathrm{r} \vee \sim \mathrm{s}, \sim \mathrm{p} \vee \mathrm{q} \vee \sim \mathrm{s}$, $\mathrm{q} \vee \mathrm{r} \vee \sim \mathrm{s}, \mathrm{q} \vee \sim \mathrm{r} \vee \sim \mathrm{s}, \sim \mathrm{p} \vee \sim \mathrm{q} \vee \sim \mathrm{s}$ that can be made simultaneously true by an assignment of the truth values to $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s, is equal to_____.

Correct Answer:
9

Solution:
1. If all p,q,r,s are true, then $\sim p \vee \sim r V \sim s$ and $\sim p \vee \sim q \vee \sim s$ are false. So 7 are true

2.If only one is false.

(i) If p is false, rest are true then all 9 statements are true, which is the maximum. So no need to check further.

$\therefore$ Answer is all 9 are true when p is false and q, r, s are true

Hence answer is 9

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29 June Morning Shift

Physics

Q. 1 Two balls $\mathrm{A\ and\ B}$ are placed at the top of $180\ \mathrm{m}$ tall tower. Ball A is released from the top at $\mathrm{t} = 0\ \mathrm{s}$. Ball B is thrown vertically down with an initial velocity $'\mathrm{u}'$ at $\mathrm{t} = 2\ \mathrm{s}$. After a certain time, both balls meet $100\ \mathrm{m}$ above the ground. Find the value of $'\mathrm{u}\ '$ in ms^{-1}. [use $g = 10\ \mathrm{ms}^{-2}$] :

Option 1:
10

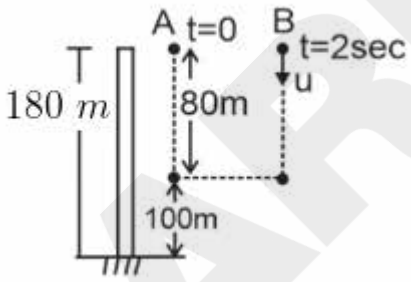
Option 2:
15

Option 3:
20

Option 4:
30

Correct Answer:
30

Solution:



Let they meet $\mathrm{t} = \mathrm{t}_{\circ}$

For A

$$80 = \frac{1}{2}\ \mathrm{g}\ \mathrm{t}_{\circ}^{2}\ \text{.....(i)}$$

For B

$$80 = \mathrm{u}\left(\mathrm{t}_0 - 2\right) + \frac{1}{2}\mathrm{g}\left(\mathrm{t}_0 - 2\right)^2\ \text{..... (ii)}$$

From (i)

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$t_\circ = 4$

$80 = u \times 2 + 5 \times 2^2$

$u = 30\ m/s$

Correct answer is **Option (4)**

Q. 2 A body of mass M at rest explodes into three pieces, in the ratio of masses $1 : 1 : 2$. Two smaller pieces fly off perpendicular to each other with velocities of $30\ ms^{-1}$ and $40\ ms^{-1}$ respectively. The velocity of the third piece will be :

Option 1:
$15\ ms^{-1}$

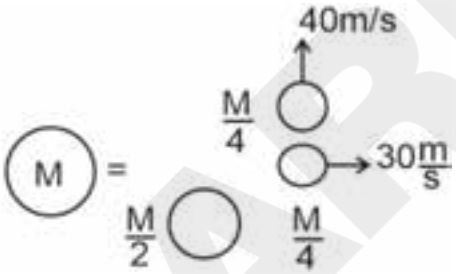
Option 2:
$25\ ms^{-1}$

Option 3:
$35\ ms^{-1}$

Option 4:
$50\ ms^{-1}$

Correct Answer:
$25\ ms^{-1}$

Solution:



Conservation of momentum,

$\frac{m}{4} \times 30i + \frac{m}{4}30J + \frac{m}{2}\vec{v} = 0$

$\tilde{v} = -15i - 20\dot{J}$

$|\tilde{v}| = \sqrt{15^2 + 20^2} = 25\ m/s$

Hence, Correct answer is **Option (2)**

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Q. 3 The activity of a radioactive material is $2.56 \times 10^{-3}\mathrm{Ci}$. If the half life of the material is 5 days, after how many days the activity will become $2 \times 10^{-5}\mathrm{Ci}$?

Option 1:
$30\ \mathrm{days}$

Option 2:
$35\ \mathrm{days}$

Option 3:
$40\ \mathrm{days}$

Option 4:
$25\ \mathrm{days}$

Correct Answer:
$35\ \mathrm{days}$

Solution:

$$\mathrm{A} = \mathrm{A}_0\, \mathrm{e}^{-\lambda \mathrm{t}}$$

$$2 \times 10^{-5} = 2.56 \times 10^{-3} \times \mathrm{e}^{-\lambda \mathrm{t}}$$

$$\bar{\mathrm{e}}^{\lambda \mathrm{t}} = \frac{1}{128} = \left[\frac{1}{2}\right]^{+7} = \frac{\mathrm{A}}{\mathrm{A}_0}$$

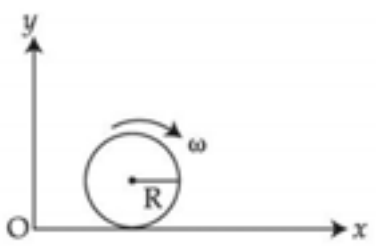
$$\text{time taken} = 7\ \text{half} - \text{life}$$

$$= 7 \times 5$$

$$= 35\ \text{days}$$

Hence, the correct answer is **Option (2)**

Q. 4 A spherical shell of $1\ \mathrm{kg}$ mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{\mathrm{a}}{3}\mathrm{R}^2\omega$. The value of a will be :



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Option 1:
2

Option 2:
3

Option 3:
5

Option 4:
4

Correct Answer:
5

Solution:

$$\tilde{L}_0 = \tilde{L}_{cm} + \tilde{L}_{\text{about cm}}$$

$$= mVR + \frac{2}{3}mR^2\omega, \quad (\text{Using} V = R\omega)$$

$$= \frac{mR^2}{1}\omega + \frac{2}{3}mR^2\omega$$

$$= \frac{5}{3}mR^2\omega$$

Hence, correct answer is **Option (3).**

Q. 5 A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by $20.0^{\circ}C$ will be :
(Given gas constant $R = 8.3 JK^{-1} - mol^{-1}$)

Option 1:
249 J

Option 2:
415 J

Option 3:
498 J

Option 4:
830 J

Correct Answer:
498 J

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Solution:
Since Isochoric process,

$$\Delta Q = nC_V \Delta T$$

$$C_V = \frac{f}{2}R$$

Helium is monoatomic gas.

Degree of freedom (f)= 3

44.8 litre at STP = 2 mole

$$\Delta Q = 2 \times \frac{3R}{2} \times 20$$

$$= 498\ J$$

Correct answer is **Option (3)**

Q. 6 A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses $1\ kg$ and $2\ kg$ are suspended respectively from its free end. Then the value of L is equal to :

Option 1:
$\sqrt{L_1\ L_2}$

Option 2:
$\frac{L_1 + L_2}{2}$

Option 3:
$2\ L_1 - L_2$

Option 4:
$3\ L_1 - 2\ L_2$

Correct Answer:
$2\ L_1 - L_2$

Solution:
$$\Delta L = \frac{FL}{AY}$$

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$$L_1 = L + \frac{1 \times g \times L}{AY} \ldots\ldots \text{(i)}$$

$$L_2 = L + \frac{2 \times g \times L}{AY} \ldots\ldots \text{(ii)}$$

$$\Rightarrow L = 2L_1 - L_2$$

Hence, Correct answer is **Option (3)**

Q. 7 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The photoelectric effect does not takes place, if the energy of the incident radiation is less than the work function of a metal.

Reason R : Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

Option 3:
$\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

Option 4:
$\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

Correct Answer:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

Solution:
For the photoelectric effect to take place the incident energy must be greater than or equal to the work function.

and its equation is given as

$$E = \phi_\circ + kE_{max}$$

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$\text{If } \mathrm{E} = \phi_{\circ}, \quad \mathrm{kE} = 0$

Hence, the correct answer is **Option (2)**

Q. 8 A particle of mass $500\mathrm{gm}$ is moving in a straight line with velocity $v = \mathrm{b}x^{5/2}$. The work done by the net force during its displacement from $\mathrm{x} = 0 \text{ to } x = 4 \text{ m}$ is : $\left(\text{ Take } \mathrm{b} = 0.25 \text{ m}^{-3/2} \text{ s}^{-1}\right)$.

Option 1:
2 J

Option 2:
4 J

Option 3:
8 J

Option 4:
16 J

Correct Answer:
16 J

Solution:
$\mathrm{v} = \mathrm{bx}^{5/2}$

Work energy theorem

$$\mathrm{W} = \Delta \mathrm{KE}$$

$$= \frac{1}{2}\mathrm{mV}_2^2 - \frac{1}{2}\mathrm{mV}_1^2$$

$$= \frac{\mathrm{m}}{2}\left[\mathrm{v}_2^2 - \mathrm{v}_1^2\right]$$

$$= \frac{0.5}{2} \times \left[\mathrm{b}^2 \times (4^{5/2})^2 - 0\right]$$

$$= \frac{0.5}{2} \times 0.25 \times 0.25 \times 4^5$$

$$= 16\mathrm{J}$$

Hence, Correct answer is **Option (2).**

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Q. 9 A charge particle moves along circular path in a uniform magnetic field in a cyclotron. The kinetic energy of the charge particle increases to 4 times its initial value. What will be the ratio of new radius to the original radius of circular path of the charge particle :

Option 1:
$1:1$

Option 2:
$1:2$

Option 3:
$2:1$

Option 4:
$1:4$

Correct Answer:
$2:1$

Solution:

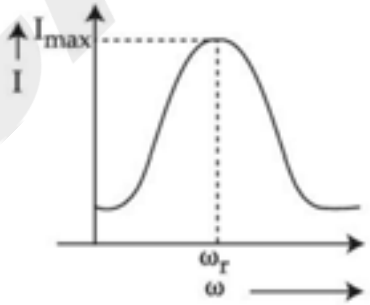
$$R=\frac{mv}{qb}=\frac{\sqrt{2mk}}{qb}$$

$$R\alpha\sqrt{k}$$

$$\text{Ratio } 2:1$$

Hence, the correct answer is **Option (3).**

Q. 10 For a series LCR circuit, I vs ω curve is shown :
(a) To the left of ω_r, the circuit is mainly capacitive.
(b) To the left of ω_r, the circuit is mainly inductive.
(c) At ω_r, impedance of the circuit is equal to the resistance of the circuit.
(d) At ω_r, impedance of the circuit is 0 .



Choose the most appropriate answer from the options given below :

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Option 1:
(a) and (d) only

Option 2:
(b) and (d) only

Option 3:
(a) and (c) only

Option 4:
(b) and (c) only

Correct Answer:
(a) and (c) only

Solution:

$$X_C = \frac{1}{\omega c}, X_L = \omega L$$

If $\omega = \omega_r$,

$$x_L = x_C$$

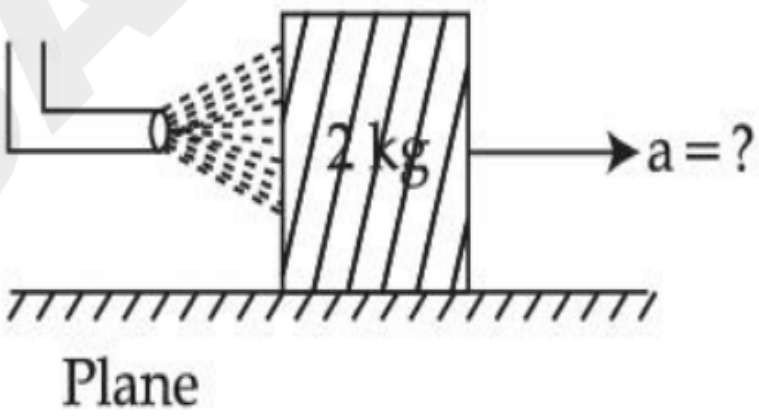
For $\omega < \omega_r$ capacitive

at $\omega = \omega_r, x_L = x_C$

$$z = \sqrt{R^2 + (x_L - x_C)^2} = R$$

Hence, correct answer is **Option (3)**

Q. 11 A block of metal weighing $2\ \text{kg}$ is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of 1kgs^{-1} and at a speed of $10\ \text{ms}^{-1}$. Then, the initial acceleration of the block, in ms^{-2}, will be :



Option 1:
3

Option 2:
6

Option 3:
5

Option 4:
4

Correct Answer:
5

Solution:

$$\mathrm{F}=\frac{\mathrm{dm}}{\mathrm{dt}} \times \mathrm{v}=\rho \mathrm{v}^2 \mathrm{a}$$

$$\mathrm{F}=\text{ mass } \times \text{ acceleration}$$

on comparing

$$1 \times 10=2 \times \text{ acceleration}$$

$$\text{acceleration}=5 \mathrm{m} / \mathrm{s}^2$$

Hence, correct answer is **Option (3)**

Q. 12 In van dar Wall equation $\left[\mathrm{P}+\frac{\mathrm{a}}{\mathrm{V}^2}\right][\mathrm{V}-\mathrm{b}]=\mathrm{RT}$; P is pressure, V is volume, R is universal gas constant and t is temperature. The ratio of constants $\frac{\mathrm{a}}{\mathrm{b}}$ is dimensionally equal to :

Option 1:
$\frac{\mathrm{P}}{\mathrm{V}}$

Option 2:
$\frac{\mathrm{V}}{\mathrm{P}}$

Option 3:
PV

Option 4:
PV^3

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Correct Answer:

PV

Solution:

From the given equation

$$[\mathrm{a}] = [\mathrm{pv}^2]$$

$$[\mathrm{b}] = [\mathrm{v}]$$

$$\left[\frac{\mathrm{a}}{\mathrm{b}}\right] = [\mathrm{pv}]$$

Hence, correct answer is **option (3)**

Q. 13 Two vectors $\vec{A}\ and\ \vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

Option 1:

$\sin^{-1}\left(\frac{3}{5}\right)$

Option 2:

$\sin^{-1}\left(\frac{1}{3}\right)$

Option 3:

$\cos^{-1}\left(\frac{3}{5}\right)$

Option 4:

$\cos^{-1}\left(\frac{1}{3}\right)$

Correct Answer:

$\cos^{-1}\left(\frac{3}{5}\right)$

Solution:

If $\overrightarrow{\mathrm{A}}$ and $\overrightarrow{\mathrm{B}}$ angle is θ then angle between $\overrightarrow{\mathrm{A}}$ and -$\overrightarrow{\mathrm{B}}$ is $180^\circ - \theta$

$$|\vec{A} + \vec{B}| = 2|\vec{A} - \vec{B}|$$

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$$\sqrt{A^2+A^2+2A^2\cos\theta}=2\sqrt{A^2+A^2-2A^2\cos\theta}$$

$$5\cos\theta=3$$

$$\theta=\cos^{-1}\left(\frac{3}{5}\right)$$

Hence, correct answer is **Option (3)**

Q. 14 The escape velocity of a body on a planet $'A'$ is 12kms^{-1}. The escape velocity of the body on another planet $'B'$, whose density is four times and radius is half of the planet $'A'$, is :

Option 1:
12kms^{-1}

Option 2:
24kms^{-1}

Option 3:
36kms^{-1}

Option 4:
6kms^{-1}

Correct Answer:
12kms^{-1}

Solution:

$$\text{Vescape }=\sqrt{\frac{2GM}{R}}.$$

$$=\sqrt{\frac{2G}{R}\times\rho\times\frac{4}{3}\pi R^3}$$

$$\text{Vescape }\alpha\ R\sqrt{\rho}$$

$$\frac{(\text{ Vescape })_B}{(\text{ Vescape })_A}=\frac{R_B\sqrt{\rho_B}}{R_A\sqrt{\rho_A}}=\frac{1}{2}\times\sqrt{4}$$

$$=1$$

Hence, the correct answer is **Option (1)**

Q. 15 At a certain place the angle of dip is 30° and the horizontal component of earth's magnetic field is 0.5G. The earth's total magnetic field (in g), at that certain place, is :

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Option 1:
$\frac{1}{\sqrt{3}}$

Option 2:
$\frac{1}{2}$

Option 3:
$\sqrt{3}$

Option 4:
1

Correct Answer:
$\frac{1}{\sqrt{3}}$

Solution:
$\mathrm{B_H} = \mathrm{B}\ \cos 30^\circ$

$\mathrm{B} = \frac{\mathrm{B_H}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}\mathrm{G}$

Hence, the correct answer is **Option (1)**

Q. 16 A longitudinal wave is represented by $x = 10 \sin 2\pi \left(\mathrm{nt} - \frac{x}{\lambda}\right)\ \mathrm{cm}$. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to :

Option 1:
2π

Option 2:
5π

Option 3:
π

Option 4:
$\frac{5\pi}{2}$

Correct Answer:
5π

Solution:

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$$x = 10 \sin 2\pi \left(nt - \frac{x}{\lambda} \right) cm$$

Particle velocity $= \frac{\partial x}{\partial t}$

maximum particle velocity $= (2\pi n) \times 10$

wave velocity=$n\lambda$

As given in the question

$$2\pi n \times 10 = n\lambda \times 4$$

$$\lambda = 5\pi$$

Hence, the correct answer is **Option (2)**

Q. 17 A parallel plate capacitor filled with a medium of dielectric constant 10 , is connected across a battery and is charged. The dielectric slab is replaced by another slab of dielectric constant 15. Then the energy of capacitor will :

Option 1:
Increase by 50%

Option 2:
Decrease by 15%

Option 3:
Increase by 25%

Option 4:
Increase by 33%

Correct Answer:
Increase by 50%

Solution:

$$U = \frac{1}{2}(kC_{\circ})V^2$$

$$\frac{U'}{U} = \frac{k'}{k} = \frac{15}{10} = 1.5$$

$$\frac{\Delta U}{U} \times 100 = \frac{U' - U}{U} \times 100$$

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$$= (1.5 - 1) \times 100$$

$$= 50\%$$

Hence, correct answer is **Option (1).**

Q. 18 A positive charge particle of 100mg is thrown in opposite direction to a uniform electric field of strength $1 \times 10^5 \text{NC}^{-1}$. If the charge on the particle is $40\mu\text{C}$ and the initial velocity is $200\ \text{ms}^{-1}$, how much distance it will travel before coming to the rest momentarily :

Option 1:
$1\ \text{m}$

Option 2:
$5\ \text{m}$

Option 3:
$10\ \text{m}$

Option 4:
$0.5\ \text{m}$

Correct Answer:
$0.5\ \text{m}$

Solution:

$$\text{a} = \frac{-\text{qE}}{\text{m}}$$

$$\text{v}^2 = \text{u}^2 + 2\text{as}$$

$$0 = 200^2 + 2 \times \left(\frac{-\text{qE}}{\text{m}}\right) \times \text{s}$$

$$-200^2 = \frac{-2 \times 40 \times 10^{-6} \times 10^5}{100 \times 10^{-6}} \times \text{S}$$

$$\text{S} = 0.5\text{m}$$

Hence, the correct answer is **Option (4)**

Q. 19 Using Young's double slit experiment, a monochromatic light of wavelength 5000Å produces fringes of fringe width $0.5\ \text{mm}$. If another monochromatic light of wavelength 6000Å is used and the separation between the slits is doubled, then the new fringe width will be :

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Option 1:
0.5mm

Option 2:
1.0mm

Option 3:
0.6mm

Option 4:
0.3mm

Correct Answer:
0.3mm

Solution:

Fringe width $(\omega) = \dfrac{\mathrm{D}\lambda}{\mathrm{d}}$

Fringe width $(\omega)\ \alpha\ \dfrac{\lambda}{\mathrm{d}}$

$$\frac{\omega'}{\omega} = \frac{\lambda'/\mathrm{d}'}{\lambda/\mathrm{d}} = \frac{\lambda'}{\lambda} \times \frac{\mathrm{d}}{\mathrm{d}'} = \frac{6000}{5000} \times \frac{\mathrm{d}}{2\mathrm{d}} = \frac{6}{10} = 0.6$$

$$\omega' = \omega \times 0.6 = 0.5 \times 0.6 = 0.3\ \text{mm}$$

Hence, the correct answer is **Option (4).**

Q. 20 Only 2% of the optical source frequency is the available channel bandwidth for an optical communicating system operating at $1000\ \text{nm}$. If an audio signal requires a bandwidth of 8kHz, how many channels can be accommodated for transmission :

Option 1:
375×10^7

Option 2:
75×10^7

Option 3:
375×10^8

Option 4:
75×10^9

Correct Answer:
75×10^7

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Solution:

$$f = \frac{v}{\lambda} = \frac{3 \times 10^8}{1000 \times 10^{-9}} = 3 \times 10^{14} \text{ Hz}$$

No of Channels,

$$= \frac{\frac{2}{100} \times 3 \times 10^{14}}{8 \times 10^3}$$

$$= 75 \times 10^7$$

Hence, correct answer is **Option (2).**

Q. 21 Two coils require 20 minutes and 60 minutes respectively to produce same amount of heat energy when connected separately to the same source. If they are connected in parallel arrangement to the same source; the time required to produce same amount of heat by the combination of coils, will be_____min .

Correct Answer:
15

Solution:

$$H = \frac{V^2}{R} \times t$$

$$H = \frac{V^2}{R_1} \times 20 = \frac{V^2}{R_2} \times 60$$

If connected in parallel.

$$Req = \frac{R_1 R_2}{R_1 + R_2}$$

$$H = \frac{v^2}{\left(\frac{R_1 R_2}{R_1 + R_2}\right)} \times \Delta t = \frac{4}{3}\frac{v^2}{R_1} \times \Delta t$$

$$[R_2 = 3R_1]$$

$$\Rightarrow \Delta t = 15$$

Hence, the correct answer is 15.

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Q. 22 The intensity of the light from a bulb incident on a surface is $0.22\ \mathrm{W/m^2}$. The amplitude of the magnetic field in this light-wave is _____$\times 10^{-9}$ T.
(Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2\ N^{-1} - m^{-2}}$, speed of light in vacuum $\mathrm{c} = 3 \times 10^8\ \mathrm{ms^{-1}}$)

Correct Answer:
43

Solution:

$$\mathrm{I} = \tfrac{1}{2}\ \varepsilon_0\ \mathrm{E_0^2} \times \mathrm{C}$$

$$\mathrm{E_0} = \mathrm{CB_0}$$

$$\mathrm{I} = \frac{1}{2} \times \varepsilon_0 \times \mathrm{C^2} \times \mathrm{B_0^2} \times \mathrm{C}$$

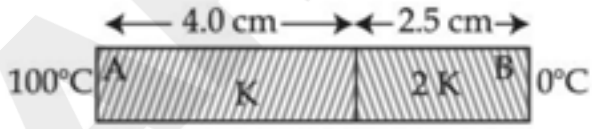
$$= \frac{1}{2}\ \varepsilon_0\ \mathrm{C^3 B_0^2}$$

$$0.22 = \frac{1}{2} \times 8.85 \times 10^{-12} \times (3 \times 10^8)^3 \times \mathrm{B_0^2}$$

$$\mathrm{B_0} = 43 \times 10^{-9}\mathrm{T}$$

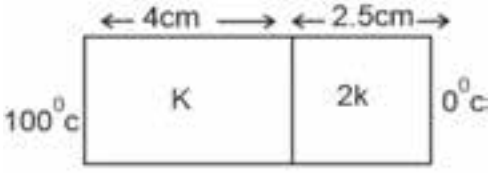
Hence, correct answer is 43.

Q. 23 As per the given figure, two plates $\mathrm{A\ and\ B}$ of thermal conductivity $\mathrm{K\ and\ 2\ K}$ are joined together to form a compound plate. The thickness of plates are $4.0\ \mathrm{cm\ and\ 2.5\ cm}$ respectively and the area of cross-section is $120\ \mathrm{cm^2}$ for each plate. The equivalent thermal conductivity of the compound plate is $\left(1 + \frac{5}{\alpha}\right)\mathrm{K}$, then the value of α will be_____.



Correct Answer:
21

Solution:



$$\text{Thermal resistance} = \mathrm{R} = \frac{1}{\mathrm{KA}}$$

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$$R_{eff} = (R_1 + R_2)$$

$$\frac{(L_1 + L_2)}{Keff \times A} = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$$

$$\frac{6.5}{Keff} = \frac{4}{k} + \frac{2.5}{2K}$$

$$Keff = \frac{13k}{10.5} = \left[1 + \frac{5}{21}\right] k$$

$$\alpha = 21$$

Hence correct answer is 21

Q. 24 A body is performing simple harmonic with an amplitude of $10\ \text{cm}$. The velocity of the body was tripled by air Jet when it is at $5\ \text{cm}$ from its mean position. The new amplitude of vibration is $\sqrt{x}\ \text{cm}$. The value of x is_____.

Correct Answer:
700

Solution:

$$v = \omega\sqrt{A^2 - x^2}$$

Initially

$$v = w\sqrt{A^2 - 5^2}$$

Given. $A = 10\ \text{cm}$

As Given in question

$$3\omega\sqrt{A^2 - 5^2} = \omega\sqrt{A'^2 - 5^2}$$

$$9 \times 75 = \left(A'\right)^2 - 25$$

$$A' = \sqrt{28 \times 25} = \sqrt{700}$$

Hence the correct answer is 700

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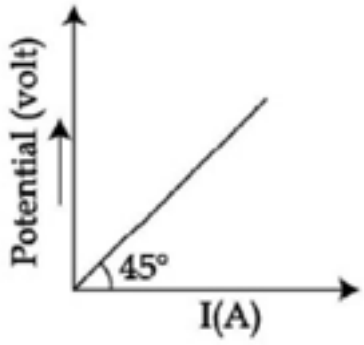
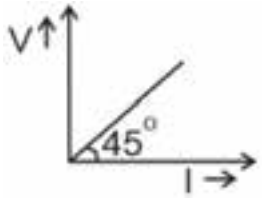
Q. 25 The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is $31.4\ \text{cm}$. The diameter of wire is measured as $2.4\ \text{cm}$. The resistivity of the given wire is measured as $x \times 10^{-3}\Omega\text{cm}$. The value of x is _____. [Take $\pi = 3.14$]



Correct Answer:
144

Solution:



$$\text{slope} = \frac{V}{I} = R = \tan 45^\circ$$

$$R = 1\Omega$$

$$R = \frac{\rho l}{A}$$

$$1 = \frac{\rho \times 31.4}{\pi r^2}$$

$$P = \frac{\pi \times (1.2\ \text{cm})^2}{31.4\ \text{cm}} = 1.44 \times 10^{-1}$$

$$x = 144$$

Hence correct answer is 144

Q. 26 300cal of heat is given to a heat engine and it rejects 225cal of heat. If source temperature is 227°C, then the temperature of sink will be_____ $^\circ\text{C}$.

Correct Answer:
102

Solution:

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$$Q_1 = W + Q_2$$
$$\eta = \frac{W}{Q_1} = \frac{300 - 225}{300} = \frac{75}{300}$$
$$\eta = 1 - \frac{T_L}{T_H} = \frac{75}{300}$$
$$\frac{T_L}{T_H} = 1 - \frac{75}{300} = \frac{3}{4}$$
$$T_L = \frac{3}{4} \times 500 = 375k$$
$$T_L = 375 - 273$$
$$= 102^{\circ}\mathrm{C}$$

Hence the correct answer is 102

Q. 27 $\sqrt{\mathrm{d_1}}$ and $\sqrt{\mathrm{d_2}}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For $\mathrm{d_1} = x\ \mathrm{d_2}$, the value of x will be_____.

Correct Answer:
3

Solution:

$$\text{Impact parameter } \alpha \cot\left(\frac{\theta}{2}\right)$$

$$\sqrt{\frac{d_1}{d_2}} = \frac{\sqrt{3}}{1}$$

$$\mathrm{d_1} = 3\mathrm{d_2}$$

$$\Rightarrow \mathrm{x} = 3$$

Hence correct answer is 3

Q. 28 A transistor is used in an amplifier circuit in common emitter mode. If the base current changes by $100\mu\mathrm{A}$, it brings a change of $10\ \mathrm{mA}$ in collector current. If the load resistance is $2\mathrm{k}\Omega$ and input resistance is $1\mathrm{k}\Omega$, the value of power gain is $x \times 10^4$. The value of x is_____.

Correct Answer:
2

Solution:

$$\text{Power gain } = \left[\frac{\Delta \mathrm{I_c}}{\Delta \mathrm{I_B}}\right]^2 \times \frac{\mathrm{R_o}}{\mathrm{R_i}}$$

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$$= \left[\frac{10^{-2}}{10^{-4}}\right]^2 \times \frac{2}{1}$$

$$= 2 \times 10^4$$

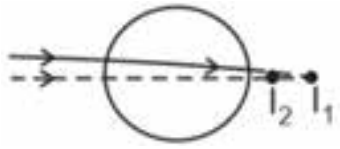
$$\mathrm{X} = 2$$

Hence correct answer is 2

Q. 29 A parallel beam of light is allowed to fall on a transparent spherical globe of diameter $30\ \mathrm{cm}$ and refractive index 1.5. The distance from the centre of the globe at which the beam of light can converge is _____mm.

Correct Answer:
225

Solution:



$1^{\mathrm{st}}\mathrm{refraction}$

$$\frac{1.5}{\mathrm{v}} - \frac{1}{\infty} = \frac{1.5-1}{+15}$$

$$\mathrm{V}_1 = 45\ \mathrm{cm}$$

For 2^{nd} refraction

$$\frac{1}{V_2} - \frac{1.5}{+15} = \frac{1-1.5}{-15}$$

$$\frac{1}{V_2} = \frac{1}{10} + \frac{1}{30} = \frac{4}{30}$$

$$V_2 = +7.5\ \mathrm{cm}$$

Distance from centre

$$= 15 + 7.5$$

$$= 22.5\mathrm{cm} = 225\mathrm{mm}$$

Hence the correct answer is 225

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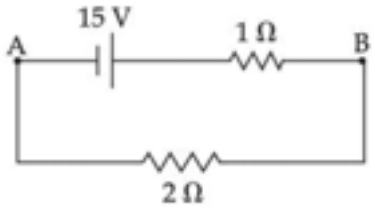
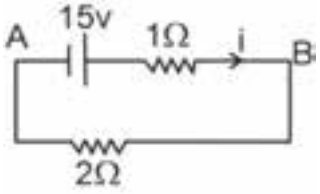
Q. 30 For the network shown below, the value of $V_B - V_A$ is _____V.



Correct Answer:
10

Solution:



$$i = \frac{15}{2+1} = 5 \text{ A}$$

$$V_B - V_A = -1 \times 5 + 15$$

$$= 10 \text{ V}$$

Hence correct answer is 10

Chemistry

Q. 1 Production of iron in blast furnace follows the following equation

$$Fe_3O_4(s) + 4CO(g) \rightarrow 3Fe(l) + 4CO_2(g)$$

when 4.640 kg of Fe_3O_4 and 2.520 kg of CO are allowed to react then the amound of iron (in g) produced is:

[Given: Molar Atomic mass $(gmol^{-1})$: Fe = 56

Molar Atomic mass $(gmol^{-1})$: O = 16

Molar Atomic mass $(gmol^{-1})$: C = 12]

Option 1:
1400

Option 2:
2200

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Option 3:
3360

Option 4:
4200

Correct Answer:
3360

Solution:

$$\text{moles of } Fe_3O_4 = \frac{4640}{232} = 20$$

$$\text{moles of CO} = \frac{2520}{28} = 90$$

$$\therefore \text{Limiting reagent is } Fe_3O_4$$

$$Fe_3O_4 + 4CO \longrightarrow 3Fe + 4CO_2$$

From reaction stoichiometry,

1 mole Fe_3O_4 produces 3 moles of Fe

$$\therefore \text{ moles of Fe produced} = 60$$

$$\therefore \text{mass of Fe produced} = 3360$$

Hence the correct answer is option 3

Q. 2 Which of the following statements are **correct** ?

(A) The electronic configuration of Cr is $[Ar]3\ d^5 4\ s^1$

(B) The magnetic quantum number may have a negative value.

(C) In the ground state of an atom, the orbitals are filled in order of their increasing energies.

(D) The total number of nodes are given by n-2.

Choose the **most appropriate** answer from the options given below:

Option 1:
(A), (C) and (D) only

Option 2:
(A) and (B) only

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Option 3:
(A) and (C) only

Option 4:
(A), (B) and (C) only

Correct Answer:
(A), (B) and (C) only

Solution:
As we have learnt,

- Cr has configuration of $[\mathrm{Ar}]4\mathrm{s}^1 3\mathrm{d}^5$

- Magnetic Quantum number can take vaue from $(-\ell)\ \mathrm{to}\ (+\ell)$

- In ground state, orbital are filled in order of their increasing energies

- Total number of nodes are given by $(\mathrm{n}-1)$

Thus statements (A) , (B) and (C) are correct

Hence the correct option is 4

Q. 3 Arrange the following in the decreasing order of their covalent character:

$(\mathrm{A})\ \mathrm{LiCl}$

$(\mathrm{B})\ \mathrm{NaCl}$

$(\mathrm{C})\ \mathrm{KCl}$

$(\mathrm{D})\ \mathrm{CsCl}$

Choose the **most appropriate** answer from the options given below:

Option 1:
$(\mathrm{A}) > (\mathrm{C}) > (\mathrm{B}) > (\mathrm{D})$

Option 2:
$(\mathrm{B}) > (\mathrm{A}) > (\mathrm{C}) > (\mathrm{D})$

Option 3:
$(\mathrm{A}) > (\mathrm{B}) > (\mathrm{C}) > (\mathrm{D})$

Option 4:
$(\mathrm{A}) > (\mathrm{B}) > (\mathrm{D}) > (\mathrm{C})$

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Correct Answer:
$(A) > (B) > (C) > (D)$

Solution:
As we have learnt,

Covalent character increases with the increase in charge density of cation

$\therefore$ Order of covalent character is

$$LiCl > NaCl > KCl > RbCl$$

Hence, the correct answer is option (3)

Q. 4 The solubility of $AgCl$ will be maximum in which of the following ?

Option 1:
$0.01\ M\ KCl$

Option 2:
$0.01\ M\ HCl$

Option 3:
$0.01\ M\ AgNO_3$

Option 4:
Deionised water

Correct Answer:
Deionised water

Solution:
KCl, HCl and $AgNO_3$ will exert a common ion effect and decrease the solubility of $AgCl$.

Thus , among the given options the solubility of $AgCl$ will be maximum in deionised water.

Hence the correct option is (4)

Q. 5 Which of the following is a **correct** statement?

Option 1:
Brownian motion destabilises sols.

Option 2:
Any amount of dispered phase can be added to emulsion without destabilising it.

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Option 3:
Mixing two oppositely charged sols in equal amount neutralises charges and stabilises colloids.

Option 4:
Presence of equal and similar charges and colloidal particles provides stability to the colloidal solution.

Correct Answer:
Presence of equal and similar charges and colloidal particles provides stability to the colloidal solution.

Solution:
As we have learnt

- Browian motion stabilizes colloidal solutions.

- Any amout of dispersion medium can be added to an emulsion without destabilising it.

- Mixing of two oppositely charged sols causes precipitation of sols.

- Presence of equal and similar charges on colloidal particle causes repulsion among them and does not let them settle or coagulate, thus providing stability to the colloidal solution

Hence correct option is 4

Q. 6 The electronic configuration of Pt (atomic number 78) is :

Option 1:
$[\mathrm{Xe}]4\mathrm{f}^{14}5\ \mathrm{d}^{9}6\ \mathrm{s}^{1}$

Option 2:
$[\mathrm{Kr}]4\mathrm{f}^{14}5\ \mathrm{d}^{10}$

Option 3:
$[\mathrm{Xe}]4\mathrm{f}^{14}5\ \mathrm{d}^{10}$

Option 4:
$[\mathrm{Xe}]4\mathrm{f}^{14}5\ \mathrm{d}^{8}6\ \mathrm{s}^{2}$

Correct Answer:
$[\mathrm{Xe}]4\mathrm{f}^{14}5\ \mathrm{d}^{9}6\ \mathrm{s}^{1}$

Solution:
Fact Based.

The electronic configuration of $\mathrm{Pt}(78)$ is $[\mathrm{Xe}]4\mathrm{f}^{14}5\mathrm{d}^{9}6\mathrm{s}^{1}$

Hence, the correct option is (1)

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Q. 7 In Isolation of which one of the following metals from their ores, the use of cyanide salt is not commonly involved?

Option 1:
Zinc

Option 2:
Gold

Option 3:
Silver

Option 4:
Copper

Correct Answer:
Copper

Solution:
Silver ad Gold are directly reached out using NaCN .

NaCN is used as a depresent in a mixture of $\mathrm{ZnS\ and\ PbS}$.

Cyanide salt is not involved in the extraction of copper.

Hence correct option is (4)

Q. 8 Which one of the following reaction indicates the reducing ability of hydrogen peroxide in basic medium?

Option 1:
$\mathrm{HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2}$

Option 2:
$\mathrm{PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O}$

Option 3:
$\mathrm{2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-}$

Option 4:
$\mathrm{Mn^{2+} + H_2O_2 \rightarrow Mn^{4+} + 2OH^-}$

Correct Answer:
$\mathrm{2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-}$

Solution:
When $\mathrm{H_2O_2}$ acts as a reducing agent, it is itself oxidised and is converted into $\mathrm{O_2}$.

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Thus, reducing nature of H_2O_2 in basic medium is shown in the reaction.

$$2MnO_4^- + 3H_2O_2 \longrightarrow 2MnO_2 + 3O_2 + 2H_2O + 2HH^-$$

Hence the correct option is (3)

Q. 9 Match **List - I** with **List - II.**

List -I **(Metal)**	**List - II** **(Emitted light wavelength(nm))**
(A) Li	(I) 670.8
(B) Na	(II) 589.2
(C) Rb	(III) 780.0
(D) Cs	(IV) 455.5

Choose the **most appropriate** answer from the options given below:

Option 1:
$(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

Option 2:
$(A) - (III), (B) - (II), (C) - (I), (D) - (IV)$

Option 3:
(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Option 4:
$(A) - (IV), (B) - (II), (C) - (I), (D) - (III)$

Correct Answer:
$(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

Solution:
The correct match of metal with emitted wavelength is given below:

$(A)\text{Li} : 670.8\ \text{nm(I)}$

$(B)\text{Na} : 589.2\ \text{nm(II)}$

$(C)\text{Rb} : 780\ \text{nm(III)}$

$(D)\text{Cs} : 455.5\ \text{nm(IV)}$

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Hence correct answer is option (1)

Q. 10 Match **List - I** with **List -II.**

List -I **(Metal)**	**List - II** **(Application)**
(A) Cs	(I) High temperature thermometer
(B) Ga	(II) Water repellent sprays
(C) B	(III) Photoelectric cells
(D) Si	(IV) Bullet proof vest

Choose the **most appropriate** answer from the options given below:

Option 1:
$(A)\text{-}(III),\ (B)\text{-}(I),\ (C)\text{-}(IV),\ (D)\text{-}(II)$

Option 2:
$(A)-(IV), (B)-(III), (C)-(II), (D)-(I)$

Option 3:
$(A)\text{-}(II),\ (B)\text{-}(III),\ (C)\text{-}(IV),\ (D)\text{-}(I)$

Option 4:
$(A)\text{-}(I),\ (B)\text{-}(IV),\ (C)\text{-}(II),\ (D)\text{-}(III)$

Correct Answer:
$(A)\text{-}(III),\ (B)\text{-}(I),\ (C)\text{-}(IV),\ (D)\text{-}(II)$

Solution:
The correct matching of metals with their uses are

$(A)Cs - \text{Photoelectric Cells}(III)$

$(B)Ga - \text{High Temperature Thermometer}(I)$

$(C)B - \text{Bullet Proof Vest}(IV)$

$(D)Si - \text{Water Repellant Sprays}(II)$

Hence the correct option is (1)

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Q. 11 The oxoacid of phosphorus that is easily obtained from a reaction of alkali and white phosphorus and has two P-H bonds, is :

Option 1:
Phosphonic acid

Option 2:
Phosphinic acid

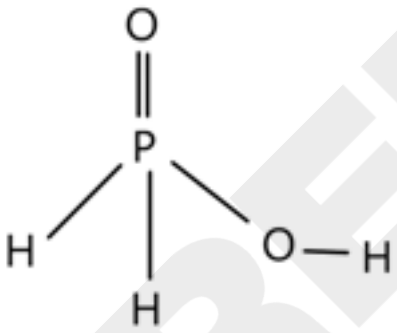
Option 3:
Pyrophosphorus acid

Option 4:
Hypophosphoric acid

Correct Answer:
Phosphinic acid

Solution:
White P disproportionates in basic medium to form $\mathrm{PH_3}$ and $\mathrm{NaH_2PO_2}$

$\mathrm{NaH_2PO_2}$ can be acidified to form $\mathrm{H_3PO_2}$



$\mathrm{H_3PO_2}$ is called as hypophosphoric acid or phosphinic acid.

Hence correct option is 2

Q. 12 The acid that is believed to be mainly responsible for the damage of Taj Mahal is

Option 1:
sulfuric acid

Option 2:
hydrofluoric acid

Option 3:
phosphoric acid

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Option 4:
hydrochloric acid

Correct Answer:
sulfuric acid

Solution:
The acid responsible for damage of Taj Mahal is H_2SO_4

$$\underset{\text{(Marble)}}{CaCO_3} + H_2SO_4 \longrightarrow CaSO_4 + H_2O + CO_2$$

Hence, the correct option is (1)

Q. 13 Two isomers 'A' and 'B' with molecular formula C_4H_8 give different products on oxidation with $KMnO_4$ in acidic medium. Isomer 'A' on reaction $KMnO_4/H^+$ results in effervescence of a gas and gives ketone. The compound 'A' is

Option 1:
But-1-ene

Option 2:
cis-But-2-ene

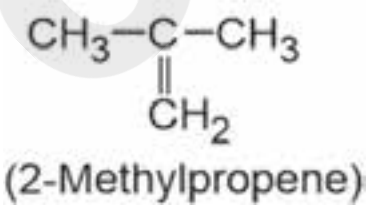
Option 3:
trans-But-2-ene

Option 4:
2-methyl propene

Correct Answer:
2-methyl propene

Solution:
Alkene (A) on reaction with $KMnO_4/H^+$ results in the formation of Ketone and effervescence of a gas CO_2.

The isomer has to be



(2-Methylpropene)

Hence the correct option is 4

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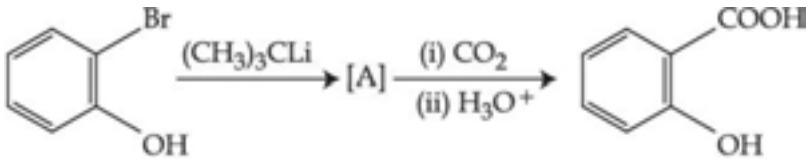
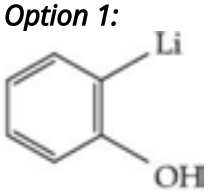
Q. 14

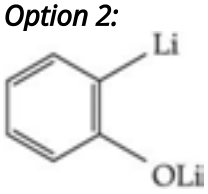


In the given conversion the compound A is:

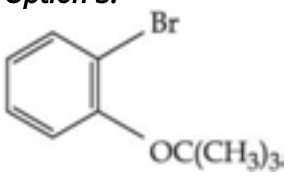
Option 1:



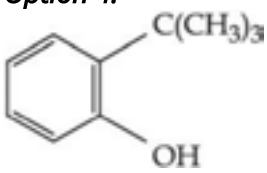
Option 2:



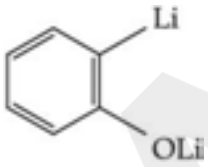
Option 3:



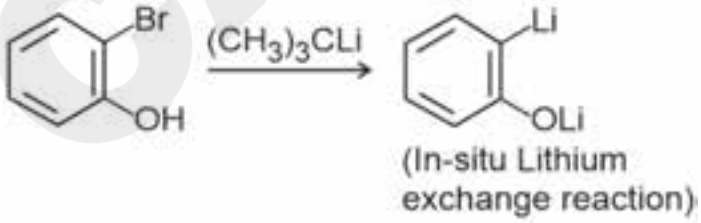
Option 4:

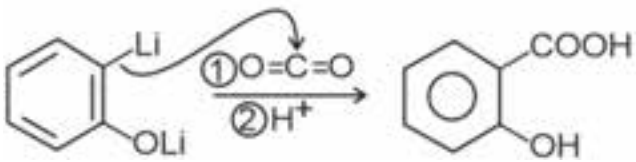


Correct Answer:



Solution:





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Hence the correct answer is option 2.

Q. 15 Given below are two statements :

Statement I : The esterification of carboxylic acid with an alcohol is a nucleophilic acyl substitution.

Statement II : Electron withdrawing groups in the carboxylic acid will increase the rate of esterification reaction.

Choose the **most appropriate** option :

Option 1:
Both **Statement I** and **Statement II** are correct.

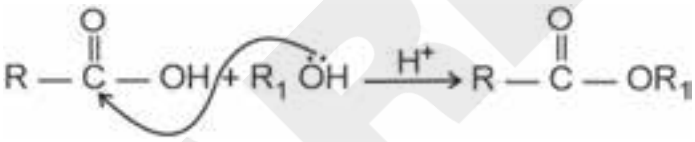
Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Both **Statement I** and **Statement II** are correct.

Solution:
Esterification is an example of nucleophilic acyl substitution



RDS in the reaction is the attack of the alcohol on the carboxylic acid and electron-withdrawing groups on the carboxylic acid increase the electrophilic nature of the acyl group.

∴ Statements I and II are both correct

Hence the correct option is 1.

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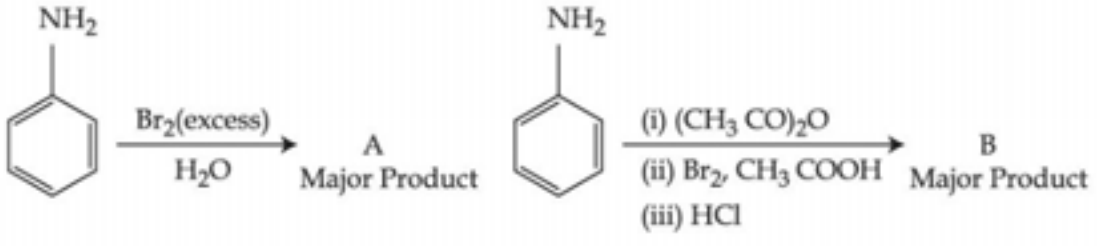
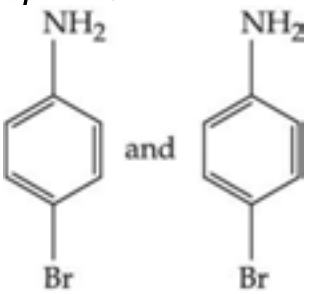
Q. 16

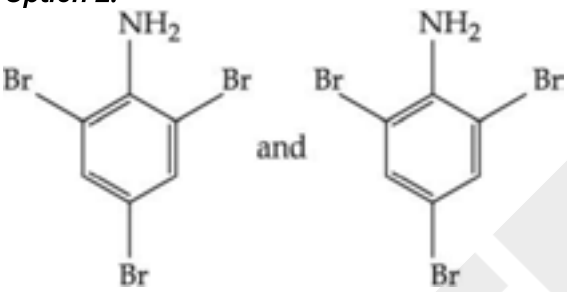


Consider the above reactions, the product A and product B respectively are

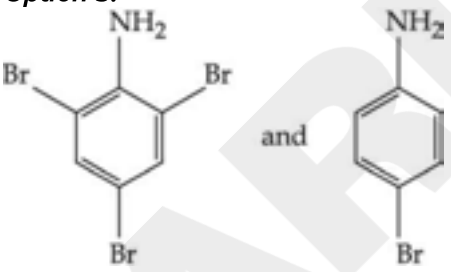
Option 1:



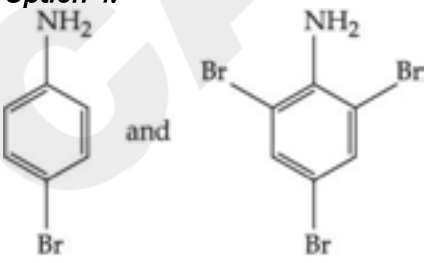
Option 2:



Option 3:



Option 4:



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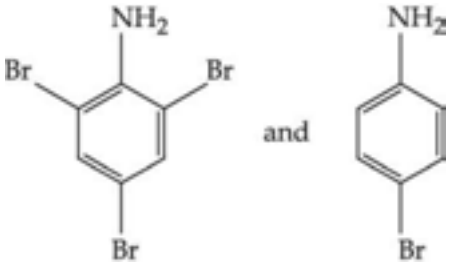
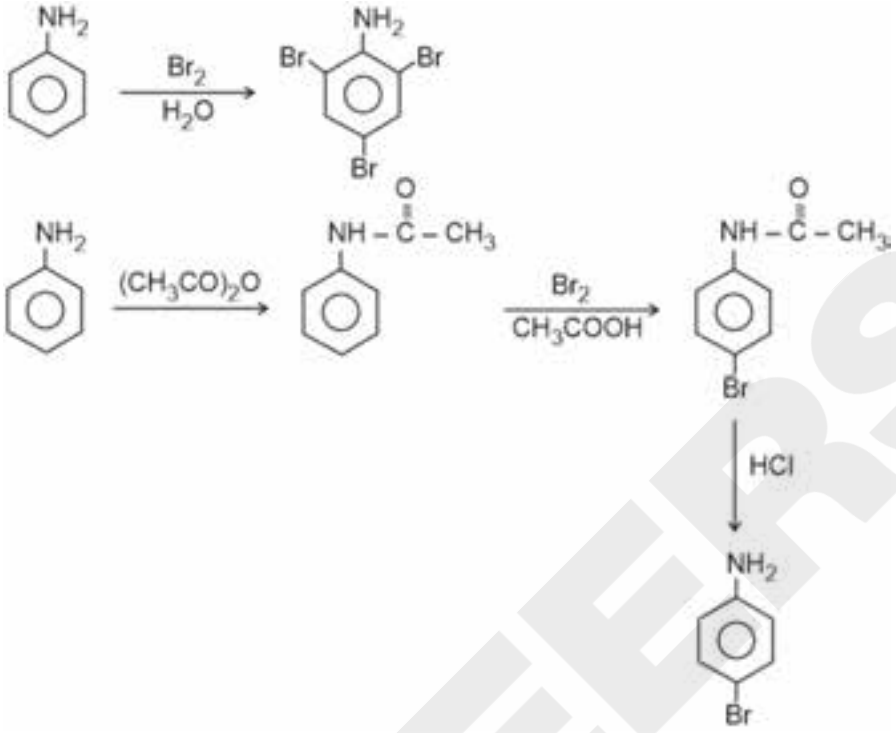
Correct Answer:



Solution:
The reactions occur as



Hence the correct option is 3.

Q. 17 The polymer, which can be stretched and retains its original status on releasing the force is

Option 1:
Bakelite.

Option 2:
Nylon 6,6.

Option 3:
Buna-N.

Option 4:
Terylene.

Correct Answer:
Buna-N.

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Solution:
Elastomers can be stretched and they regain their original status when the force is released.

Out of the given polymers, Buna-N is synthetic rubber and is an elastomer.

Hence, correct option is 3.

Q. 18 Sugar moiety in DNA and RNA molecules respectively are

Option 1:
β-D-2-deoxyribose, β-D-deoxyribose.

Option 2:
β-D-2-deoxyribose, β-D-ribose

Option 3:
β-D-ribose, β-D-2-deoxyribose.

Option 4:
β-D-deoxyribose, β-D-2-deoxyribose.

Correct Answer:
β-D-2-deoxyribose, β-D-ribose

Solution:
DNA contains $\beta - \mathrm{D} - 2 - \text{deoxyribose}$ while RNA contains $\beta - \mathrm{D} - \text{ribose}$

Hence correct option is (2)

Q. 19 Which of the following compound does not contain sulfur atom?

Option 1:
Cimetidine

Option 2:
Ranitidine

Option 3:
Histamine

Option 4:
Saccharin

Correct Answer:
Histamine

Solution:

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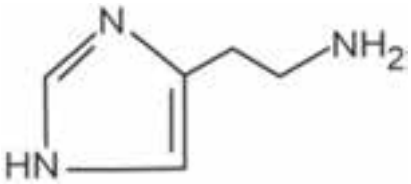
Histamine is an Imidazole derivative and does not contain Sulphur atoms.



Hence the correct option is 3.

Q. 20 Given below are two statements.

Statement I : Phenols are weakly acidic.

Statement II : Therefore they are freely soluble in NaOH solution and are weaker acids than alcohols and water.

Choose the **most appropriate** option :

Option 1:
Both **Statement I** and **Statement II** are correct.

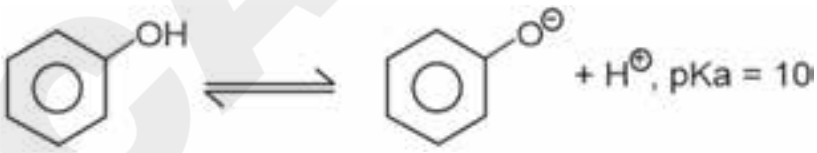
Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Statement I is correct but **Statement II** is incorrect.

Solution:
Phenol has pka of 10 and is weakly acidic



Phenols are freely soluble in NaOH but they are stronger acids than alcohols or water.

$\therefore$ Statement I is correct while Statement II is incorrect.

Hence correct option is 3.

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Q. 21 Geraniol, a volatile organic compound, is a componenet of rose oil. The density of the vapour is $0.46\ \mathrm{gL^{-1}}$ at $257^{\circ}\mathrm{C}$ and $100\ \mathrm{mm\ Hg}$. The molar mass of geraniol is _____$\mathrm{gmol^{-1}}$. (Nearest Integer)

$[\text{Given : } \mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}\ ]$

Correct Answer:
152

Solution:
As we have learnt ,

$$\mathrm{P} = \frac{\mathrm{dRT}}{\mathrm{M_0}}$$

$$\Rightarrow \frac{100}{760} = \frac{0.46 \times 0.082 \times 530}{\mathrm{M_0}}$$

$$\Rightarrow \quad \mathrm{M_0} = 151.9 \simeq 152.$$

Hence correct answer is 152

Q. 22 $17.0\ \mathrm{g\ of\ NH_3}$ completely vapourises at $-33.42^{\circ}\mathrm{C}$ and 1 bar pressure and the enthalpy change in the process is $23.4\ \mathrm{kJ\ mol^{-1}}$. The enthalpy change for the vapourisation of $85\ \mathrm{g\ of\ NH_3}$ under the same conditions is _____kJ.

Correct Answer:
117

Solution:
Given,

$1\ \mathrm{mole\ NH_3}$ requires $23.4\ \mathrm{kJ\ mol^{-1}}$

$\therefore 5\ \mathrm{mole\ NH_3}$ under identical conditions require $23.4 \times 5\ \mathrm{kJ} = 117\ \mathrm{kJ}$

Hence answer is 117

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Q. 23 1.2 mL of acetic acid is dissolved in water to make 2.0 L of solution. The depression in freezing point observed for this strength of acid is $0.0198^{\circ}\mathrm{C}$. The percentage of dissociation of the acid is _____.(Nearest integer)

[Given : Density of acetic acid is $1.02\ \mathrm{g\ mL^{-1}}$

Molar mass of acetic acid is $60\ \mathrm{g\ mol^{-1}}$

$\mathrm{K}_f\,(\mathrm{H_2O}) = 1.85\ \mathrm{K\ kg\ mol^{-1}}$]

Correct Answer:
5

Solution:
$\text{mass of acetic acid} = 1.02 \times 1.2\text{g} = 1.224\ \text{g}$

$\therefore \text{moles of acetic acid} = 0.0204$

$\therefore\ \text{molality} = \dfrac{0.0204}{2} = 0.0102$

Now,

$\Delta \mathrm{T_f} = \mathrm{iK_f m}$

$\Rightarrow 0.0198 = \mathrm{i}(1.85)(0.0102)$

$\Rightarrow \mathrm{i} = 1.05$

$\Rightarrow 1 + \alpha = 1.05$

$\Rightarrow \alpha = 0.05$

$\Rightarrow \%\ \alpha = 5$

Hence answer is 5

Q. 24 A dilute solution of sulphuric acid is electrolysed using a current of 0.10 A for 2 hours to produce hydrogen and oxygen gas. The total volume of gases produced at STP is _____ $\mathrm{cm^3}$. (Nearest integer)

[Given: Faraday constant $\mathrm{F} = 96500\ \mathrm{C mol^{-1}}$ at STP,
, molar volume of an ideal gas is $22.7\ \mathrm{L\ mol^{-1}}$]

Correct Answer:
127

Solution:

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$C : 2H^+ + 2e^- \rightarrow H_2$

$A : H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$

$\therefore$ For every 2 mole of e^-, 1 mole of H_2 and 0.5 mole of O_2 are liberated.

$\therefore$ 1 mole e^- liberates 0.75 mole of gases

Now,

Charge passed $= 0.1 \times 2 \times 60 \times 60$

$= 720\ C$

$\therefore$ moles of electron $= \frac{720}{96500} = 7.46 \times 10^{-3}$

$\therefore$ volume of gases $= 22.7 \times 0.75 \times 7.46 \times 10^{-3}$

$= 0.127\ L$

$= 127 cm^3$

Hence answer is 127

Q. 25 The activation energy of one of the reactions in a biochemical process is $532611\ J\ mol^{-1}$. When the temperature falls from $310\ K$ to $300\ K$, the change in rate constant observed is $k_{300} = x \times 10^{-3} k_{310}$. The value of x is _____.

[Given: $\ln 10 = 2.3$

$R = 8.3\ J\ K^{-1}\ mol^{-1}$]

Correct Answer:
1

Solution:
As we have learnt,

$K = A \exp\left(-\frac{Ea}{RT}\right)$

$\therefore K_{300} = A \exp\left(-\frac{Ea}{300R}\right)$

$K_{310} = A \exp\left(\frac{-Ea}{310R}\right)$

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$$\therefore \frac{K_{300}}{K_{310}} = \exp\left\{\frac{Ea}{R}\left(\frac{1}{310} - \frac{1}{300}\right)\right\}$$

$$= \exp\left\{\frac{-532611}{8.3} \times \frac{10}{310 \times 300}\right\}$$

$$= \exp(-6.9)$$

$$= e^{-6.9}$$

$$= \left(e^{-2.3}\right)^3$$

$$= 10^{-3}$$

$$\therefore K_{300} = 10^{-3} \times K_{310}$$

$$\therefore x = 1$$

Hence answer is 1

Q. 26 The number of terminal oxygen atoms present in the product B obtained from the following reaction is _____.

$$FeCr_2O_4 + Na_2CO_3 + O_2 \rightarrow A + Fe_2O_3 + CO_2$$

$$A + H^+ \rightarrow B + H_2O + Na^+$$

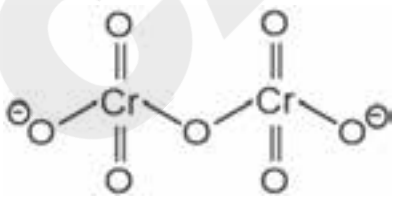
Correct Answer:
6

Solution:
The reactions occur as

$$FeCr_2O_4 + Na_2CO_3 + O_2 \longrightarrow Na_2CrO_4 + Fe_2O_3 + CO_2$$

$$Na_2CrO_4 + H^+ \longrightarrow Na_2Cr_2O_7 + H_2O$$

Thus B is $Cr_2O_7^{2-}$



It contains 6 terminal oxygen atoms

Hence answer is 6

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Q. 27 An acidified manganate solution undergoes disproportionation reaction. The spin-only magnetic moment value of the product having manganese in higher oxidation state is _____B.M (Nearest integer)

Correct Answer:
0

Solution:
Disproportionation of $\mathrm{MnO_4}^{2-}$ is given as

$$\underset{}{\mathrm{MnO_4^{2-}}} \longrightarrow \underset{(+7)}{\mathrm{MnO_4^-}} + \underset{(+4)}{\mathrm{MnO_2}}$$

Mn^{+7} has d^0 configuration and has no unpaired electrons.

Hence answer is 0

Q. 28 Kjeldahl's method was used for the estimation of nitrogen in an organic compound. The ammonia evolved from 0.55 g of the compound neutralised $12.5\ \mathrm{mL\ of\ 1M\ H_2SO_4}$ solution.

The percentage of nitrogen in the compound is _____.(Nearest integer)

Correct Answer:
64

Solution:
$$\mathrm{meq\ of\ H_2SO_4} = 25$$

$$\therefore \mathrm{meq\ of\ NH_3} = 25 = \mathrm{m\ moles\ of\ NH_3}$$

$$= \mathrm{m\ moles\ of\ N}$$

$$\therefore\ \mathrm{mass\ of\ N} = 25 \times 10^{-3} \times 14$$

$$= 0.35$$

$$\therefore \%\ \mathrm{by\ weight\ of\ N} = \frac{0.35}{0.55} \times 100$$

$$= 63.6\%$$

Hence answer is 64

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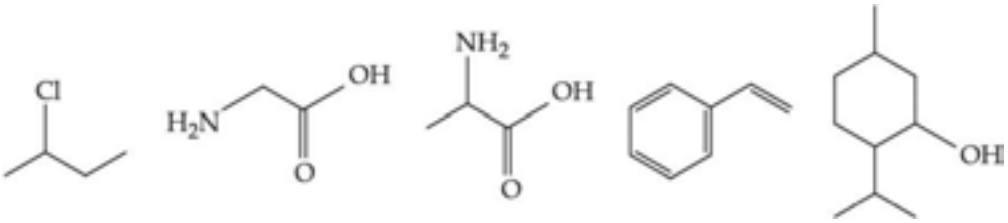
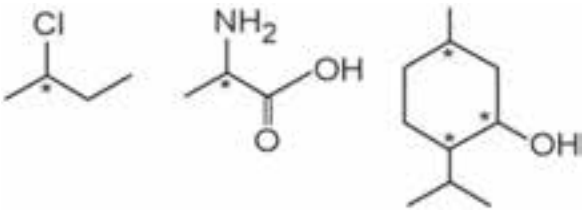
Q. 29 Observe structures of the following compounds



The total number of structures / compounds which possess asymmertic carbon atoms is _____.

Correct Answer:
3

Solution:
The compounds containing chiral centers among the given ones are



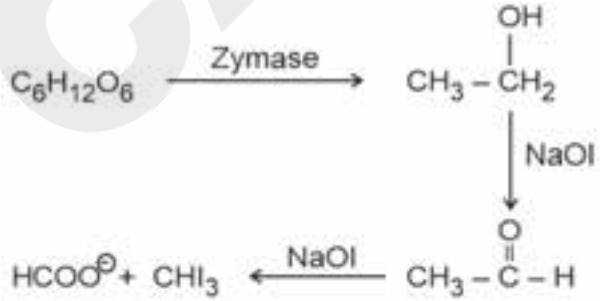
Hence answer is 3

Q. 30 $C_6H_{12}O_6 \xrightarrow{\text{Zymase}} A \xrightarrow[\Delta]{\text{NaOI}} B + CHI_3$

The number of carbon atoms present in the product B is _____.

Correct Answer:
1

Solution:
Zymase converts Glucose into Ethanol and CO_2



So "B" is sodium formate and it contains 1 carbon atom

Hence answer is 1

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Maths

Q. 1 The probability that a randomly chosen 2×2 matrix with all the entries from the set of first 10 primes, is singular, is equal to :

Option 1:

$\frac{133}{10^4}$

Option 2:

$\frac{18}{10^3}$

Option 3:

$\frac{19}{10^3}$

Option 4:

$\frac{271}{10^4}$

Correct Answer:

$\frac{19}{10^3}$

Solution:

Let the matrix A be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

If A is singular then $|\mathrm{A}| = 0$

$\Rightarrow \mathrm{ad} - \mathrm{bc} = 0$

$\Rightarrow \mathrm{ad} = \mathrm{bc}$

Case I : $\mathrm{a} = \mathrm{d}$

a can take 10 values.

d can take 1 value which equals a

Now for $\mathrm{ad} = \mathrm{bc}$ to hold

$\mathrm{bc} = \mathrm{a}^2 \Rightarrow \mathrm{b} = \mathrm{a}, \mathrm{c} = \mathrm{a}(\text{as a is prime})$

$\therefore$ only one option for b and c

Total such matrices $= 10 \times 1 \times 1 \times 1 = 10$

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Case II : $a \neq d$

In this case a can take 10 values and d can take 9 values.

Now for $ad = bc, \quad (b = a \text{ and } c = d)$

or $\quad (b = d \text{ and } c = a)$

$\therefore \quad 10 \times 9 \times 2 = 180$

$$\text{favourable } = 10 + 180 = 190$$

$$\text{Total matrices} = 10^4$$

$$\text{Probability} = \frac{190}{10^4} = \frac{19}{10^3}$$

Hence answer is option 3.

Q. 2 Let the solution curve of the differential equation $x\frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}, y(1) = 3$ be $y = y(x)$. Then $y(2)$ is equal to :

Option 1:
15

Option 2:
11

Option 3:
13

Option 4:
17

Correct Answer:
15

Solution:

$$x\frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{y^2 + 16x^2}}{x}$$

Let $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

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$$\Rightarrow v + x\frac{dv}{dx} = \frac{vx + \sqrt{v^2x^2 + 16x^2}}{x}$$

$$\Rightarrow v + x\frac{dv}{dx} = v + \sqrt{v^2 + 16}$$

$$\Rightarrow \frac{dv}{\sqrt{v^2 + 16}} = \frac{dx}{x}$$

Integrating

$$\Rightarrow \ln\left|v + \sqrt{v^2 + 16}\right| = \ln x + \ln k$$

$$\Rightarrow v + \sqrt{v^2 + 16} = kx$$

$$\Rightarrow \frac{y}{x} + \sqrt{\frac{y^2}{x^2} + 16} = kx$$

$$\text{As } y(1) = 3$$

$$\Rightarrow 3 + \sqrt{9 + 16} = k$$

$$\Rightarrow k = 8$$

$$\text{For } x = 2$$

$$\frac{y}{2} + \sqrt{\frac{y^2}{4} + 16} = 8 \times 2$$

$$\Rightarrow y + \sqrt{y^2 + 64} = 32$$

$$\Rightarrow \sqrt{y^2 + 64} = -y + 32$$

$$\Rightarrow y^2 + 64 = y^2 + 1024 - 64y$$

$$\Rightarrow 64y = 960$$

$$\Rightarrow y = 15$$

Hence answer is option 1

Q. 3 If the mirror image of the point $(2, 4, 7)$ in the plane $3x - y + 4z = 2$ is (a, b, c), then $2a + b + 2c$ is equal to :

Option 1:

54

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Option 2:
50

Option 3:
-6

Option 4:
-42

Correct Answer:
-6

Solution:
Using formula for mirror image

$$\frac{a-2}{3}=\frac{b-4}{-1}=\frac{c-7}{4}=\frac{-2(6-4+28-2)}{9+1+16}$$

$$\frac{a-2}{3}=\frac{b-4}{-1}=\frac{c-7}{4}=-\frac{1}{13}(28)$$

$$a=2-\frac{28\times 3}{13}=\frac{-58}{13}$$

$$b=4+\frac{28}{13}=\frac{80}{13}$$

$$c=7-\frac{28\times 4}{13}=-\frac{21}{13}$$

$$\therefore 2a+b+2c=\frac{-116}{13}+\frac{80}{13}-\frac{42}{13}$$

$$=-\frac{78}{13}=-6$$

Hence answer is option 3.

Q. 4 Let $f:\mathbf{R}\to\mathbf{R}$ be a function defined by :

$$f(x)=\begin{cases}\max\limits_{t\le x}\{t^3-3t\} & ; \quad x\le 2\\ x^2+2x-6 & ; \quad 2<x<3\\ [x-3]+9 & ; \quad 3\le x\le 5\\ 2x+1 & ; \quad x>5\end{cases}$$

where $[t]$ is the greatest integer less than or equal to t. Let m be the number of points where f is not differentiable and $I=\int_{-2}^{2} f(x)dx$. Then the ordered pair (m, I) is equal to :

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Option 1:

$\left(3, \frac{27}{4}\right)$

Option 2:

$\left(3, \frac{23}{4}\right)$

Option 3:

$\left(4, \frac{27}{4}\right)$

Option 4:

$\left(4, \frac{23}{4}\right)$

Correct Answer:

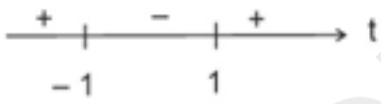
$\left(4, \frac{27}{4}\right)$

Solution:

For $x \leq 2$

Let $g(t) = t^3 - 3t$

$g'(t) = 3t^2 - 3 = 3\left(t^2 - 1\right) = 3(t+1)(t-1)$

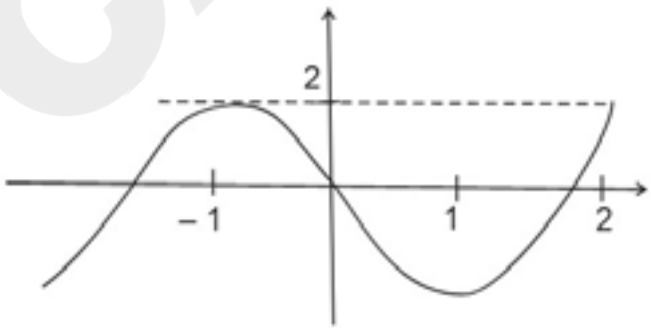


$g(-1) = -1 + 3 = 2$

$g(0) = 0$

$g(1) = 1 - 3 = -2$

$g(2) = 8 - 6 = 2$



$\therefore \max_{t \leq x} \left\{t^3 - 3t\right\}$ is

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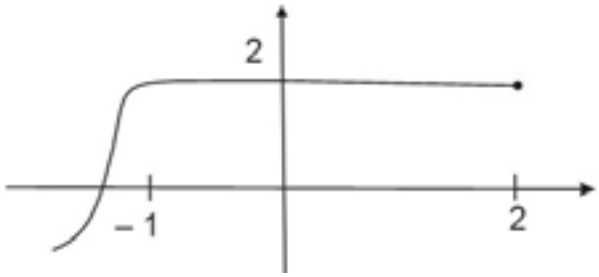

$\therefore$ Differentiable in this interval

For $2 < x < 3$

$y = x^2 + 2x - 6$

$\text{At } x = 2^+; y = 4 + 4 - 6 = 2$

$f'(x) = 2x + 2$

$f'\left(2^+\right) = 4 + 2 = 6$

$\therefore f'\left(2^-\right) \neq f'\left(2^+\right)$

$\therefore \text{Non-differentiable at } x = 2$

$f\left(3^-\right) = 9 + 6 - 6 = 9$

$f'\left(3^-\right) = 6 + 2 = 8$

For $3 \leqslant x \leqslant 5$

$y = [x - 3] + 9 = [x] - 3 + 9 = [x] + 6$

$\text{For } 3 \leq x < 4, y = 3 + 6 = 9$

$f'\left(3^+\right) = 0 \neq f'\left(3^-\right) \Rightarrow$ non-differentiable at x=3

For $4 \leq x < 5, y = 4 + 6 = 10$

Discontinuous at $x = 4 \Rightarrow \text{non} - \text{differentiable at } x = 4$

$f'\left(5^{-1}\right) = 0$

$\& f\left(5^-\right) = 10$

For $x > 5$

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$f\left(5^{+}\right)=10+1=10$

$\therefore$ Discontinuous and thus non-differentiable at $x=5$

$\therefore m=4 \quad (x=2,3,4,5)$

$$I=\int_{-2}^{2} f(x)dx$$

$$=\int_{-2}^{-1}\left(x^3-3x\right)dx+\int_{-1}^{2}2dx$$

$$=\frac{x^4}{4}\Big|_{-2}^{-1}-\frac{3}{2}x^2\Big|_{-2}^{-1}+2x|_{-1}^{2}$$

$$=\frac{1}{4}-4-\frac{3}{2}(1-4)+2(2+1)$$

$$=\frac{1}{4}-4+\frac{9}{2}+6$$

$$=\frac{1-16+18+24}{4}$$

$$=\frac{27}{4}$$

Hence answer is option 3

Q. 5 Let $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ where $\alpha,\beta\in\mathbf{R}$, be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$, then the value of $\alpha+\beta$ is equal to :

Option 1:
3

Option 2:
4

Option 3:
5

Option 4:
6

Correct Answer:
3

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Solution:

Projection of $\vec{a}$ on $\vec{c} = \dfrac{\vec{a}\cdot\vec{c}}{|\vec{c}|}$

$$= \frac{\alpha+6+2}{\sqrt{1+2^2+2^2}}$$

$$= \frac{8+\alpha}{3}$$

Given $\dfrac{8+\alpha}{3} = \dfrac{10}{3} \Rightarrow \alpha = 2$

$$\vec{b}\times\vec{c} = \begin{vmatrix} i & j & k \\ 3 & -\beta & 4 \\ 1 & 2 & -2 \end{vmatrix}$$

$$= i(2\beta - 8) - j(-6-4) + k(6+\beta)$$

Comparing with given value

$$2\beta - 8 = -6 \Rightarrow 2\beta = 2 \Rightarrow \beta = 1$$

$$\alpha + \beta = 2 + 1 = 3$$

Hence answer is option 1

Q. 6 The area enclosed by $y^2 = 8x$ and $y = \sqrt{2}x$ that lies outside the triangle formed by $y = \sqrt{2}x, x = 1, y = 2\sqrt{2},$ is equal to:

Option 1:

$\dfrac{16\sqrt{2}}{6}$

Option 2:

$\dfrac{11\sqrt{2}}{6}$

Option 3:

$\dfrac{13\sqrt{2}}{6}$

Option 4:

$\dfrac{5\sqrt{2}}{6}$

Correct Answer:

$\dfrac{13\sqrt{2}}{6}$

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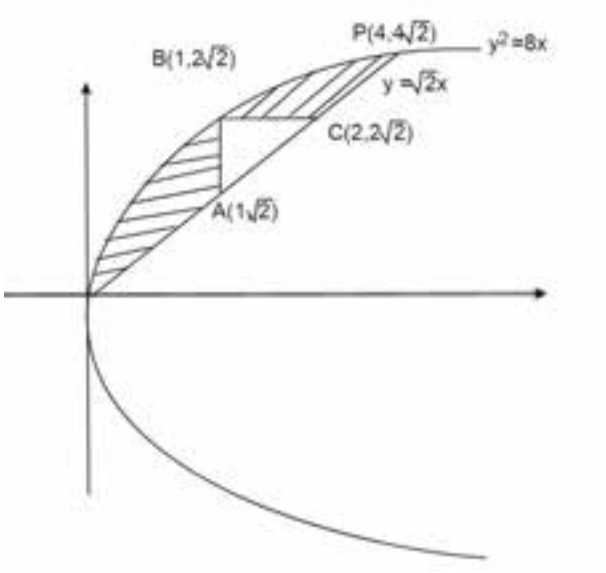
Solution:



Point of intersection of $\mathrm{y} = \sqrt{2}\mathrm{x}$ and $\mathrm{y}^2 = 8\mathrm{x}$

$\Rightarrow (\sqrt{2}\mathrm{x})^2 = 8\mathrm{x} \Rightarrow 2\mathrm{x}^2 = 8\mathrm{x} \Rightarrow \mathrm{x} = 0, 4$

$\Rightarrow \mathrm{y} = \sqrt{2} \cdot 4 = 4\sqrt{2}$

$\text{At } \mathrm{x} = 1 \Rightarrow \mathrm{y}^2 = 8\mathrm{x} \Rightarrow \mathrm{y}^2 = 8 \Rightarrow \mathrm{y} = 2\sqrt{2}$

We need shaded area

$\left(\text{As } \mathrm{y}^2 = 8\mathrm{x} \Rightarrow \mathrm{y} = 2\sqrt{2\mathrm{x}} \right)$

$= \int_0^4 (2\sqrt{2\mathrm{x}} - \sqrt{2}\mathrm{x})\mathrm{dx} - \frac{1}{2} \cdot \sqrt{2} \cdot 1$

$= 2\sqrt{2} \cdot \frac{\mathrm{x}^{3/2}}{3/2}\Big|_0^4 - \sqrt{2} \cdot \frac{\mathrm{x}^2}{2}\Big|_0^4 - \frac{1}{\sqrt{2}}$

$= \frac{8\sqrt{2}}{3} - \frac{1}{\sqrt{2}} = \frac{16\sqrt{2} - 3\sqrt{2}}{6} = \frac{13\sqrt{2}}{6}$

Hence the answer is option 3.

Q. 7 If the system of linear equations

$$2x + y - z = 7$$
$$x - 3y + 2z = 1$$
$$x + 4y + \delta z = \mathrm{k}, \text{ where } \delta, \mathrm{k} \in \mathbf{R}$$

has infinitely many solutions, then $\delta + \mathrm{k}$ is equal to :

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Option 1:
-3

Option 2:
3

Option 3:
6

Option 4:
9

Correct Answer:
3

Solution:
For infinite solutions

$$\Delta = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 1 & 4 & \delta \end{vmatrix} = 0$$

$\Rightarrow 2(-3\delta - 8) - 1(\delta - 2) - 1(4 + 3) = 0$

$\Rightarrow -6\delta - 16 - \delta + 2 - 7 = 0$

$\Rightarrow -7\delta = 16 + 7 - 2$

$\Rightarrow -7\delta = 21$

$\Rightarrow \delta = -3$

Also

$\Delta_3 = 0$

$$\begin{vmatrix} 2 & 1 & 7 \\ 1 & -3 & 1 \\ 1 & 4 & k \end{vmatrix} = 0$$

$\Rightarrow 2(-3\mathrm{k} - 4) - 1(\mathrm{k} - 1) + 7(4 + 3) = 0$

$\Rightarrow -6\mathrm{k} - 8 - \mathrm{k} + 1 + 49 = 0$

$\Rightarrow -7\mathrm{k} = -42$

$\Rightarrow \mathrm{k} = 6$

$\therefore \delta + \mathrm{k} = 3$

Hence the answer is option 2

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Q. 8 Let $\alpha \text{ and } \beta$ be the roots of the equation $x^2 + (2i - 1) = 0$. Then, the value of $\left|\alpha^8 + \beta^8\right|$ is equal to :

Option 1:
50

Option 2:
250

Option 3:
1250

Option 4:
1500

Correct Answer:
50

Solution:
$x^2 = 1 - 2i$

$\text{As } \alpha, \beta \text{ are its roots}$

$\alpha^2 = 1 - 2i \quad \& \quad \beta^2 = 1 - 2i \Rightarrow \alpha^2 = \beta^2$

$\Rightarrow \alpha^8 = \beta^8$

$\therefore \left|\alpha^8 + \beta^8\right|$

$= \left|2\alpha^8\right|$

$= 2\left|\alpha^2\right|^4$

$= 2(\sqrt{1+4})^4$

$= 2 \cdot 5^2$

$= 50$

Hence correct answer is option 1.

Q. 9 Let $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$ be such that $(p \wedge q)\Delta((p \vee q) \Rightarrow q)$ is a tautology. Then Δ is equal to:

Option 1:
$\wedge$

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Option 2:

$\vee$

Option 3:

$\Rightarrow$

Option 4:

$\Leftrightarrow$

Correct Answer:

$\Rightarrow$

Solution:

p	q	$p \wedge q$	$p \vee q$	$(p \vee q) \Rightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Looking at third ad fifth coloums, given statement will be a tautology only when Δ is $\Rightarrow$

Hence the correct answer is option 3.

Q. 10 Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all $i, j = 1, 2, 3$. Then, the matrix $A^2 + A^3 + \ldots + A^{10}$ is equal to :

Option 1:

$\left(\frac{3^{10} - 3}{2}\right) A$

Option 2:

$\left(\frac{3^{10} - 1}{2}\right) A$

Option 3:

$\left(\frac{3^{10} + 1}{2}\right) A$

Option 4:

$\left(\frac{3^{10} + 3}{2}\right) A$

Correct Answer:

$\left(\frac{3^{10} - 3}{2}\right) A$

Solution:

Forming matrix A using given aij formula

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$$A = \begin{bmatrix} 1 & 2 & 2^2 \\ 2^{-1} & 1 & 2 \\ 2^{-2} & 2^{-1} & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 & 2^2 \\ 2^{-1} & 1 & 2 \\ 2^{-2} & 2^{-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2^2 \\ 2^{-1} & 1 & 2 \\ 2^{-2} & 2^{-1} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 12 \\ 3 \cdot 2^{-1} & 3 & 6 \\ 3 \cdot 2^{-2} & 3 \cdot 2^{-1} & 3 \end{bmatrix} = 3\ A$$

$\therefore A^2 = 3A$

$\Rightarrow A^3 = A^2 \cdot A = 3A \cdot A = 3A^2 = 3 \cdot 3A = 3^2A$

$\Rightarrow A^4 = A^2 \cdot A^2 = 3A \cdot 3A = 3^2A^2 = 3^2 \cdot 3A = 3^3A$

Similarly $A^5 = 3^4A, \cdots A^{10} = 3^9A$

$\therefore A^2 + A^3 + \ldots + A^{10} = A\left(3 + 3^2 + \cdots + 3^9\right)$

$$= \frac{3\left(3^9 - 1\right)}{3 - 1}A$$

$$= \left(\frac{3^{10} - 3}{2}\right) A.$$

Hence the correct answer is option 1.

Q. 11 Let a set $A = A_1 \cup A_2 \cup \ldots \cup A_{k'}$ where $A_i \cap A_j = \phi$ for $i \neq j, 1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i, 1 \leq i \leq k\}$. Then, R is :

Option 1:
reflexive, symmetric but not transitive

Option 2:
reflexive, transitive but not symmetric

Option 3:
reflexive but not symmetric and transitive

Option 4:
an equivalence relation

Correct Answer:
an equivalence relation

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Solution:
As R contains all ordered pairs of $\mathrm{A_1, A_2, A_3, \ldots A_k}$.

So it must be an equivalence relation .

Hence the correct answer is option 4.

Q. 12 Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 2a_{n+1} - a_n + 1$ for all $n \geqslant 0$.
Then, $\sum_{n=2}^{\infty} \frac{a_n}{7^n}$ is equal to:

Option 1:
$\frac{4}{343}$

Option 2:
$\frac{7}{216}$

Option 3:
$\frac{8}{343}$

Option 4:
$\frac{49}{216}$

Correct Answer:
$\frac{7}{216}$

Solution:
$a_0 = a_1 = 0$

$a_{n+2} = 2a_{n+1} - a_n + 1$

$\Rightarrow \frac{a_{n+2}}{7^{n+2}} = 2\frac{a_{n+1}}{7^{n+2}} - \frac{a_n}{7^{n+2}} + \frac{1}{7^{n+2}}$

$\Rightarrow \frac{a_{n+2}}{7^{n+2}} = \frac{2}{7} \cdot \left(\frac{a_{n+1}}{7^{n+1}}\right) - \frac{1}{7^2}\left(\frac{a_n}{7^n}\right) + \frac{1}{7^{n+2}}$

Applying summation from $n = 0$ to infinity

$\Rightarrow \sum_{n=0}^{\infty} \frac{a_{n+2}}{7^{n+2}} = \frac{2}{7}\sum_{0}^{\infty} \frac{a_{n+1}}{7^{n+1}} - \frac{1}{49}\sum_{0}^{\infty} \frac{a_n}{7^n} + \sum_{0}^{\infty} \frac{1}{7^{n+2}}$ .....(i)

Now

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$$\sum_{0}^{\infty}\frac{a_{n+2}}{7^{n+2}}=\frac{a_2}{7^2}+\frac{a_3}{7^3}+\cdots=S\ (let)$$

And $$\sum_{0}^{\infty}\frac{a_{n+1}}{7^{n+1}}=\frac{a_1}{7}+\frac{a_2}{7^2}+\frac{a_3}{7^3}+\cdots$$

$$=\frac{a_2}{7^2}+\frac{a_3}{7^3}+\cdots\cdot(\text{as } a_1=0)$$

$$=S$$

And $$\sum_{0}^{\infty}\frac{a_n}{7^n}=\frac{a_0}{1}+\frac{a_1}{7}+\frac{a_2}{7^2}+\cdots$$

$$=\frac{a_2}{7^2}+\frac{a_3}{7^3}+\cdots=S$$

From (i)

$$\Rightarrow S=\frac{2}{7}S-\frac{1}{49}S+\left(\frac{1}{7^2}+\frac{1}{7^3}+\cdots\right)$$

$$\Rightarrow S-\frac{2}{7}S+\frac{1}{49}S=\frac{1}{49}\cdot\frac{1}{1-\frac{1}{7}}$$

$$\Rightarrow \frac{(49-14+1)S}{49}=\frac{1}{7\cdot 6}$$

$$\Rightarrow S=\frac{49}{7\cdot 6\cdot 36}$$

$$\Rightarrow S=\frac{7}{216}$$

Hence the correct answer is option 2

Q. 13 The distance between the two points A and A' which lie on $y=2$ such that both the line segments AB and $A'B$ (where B is the point $(2,3)$) subtend angle $\frac{\pi}{4}$ at the origin, is equal to :

Option 1:

10

Option 2:

$\frac{48}{5}$

Option 3:

$\frac{52}{5}$

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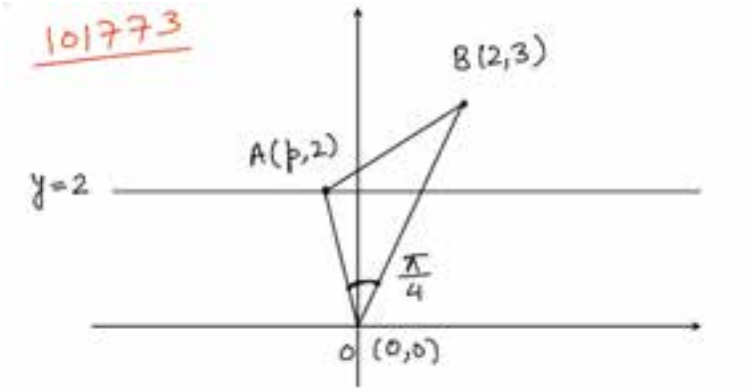
Option 4:

3

Correct Answer:

$\frac{52}{5}$

Solution:



Let A be$(p, 2)$

$$\text{Slope of OA} = m_1 = \frac{2}{p}$$

$$\text{Slope of OB} = m_2 = \frac{3}{2}$$

$$\text{Angle between these} = \frac{\pi}{4}$$

$$\therefore \tan\left(\frac{\pi}{4}\right) = \left|\frac{\frac{3}{2} - \frac{2}{p}}{1 + \frac{3}{2} \cdot \frac{2}{p}}\right|$$

$$\Rightarrow 1 = \left|\frac{3p - 4}{2p + 6}\right|$$

$$\Rightarrow \frac{3p - 4}{2p + 6} = 1 \text{ or } \frac{3p - 4}{2p + 6} = -1$$

$$\Rightarrow 3p - 4 = 2p + 6 \text{ or } 3p - 4 = -2p - 6$$

$$\Rightarrow p = 10 \quad \text{or } 5p = -2$$

$$\Rightarrow p = 10 \text{ or } p = -\frac{2}{5}$$

$$\therefore AA' = 10 + \frac{2}{5} = \frac{52}{5}$$

Hence the correct answer is option 3.

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Q. 14 A wire of length $22\ \mathrm{m}$ is to be cut into two pieces. One of the pieces is to be made into a square and the other into an equilateral triangle. Then, the length of the side of the equilateral triangle, so that the combined area of the square and the equilateral triangle is minimum, is :

Option 1:

$\frac{22}{9+4\sqrt{3}}$

Option 2:

$\frac{66}{9+4\sqrt{3}}$

Option 3:

$\frac{22}{4+9\sqrt{3}}$

Option 4:

$\frac{66}{4+9\sqrt{3}}$

Correct Answer:

$\frac{66}{9+4\sqrt{3}}$

Solution:

Let the side of square be x and side of triangle be z

So $4\mathrm{x}+3\mathrm{z}=22$

$\Rightarrow \mathrm{x}=\frac{22-3\mathrm{z}}{4}$ .....(i)

$\text{combined area } \mathrm{A}=\mathrm{x}^2+\frac{\sqrt{3}}{4}\mathrm{z}^2$

$=\frac{1}{16}(22-3\mathrm{z})^2+\frac{\sqrt{3}}{4}\mathrm{z}^2$

$\frac{\mathrm{dA}}{\mathrm{dz}}=0 \text{ for max/min area}$

$\Rightarrow -\frac{3}{8}(22-3\mathrm{z})+\frac{\sqrt{3}}{2}\mathrm{z}=0$

$\Rightarrow \quad \frac{\sqrt{3}}{2}\mathrm{z}=\frac{3}{8}(22-3\mathrm{z})$

$\Rightarrow \quad 4\mathrm{z}=22\sqrt{3}-3\sqrt{3}\mathrm{z}$

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$$\Rightarrow \quad z=\frac{22\sqrt{3}}{4+3\sqrt{3}}=\frac{66}{4\sqrt{3}+9}$$

Hence the correct answer is option 2

Q. 15 The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is :

Option 1:

$\mathbf{R}-\left\{-\frac{1}{2},\frac{1}{2}\right\}$

Option 2:

$(-\infty,-1]\cup[1,\infty)\cup\{0\}$

Option 3:

$\left(-\infty,\frac{-1}{2}\right)\cup\left(\frac{1}{2},\infty\right)\cup\{0\}$

Option 4:

$\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$

Correct Answer:

$\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$

Solution:

For $\cos^{-1}$ to be defined

$$-1\leq\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\leqslant 1$$

$$\Rightarrow -\frac{\pi}{2}\leqslant\sin^{-1}\left(\frac{1}{4x^2-1}\right)\leqslant\frac{\pi}{2}$$

$$\Rightarrow \quad -1\leqslant\frac{1}{4x^2-1}\leq 1$$

$$\Rightarrow -1\leq\frac{1}{4x^2-1}\leq 1$$

$$\Rightarrow -1\leq\frac{1}{4x^2-1}\text{ and }\frac{1}{4x^2-1}\leq 1$$

$$\Rightarrow \frac{1}{4x^2-1}+1\geqslant 0\text{ and }\frac{1}{4x^2-1}-1\leq 0$$

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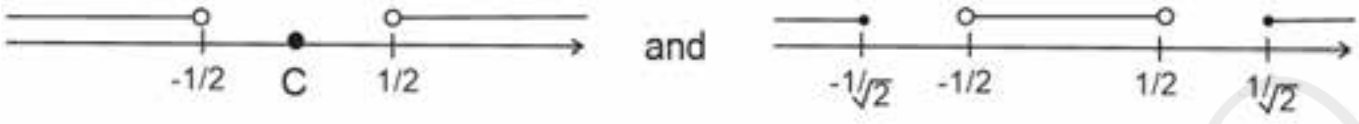
$$\Rightarrow \frac{4x^2}{(2x+1)(2x-1)} \geqslant 0 \text{ and } \frac{2-4x^2}{(2x+1)(2x-1)} \leq 0$$

$$\Rightarrow \frac{4x^2}{(2x+1)(2x-1)} \geqslant 0 \text{ and } \frac{(\sqrt{2}x+1)(\sqrt{2}x-1)}{(2x+1)(2x-1)} \geqslant 0$$



Taking intersection

Domain is $\left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

Hence the correct answer is option 4.

Q. 16 If the constant term in the expansion of $\left(3x^3 - 2x^2 + \frac{5}{x^5}\right)^{10}$ is $2^k \cdot l$, where l is an odd integer, then the value of k is equal to :

Option 1:
6

Option 2:
7

Option 3:
8

Option 4:
9

Correct Answer:
9

Solution:

$$\left(3x^3 - 2x^2 + \frac{5}{x^5}\right)^{10}$$

$$= \frac{1}{x^{50}}\left(3x^8 - 2x^7 + 5\right)^{10}$$

Constant term in this expression

$$= \text{ coefficient of } x^{50} \text{ in } \left(3x^8 - 2x^7 + 5\right)^{10}$$

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General term in $\left(3x^8-2x^7+5\right)^{10}$ is $\frac{10}{a!b!c!}\left(3x^8\right)^a\left(-2x^7\right)^b(5)^c$

$=\frac{10}{a!b!c!}3^a\cdot(-2)^b5^c\cdot(x)^{8a+7b}$

$\therefore \quad 8a+7b=50$ and $a+b+c=10$ and $a,b,c\in W$

There is only one such set of values $a=1,b=6,c=3$

$\therefore$ Constant term is

$\frac{10!}{6!3!}3^1(-2)^6(5)^3$

$=\frac{10\cdot9\cdot8\cdot7}{5}\cdot3^1\cdot2^6\cdot5^3$

$=2^9\cdot5^4\cdot21$

$k=9$

Hence the correct answer is option 4.

Q. 17 $\int_0^5\cos\left(\pi\left(x-\left[\frac{x}{2}\right]\right)\right)dx,$

where [t] denotes greatest integer less than or equal to t, is equal to :

Option 1:
-3

Option 2:
-2

Option 3:
2

Option 4:
0

Correct Answer:
0

Solution:
For $0<x<2,\left[\frac{x}{2}\right]=0$

For $2<x<4,\quad\left[\frac{x}{2}\right]=1$

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For $4 < x < 5, \quad \left[\frac{x}{2}\right] = 2$

$$\therefore \int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right) dx$$

$$= \int_0^2 \cos(\pi x)dx + \int_2^4 \cos(\pi(x-1))dx + \int_4^5 \cos(\pi(x-2))dx$$

$$= \frac{\sin \pi x}{\pi}\Big|_0^2 + \int_2^4 \cos(\pi x - \pi)dx + \int_4^5 \cos(\pi x - 2\pi)dx$$

$$= 0 + \frac{\sin(\pi x - \pi)}{\pi}\Big|_2^4 + \frac{\sin(\pi x - 2\pi)}{\pi}\Big|_4^5$$

$$= 0 + (0-0) + (0-0)$$

$$= 0$$

Hence the correct answer is option 4.

Q. 18 Let PQ be a focal chord of the parabola $y^2 = 4x$ such that it subtends an angle of $\frac{\pi}{2}$ at the point $(3, 0)$. Let the line segment PQ be also a focal chord of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$. If e is the eccentricity of the ellipse E, then the value of $\frac{1}{e^2}$ is equal to :

Option 1:
$1 + \sqrt{2}$

Option 2:
$3 + 2\sqrt{2}$

Option 3:
$1 + 2\sqrt{3}$

Option 4:
$4 + 5\sqrt{3}$

Correct Answer:
$3 + 2\sqrt{2}$

Solution:

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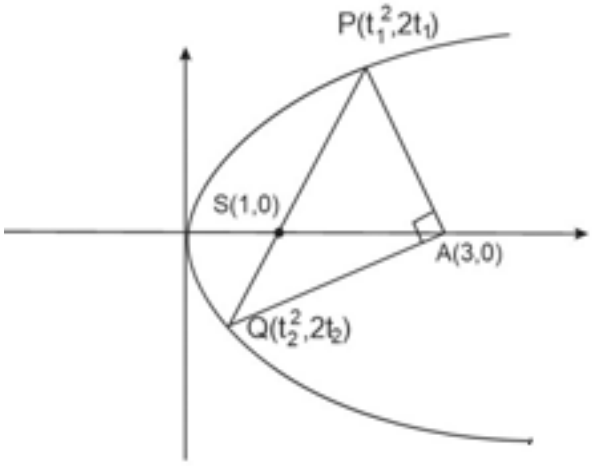

$$\text{Slope of PA} = \frac{2t_1}{t_1^2 - 3}$$

$$\text{Slope of QA} = \frac{2t_2}{t_2^2 - 3}$$

Now these are perpendicular

$$\Rightarrow \quad \frac{2t_1}{t_1^2 - 3} \cdot \frac{2t_2}{t_2^2 - 3} = -1$$

$$\Rightarrow \quad 4t_1t_2 = -1\left(t_1^2 - 3\right)\left(t_2^2 - 3\right)$$

$$\text{Using } t_1t_2 = -1 \text{ for focal chord}$$

$$\Rightarrow -4 = -1\left(t_1^2 - 3\right)\left(\frac{1}{t_1^2} - 3\right)$$

$$\Rightarrow \left(t_1^2 - 3\right)\left(\frac{1}{t_1^2} - 3\right) = 4$$

$$\Rightarrow 1 - \frac{3}{t_1^2} - 3t_1^2 + 9 = 4$$

$$\Rightarrow 3t_1^2 + \frac{3}{t_1^2} = 6$$

$$\Rightarrow 3\left(t_1^2 + \frac{1}{t_1^2} - 2\right) = 0$$

$$\Rightarrow \left(t_1 - \frac{1}{t_1}\right)^2 = 0$$

$$\Rightarrow t_1 = \frac{1}{t_1}$$

$$\Rightarrow t_1^2 = 1$$

$$\Rightarrow t_1 = 1 \text{ or } -1$$

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$\therefore \mathrm{P}(1,2), \mathrm{Q}(1,-2)$

$\therefore \mathrm{PQ}$ is also the focal chord and latus rectum of ellipse

$\mathrm{P}(1,2)$ is end point ofn latus rectum

$\therefore \mathrm{ae}=1 \text{ and } \frac{\mathrm{b}^2}{\mathrm{a}}=2$

$\mathrm{a}=\frac{1}{\mathrm{e}} \text{ and } \mathrm{b}^2=2\mathrm{a}$

$\text{Now } \mathrm{e}^2=1-\frac{\mathrm{b}^2}{\mathrm{a}^2}$

$\mathrm{e}^2=1-\frac{2\mathrm{a}}{\mathrm{a}^2}$

$\mathrm{e}^2=1-\frac{2}{\mathrm{a}}$

$\mathrm{e}^2=1-2\mathrm{e}$

$\mathrm{e}^2+2\mathrm{e}-1=0$

$\mathrm{e}=\frac{-2+\sqrt{4+4}}{2}$ (negative rejected)

$=\frac{-2+2\sqrt{2}}{2}$

$=\sqrt{2}-1$

$\therefore \frac{1}{\mathrm{e}^2}=\frac{1}{(\sqrt{2}-1)^2}=\frac{1}{(3-2\sqrt{2})} \cdot \frac{(3+2\sqrt{2})}{(3+2\sqrt{2})}$

$=3+2\sqrt{2}$

Hence the correct answer is option 2.

Q. 19 Let the tangent to the circle $\mathrm{C}_1: x^2+y^2=2$ at the point $\mathrm{M}(-1,1)$ intersect the circle $\mathrm{C}_2:(x-3)^2+(y-2)^2=5$, at two distinct points $\mathrm{A} \text{ and } \mathrm{B}$. If the tangents to C_2 at the points $\mathrm{A} \text{ and } \mathrm{B}$ intersect at N, then the area of the triangle ANB is equal to :

Option 1:

$\frac{1}{2}$

Option 2:

$\frac{2}{3}$

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Option 3:

$\frac{1}{6}$

Option 4:

$\frac{5}{6}$

Correct Answer:

$\frac{1}{6}$

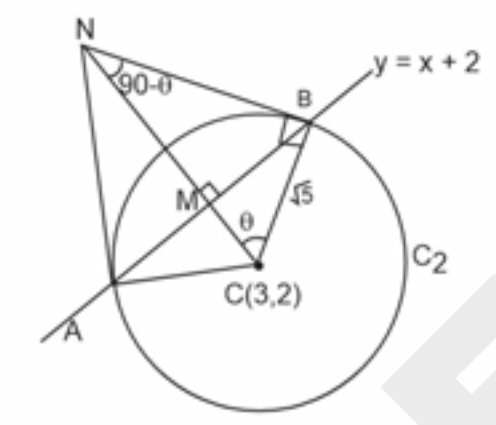
Solution:

Using $T = 0$, equation of tangent at $M(-1, 1)$ to $x^2 + y^2 = 2$ is

$xx_1 + yy_1 = 2$

$-x + y = 2$

$\Rightarrow y = x + 2$



$$CM = \frac{|2 - 3 - 2|}{\sqrt{1+1}} = \frac{3}{\sqrt{2}}$$

$$\text{Now } \cos\theta = \frac{CM}{BC} = \frac{3}{\sqrt{2} \cdot \sqrt{5}} = \frac{3}{\sqrt{10}}$$

$$\text{And} \quad BM = \sqrt{CB^2 - CM^2}$$

$$= \sqrt{5 - \frac{9}{2}}$$

$$= \frac{1}{\sqrt{2}}$$

$$\text{Area of } \triangle ANB = 2 \text{ area of } \triangle NMB$$

$$= 2 \times \frac{1}{2} \times MB \cdot MN$$

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$$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\text{MB}}{\tan(90-\theta)}\right)$$

$$= \frac{1}{2} \cdot \tan\theta$$

$$= \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Hence the correct answer is option 3.

Q. 20 Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively.

If the mean and variance of the first 4 observation are $\frac{7}{2}$ and a respectively, then $(4a + x_5)$ is equal to:

Option 1:

13

Option 2:

15

Option 3:

17

Option 4:

18

Correct Answer:

15

Solution:

$$\text{Mean } = \frac{24}{5}$$

$$\Rightarrow \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{24}{5}$$

$$\Rightarrow \quad x_1 + x_2 + x_3 + x_4 + x_5 = 24 \qquad \text{.....(i)}$$

$$\text{Mean of } x_1, x_2, x_3, x_4 = \frac{7}{2}$$

$$\Rightarrow \quad \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 14 \qquad \text{.....(ii)}$$

$$\Rightarrow x_5 = 10 \qquad \text{[from (i) and (ii)]}$$

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$$\text{Variance } = \frac{194}{25}$$

$$\Rightarrow \frac{x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2}{5} - \frac{576}{25} = \frac{194}{25}$$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 = 154$$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 = 54 \ (\text{as } x_5 = 10)$$

$$\therefore \text{ Variance of } x_1, x_2, x_3, x_4$$

$$\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2}{4} - \frac{49}{4} = a$$

$$\Rightarrow a = \frac{54}{4} - \frac{49}{4}$$

$$\Rightarrow a = \frac{5}{4}$$

$$\therefore 4a + x_5 = 5 + 10 = 15$$

Hence the correct answer is option 2.

Q. 21 Let $S = \{z \in C : |z - 2| \leq 1, z(1 + i) + \bar{z}(1 - i) \leq 2\}$. Let $|z - 4i|$ attains minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. If $5\left(|z_1|^2 + |z_2|^2\right) = \alpha + \beta\sqrt{5}$, where α and β are integers, then the value of $\alpha + \beta$ is equal to_____.

Correct Answer:
26

Solution:
$|z - 2| \leq 1 \Rightarrow$ inside region of circle with centre $(2, 0)$ and radius 1

$$z(1 + i) + \bar{z}(1 - i) \leq 2$$

$$\text{Put } z = x + iy$$

$$(x + iy)(1 + i) + (x - iy)(1 - i) \leq 2$$

$$\Rightarrow x + ix + iy - y + x - ix - iy - y \leq 2$$

$$\Rightarrow x - y \leq 1$$

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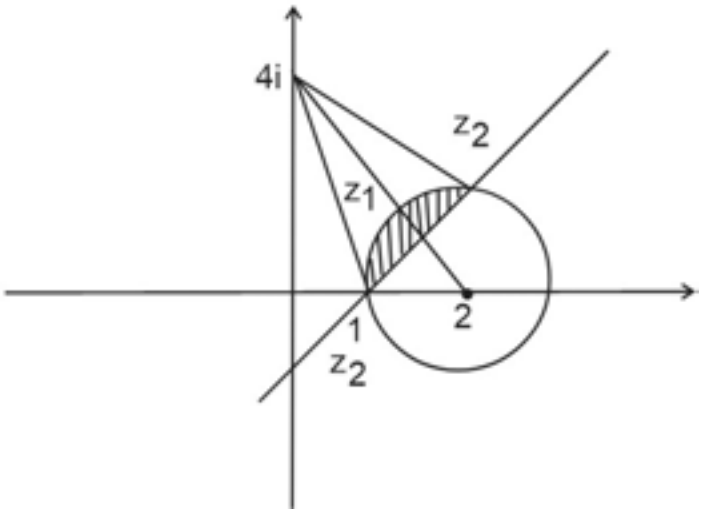

Line joining $(0, 4)$ and $(2, 0)$

$$\frac{x}{2} + \frac{y}{4} = 1 \Rightarrow 2x + y = 4$$

For z_1 solve, $(x-2)^2 + y^2 = 1$ and $2x + y = 4$

$$(x-2)^2 + (4-2x)^2 = 1$$

$$x^2 - 4x + 4 + 16 + 4x^2 - 16x = 1$$

$$5x^2 - 20x + 19 = 0$$

$$x = \frac{20 \pm \sqrt{20}}{10} = \frac{20 \pm 2\sqrt{5}}{10} = 2 \pm \frac{1}{\sqrt{5}}$$

$$x = 2 - \frac{1}{\sqrt{5}}$$

$$y = 4 - 2x$$

$$= 4 - 2\left(2 - \frac{1}{\sqrt{5}}\right)$$

$$= 4 - 4 + \frac{2}{\sqrt{5}} = \frac{2}{\sqrt{5}} \Rightarrow z_1 = \left(2 - \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

$$|z_1|^2 = \left(2 - \frac{1}{\sqrt{5}}\right)^2 + \frac{4}{5} = 4 + \frac{1}{5} - \frac{4}{\sqrt{5}} + \frac{4}{5} = 5 - \frac{4}{\sqrt{5}}$$

$$z_2 = 1 \Rightarrow |z_2|^2 = 1$$

$$\therefore 5\left(|z_1|^2 + |z_2|^2\right) = 5\left(5 - \frac{4}{\sqrt{5}} + 1\right)$$

$$= 5\left(6 - \frac{4}{\sqrt{5}}\right)$$

$$= 30 - 4\sqrt{5}$$

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$\Rightarrow \alpha = 30, \beta = -4$

$\Rightarrow \alpha + \beta = 26$

Hence answer is 26

Q. 22 Let $y = y(x)$ be the solution of the differential equation
$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}, 0 < x < \pi/2$ with $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}$.
If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to_____.

Correct Answer:
2

Solution:
$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}$$

Now to get I.F, first find

$$\int \frac{\sqrt{2}}{2\cos^4 x - \cos 2x}dx$$

$$= \int \frac{\sqrt{2}}{2\cos^4 x - 2\cos^2 x + 1}dx$$

$$= \int \frac{\sqrt{2}}{2\cos^2 x(\cos^2 x - 1) + 1}dx$$

$$= \int \frac{\sqrt{2}}{-2\sin^2 x\cos^2 x + 1}dx$$

$$= \int \frac{2\sqrt{2}}{2 - \sin^2(2x)}dx$$

Dividing by $\cos^2 2x$

$$= \int \frac{2\sqrt{2}\sec^2 2x}{2\sec^2 2x - \tan^2(2x)}dx$$

$$= \int \frac{2\sqrt{2}\sec^2 2x}{2 + \tan^2(2x)}dx$$

Let $\tan 2x = t \Rightarrow 2\sec^2 2xdx = dt$

$$= \int \frac{\sqrt{2}dt}{t^2 + 2} = \tan^{-1}\left(\frac{t}{\sqrt{2}}\right)$$

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$$= \tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)$$

$$\therefore \text{ I.F. } = e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)}$$

Solution of equation is

$$y \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} = \int x e^{\tan^{-1}(\sqrt{2}\cot 2x)} \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} dx$$

$$y \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} = \int x e^{\pi/2} dx$$

$$(\text{As } \sqrt{2}\cot x \cdot \frac{1}{\sqrt{2}}\tan x = 1\ )$$

$$\Rightarrow y = \frac{x^2}{2} e^{\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} + ce^{-\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)}$$

$$\Rightarrow y = \frac{x^2}{2} e^{\frac{\pi}{2} - \cot^{-1}(\sqrt{2}\cot 2x)} + ce^{-cot^{-1}(\sqrt{2}\cot 2x)}$$

$$\Rightarrow y = \frac{x^2}{2} e^{\tan^{-1}(\sqrt{2}\cot 2x)} + ce^{-\cot^{-1}(\sqrt{2}\cot 2x)}$$

$$\text{As } y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} \Rightarrow \frac{\pi^2}{32} = \frac{\pi^2}{32} + c \cdot e^{-\pi/2} \Rightarrow c = 0$$

$$\text{Put } x = \frac{\pi}{3}$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18} e^{\tan^{-1}\left(-\frac{\sqrt{2}}{\sqrt{3}}\right)} \Rightarrow \alpha = \sqrt{\frac{2}{3}} \Rightarrow 3\alpha^2 = 2$$

Hence answer is 2

Q. 23 Let d be the distance between the foot of perpendiculars of the points $P(1, 2, -1)$ and $Q(2, -1, 3)$ on the plane $-x + y + z = 1$. Then d^2 is equal to_____.

Correct Answer:
26

Solution:
$P(1, 2, -1)$

Let $P'(a, b, c)$ be its foot of perpendicular

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$$\frac{a-1}{-1}=\frac{b-2}{1}=\frac{c+1}{1}=\frac{-(-1+2-1-1)}{3}=\frac{1}{3}$$

$$\Rightarrow a=-\frac{1}{3}+1=\frac{2}{3}$$

$$b=2+\frac{1}{3}=\frac{7}{3}$$

$$c=\frac{1}{3}-1=-\frac{2}{3}$$

$$\therefore P'\left(\frac{2}{3},\frac{7}{3},-\frac{2}{3}\right)$$

Let $Q'(x,y,z)$ be foot of perpendicular of Q

$$\frac{x-2}{-1}=\frac{y+1}{1}=\frac{z-3}{1}=\frac{-(-2-1+3-1)}{3}=\frac{1}{3}$$

$$x=2-\frac{1}{3}=\frac{5}{3}$$

$$y=-1+\frac{1}{3}=-\frac{2}{3}$$

$$Z=3+\frac{1}{3}=\frac{10}{3}$$

$$Q'\left(\frac{5}{3},-\frac{2}{3},\frac{10}{3}\right)$$

$$\therefore d^2=P'Q'^2=\left(\frac{5}{3}-\frac{2}{3}\right)^2+\left(-\frac{2}{3}-\frac{7}{3}\right)^2+\left(\frac{10}{3}+\frac{2}{3}\right)^2$$

$$=1+9+16$$

$$=26$$

Q. 24 The number of elements in the set
$S=\left\{\theta\in[-4\pi,4\pi]:3\cos^2 2\theta+6\cos 2\theta-10\cos^2\theta+5=0\right\}$ is_____.

Correct Answer:
32

Solution:
$3\cos^2 2\theta+6\cos 2\theta-10\cos^2\theta+5=0$

$\Rightarrow 3\cos^2 2\theta+6\cos 2\theta-5(1+\cos 2\theta)+5=0$

$\Rightarrow \ 3\cos^2 2\theta+\cos 2\theta=0$

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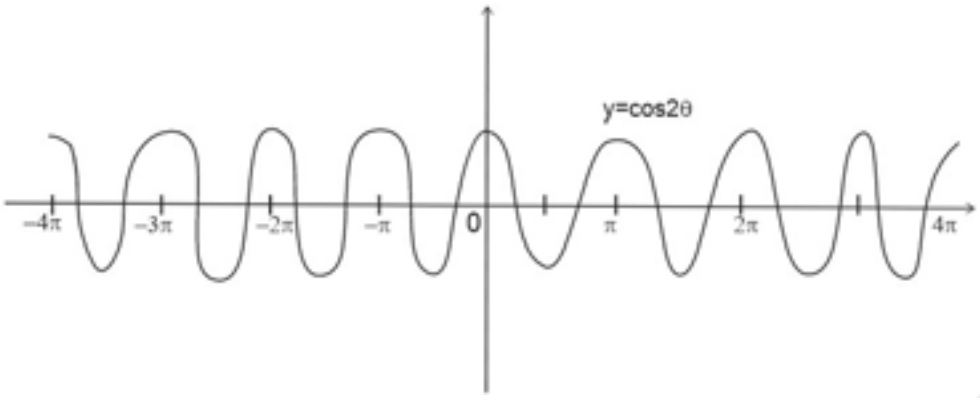
$\Rightarrow \cos 2\theta(3\cos 2\theta + 1) = 0$

$\Rightarrow \cos 2\theta = 0 \text{ or } \cos 2\theta = -\frac{1}{3}$



$\cos 2\theta = 0$ has 16 solutions and $\cos 2\theta = -\frac{1}{3}$has 16 solutions

$\therefore$ Total 32 solutions

Hence the answer is 32

Q. 25 The number of solutions of the equation $2\theta - \cos^2\theta + \sqrt{2} = 0 \text{ in } \mathbf{R}$ is equal to_____.

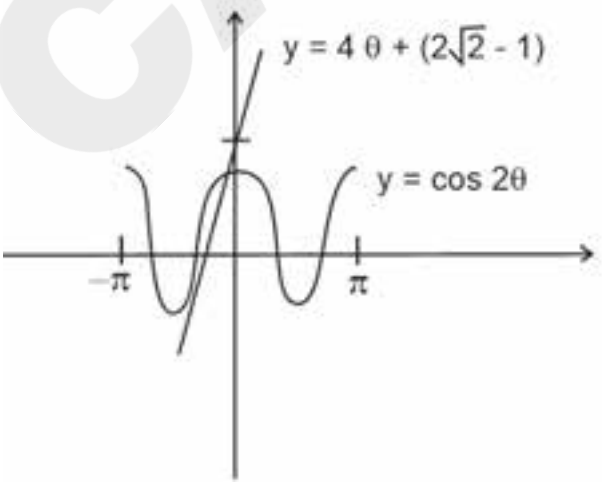
Correct Answer:
1

Solution:
$2\theta - \cos^2\theta + \sqrt{2} = 0$

$\Rightarrow 4\theta + 2\sqrt{2} = 2\cos^2\theta$

$\Rightarrow 4\theta + 2\sqrt{2} = 1 + \cos 2\theta$

$\Rightarrow 4\theta + (2\sqrt{2} - 1) = \cos 2\theta$



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$\therefore$ only one solution

Hence answer is 1

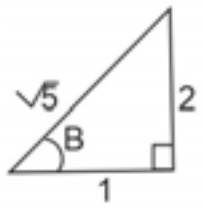
Q. 26 $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right)$ is equal to _____.

Correct Answer:
29

Solution:

$\text{Let } \tan^{-1}\left(\frac{1}{2}\right)=\text{A and } \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)=\text{B}$

Using interconversion



$$\text{B}=\tan^{-1}(2)=\cot^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{2}-\text{A}$$

$$\therefore 50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)$$

$$=50\tan(3\text{A}+2\text{B})$$

$$=50\tan\left(3\text{A}+2\left(\frac{\pi}{2}-\text{A}\right)\right)$$

$$=50\tan(\pi+\text{A})$$

$$=50\tan\text{A}$$

$$=50\tan\left(\tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$=50\cdot\frac{1}{2}$$

$$=25 \qquad \text{.....(i)}$$

$$\text{Now let } \tan^{-1}(2\sqrt{2})=\text{c}\Rightarrow\tan\text{c}=2\sqrt{2}$$

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$$\Rightarrow \quad \frac{2\tan\left(\frac{c}{2}\right)}{1-\tan^2\left(\frac{c}{2}\right)} = 2\sqrt{2}$$

$$\Rightarrow \quad 2\tan\left(\frac{c}{2}\right) = 2\sqrt{2} - 2\sqrt{2}\tan^2\left(\frac{c}{2}\right)$$

$$\Rightarrow \sqrt{2}\tan^2\left(\frac{c}{2}\right) + \tan\left(\frac{c}{2}\right) - \sqrt{2} = 0$$

$$\Rightarrow \tan\left(\frac{c}{2}\right) = \frac{-1+\sqrt{1+8}}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right) = \frac{1}{\sqrt{2}}$$

$$\therefore \quad 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right) = 4$$

$$\therefore \text{ Required expression} = 25 + 4 = 29$$

Hence answer is 29

Q. 27 Let $c, k \in \mathbf{R}$. If $f(x) = (c+1)x^2 + (1-c^2)\,x + 2\mathbf{k}$ and $f(x+y) = f(x) + f(y) - xy$, for all $x, y \in \mathbf{R}$, then the value of $|2(f(1) + f(2) + f(3) + \ldots\ldots\ldots + f(20))|$ is equal to_____.

Correct Answer:
3395

Solution:

$f(x+y) = f(x) + f(y) - xy,$ .....(i)

$\text{Put } y = 0 : f(x) = f(x) + f(0) \Rightarrow f(0) = 0$

$\text{From } f(x) = (c+1)x^2 + \left(1-c^2\right)x + 2k$

$\text{Put } x = 0 : 0 = 0 + 0 + 2k \Rightarrow k = 0$

$\text{Using } f(x+y) = f(x) + f(y) - xy$

$(c+1)(x+y)^2 + \left(1-c^2\right)(x+y)$

$= (c+1)x^2 + \left(1-c^2\right)x + (c+1)y^2 + \left(1-c^2\right)y - xy$

$\Rightarrow 2xy(c+1) = -xy$

$\Rightarrow 2(c+1) = -1$

$\Rightarrow c = -\frac{3}{2}$

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$$\therefore f(x) = -\frac{x^2}{2} - \frac{5}{4}x$$

$$\therefore |\ 2(f(1) + f(2) + \ldots + f(20)\ |$$

$$= |2\left(-\frac{1}{2}(1)^2 - \frac{5}{4}(1) - \frac{1}{2}(2)^2 - \frac{5}{4}(2) \ldots\ldots\ldots - \frac{1}{2}(20)^2 - \frac{5}{4}(20)\right)|$$

$$= \left|2\left(-\frac{1}{2}\sum_{1}^{20} n^2 - \frac{5}{4}\sum_{1}^{20} n\right)\right|$$

$$= \left|2\left(-\frac{1}{2}\frac{20.21 \cdot 41}{6} - \frac{5}{4}\frac{.20.21}{2}\right)\right|$$

$$= \left|-\left(\frac{20 \cdot 21 \cdot 41}{6} + \frac{5}{2} \cdot \frac{20 \cdot 21}{2}\right)\right|$$

$$= |-(2870 + 525)|$$

$$= 3395$$

Hence the answer is 3395

Q. 28 Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a > 0,\ b > 0$, be a hyperbola such that the sum of lengths of the transverse and the conjugate axes is $4(2\sqrt{2} + \sqrt{14})$. If the eccentricity H is $\frac{\sqrt{11}}{2}$, then the value of $a^2 + b^2$ is equal to_____

Correct Answer:
88

Solution:
Given $2a + 2b = 4(2\sqrt{2} + \sqrt{14})$

$a + b = 2(2\sqrt{2} + \sqrt{14})$ .....(i)

And $e^2 = 1 + \frac{b^2}{a^2}$

$\frac{11}{4} = 1 + \frac{b^2}{a^2}$

$\Rightarrow \frac{b^2}{a^2} = \frac{7}{4}$

$\Rightarrow b = \frac{\sqrt{7}}{2}a$

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Using this in (i)

$$\left(\frac{\sqrt{7}+2}{2}\right) \mathrm{a} = 2\sqrt{2}(2+\sqrt{7})$$

$$\mathrm{a} = 4\sqrt{2}$$

$$\Rightarrow \mathrm{a}^2 = 32$$

$$\Rightarrow \mathrm{b}^2 = \frac{7}{4}\mathrm{a}^2 = 56$$

$$\Rightarrow \mathrm{a}^2 + \mathrm{b}^2 = 32 + 56 = 88$$

Hence answer is 88

Q. 29 Let $\mathrm{P}_1 : \vec{\mathrm{r}} \cdot (2\hat{i} + \hat{j} - 3\hat{k}) = 4$ be a plane. Let P_2 be another plane which passes through the points $(2, -3, 2), (2, -2, -3)$ and $(1, -4, 2)$. If the direction ratios of the line of intersection of P_1 and P_2 be $16, \alpha, \beta$, then the value of $\alpha + \beta$ is equal to_____.

Correct Answer:
28

Solution:
Let $\mathrm{A}(2,-3,2)$, $\mathrm{B}(2,-2,-3)$, $\mathrm{C}(1,-4,2)$

$$\vec{n_1} = 2\mathrm{i} + \mathrm{j} - 3\mathrm{k}.$$

$$\vec{n}_2 = \text{ vector perpendicular to } 2^{\text{nd}} \text{ plane}$$

$$= \overrightarrow{AB} \times \overrightarrow{BC}$$

$$= \begin{vmatrix} i & j & k \\ 0 & 1 & -5 \\ -1 & -2 & 5 \end{vmatrix}$$

$$= (5-10)\mathrm{i} - (-5)\mathrm{j} + (1)\mathrm{k}$$

$$= -5\mathrm{i} + 5\mathrm{j} + \mathrm{k}.$$

Directio ratios of line of intersection

$$= \vec{n}_1 \times \vec{n}_2$$

$$= \begin{vmatrix} i & j & k \\ 2 & 1 & -3 \\ -5 & 5 & 1 \end{vmatrix}$$

$$= (16)\mathrm{i} - (-13)\mathrm{j} + (15)\mathrm{k}$$

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$= 16\mathrm{i} + 13\mathrm{j} + 15\mathrm{k}$

$\therefore \quad \alpha = 13, \beta = 15$

$\Rightarrow \alpha + \beta = 28$

Hence answer is 28

Q. 30 Let $\mathrm{b}_1\ \mathrm{b}_2\ \mathrm{b}_3\ \mathrm{b}_4$ be a 4-element permutation with $\mathrm{b}_i \in \{1, 2, 3, \ldots\ldots, 100\}$ for $1 \leq i \leq 4$ and $\mathrm{b}_i \neq \mathrm{b}_j$ for $i \neq j$, such that either $\mathrm{b}_1,\ \mathrm{b}_2,\ \mathrm{b}_3$ are consecutive integers or $\mathrm{b}_2,\ \mathrm{b}_3,\ \mathrm{b}_4$ are consecutive integers. Then the number of such permutations $b_1 b_2 b_3 b_4$ is equal to_____.

Correct Answer:
18915

Solution:
1. If $\mathrm{b}_1, \mathrm{b}_2, \mathrm{b}_3$ are consecutive, then

(i) b_1 can take any value from $1, 2, 3, \ldots, 98$

(ii) b_2 and b_3 are automatically fixed

(iii) b_4 can take any of the remaining 97 values

$\therefore 98 \times 97 = 9506$

2. If $\mathrm{b}_2, \mathrm{b}_3, \mathrm{b}_4$ are consecutive, then

(i) b_4 can take any value from $3, 4, 5, \ldots, 100$

(ii) b_2 and b_3 are automatically fixed

(iii) b_1 can take any of the remaining 97 values

$\therefore 98 \times 97 = 9506$

3. Cases common in 1. and 2. (i.e, when $\mathrm{b}_1, \mathrm{b}_2, \mathrm{b}_3, \mathrm{b}_4$ are consecutive)

(i) b_1 can take any value from $1, 2, 3, \ldots\ldots, 97$

(ii) $\mathrm{b}_2, \mathrm{b}_3, \mathrm{b}_4$ are automatically fixed

$\therefore \ \mathrm{Total} \ = 9506 + 9506 - 97$

$= 18915$

Hence answer is 18915

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29 June Evening Shift

Physics

Q. 1 A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be :

Option 1:
$10\mathrm{m}$

Option 2:
$20\mathrm{m}$

Option 3:
$30\mathrm{m}$

Option 4:
$40\mathrm{m}$

Correct Answer:
$30\mathrm{m}$

Solution:

$$\mathrm{s}=\frac{1}{2}\mathrm{at}^2=10\ \mathrm{m}$$

$$\mathrm{In(2t)time}$$

$$\mathrm{s}=\frac{1}{2}(\mathrm{a})(2\mathrm{t})^2=40\ \mathrm{m}$$

$$\mathrm{distance\ in\ time\ t}\rightarrow 2\mathrm{t}$$

$$=40-10=30\ \mathrm{m}$$

Hence correct option is 3.

Q. 2 At what temperature a gold ring of diameter $6.230\mathrm{cm}$ be heated so that it can be fitted on a wooden bangle of diameter $6.241\mathrm{cm}$? Both the diameters have been measured at room temperature $(27^{\circ}\mathrm{C})$.

$(\mathrm{Given:\ coefficient\ of\ linear\ thermal\ expansion\ of\ gold}\ \alpha_{\mathrm{L}}=1.4\times10^{-5}\ \mathrm{K}^{-1}\ )$

Option 1:
$125.7^{\circ}\mathrm{C}$

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Option 2:
$91.7^{\circ}\mathrm{C}$

Option 3:
$425.7^{\circ}\mathrm{C}$

Option 4:
$152.7^{\circ}\mathrm{C}$

Correct Answer:
$152.7^{\circ}\mathrm{C}$

Solution:

$$\Delta \mathrm{D} = \mathrm{D}\alpha\Delta\mathrm{T}$$

$$\Delta \mathrm{T} = \frac{\Delta \mathrm{D}}{\mathrm{D}\alpha}$$

$$\Delta \mathrm{D} = 6.241 - 6.230$$

$$= 0.011 \text{ cm}$$

$$\Delta \mathrm{T} = \frac{0.011 \times 10^{2}}{6.230 \times 10^{-2} \times 1.4 \times 10^{-5}}$$

$$\Delta \mathrm{T} = 126.11^{\circ}\mathrm{C}.$$

$$\mathrm{T_f} = \mathrm{T} + \Delta \mathrm{T}$$

$$= 27 + 126.11$$

$$= 153.11^{\circ}\mathrm{C}$$

Hence correct answer is option 4

Q. 3 Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is :

Option 1:
$\mathrm{x} = \mathrm{d}$

Option 2:
$\mathrm{x} = \frac{\mathrm{d}}{2}$

Option 3:
$\mathrm{x} = \frac{\mathrm{d}}{\sqrt{2}}$

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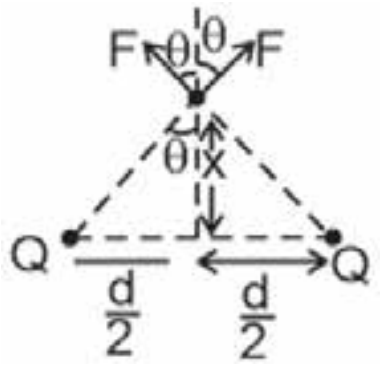
Option 4:

$$x = \frac{d}{2\sqrt{2}}$$

Correct Answer:

$$x = \frac{d}{2\sqrt{2}}$$

Solution:



$$\text{Fnet } = 2\text{F}\cos\theta$$

$$= \frac{2 \times \text{K} \times \text{Qq}}{\left[\left(\frac{\text{d}}{2}\right)^2 + \text{x}^2\right]} \times \frac{\text{x}}{\sqrt{\left(\frac{\text{d}}{2}\right)^2 + \text{x}^2}}$$

$$= \frac{2\text{KQqx}}{\left(\frac{\text{d}^2}{4} + \text{x}^2\right)^{3/2}}$$

For F_{net} to be maximum

$$\frac{\text{d}(\text{ Fnet })}{\text{dx}} = 0$$

On solving we get

$$x = \frac{d}{2\sqrt{2}}$$

Hence the correct option is 4

Q. 4 The speed of light in media $'\text{A}'$ and $'\text{ B }'$ are 2.0×10^{10} cm/s and 1.5×10^{10}cm/s respectively. A ray of light enters from the medium B to A at an incident angle $'\theta\ '$. If the ray suffers total internal reflection, then

Option 1:

$$\theta = \sin^{-1}\left(\frac{3}{4}\right)$$

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Option 2:

$\theta > \sin^{-1}\left(\frac{2}{3}\right)$

Option 3:

$\theta < \sin^{-1}\left(\frac{3}{4}\right)$

Option 4:

$\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Correct Answer:

$\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Solution:

$$n_A = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$$

$$n_B = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$$

For TIR

$\theta > i_c$

$\theta > \sin^{-1}\left(\frac{1.5}{2}\right)$

$\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Hence correct option is 4

Q. 5 In the following nuclear reaction,

$$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$$

Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be _____

Option 1:

174 and 71

Option 2:

174 and 69

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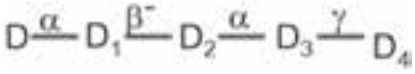
Option 3:
172 and 69

Option 4:
172 and 71

Correct Answer:
174 and 71

Solution:



$\alpha = {}^{4}He_{2}$

$\bar{\mathrm{B}} \rightarrow \text{electron.}$

$\mathrm{D}_1 = {}^{182-4}\,\mathrm{D}_{74-2} = {}^{178}\,\mathrm{D}_{72}$

$\mathrm{D}_2 = {}^{178}\,\mathrm{D}_{73}$

$\mathrm{D}_3 = {}^{174}\,\mathrm{D}_{71}$

$\mathrm{D}_4 = \mathrm{D}_3 = {}^{174}\,\mathrm{D}_{71}$

Hence the correct option is 1.

Q. 6 The electric field at a point associated with a light wave is given by

$$\mathrm{E} = 200\left[\sin\left(6\times10^{15}\right)\mathrm{t} + \sin\left(9\times10^{15}\right)\mathrm{t}\right]\ \mathrm{Vm}^{-1}$$

$$\text{Given}:\ \mathrm{h} = 4.14\times10^{-15}\mathrm{eVs}$$

If this light falls on a metal surface having a work function of $2.50\mathrm{eV}$, the maximum kinetic energy of the photoelectrons will be

Option 1:
1.90 eV

Option 2:
3.27 eV

Option 3:
3.60 eV

Option 4:
3.42 eV

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Correct Answer:

$3.42\ \text{eV}$

Solution:

$\text{E} = 200\left[\sin\left(6 \times 10^{15}\text{t}\right) + \sin\left(9 \times 10^{15}\text{t}\right)\right]$

Freqneny of EM waves are

$= \frac{6}{2\pi} \times 10^{15} \text{ and } \frac{9}{2\pi} \times 10^{15}$

Energy of one photon of these waves

$\text{E}_1 = \text{hf}_1 = 4.14 \times 10^{-15} \times \frac{6}{2\pi} \times 10^{15}$

$= 3.95\text{eV}$

$\text{E}_2 = \text{hf}_2 = 4.14 \times 10^{-15} \times \frac{9}{2\pi} \times 10^{15}$

$= 5.93\text{eV}$

Energy of maximum kinetic

$\text{E} = \text{W}_0 + \text{KE}_{\text{max}}$

$\text{KE}_{\text{max}} = 5.93 - 2.50$

$= 3.43\text{eV}$

Hence the correct option is 4

Q. 7 A capacitor is discharging through a resistor R. Consider in time t_1, the energy stored in the capacitor reduces to half of its initial value and in time t_2, the charge store reduces to one eighth of its initial value. The ratio t_1/t_2 will be

Option 1:

$1/2$

Option 2:

$1/3$

Option 3:

$1/4$

Option 4:

$1/6$

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Correct Answer:
$1/6$

Solution:
For discharging cpapaitor

$$q = q_0 e^{\frac{-t}{Rc}}$$

$$u = \frac{q^2}{2c} = \frac{q_0^2}{2c} e^{-\frac{2t}{Rc}}$$

If energy become half, the charge becomes $\frac{1}{\sqrt{2}}$ times

$$\frac{q_0}{\sqrt{2}} = q_0 e^{-t/\tau}$$

$$e^{-t/\tau} = \frac{1}{\sqrt{2}} \quad \text{.....(1)}$$

$$e^{-t_1/\tau} = \left(\frac{1}{2}\right)^{\frac{1}{2}}$$

similarly at t_2 charge becomes $\frac{1}{8}$ times

$$e^{-t_2/\tau} = \left(\frac{1}{2}\right)^{3}$$

$$\frac{t_1}{t_2} = \frac{1}{6}$$

Hence the correct option is 4

Q. 8 Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2 , if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

Option 1:
$W_1 < W_2 < W_3$

Option 2:
$W_2 < W_3 < W_1$

Option 3:
$W_3 < W_1 < W_2$

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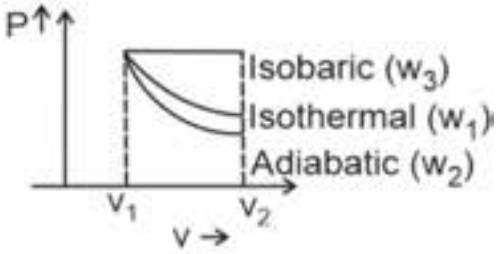
Option 4:

$W_2 < W_1 < W_3$

Correct Answer:

$W_2 < W_1 < W_3$

Solution:



area under P-V graph give work done.

So $W_3 > W_1 > W_2$

Hence the correct option is 4

Q. 9 Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is $300\ \mu T$. The equal current flowing in the two conductors is :

Option 1:

30A in the same direction.

Option 2:

30A in the opposite direction.

Option 3:

60A in the opposite direction.

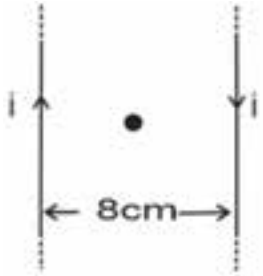
Option 4:

300A in the opposite direction.

Correct Answer:

30A in the opposite direction.

Solution:



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As $\mathrm{Bnet} \neq 0$ so currents are in the opposite direction

$$\frac{\mu_o i}{2\pi \times 4\mathrm{cm}} \times 2 = 30 \times 10^{-6}$$

$$i = \frac{30 \times 10^{-6} \times 2\pi \times 4 \times 10^{-2}}{2 \times 4\pi \times 10^{-7}}$$

$$i = 30\ \mathrm{A}$$

Hence the correct option is 2

Q. 10 The time period of a satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to three times its previous value, then approximate new time period of the satellite will be

Option 1:
$40\ \mathrm{hours}$

Option 2:
$36\ \mathrm{hours}$

Option 3:
$30\ \mathrm{hours}$

Option 4:
$25\ \mathrm{hours}$

Correct Answer:
$36\ \mathrm{hours}$

Solution:
$T^2 \alpha R^3$

$$\frac{T_1^2}{T_2^2} = \left(\frac{R_1}{R_2}\right)^3$$

$$\left(\frac{7}{T_2}\right)^2 = \left(\frac{R}{3R}\right)^3$$

$$\frac{49}{T_2^2} = \frac{1}{27}$$

$$T_2^2 = 49 \times 27$$

$$T_2 = \sqrt{49 \times 27} = 7 \times 3\sqrt{3}$$

$$= 36\ \mathrm{hrs}$$

Hence correct answer is option 2

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Q. 11 The TV transmission tower at a particular station has a height of $125\ \mathrm{m}$. For doubling the coverage of its range, the height of the tower should be increased by

Option 1:
$125\ \mathrm{m}$

Option 2:
$250\ \mathrm{m}$

Option 3:
$375\ \mathrm{m}$

Option 4:
$500\ \mathrm{m}$

Correct Answer:
$375\ \mathrm{m}$

Solution:
$\mathrm{Range(R)} = \sqrt{2\mathrm{hR_e}}$

$\mathrm{h}\alpha\mathrm{R}^2$

$\mathrm{h'} = 4\mathrm{h}$

$= 4 \times 125$

$= 500\ \mathrm{m}$

$\Delta\mathrm{h} = \mathrm{h'} - \mathrm{h} = 500 - 125$

$= 375\ \mathrm{m}$

Hence correct option is 3

Q. 12 The motion of a simple pendulum executing S.H.M. is represented by the following equation.

$\mathrm{y} = \mathrm{A}\sin(\pi\mathrm{t} + \phi)$, where time is measured in second.

The length of pendulum is

Option 1:
$97.23\ \mathrm{cm}$

Option 2:
$25.3\ \mathrm{cm}$

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Option 3:
$99.4\ \text{cm}$

Option 4:
$406.1\ \text{cm}$

Correct Answer:
$99.4\ \text{cm}$

Solution:
$y = A\sin(\pi t + \phi)$

$\omega = \pi$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2$

$T = 2\pi\sqrt{\frac{L}{g}} = 2$

$\sqrt{\frac{L}{g}} = \frac{1}{\pi}$

$\frac{L}{g} = \frac{1}{\pi^2}$

$L = 99.4\ \text{cm}$

Hence the correct option is 3

Q. 13 A vessel contains 16g of hydrogen and 128g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

Option 1:
72×10^5

Option 2:
32×10^5

Option 3:
27×10^4

Option 4:
54×10^4

Correct Answer:
27×10^4

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Solution:

$$\text{Total no. of moles}$$

$$n = n_{H_2} + n_{O_2}.$$

$$= \frac{16}{2} + \frac{128}{32}$$

$$= 12 \text{ moles}$$

$$\text{using } PV = nRT$$

$$V = \frac{nRT}{P} = \frac{12 \times 8.314 \times 273.15 \text{ m}^3}{10^5}$$

$$= 0.27 \text{ m}^3$$

$$= 27 \times 10^4 \text{ cm}^3$$

Hence correct option is 3

Q. 14 Given below are two statements :

Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

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Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Electric force can perform work on charge particles but work done by magnetic force on charge particles is always zero.

Hense statement I is true.

The magnetic force acting on the moving charged particle along the direction of the magnetic field is zero.

So Hense statement II is false.

Hence the correct option is 3

Q. 15 A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below.

The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is.$\left(\text{ Given } g = 10 \text{ ms}^{-2}. \right)$

Option 1:
1 ms^{-2}

Option 2:
$1/5 \text{ ms}^{-2}$

Option 3:
$4/5 \text{ ms}^{-2}$

Option 4:
$8/11 \text{ ms}^{-2}$

Correct Answer:
$8/11 \text{ ms}^{-2}$

Solution:

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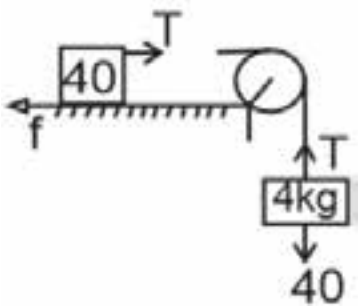


$$f_{lim} = \mu N$$

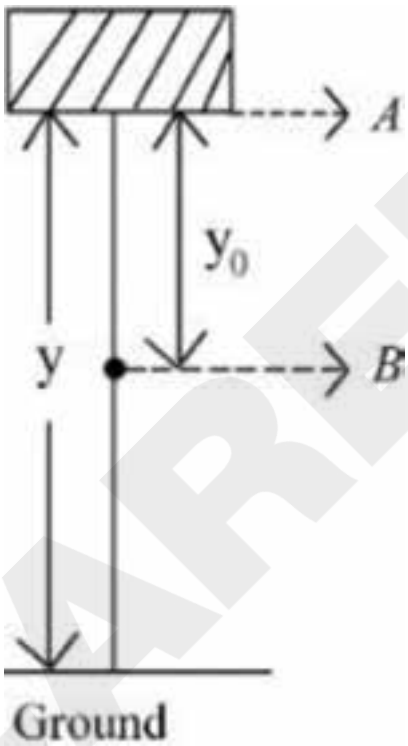
$$= 0.02 \times 400$$

$$= 8 \text{ N}$$

$$a = \frac{40-8}{40+4} = \frac{32}{44} = \frac{8}{11} \text{ m/s}^2$$

Hence the correct option is 4

Q. 16 In the given figure, the block of mass *m* is dropped from the point *'A'*. The expression for kinetic energy of block when it reaches point '*B*' is



Option 1:

$$\frac{1}{2}mg\,y_0^2$$

Option 2:

$$\frac{1}{2}mg\,y^2$$

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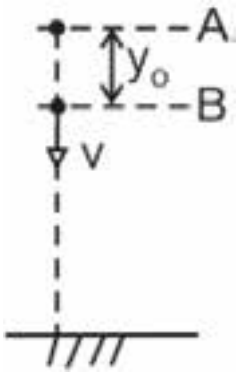
Option 3:
$mg\left(y-y_0\right)$

Option 4:
mgy_0

Correct Answer:
mgy_0

Solution:



$v^2=u^2+2gy_0$

$v^2=2gy_0$

$KE=\frac{1}{2}mv^2=\frac{m}{2}\times 2gy_0$

$=mgy_0$

Hence correct option is 4

Q. 17 A block of mass M placed inside a box descends vertically with acceleration 'a'.

The block exerts a force equal to one-fourth of its weight on the floor of the box.

The value of 'a' will be

Option 1:
$\frac{g}{4}$

Option 2:
$\frac{g}{2}$

Option 3:
$\frac{3g}{4}$

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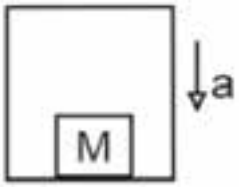
Option 4:

g

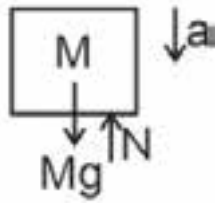
Correct Answer:

$\frac{3g}{4}$

Solution:



FBD wrt ground frame



$Mg - N = Ma$

$N = Mg - Ma - \ (i)$

$\text{Given } N = \frac{Mg}{4} - \ (ii)$

$Ma = Mg - \frac{Mg}{4} = \frac{3Mg}{4}$

$a = \frac{3g}{4}$

Hence correct option is 3

Q. 18 If the electric potential at any point $(x, y, z)m$ in space is given by $V = 3x^2$ volt.

The electric field at the point $(1, 0, 3)m$ will be:

Option 1:

$3\,Vm^{-1}$, directed along positive x-axis.

Option 2:

$3\,Vm^{-1}$, directed along negative x-axis.

Option 3:

$6\,Vm^{-1}$, directed along positive x-axis.

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Option 4:

$6\,\mathrm{Vm}^{-1}$, directed along negative x-axis.

Correct Answer:

$6\,\mathrm{Vm}^{-1}$, directed along negative x-axis.

Solution:

$$\mathrm{V} = 3\mathrm{x}^2$$

$$\mathrm{E} = \frac{-\mathrm{dv}}{\mathrm{dx}} = -6\mathrm{x}$$

$$\vec{E} = -6x\hat{i}$$

at $(1, 0, 3)$

$$\vec{E} = -6\hat{i}$$

Hence correct option is 4

Q. 19 The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω. The value of internal resistance of each cell is

Option 1:

2Ω

Option 2:

4Ω

Option 3:

6Ω

Option 4:

8Ω

Correct Answer:

2Ω

Solution:

Figure 1

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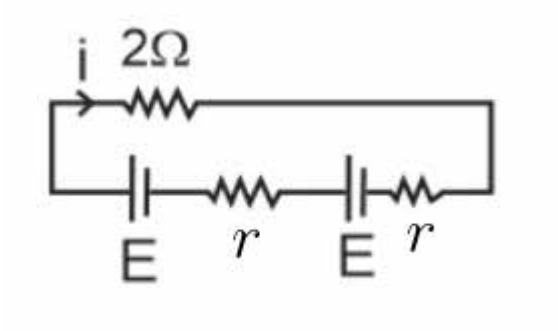
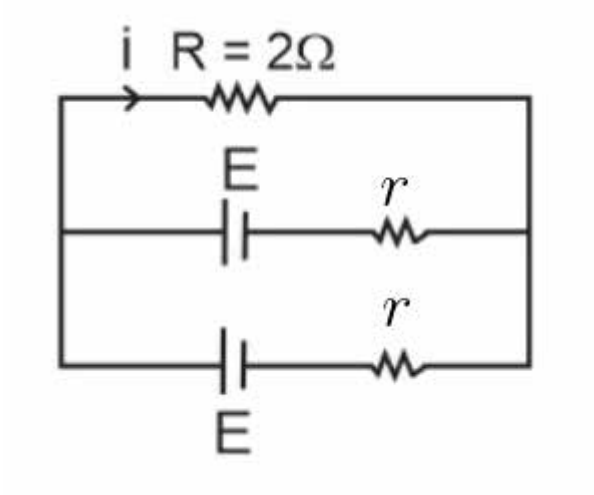


Figure 2



For fig(1)

$$i = \frac{2E}{2 + 2r} = (i)$$

From fig(2)

$$i = \frac{E}{2 + \frac{r}{2}} = (ii)$$

$$(i) = (ii)$$

$$\frac{2E}{2(1 + r)} = \frac{2E}{4 + r}$$

$$4 + r = 2 + 2r$$

$$4 - 2 = r$$

$$r = 2\Omega$$

Hence the correct answer is option 1

Q. 20 A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball?

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Option 1:
$25\ \mathrm{m}$

Option 2:
$50\ \mathrm{m}$

Option 3:
$100\ \mathrm{m}$

Option 4:
$200\ \mathrm{m}$

Correct Answer:
$50\ \mathrm{m}$

Solution:

$$\mathrm{R_{max}} = \frac{\mathrm{u^2}}{\mathrm{g}} = 100$$

$$\mathrm{u^2} = 1000 - (\mathrm{i})$$

$$\mathrm{H_{max}} = \frac{\mathrm{u^2}}{2\mathrm{g}} = \frac{1000}{2 \times 10} = 50\ \mathrm{m}$$

Hence the correct option is 2

Q. 21 The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (-0.05)cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be _____$\times 10^{-2}\ \mathrm{cm}$.

Correct Answer:
180

Solution:
Since zero error is negative we will add $0.05\mathrm{cm}$.

Corrected reading

$$= 1.7\ \mathrm{cm} + 5 \times 0.1\ \mathrm{mm} + 0.05\ \mathrm{cm}$$

$$= 180 \times 10^{-2}\ \mathrm{cm}$$

Hence answer is 180

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Q. 22 A small spherical ball of radius 0.1 mm and density $10^4\ \text{kg m}^{-3}$ falls freely under gravity through a distance *h* before entering a tank of water. If, after entering the water the velocity of ball does not change and it continue to fall with same constant velocity inside water, then the value of *h* will be _____m

$(\text{Given } g = 10\ \text{ms}^{-2}, \text{ viscosity of water } = 1.0 \times 10^{-5}\ \text{N} - \text{sm}^{-2}\ ).$

Correct Answer:
20

Solution:
$\text{terminal speed} = \sqrt{2gh}$

$\frac{2}{9}r^2\frac{(\sigma-\rho)g}{\eta} = \sqrt{2gh}$

$\frac{2}{9} \times \left(1 \times 10^{-4}\right)^2 \times \frac{[10^4 - 10^3] \times 10}{1 \times 10^{-5}} = \sqrt{2 \times 10 \times h}$

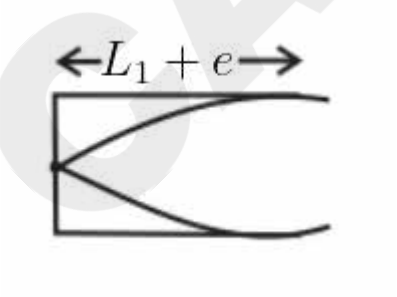
$h = 20\ \text{m}$

Hence answer is 20

Q. 23 In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first reasonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is $336\ \text{ms}^{-1}$.The third resonance is observed when the air column has a length of _____cm

Correct Answer:
104

Solution:



$400 = \frac{V}{4\,(L_1 + e)} - (i)$

$400 = \frac{5V}{4\,(L_2 + e)} - (ii)$

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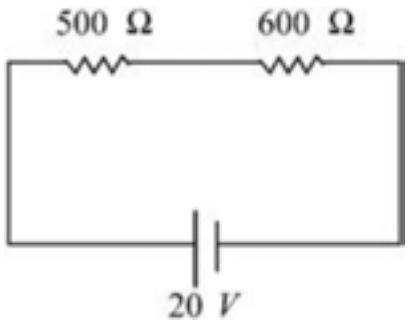
$$L_1 + e = \frac{\lambda}{4} = 21 \text{ cm}$$

$$L_2 + e = \frac{5\lambda}{4} = 105 \text{ cm}$$

$$e = 1 \text{ cm}, \quad L_2 = 104\text{cm}$$

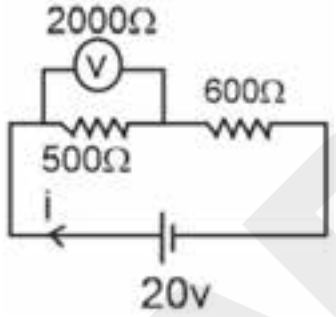
Hence answer is 104

Q. 24 Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance $2000\ \Omega$ is used to measure the potential difference across $500\ \Omega$ resister, the reading of the voltmeter will be _____V



Correct Answer:
8

Solution:



$$\text{Reff} = \frac{500 \times 2000}{500 + 2000} + 600$$

$$= 1000\Omega$$

$$i = \frac{20}{1000} \text{ A}$$

Reading of voltmeter

$$= i \times 400$$

$$= \frac{20 \times 400}{1000} = 8 \text{ volt}$$

Hence correct answer is 8

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Q. 25 A potential barrier of $0.4\ \mathrm{V}$ exists across a p-n junction. An electron enters the junction from the n-side with a speed of $6.0 \times 10^5\ \mathrm{ms}^{-1}$. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5\ \mathrm{ms}^{-1}$ the value of x is _____.

(Given mass of electron $= 9 \times 10^{-31}$ kg, charge on electron $= 1.6 \times 10^{-19}$C.)

Correct Answer:

14

Solution:

$\mathrm{KE_i} = \mathrm{KE_f} +$ Potential bavner energy

$$\frac{1}{2}\mathrm{mv}^2 = \frac{1}{2} \times \mathrm{m} \times \left(6 \times 10^5\right)^2 - 0.4\mathrm{eV}$$

$$\frac{1}{2} \times 9 \times 10^{-31} \times \mathrm{v}^2 = \frac{9 \times 10^{-31}}{2} \times 36 \times 10^{10} - 0.4 \times 1.6 \times 10^{-19}$$

$$\mathrm{V} = \frac{14}{3} \times 10^5\ \mathrm{m/s}$$

$$\mathrm{x} = 14$$

Hence answer is 14

Q. 26 The displacement current of $4.425\mu\mathrm{A}$ is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of $10^6\ \mathrm{Vs}^{-1}$.The area of each plate of the capacitor is $40\ \mathrm{cm}^2$. The distance between eeach plate of the capacitor is $\mathrm{x} \times 10^{-3}\ \mathrm{m}$. The value of x is,

(Permittivity of free space, $\mathrm{E_0} = 8.85 \times 10^{-12}\mathrm{C}^2\ \mathrm{N}^{-1}\ \mathrm{m}^{-2}$)_____

Correct Answer:

8

Solution:

$$\mathrm{I_d} = \varepsilon_o \frac{\mathrm{d}\phi_\mathrm{E}}{\mathrm{dt}}$$

$$\phi = EA$$

$$\phi \times \mathrm{d} = \mathrm{E} \times \mathrm{A} \times \mathrm{d}$$

$$\phi = \frac{\mathrm{E}}{\mathrm{d}} \times \mathrm{A} \times \mathrm{d} = \frac{\mathrm{V} \times \mathrm{A}}{\mathrm{d}}$$

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$$i_d = \varepsilon_0 \times \frac{dv}{dt}\frac{A}{d} = C\frac{dv}{dt}$$

$$i_d = \frac{\varepsilon_0 A}{d} \times \frac{dv}{dt}$$

$$4.425 = \frac{8.85 \times 10^{-12} \times 40 \times 10^{-4} \times 10^6}{d}$$

$$d = 8 \times 10^{-3}\ m$$

$$x = 8$$

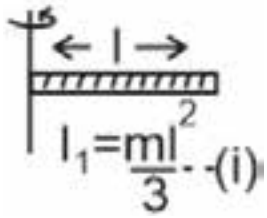
Hence the correct answer is 8

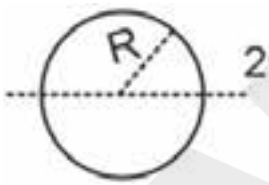
Q. 27 The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2. If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be _____.

Correct Answer:

8

Solution:





$$2\pi R = l$$

$$I = \frac{mR^2}{2} = \frac{m}{2} \times \frac{l^2}{4\pi^2} = \frac{ml^2}{8\pi^2}$$

$$\frac{I_1}{I_2} = \frac{ml^2/3}{ml^2/8\pi^2} = \frac{8\pi^2}{3}$$

$$x = 8$$

Hence answer is 8

Q. 28 The half life of a radioactive substance is 5 years. After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value, The value of x is _____

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Correct Answer:
20

Solution:
$N = N_0 e^{-\lambda t}$

$\frac{6.25}{100} = e^{-\lambda t}$

$e^{-\lambda t} = \frac{1}{16} = \left(\frac{1}{2}\right)^4$

$t = 4t_{\frac{1}{2}} = 20 \text{ yrs.}$

Hence answer is 20

Q. 29 In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength is the light will be _____nm.

Correct Answer:
600

Solution:
$\beta = \frac{D\lambda}{d}$

$\Delta\beta = \frac{\lambda}{d}\Delta D$

$3 \times 10^{-3} \text{cm} = \frac{\lambda}{1\text{mm}} \left(5 \times 10^{-2} \text{ m}\right)$

$\lambda = \frac{3 \times 10^{-8}}{5 \times 10^{-2}} = 600 \text{ nm}$

Hence answer is 600

Q. 30 An inductor of 0.5mH, a capacitor of $200\ \mu\text{F}$ and a resistor of $2\ \Omega$ are connected in series with a $220\ \text{V}$ ac source. If the current is in phase with the emf, the frequency of ac source will be _____$\times 10^2$ Hz

Correct Answer:
5

Solution:

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Current will be in phase with emf when

$$x_L = x_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.5 \times 10^3 \times 200 \times 10^{-6}}}$$

$$\omega = \frac{10^4}{\sqrt{10}} \text{rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{10^4}{2\pi \times \sqrt{10}}$$

$$f = 500 \text{ Hz}$$

Hence answer is 5

Chemistry

Q. 1 Using the rules for significant figures, the correct answer for the expression $\frac{0.02858 \times 0.112}{0.5702}$ will be

Option 1:
0.005613

Option 2:
0.00561

Option 3:
0.0056

Option 4:
0.006

Correct Answer:
0.00561

Solution:
The correct answer is rounded off to same number of significant figures as present in the number with least significant figure

In the given case, number with least significant figure is 0.112 (3 significant figures).

Thus, the result will also contain only 3 significant figures.

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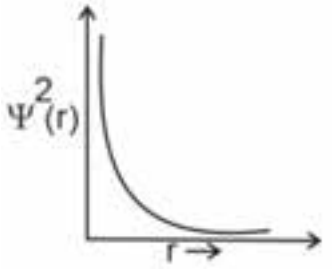
$$\frac{0.02858 \times 0.112}{0.5702}$$

$$= 0.00561(3\ \text{S.F})$$

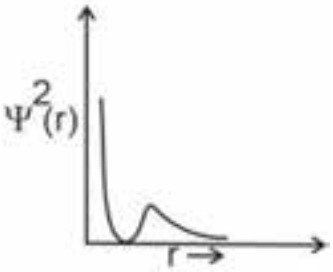
Hence correct option is 2

Q. 2 Which of the following is the correct plot for the probability density $\Psi^2(\mathrm{r})$ as a function of distance 'r' of the electron from the nucleus for 2s orbital?

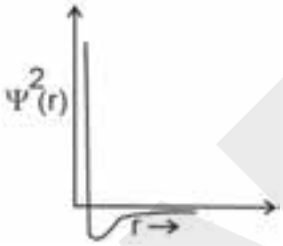
Option 1:



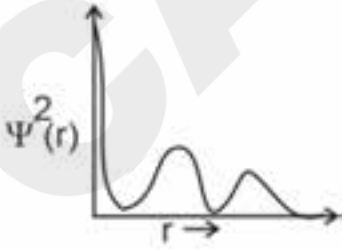
Option 2:



Option 3:



Option 4:



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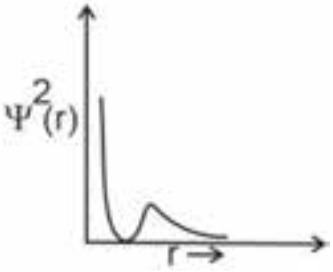
Correct Answer:



Solution:
For 2s Orbital,

$$\mathrm{n} = 2,\ \ell = 0$$

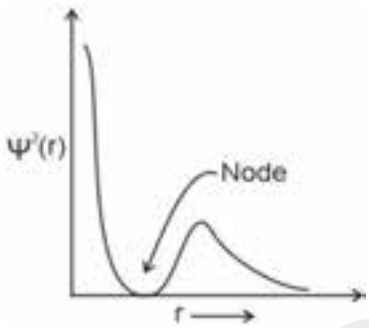
Number of radial nodes

$$= \mathrm{n} - \ell - 1$$

$$= 2 - 0 - 1 = 1$$

And value of ψ^2 is always positive.

So the plot of ψ^2 vs r is



Hence the correct option is 2.

Q. 3 Consider the species $\mathrm{CH_4}, \mathrm{NH_4^+}$ and $\mathrm{BH_4^-}$. Choose the correct option with respect to the three species.

Option 1:
They are isoelectronic and only two have tetrahedral structures.

Option 2:
They are isoelectronic and all have tetrahedral structure.

Option 3:
Only two are isoelectronic and all have tetrahedral structures.

Option 4:
Only two are isoelectronic and only two have tetrahedral structures.

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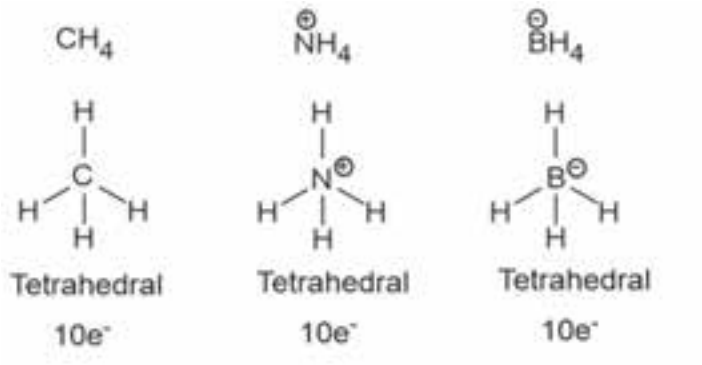
Correct Answer:
They are isoelectronic and all have tetrahedral structure.

Solution:



So, They are isoelectronic and all have tetrahedral structures.

Hence correct answer is option 2.

Q. 4 4.0 moles of argon and 5.0 moles of $\mathrm{PCl_5}$ are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The $\mathrm{K_p}$ for the reaction is

$$[\text{ Given: } \mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}]$$

Option 1:
2.25

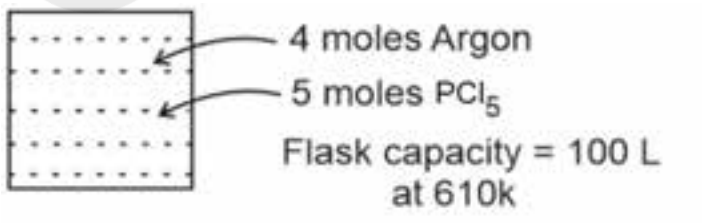
Option 2:
6.24

Option 3:
12.13

Option 4:
15.24

Correct Answer:
2.25

Solution:



Total pressure inside the flask

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$$P_{Total} = \frac{9 \times 0.082 \times 610}{100} = 4.5 \text{ atm}$$

$$P_{PCl_5} = \frac{5}{9} \times 4.5 = 2.5 \text{ atm}$$

$$P_{Ar} = \frac{4}{9} \times 4.5 = 2 \text{ atm}$$

Now,

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2$$

$t = 0$	2.5	0	0
$t = t_{eq}$	$2.5 - p$	p	p

$$P_{Total} \text{ at equilibrim } = 2.5 - p + p + p + p_{Ar} = 6$$

$$p = 1.5$$

$$k_p = \frac{1.5 \times 1.5}{2.5 - 1.5} = 2.25$$

Hence the correct answer is option 1

Q. 5 A $42.12\%(w/v)$ solution of $NaCl$ causes precipitation of a certain sol in 10 hours. The coagulating value of $NaCl$ for the sol is

$[\text{Given: Molar mass : } Na = 23.0 \text{ g mol}^{-1}; Cl = 35.5 \text{ g mol}^{-1}]$

Option 1:

36 mmol L^{-1}

Option 2:

36 mol L^{-1}

Option 3:

1440 mol L^{-1}

Option 4:

1440 mmolL^{-1}

Correct Answer:

1440 mmolL^{-1}

Solution:

$42.12\%(w/v)NaCl$ solution means

$42.12 \text{ g NaCl is present in 100 mL solution}$

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$$\therefore \text{ Molarity of NaCl solution } = \frac{42.12}{58.5 \times 100} \times 1000$$

$$= 7.2\text{M}$$

$7.2 \text{ mol NaCl present in } 1 \text{litre solution}$

$7200 \text{ mmol NaCl present in } 1 \text{ litre solution}$

Now, 7200 mmol/L NaCl causes precipitation of a certain solution in 10 hr

NaCl electrolyte in milli moles per litre required to cause precipitation of a sol in two hours
$= \frac{7200}{10} \times 2 = 1440 \text{ mmol L}^{-1}$

Hence correct answer is option 4

Q. 6 Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R: The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the ***correct*** answer from the options given below.

Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

Option 4:
A is not correct but **R** is correct.

Correct Answer:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Solution:
As we have learnt,

$\text{Ionization energy: O} < \text{N}$

This can be justified by

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(i) Oxygen has a p^4 configuration and two out of the four '2p' electrons must occupy same orbital. So they experience more electron-electron repulsion.

(ii) This is because of the extra stability of the half filled p^3 configuration of Nitrogen

Hence, the correct answer is option (1)

Q. 7 Match **List I** with **List II.**

List I Ore	List II Composition
A. Siderite	I. $FeCO_3$
B. Malachite	II. $CuCO_3 \cdot Cu(OH)_2$
C. Sphalerite	III. ZnS
D. Calamine	IV. $ZnCO_3$

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-I, B-II, C-IV, D-III

Correct Answer:
A-I, B-II, C-III, D-IV

Solution:
The correct matching of the ore and composition are given below

$$\begin{aligned} &\text{ore} && \text{composition} \\ &(\text{A})\text{Siderite} && \rightarrow \quad (\text{I})FeCO_3 \\ &(\text{B})\text{Malachite} && \rightarrow \quad (\text{II})CuCO_3 \cdot Cu(OH)_2 \\ &(\text{C})\text{Sphalerite} && \rightarrow \quad (\text{III})ZnS \end{aligned}$$

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$(D)\text{Calamine} \rightarrow (IV)ZnCO_3$

Hence, the correct answer is option (1)

Q. 8 Given below are two statements.

Statement I: In $CuSO_4 \cdot 5H_2O$, Cu-O bonds are present.

Statement II: In $CuSO_4 \cdot 5H_2O$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the ***correct*** answer from the options given below

Option 1:
Both Statement I and Statement II are correct.

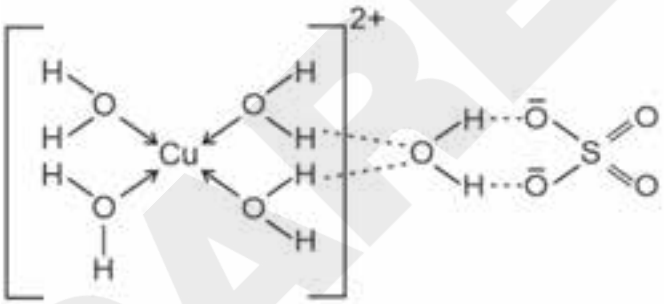
Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:



Statement I is correct

As evident from the structure, $CuSO_4 \cdot 5H_2O$ contains $Cu - O$ bonds.

Statement II is incorrect

In $CuSO_4 \cdot 5H_2O$, ligand coordinating with $Cu(II)$ ion is $'O'$ based ligand and not $'S'$ based ligand.

Statement I is correct but Statement II is incorrect.

Hence the correct answer is an option (3).

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Q. 9 Amongst baking soda, caustic soda and washing soda, carbonate anion is present in

Option 1:
washing soda only.

Option 2:
washing soda and caustic soda only.

Option 3:
washing soda and baking soda only.

Option 4:
baking soda, caustic soda washing soda.

Correct Answer:
washing soda only.

Solution:

$\text{Caustic Soda} \rightarrow \text{NaOH}$

$\text{Baking soda} \rightarrow \text{NaHCO}_3$

$\text{Washing Soda} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

Carbonate ion present in washing soda only

Hence correct answer is option (1)

Q. 10 Number of lone pair(s) of electrons on central atom and the shape of BrF_3 molecule respectively, are

Option 1:
$0,$ triangular planar.

Option 2:
$1,$ pyramidal.

Option 3:
$2,$ bent T-shape.

Option 4:
$1,$ bent T-shape.

Correct Answer:
$2,$ bent T-shape.

Solution:

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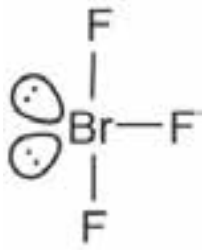

$\text{Hybridisation } = \text{sp}^3\,\text{d}$

$\text{Shape} = \text{Bent T} - \text{Shape}$

$\text{Number of lone pairs of electrons on central atom} = 2$

Hence the correct answer is option (3)

Q. 11 Aqueous solution of which of the following boron compounds will be strongly basic in nature?

Option 1:
NaBH_4

Option 2:
LiBH_4

Option 3:
B_2H_6

Option 4:
$\text{Na}_2\ \text{B}_4\text{O}_7$

Correct Answer:
$\text{Na}_2\ \text{B}_4\text{O}_7$

Solution:
Aqueous solution of Borax is basic due to the formation of NaOH upon hydrolysis

$$\text{Na}_2\ \text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{BO}_3 + \underset{\text{(Strong base)}}{2\ \text{NaOH}}$$

Hence the correct answer is option (4)

Q. 12 Sulphur dioxide is one of the components of polluted air. SO_2 is also a major contributor to acid rain. The correct and complete reaction to represent acid rain caused by SO_2 is

Option 1:
$2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$

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Option 2:

$SO_2 + O_3 \rightarrow SO_3 + O_2$

Option 3:

$SO_2 + H_2O_2 \rightarrow H_2SO_4$

Option 4:

$2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

Correct Answer:

$2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

Solution:

Complete reaction to represent acid rain caused by SO_2

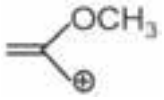
$$2SO_2 + O_2 + 2H_2O \longrightarrow 2H_2SO_4$$

$$(\text{acid rain})$$

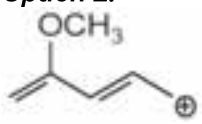
Hence the correct answer is option 4

Q. 13 Which of the following carbocations is most stable?

Option 1:



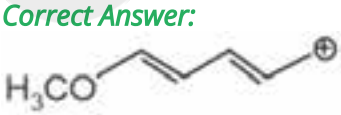
Option 2:



Option 3:

Option 4:

Correct Answer:



Solution:

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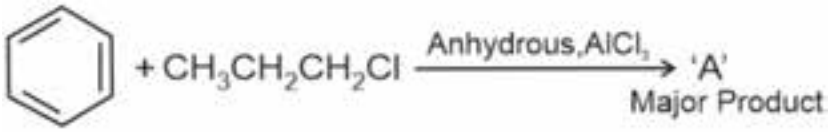
Most stable carbocation due to linear conjugation

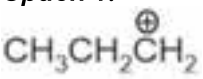
Hence the correct answer is option 4.

Q. 14

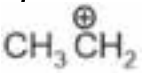


The stable carbocation formed in the above reaction is

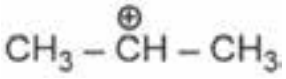
Option 1:



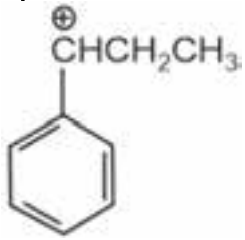
Option 2:



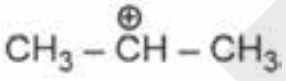
Option 3:



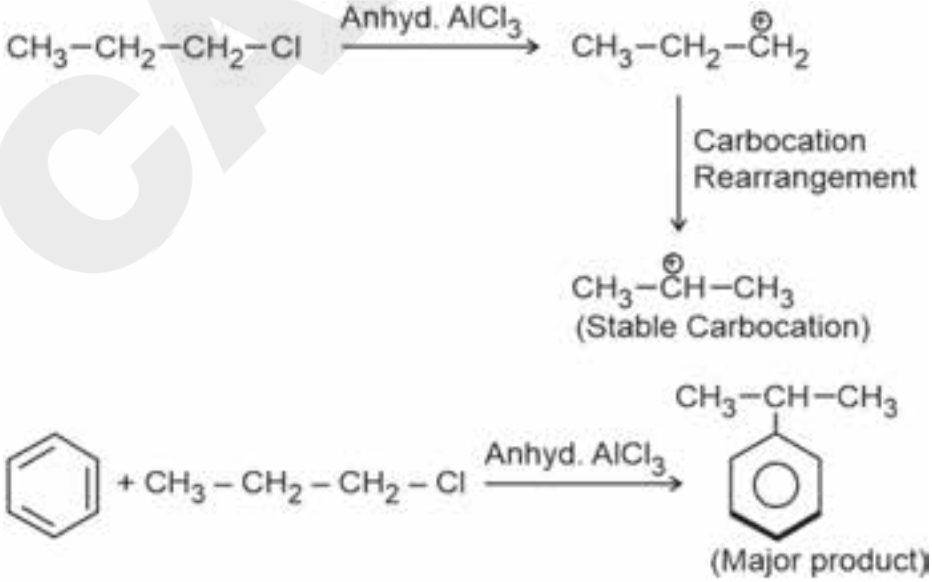
Option 4:



Correct Answer:



Solution:



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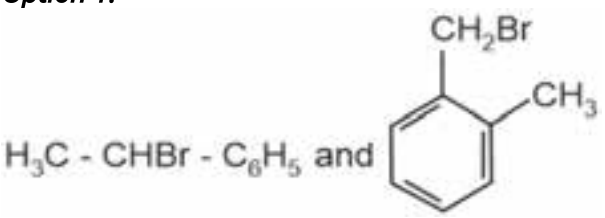
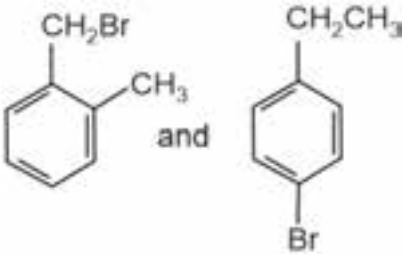
Hence the correct answer is option 3.

Q. 15 Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition $\mathrm{C},\ 52.2\%; \mathrm{H},\ 4.9\%$ and $\mathrm{Br},\ 42.9\%$ gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with $\mathrm{KMnO_4}$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $\mathrm{AgNO_3}$. Isomer 'A' and 'B' are, respectively

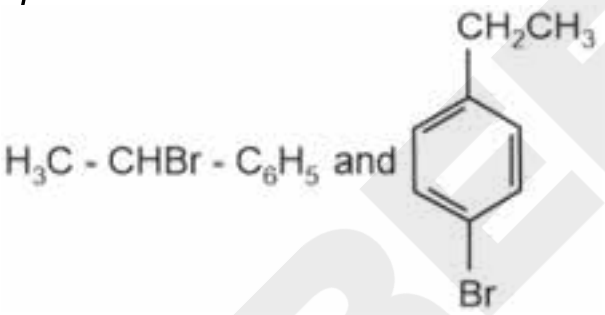
Option 1:



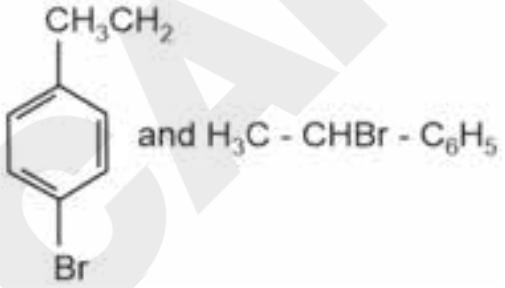
Option 2:



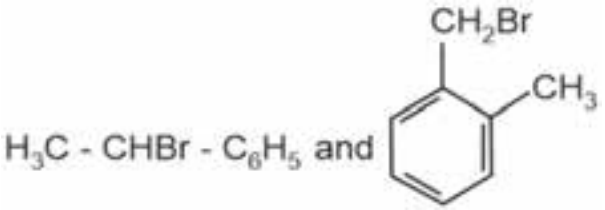
Option 3:



Option 4:



Correct Answer:



Solution:

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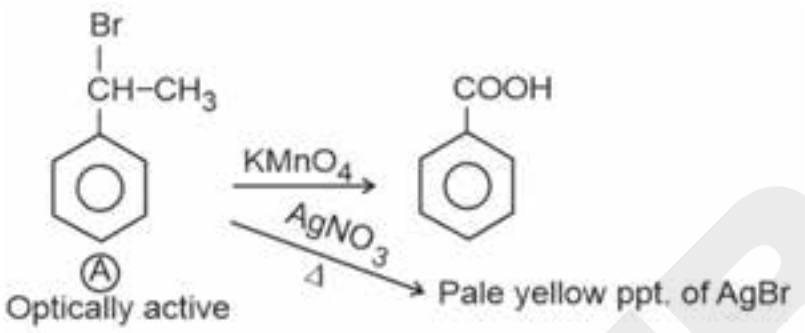
Find molecular formula

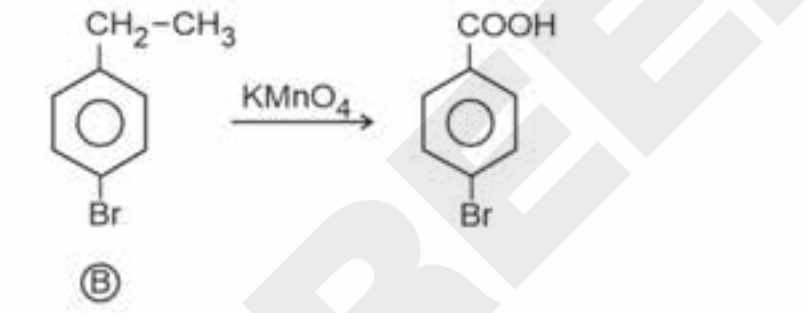
Elements	C	H	Br
Mass(g)	52.2	4.9	42.9
Number of moles of atoms	$\frac{52.2}{12}$	$\frac{4.9}{1}$	$\frac{42.9}{79.9}$
Ratio of moles	4.35	4.9	0.54
Simplest ratio of moles	$\frac{4.35}{0.54}=8$	$\frac{4.9}{0.54}=9$	$\frac{0.54}{0.54}=1$

$\text{Empirical formula } = C_8H_9Br$

$\text{Given, molar mass } = 184 \text{ gmol}^{-1}$

$\text{So, Molecular formula } = C_8H_9Br$





Hence correct answer is option 3

Q. 16 In Friedel-Crafts alkylation of aniline, one gets

Option 1:
alkylated product with ortho and para substitution.

Option 2:
secondary amine after acidic treatment.

Option 3:
an amide product.

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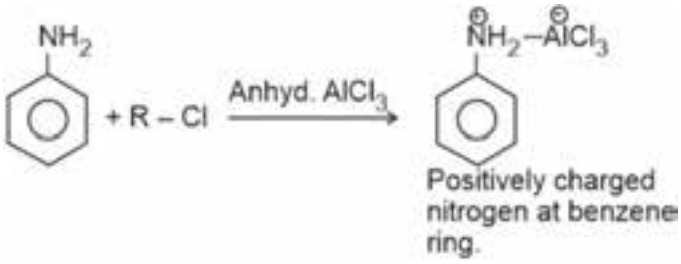
Option 4:
positively charged nitrogen at benzene ring.

Correct Answer:
positively charged nitrogen at benzene ring.

Solution:
Friedel- Crafts alkylation of aniline⟶



Positively charged nitrogen at benzene ring.

Hence the correct answer is option 4

Q. 17 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: Dacron is an example of polyester polymer.

Reason R: Dacron is made up of ethylene glycol and terephthalic acid monomers.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

Option 4:
A is not correct but **R** is correct.

Correct Answer:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Solution:

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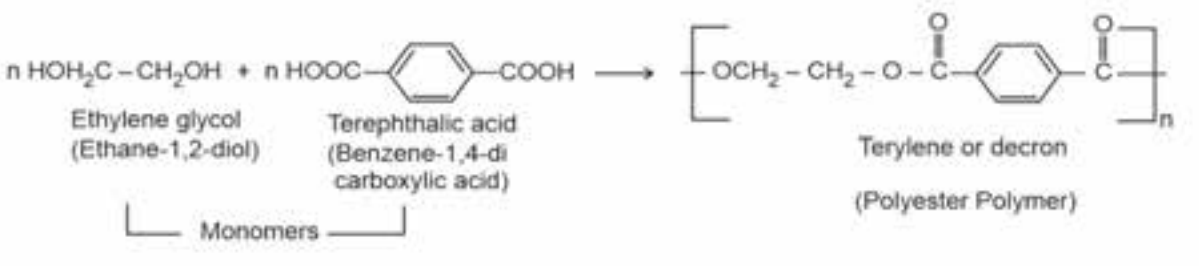

Hence the correct answer is option 1

Q. 18 The structure of protein that is unaffected by heating is

Option 1:
secondary structure

Option 2:
tertiary structure

Option 3:
primary structure

Option 4:
quaternary structure

Correct Answer:
primary structure

Solution:
Primary structure of protein is unaffected by heating.

All other structures (secondary, tertiary and quaternary) are affected due to heat. This is called as denaturation of Protein.

Hence the correct answer is option (3)

Q. 19 The mixture of chloroxylenol and terpineol is an example of

Option 1:
antiseptic

Option 2:
pesticide

Option 3:
disinfectant

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Option 4:
narcotic analgesic

Correct Answer:
antiseptic

Solution:
As we have learnt,

Dettol is a mixture of Chloroxylenol and Terpineol.

Dettol is an antiseptic

Hence the correct answer is option (1)

Q. 20 A white precipitate was formed when $\mathrm{BaCl_2}$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is

Option 1:
$\mathrm{I^-}$

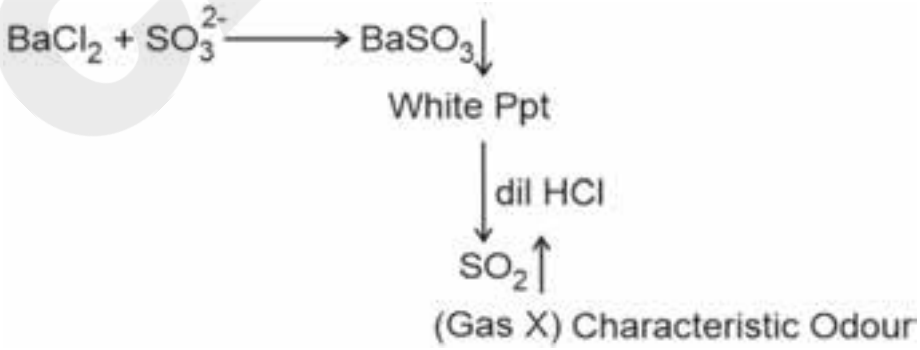
Option 2:
$\mathrm{SO_3^{2-}}$

Option 3:
$\mathrm{S^{2-}}$

Option 4:
$\mathrm{NO_2^-}$

Correct Answer:
$\mathrm{SO_3^{2-}}$

Solution:
The given conditions are satisfied by the Sulphite anion $(\mathrm{SO_3^{2-}})$ as represented below :



Hence, the correct answer is option (2)

Q. 21 A box contains 0.90 g of liquid water in equilibrium with water vapour at $27^{\circ}C$. The equilibrium vapour pressure of water at $27^{\circ}C$ is 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be _____liter. [nearest integer]

$(\text{Given: R} = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1})$

(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)

Correct Answer:
29

Solution:
Given, all of the water vapor evaporates upon increasing the volume.

So, amount of water vapour formed will be 0.9 g and the pressure in the container will be 32 Torr.

Therefore, from ideal gas equation, we have

$$PV = nRT$$

$$V = \frac{nRT}{P} = \frac{0.90 \times 0.082 \times 300 \times 760}{18 \times 32} = 29.21 \text{ L}$$

$\simeq 29\text{L}$ (nearest integer)

Hence answer is 29

Q. 22 2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is $' - x' J$. The value of 'x' is ____.[nearest integer]

$(\text{Given: atomic mass of N} = 14 \text{ g mol}^{-1} \text{ and of O} = 16 \text{ g mol}^{-1}.$

$\text{Molar heat capacity of } N_2O \text{ is } 100 \text{ J K}^{-1} \text{ mol}^{-1})$

Correct Answer:
195

Solution:

$$\text{moles of } N_2O = \frac{2.2}{44} = 0.05$$

$$\Delta H = nC_p\Delta T = 0.05 \times 100(270 - 310)$$

$$= 0.05 \times 100(-40) = -200 \text{ J}$$

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$$W = -P_{ext}\Delta V$$

$$= -1 \times \left(\frac{167.75 - 217.1}{1000}\right) \times 101.3 \text{ J}$$

$$W = +5 \text{ J}$$

$$\Delta U = q_p + W$$

$$\Delta U = -200 + 5 = -195 \text{ J}$$

$$\text{value of 'x' is } 195$$

Hence the answer is 195

Q. 23 Elevation in boiling point for 1.5 molal solution of glucose in water is 4 K. The depression in freezing point for 4.5 molal solution of glucose in water is 4 K. The ratio of molal elevation constant to molal depression constant (K_b/K_f) is _____.

Correct Answer:
3

Solution:
As we have learnt,

$$\Delta T_b = K_b\, m$$

$$\Delta T_f = K_f\, m$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b m}{K_f m} \Rightarrow \frac{4}{4} = \frac{K_b \times 1.5}{K_f \times 4.5}$$

$$\frac{K_b}{K_f} = 3$$

Hence the answer is 3

Q. 24 The cell potential for the given cell at 298 K

$$Pt\,|H_2(g, 1bar)|\,H^+(aq)||Cu^{2+}(aq)\,|\,Cu(s)$$

is 0.31 V. The pH of the acidic solution is found to be 3, whereas the concentration of Cu^{2+} is $10^{-x}M$. The value of x is ___.

$$\left(\text{Given: } E^{\ominus}_{Cu^{2+}/Cu} = 0.34 \text{ V and } \frac{2.303RT}{F} = 0.06 \text{ V}\right)$$

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Correct Answer:
7

Solution:
The reaction occuring in the given cell is

$$\mathrm{H_2(g) + Cu^{2+}\ (aq.)\ \longrightarrow 2H^{+}(aq.)\ + Cu(s)}$$

Applying the Nernst equation, we have

$$0.31 = 0.34 - \frac{0.06}{2}\log\frac{[\mathrm{H^+}]^2}{[\mathrm{Cu^{2+}}]}$$

$$0.31 = 0.34 - \frac{0.06}{2}\log\frac{[10^{-3}]^2}{[\mathrm{Cu^{2+}}]}$$

$$[\mathrm{Cu^{2+}}] = 10^{-7}$$

$$\mathrm{x} = 7$$

Hence the answer is 7

Q. 25 The equation

$$\mathrm{k} = \left(6.5 \times 10^{12}\ \mathrm{s^{-1}}\right) \mathrm{e}^{-26000\ \mathrm{K/T}}$$

is followed for the decomposition of compound A. The activation energy for the reaction is ___$\mathrm{kJmol^{-1}}$.[nearest integer]

$(\text{Given: } \mathrm{R} = 8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}\ )$

Correct Answer:
16

Solution:
$$\mathrm{K} = \mathrm{Ae}^{-\mathrm{Ea/RT}}$$

$$= \left(6.5 \times 10^{12}\mathrm{s^{-1}}\right) \mathrm{e}^{-26000\mathrm{K/T}}$$

$$\frac{\mathrm{E_a}}{\mathrm{R}} = 26000$$

$$\mathrm{E_a} = 26000 \times 8.314\ \mathrm{J/mol} = 216.164\ \mathrm{kJ/mol}$$

$$\simeq 216\ \mathrm{kJ/mol}$$ (nearest integer)

Hence the answer is 216

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Q. 26 Spin only magnetic moment of $[MnBr_6]^{4-}$ is _____ B.M.(round off to the closest integer)

Correct Answer:
6

Solution:
$[MnBr_6]^{4-} \Rightarrow Mn^{2+}$ with weak field ligard

$Mn^{2+} \rightarrow t_{2g}^{1,1,1} e_g^{1,1}$

$\mu_S = \sqrt{n(n+2)}$ B.M.

$\mu_s = \sqrt{5(5+2)}$ B.M.

$\mu_s = \sqrt{35}$ B.M.

$\mu_s = 5.91$ B.M.

Hence the answer is 6 when rounded off to nearest integer.

Q. 27 For the reaction given below :

$$CoCl_3 \cdot xNH_3 + AgNO_3(aq) \rightarrow$$

If two equivalents of $AgCl$ precipitate out, then the value of x will be _____.

Correct Answer:
5

Solution:
$CoCl_3 \cdot xNH_3 + AgNO_3 \longrightarrow AgCl \downarrow$

Given that 1 equivalent of the complex precipitates 2 equivalents of $AgCl$

Thus, the given complex has two Chloride ions outside of the co-ordination sphere.

So, the complex is

$$CoCl_3 \cdot 5NH_3 = [Co(NH_3)_5Cl]\,Cl_2$$

The reaction with $AgNO_3$ is given below :

$$[Co\,(NH_3)_5\,Cl]\,Cl_2 + AgNO_3 \longrightarrow \underset{(2\text{ eq.})}{AgCl} \downarrow$$

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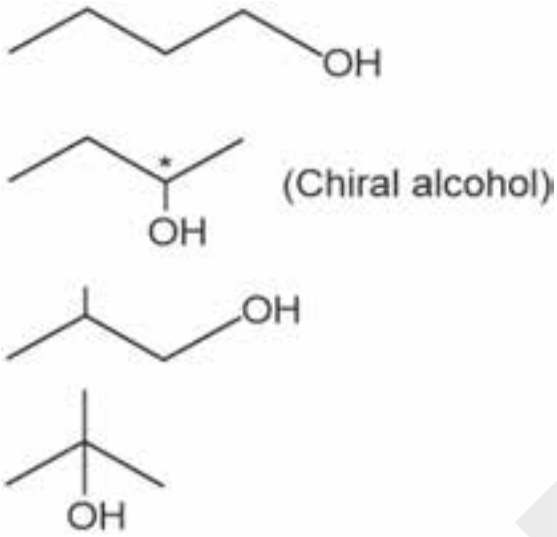
Thus the value of $x = 5$

Hence answer is 5

Q. 28 The number of chiral alcohol (s) with molecular formula $C_4H_{10}O$ is _____.

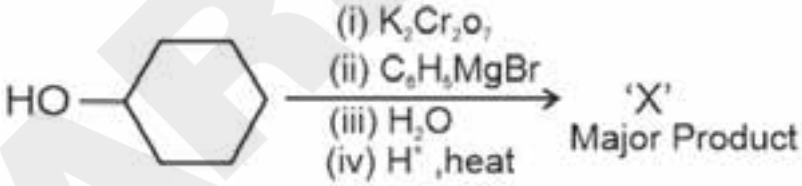
Correct Answer:
1

Solution:
The number of possible alcohol structures with molecular formula $C_4H_{10}O$ are



Hence answer is 1

Q. 29 In the given reaction,



the number of sp^2 hybridised carbon (s) in compound 'X' is _____.

Correct Answer:
8

Solution:

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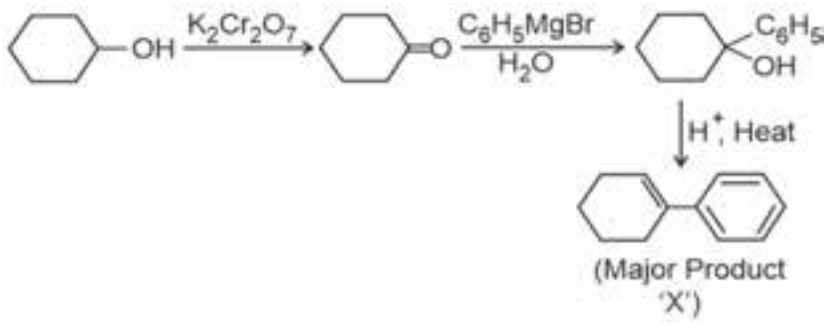
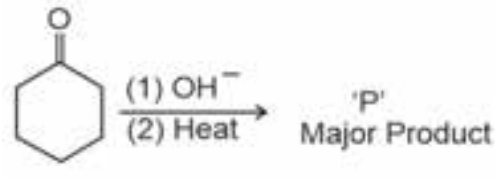


The number of sp^2 hybridized carbon(s) in compound 'x' is 8.

Hence the answer is 8.

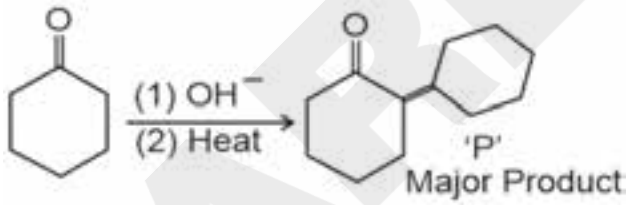
Q. 30 In the given reaction,



The number of π electrons present in the product 'P' is _____.

Correct Answer:
4

Solution:
Aldol condensation reaction



Number of $\pi-\mathrm{electrons}$ present in the product 'P' is 4

Hence, Answer is 4.

Maths

Q. 1 Let α be a root of equation $1+\mathrm{x}^2+\mathrm{x}^4=0$. Then the value of $\alpha^{1011}+\alpha^{2022}-\alpha^{3033}$ is equal to:

Option 1:
1

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Option 2:
α

Option 3:
$1+\alpha$

Option 4:
$1+2\alpha$

Correct Answer:
1

Solution:
$1+\mathrm{x}^2+\mathrm{x}^4=0$

$\text{Let } \mathrm{x}^2=\mathrm{t}$

$\Rightarrow \mathrm{t}^2+\mathrm{t}+1=0$

$\Rightarrow \mathrm{t}=\omega \text{ or } \omega^2$

$\Rightarrow \mathrm{x}^2=\omega \text{ or } \omega^2$

$\Rightarrow \alpha^2=\omega \text{ or } \omega^2$

$\therefore \alpha^{1011}+\alpha^{2022}-\alpha^{3033}$

$=\alpha\left(\alpha^2\right)^{505}+\left(\alpha^2\right)^{1011}-\alpha\left(\alpha^2\right)^{1516} \cdots (\mathrm{i})$

$=\alpha \cdot \omega^{505}+\omega^{1011}-\alpha \cdot \omega^{1516}$

$=\alpha\omega+1-\alpha\omega$

$=1$

We get the same answer by putting $\alpha^2=\omega^2$ in (i)

Hence answer is option 1.

Q. 2 Let $\arg(\mathrm{z})$ represents the principal argument of the complex number z. Then $|\mathrm{z}|=3$ and $\arg(\mathrm{z}-1)-\arg(\mathrm{z}+1)=\frac{\pi}{4}$ intersects

Option 1:
exactly at one point

Option 2:
exactly at two point

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Option 3:
nowhere

Option 4:
at infinitely many points

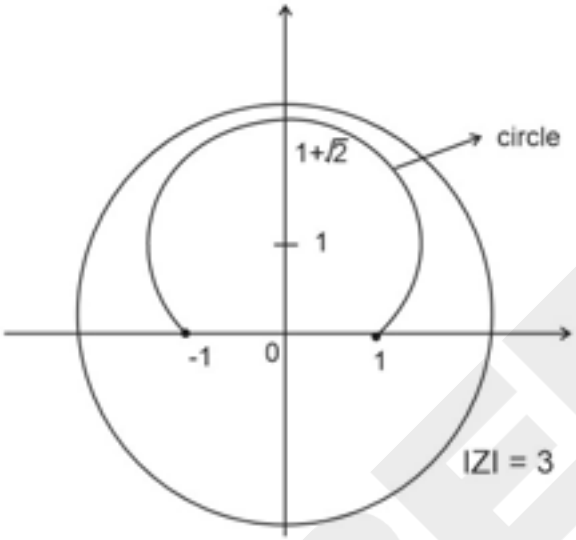
Correct Answer:
nowhere

Solution:

$$\arg(z-1)-\arg(z+1)=\frac{\pi}{4}$$

$$\Rightarrow \arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{4}$$

$\Rightarrow$ Major arc of a circle sutending $\frac{\pi}{4}$ at circumference.



So, they do not intersect.

Hence the correct answer is option 3

Q. 3 Let $A=\begin{pmatrix}2 & -1\\ 0 & 2\end{pmatrix}$. If $B=I-{}^5C_1(\text{adj}\,A)+{}^5C_2(\text{adj}\,A)^2-\ldots-{}^5C_5(\text{adj}\,A)^5$ then the sum of all elements of the matrix B is

Option 1:
-5

Option 2:
-6

Option 3:
-7

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Option 4:

-8

Correct Answer:

-7

Solution:

As I and A are commutative in multiplication $(\mathrm{IA} = \mathrm{AI})$, so they follow all identities.

$$\mathrm{B} = (\mathrm{I} - \operatorname{adj} \mathrm{A})^5$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \right)^5$$

$$= \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}^5 \quad \cdots (\mathrm{i})$$

$$\text{Now let } \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} = \mathrm{P}$$

$$\therefore \mathrm{P}^2 = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\mathrm{P}^4 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$\mathrm{B} = \mathrm{P}^5 = \mathrm{P}^4 \cdot \mathrm{P} = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -5 \\ 0 & -1 \end{bmatrix}$$

Sum of its elements $= -7$

Hence answer is option 3

Q. 4 The sum of the infinite series $1 + \frac{5}{6} + \frac{12}{6^2} + \frac{22}{6^3} + \frac{35}{6^4} + \frac{51}{6^5} + \frac{70}{6^6} + \ldots$ is equal to :

Option 1:

$\frac{425}{216}$

Option 2:

$\frac{429}{216}$

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Option 3:

$\frac{288}{125}$

Option 4:

$\frac{280}{125}$

Correct Answer:

$\frac{288}{125}$

Solution:

$\text{Let } S = 1 + \frac{5}{6} + \frac{12}{6^2} + \frac{22}{6^3} + \frac{35}{6^4} + \cdots$

$\therefore \quad \frac{S}{6} = \frac{1}{6} + \frac{5}{6^2} + \frac{12}{6^3} + \frac{22}{6^4} + \cdots$

Subtracting these

$\frac{5S}{6} = 1 + \frac{4}{6} + \frac{7}{6^2} + \frac{10}{6^3} + \frac{13}{6^4} + \cdots$

$\frac{5S}{6} \cdot \frac{1}{6} = \frac{1}{6} + \frac{4}{6^2} + \frac{7}{6^3} + \frac{10}{6^4} + \cdots$

Subtracting

$\frac{5S}{6}\left(1 - \frac{1}{6}\right) = 1 + \frac{3}{6} + \frac{3}{6^2} + \frac{3}{6^3} + \cdots$

$\Rightarrow \frac{25S}{36} = 1 + \frac{3}{6}\frac{1}{\left(1 - \frac{1}{6}\right)} = 1 + \frac{3}{5}$

$\Rightarrow \frac{25S}{36} = \frac{8}{5}$

$\Rightarrow S = \frac{288}{125}$

hence the answer is 3.

Q. 5 The value of $\lim\limits_{x \to 1} \frac{(x^2 - 1)\sin^2(\pi x)}{x^4 - 2x^3 + 2x - 1}$ is equal to:

Option 1:

$\frac{\pi^2}{6}$

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Option 2:

$\frac{\pi^2}{3}$

Option 3:

$\frac{\pi^2}{2}$

Option 4:

π^2

Correct Answer:

π^2

Solution:

$$\lim_{x\to 1}\frac{(x^2-1)\sin^2(\pi x)}{x^4-2x^3+2x-1}$$

$$=\lim_{x\to 1}\frac{(x^2-1)\sin^2(\pi x)}{(x^4-1)-2x(x^2-1)}$$

$$=\lim_{x\to 1}\frac{(x^2-1)\sin^2(\pi x)}{(x^2-1)(x^2+1-2x)}$$

$$=\lim_{x\to 1}\frac{\sin^2(\pi x)}{(x-1)^2}$$

$$\text{Let } x=1+h$$

$$\lim_{h\to 0}\frac{\sin^2(\pi+\pi h)}{h^2}$$

$$=\lim_{h\to 0}\frac{\sin^2(\pi h)}{h^2}\cdot\frac{\pi^2}{\pi^2}$$

$$=\pi^2$$

Hence answer is option 4

Q. 6 Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=(x-3)^{n_1}(x-5)^{n_2}, n_1, n_2\in N$. Then, which of the folllowing is NOT true?

Option 1:

For $n_1=3, n_2=4$, there exists $\alpha\in(3,5)$ where f attains local maxima.

Option 2:

For $n_1=4, n_2=3$, there exists $\alpha\in(3,5)$ where f attains local minima.

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Option 3:
For $n_1 = 3, n_2 = 5$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

Option 4:
For $n_1 = 4, n_2 = 6$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

Correct Answer:
For $n_1 = 3, n_2 = 5$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

Solution:

$$f(x) = (x-3)^{n_1}(x-5)^{n_2}$$

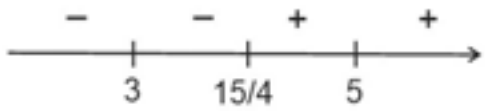
$$f'(x) = n_1(x-3)^{n_1-1}(x-5)^{n_2} + n_2(x-3)^{n_1}(x-5)^{n_2-1}$$

$$= (x-3)^{n_1-1} \cdot (x-5)^{n_2-1} \left[\; n_1(x-5) + n_2(x-3) \; \right]$$

$$= (x-3)^{n_1-1} \cdot (x-5)^{n_2-1} \left[\; (n_1+n_2)x - (5n_1+3n_2) \; \right]$$

For option 3

$$f'(x) = (x-3)^2(x-5)^4(8x-30)$$



x = 15/4 is a local minima

(Using first order derivative test)

Therefore option 3 is wrong.

We can similarly check other options.

Hence answer is option 3

Q. 7 Let f be a real valued continuous function on $[0, 1]$ and $f(x) = x + \int_0^1 (x-t)f(t)dt$.
Then which of the following points (x, y) lies on the curve $y = f(x)$?

Option 1:
$(2, 4)$

Option 2:
$(1, 2)$

Option 3:
$(4, 17)$

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Option 4:

$(6, 8)$

Correct Answer:

$(6, 8)$

Solution:

$$f(x) = x + x\int_0^1 f(t)dt - \int_0^1 tf(t)dt$$

$$= x + Ax - B = (A + 1)x - B$$

$$\text{where } A = \int_0^1 f(t)dt, B = \int_0^1 tf(t)dt$$

$$A = \int_0^1 f(t)dt$$

$$\Rightarrow A = \int_0^1 ((A + 1)t - B)dt$$

$$\Rightarrow A = (A + 1)\frac{t^2}{2} - Bt|_0^1$$

$$\Rightarrow A = \frac{A + 1}{2} - B$$

$$\Rightarrow \frac{A}{2} = \frac{1}{2} - B$$

$$\Rightarrow A = 1 - 2B \quad \text{.....(i)}$$

$$\text{Also } B = \int_0^1 tf(t)dt$$

$$B = \int_0^1 t((A + 1)t - B)dt$$

$$B = \int_0^1 \left((A + 1)t^2 - Bt\right)dt$$

$$B = (A + 1)\frac{t^3}{3} - \frac{Bt^2}{2}\Big|_0^1$$

$$B = \frac{A + 1}{3} - \frac{B}{2}$$

$$6B = 2A + 2 - 3B$$

$$9B = 2A + 2 \quad \text{.....(ii)}$$

From (i) and (ii)

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$$B = \frac{4}{13}, A = \frac{5}{13}$$

$$\therefore \quad f(x) = (A+1)x - B$$

$$= \frac{18}{13}x - \frac{4}{13}$$

Checking option 4 lies on it

Hence answer is option 4

Q. 8 If $\int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2}\right) dx = \int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2}\right) dy + \int_1^2 \left(2 - \frac{y^2}{2}\right) dy + I$ then I equals

Option 1:

$$\int_0^1 \left(1 + \sqrt{1 - y^2}\right) dy$$

Option 2:

$$\int_0^1 \left(\frac{y^2}{2} - \sqrt{1 - y^2} + 1\right) dy$$

Option 3:

$$\int_0^1 \left(1 - \sqrt{1 - y^2}\right) dy$$

Option 4:

$$\int_0^1 \left(\frac{y^2}{2} + \sqrt{1 - y^2} + 1\right) dy$$

Correct Answer:

$$\int_0^1 \left(1 - \sqrt{1 - y^2}\right) dy$$

Solution:

$$\int_0^2 \sqrt{2x} dx$$

$$\text{Let } 2x = t^2 \Rightarrow 2dx = 2tdt$$

$$= \int_0^2 t(tdt)$$

$$= \int_0^2 t^2 dt$$

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$$\text{And } \int_0^2 \sqrt{2x - x^2}dx$$

$$= \int_0^2 \sqrt{-(x^2 - 2x)}dx$$

$$= \int_0^2 \sqrt{1 - (x - 1)^2}dx$$

$$\text{Let } x - 1 = t$$

$$= \int_{-1}^1 \sqrt{1 - t^2}dt$$

$$= 2\int_0^1 \sqrt{1 - t^2}dt \text{ (even function)}$$

$$\therefore I = \int_0^2 (\sqrt{2x})dx - \int_0^2 \sqrt{2x - x^2}dx - \int_0^1 dy$$

$$+ \int_0^1 \sqrt{1 - y^2}dy + \int_0^1 \frac{y^2}{2}dy - \int_1^2 2dy + \int_1^2 \frac{y^2}{2}dy$$

$$= \int_0^2 y^2dy - 2\int_0^1 \sqrt{1 - y^2}dy - \int_0^1 dy + \int_0^1 \sqrt{1 - y^2}dy$$

$$+ \int_0^2 \frac{y^2}{2}dy - 2\int_1^2 dy$$

$$= \int_0^2 \left(y^2 + \frac{y^2}{2}\right)dy - \int_0^1 \sqrt{1 - y^2}dy - \int_0^1 dy - 2\int_1^2 dy$$

$$= \frac{y^3}{2}\Big|_0^2 - \int_0^1 \sqrt{1 - y^2}dy - y|_0^1 - 2y|_1^2$$

$$= 4 - \int_0^1 \sqrt{1 - y^2}dy - 1 - 2$$

$$= 1 - \int_0^1 \sqrt{1 - y^2}dy$$

$$= \int_0^1 \left(1 - \sqrt{1 - y^2}\right)dy.$$

Hence the answer is option 3

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Q. 9 If $y = y(x)$ is the solution of the differential equation $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$ and $y(0)=0$, then $6\left(y'(0)+\left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ is equal to

Option 1:
2

Option 2:
-2

Option 3:
-4

Option 4:
-1

Correct Answer:
-4

Solution:

$$\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$$

$$\left(1+e^{2x}\right)dy=-2\left(1+y^2\right)e^x dx$$

$$\frac{dy}{1+y^2}=-\frac{2e^x dx}{1+e^{2x}}$$

$$\text{Let } e^x=t \Rightarrow e^x dx=dt$$

$$\frac{dy}{1+y^2}=-2\frac{dt}{1+t^2}$$

Integrating

$$\tan^{-1}(y)=-2\tan^{-1}(t)+c$$

$$\tan^{-1}(y)=-2\tan^{-1}\left(e^x\right)+c$$

$$\text{As } y(0)=0$$

$$0=-2\cdot\frac{\pi}{4}+c \Rightarrow c=\frac{\pi}{2}$$

$$\therefore \tan^{-1}y=-2\tan^{-1}\left(e^x\right)+\frac{\pi}{2} \quad \text{.....(1)}$$

From given equation putting $x=0, y=0$

$$\left(1+e^0\right)\cdot y'(0)+2(1+0)\cdot e^0=0$$

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$2 \cdot y'(0) + 2 = 0$

$y'(0) = -1$

$\text{Putting } x = \log(\sqrt{3}) \text{ in (i)}$

$\tan^{-1} y = -2\tan^{-1}(\sqrt{3}) + \frac{\pi}{2}$

$\tan^{-1} y = -\frac{2\pi}{3} + \frac{\pi}{2}$

$\tan^{-1} y = -\frac{\pi}{6}$

$y = -\frac{1}{\sqrt{3}}$

$\therefore 6\left(y'(0) + (y(\log\sqrt{3}))^2\right)$

$= 6\left(-1 + \frac{1}{3}\right)$

$= 6\left(-\frac{2}{3}\right)$

$= -4$

Hence answer is option 3

Q. 10 Let $P : y^2 = 4ax, a > 0$ be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line $y = 3x + 5$ touch the parabola P at A and B. Then the value of a for which A, B and S are collinear is

Option 1:
8 only

Option 2:
2 only

Option 3:
$\frac{1}{4}$ only

Option 4:
any $a > 0$

Correct Answer:
any $a > 0$

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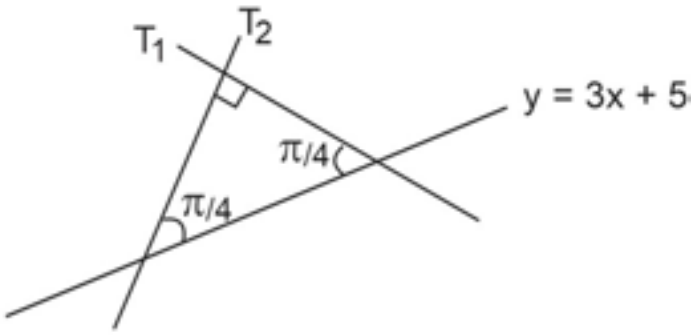
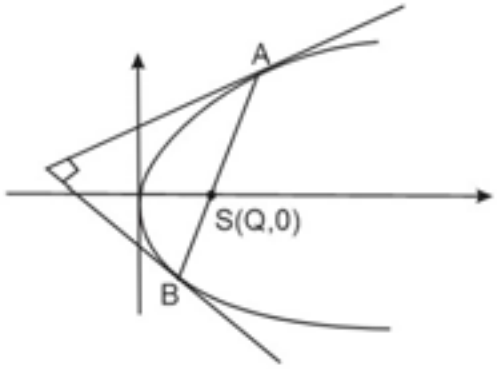
Solution:



Tangents $T_1 \text{ and } T_2$ make angle $\frac{\pi}{4}$ with line $y = 3x + 5$ means that tangents are perpendicular



Using property of parabola, point of contact of perpendicular tangents always form a focal chord. So $A, S \text{ and } B$ will always be collinear for any value of $a > 0$.

Hence the answer is option 4

Q. 11 Let a triangle ABC be inscribed in the circle $x^2 - \sqrt{2}(x + y) + y^2 = 0$ such that $\angle BAC = \frac{\pi}{2}$. If the length of side AB is $\sqrt{2}$, then the area of the ΔABC is equal to:

Option 1:
1

Option 2:
$(\sqrt{6} + \sqrt{3})/2$

Option 3:
$(3 + \sqrt{3})/4$

Option 4:
$(\sqrt{6} + 2\sqrt{3})/4$

Correct Answer:
1

Solution:
$x^2 + y^2 - \sqrt{2}x - \sqrt{2}y = 0$

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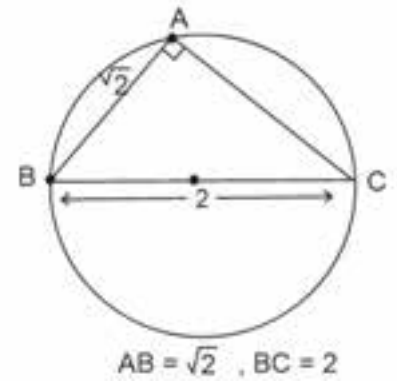
$2g = -\sqrt{2} \Rightarrow g = -\frac{1}{\sqrt{2}}$

$2f = -\sqrt{2} \Rightarrow f = -\frac{1}{\sqrt{2}}$

$\text{Radius } r = \sqrt{\frac{1}{2} + \frac{1}{2} - 0} = 1$

As $\angle ABC = \frac{\pi}{2}$, so BC is diameter,



$\therefore \angle ABC = \frac{\pi}{4} \Rightarrow \angle ACB = \frac{\pi}{4}$

$\Rightarrow AC = \sqrt{2}$

$\text{Area of } \triangle ABC = \frac{1}{2} \cdot AB \cdot AC = \frac{1}{2} \cdot \sqrt{2} \cdot \sqrt{2} = 1$

Hence answer is option 1

Q. 12 Let $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$ for some $p, q \in \mathbb{R}$. The shortest distance of the plane from the origin is:

Option 1:

$\sqrt{\frac{3}{109}}$

Option 2:

$\sqrt{\frac{5}{142}}$

Option 3:

$\frac{5}{\sqrt{71}}$

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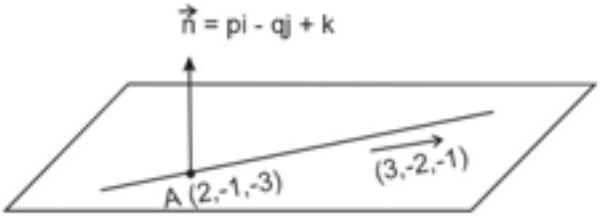
Option 4:

$\frac{1}{\sqrt{142}}$

Correct Answer:

$\sqrt{\frac{5}{142}}$

Solution:



For line to lie on the plane, point $A(2, -1, -3)$ lying on line should lie on plane as well

$\therefore \quad 2p + q - 3 = 5 \Rightarrow 2p + q = 8$ .....(i)

Also vector parallel to line $(3i - 2j - k)$ should be perpendicular to normal vector $\vec{n}$ of plane $(pi - qj + k)$

$\therefore (3i - 2j - k) \cdot (pi - qj + k) = 0$

$\Rightarrow 3p + 2q - 1 = 0$

$\Rightarrow 3p + 2q = 1 \quad .... (ii)$

From (i) and (ii)

$p = 15, q = -22$

$\therefore$ plane is $15p + 22y + z - 5 = 0$

Distance from origin $(0, 0, 0)$

$= \frac{|-5|}{\sqrt{15^2 + 22^2 + 1^2}}$

$= \frac{5}{\sqrt{710}}$

$= \sqrt{\frac{5}{142}}$

hence answer is option 2

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Q. 13 The distance of the origin from the cetroid of the triangle whose two sides have the equations $x-2y+1=0$ and $2x-y-1=0$ and whose orthocentre is $\left(\frac{7}{3},\frac{7}{3}\right)$ is:

Option 1:

$\sqrt{2}$

Option 2:

2

Option 3:

$2\sqrt{2}$

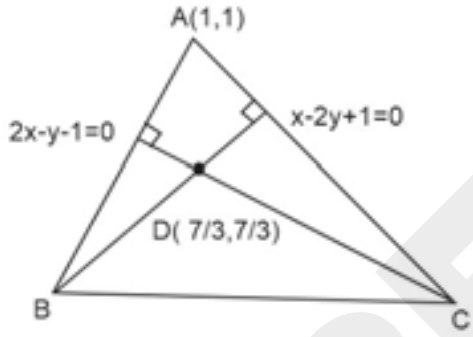
Option 4:

4

Correct Answer:

$2\sqrt{2}$

Solution:



Altitude BD is perpendicular to $AC \quad (x-2y+1=0)$

$\therefore$ Let BD is $2x+y+\lambda=0$

It passes through D

$\therefore \frac{14}{3}+\frac{7}{3}+\lambda=0$

$\Rightarrow \lambda=-7$

$\therefore$ BD is $2x+y-7=0$

Solving it with equation of AB we get $B(2,3)$

Similarly we can get coordinates of C as $(3,2)$

Centroid of ΔABC

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$= \left(\frac{1+2+3}{3}, \frac{1+3+2}{3}\right)$

$= (2, 2)$

Distance of centroid from origin

$= \sqrt{4+4} = 2\sqrt{2}$

Hence answer is option 3

Q. 14 Let Q be the mirror image of the point $\mathrm{P}(1, 2, 1)$ with respect to the plane $\mathrm{x} + 2\mathrm{y} + 2\mathrm{z} = 16$. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda(\hat{i} + \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$. Then, which of the following points lies on T

Option 1:
$(2, 1, 0)$

Option 2:
$(1, 2, 1)$

Option 3:
$(1, 2, 2)$

Option 4:
$(1, 3, 2)$

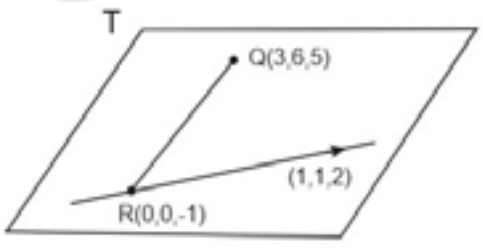
Correct Answer:
$(1, 2, 1)$

Solution:
Let coordinate of Q in $\mathrm{x} + 2\mathrm{y} + 2\mathrm{z} = 16$ be $(\mathrm{x}_1, \mathrm{y}_1, \mathrm{z}_1)$

$$\therefore \frac{\mathrm{x}_1 - 1}{1} = \frac{\mathrm{y}_1 - 2}{2} = \frac{\mathrm{z}_1 - 1}{2} = \frac{-2(1+4+2-16)}{1+2^2+2^2}$$

$$\Rightarrow \frac{\mathrm{x}_1 - 1}{1} = \frac{\mathrm{y}_1 - 2}{2} = \frac{\mathrm{z}_1 - 1}{2} = 2$$

$$\Rightarrow \mathrm{x}_1 = 3, \quad \mathrm{y}_1 = 6, \quad \mathrm{z}_1 = 5$$



Vector $\vec{b}$ parallel to given line is $\mathrm{i} + \mathrm{j} + 2\mathrm{k}$

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$\mathrm{R}(0,0,-1)$ lies on line, so $\overrightarrow{RQ}$ is also parallel to plane

$\overrightarrow{RQ} = 3\mathrm{i} + 6\mathrm{j} + 6\mathrm{k}$

$\overrightarrow{n}$ of plane $= (\mathrm{i} + \mathrm{j} + 2\mathrm{k}) \times (3\mathrm{i} + 6\mathrm{j} + 6\mathrm{k})$

$$= \begin{vmatrix} i & j & k \\ 1 & 1 & 2 \\ 3 & 6 & 6 \end{vmatrix}$$

$$= -6\mathrm{i} - \mathrm{j}(0) + \mathrm{k}(3)$$

$$= -6\mathrm{i} + 3\mathrm{k}$$

Equation of plane

$-6(\mathrm{x} - 0) + 3(\mathrm{z} + 1) = 0$

$\Rightarrow -6\mathrm{x} + 3\mathrm{z} + 3 = 0$

$\Rightarrow -2\mathrm{x} + \mathrm{z} + 1 = 0$

Checking options,option 2 lies on this plane

Hence answer is option 2

Q. 15 Let $\mathrm{A}, \mathrm{B}, \mathrm{C}$ be three points whose position vectors respectively are

$\vec{a} = \hat{i} + 4\hat{j} + 3\hat{k}$
$\vec{b} = 2\hat{i} + \alpha\hat{j} + 4\hat{k}, \alpha \in \mathbb{R}$
$\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$

If α is the smallest positive integer for which $\vec{a}, \vec{b}, \vec{c}$ are non-collinear, then the length of the median in $\Delta\mathrm{ABC}$, through A is:

Option 1:
$\frac{\sqrt{82}}{2}$

Option 2:
$\frac{\sqrt{62}}{2}$

Option 3:
$\frac{\sqrt{69}}{2}$

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Option 4:

$\frac{\sqrt{66}}{2}$

Correct Answer:

$\frac{\sqrt{82}}{2}$

Solution:

$\text{Let A}(\text{i}+4\text{j}+3\text{k}), \text{B}(2\text{i}+\alpha\text{j}+4\text{k}) \text{ and C}(3\text{i}-2\text{j}+5\text{k})$

If A, B and C are collinear, then

$\overrightarrow{AC} = \lambda(\overrightarrow{BC}) \quad \text{for any } \lambda \in \text{R}$

$\Rightarrow (2\text{i}-6\text{j}+2\text{k}) = \lambda(\text{i}+(-2-\alpha)\text{j}+\text{k})$

$\Rightarrow 2 = \lambda, -6 = \lambda(-2-\alpha), \quad 2 = \lambda$

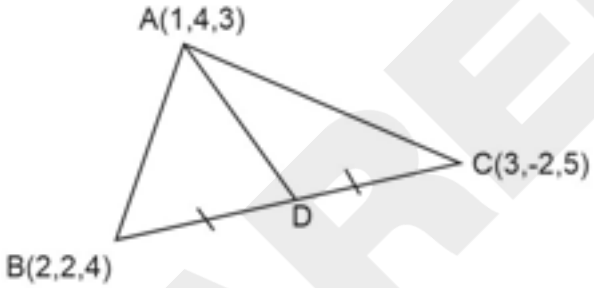
$\Rightarrow -6 = 2(-2-\alpha)$

$\Rightarrow -3 = -2-\alpha$

$\Rightarrow \alpha = 1$

$\therefore$ For $\alpha \neq 1, \text{A}, \text{B}, \text{C}$ are non-collinear

Smallest positive integral $\alpha = 2$



D using mid-point formula is

$\left(\frac{5}{2}, 0, \frac{9}{2}\right)$

$\text{AD} = \sqrt{\left(\frac{5}{2}-1\right)^2 + (0-4)^2 + \left(\frac{9}{2}-3\right)^2}$

$= \sqrt{\frac{9}{4}+16+\frac{9}{4}}$

$= \sqrt{\frac{9+64+9}{4}}$

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$= \sqrt{\frac{82}{2}}$

Hence answer is option 1

Q. 16 The probability that a relation R from $\{\mathrm{x}, \mathrm{y}\}$ to $\{\mathrm{x}, \mathrm{y}\}$ is both symmetric and transitive, is equal to

Option 1:
$\frac{5}{16}$

Option 2:
$\frac{9}{16}$

Option 3:
$\frac{11}{16}$

Option 4:
$\frac{13}{16}$

Correct Answer:
$\frac{5}{16}$

Solution:
The possible relations that are both symmetric and transitive are

$\phi, \{(\mathrm{x}, \mathrm{x})\}, \{(\mathrm{y}, \mathrm{y})\}, \{(\mathrm{x}, \mathrm{y}), (\mathrm{y}, \mathrm{x})\}$

$\{(\mathrm{x}, \mathrm{x}), (\mathrm{y}, \mathrm{y}), (\mathrm{x}, \mathrm{y}), (\mathrm{y}, \mathrm{x})\}$

$\text{Total number of relations } = 2^{2\cdot 2} = 16$

$\therefore \quad \text{Probability } = \frac{5}{16}$

Hence answer is option 1

Q. 17 The number of values of $\mathrm{a} \in \mathbb{N}$ such that the variance of $3, 7, 12, \mathrm{a}, 43 - \mathrm{a}$ is a natural number is :

Option 1:
0

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Option 2:
2

Option 3:
5

Option 4:
infinite

Correct Answer:
0

Solution:

$$\text{Mean } = \frac{3+7+12+a+(43-a)}{5}$$

$$= 13$$

$$\text{Variance} = \frac{(3-13)^2+(7-13)^2+(12-13)^2+(a-13)^2+(43-a-13)^2}{5}$$

$$= \frac{100+36+1+a^2+169-26a+900+a^2-60a}{5}$$

$$= \frac{1206+2a^2-86a}{5}$$

$$= \frac{2\left(a^2-43a+603\right)}{5}$$

Now we can check that using any $a \in N$, unit digit of $a^2-43a+603$ can never be 0 or 5

So numerator is not a multiple of 5 and variance, thus can never be a natural number.

Hence answer is option 1.

Q. 18 From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is $60°$. The pole sutends an angle $30°$ at the top of the tower. Then the height of the tower is :

Option 1:
$15\sqrt{3}$

Option 2:
$20\sqrt{3}$

Option 3:
$20+10\sqrt{3}$

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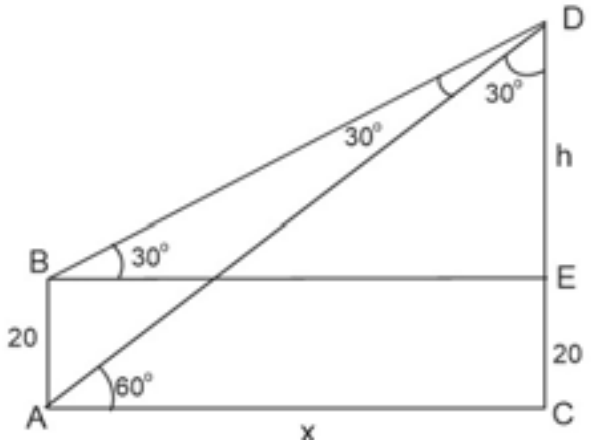
Option 4:

30

Correct Answer:

30

Solution:



Let AB be the pole and CD be the tower

$\text{As } \angle DAC = 60^\circ \Rightarrow \angle ADC = 30^\circ$

$\text{As } \angle BDE = 60^\circ \Rightarrow \angle DBE = 30^\circ$

$\text{In } \triangle DCA, \frac{(h+20)}{x} = \tan 60^\circ$

$\Rightarrow x = \frac{h+20}{\sqrt{3}}$

$\text{In} \triangle DEB, \quad \frac{h}{x} = \tan 30^\circ$

$\Rightarrow x = \sqrt{3}h$

Equating x

$\frac{h+20}{\sqrt{3}} = \sqrt{3}h$

$h + 20 = 3h$

$2h = 20$

$h = 10$

$\therefore \quad CD = h + 20 = 30 \text{ m}$

Hence the answer is option 4

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Q. 19 Negation of the Boolean statement $(p \vee q) \Rightarrow ((\sim r) \vee p)$ is equivalent to

Option 1:
$p \wedge (\sim q) \wedge r$

Option 2:
$(\sim p) \wedge (\sim q) \wedge r$

Option 3:
$(\sim p) \wedge q \wedge r$

Option 4:
$p \wedge q \wedge (\sim r)$

Correct Answer:
$(\sim p) \wedge q \wedge r$

Solution:
$(p \vee q) \Rightarrow ((\sim r) \vee p)$

As negation of $p \rightarrow q$ is $p \wedge \sim r$

$\therefore$ Negation of given statement is

$(p \vee q) \wedge \sim ((\sim r) \vee p)$

$\equiv (p \vee q) \wedge (r \wedge \sim p)$

Using Venn diagram techique

$\equiv (\sim p) \wedge q \wedge r$

Hence answer is option 3

Q. 20 Let $n \geq 5$ be an integer. If $9^n - 8n - 1 = 64\alpha$ and $6^n - 5n - 1 = 25\beta$, then $\alpha - \beta$ is equal to

Option 1:
$1 + {}^nC_2(8-5) + {}^nC_3\left(8^2 - 5^2\right) + \ldots + {}^nC_n\left(8^{n-1} - 5^{n-1}\right)$

Option 2:
$1 + {}^nC_3(8-5) + {}^nC_4\left(8^2 - 5^2\right) + \ldots + {}^nC_n\left(8^{n-2} - 5^{n-2}\right)$

Option 3:
${}^nC_3(8-5) + {}^nC_4\left(8^2 - 5^2\right) + \ldots + {}^nC_n\left(8^{n-2} - 5^{n-2}\right)$

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Option 4:

$^{n}C_{4}(8-5)+{}^{n}C_{5}\left(8^{2}-5^{2}\right)+\ldots+{}^{n}C_{n}\left(8^{n-3}-5^{n-3}\right)$

Correct Answer:

$^{n}C_{3}(8-5)+{}^{n}C_{4}\left(8^{2}-5^{2}\right)+\ldots+{}^{n}C_{n}\left(8^{n-2}-5^{n-2}\right)$

Solution:

$64\alpha=9^{n}-8n-1$

$64\alpha=(1+8)^{n}-8n-1$

$64\alpha=1+8n+{}^{n}C_{2}8^{2}+{}^{n}C_{3}8^{3}+\ldots\ldots+{}^{n}C_{n}8^{n}-8n-1$

$64\alpha={}^{n}C_{2}\cdot 8^{2}+{}^{n}C_{3}\cdot 8^{3}+\ldots+{}^{n}C_{n}8^{n}$

$\alpha={}^{n}C_{2}+{}^{n}C_{3}\cdot 8+\ldots+{}^{n}C_{n}\cdot 8^{n-2}$

And

$25\beta=6^{n}-5n-1$

$=(1+5)^{n}-5n-1\quad=1+5n+{}^{n}C_{2}5^{2}+{}^{n}C_{3}5^{3}+\cdots\cdots+{}^{n}C_{n}5^{n}-5n-1$

$={}^{n}C_{2}\cdot 5^{2}+{}^{n}C_{3}5^{3}+\ldots+{}^{n}C_{n}\cdot 5^{n}$

$\Rightarrow\beta={}^{n}C_{2}+{}^{n}C_{3}5+{}^{n}C_{4}5^{2}+\ldots+{}^{n}C_{n}5^{n-2}$

$\therefore\alpha-\beta={}^{n}C_{3}(8-5)+{}^{n}C_{4}\left(8^{2}-5^{2}\right)+\cdots+{}^{n}C_{n}\left(8^{n-2}-5^{n-2}\right)$

Hence answer is option 3

Q. 21 Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k},\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}+(\vec{b}\times\vec{c})=\overrightarrow{0}$ and $\vec{b}\cdot\vec{c}=5$. Then, the value of $3(\vec{c}\cdot\vec{a})$ is equal to _____.

Correct Answer:

0

Solution:

$\vec{a}+(\vec{b}\times\vec{c})=\overrightarrow{0}$

$\Rightarrow\vec{b}\times\vec{c}=-\vec{a}$

$\Rightarrow\vec{a}\perp\vec{c}$

$\Rightarrow\vec{a}\cdot\vec{c}=0$

$\Rightarrow 3(\vec{a}\cdot\vec{c})=0$

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Hence answer is 0

Q. 22 Let $y = y(x), x > 1$ be the solution of the diffrential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha + 1}{\beta e^\alpha}$ then the value of $\alpha + \beta$ is equal to

Correct Answer:
14

Solution:

$$\frac{dy}{dx} + \left(\frac{2x}{x-1}\right)y = \frac{1}{(x-1)^2}$$

$$I \cdot F \cdot = e^{\int \frac{2x}{x-1}dx}$$

$$\text{Now } \int \frac{2x}{x-1}dx$$

$$= 2\left[\int \frac{x-1}{x-1}dx + \int \frac{1}{x-1}dx\right]$$

$$= 2[x + \log(x-1)]$$

$$\therefore I.F = e^{2x+\log(x-1)^2}$$

$$= (x-1)^2 \cdot e^{2x}$$

$$\therefore y \cdot (x-1)^2 e^{2x} = \int \frac{1}{(x-1)^2} \cdot (x-1)^2 e^{2x} dx$$

$$\Rightarrow y(x-1)^2 e^{2x} = \frac{e^{2x}}{2} + c$$

$$\Rightarrow y = \frac{1}{2(x-1)^2} + \frac{c}{e^{2x} \cdot (x-1)^2}$$

$$\text{As } y(2) = \frac{1+e^4}{2e^4}$$

$$\Rightarrow \frac{1+e^4}{2e^4} = \frac{1}{2} + \frac{c}{e^4}$$

$$\Rightarrow \frac{1+e^4}{2e^4} = \frac{e^4+2c}{2e^4}$$

$$\Rightarrow c = \frac{1}{2}$$

Putting $x = 3$

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$$y(3) = \frac{1}{2 \cdot 4} + \frac{1}{2e^6 \cdot (4)}$$

$$= \frac{e^6 + 1}{8e^6}$$

$\therefore \quad \alpha = 6, \ \beta = 8 \Rightarrow \alpha + \beta = 14$

Hence answer is 14

Q. 23 Let $3, 6, 9, 12, \ldots$ upto 78 terms and $5, 9, 13, 17, \ldots$ upto 59 terms be two series. Then the sum of the terms common to both the series is equal to _____

Correct Answer:
2223

Solution:
Both given series are $AP's$

$3, 6, 9, 12, \ldots$

$a = 3, d = 3, n = 78.$

$5, 9, 13, 17, \ldots$

$a' = 5, d' = 4, n' = 59$

Common terms to $2AP's.$ also form an AP

First term of common AP, $A = 9$

Common Difference $D = lcm(d, d')$

$= lcm(3, 4)$

$= 12$

Let Nth term of this common AP be its last term

Last term of first $AP = 3 + (78 - 1)3$

$= 234$

Last term of second $AP = 5 + (59 - 1)4$

$= 237$

$\therefore \quad A + (N - 1)D \leqslant 234$

$9 + (N - 1)12 \leqslant 234$

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$$N - 1 \leq \frac{225}{12}$$

$$\Rightarrow \quad N \leqslant 19.75$$

$$\therefore \quad N = 19$$

Sum of common AP

$$= \frac{19}{2}[2 \cdot 9 + (19-1)12]$$

$$= 19[9 + 18 * 6]$$

$$= 2223$$

Hence answer is 2223

Q. 24 The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0, 10)$ is _____

Correct Answer:
4

Solution:

$$\sin x = \cos^2 x$$

$$\Rightarrow \sin x = 1 - \sin^2 x$$

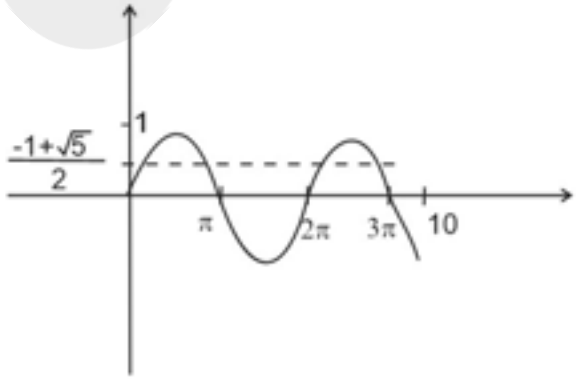
$$\Rightarrow \sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$= \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}$$

As only $-1 \leq \frac{-1+\sqrt{5}}{2} \leq 1$

$$\therefore \quad \sin x = \frac{-1+\sqrt{5}}{2}$$



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So 4 solutions

Hence answer is 4

Q. 25 For real numbers $a, b(a > b > 0)$ let

$$\text{Area}\left\{(x, y) : x^2 + y^2 \leq a^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \geq 1\right\} = 30\pi$$

and

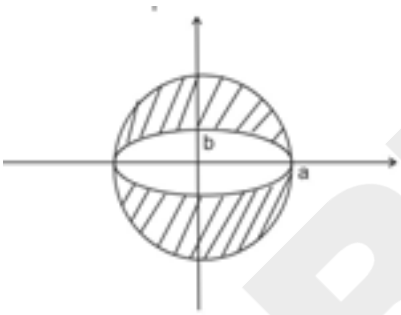
$$\text{Area}\left\{(x, y) : x^2 + y^2 \geq b^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\right\} = 18\pi$$

Then the value of $(a - b)^2$ is equal to _____.

Correct Answer:
12

Solution:
$x^2 + y^2 \leq a$ represents region inside circle $x^2 + y^2 = a^2$ and

$\frac{x^2}{a^2} + \frac{y^2}{b^2} \geqslant 1$ represents region outside the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



First area $= 30\pi =$ area of circle - area of ellipse

$= \pi a^2 - \pi ab$

$\Rightarrow 30\pi = \pi a^2 - \pi ab$

$\Rightarrow a^2 - ab = 30$ .....(i)

Similarly second area $= 18\pi = \pi ab - \pi b^2$

$\Rightarrow \quad ab - b^2 = 18$ .....(ii)

Subtracting (i) - (ii)

$\Rightarrow a^2 - ab - ab + b^2 = 30 - 18$

$\Rightarrow (a - b)^2 = 12$

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Hence answer is 12

Q. 26 Let f and g be twice differentiable even functions on $(-2,2)$ such that $f\left(\frac{1}{4}\right)=0, f\left(\frac{1}{2}\right)=0, f(1)=1$ and $g\left(\frac{3}{4}\right)=0, g(1)=2$. Then, the minimum number of solutions of $f(x)g''(x)+f'(x)g'(x)=0$ in $(-2,2)$ is equal to _____.

Correct Answer:
4

Solution:
Let $h(x)=f(x)\cdot g'(x)$

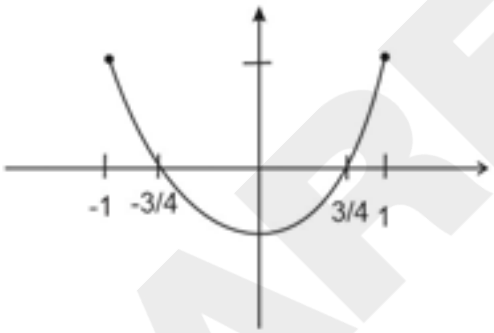
So $h(x)$ is continuous and differentiable.

Now as $f\left(\frac{1}{4}\right)=f\left(\frac{1}{2}\right)=0$ and $f(x)$ is even function, so $f\left(\frac{-1}{4}\right)=f\left(\frac{-1}{2}\right)=0$ which means $h\left(-\frac{1}{2}\right)=h\left(-\frac{1}{4}\right)=h\left(\frac{1}{4}\right)=h\left(\frac{1}{2}\right)=0$

So $-\frac{1}{2},-\frac{1}{4},\frac{1}{4},\frac{1}{2}$ are 4 roots of $h(x)$

Also $g'(0)=0$ ($g(x)$ is even diffrential function $\Rightarrow h(0)=0$

From minimum roots of $h(x), g(x)$ may look something like this



$\therefore h(x)$ has minimum 5 roots

$-\frac{1}{2},-\frac{1}{4},0,\frac{1}{4},\frac{1}{2}$

So, $h'(x)=f'g'+fg''$ has at least 4 roots

Hence answer is 4

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Q. 27 Let the coefficient of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}}-\frac{1}{x^{\frac{1}{5}}}\right)^{15}, x>0$, be m and n respectivly. If r is a positive integer such that $mn^2 = {}^{15}C_r.2^r$, then the value of r is equal to _____.

Correct Answer:
5

Solution:

$$\text{General term} = {}^{15}C_r\left(2x^{\frac{1}{5}}\right)^{15-r}\left(-\frac{1}{x^{\frac{1}{5}}}\right)^r$$

$$= {}^{15}C_r \cdot 2^{15-r}(-1)^r \cdot x^{3-\frac{r}{5}-\frac{r}{5}}$$

For coefficient of x^{-1}

$$3-\frac{2r}{5}=-1 \Rightarrow \frac{2r}{5}=4 \Rightarrow r=10$$

For coefficient of x^{-3}

$$3-\frac{2r}{5}=-3 \Rightarrow \frac{2r}{5}=6 \Rightarrow r=15$$

$\therefore m = {}^{15}C_{10} \cdot 2^5, n=-1$

$\therefore mn^2 = {}^{15}C_{10} \cdot 2^5 = {}^{15}C_5 \cdot 2^5$

$\therefore r = 5$

Hence answer is 5

Q. 28 The total number of four digit numbers such that each of first three digits is divisible by the last digit, is equal to _____.

Correct Answer:
1086

Solution:

$abcd$

1. When $d=1$, a can take any value from 1 to 9 and b and c can take any value from 0 to 9

$\therefore \quad 9\times10\times10=900$ such numbers

2. $d=2, a$ can be $2,4,6,8, b$ and c can be $0,2,4,6,8$

$\therefore 4\times5\times5=100$

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3. $\mathrm{d}=3, \mathrm{a}=3/6/9, \ \mathrm{b} \ \text{and} \ \mathrm{c}=0/3/6/9$

$\therefore 3 \times 4 \times 4=48$

4. $\mathrm{d}=4, \quad \mathrm{a}=4/8, \mathrm{b} \text{ and } \mathrm{c}=0/4/8$

$\therefore 2 \times 3 \times 3=18$

5. $\mathrm{d}=5, \mathrm{a}=5, \mathrm{b} \text{ and } \mathrm{c}=5/0$

$\therefore \mathrm{d}=1 \times 2 \times 2=4$

6. $\mathrm{d}=6, \mathrm{a}=6, \quad \mathrm{b} \text{ and } \mathrm{c}=6/0$

$1 \times 2 \times 2=4$

Similarly for $\mathrm{d}=7,8,9 \Rightarrow \ \text{numbers} \ =4,4,4$

$\text{Total}=900+100+48+18+4+4+4+4+4=1086$

Hence answer is 1086

Q. 29

Let $\mathrm{M}=\begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number and $\mathrm{N}=\sum_{\mathrm{k}=1}^{49} \mathrm{M}^{2\mathrm{k}}$. If $\left(\mathrm{I}-\mathrm{M}^2\right) \mathrm{N}=-2\mathrm{I}$, then the positive integral value of α is _____.

Correct Answer:

1

Solution:

$\mathrm{M}=\begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$

$\mathrm{M}^2=\begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}\begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$

$=\begin{bmatrix} -\alpha^2 & 0 \\ 0 & -\alpha^2 \end{bmatrix}$

$=-\alpha^2 \cdot \mathrm{I}$

$\therefore \mathrm{M}^4=\mathrm{M}^2 \cdot \mathrm{M}^2=\alpha^4 \mathrm{I}$

$\mathrm{M}^6=\mathrm{M}^4 \cdot \mathrm{M}^2=-\alpha^6 \mathrm{I}$

.....

.....

$\therefore \mathrm{N}=\mathrm{M}^2+\mathrm{M}^4+\mathrm{M}^6+--+\mathrm{M}^{98}$

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$= -\alpha^2 I + \alpha^4 I - \alpha^6 I \ldots\ldots 49 \text{ terms}$

$= \left(-\alpha^2 + \alpha^4 - \alpha^6 - \ldots\right) I$

$= -\alpha^2 \left(\frac{(-\alpha^2)^{49} - 1}{-\alpha^2 - 1}\right) I$

Given $\left(I - M^2\right) N = -2I$

$\Rightarrow \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -\alpha^2 & 0 \\ 0 & -\alpha^2 \end{bmatrix}\right) \left(\frac{\alpha^2 (-\alpha^{98} - 1)}{(\alpha^2 + 1)}\right) I = -2I$

$\Rightarrow \left(1 + \alpha^2\right) I \cdot \frac{\alpha^2 (-\alpha^{98} - 1)}{(\alpha^2 + 1)} = -2I.$

$\Rightarrow \quad \alpha^2 \left(-\alpha^{98} - 1\right) = -2$

Only $\alpha = 1$ satisfies it

Hence answer is 1

Q. 30 Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

Correct Answer:
18

Solution:
$f(g(x)) = 8x^2 - 2x$

$g(f(x)) = 4x^2 + 6x + 1$

Let $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$

$f(g(x)) = a(dx + e)^2 + b(dx + e) + c$

$= ad^2x^2 + 2adex + ae^2 + bdx + be + c$

$= ad^2x^2 + (2ade + bd)x + \left(ae^2 + be + c\right)$

$g(f(x)) = d\left(ax^2 + bx + c\right) + e$

$= adx^2 + bdx + (cd + e)$

Comparing

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$ad = 4, ad^2 = 8$

Dividing $d = 2 \Rightarrow a = 2$

And $bd = 6 \Rightarrow b = 3$

And $2ade + bd = -2$

$8e + 6 = -2 \Rightarrow e = -1$

And $cd + e = 1 \Rightarrow 2c - 1 = 1 \Rightarrow c = 1$

$\therefore f(x) = 2x^2 + 3x + 1, g(x) = 2x - 1$

$\therefore f(2) + g(2) = (8 + 6 + 1) + (4 - 1)$

$= 15 + 3$

$= 18$

Hence answer is 18

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